

ADULT AUDIOLOGIC REHABILITATION

THIRD EDITION

JOSEPH J. MONTANO
JACLYN B. SPITZER

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FOREWORD TO THE THIRD EDITION

The third edition of the Montano and Spitzer text on auditory rehabilitation comes at an opportune time. Many recent or current developments have the potential to affect hearing health care. These include such things as self-administered hearing tests, hearing aid self-fitting software, over-the-counter hearing aids, and telemedicine. Some may see these developments as a threat to the profession of audiology, but others are in the process of adapting to, and taking advantage of, new and emerging possibilities. What a great time for introspection and planning! And what a great opportunity for renewed and increased attention to aspects of audiologic management that go beyond sensory management! The provision of the best possible hearing capacity is obviously an essential first step. But the concept of automatic “trickle down” to optimal function, activity, participation, and quality

of life is untenable. Hearing aids do not restore normal hearing, nor do they alleviate the contributions of other factors such as loss of visual acuity, mild cognitive impairment, personality, adaptability, self-efficacy, lifestyle, and aging. This raises the issue of the contrast between a client-centered and a hearing-centered approach—helping people who have a hearing loss rather than treating hearing loss in people. Many of the authors represented here have been making these points for a long time. Others are providing insights to those changes that have the potential to impact audiological practice. If and when the profession seeks to identify itself as the provider of holistic hearing health care (and establishes a way to fund the expanded services), this comprehensive and forward-looking text will serve as an indispensable guide.

—Arthur Boothroyd, PhD

FOREWORD TO THE SECOND EDITION

Audiologic rehabilitation involves several stages: Identification and evaluation of hearing loss, treatment, and post-treatment care, emphasizing the point that audiologic rehabilitation involves more than just the final stage. It is unfortunate that the term *audiologic rehabilitation* is often misinterpreted as referring only to this final stage. It is important to bear in mind the broad scope of audiologic rehabilitation and that the goal of each stage in the process is in fact improved communication. The benefits of audiologic rehabilitation can be maximized by means of early identification, accurate evaluation, effective treatment, and efficient post-treatment rehabilitation.

Whereas identification, evaluation, and treatment are important stages in audiologic rehabilitation, they are only part of the process. Many audiologists have focused their efforts on the early stages of audiologic rehabilitation with the result that the post-treatment stage is neglected to a large extent. The neglect is significantly greater for adults than for children. This may be because of the difficulty and effort required for effective post-treatment rehabilitation, or because the early stages of the rehabilitation process show improvements more rapidly with less effort; that is, more bang for the buck, or simply more bucks. Whatever the reason, the post-treatment stage of the rehabilitation process has not received the attention it deserves. Rehabilitation that ends at the conclusion of the treatment stage is rehabilitation incomplete.

The previous foreword by my esteemed colleague, Mark Ross, describes the importance of and commitment to post-treatment rehabilitation in a previous, well-funded era. In the years that followed, the inevitable competition for diminishing resources resulted in the post-treatment stage of audiologic rehabilitation receiving less attention and less effort than it deserves. Fortunately, there is a cadre of dedicated researchers and clinicians who have maintained the emphasis on rehabilitation in all stages of the rehabilitation process and have carried the field forward, even with limited resources. This volume by Dr. Joseph Montano and Dr. Jaclyn Spitzer, now in its second edition, stands out as a beacon in a rough sea of competing and often misguided priorities. The con-

tributors to the volume are leading authorities who have pioneered new approaches in the field. Many of the contributors are from countries other than the United States, thereby providing an international perspective to the volume.

The breadth of coverage is substantial, as reflected by the large number of chapters on the various topics relevant to adult audiologic rehabilitation. The first section of the volume begins, sensibly, with a brief outline of the book's goals and scope, followed by an historical review that places adult audiologic rehabilitation in perspective. The next several chapters provide important background material on the nature of adult audiologic rehabilitation, the classification of functioning and its relevance to the rehabilitation process, and the psychological effects of social stigma in the target population. The last-mentioned chapter is particularly important to understand the multidimensional nature of the rehabilitation process. It is not merely hearing loss. It is hearing loss compounded by the sequelae of hearing loss.

Assessment and verification are particularly important issues and are dealt with in the second section of the volume. Self-assessment is a critical aspect of the evaluation process, and this section contains several chapters on different aspects of this broad area. Although self-assessment is inherently subjective, the client's perception of the efficacy of the rehabilitation program is a major factor affecting the success or failure of the program. In this context, perception is the ultimate reality. The last chapter in this section deals with the particularly important issue of verification.

The third section of the volume, The Rehabilitative Toolbox: Therapeutic Management, is by far the largest and provides excellent coverage of the substantial armamentarium of tools available to the rehabilitative audiologist. The chapters in this section provide valuable information on the merits of the various rehabilitative techniques and the considerations that need to be taken into account in their implementation. There is no single best technique. An understanding of the capabilities and shortcomings of each technique is needed to select a technique, or

combination of techniques, that is most appropriate for each client. It is thus valuable to have broad coverage of the large array of rehabilitative techniques in a single volume. This is not a cookbook. Each chapter deserves careful reading to appreciate the capabilities of each technique and its potential value for each client.

The last section of the volume deals with special issues and expanding the scope of audiologic rehabilitation. The opening chapter in this section deals with evidence-based research. This is a rapidly growing area of research with important implications for funding and the allotment of resources. Evidence-based research with positive results will do much to secure the viability of audiologic rehabilitation in a future of increased competition for limited resources. The next chapter in this section reflects the remarkable advances in the field in that audiologic rehabilitation has expanded to include appreciation of music in addition to speech understanding for adults with cochlear prostheses. The field has also expanded to include a broader view of vocational issues as dis-

cussed in the chapter dealing with this topic. The chapters on older adults and on auditory and cognitive processing reflect the growing recognition of the importance of age-related issues in audiologic rehabilitation and the importance of cognitive processing in addressing age-related auditory deficits. Tinnitus is a major problem with no simple solution, though progress is being made on treatment. The chapter on this topic provides insightful coverage of this problem and current rehabilitative techniques. The last chapter, logically, deals with current and future research needs.

This is an impressive volume containing a wealth of information. The range of topics and their relevance to practical issues in adult audiologic rehabilitation is impressive. The chapters deserve to be read and reread to gain an in-depth appreciation of the multidimensional considerations to be taken into account to maximize the efficacy of adult audiologic rehabilitation. Read, understand, and make good use of the valuable information contained in this substantive volume.

—Harry Levitt

FOREWORD TO THE FIRST EDITION

I had my first experience with the concept and practice of audiologic rehabilitation (AR) when I was admitted as a patient to the AR program at Walter Reed Army Hospital in January 1952. Essentially, this was the type of program from which the profession of audiology would later emerge (see McCarthy, this volume). For the 2 months I was there, I was exposed to the best clinical practices that existed at the time. Classes were conducted by former teachers of the deaf, speech correctionists, and lipreading teachers (the term *speechreading* had not yet been coined). For the most part, the lessons consisted of various speechreading and visual training exercises, supplemented by some auditory training sessions (following a classic Carhart approach). These two activities basically defined AR at the time. Also included in the program were occasional didactic lectures on various topics (e.g., the audiogram, anatomy and physiology of the auditory system, etc.). Appointments for hearing aid fittings and follow-ups were spaced throughout the 2-month period. The hearing aids provided to us were monaural body-worn vacuum tube aids, and we were told we were lucky to have them. Patients just a few years earlier were issued a duo-pack hearing aid, which required separate packs for the battery and the amplifier, with a rather large wire connecting the two.

In retrospect, although we couldn't really appreciate it at the time, those of us who were able to participate in that program were indeed very lucky to be where we were, receiving the kind of care we did. I don't think any of us fully understood the reality of what a hearing loss would mean in our lives. It was very easy to underestimate the full and eventual impact of a hearing loss. At any rate, in that setting and at that time, the issues facing someone with an "invisible" hearing loss paled in comparison to what we saw of the men with combat injuries at the main hospital. Complaints and self-pity were simply not acceptable reactions. While we didn't choose to attend the program, being there turned out to be a turning point, at least in my life.

In the company of others, I learned to accept myself and the reality of my hearing loss, perhaps the most significant goal in any AR program. While

this was not an explicit objective of the program, it arose naturally as a consequence of being somewhat sequestered for 2 months with a group of young men with similar problems. Later, as a professional audiologist, I thought of that program as a kind of AR Camelot—the ultimate model to which we could aspire but never really attain. Still, the question arises: Are there any lessons we could take from our early romance in "Camelot" and apply them in this day and age? I think there are.

The first is that the hearing aid selection procedures, primitive as they were, were viewed as an ongoing process (as I recall, one appointment a week was scheduled over an 8-week period) and completely integrated with the rest of the AR program. Questions about hearing aids, difficult listening situations, speechreading issues, and so forth could arise—formally or informally—throughout the day. From a conceptual point of view, hearing aids were not separated from all the rest of the AR activities that took place. This is somewhat at variance with how AR is now generally practiced. Currently, hearing aids are selected and several follow-up appointments are scheduled. But the need for, and provision of, any other type of AR service is completely happenstance—sometime yes, sometimes no, sometimes this, and sometimes that. Not so in Camelot; all was integrated into a single curriculum.

The second important lesson I think we can take with us is that the group experience was perhaps the most valuable benefit of the program. I doubt this was an explicit goal of the program. It was, rather, a fallout benefit that proved to be profoundly effective despite being inadvertent. Just the fact of being part of a group, of sharing our experiences and concerns with other young men in the same situation, helped us to accept the reality of our condition. We would joke with one another about "being on the air," to the point where the daily use of a hearing aid was common—and expected. As audiologists, we are now keenly aware of the advantages of group management; the fact that I can still feel its impact some 56 years later is a testament to the power, and effectiveness, of a group program.

A third observation regarding the Walter Reed program is that it wasn't voluntary. We were simply transferred there, with no option to refuse. But I think it worked. Many men who in other settings would have to be dragged kicking and screaming into a therapy room, or more realistically simply wouldn't go, were enrolled and then benefited from the inclusion in spite of themselves. Clearly, this is not exactly a formula that can be applied in any other kind of situation! We don't draft people into an AR program. It does suggest, however, that even people who are reluctant to engage in a therapy program can be helped, as long as they can somehow be convinced or cajoled to participate.

I do believe that this can be somewhat achieved if a short-term group AR program were considered to be a routine component of the hearing aid selection process. In other words, just as a client's hearing status is expected to be evaluated by an audiologist prior to the hearing aid fitting, so too should such a program be instituted subsequent to a hearing aid selection. People would be free to ignore the recommendation, of course—we are not a totalitarian society—but, for example, how many people ignore their orthopedist's suggestion for a course of physical therapy after a surgical procedure? I think we can agree, not very many. Clearly, the perception of need is completely different in these two situations, which I believe is due to a society that trivializes the possible consequences of a hearing loss and underestimates what can be achieved with an AR program. Even a short-term, post-hearing-aid-selection AR program can pay big dividends for many people with hearing loss.

What was provided in the Walter Reed AR program has not been, and could not be, replicated in its entirety anywhere else. At the time it was developed, during World War II, nothing and no expense was too much for "the boys." In retrospect, however, it now seems like overkill; it is likely that the same results could have been achieved with a somewhat truncated program, or one constituted a bit differently. But we have no way of knowing now. We have no objective evidence of the program's efficacy (though I would gladly provide my personal testimonial regarding its effectiveness). We didn't have the scholarly insights and information on AR that exist currently. During that era, AR meant being fit monaurally with a body-worn, vacuum tube hearing aid and attending speechreading and occasional auditory training classes—period. Now, on the other hand, we have a comprehensive body of information applicable and relevant to the entire AR process. And, it seems, just about all of it is included in this book.

When Dr. Montano and Dr. Spitzer forwarded the detailed outline to me, my initial response was, "Wow!" I jumped at the opportunity they gave me to write the foreword. I have spent my entire professional life dealing one way or another with AR, but until I saw the outline of topics contained in this book, I never really appreciated just how much audiology has changed and developed since the early days. The authors selected to write the chapters are a compilation of the best and the brightest that the profession has to offer. The book is divided into four logical sections, each of which deals comprehensively with a different aspect of AR.

The first section of the book lays the groundwork. In this introductory portion, AR itself is precisely defined (we will see that it involves more than is at first apparent) and placed in an historical framework. Perhaps because of my own history, I find this a very appealing topic; we really don't know where we are or where we're going unless we can appreciate where we've been. This chapter does that for us. It provides a frame of reference we can use when we examine any efforts in this area, our own as well as those of others.

The broad area of assessment is covered in the second section. One convincing indicator of how far the profession has progressed is that three of the topics in this section—self-assessment, quality of life, and implantable technologies—did not even exist 50 years ago. The fact that hearing-impaired people could, and should, personally rate the communicative and social effects of their own hearing loss seems obvious to us now, but not so years ago. Now, on the other hand, self-assessment scales are a component in every "best practice" recommendation.

Insofar as quality of life is concerned, of course people years ago were aware that a hearing loss could affect it, but the idea that quality of life considerations could and should be formally assessed never seemed to arise. Now, as the profession finds it necessary to justify the expense of hearing aids and therapeutic procedures to third-party payers, the positive impact of our management efforts have to be quantitatively demonstrated. The administrators who manage the purse strings will not be satisfied with only our personal assurance that some procedure or device is helpful. They want to see the evidence before any financial outlay is approved.

Also included in this second section is a chapter on hearing aids, an area that is clearly a central component of any AR effort. We've come a long way since the original Carhart procedure for the comparative evaluation of hearing aids, abetted by the ubiquitous

question, “How does that sound?” As this chapter indicates, the vast improvements in technology that have occurred have been accompanied by concurrent developments in assessment techniques. In other words, it’s not enough to point to some new and impressive technology as an indicator of progress; it is also necessary to corroborate its merits in behavioral terms. Not every technical advance is accompanied by listening improvements.

The third section constitutes the bulk of the book. In it, we see not only the traditional concepts of speechreading and auditory training upgraded and cloaked in modern dress, but also full coverage of the relevant psychosocial issues (e.g., counseling, stigma, the group process, consumer advocacy) and specific therapeutic techniques (e.g., music therapy, repair strategies). The inclusion of these areas is another reminder of how our approach to AR has changed since the early days. The chapter on assistive listening technology reminds us how much communication can be enhanced by the direct transmission of a talker’s voice to a hard of hearing listener. I know that whenever I use an assistive listening device, I am reminded anew of the great help that these systems can provide someone with a hearing loss—and how much they are underemployed by people with hearing loss.

A review of special issues in AR is provided for us in the last section. New information and insights have given rise to additional areas of professional responsibility. One of these is the manifestation, evaluation and management of people with central auditory processing (CAP) problems. The case is made that this is a rehabilitation issue that the profession of audiology must address. The evidence presented in this section will help lay the necessary evidential and conceptual groundwork for the involvement of audiologists in this area. Finally, lest we forget that human beings are our core concern, the rehabilitation of older adults is discussed in this section. We are an increasingly aging society, and older people present issues that may, and probably will, differ in some respect from those observed in younger people with hearing loss.

In brief, this book presents an impressive display of the concepts and content areas that now constitute AR. The people selected to write the various chapters are well known in their own countries and internationally. What they have presented is the current state of the art—a compilation of information, insights, practices and concepts that were unheard of when I was a patient, and even later when I started practicing as an audiologist. As a body of knowledge, audiologic rehabilitation has been a growing real-

ity. Still, in spite of all that has changed, we should remind ourselves what has not changed, and that is the impact of a hearing loss upon the life and well-being of the afflicted person. We are still going to see the same reactions from people that we saw years ago, from denial to isolation. And it is still going to take conscientious, caring, and competent clinicians to provide the services, so well documented in these pages, that these people need.

The challenge that now confronts the profession is to employ this vast body of knowledge for the benefit of adults with hearing loss. Knowing more does not mean we are doing more. Certainly knowledge is a prerequisite to action, which this book amply provides us, but somehow this knowledge has to find its way to people. There is still an inadequate public appreciation of the consequences of this “invisible” handicap. Impaired hearing is still more often a subject for so-called “humor” than for empathy and assistance. A public understanding of the potential consequences of a hearing loss—realizing that it is not a joke—is *the* prerequisite for hearing loss to receive the same kind of public support that other disabilities now receive via third-party payments. The kind of public support I envision goes beyond support for various kinds of devices and would include individual, group, and self-administered (possibly home-based) AR therapies. Support groups, too, like the Hearing Loss Association of America (HLAA) can play a significant role in the broader goal of helping people live with a hearing loss.

Ironically, it seems that the latest major development in hearing rehabilitation—cochlear implants—seems to be stimulating a modern resurgence of the traditional AR therapies. There is a recognition that new implant users require more than the device itself. While most of this recent interest seems focused on helping these people to adjust to the new and strange auditory sensations produced by a cochlear implant (i.e., auditory training), there also appears to be an increased appreciation of AR as a concept that potentially applies to all people with hearing loss. In my judgment, the fact that medical profession is now involved with implants, to an extent they never were with hearing aids, has provided additional impetus for follow-up AR therapy. A surgeon “prescribing” a course of therapy (probably conducted by the audiologist or speech-language pathologist) imbues the process with an authority that currently is lacking when a nonphysician makes the same recommendation. Like it or not, it is the reality.

There is one final comment I’d like to make about AR: It can be fun. Plus, it is an area for which

audiologists and speech-language pathologists bear the uncontested, primary professional responsibility. It affords these professionals an opportunity to interact with their clients on a more personal, human level. Now they can switch their focus from the hearing loss to the hearing problem, away from the audio-

gram and to the human being with the hearing loss. It can be a very rewarding, and sometimes surprisingly enjoyable experience. I believe that professionals so involved will find that they now enjoy their work even more than they did before. So jump in . . . the water's fine.

—Mark Ross

INTRODUCTION: GOALS AND TOPICS

Since the publication of the second edition of this text, the audiology landscape has shifted dramatically due to changes in legislation, technology, and practice patterns. Audiologists are now adapting to over-the-counter (OTC) and self-fitting hearing aids in the marketplace, increases in e-technologies, and growing evidence of the benefit of personalized health care. As a result, these changes became the motivation for this third edition.

It became apparent in October 2015, with the release of the report of the President's Council of Advisors on Science and Technology (PCAST, 2015), that a tide of change was coming due to the critical nature of their findings. The report focused on untreated hearing loss in older adults. It acknowledged that this is a substantial national problem in the United States and that hearing loss represents a threat for increased social isolation, dementia, and related problems. In their opinion, the cost and distribution models for hearing aids were barriers to access.

Following the PCAST report, the Food and Drug Administration (FDA) held a hearing (April 21, 2016) on streamlining good manufacturing practices (GMPs) for hearing aids. They were seeking expert input from professionals in hearing health care about self-diagnosis of hearing loss, designating a category for hearing aids for mild to moderate hearing loss, and implications of OTC products. Reports were provided by representatives from such organizations as the American Academy of Audiology, American Speech-Language-Hearing Association, Academy of Doctors of Audiology, American Academy of Otolaryngology/Head and Neck Surgery, International Hearing Society, Hearing Loss Association of America, and others. These sources emphasized that hearing instruments should be provided in a context of patient counseling and aural/audiologic rehabilitation (AR).

In June of the same year, the National Academy of Sciences, Engineering, and Medicine released a publication entitled "Hearing Health Care for Adults: Priorities for Improving Access and Affordability" (National Academy of Sciences, Engineering, and Medicine, 2016). Their recommendations included

advocating improved insurance coverage for hearing aids and, most notably, Medicare coverage for AR, with reimbursement for these services to audiologists.

These reports and recommendations form the backdrop for the Over the Counter Hearing Aid Act of 2017 (as part of the FDA Reauthorization Act of 2017). The latter act defines an OTC product, specifies the intended users (adults over 18 years of age with mild-to-moderate hearing loss), and customization of devices. The law requires that the FDA begin the process of developing regulations for these products. At the time of this publication, specific guidelines for OTC hearing aids are not yet available.

So, what are the implications for audiologists? Some believe there is a risk that safe hearing aid dispensing may come with the loss of protections for the person with hearing loss who needs medical treatment and careful audiologic management. Some fear a disruption of dispensing practices and patterns with increased competition from various nonaudiologic sources. Many audiologists are concerned that potential hearing aid users will attempt to use an OTC device and, experiencing frustration or poor fit, will abandon pursuit of amplification. Yet others embrace the anticipated changes and see an opportunity to provide increased rehabilitative services. These trends force audiologists to consider our role in the provision of hearing health care and to recognize our importance in the delivery of rehabilitative services to manage the consequences of hearing loss. It is from the standpoint of the latter that we opted to develop the third edition of this textbook.

The seeds for the first edition were planted in 2003, when it was learned that Jerry Alpiner and Patti McCarthy were not intending to prepare a revised edition of their classic audiologic rehabilitation (AR) textbook, *Rehabilitative Audiology: Children and Adults*. The Alpiner, then subsequent Alpiner/McCarthy, textbook had been a staple of AR graduate education in audiology and speech-language pathology since the 1970s. Its absence would indeed create a void in the education of audiologists. Having become a doctoral-level profession with advanced curricula, audiology needs textbooks reflecting that complexity.

This book was designed to go beyond an introduction to AR and explore the breadth and depth of audiology. It is intended to present theory, supported by evidence, with practical applications for both the academician and clinician.

With this background in mind, we decided to tackle the creation of an advanced AR textbook with an international perspective. While Alpiner and McCarthy addressed both children and adults in their book, we felt that, given the changing milieu with technological developments and demographic shifts in society, it was appropriate to focus the scope of this book solely on adults.

The third edition of this text is a reflection of the feedback we have received in the last 4 to 5 years about new areas of content that are important to doctoral students and audiologists in the field. It is in line with our commitment to promoting greater engagement by our field in rehabilitation, especially in light of the changing health care delivery system. Once again, we have organized the text in the following sections: (1) Developing a Knowledge Base: Introduction and Background; (2) Building the AR Plan; (3) The Rehabilitative Toolbox: Therapeutic Management; and (4) Expanding the Scope of AR: Special Issues.

Our concept of a fully functional doctoral-level audiologist required a sophisticated, in-depth background of information not sufficiently addressed in previous introductory textbooks. Therefore, we chose to include topic areas that are not only clinical but that also delve into the underlying issues surrounding hearing loss in adults. We need a long-term perspective to define the subject matter, as explained by Joe Montano, and have insight into its history, as fully described by McCarthy and Alpiner. Once again, we gathered together expert opinions and research-based formulations in chapters on such significant aspects of living with hearing loss as activity and participation, stigma, impact on quality of life, motivation for change and improving patient adherence, and interaction with communication partners. Crucial elements of service provision in AR are covered in chapters dedicated to clinical utility of self-assessment, counseling, self-efficacy, auditory and visual speech perception, auditory training, and auditory and cognitive processing. We welcome J.-P. Gagné, Mary Beth Jennings, Ken Southall, Harvey Abrams, Terri Chisolm, Anna Marie Jilla, Sue Ann Erdman, Kathy Cienkowski, John Greer Clark, Geoff Plant, Kathy Pichora-Fuller, and Sherri Smith with excellent chapters on these topics. We chose to include information on dimensions of service that are not as

widespread in their distribution in the field, such as communication partner therapy, group processes, tinnitus management, and vocational considerations. These are represented by chapters by Jill Preminger, Laura Galloway, Chris Lind, Craig Newman, Sharon Sandridge, Sophia Kramer, and Theo Goverts. The influence of technology is seen in chapters relating to hearing assistive technology systems and assessment for implantable devices. Linda Thibodeau, Jaclyn Spitzer, Dean Mancuso, and Rachel Scheperle have provided insights on these areas. We believe that our field is in critical need of research and literature support of AR practices and efficacy, and we have included expert guidance on evidence-based practice as provided by Louise Hickson and Barbra Timmer.

We welcome new chapters and new authors to this third edition that include hearing health-seeking behaviors by Gaby Saunders, social factors in hearing aids by Gurjit Singh and Stef Launer, online multimedia educational services by Mel Ferguson, family-centered care by Nerina Scarinci, use of patient narratives by Abbey Berg and Emilie Zaslow, auditory and visual speech perception by Nancy Tye-Murray, and e-health by Alessia Paglialonga.

An underlying concept of this text is that AR is an expanding aspect of our discipline. Changes are taking place not only in our credentials and in our knowledge base, but also in our scope of practice as we define it. The scope of rehabilitative services is inclusive rather than exclusive. The role of technology in the current practice of AR is apparent, but, as we see in our text, it should be only the means, rather than the end, to the larger picture of rehabilitative planning for individual patients. Learning about and applying technology is intriguing and seductive, and we must always be aware of the potential for mistaking recommendations for devices as an endpoint in formulating plans. As we remain sensitive to this sometime temptation, the reader will note what a powerful emphasis there is throughout this text on the information necessary to develop excellence in counseling and developing a client-centered approach to AR.

The topics we chose to include in this text are ones we believed would represent the current state of AR and perhaps shape its future. Limits in a book's length, time constraints, and author availability precluded the inclusion of some topics that we nonetheless recognize as important. In particular, balance and vestibular rehabilitation was not addressed, and we believe it is a topic of importance for the future. In addition, this is one of those areas where our scope

of practice is currently not clear, with other professionals involved in the day-to-day management of the dizzy patient. We also chose not to include material on specific devices, such as hearing aids and cochlear implants, as we believed that entire texts have been devoted to these topics, and that the concepts and methodologies we explore are the true focus of AR.

Audiologic rehabilitative services are the foundation of our profession. We could debate how this priority is reflected in many audiologic practices, but it is not one entity. It is not just a hearing test. It is not hearing aids or cochlear implants. It is an area whose importance does not appear to be recognized by the broad health care establishment. It is up to us to advance the provision of audiologic services by offering the best practices, as reflected in this textbook. We do not wish the fate of audiologic rehabilitative services—nor of our consumers—to be placed in the hands of others. This book should be a rallying cry to all practicing and student audiologists to define

ourselves and provide the highest possible quality of services to people with hearing loss and their families.

—Joseph J. Montano and Jaclyn B. Spitzer

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—Joseph J. Montano

My thanks to the many patients and colleagues who taught me and gave me insight from their experiences. I hope to continue to work for the improvement of persons with hearing loss.

—Jaclyn B. Spitzer

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This book is dedicated to my wife Patricia and our children and grandchildren: Katie, Daniel, Lauren, Laura, and Andrew; A. J., Sophia, Alex, Gabby, Cristian, and Joey, who, somehow, put up with my antics and continue to love me.

—Joe Montano, AKA Popi

I dedicate this book to Elaine Robinson, whose frustration and tears burned me, and whose strength taught me. Further, to Harriet and Victor Robinson, who also lived the stories and, sometimes, pain and joy of Elaine; and to Raquel, who knew her grandmother all too little. I am glad you endured all with me.

—Jaclyn Spitzer

Part I

DEVELOPING A KNOWLEDGE BASE: INTRODUCTION AND BACKGROUND

1

HISTORY OF ADULT AUDIOLOGIC REHABILITATION

Understanding the Past to Shape the Future

Patricia A. McCarthy and Jerome G. Alpiner

In the words of Lord Byron, “The best prophet of the future is the past.” Those words portend well for the future of audiology rehabilitation as the past has been dynamic, innovative, and patient centered. Audiology did not emerge as a health care profession until the 1940s. Yet in seven decades, the contributions of audiology rehabilitation (AR) to the lives of individuals with hearing loss and their families has been significant and impactful. As such, there is every reason to believe that the future of AR will continue on that trajectory.

What can be learned by examining the history of adult AR? It is a well-accepted maxim that understanding the past is fundamental to understanding the present. Furthermore, our future practice can be shaped by reflecting on our roots. The aging of the Baby Boom generation has presented new challenges and produced increased demand for effective management of adult hearing loss. As such, understanding our history can provide the baseline on which to grow as AR continues to evolve into a sophisticated, respected 21st-century, evidence-based treatment process. Figure 1–1 provides a timeline to illustrate the history and development of our AR practices.

Indeed, while science, technology, and techniques continue to reach new heights in the 21st century, the *raison d’être* for adult AR has remained the

same; that is, improvement of human communication. In 1973, Toubbeh validated this goal of AR by stating, “Human communication is action, it is culture, it is the *history* of man, it is the fabric of all societies, its absence negates man’s existence.” This impressive quote, made almost 50 years ago, reminds us that improvement of human communication in those with hearing loss is a vital pursuit for the individual as well as society. Individuals, society, and technology have changed dramatically and quickly; the fundamental purpose and goals of AR have remained steadfast as the 21st century unfolds.

Certainly, the knowledge and skills used in the contemporary practice of adult AR are not the result of magical events but the product of building a profession by complementing each generation’s progress. In this chapter, therefore, we explore the contributions of these previous generations and consider their impact on the present and future practice of adult AR.

What to Call It?

Over the years, the terminology used to describe the rehabilitation process has assumed many names thus

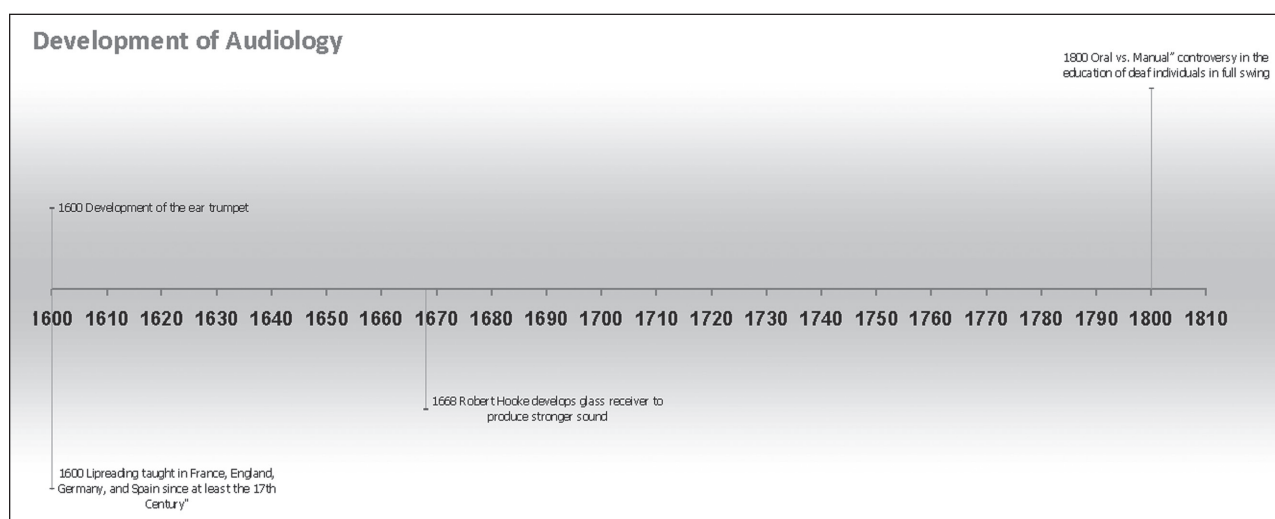


Figure 1-1. Timeline for development of audiology (1600–1810).

leading to confusion among the public, the profession and third-party payers. Through the years there have been various squabbles, so to speak, about what the identifying term should be. Some terminology examples include aural rehabilitation, rehabilitative audiology, auditory rehabilitation, audiologic rehabilitation, lipreading, speechreading, auditory training, and hearing therapy. Titles of textbooks, journal articles, and professional presentations verify this lack of agreement.

Much of the disagreement about terminology centers around use of the term *aural rehabilitation*, which implies a process focused on hearing per se. The argument posits that the process of improving the communication of adults with hearing loss is a holistic approach that extends well beyond simply improving hearing. For example, strictly speaking, even lipreading would not be part of “aural” rehabilitation because it is primarily a visual process. Therefore, the quest to use a more representative term than aural rehabilitation has ensued.

It was not until World War II when a military aural rehabilitation program was started that the term *audiology* came into existence. At that point in history, audiology was defined by the Veterans Administration (VA) (Newby, 1964). The VA defined an audiologist as one concerned with the assessment of hearing as well as habilitation and rehabilitation of children and adults with hearing impairment. Even at that time, there was concern about the lack of audiologists to provide services in aural rehabilitation. At the same time, there appeared to be a division between educational services and the provision of audiology

services for individuals with hearing impairment. There also was concern about emphasis on audiology diagnostic services rather than on aural rehabilitation. Despite this, the term aural rehabilitation continued to be used widely to describe a process distinct from diagnostic audiology. (*Editor's note:* For further information about the terminology of AR, the reader is referred to Chapter 2 of this text.)

The American Speech-Language-Hearing Association (1984) defined aural rehabilitation as referring to “services and procedures for facilitating receptive and expressive communication in individuals with hearing impairment. These services and procedures are intended for those persons who demonstrate a loss of hearing sensitivity” (p. 1). This definition is not so much what aural rehabilitation is but refers more to the procedures performed. The American Speech-Language-Hearing Association then describes specific procedures that can be found in the guidelines. The main categories are (a) identification and evaluation of sensory capabilities; (b) interpretation of results, counseling, and referral; (c) intervention for communication difficulties; (d) reevaluation of the client’s status; and (e) evaluation and modification of the intervention program.

One major reason for describing procedures as well as training requirements for aural rehabilitation is the interdisciplinary nature of the field whereby speech-language pathologists can perform these activities without a Certificate of Clinical Certification in Audiology (CCC-A) or an audiology license. In fact, there are some Medicare and third-party payer guidelines that will only reimburse speech-language

pathologists, not audiologists, for services typically considered under the rubric of aural rehabilitation (e.g., speechreading). In some ways, audiologists see this as a turf battle in which another profession has co-opted part of their scope of practice without the requisite training. It could be argued that if the process were called “audiologic rehabilitation,” it might be more clearly seen as requiring the expertise of an audiologist. (*Editor’s note:* For more information on defining AR and reimbursement issues, the reader is referred to Chapter 2.)

Further confusion has arisen with the various titles used by professionals engaged in rehabilitation with adults with hearing loss. For example, many professionals want to be called audiologists as an umbrella title that encompasses the entire scope of practice including aural rehabilitation. Others have preferred to use an adjective that specifies his/her primary expertise in the profession such as “rehabilitation audiologist” or “diagnostic audiologist.” These terms typically have no basis in credentialing. Furthermore, insurance companies are more apt to reimburse when procedures are listed and the professional has certain credentials.

Terminology also can be confusing to the public and raises some interesting questions. Does aural rehabilitation include deaf individuals who might use sign language as their primary mode of communication? Do some persons hear the word *oral* (for *aural*) and think of a dentist? Does the consumer with hearing loss even know what the term *aural rehabilitation* encompasses? One speech and hearing clinic ran an advertisement in a newspaper indicating that it provided speechreading therapy. A respondent to the ad called to request an appointment for a class in speed-reading. To a certain degree, terminology has caused somewhat of an identity crisis!

While there is no right answer as to what the process should be correctly called, audiologists in the 21st century are increasingly using the term *audiologic rehabilitation*. This term gives ownership to the profession of audiology and encompasses the vast scope of the rehabilitation process. Audiologists have an extensive armamentarium of knowledge, skills, and tools that prepare them to engage successfully in the rehabilitation process. As discussed throughout this textbook, the rehabilitation process does not begin and end with surgery (e.g., cochlear implants, implantable hearing aids, etc.) or the dispensing of amplification (e.g., hearing aids, hearing assistive technology, etc.). As such, we strongly suggest use of the term audiologic rehabilitation in current practice.

From the Ear Trumpet to Lipreading (Early Origins)

In our technosophisticated world, the idea of using ear trumpets as amplification is archaic and amusing. Yet ear trumpets were creative developments to increase speech intensity initially used by sailors and others who needed to communicate at distances. Ear trumpets were later used to help persons with hearing impairment communicate better. One of the earliest descriptions of ear trumpets (Figure 1–2) was by Jean Leurechon in his 1624 book, *Recreation Mathematique*.

Ear trumpets were generally custom made, and their use as hearing aids began around 1800 (Berger, 1984.) Hearing aid trumpet styles and prices varied, as did the rate of success by the user. (It is interesting that the same could be said of hearing aids in the 21st century!) Technology in the 17th century was even of concern in the development of the ear trumpet. Robert Hooke in 1668 worked toward improving hearing by developing a glass receiver, which produced stronger sound (Berger, 1984). He also produced two receivers: one which was latten, and conical in nature; the other of glass and round. Both were sharp at one end. The former produced a stronger sound. It appears that, starting with ear trumpets, amplification became a significant factor in helping individuals with hearing impairment.



Figure 1–2. Early hearing device.

Even before the emergence of ear trumpet amplification, lipreading played an important role in the rehabilitation and education of individuals with hearing loss. Lipreading was taught in various countries such as France, England, Germany, and Spain since at least the 17th century. Children and adults received instruction through both analytic and synthetic methods. The analytic approach emphasized the learning of individual sounds on the lips; the synthetic approach had a more broad-based communication orientation. Some of the prominent names in the development of these approaches include Bruhn (1955), Bunker (1961), Nitchie (1950), and the Kinzie sisters (1920). Electronic amplification technology did not exist in those days. Therefore, lipreading served as one of the only viable methods for teaching the “deaf and dumb” (Peet, 1851). For a comprehensive and fascinating review of lipreading the reader is referred to *The Story of Lipreading* by DeLand (1931).

By the 19th century, the oral versus manual controversy in the education of deaf individuals was in full swing. Advocates of oralism relied on lipreading as a major communication mode so that “deaf” individuals could learn to function in a hearing society. Advocates of manualism decried the poor verbal and written language skills of deaf individuals who relied purely on oralism. They argued that use of American Sign Language would allow deaf individuals to develop language and communication skills. By the mid-20th century, however, much of the passion for this controversy dissipated with the development of powerful, miniaturized hearing aids brought about by the development of the transistor in 1949 (Scientific, Inc., 1999). Adults with moderate to severe hearing losses who had previously functioned as deaf individuals could now benefit from both powerful hearing aids and lipreading.

While lipreading schools and teachers have become a thing of the past, the role of visual cues in communication cannot be underemphasized in adult AR in the 21st century. Best-practice hearing aid fitting protocols stress the synergistic benefits of bisen-sory stimulation that result from amplification and the use of visual cues in communication. (*Editor’s note:* In keeping with the authors’ discussion of consumers confusing terminology, the term *lipreading* is often used interchangeably with the term *speechreading*. While *speechreading* is meant to be more inclusive, both identifiers refer to visual communication. The reader is referred to Chapter 14 in this text for a review of the contemporary research related to lipreading and visual communication.)

World War II Era: Birth of Audiology

If they were still alive, those responsible for the creation of audiology as a profession might be chagrined that some 60 years later, audiologists have strayed a bit from our rehabilitative roots. Indeed, the profession was created in response to the needs of the World War II “aural casualties requiring rehabilitative measures” (Truex, 1944). In response to these needs, both the U.S. Army and Navy developed aural rehabilitation centers: three by the Army (Deshon, Borden, and Hoff General Hospitals) and one by the Navy (Philadelphia Naval Hospital) (Spencer, 1946; Truex, 1944). Raymond Carhart, the acknowledged father of audiology, served in the Army Medical Corps (1944–1946) as the director of the Acoustics Clinics in the Veterans Rehabilitation Center of Deshon General Hospital before beginning the first academic program in audiology at Northwestern University in 1947 (Olsen, Rose, & Hedgecock, 2003) (Figure 1–3).

The purpose of the “listening clinics” was to allow people to communicate effectively with others in their environment (Spencer, 1946). The rehabilitative process began with a hearing examination and fitting of the most satisfactory hearing instrument. But this was truly just the beginning of the 8-week process. An intense course of instruction was designed for each patient to reorient him to sounds transmitted by the hearing instrument. Use of the term *reorient* suggests an understanding that fitting the hearing aid was only the first step in the process. Special instruction involved cerebral reorientation to the auditory stimuli newly introduced with hearing aid use (Truex, 1944). The curriculum included daily individual and group lipreading instruction; counseling regarding personal, economic, or domestic problems; maintenance of contacts with the patient’s family; and bridging the gap between the veteran’s facility and civilian agencies and clinics that would provide follow-up rehabilitation post-discharge (Truex, 1944). The fact that a veteran with hearing impairment might need “additional rehabilitative measures” beyond the 8-week course is evidence that rehabilitative measures were respected and essential in the 1940s (Newby, 1964). Almost seven decades later, just a few hours, not weeks, of post-hearing aid fitting rehabilitation is standard.

Hearing aid satisfaction and outcome measurement were critical parts of the rehabilitation process. The Army and Navy programs administered satisfaction questionnaires and reported that “94% of the

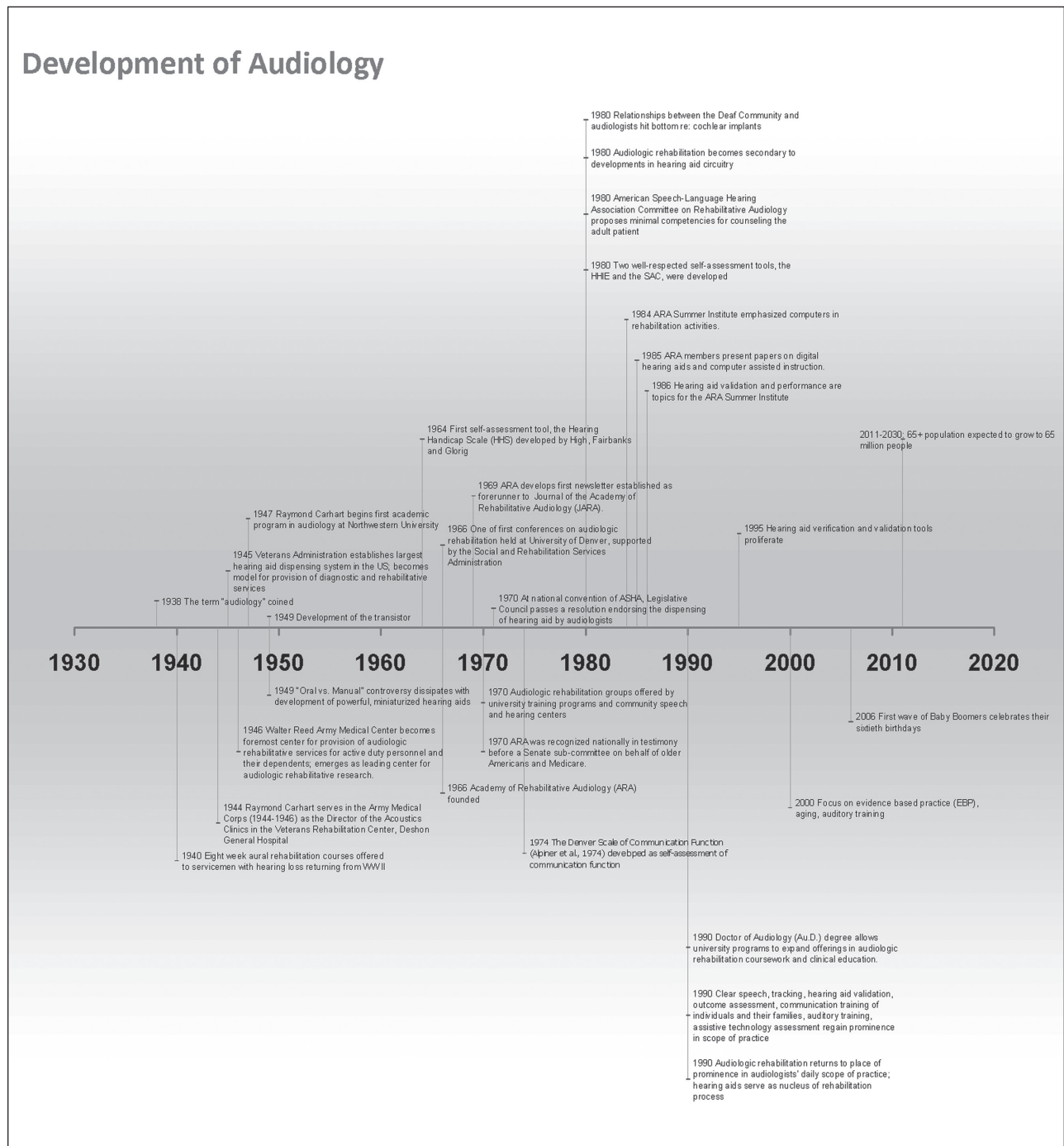


Figure 1-3. Timeline for development of audiology (after 1930).

boys" were still using their hearing aids 3 months to 2 years post-fitting (Spencer, 1946). Captain Grant Fairbanks reported that more than 350 hearing aid prescriptions were possible with various combinations of microphones, receivers, amplifiers, and

batteries. Yet there still were complaints like "I can hear everything in the room except the guy I'm trying to listen to" (Spencer, 1946), thus supporting the adage that the more things change, the more they stay the same.

Even in the 1940s, unrealistic hearing aid expectations were prevalent as some patients wanted “miracles from their hearing aids” (Spencer, 1946). Auditory training (“expertly supervised listening”) was designed to help patients adjust to electronic amplification and to improve sound perception and discrimination. Groups of six to 10 participated in Army and Navy listening clinics that provided a variety of sound experiences including phoneme discrimination tasks, music, radio programs, and speeches. Attention was also given to vocational/occupational issues presented by hearing loss. (*Editor’s note: For more information on the stigma associated with hearing loss, the reader is referred to Chapter 4.*)

Motivation for engaging in the rehabilitation process was an issue for these early military programs. To combat the pessimism and fatigue of soldiers and sailors with hearing impairment returning from World War II, some interesting motivational methods were employed. For example, an “orientation girl” was used to brief the new patient and because he could not “hear the bugle,” a “pretty WAC” would wake him in the morning. Furthermore, lipreading teachers were mostly women who were “easier to look at” and “vocally better groomed” (Spencer, 1946). Certainly, recognizing the role of motivation was laudable despite the rather sexist methods employed!

Compared to the rehabilitative processes offered in the 1940s, what new concepts or strategies have emerged in the past six decades? The aural rehabilitation centers designed by the Army and Navy addressed amplification, adjustment, and “cerebral reorientation” to hearing aids, motivation, expectations, self-assessment, auditory training, lipreading, counseling, family support, psychosocial and vocational issues, group sessions, and the need for follow-up rehabilitation. The new hearing aid user in the early 21st century who is receiving all of those services is indeed rare.

Post-World War II: Walter Reed Army Medical Center and Veterans Administration

The provision of AR services during and post-World War II established it as a priority for service personnel and veterans with hearing loss. Since that time, hearing loss has continued as a major health concern during both peace and war times. As such, many of the major trends in AR have emerged as the result of the efforts of the Veterans Administration (VA, cur-

rently the Department of Veterans Affairs) and the Walter Reed Army Medical Center (WRAMC). The VA established the largest hearing aid dispensing system in the United States and became the model for provision of diagnostic and rehabilitative services to eligible veterans with hearing loss. The WRAMC became the foremost center in the provision of audiologic rehabilitative services for active duty personnel and their dependents and emerged as the leading center for audiologic rehabilitative research. In the 21st century, these agencies continue to lead the way in research and development of rehabilitative technology, methods, amplification, service delivery, and outcome measurement.

“What’s Happening” in the 1960s

American adult AR was enhanced in 1966 with the emergence of the Academy of Rehabilitative Audiology (ARA). Prior to this time, there had been no major effort to establish a formal organization to emphasize the rehabilitation aspect of audiology. The name itself helped to institutionalize the concept of “audiologic” in lieu of “aural” rehabilitation.

The impetus for the development of the ARA came from a group of audiologists, led by John O’Neill and Herbert Oyer, who were concerned about the lack of interest in aural rehabilitation programs, including in the training of university students. As a result of this concern, the ARA was inaugurated in 1966 with assistance from the Federal Office of Vocational Rehabilitation in Washington, D.C. The purpose of the academy was to generate interest in AR and offer a mechanism for interest and participation in this area of the profession.

Interestingly, various issues that have faced AR since its inception mirror the activities of the ARA. There seemed to be turf concerns regarding the ownership of audiological rehabilitation between rehabilitation and clinical audiologists. To track the development of aural rehabilitation, the ARA approved a position for an archivist. In her role as archivist, Skalbeck (1984) provided a rationale for the transition of AR through the years by invoking Santayana’s warning that those who do not know history are destined to repeat it. She cautioned audiologists to review the past and plan for the future of AR.

Audiology continued to grow at a logarithmic pace during the years from 1966, particularly in the development of diagnostic techniques. Unfortunately, there appeared to be a continuing decline in inter-

est related to the remediation aspects of audiology. Some colleagues even took a position that it was more important to test than to engage in AR. There seemed to be a widespread attitude that if rehabilitation was ignored, the use of diagnostic equipment and techniques would become more professional and valued. Surveys of audiology journals illustrated the trend of publishing many more diagnostic articles than rehabilitative.

Institute on Aural Rehabilitation

One of the early conferences on AR was held at the University of Denver in 1966 and supported by the Social and Rehabilitation Services Administration with Jerome Alpiner serving as project director (Alpiner, 1966). Representatives from throughout the United States attended. The institute focused on significant areas of AR for audiologists and helped define and encourage their role in the rehabilitation process.

This was a landmark and exciting conference for advocates of AR. It is worth noting that this event was held several years *before* the American Speech-Language-Hearing Association approved the dispensing of hearing aids by audiologists. This was a time in which audiologists were excited about developing ways to help those adults with hearing loss after patients purchased aids from dispensers (referred to as hearing aid dealers at that time). Some of the relevant topics included lipreading, auditory training, social and psychological factors, vocational programs, closed-circuit and commercial television training, and research in aural rehabilitation. An interesting exercise would be to survey today's audiologists to determine how many, if any, of the above procedures currently are provided in their practices as part of the hearing aid fitting or post-fitting follow-up.

An ARA newsletter was established prior to the development of the *Journal of the Academy of Rehabilitative Audiology (JARA)*. Three articles appeared in the April 1969 issue. Brainerd (1969) reported on an investigation of the relationship between performance on a filmed lipreading test and analysis of the visual environment. Lovering (1969) reported on visual acuity and lipreading performance and reported that lipreading scores would degenerate as a function of visual distraction. Kitchen (1969) studied the relationship of visual synthesis to lipreading performance. Oyer (1969) reported the establishment of a laboratory at Michigan State University that was

solely for the purpose of scientific studies in aural rehabilitation. And so it was! AR was on its way as an identified, valued aspect of the profession.

The 1970s: Things Are a Changin'

Probably one of the most significant decisions ever to be made by the American Speech-Language-Hearing Association was to allow audiologists to dispense hearing aids for profit. At that time, one of the co-authors (Alpiner) was on the the American Speech-Language-Hearing Association executive board and recalls that the exchanges were emotional, to the extent that one member indicated a resignation would be forthcoming if the resolution passed. It passed; he did not resign.

Prior to the American Speech-Language-Hearing Association's decision to permit the dispensing of hearing aids, there was a "Fourth of July" approach in which both sides were quite adamant, emphatically holding a position that audiologists should not be able to dispense hearing aids. Siegenthaler (1972) took the position that audiologists should not dispense hearing aids. Several of his major points at the time included:

1. The clinical audiologist, who was free from maintaining his/her income by the sale of hearing aids, was in a proper position to offer services (AR), while another group (hearing aid dealers) could dispense hearing aids on the basis of the audiologist's recommendations.
2. If audiologists dispensed hearing aids, some predicted the demise of hearing aid dealers as well as audiologists. Audiologists would lose subsidies from universities and government agencies over a period of time. Meanwhile, hearing aid dealers would lose their business because audiologists would go into private practice and dispense aids from their offices.
3. Audiology technicians will be trained to do basic audiometrics submitting results, for example, to otolaryngologists. Technicians might even be franchised to dispense hearing aids. It was pointed out that some otolaryngologists already trained their nurses to do the basic testing. Siegenthaler reported that the Rehabilitation Services Administration (RSA) had awarded a grant to the National Association of Speech and Hearing Agencies to train armed services medical corpsmen to be audiometric technicians.

4. The issues were complex and somewhat confusing regarding who would perform the various audiology procedures.

Harford (1972) took the position in favor of dispensing of hearing aids by audiologists. At the national convention of the American Speech-Language-Hearing Association in 1971, the legislative council passed a resolution endorsing the dispensing of hearing aid by audiologists. The resolution was then submitted to the executive board. A statement of rationale was presented to answer the basic question: "Why should the ASHA change its Code of Ethics to allow the dispensing of hearing aids by its members who are audiologists if they wish to do so?" A brief review of these responses follows:

1. There was an increased concern in the 1960s and 1970s that audiologists were not able to fulfill their obligations to the hearing-impaired population. There was a feeling that the hearing delivery system should be expanded to dispensing hearing aids to provide comprehensive AR services.
2. There needed to be a more efficient system for hearing aid delivery beyond recommending the aids to be fit by hearing aid dealers.
3. A number of audiologists had given up their American Speech-Language-Hearing Association certification to become directly involved in the hearing aid fitting process. This was an extremely significant move by some audiologists.
4. A number of audiologists felt they would be better able to research the effectiveness of hearing aids and other wearable amplification devices as part of the AR process.

Changes in the American Speech-Language-Hearing Association Code of Ethics influenced the training of audiologists, the breadth of the job market, the financial implications of the profession, and service delivery to adults with hearing impairment in the United States. Had this change in the code of ethics not occurred, it is arguable that the profession of audiology would not exist in the 21st century.

The 1970s were, in many ways, a decade when AR practice patterns were established and AR was respected as integral part of audiology practice. In 1973, Garwood, Bergman, Dixon, and Haspiel reported on the roles assumed by audiologists and concluded that the aural rehabilitation process was the one common thread that should hold audiologists together. Regardless of roles assumed by audiologists as administrators, clinicians, academicians, private practitioners, or government employees, rehabilita-

tion of individuals with hearing loss was the uniting goal of the profession.

The emphasis on rehabilitation was particularly evident in university training programs. In the 1970s, most AR groups were offered by university training programs and community speech and hearing centers. The usual format was for 6-week programs (once a week) with groups of about six or seven individuals. Most university programs were either free or required a nominal charge that probably did not cover the overhead of the clinic. Group sessions were supervised by university instructors, with clinical practicum students actively involved in this learning model. The typical format for these group sessions generally included lipreading, auditory training, and some hearing aid orientation. Counseling was included to allow participants to engage in dialogue regarding their communication situations. Community programs generally charged, although nominally, and the services provided were similar to the university programs.

Interest in outcome measurement was prevalent in the 1970s, although that terminology was not yet in the audiologist's lexicon. Pre- and post-lipreading tests were typically given to document improvement accrued as a result of group AR. The limitations of lipreading testing often made this an exercise in frustration given ceiling effects and the inherent difficulty of creating a reliable, valid assessment of visual communication ability. However, a 1977 study by Binnie suggested there were positive outcomes associated with these group rehabilitation sessions. Individuals enrolled in a university AR group at Purdue University were given a lipreading pretest during the first group session. At the end of the program, these same participants were given a lipreading post-test. Not surprisingly, there was little measured improvement in lipreading skills. In addition, participants also were given a questionnaire probing whether they felt their communication had improved as a result of these group sessions. Interestingly, the participants overwhelmingly responded that their communication and their confidence in communicating had improved dramatically. So, despite no measurable changes in lipreading ability, participants in group AR self-reported marked improvement in communication abilities. These intriguing results suggested that something positive other than improvement in lipreading skills was occurring as the result of the rehabilitation process.

Even in the 1970s, audiologists were frustrated by adults with hearing loss who could improve their communication efforts via rehabilitation but chose not to do so. Based on their observations that adults

with hearing loss frequently did not follow through with AR recommendations, Oyer et al. (1976) sought to determine some of the reasons for this lack of follow-up. A questionnaire was sent to 45 adults who had been evaluated and advised to participate in rehabilitation programs yet chose not to participate. A review of their results and insights into audiologic service delivery suggests their conclusions differ little from what we hear in the 21st century. Some examples include:

1. Audiologists are unable to demonstrate the worth of AR and the changes that it can bring about.
2. The flexibility in scheduling AR programs for working people should be studied.
3. Audiologists do not differentiate and place in proper perspective the relative values of amplification derived from a hearing aid and the further refinements to be achieved through auditory training and lipreading.
4. Perhaps audiologists need to enlist the support of family members, friends, or other significant persons to aid in encouraging the hearing impaired to participate in AR. It is possible that insufficient attention is given to familiarizing these people with the communication problem, the limitations of hearing aids, and the difficulty in hearing aid adjustment.

Forty years later, the challenges of engaging adults with hearing loss in the AR process have become even more complex. As in the 1970s, audiologists often short-sell the value and impact of the rehabilitation process. For example, few audiologists involved in hearing aid fittings offer group rehabilitation sessions despite the evidence that suggests this model reduces hearing handicap (Preminger, 2003) and is cost effective (Northern & Beyer, 1999). Furthermore, those audiologists who do offer group sessions are challenged to find ways to encourage individuals to attend. In many ways, adult group rehabilitation as a treatment model is less accepted in the 21st century than it was in the 1970s. Furthermore, while giving lip service to the value of rehabilitation, audiologists three decades later often fail to convince individuals of the value of rehabilitation, continue to report that they do not have the time to offer rehabilitation, and appear to be unconvinced of its value in the hearing aid fitting process despite compelling evidence to the contrary (Chisolm et al., 2007).

During the 1970s, the ARA continued to emphasize AR as a flexible product that could be infused into any type of audiologic practice. For the first time, the ARA was recognized nationally to provide

testimony before a Senate subcommittee on behalf of older Americans and Medicare. Furthermore, Hardick (1976) provided expert testimony to the ENT Committee of the Food and Drug Administration (FDA) on standards for auditory training. A significant recommendation was to license providers of auditory training the same as hearing aid dispensers. During this time, initial discussions began regarding the need for audiologists to be licensed to make them more “recognized” and accountable.

New Approach in the 1970s and 1980s: Self-Assessment Tools

One day in the 1970s, Jerome Alperin, an author of this chapter (then a university professor), posed a question to his doctoral students: “When audiologists make recommendations to patients, do they really objectively know that individual’s communication function, aside from the numerical evaluation scores?” The lively, ensuing discussion encompassed such issues as how the patient got along at home, at work, in social contacts, and in all everyday living situations. The discussion, as well as others that were occurring among those interested in AR, focused on the growing belief that audiology test scores were not the only index to determine the rehabilitative needs of adults with hearing loss. Certainly, obtaining numerical audiologic data was one critical part of the rehabilitation process, but the next step was to think of the ways to assess what impact the hearing loss had on the individual’s life. As such, an emerging focus of the 1970s was how to assess communication function beyond the audiogram.

There were initial attempts to address this issue as early as the 1940s. Davis designed the Social Adequacy Index in 1948 to examine the relationship between speech reception thresholds and discrimination scores. It was felt that more information about the connection between hearing loss and understanding speech was needed. High, Fairbanks, and Glorig developed the Hearing Handicap Scale (HHS) in 1964 to assess the effects of hearing loss on everyday living activities. Although a groundbreaking effort, the HHS was criticized for its lack of validity and failure to address vocational and psychological effects of hearing loss. However, these early pioneers provided impetus for the numerous assessment procedures that followed in the 1970s.

As researchers began to develop self-assessment tools to identify the effects of hearing loss on the individual, it became clear that audiologists approached

self-assessment in many different ways. Some were interested in using the procedures for research purposes. Others focused on assessment results as outcome measures by comparing a patient's performance pre- and post-audiologic rehabilitation. Tools were developed for specific populations including the elderly. Noble and Atherley (1970) devised the Hearing Measurement Scale (HMS) to assess handicap resulting from industrial noise exposure. Interestingly, the seven subcategories in the HMS laid the groundwork for future self-assessment tools developed over the next three decades. They include:

1. Speech-hearing
2. Acuity for nonspeech sound
3. Localization
4. Reaction to handicap
5. Speech distortion
6. Tinnitus
7. Personal opinion of hearing loss

The Denver Scale of Communication Function (DSCF) (Alpiner et al., 1974) approached self-assessment from a communication function standpoint. It was designed to help the audiologist make a subjective assessment of communication attitudes of adults with acquired hearing loss for more effective rehabilitative planning. It is based on the assumption that lipreading tests failed to measure improvement in communication function. Responses were made using a 7-point scale from agree to disagree. Some examples of items in the DSCF include:

The members of my family are annoyed with my hearing loss.

People sometimes avoid me because of my hearing loss.

Since I have trouble hearing, I hesitate to meet new people.

Hutton (1980) developed the Hearing Problem Inventory in Atlanta to help the audiologist design an individualized rehabilitative program for patients. The instructions to clients stated, "We want you to know about you and your problems. In this way we can do a better job of solving your specific problems." Interestingly, Hutton was one of the first to determine that a critical domain in assessing hearing aid satisfaction is hearing aid wear time. Hours of hearing aid use has since been included in such self-assessment tools as the International Outcomes Inventory (Cox & Alexander, 2002) and the Glasgow Hearing Aid Benefit Profile (GHABP) (Gatehouse, 1999).

Sanders devised another variation of self-assessment (1975). Three profile questionnaires probe communicative performance in home environments, occupational environments, and social environments, with a 4-point response scale format (*little or no difficulty to great difficulty*). A useful feature of these scales is that they assess how often each particular situation occurs for the individual thus allowing relevant responses to each item. Interestingly, more than 20 years later, Gatehouse (1999) incorporated this concept into the GHABP by asking "Does this situation happen in your life?" for each item.

A different twist on self-assessment, incorporating a family member's responses, was developed in the late 1970s by this chapter's authors. The McCarthy-Alpiner Scale of Hearing Handicap (M-A Scale) (1980) was designed to assess the psychological, social, and vocational effects of hearing loss from the perspective of the individual with hearing loss compared to a family member, allowing the audiologist to examine responses for counseling purposes (McCarthy & Alpiner, 1983). Use of this scale provides a platform for determining the attitudes of family members that can contribute to a counseling-based rehabilitation plan.

Two well-respected self-assessment tools, the Hearing Handicap Inventory for the Elderly (HHIE) and the Self-Assessment of Communication (SAC), were developed in the early 1980s. The HHIE and the screening version (HHIE-S) (Ventry & Weinstein, 1982, 1983) include items that assess the social and emotional handicapping effects of hearing loss. Even today, they are used widely in research and clinical practice. The 10-item HHIE-S, in particular, has been used not only as a screening instrument and rehabilitative assessment tool for the elderly, but also to document handicap reduction with hearing aid use and as a valid outcome measurement tool. The SAC, another screening tool, was designed to measure the communication ability of hearing-impaired adults, and has a companion scale, the Significant Other Assessment of Communication (SOAC) (Schow & Nerbonne, 1982). Many modifications of both of these tools have been published over the last few decades including the Hearing Handicap Inventory for Adults (HHIA) (Newman, Weinstein, Jacobson, & Hug, 1991) and the adaptation of the SAC/SOAC for use with adolescents (Elkayam & English, 2003).

Another important development in the realm of self-assessment was the Communication Profile for the Hearing Impaired (CPHI) developed by Demorest and Erdman (1987). This profile provides an in-depth analysis of adjustment to hearing loss in four main

areas: communication performance, communication environment, communication strategies, and personal adjustment. The CPHI was originally standardized on more than 900 active and retired military personnel and was later standardized further with a variety of clinical population (Erdman & Demorest, 1998).

As the concept of the patient self-assessing his/her difficulties caught on as an adjunct to the audiometric evaluation, self-assessment tools have continued to proliferate. Many of the originally developed assessment tools are still used while new tools have been developed to reflect the trends of the day (e.g., hearing aid benefit and satisfaction, quality of life, etc.). Since 1990, Robyn Cox has consistently contributed some of the most relevant self-assessment tools used in audiology. These include the Abbreviated Profile of Hearing Aid Benefit (APHAB) (Cox & Alexander, 1995), the Expected Consequences of Hearing Aid Ownership (ECHO) (Cox & Alexander, 2000), and the Satisfaction with Amplification in Daily Living (SADL) (Cox & Alexander, 2001). The previously mentioned GHABP (Gatehouse, 1999), the Client Oriented Scale of Improvement (COSI) (Dillon et al., 1999), and the International Outcome Inventory for Hearing Aids (IOI-HA) (Cox, Stephens, & Kramer, 2002) represent the next generation of self-assessment tools that followed the initial efforts of the 1970s. (*Editors' note:* For more information on self-assessment, the reader is referred to Chapter 6 of this text.)

The Traditional Approach Gives Way to Counseling

By the 1970s and 1980s, the traditional approach to AR had changed little since the early days (McCarthy & Culpepper, 1987). As a result, it is fair to say that not all audiologists or hearing aid vendors were using the traditional approach as an integral part of their hearing aid fittings. The role of rehabilitation was becoming secondary to the amazing developments in hearing aid circuitry that were emerging.

This attitude was widespread and exemplified in this anecdote. At that time, a local hearing aid vendor was invited to speak to an audiology class about the hearing aid fitting process. (Recall the model of the audiologist doing the hearing aid evaluation [HAE] and the hearing aid vendor "selling" the device.) The vendor agreed to discuss hearing aids but was negative on the role of the rehabilitation process. His rationale was that if one discussed anything other than the hearing aid itself, the patient would think the hear-

ing aids were "not perfect." Unfortunately, this attitude still exists to some degree in the 21st century. The expectations for hearing aids by many consumers with hearing loss are too high, often as a result of marketing and advertising. People with hearing loss expect a miracle where the emphasis on the need for rehabilitation to improve communication is ignored or given little attention.

The decline of the traditional approach was also influenced by audiologists' frustration with lack of reimbursement for these services. The traditional approach included a comprehensive audiologic evaluation, an assessment of lipreading function, planning for group and/or individual therapy sessions, implementation of therapy, post-therapy evaluation, prognosis, and recommendations. Although audiologists could be reimbursed for hearing aids (typically as a self-pay prosthetic device), they could not be reimbursed by a third-party payer for hearing aids or such rehabilitative services as lipreading or auditory training. Furthermore, only speechreading was identified as a reimbursable service by Medicare; even then, only speech-language pathologists were specified as providers (Madell & Montano, 2000). Although the codes have been rewritten recently, reimbursement for audiologic services, with the exception of rehabilitation following cochlear implantation, remains elusive today.

Another contributing factor to the decline of the traditional approach to AR was the lack of candidacy or discharge criteria needed for decision making in the treatment process. Obviously not all patients are candidates for AR and for those who are candidates, how much rehabilitation is enough? In 1981, Tulko and Santore suggested a method for determining candidacy based on an assessment of the patient's communication ability at the time of the evaluation. Interestingly, their method did not catch on nor was it validated. Yet, several decades later, determining candidacy for AR remains a dilemma for audiologists. Furthermore, development of discharge criteria is a critical need in the 21st century if AR is to become a reimbursable service in the future.

With the decline in the traditional approach, audiologists began to embrace a counseling approach to rehabilitation in the 1980s. Vernon and Oettinger (1989) argued that the term *hearing loss* had true significance because the longer an individual experiences a hearing impairment, the more it is perceived as a true "loss." Audiologists began to realize that they often were the initial "listeners" to patients experiencing a loss of hearing. Consequently, audiologists began to address the psychological, emotional, social,

and vocational implications of hearing loss and to realize they had the responsibility to determine the extent of these issues for the individual and his/her family. Sanders (1975) helped to define the audiologist's role in counseling by suggesting that it can be divided into two areas: informational and personal-adjustment counseling. He described informational counseling as "when the audiologist provides the client with an understanding of the hearing impairment, its consequences and the role of amplification." Personal adjustment counseling was described as "when the audiologist assists the client in finding a solution to her/his problem and achieving independence."

During the 1980s, there was a proliferation of books, articles, and conferences proffered to upgrade audiologists' skills and knowledge of the theory, content, and action of the counseling process (Wylde, 1987). Counseling assumed such importance within the profession that the American Speech-Language-Hearing Association Committee on Rehabilitative Audiology (1980) proposed minimal competencies and described four objectives for counseling the adult patient: (a) enhancement of the individual's welfare; (b) assistance in the resolution of pertinent problems; (c) stimulation and motivation to achieve; and (d) improvement of self-concept and social relationships.

Despite the upsurge in interest in counseling in the 1980s as part of the rehabilitation process, not all audiologists embraced it as a requisite part of the scope of practice. There was a pervasive feeling among many audiologists that engaging in diagnostic evaluations with sophisticated equipment was higher level than engaging in the counseling process. Furthermore, some audiologists who wished to provide counseling did not because it was not reimbursable. Consequently, the "time as money" factor ruled out counseling for many patients.

1980s and 1990s: The Deaf Community and Audiology

It is perhaps ironic that audiologists, as hearing health professionals, traditionally have known so little about deafness as a culture. Prevalent attitudes of the Deaf community towards audiologists in the 1980s can be seen in the work of Rohland and Meath-Lang (1984), who analyzed essays written by 193 Deaf individuals. They found a positive attitude toward audiologists when they were able to use sign language. However,

many essays expressed negative feelings towards audiologists in the form of anger, mistrust, and misperceptions about the reasons for being tested. Audiologists generally were viewed in a negative light.

Relationships between the Deaf community and audiologists have had a history of mistrust and frustration for many years but hit bottom during the 1980s and 1990s, when cochlear implants were first approved by the FDA. Many Deaf individuals, as well as their family members, took the position of desiring to live in a Deaf culture (note capitalization of "Deaf") and viewed cochlear implants as a threat to that culture. American Sign Language (ASL) was represented as the "natural" language of the Deaf and a symbol of the culture. Initial responses to the new technology caused suspicion that audiologists and physicians were trying to "medicalize" deafness and wipe out the Deaf culture.

During the 1980s and 1990s, the Deaf community and audiologists were moving on diverse paths. As audiologists became experts in cochlear implant evaluations, programming, and rehabilitation, the Deaf community embarked on somewhat of a civil rights movement. The Deaf demanded that they no longer be treated like second-class citizens in every aspect of their lives. For example, they demanded that for the first time in history, the president of Gallaudet University must be a member of the Deaf community. Given these developments, it is not surprising that relationships between the Deaf and audiologists were strained during this period. With the new millennium, however, the anger and suspicion on both sides dissipated and were replaced by cooperation and increased trust. These issues are illustrated beautifully in a poignant, true-life film titled *Sound and Fury* (Aronson, 2000), a 2001 Academy Award nominee for Best Documentary. *Sound and Fury: Six Years Later* (Aronson, 2006) further documents the family's struggles with the decision to have their children implanted and, ultimately, themselves.

1980s and 1990s: ARA Contributions

During the last two decades of the 20th century, the ARA continued its efforts to strengthen AR as a professional entity with credibility. A review of some of ARA activities of this period suggests the ARA was leading the way with innovations in AR. For example, in the early 1980s, an ARA-sponsored the American Speech-Language-Hearing Association Convention

presentation focused on the training necessary to provide AR services. As early as 1984, the ARA Summer Institute emphasized computers in rehabilitation activities. By 1985, ARA members were preparing papers on digital hearing aids and computer-assisted instruction. Amplification, validation, and performance were topics for the ARA Summer Institute in 1986. In 1988, a major ARA presentation dealt with digital signal processing of video signals. These presentations certainly demonstrate that AR had progressed beyond just lipreading and auditory training.

By the end of the 20th century, the ARA annual meeting was no longer planned as a summer institute, but was held in the fall to attract a larger audience. ARA leadership focused on putting together meeting programs with speakers from diverse backgrounds who would bring fresh and alternative perspectives to the AR process. Thus, the ARA evolved as a reflection of the changing face of AR by becoming an inclusive, eclectic forum for the exchange of ideas that has broadened the definition of AR and strengthened it as a clinical process.

1990s: “Rehab” Returns

By the 1990s, AR had returned to a place of prominence in many audiologists’ daily scope of practice. Mueller (1998) summarized this evolution by noting that audiologists were beginning to recognize that hearing aid fitting is an integral part of rehabilitative audiology and not a separate process.

This return to its roots was precipitated by several developments in the profession. First, despite amazing developments in amplification technology, audiologists recognized that hearing aids were not a magic cure-all for communication problems caused by hearing impairment. As such, audiologists acknowledged that new hearing aid users could benefit from services that transcend what hearing aids alone provide (Carmen & Ross, 2000). Second, with the advent of high-end digital amplification, meeting patients’ expectations became more important than meeting targets during hearing aid fittings. Assessing patients’ expectations before engaging in the hearing aid fitting process was seen as preemptive. Third, the importance of rehabilitation with cochlear implants suggested commensurate rehabilitation might be warranted for new hearing aid users. Gagné and Tye-Murray (1994, p. 7) summarized this by stating, “Adult audiology rehabilitation has evolved consid-

erably in recent years. Changes in the services provided to clients have been influenced by technological advances, clinical experience, and a growing body of research literature in our field as well as in related disciplines.”

Graduate coursework in the 1990s reflected the increasing knowledge and skills necessary to provide AR. By the 1990s, a typical course syllabus for an adult rehabilitative audiology course included these course objectives: The student will demonstrate (a) A functional understanding of the full scope of AR in adults; (b) knowledge of self-assessment tools and interpretation; (c) knowledge of the psychosocial effects of hearing loss; and (d) knowledge of amplification and assistive listening device assessment, evaluation, and technology. Furthermore, the newly developed Doctor of Audiology (AuD) allowed university programs to expand their offerings in AR coursework and clinical education.

With the hearing aid as the nucleus of the AR process, audiologists in the 1990s expanded their repertoire of rehabilitative services. These included clear speech, speech tracking, hearing aid validation, outcome assessment, communication training of individuals and their families, auditory training, assistive technology assessment, fitting and evaluation, vestibular rehabilitation, tinnitus management, rehabilitation following implantation, and counseling.

End of the 20th Century: Focus on the Aging Population

Three large-scale demographic population trends in the late 20th century required audiologists to rethink the role of AR with the aging population. First, the increased longevity of the World War II generation meant that many individuals were living well into their 80s and 90s. For example, in 1980 the over 85-year-old population in the United States was 2.2 million; by 2000, that population was 4.3 million and expected to increase over the next several decades (U.S. Department of Health and Human Services, 2004). Second, the number of Baby Boomers aging into “Senior Boomers” in the early 21st century was staggering. In fact, by 2006 the first wave of Baby Boomers had celebrated their 60th birthdays. Between 2011 and 2030, the 65+ population was expected to grow to 65 million people, representing a 75% increase (Hodge, 2006). Third, the aging American population was becoming more racially and ethnically diverse

than ever before (McCarthy, 2006). For example, the older Hispanic population was estimated to increase from about 2 million in 2003 to 8 million in 2030 (He, Sengupta, Velkoff, & DeBarros, 2005). Consequently, by the end of the 20th century, audiologists were faced with dramatically changing caseloads that would require innovative approaches to meet the needs of this expanding population.

As the American population "grayed," audiologists began to recognize that the normal aging process could undermine the effectiveness of AR (McCarthy & Culpepper, 1987). Audiologists understood that anatomic, physiologic, social, and psychological changes of aging were negatively impacting methodologies that were successful with the "younger" population. Furthermore, hearing aid success with some aging patients could be limited by visual problems, psychosocial and cognitive issues, and touch sensitivity. Audiologists began to learn that money, mobility, and family relationships were part of the AR process. Consequently, it became clear that assessment tools were needed that would provide a holistic profile of aging patients that would encompass these variables.

Alpiner developed one of the first tools that focused on the needs of aging individuals. The Denver Scale of Communication Function for Senior Citizens Living in Retirement Centers was designed to make each item in the scale relevant to the individual by posing a primary question followed by a probe effect and an exploration effect (Zarnoch & Alpiner, 1977). The scale was developed in response to concerns about the effects of the environment on the communication of aging individuals living in extended care facilities (Alpiner, 1982). A novel aspect of this scale was the checklist for identification of environmental noise in a nursing home setting, including noise levels and architectural obstructions, both internal and external, for each area of the nursing home. The goal was to create a favorable listening environment for residents by eliminating or reducing noise. A second checklist evaluated the physical environment for excessive ambient noise, personal safety, and furniture arrangement.

The HHIE, discussed earlier in this chapter, assessed both social and emotional effects of hearing impairment in the noninstitutionalized older person (Ventry & Weinstein, 1982). Alpiner and Baker (1981) developed the Communication Assessment Procedure for Seniors (CAPS) to evaluate attitudes and communication status in specific situations. This interview-type scale also probed the aging individual's interest in rehabilitation.

Another innovative approach to AR for the aging population was the establishment of an Elderhostel focusing on hearing loss. Elderhostels had been in existence since 1975 and typically provided learning opportunities for seniors in a residential setting. In the 1980s, an Elderhostel at Gallaudet University was developed specifically for older adults with hearing impairment (Bally & Kaplan, 1988). The Gallaudet program provided information about hearing loss for seniors with hearing impairment and communication partners, adjustment and coping strategies, consumerism and self-help, and an individualized AR program. This unique program allowed in-depth AR as well as peer and family support.

For decades, audiologists had used the term *phonemic regression* (Gaeth, 1948) to describe the inordinately poor speech recognition abilities of presbycusis patients. By the 1990s, a debate ensued regarding the factors that contributed to this poor speech perception. Three hypotheses were used to explain the speech understanding problems of the elderly (Humes, 1996). The peripheral hypothesis suggested that the speech recognition problems of the elderly could be explained by aging changes in the auditory periphery. The central auditory hypothesis focused on the structural and functional changes in the auditory pathways of the brainstem. The cognitive hypothesis suggested that cognitive slowing and deficits contributed to the poor speech understanding of the elderly population. A number of studies were published that provided evidence for each of these hypotheses.

Audiologists had a vested interest in this research because of the implications for rehabilitation of the aging presbycusis population. For example, the efficacy of putting hearing aids on elderly patients with central auditory processing problems was debated in the 1990s. Some studies suggested that older individuals with central auditory processing disorder would not benefit from amplification (Chmiel & Jerger, 1996). Furthermore, estimates of the prevalence of CAPD in the elderly population were as high as 70% for individuals over 60 years old (Stach, Spretnjak, & Jerger, 1990). Others, however, argued that diagnosing CAPD in the aging population was difficult because of these intervening variables and lack of a reliable sensitive test battery. By the early 21st century, it was well accepted that each of these complex factors could variably influence the speech perception abilities of aging individuals with hearing loss. Further, the relationship between aging, hearing loss, and cognitive moved to the forefront of multidisciplinary research.

The Early 21st Century: Training the Ear and the Brain?

In 1994, Carl Binnie made three predictions to provide a forecast for the evolution of AR in the 21st century. He projected that the population of the United States would grow to 280 million persons, with about 24 million to 28 million adults with hearing loss. The second prediction was that by the year 2000, 75% of all hearing instruments would be digitally programmable, allowing audiologists to adjust signal processing, and computer-based instruction would assume a significant role in AR. Clearly, his spot-on predictions were all realized.

Binnie's third prediction regarding computer-based instruction was particularly prescient, as seen in the widespread development and use of auditory training computer software in AR. This trend in AR was facilitated by three separate developments. First, a burgeoning body of research began to identify the intertwined roles of audition and cognitive processing particularly with the aging population. Second, discoveries in neuroscience began to demonstrate plasticity in the auditory system and brain as the result of training. Third, the use of personal computers and user-friendly software programs became an important, viable part of life in the 21st century. Concurrent with these developments, audiologists were increasingly recognizing the need for rehabilitation to support adjustment to high tech hearing aids. As such, auditory training was viewed with a new 21st-century perspective and was once again embraced as an important component of the hearing aid fitting protocol.

By the end of the 20th century, research on the role of cognition in speech perception showed that listening is not only a bottom-up process, but also a top down process that uses considerable cognitive resources. Ears allow us to hear passively, whereas the brain enables us to actively use what is heard (Pichora-Fuller, 2008). Given the cognitive changes that occur with aging (e.g., working memory, attention, reaction time, etc.), speech perception in aging adults with hearing loss has become the topic of ongoing research. Pichora-Fuller and Singh (2006) hypothesized that the extra cognitive effort required for complex speech perception tasks may be more difficult for older adults with lower cognitive abilities as compared to those with more cognitive resources. If listening effort increases for aging adults with hearing loss, fewer cognitive resources may be avail-

able for other cognitive processes such as retention and comprehension (Pichora-Fuller, 2008). Reduced cognitive processing ability as the result of hearing loss in aging individuals has rehabilitative implications. Consequently, the relationship between cognitive processing and listening continues to be a major research focus for cognitive psychologists and audiologists with a major focus on the aging population. (*Editors' note:* For further information about audition and cognitive processing, the reader is referred to Chapter 26 in this text.)

As more has been learned about the role of the brain in listening in the early 21st century, convincing research is demonstrating the plasticity of the brain as well. Animal research has provided evidence of central auditory system plasticity for decades. More recent research from studies of cortical activity in adults with hearing loss is providing evidence of cortical changes that parallel the findings from animal research (Neumann, 2005). Furthermore, increasing evidence is showing that changes in the auditory system can accrue as the result of auditory training. For example, Tremblay, Shahin, and Picton (2009) concluded that both stimulus-specific and general effects of auditory training could be measured in adults and further suggested that the N1 response might be a predictor of capacity for improvement. As research provides further physiologic evidence of changes in the auditory system as the result of training, questions about the effects of auditory training with adults with hearing loss both in the long term and the short term will be of paramount interest to audiologists.

Concurrent with research focused on cognition and neural plasticity, software developments in the 21st century and the widespread prevalence of personal computers have allowed auditory training to be transformed into a process Carhart would not have recognized in the 1940s. Since Carhart's time, audiologists have recognized that new hearing aid users face a daunting challenge when relearning to understand speech with amplification (Pichora-Fuller, Schneider, & Daneman, 1995). As such, computerized auditory training programs represent a 21st-century approach to providing traditional intervention. Some of the more established training programs like the Listening and Communication Enhancement (LACE) (Sweetow, 2005) have been incorporated clinically while other programs continue to collect data. At least 20 computerized auditory training programs are currently available for adults with hearing loss, each with unique characteristics. These computerized programs are designed to be individualized, self-paced, cognitively

challenging, and often completed at home. Evidence regarding the efficacy of these programs is ongoing but showing promise.

The synergy of the above-discussed developments has provided the impetus for the development of these auditory/cognitive training programs that address the unique needs of the aging population with hearing loss. Contemporary computerized auditory training programs reflect our increasing knowledge of brain plasticity, the role of cognition in listening, and the effects of aging. While auditory training was overlooked for decades, it now appears to have the potential to improve individual overall performance and satisfaction with amplification (Kricos & McCarthy, 2007). As smartphone applications (apps), tablets, and future e-delivery systems are developed, auditory training is more accessible than ever for adults with hearing loss.

21st-Century Focus on Patient-Centered Care

Twenty-first-century advancements in AR have not only been technological, but patient-centered as well. A global initiative sponsored by the Ida Institute has emphasized the role of counseling and motivational tools in the rehabilitation process. The Ida Institute is an independent, nonprofit organization located in Denmark and funded by a grant from the Oticon Foundation (Ida Institute, 2012). It is organized around the concept of collaborative learning and embraces a multidisciplinary approach to exploring the human dynamics associated with hearing loss.

The Ida Institute uses a “bottom-up” approach to create and share innovative, actionable knowledge that addresses the psychological and social challenges of hearing loss. Its seminars invite members from the audiology community and related disciplines to come together to advance thinking about hearing loss. It offers seminars, workshops, an e-learning lab, an interactive website, educational materials, and programs. The purpose of all of these initiatives is to develop insights and strategies that can be implemented in clinical and educational settings. In addition, the Ida Institute creates practical, easy-to-use tools to enable hearing care professionals to open a dialogue and develop a partnership with their patient.

The Ida Institute has taken a global, creative approach to AR by focusing on patient-centered care as the core of the rehabilitation process. In many ways, the work of the Ida Institute has identified the early

values of AR and updated them for contemporary, 21st-century practice. By emphasizing an understanding of the human dynamics associated with hearing loss, the Ida Institute has already taken a place in the history of AR as it assists audiologists in developing insightful approaches to working with persons with hearing loss.

21st-Century Changes in Delivery Models

Science, technology, and innovative rehabilitative strategies are hallmarks of the significant, positive changes in AR that have emerged in the first decade of the 21st century. Audiologists are embracing these ongoing advances and incorporating them into meaningful, efficacious rehabilitative tools. But how individuals receive their hearing health care has changed remarkably in at least three significant domains: the use of audiology assistants, the availability of personal amplification over-the-counter devices, and the availability of audiology in big-box stores.

The use of audiology assistants historically had been controversial as cogent arguments were developed on both sides of the issue. Those opposed to the use of audiology assistants feared that audiologists would be replaced by assistants whom employers viewed as “cheaper labor.” Proponents countered that using trained assistants would allow audiologists to increase their patient loads by freeing them from routine diagnostic and rehabilitative measures that could be delegated and supervised. As the discussion ensued, audiologists increasingly began to see the value of incorporating assistants into their practices. As such, audiologists who utilized trained assistants reported increased productivity, reduced wait times and costs, and enhanced patient satisfaction (Kasewurm, 2006).

As this model of hearing health delivery has become common practice, training and certification of audiology assistants have become the major focus. As such, state laws, regulations, and professional association guidelines vary, as do the tasks delegated to audiology assistants (ASHA, 2018). But any argument as to the benefit of using these support personnel has ceased. Indeed, the use of audiology assistants is a delivery model that appears to be a critical part of the rehabilitation process, particularly with hearing aid fittings and follow-up.

Technologic advances in amplification have triggered a change in the types of devices available to

those with hearing impairment in the 21st century. Whereas the development of personal sound amplification products (PSAPs) is ostensibly not for consumers with hearing loss, the reality is that PSAPs have become an option. As such, PSAPs have produced both opportunities and threats to audiologists and the patients they serve. PSAPs use much of the same technology as hearing aids but are not considered as hearing aids; consequently, they are not regulated by the FDA. While wearable PSAPs look like hearing aids and have similar component parts, they can be purchased by the consumer without medical clearance or an audiology evaluation. As such, there is considerable potential for consumer confusion regarding the differences in regulated hearing aids and PSAPs.

PSAPs have introduced no shortage of controversy as to their role in the adult AR process. While PSAPs are affordable and accessible, their lack of regulation has raised red flags about their potential safety. The electroacoustic properties, including not only gain and output but also distortion levels, vary and are inconsistent across models. PSAPs, therefore, present as a two-pronged concern, as individuals with hearing loss may not only self-diagnose hearing loss, they may also acquire PSAPs that are inappropriate for their hearing loss. Further, if the PSAP does not provide adequate and appropriate amplification, the consumer may not recognize the benefits of professionally fit amplification.

Some audiologists have taken a “glass half full” approach to PSAPs and view them as one more device on the continuum of amplification devices available to patients. In the vernacular, PSAPs have been referred to as “gateway” devices that will ultimately lead consumer with mild hearing loss to the use of appropriately fitted hearing aids. Indeed, some audiologists have incorporated PSAPs into their practices.

PSAPs are not the center of the AR process but rather represent a cure-all for communication problems caused by hearing loss. Audiologists know the hearing aid is only the nucleus of the AR process surrounding the hearing aid fitting. Indeed, the amount and quality of rehabilitative follow-up can often make or break the success use of amplification. Therefore, there is little reason to think that consumers’ use of PSAPs requires no rehabilitative follow-up.

The current PSAP self-purchase model ignores the essential value of AR and consequently is likely to be only a temporary solution for the adults with hearing loss who purchase them. If history predicts the future, PSAPs may go the way of disposable hearing aids. We have learned that consumers with hearing loss want not only high-tech devices, but also “high-

touch” devices in dealing with the communication problems resulting from their hearing loss.

The advent of the big-box chain stores has changed the way consumers in the early 21st century purchase food, drink, clothing, and household products, and has allowed these stores to also offer hearing aids and audiology services. The convenience and cost of their audiology services has made big-box stores attractive to consumers with hearing loss. While there are a number of big-box retailers, this recent development in hearing aid delivery is referred to as the “Costco effect” (Lee, Barrett, & Samuel, 2017). Indeed, Costco alone has acquired a large market share of hearing aid sales with per-unit sales growth 3 to 5 times greater than seen with other practice models (Strom, 2015). This rapid growth in big-box hearing aid dispensing has been perceived as a threat to audiologists, particularly those involved in private practices and clinics that focus on hearing aid dispensing and AR.

Big-box hearing aid service delivery is a relatively new phenomenon that audiologists are just beginning to face. It is easy to visualize the Costco effect as a threat that will negatively change the practice of hearing aid dispensing. However, viewing this development through an opportunity lens, some are projecting that Costco could actually expand the hearing aid market rather than shrink the share of private practices and clinics that dispense hearing aids. Indeed, Lee, Barrett, and Samuel posit that the accessibility of hearing aids at Costco may move people with hearing loss to seek out professionals outside the big-box store. Further, they suggest a strategy of treating Costco as an ally and referral source rather than a competitor. Clearly, the big-box strategy depends on sales volume; consequently, the range of services offered is narrow when compared to the scope of practice of audiology.

Viewing the Costco effect as an opportunity is consistent with what the history of audiology has shown us. In the 1990s when audiologists assumed a device-driven philosophy, hearing aid sales did not soar. Instead, audiologists learned of their own value when the device is only a part of the rehabilitation process. As we look back at our profession in the future, perhaps we will be able to see how the threat of big-box hearing aid dispensing compelled audiologists to focus on their education, scope of practice, and patient-centered care rather than sales-centered numbers.

On the horizon is the development of the over-the-counter (OTC) model of hearing aid delivery. As a result of the FDA Reauthorization Act of 2017 (H.R.

2430), the FDA has been charged with the development of standards for the safety and delivery of OTC hearing aids designed for adults with mild to moderate hearing loss. At the time of this writing, it is still unclear how this new amplification delivery model will affect audiologists and consumers with hearing impairment. However, looking back through history, it is doubtful that the user of OTC devices will obtain optimal results without the contribution of AR.

Conclusions

Can we predict the future of AR with the same accuracy as Binnie did in 1994? Perhaps not. But to paraphrase Lord Byron, our past is the best predictor of how audiologic rehabilitation will fare in this quickly changing new world of technology, market-driven economies, a shrinking global environment, and changing demographics. Despite the challenges and threats to AR, it remains the most important part of what audiologists do and is at the heart of patient-centered care. The historical review provided in this chapter reinforces the belief that the strong rehabilitative roots of audiology will help us shape our future. Returning to a patient-centered rehabilitative profession with a holistic focus on improved communication, rather than a device-centered profession, will continue to ensure that audiology will meet the hearing health care needs and expectations of our patients well into the 21st century.

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2

DEFINING AUDIOLOGIC REHABILITATION

Joseph J. Montano

Should one query audiologists about the provision and definition of audiologic rehabilitation (AR), it would soon become apparent that perceptions vary greatly. Some may deny they provide AR services because their clinical function is purely diagnostic, while others may believe that everything they do as an audiologist is rehabilitative, including the assessment of hearing. AR seems to have an ambiguous nature and, while most published definitions imply it is an inclusive process, most practitioners view it as exclusive: a specialty area distinctly different from diagnostic audiology. It is not uncommon to hear audiologists say they do not provide AR services, even when they dispense amplification and hearing assistive technology systems (HATS). As a result, AR is often thought of as being restricted to the clinical provision of services such as auditory training and speechreading. While these procedures are certainly components of the AR process, they do not define it.

These are changing times in the audiology profession. As service delivery models change, many audiologists are seeking new identities. Procedures once thought of as staples to practice are being encroached upon by various entities. If ever there was a time when AR needs to be defined and practiced, it is now. Since the publication of the previous edition of this text, technological developments in amplification such as products like hearables, wearables, self-fitting hearing aids and, most notably, over-the-counter hearing aids (OTCs) have begun to flourish. Perhaps, then, it is time for audiologists to return to their roots

and make AR not a specialized service, but rather a standard of practice.

The origin of the field of audiology is largely believed to be a direct outcome of the services provided to military personnel suffering from noise-induced hearing loss during World War II (Alpiner & McCarthy, 2014; Bergman, 2002; Ross, 1997). The programs that developed were rich in AR services that included hearing aids, counseling, auditory training, and speechreading. For a number of reasons, as the profession evolved, emphasis shifted from rehabilitation to diagnostics. AR seemed to have gotten lost in the excitement of new technological developments such as auditory brain stem response (ABR) and otoacoustic emissions (OAEs). Even in academic programs, AR classes were usually limited to one, or perhaps two, and seemed to focus primarily on speechreading and auditory training (Ross, 1997). Although AR diminished in popularity, acknowledgment of its importance remained. The provision of AR services, in fact, is one of the most relevant characteristics that distinguish the practice of audiologists from hearing aid specialists in the amplification arena. Ross goes on to state, "When it becomes necessary for us to justify our existence as a unique profession to various health, education and governmental agencies, we never fail to claim this activity [AR] as our own" (1997, p. 14).

Ross' words could not have been truer when, in the spring of 2016, in response the President's Council of Advisors on Science and Technology's report on

age-related hearing loss (PCAST, 2015), the Food and Drug Administration (FDA) held a series of meetings on streamlining regulations for good manufacturing practices for hearing aids (FDA, 2016). Representatives from various professional and consumer organizations such as American Speech-Language-Hearing Association (ASHA), American Academy of Audiology (AAA), American Academy of Otolaryngology—Head and Neck Surgery (AAO-HNS), and Hearing Loss Association of America (HLAA) presented their opinions on OTC hearing aids and included statements stressing the importance of AR in the hearing aid attainment process. There seemed to be no disagreement as to the importance of rehabilitative efforts in the management of hearing loss. The challenge for all of us, then, is to increase the inclusion of AR in audiology practice.

Many practitioners still believe AR to be the roots of our field and thus have pursued research and practice in this area. Alpiner and McCarthy (2014; also see Chapter 1 in this edition) discuss the evolution of AR in the field of audiology. As we study the history of AR from its origins through the establishment of professional organizations such as the Academy of Rehabilitative Audiology (ARA), ASHA and the latter's Aural Rehabilitation and its Instrumentation Special Interest Group (SIG 7), and the evolution of professional organizations like the AAA and the Academy of Doctors of Audiology (ADA), we are able to understand better the development of the definitions that provide important clinical direction.

Perhaps reviewing the existing definitions of AR can shed some light on the process and attempt to unify the perceptions of audiologists. With this in mind, the purpose of this chapter is to explore the variety of definitions of AR and identify barriers to the provision of AR services.

Who Provides Services in AR?

Many consider AR to be the true marriage of audiology and speech-language pathology, particularly with respect to service delivery for children with hearing loss. Membership in both ARA and ASHA SIG 7 consists of both audiologists and speech-language pathologists (SLPs), with many holding degrees and certification in both professions. Given the nature of AR, the roles of SLPs and audiologists can be complementary and cooperative (ASHA, 2001). This overlap, however, may have contributed to confusion and misunderstanding among many audiologists. AR

falls within the scope of practice of both professional groups (ASHA, 2018) and although knowledge and skills may have been delineated (ASHA, 2001), third-party reimbursement for service provision in AR is distinctly different. Audiologists are rarely, if ever, reimbursed for providing these services. In fact, it is often cited as the reason audiologists do not provide AR. Within the realm of third-party reimbursement, many AR services are covered only when performed by SLPs because this field of practice is classified as both diagnostic and rehabilitative while audiology is considered an exclusively diagnostic profession through the Centers for Medicare and Medicaid Services (CMS) (<https://www.cms.gov>).

Strides have been made to improve the reimbursement of AR services for audiologists, and as a result, the current procedural terminology (CPT) codes were established in 2006. Kander and White (2006) described the following codes: 92626: evaluation of auditory rehabilitation status, first hour; 92627: each additional 15 minutes, on same day as 92626; 92630: auditory rehabilitation of children; and 92633: auditory rehabilitation of adults. The codes developed for adults are more specifically aimed at adult cochlear implant recipients.

Reimbursement for audiology services provided under Medicare is limited to diagnostic services only, with limited coverage for treatment. Procedures for auditory training and speechreading are still not covered under Medicare for audiologists, even though the SLP can apply for reimbursement for these services by using the treatment code of 92507. Efforts are continuing to expand Medicare coverage for audiology services that would include the provision of AR and are included as part of the current ASHA Public Policy Agenda (ASHA, 2019a).

Not only is there confusion about who provides AR services, but basic nomenclature is also at issue. What should this process be called? Alpiner and McCarthy (2014) discuss the terminology paradox that is currently present in our field. The terms *aural rehabilitation*, *audiologic rehabilitation* and, most recently, *auditory rehabilitation*, are often used interchangeably. SLPs prefer to use the term *aural rehabilitation*, while audiologists have more consistently referred to these services as *audiologic rehabilitation*. One need only refer to the ASHA Preferred Practice Patterns (PPP) for the Professions of Speech-Language Pathology (ASHA, 2004) and Audiology (ASHA, 2006) to illustrate the difference. PPPs for speech-language pathology refer to services provided to individuals with hearing loss as aural rehabilitation; meanwhile, the audiology PPPs use the term *audiologic rehabilitation*.

tation. It is for this reason that the acronym AR was employed in the ASHA(2001) knowledge and skills publication on aural/audiologic rehabilitation. Currently, within the ASHA Practice Portal (2019b), adult aural rehabilitation is identified as an area of practice for both audiologists and SLPs with practice delineations provided.

What Is AR?

This question leads to a discussion of defining AR. A review of the literature reveals an abundance of definitions that seem to vary significantly historically. Gagné and Jennings (2009) reported that, while some definitions stress the activities associated with the rehabilitation of hearing loss, others focus on the reduction of associated disabilities and handicaps. The emphasis of AR has shifted from procedure-specific to a more patient-centered concept. Schow et al. (1993) reported on a trend of decreasing traditional treatments in AR (auditory training and speechreading), but an increasing perception by audiologists that their clinical duties included both diagnostic and rehabilitative services. Prendergast and Kelley (2002) postulated that this change was primarily due to a shift from traditional procedural AR approaches toward services designed to address emotional and social aspects of hearing loss. This belief can be substantiated when we compare the definitions used to describe AR throughout the years.

In 1971, Sanders published the first edition of his seminal book, *Aural Rehabilitation*. This text, along with subsequent editions, introduced a generation of audiologists to AR. The common thread throughout the chapters of this book was the importance of an individual's overall communication ability. He reported that approaching assessment from a theoretical framework that encouraged rehabilitation was just as critical for individuals with mild to moderate hearing loss as it was for those with severe to profound loss. This was a change in thinking at the time since the lion's share of AR services were offered only to those with more severe hearing impairments. Assessing individual areas of strengths and weaknesses would help identify abilities that were in need of improvement, making it possible to "circumvent the weaknesses through the development of the use of compensatory channels" (Sanders, 1971, p. 5). Included in this text were chapters dedicated to various topic areas: auditory and visual perception, amplification and hearing aids, auditory training,

visual communication training, and the integration of vision and audition. Sanders stressed the importance of not limiting our services to just the assessment of hearing.

ASHA (1984) published a position statement that discussed the definition of, and competencies for, aural rehabilitation. Developed by the Committee on Rehabilitative Audiology, the paper revisited the American Speech-Language-Hearing Association's 1974 legislative council resolution that supported the audiologist as the primary provider of AR services and their role in the supervision of such services. It was felt that in practice, however, the SLP was providing a significant proportion of AR services for individuals with hearing loss. As a result, the committee created a new definition of AR and included the components necessary for service provision: "Aural rehabilitation refers to services and procedures for facilitating adequate receptive and expressive communication in individuals with hearing impairment" (ASHA, 1984, p. 23). The authors went on to describe the elements necessary to accomplish the goals of AR. These included evaluation of sensory capabilities, fitting of auditory and sensory aids, counseling, and referral. The importance of counseling both the individual and family regarding the impact of hearing loss began to become a prominent message in the definition of AR.

While this position statement acknowledged the interdisciplinary nature of AR, it did nothing to help delineate the difference between the audiologists and SLPs in service delivery. In fact, it appeared that one of its purposes was to negate the audiologists' primary role in AR that had been established earlier in 1974. Although perhaps unintentional, the use of the terms *expressive* and *receptive communication* within the core context of the definition, while certainly accurate, may have created the appearance that AR is more directly associated with speech-language pathology. The terminology, while not exclusive to speech and language, is often associated with child language development or adult neurologic language processes. The paper appeared at a time when audiologists were seeking a more independent professional identity and was probably indirectly responsible for the increased popularity of the label *audiologic rehabilitation*.

Not long after the publication of this position paper, the creation of Special Interest Divisions of the ASHA (now referred to as Special Interest Groups or SIGs) became a reality. These divisions were established to meet member needs for specialized areas of professional practice. Division 7 (Aural Rehabilitation and its Instrumentation) was among the first

groups developed. The Steering Committee of Special Interest Division 7 (1992) contributed an article to the ASHA magazine that focused attention on some of the beliefs of their Division and stated simply that "Audiologic rehabilitation was Audiology" (p. 18). The implication, of course, was that the entire field of audiology was designed to be rehabilitative in nature. Their definition includes the importance of the impact of hearing loss on function within the context of the family and environment. These concepts would soon become primary objectives for AR.

Erdman (1993, 2000) stated that the "ultimate goal of rehabilitative audiology is to facilitate adjustment to the auditory and non-auditory consequences of hearing impairment" (p. 374). This brief description of AR begins to show the shifting emphasis in rehabilitation models from procedure-specific definitions to more function-driven descriptors. Here, the author places equal value on the auditory (e.g., hearing loss degree and nature, speech perception, etc.) and non-auditory (e.g., emotional, psychosocial, vocational, etc.) implications associated with the loss of hearing ability and goes on to discuss the importance of counseling in the rehabilitation process. In fact, Erdman (2000) remarks that counseling is the "essence of successful rehabilitation" (p. 435) of AR.

It became important that our profession begin to expand its definition of AR beyond the evaluation and management of hearing loss to include the impact of psychosocial functioning (Ross, 1997). The increase in popularity of self-assessment measurement tools, such as the Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982), the Hearing Performance Inventory (Giolas et al., 1979), and the Communication Profile for the Hearing Impaired (CPHI) (Demorest & Erdman, 1987), provided the audiologist with an arsenal that could be used to quantify the impact of adult hearing loss. The measures could yield a basic classification function; for example, the HHIE could be used to determine a level of hearing handicap or, more broadly, the CPHI could be used to provide a profile that is an effective counseling tool. Regardless of measurement focus, the direction of assessment for AR was tending to shift more toward the implications of a loss of hearing on numerous domains of function.

Ross (1997) reported that the process of AR includes "any device, procedure, information, interaction, or therapy which lessens the communicative and psychosocial consequences of a hearing loss" (p. 19). Like Erdman (1993), Ross seems to place equal emphasis on issues related to the auditory and psychosocial aspects of functioning. Without specifying

further, he relegates any procedure appropriate to lessening the impact of hearing loss on communication function to AR. Certainly, one could include traditional activities such as speechreading and auditory training as treatment components, but one could just as easily consider counseling, vocational assessment, and family intervention.

Gagné (1998) continued to emphasize function over procedure when he defined AR as the process designed to "eliminate or reduce the situations of handicap experienced by individuals who have a hearing impairment and by persons with normal hearing who interact with those individuals" (p. 70). The author was quite specific about the inclusion of others within the communication environment of the individual with hearing impairment, an approach that can be directly related to a landmark publication under development at the time by the World Health Organization (WHO).

The World Health Organization and Its Impact on the Definition of AR

In 1980, the WHO published its first classification of disability with the International Classification of Impairment, Disability, and Handicap (ICIDH). It attempted to unify related terminology in the realm of handicap and disability. This original paper identified "impairment" as having an impact on the organ level and "disability" as related to activities and handicap referring to one's role in society. During the late 1990s, WHO began to revisit this classification system and subsequently published the International Classification of Functioning, Disability, and Health, commonly referred to as the ICF, in 2001. The impact of the WHO classifications is covered in depth in Chapter 3 of this text.

The ICF described both health and health-related domains in relationship to the body, the individual, and to society at large. The primary areas of concern within the ICF are body functions and structures and what is referred to as activities and participations. The ICF describes "what a person with a disease or disorder does do or can do." While the ICIDH seemed to highlight the negative consequences of impairment, the ICF emphasizes the positive (Boothroyd, 2007).

The influence of the WHO ICF can be directly observed in a publication prepared by the ASHA Working Group on Audiologic Rehabilitation (ASHA, 2001). This group was charged with the task of updating the ASHA's 1984 position statement (discussed earlier) on the definition of, and competencies for, AR.

The interdisciplinary nature of AR was evident in the group membership, which consisted of both audiologists and SLPs. Unlike the 1984 document, this group decided to directly address the issues related to the provision of services by two distinct professions and, essentially, created two documents within one.

The format of the 2001 publication consists of an introduction that defined AR and provided background and history. The specific areas of basic knowledge and specific knowledge and skills are divided into two sections: (a) knowledge and skills for audiologists providing AR services, and (b) AR knowledge and skills for SLPs. The paper specifically delineates the roles of the two professions; while collaborative, these roles are distinctly different.

The authors emphasize that AR no longer refers to simply procedure-specific treatments such as auditory training, speechreading, or even hearing aid dispensing, but rather, it is a broad process with tentacles reaching into all aspects of a person's functioning. With this in mind, the working group proposed a definition of AR: "Audiologic/aural rehabilitation (AR) is an ecological, interactive process that facilitates one's ability to minimize or prevent the limitations and restrictions that auditory dysfunctions can impose on well-being and communication, including interpersonal, psychosocial, educational, and vocational functioning" (p. 2).

The WHO ICF (2001) terminology includes descriptors such as activity and activity limitations, and participation and participation restrictions. In addition, this publication stresses the importance of contextual factors, such as environmental and personal influences, on an individual's ability to function with an impairment. Inclusion of such terms as *ecological*, *interactive*, *minimize or prevent limitations* (activity), and *restrictions* (participation) reinforce the impact of the ICF on this ASHA (2001) document.

The knowledge and skills for audiologists providing AR services addressed by ASHA (2001) include areas of general knowledge, such as psychology, human growth and development, cultural and linguistic diversity, and quantitative research methodologies. Within the area of basic communication processes, those recognized are knowledge of anatomy, physiology, speech and hearing sciences, linguistics, psycholinguistics, and dynamics of interpersonal skills.

Table 2-1 includes the special areas of knowledge and skills identified in the ASHA (2001). The skills described are meant to educate the audiologist on the specific areas one should develop to practice AR. As is evident through a review of this table, the list is

quite inclusive and does not distinguish between services for children and adults. A similar presentation of information was developed for SLPs, but will not be covered in the context of this chapter.

Building on this definition, the ASHA (2006) referred to AR in its preferred practice patterns for audiology as

a facilitative process that provides intervention to address the impairments, activity limitations, participation restrictions and possible environmental and personal factors that may affect the communication, functional health and well-being of persons with hearing impairment by others who participate with them in those activities. (Section 15, p. 40)

Boothroyd (2007) advocated a holistic approach to AR. He defined this as "the reduction of hearing loss-induced deficits of function, activity, participation and quality of life through a combination of sensory management, instruction and perceptual training and counseling" (p. 63). The WHO ICF is acknowledged as a major influence on the purpose of his paper. As was evident in the ASHA (2001) definition, specific terminology (e.g., function, activity, participation) directly aligns Boothroyd's philosophy with the trend in the profession to an expanded view of AR: a shift from primarily procedural activities to the therapeutic process aimed at reducing the impact of hearing loss on function (ASHA, 2001; Erdman, 1993; Gagné, 1998; Ross, 1997).

A New Definition of Audiologic Rehabilitation

As we continue to expand the scope of services we include in the process of AR, our definitions will need to be reviewed and revised to allow for changes occurring in our professional practices. The following, which appeared in the second edition of this text, is what is believed to represent the current state of AR: "AR is a person-centered approach to assessment and management of hearing loss that encourages the creation of a therapeutic environment conducive to a shared decision process, which is necessary to explore and reduce the impact of hearing loss on communication, activities, and participations" (Montano, 2014, p. 27). As we review trends in audiology practice, there is a growing body of evidence to expand the concept of person-centered care to family-centered care. The inclusion of family in the rehabilitation process has been shown to improve

Table 2-1. Special Areas of Knowledge and Skills for Audiologists Providing Audiologic Rehabilitation (AR) Services

Knowledge	Skills
Auditory System Function and Disorders	Identify and describe disorders of auditory function
Developmental Status, Cognition, and Sensory Perception	Assess client's preferred mode of communication; verify visual acuity; identify need for assessment of cognitive skills, sensory perception, developmental delays, academic achievement and literacy; determine need for referral
Audiologic Assessment Procedures	Case history; otoscopic exam; conduct/interpret behavioral, physiologic or electrophysiologic evaluations; administer self-report measures; conduct and assess for APD; ¹ identify need for referral
Speech-Language Assessment Procedures	Identify need for and perform screenings; describe effects of hearing loss on speech and language development; provide appropriate measures of speech, language and voice production; administer and interpret measurements of communication skills in auditory, visual, auditory-visual, and tactile modalities
Evaluation and Management of Devices and Technologies	Electroacoustic characteristic of devices and technologies; describe, perform, and interpret behavioral/psychophysical aided measures; conduct appropriate fittings; monitor fittings; perform routine visual; troubleshoot common causes of device malfunctioning; describe effects of devices on communication and psychosocial functioning; plan and implement program of orientation; conduct routine assessment of adjustment to and effective use of amplification; monitor outcomes
Effects of Hearing Impairment on Functional Communication	Identify and evaluate situational expressive and receptive communication needs; identify environmental factors that affect communication performance; identify the effect of interpersonal relations on communication function
Effects of Hearing Impairment on Psychosocial, Educational, and Occupational Functioning	Describe/evaluate impact of hearing loss on psychosocial development and functioning; describe systems and methods of educational programming; describe/evaluate impact of loss on vocational status; identify effects of hearing problems on the marital dyad and family dynamics; identify need for and provide for counseling in relation to hearing impairment and communication difficulties; provide assessment of family members' perception of and reactions to communication difficulties
AR Case Management	Use effective interpersonal communication in interviewing and interacting with clients and family; describe client-centered, behavioral, cognitive, and integrative theories and methods of counseling and their relevance in AR; provide appropriate individual and group adjustment counseling; provide auditory, visual, and auditory-visual communication training; provide training in communication strategies; provide appropriate expressive communication training; provide appropriate technological intervention; provide appropriate intervention for management of vestibular disorders; develop and implement an intervention plan; develop and implement a system for measuring and monitoring outcomes
Interdisciplinary Collaboration and Public Advocacy	Collaborate effectively as part of multidisciplinary teams; plan and implement in-service and public information programs; plan and implement parent education programs; advocate implementation of public law in educational, occupational, and public settings; make appropriate referrals to consumer-based organizations
Hearing Conservation/Acoustic Environments	Plan and implement programs for prevention of hearing impairment; identify need for and provide appropriate hearing protection devices; monitor the effect of environmental conditions; measure and evaluate environmental acoustic conditions

¹APD = Auditory Processing Disorder.

Source: Adapted from the American Speech-Language-Hearing Association (2001).

health-related and audiology outcomes (Laplante-Lévesque et al., 2010; Meyer et al., 2014; Rathert et al., 2013; Singh & Launer, 2016). As a result, a slight modification to the Montano (2014) definition of AR would be the influence of family-centered care rather than limiting it to patient-centered care.

Montano (2011) emphasized the importance of relationship building in the AR process. He believed too much emphasis had been placed on technology and instrumentation, with the person actually living with hearing loss playing an almost secondary role. Figure 2-1 illustrates a technocentric model of audiology service delivery. This model focused attention on the hearing loss with the services provided revolving around the technology: the audiometer to assess the degree and nature of the loss, hearing aids to improve perception of sound and speech, hearing aid orientation to ensure the patient is able to successfully manage the hearing loss, real-ear verification, and, finally, technological accessories such as streamers and HATS to increase communication. This model appeared to

concern itself more with the results appearing on the audiogram than the impact of those results on patient function.

A patient-centered approach to treatment is illustrated in Figure 2-2. This model had counseling at its core and emphasized a cooperative patient relationship in the services provided. Montano (2012) acknowledged the contributions of technology even in this AR approach, but believed they should be considered tools to assist in the patient's ability to adjust to hearing loss rather than the sole purpose of the professional interaction. The figure highlights areas of assessment such as the patient history, recommending it be performed as a narrative rather than a group of closed-set intake questions. The patient narrative is an effective way to establish a rapport with patients and is a major component of patient-centered treatment (Erdman, 2014; Gagné & Jennings, 2011).

The model continues to highlight more interactive treatment processes, including the use of self-assessment, discussion of communication strategies,

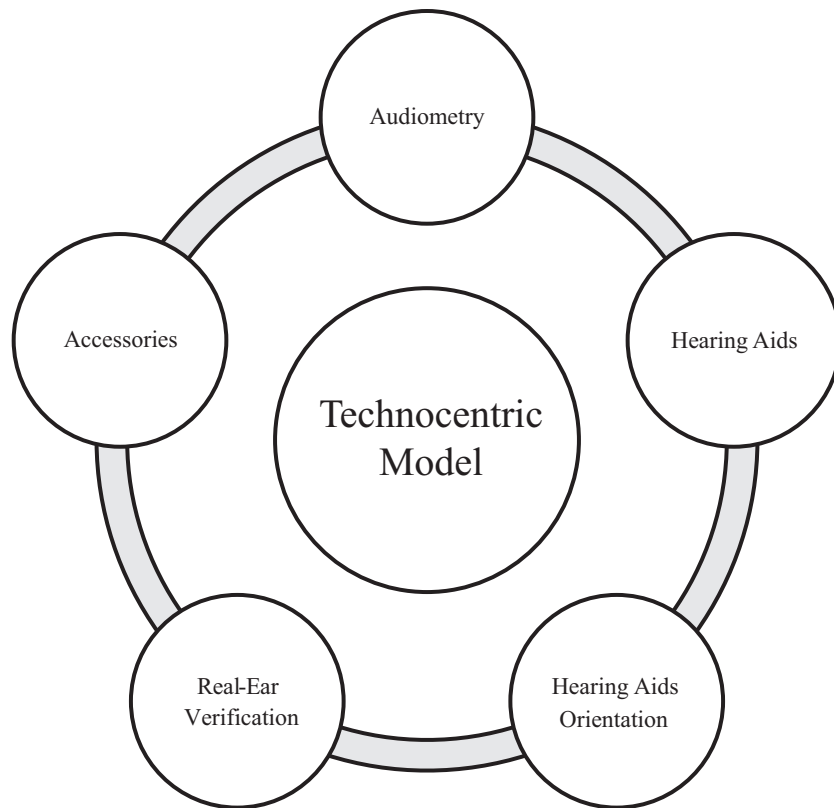


Figure 2-1. The technocentric model of audiology practice. *Source:* Montano, J. J. (2011). Adapted from "Building relationships: An important component to the aural rehabilitation process." *ENT and Audiology*, 20(4), 91–92.

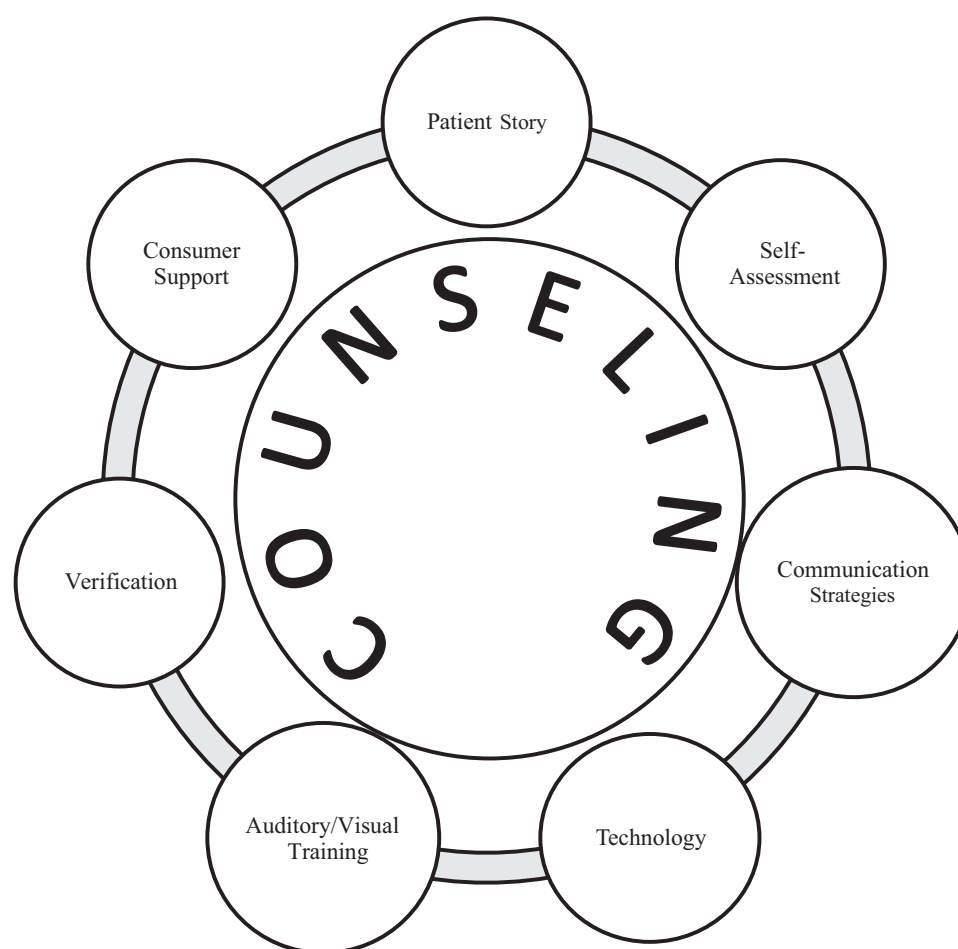


Figure 2-2. The person-centered model of audiology practice. *Source:* Montano, J. J. (2011). Adapted from “Building relationships: An important component to the aural rehabilitation process.” *ENT and Audiology*, 20(4), 91–92.

auditory-visual communication, and consumer support. Although emphasizing person-to-person interaction, it does not diminish the importance of amplification technology and verification of hearing aid performance.

Where and When Is AR Found?

AR programs have traditionally been found in settings such as university programs and Department of Veterans Affairs medical centers that not only provide clinical services, but have also been responsible for much of the published research in this area. Even though AR has been shown to be cost-effective (Abrams et al., 2002) and is identified as a desired ser-

vice to consumers (Pope & Stika, 2009), its availability in most practice settings is limited.

AR can, however, be effectively implemented in a variety of audiology employment sectors with the use of some creative programming and clinical intuition. Madell and Montano (2000) discussed the inclusion of AR services in a variety of work settings and provided suggestions for program development. In particular, these authors encouraged the provision of AR in locations like private practice and small clinical settings, hospitals, and long-term care facilities. Jessen (2019) developed a 5 Keys AR program, which she has marketed to audiologists, encouraging the use of AR strategies within private practice settings.

Warner-Czyz (2000) provided suggestions for program designs for group AR that can be successfully implemented in hospitals, small facilities, and

private practice offices. She postulated that the lack of AR services in these settings is primarily due to constraints such as time and financial concerns, lack of interest among consumers, and lack of information and confidence in the provision of services by audiologists. The Group Rehabilitation Online Utility Pack (GROUP) developed through the Ida Institute (2012) provides audiologists with an interactive resource for developing and implementing group AR programs in their practices. (*Editor's note:* For more information on GROUP and group AR, the reader is referred to Chapter 18.)

Jennings and Head (1994) discussed an ecological approach to service provision for long-term care. In their model, the audiologist serves a multipurpose role, and programming includes the individual, along with the family, staff, and the environment. Montano (2001) encouraged the use of creative in-service programming as a means to educate staff that work closely with nursing home residents with hearing loss. In such locations, it is unusual for staffing to include a full-time audiologist. Commonly, an audiologist is only available on either a part-time or consultation basis, if at all. It is therefore necessary to recruit into the process the existing staff who work with these clients on a day-to-day basis. This can be done in training sessions that take place with small hands-on groups working directly with the client, rather than in the classroom. Management of hearing aids, troubleshooting techniques, and application of communication strategies are all important topics for staff training. In addition, Montano (2001) encouraged discussion of the psychosocial impact that hearing loss may have on the residents in a nursing home. The use of hearing loss simulation software, such as Hearing Loss and Prosthesis Simulator HELPS (Sensimetrics, 2006), can provide a means for the caregiving staff to understand the potential issues their patients can face with hearing loss.

Specialized hearing and speech centers, such as the Center for Hearing and Communication (formerly the League for the Hard of Hearing), have long provided AR services for their communities. If it is not feasible to offer these services directly in one's practice, it becomes the responsibility of the audiologist to make the necessary and appropriate referrals for the service when available. Programs like the Gallaudet peer mentoring program (Bally & Bakke, 2007) have been created to train qualified consumers (i.e., individuals who are hard of hearing or deaf) as audiologist extenders in pertinent areas related to the management of hearing loss. With this knowl-

edge, audiologist extenders are capable of becoming an important resource for the audiologist and can be of assistance to others who are beginning to adjust to hearing loss. Many of the individuals who have begun to participate in this program are members of the HLAA. As a result, the trained persons can serve an additional purpose by educating individuals on the existence of self-help groups and can be a source of information on coping strategies for people with hearing loss. Barlow et al. (2007) reported that individuals with late onset of deafness reported mixed response to the services provided by health care providers, but were enthusiastic about and greatly valued the peer support and resources they gained from programs offered by other individuals who are deaf or hard of hearing. It is interesting to note, however, that the fact that such a peer mentoring program has been established suggests that it has become necessary to address an unmet need for consumers with hearing loss. This may be an example of the audiologist being constrained by schedules and lack of resources and the consumer needing to take up the slack.

Hospital settings can often prove to be a challenging environment for the establishment of AR programs (Tye-Murray et al., 1994). With the major focus on medical diagnostic audiology, therapeutic intervention for hearing loss is often unavailable. Given the interdisciplinary nature of the hospital environment, however, it could actually be an ideal setting for AR. Hospitals strive to provide state-of-the-art care for their patients and the availability of technologically advanced amplification and HATS can fit nicely into a modern outpatient facility. In addition, the use of HATS for inpatients with hearing loss provides an essential AR element to care. One resource that is particularly helpful, if establishing a hospital-based program, is the availability of referral personnel such as psychologists, social workers, and vocational rehabilitation specialists.

Even when AR services are not offered in a practice, it becomes our responsibility to make the appropriate referrals as necessary. This may mean sending patients to local SLPs who specialize in AR, universities, or even other practices where the services are offered. With a growing trend toward telepractice, AR may be available from distant locations, allowing patients to participate easily from their home or even within a practice setting. Current license restrictions prevent easy access to services in other states, but efforts are underway to ease licensure restrictions.

Regardless of the work environment, AR should and can be a part of any audiology practice. The

extent of services offered will, of course, depend largely on the work environment and the resources available. At any rate, when one attempts to identify a common thread that runs through the AR definitions previously discussed, counseling appears at the forefront. The counseling process is all-inclusive and begins the moment the audiologist first greets his patient in the waiting room. If one keeps in mind the need to help patients adjust to hearing loss and reduce the limitations and restrictions associated with it, then counseling should take a prominent position in one's practice.

How Can We Provide AR Services?

The ability to provide services in AR may be directly associated with the delivery model used in practice. Erdman et al. (1994) characterized the service delivery models used in audiology as either medical or rehabilitation models; more recently, Erdman (2014) conceptualized these models as biomedical or biopsychosocial. The medical/biomedical model is considered a top-down approach where the clinician provides the assessment and determines the audiologic diagnosis and treatment options. In this model, the clinician is characterized as doing something "to" the patient and makes the decisions necessary for proper rehabilitation. The client's role is passive and his/her actions are directed by the clinician. It assumes that it is the clinician who knows what is best for the client and is based on the premise that hearing loss is disease- or pathology-oriented. This model is most often associated with acute conditions.

The rehabilitation/biopsychosocial model of service delivery takes a more horizontal approach and is characterized as more interactive and facilitative. Here, the clinician helps identify problems and works with the client toward resolution. This model is most often considered when dealing with chronic conditions and assumes the client's needs and perceptions will influence treatment strategies. In this model, the client takes an active role in his/her rehabilitation and its focus shifts toward functional performance, including activities and participations.

Providing AR services in clinical practice frequently involves more than the identification of the problem and perhaps provision of amplification systems to improve auditory performance. Hearing loss often contains many layers in need of treatment. Providing a hearing aid may simply address one area of need (audibility), without satisfactorily dealing with

the imbedded problems associated with adjustment. In order to provide the necessary comprehensive remediation, multiple aspects of program provision may be necessary. Hearing aid delivery may fall short without the provision of hearing aid orientation. Communication strategies, speechreading, and auditory training may enhance communication performance. Access to peer support groups such as through the HLAA may add an important dimension to an AR program,

The effectiveness of many programs in AR has been evaluated and audiologists are now beginning to develop an arsenal of evidenced-based practice studies to justify their services. Reviews of practice areas such as counseling-based groups (Hawkins, 2005), amplification (Cox, 2005), individual auditory training (Sweetow & Palmer, 2005), and most recently, a summary of a variety of adult AR services (Hickson, 2009) are continuing to provide this important foundation. Evidence supporting the effectiveness of our services will help further the quest to find acknowledgment of the importance of AR by third-party payers. Wong and Hickson (2012) published an entire text dedicated to evidenced-based practice in audiology with emphasis on the intervention and treatment.

Implementation can be difficult in many of our work environments, but even in busy hospital clinics or private practice, creative solutions can be developed to provide AR for patients. Tye-Murray et al. (1994) provided a number of suggestions for AR program provision including home training, client libraries containing relevant reading materials and DVDs, and established assistive device centers where technology can be displayed and demonstrated.

Sweetow and Sabes (2007) identified AR as a growing area of interest among audiologists. Computer-based programs that can assist the audiologist in the provision of individualized auditory/auditory-visual training are also attracting attention, as illustrated by the popularity of a variety of programs: CasperSent (Boothroyd, 1987), an example of computer-assisted speech perception testing and training at the sentence level; CAST, a computer-aided speechreading training program (Pichora-Fuller & Benguerel, 1991); LACE, a tool for listening and communication enhancement (Sweetow & Sabes, 2006); Seeing and Hearing Speech, a lipreading software (Sensimetrics, 2008); and ReadMyQuips (Levitt et al., 2011). More recently, Tye-Murray (2016) introduced a computer-based AR program called cLEAR (customized learning: Exercises for Aural Rehabilitation) that provides the user with a gamelike structure for auditory training. With the growing emphasis on family-

centered care, cLEAR can include a family member in the rehabilitation process by allowing their voice to be used for the training materials. This can be particularly useful as the training voice will be a communication partner of the person with hearing loss.

Spitzer (2000) identified emerging trends that she believed would have an impact on the provision of AR services. Technological developments influence the direction of the services provided, not only in amplification options and cochlear implants, but also in assessment procedures, especially the focus on auditory processing. The relationship between cognitive processing and impaired auditory function, and its importance to future developments of treatment and technology, are at the heart of an editorial presented by Pichora-Fuller (2007).

The impact of the environment on communication function is a growing concern in AR programs. Models have been established that take an ecological approach to treatment (Nobel & Héту, 1994; Jennings & Head, 1994). Madell and Montano (2000) described a process that recommends identifying the individual with hearing loss and understanding the implications. Furthermore, the barriers to communication must be analyzed within the context of their environments and interactions with communication partners. Kramer et al. (2005) encouraged the use of significant others in a home education program as a component of the rehabilitation process for older adults with hearing loss. The emphasis of these programs is on developing skills to cope with the impact of the hearing loss on communication function within the environment, a concept closely aligned with WHO ICF.

As we look to the literature, it is apparent that AR has evolved over the years and is no longer viewed as simply individual therapeutic services provided to a person with hearing loss. There are many opportunities for clinicians to provide AR, whether it is within their own work setting, in community-based programs, or even at home through computer-generated educational software. Acknowledgment and understanding of issues surrounding successful adjustment to hearing loss will help guide the audiologist in the development of appropriate objectives for treatment.

Conclusions

As we begin to develop new AR programs, or perhaps refine the ones already in existence, we tend to look toward accepted definitions to guide us. We search for meaning, guidance, and clarity as we establish

program goals and objectives. The role of the audiologist in the management of adults with hearing loss continues to evolve. Changes in the health delivery process, technological advances, and the recognition of the impact of hearing loss on function contribute to an ever-changing canvas. Now more than ever, AR needs to be a standard for audiology practice, not a second thought. As audiologists, we must be prepared to meet the challenges of our health system and continue to develop the programs to help patients and their communication partners live with hearing loss. Additionally, we need to take an active role in fostering a regulatory environment that recognizes the importance of audiologist-delivered AR.

The definition of AR has progressed and developed along with the culture of science and health. The breadth of services has expanded and the impact of hearing loss is better understood. Definitions are dynamic and will continue to shift and change. In our role as audiologists, we are directly responsible for the future of AR. Along with our consumers who have hearing impairment, we are providing the next generation with the basis for descriptions used to define our profession. We look to these definitions for guidance; many of our decisions are based on their existence. Therefore, embrace what we know about AR and look ahead to where it will be tomorrow.

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3

THE INTERNATIONAL CLASSIFICATION OF FUNCTIONING

Implications and Applications to Audiologic Rehabilitation

Jean-Pierre Gagné, Mary Beth Jennings, and Kenneth Southall

The main goal of the present chapter is to describe conceptual frameworks of health that can be used to conceive, describe, apply/conduct, and analyze audiologic rehabilitation (AR) services. First, we make the case that adopting and applying a conceptual framework is a natural and essential component of AR. Second, we discuss some shortcomings of applying a medical (curative) model of health as a conceptual framework for rehabilitation. Third, we describe two conceptual frameworks that have been effectively applied to the rehabilitation sciences. In both cases the models are based on classification systems adopted by the World Health Organization (WHO). They are the International Classification of Impairment, Disabilities, and Handicaps (ICIDH; WHO, 1980); and the International Classification of Functioning, Disability, and Health (commonly referred to as the ICF; WHO, 2001). Fourth, we illustrate how the ICF (WHO, 2001) can be used to guide clinical intervention services and evaluative research in AR.

What Are Conceptual Frameworks? Why Are They Useful?

The way we perceive a given phenomenon is influenced by how we view the world. The knowledge of an individual is usually organized and explained according to the relevant practices of a given society. Stated in another way, the relevant practices of a given society are used to describe and explain phenomena that exist or are thought to exist. Depending on their robustness and accuracy, those practices are considered to constitute theories, models, or in some cases conceptual frameworks.

In academia, most disciplines are driven by concepts, theories, or models. For example, many physicists use the Big Bang theory to explain how our universe was created and how it is evolving as a function of time. Acousticians choose to describe and analyze sounds based on three different parameters:

namely, duration, intensity, and frequency. There exist models of communication that describe and explain the rules that people intuitively use when they interact with one another.

Theories, models, and frameworks constitute the underlying bases on which science is organized and upon which science will progress. They are used to organize, describe, and investigate elements of knowledge and the constituents of a concept of interest. For example, Gagné, Southall, and Jennings (2009) described a theoretical model (Major & O'Brien, 2005) that can be used to describe and explain how the phenomenon of stigma may operate in people who have a hearing loss. Conceptual models provide precise and comprehensive definitions of concepts. Hyde and Riko (1994, p. 347) argued that "terminology is more than labels; it reflects and affects its underlying constructs and it provides a vehicle for debate and research." Moreover, conceptual models provide the basis on which various people from different backgrounds can have common understanding and perspective. Conceptual models provide the foundation on which a phenomenon is conceived, described, analyzed, understood, and explained. They constitute the starting point from which research questions are identified and hypotheses are tested. In rehabilitative audiology (as in all other health disciplines), conceptual models guide the way clinical services are organized, designed, and dispensed. Furthermore, these frameworks influence the type of research that takes place within a discipline, as well as how that research is organized and conducted.

From a clinical perspective, the conceptual framework adopted will determine how AR is conceived and perceived. This framework will influence how we define AR, and will govern rehabilitation services selected and provided. Moreover, it will influence how we evaluate the effects and benefits of the services provided.

From a research perspective, the conceptual framework chosen will determine the research issues to be addressed. In turn, the research question addressed or the hypothesis tested will influence the experimental paradigm used and the type of data collected and analyses to be performed to test the hypotheses formulated. Moreover, it will influence the conclusions that are drawn from research investigations.

Imagine two audiologists who adhere to different conceptual frameworks of AR. The conceptual framework adopted by one audiologist may lead them to design a program that aims to eliminate a client's hearing loss. Based on a different conceptual

framework, the other professional's goal may be to reduce or eliminate the deleterious effects of hearing loss on the client's everyday life activities. Under such circumstances, it is likely that, for the same client, the treatment program selected by one audiologist will be different from the program selected by the other professional. Furthermore, it is very likely that the research methodology used, as well as the method and criteria employed to evaluate the success of their intervention program, will differ across the two professionals. As unlikely as it may seem, in AR, as in other rehabilitation sciences, dramatically different conceptual frameworks have been used to guide the types of rehabilitation services provided to people with hearing loss and to quantitatively and qualitatively describe the outcomes of those services. Needless to say, for a discipline to progress and improve, the conceptual frameworks underlying that discipline must be appropriate, realistic, and valid. Over the years, the types of conceptual frameworks used to characterize AR have evolved.

A Medical Model of Health

Given its long history, its importance, and the overwhelming presence of medicine in Western societies, the medical model has been the predominant conceptual framework of health used in all health-related disciplines. The medical model is grounded in causal logic. Health care professionals aim to identify and explain a patient's symptoms based on what causes them. Stated simplistically, in a medical model of health, patients have symptoms that are caused by diseases or impairments. Based on the symptoms, a remedy or treatment is selected and applied (e.g., the prescription of medication, surgery, a program of exercises, dietary regime, etc.). A treatment is considered successful if, after its administration, the symptoms, the disease, or the impairment disappears. As a result of the treatment, the patient is cured and the person reestablishes the condition of health held before consulting the health care professional (Duchan, 2004). This conceptualization of health is very "body oriented." This model is effective when the treatment program is at the level of the cell, organ, or body structure. For example, health problems often require the elimination of a virus (e.g., an organism that causes a cold, the flu, or a childhood disease), the removal of a body part (e.g., tonsils, gallstones, tumors), or the repair of a body structure (e.g., fractured leg or arm) to restore "normal" function. In a

medical model of health, the goal of a treatment program is to “cure” the patient (i.e., to restore normal biological functioning or to minimize the impact of the patient’s symptoms; Duchan, 2004). This conception of health promotes a view of pathology as an entity in isolation from the affected individual. Such a view of health has limited relevance for chronic, progressive, and irreversible diseases such as sensorineural hearing loss (Hyde & Riko, 1994).

The health issues addressed by professionals who provide rehabilitation services are very different from the acute health care issues that can be conceptualized within a medical model. In the domain of rehabilitation, the health problems of the people seeking help are usually chronic. Often, the health problem is irreversible and cannot be resolved in a short period of time. Consequently, people with chronic health problems have to learn to cope with the effects of their health condition and modify their lifestyle. Often, the chronic nature of the health condition will have deleterious effects on non-body-related dimensions of the individual’s personal life (e.g., at the psychological, social, economic, and levels, as well as the level of leisure activities) and the social integration of that person in society.

It may not be appropriate to use a medical model to conceptualize the health and treatment needs of people who have chronic disorders such as a permanent sensorineural hearing loss. At the present time, beyond the services that might initially be provided to the person with a permanent hearing loss (e.g., injection of cortisone), there is not much that can be done medically to restore the person’s hearing abilities. Although a number of helpful rehabilitation services may be provided, none of those treatments are likely to cure the hearing loss. Thus, viewed from the perspective of the medical model of health, it is difficult to imagine treatment programs that would be shown to be effective. That is, at the present time, the availability of treatment programs designed to eliminate sensorineural hearing loss are limited (although some progress is being made with respect to treatments that regenerate hair cells at the level of the cochlea). The person has a hearing loss before the rehabilitation services are provided and will continue to have a hearing loss after having completed the rehabilitation program. Hence, if the criterion used to evaluate the benefits provided by a rehabilitation program consists of evaluating aspects of hearing impairment, it is unlikely that the program will be shown to be successful (Gagné, 1998, 2000; Gagné, McDuff, & Getty, 1999).

Although it is never explicitly stated, it can be argued that, in the past, rehabilitation services

and evaluation research in AR were designed and evaluated exclusively according to a medical model of health. For example, several decades ago, the unstated goal of fitting hearing aids was to restore normal hearing acuity. The appropriateness of a hearing aid fitting was evaluated according to the results of the aided audiogram or measures of functional aided hearing. An intervention program (i.e., fitting hearing aids) was deemed to be successful if the aided auditory detection thresholds were within the audiometric limits of normal hearing (Olsen, Hawkins, & Van Tasell, 1987; Skinner, 1988). The unstated premise here was that the hearing aids would cure the hearing loss. Similarly, the efficacy of a speechreading training program was evaluated by comparing the speechreading proficiency of a participant before and after the speechreading program was administered (Binnie, 1977). The underlying assumption was that the speech perception problems associated with hearing loss would be cured if the participant displayed improvements in speechreading proficiency (or audiovisual speech-perception proficiency) based on post-treatment speech perception tests administered in a laboratory setting. The results of these investigations provided little information on how different (hopefully better) the person’s speech perception proficiency or conversational fluency was while accomplishing his or her everyday living activities.

Based on the conceptual models of health that were available and used at the time, it is not surprising that the results of evaluative research investigations generally were not successful in demonstrating the benefits of AR. The underlying premise of the medical model, and the type of outcome measures typically used to evaluate benefit (both clinically and in research projects), were not suitable to the goals of rehabilitation. It was not until the 1980s that other conceptual models of health were developed and applied to rehabilitation sciences.

The International Classification of Impairment, Disability, and Handicap

According to the WHO (1948), health is a “state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.” This view extends the conception of health beyond the level of body parts and body functions. In 1980, the WHO proposed an international classification of health that attempted to describe encapsulate its conception of health and well-being. A main objec-

tive of the ICIDH (WHO, 1980) was to propose a generic model of health and rehabilitation that would be applicable to and internationally accepted by all forms of rehabilitation services, regardless of the discipline. The ICIDH (WHO, 1980) considers the effects that diseases and disorders may have at the organic level (impairments), at the level of the individual in real-life settings (disabilities), and at the sociocultural level (handicaps). The WHO (1980) definitions of impairment, disability, and handicap are provided in Table 3-1. A visual representation of the ICIDH (WHO, 1980) conceptual framework is shown in Figure 3-1.

According to this framework, hearing *impairments* are dysfunctions in body structures or body functions that are measurable in the laboratory or clinic. For example, an individual's audiogram may reveal the presence of elevated bone conduction detection thresholds. Or, the result of laboratory experiments may demonstrate that an individual has broader than normal auditory psychophysical tuning curves. The results of both tests are indicative of a hearing impairment attributable to some pathology in the hearing system. A *disability* is defined as a restriction or inability, resulting from an impairment, to perform an activity in the manner or within the range considered normal for a human being. Examples of hearing disabilities include having poor auditory localization skills and poor speech-perception performances in quiet or in noise. People with hearing impairment

Disease



Figure 3-1. The relationship between the different components of ICIDH (WHO, 1980).

who perform less well than matched peers with normal hearing on tasks of auditory sound localization or on speech perception tests would be deemed to display specific hearing disabilities. A *handicap* is a disadvantage caused by an impairment or a disability that prevents or limits a person from fulfilling the role that would otherwise be considered normal for that individual, given the sociocultural environment in which the person lives. From an audiology perspective, handicaps are nonauditory problems that result from hearing impairment or disability. For example, a specific job may require that workers be in regular verbal communication with each other, even though the level of noise in the work setting is very high. A person with a hearing loss may experience problems in that work setting if that individual has more difficulties than the other workers understanding speech in noise (i.e., a speech perception in noise disability). In this example, the person with hearing loss would be deemed to have an occupational handicap.

The following example illustrates how the domains of impairment, disability, and handicap apply to AR. Meningitis, a *disease*, may damage the inner and outer hair cells at the level of the cochlea, causing an auditory impairment (a severe bilateral sensorineural hearing loss). The impairment may cause the person to experience some hearing disabilities, including poorer performance on speech-perception tasks than peers who have normal hearing. The hearing impairment and associated disabilities may cause handicaps that limit or prevent that person from fulfilling social roles that would be considered normal for that individual. For example, that person may have work-related handicaps due to: the inability to converse on the telephone; difficulties communicating with one other individual in a noisy work environment; or difficulty in taking part in meetings in which several persons are involved. The same person may have leisure-related handicaps due to: the inability to communicate by telephone with friends; difficulty communicating with others in noisy environments such as restaurants; the inability to watch the evening news on the television (because it is not possible to understand the audio signal unless the

Table 3-1. Definition of Impairment, Disability, and Handicap According to the International Classification of Impairments, Disorders, and Handicaps (ICIDH: WHO, 1980)

Impairment:	any loss or an abnormality of a psychological, or anatomical structure or function.
Disability:	any restriction or inability (resulting from an impairment) to perform an activity in the manner or within the range considered normal for a human being.
Handicap:	any disadvantage for a given individual, resulting from an impairment or a disability, that limits or prevents the fulfillment of a role that is normal (depending on age, sex, and social and cultural factors) for that individual.

volume is set very loud or unless the audio signal is amplified); or an inability to take part in bird-watching activities (because the person cannot hear the bird songs or localize where they are coming from).

It is important to recognize that there is not a direct relationship between the domains of impairment, disability, and handicap. An impairment may not always result in a disability, and a disability does not necessarily result in a handicap (Hyde & Riko, 1994). Furthermore, two persons may have the same type and degree of hearing loss but experience differing types or degrees of handicap. Consider the hypothetical case of Tom and Jerry, who have a similar hearing loss. Their hearing loss may make it difficult to take part in conversations that involve two or more participants. This disability (e.g., difficulty conversing in noise) may constitute a work-related handicap for Tom because his job requires that he meet regularly with coworkers to establish their weekly sales objectives. Jerry, on the other hand, may be a mail carrier, a job that does not require him to participate in group meetings. Thus, having difficulty conversing in noise may not be an occupational handicap for Jerry. Further, two persons with different hearing loss may have the same handicap. For example, although Charlie's hearing loss may be less severe than Tom's and Jerry's, he may have the same hearing disability (difficulty conversing in noise). Moreover, that hearing disability may constitute an occupational handicap because Charlie is a waiter in a poorly lit and noisy sports bar where he must interact with his customers.

It is important to note that, according to this conceptual framework, people with hearing loss do not have a hearing handicap; however, they may have a handicap due to their hearing loss. Similarly, a person with a hearing loss is not hearing handicapped; however, in some situations, the person may experience a handicap because of the hearing loss (Stephens & Héту, 1991). Although it not stated explicitly, according to the ICIDH (WHO, 1980) framework, a handicap is the result of an interaction between, on the one hand, impairments and disabilities and, on the other hand, the particular sociocultural and physical environment in which an activity or an event takes place (Stephens & Héту, 1991). For example, Natasha, a school crossing guard with a mild to moderate hearing loss may have difficulty localizing sound in space. This disability constitutes a work-related handicap when she directs traffic at a busy street intersection. However, the same disability may not constitute a handicap when she converses with children while standing on the sidewalk. A useful reference, Stephens and Héту (1991) described how the concepts of

impairment, disability, and handicap apply to rehabilitative audiology.

At the time that it was proposed, the ICIDH (WHO, 1980) constituted a major breakthrough for all disciplines of rehabilitation. By extending the concept of health beyond the domains of disorders and impairments, the ICIDH framework provided rehabilitation sciences with an opportunity to develop a different conceptualization of itself and to redefine its goals. Consistent with the ICIDH model (WHO, 1980), the goal of AR can be defined as the alleviation or reduction of hearing disabilities and handicaps encountered by individuals with hearing loss (Gagné & Jennings, 2008). Within this perspective, AR services could be helpful for people with a chronic hearing loss. Whereas, AR cannot cure permanent hearing impairment, some programs can be designed to reduce or eliminate hearing disabilities and handicaps. For example, within a medical (curative) model of health, the stated or implicit goal of providing a client with hearing aids is to restore the client's impaired hearing abilities (e.g., restoring auditory detection thresholds to normal levels). However, within the perspective of the ICIDH (WHO, 1980), the goal of providing the client with hearing aids may be to eliminate or reduce hearing disabilities (e.g., improving the detection of acoustic alerting signals, improving speech understanding in quiet and/or in noise, making it possible to localize voices or warning signals in space) or to reduce situations of handicap attributable to the hearing loss (e.g., maintaining one's occupation even though it requires conversing on the telephone; continued appreciation of one's leisurely activities such as playing Scrabble with friends or watching sports programs on television). By defining the goals of rehabilitation intervention according to disabilities and handicaps, the ICIDH makes it possible to evaluate the efficacy and the effectiveness of specific types of services or programs provided to people with specific impairments and specific needs, defined in terms of disabilities and handicap. For example, how successful is the treatment (e.g., a visual alerting device) in making the client aware that someone is ringing the doorbell? Or, how helpful is the treatment (e.g., earphones connected to a personal infrared amplification system) in enabling the client to understand the television? Conceiving AR from an ICIDH (WHO, 1980) perspective makes it possible to assess the impacts of rehabilitation programs. Also, it provides insights into identifying the types of programs that are most efficacious to manage specific disabilities and handicaps. For example, using a single hearing aid may constitute an efficacious treatment program if the

goal of the intervention program is to improve speech comprehension in quiet. The same treatment program may not be as efficacious if the goal of the program is to improve sound localization in a noisy work environment. More importantly, the ICIDH (WHO, 1980) made it possible to demonstrate that, by reducing disabilities and handicaps, the quality of life of people with hearing loss could be improved without altering the person's disorder or impairment (Mulrow, Aguilar, Endicott, Velez, & Tuley, 1990; Mulrow, Tuley, & Aguilar, 1992).

Notwithstanding its contribution to rehabilitation sciences, there remained some confusion among researchers and professionals concerning the concepts addressed in the ICIDH model. For example, some people had difficulty establishing the line of demarcation between elements that were in the domain of impairments and those that were in the domain of disabilities (Stephens & Héту, 1991). For example, the inability to detect a 1000-Hz pure tone presented at 40 dB HL via loudspeakers in an audiometric test booth may be taken as a measure of hearing impairment. Does the inability to detect complex tonal signals such as an FM signal centered at 1000 Hz (e.g., a signal that resembles an ambulance warning signal) in the same test setting constitute a measure of impairment or disability? Some professionals considered performance on a speech perception test administered in noise as a measure of impairment (a diagnostic sign of a cochlear hearing loss), whereas others considered the performance on the same task a measure of hearing disability (Stephens & Héту, 1991). Similarly, there was some confusion between what was considered a disability and what was considered a handicap. For example, does having difficulty understanding speech on the telephone constitute a hearing disability or does it constitute a handicap?

Over the years, some shortcomings of the ICIDH model were identified. For example, according to Figure 3-1, which represents a schematized representation of the ICIDH, the model is unidirectional (Fratalli, 1998; Threats, 2006). Specifically, disorders may lead to impairments, which may lead to disabilities. Furthermore, according to the model, impairments and disabilities may lead to handicaps. However, the model does not account for situations whereby disabilities may lead to impairments or where handicaps may lead to disabilities or impairments. In some circumstances, those possibilities may occur. For example, someone may have a hearing impairment (elevated hearing detection thresholds), which may lead to hearing disabilities (e.g., understanding speech in quiet and in noise). The hearing disability

may lead to a social integration handicap—because of the difficulties associated with understanding speech, the person avoids social interactions with others. Over a period of time, the social integration handicap (staying at home alone, isolating oneself from others) may bring the person to suffer from psychological depression. How does the ICIDH model account for this situation? In this instance, some people may claim the depression constitutes a secondary handicap (Stephens & Héту, 1991). Others would report that the handicap (social isolation) led to the development of a new disorder; namely, depression.

Another criticism of the ICIDH model was that some elements known to influence disabilities and handicaps could not be accounted for in the ICIDH framework. A case in point, it has been demonstrated that the physical and social environment are involved in the handicap creation process (Fougeyrollas, Bergeron, Cloutier, Côté, & St. Michel, 1998; Fougeyrollas & Majeau, 1991; Noble & Héту, 1994). For example, it is well known that the physical environment (e.g., noise, reverberation, and poor illumination) will influence the magnitude of disability and handicap experienced by people with a sensorineural hearing loss. The social environment may also alter the degree of handicap experienced by an individual. For example, to minimize a leisure-related handicap, it would be acceptable for someone with a hearing loss to ask conversational partners to take turns and speak one at a time when playing bridge to optimize the ability to understand speech. However, socially, it is less acceptable to make the same request (asking the people who are sitting close by to speak one at a time) at a large sporting event such as a football game. Recognizing the intricate interaction between the person with hearing loss and the physical and social environment in which a specific activity or event takes place, Noble and Héту (1994) proposed that the concepts of hearing disability and handicap always be considered within an ecologic approach. Specifically, they described how a handicap should be defined according to the interaction that exists between the person and the environment. As illustrated in an above-mentioned example, the disabilities and handicaps experienced by Natasha, the school crossing guard, are situation specific and contextually determined. Investigators in other rehabilitation disciplines also recognized the importance of the environment in describing disability and handicap (Fougeyrollas et al., 1998; Fougeyrollas & Majeau, 1991; WHO, 1997, 1999, 2001).

There is evidence that personal attributes of the individual with an impairment may also influence the

degree of disability and handicap that is experienced. For example, a person who has a very outgoing personality may request communication partners to apply specific communication strategies to facilitate speech understanding (e.g., requesting conversational partners to use clear speech). However, a shy person may refrain from making the same request. Hence, under similar communicative settings, the latter person is likely to display a greater amount of disability or handicap. In certain cultures, it is not acceptable for women to make eye contact when they interact with men. Hence, for those women, the use of visual speech cues such as speechreading to improve their speech understanding proficiency is not possible. In this instance, a person's cultural background may influence the degree of disability and handicap experienced in a given communication setting. Given these shortcomings associated with the ICIDH classification system, the WHO undertook a major revision of its conceptual framework of health.

The International Classification of Functioning, Disability, and Health

After exploring different variations of conceptual frameworks (WHO, 1997, 1991, 2001), the WHO formally proposed a revised classification system: the International Classification of Functioning, Disability, and Health, most commonly referred to as the International Classification of Functioning or ICF (WHO, 2001). This classification system can be applied to all rehabilitation sciences, and is accepted and recognized internationally (e.g., Peterson, 2005; Peterson &

Rosenthal, 2005; Smiley, Threats, Mowry, & Peterson, 2005; Threats & Worrall, 2004). Thus, an important advantage of the ICF (WHO, 2001) is that it can be used to compare the results of rehabilitation research as well as aspects of clinical services across disciplines and across countries. Like its predecessor, the ICF (WHO, 2001) changes how intervention services are conceived, organized, and dispensed. Given the importance of this classification system, a description of the model follows.

The overall aim of the ICF (WHO, 2001) is to provide a common framework for describing health-related states and, specifically, for understanding the dimensions of health and functioning in all domains of health. The main elements of the ICF (WHO, 2001) are displayed in Figure 3–2 and defined in Table 3–2. The ICF has two parts, each with two components that can be expressed both in positive and in negative terms. The first part incorporates aspects of functioning and disability. This part includes body functions and body structures as well as activities and participation. The second part includes contextual factors, specifically environmental factors and personal factors. According to the ICF (WHO, 2001), an individual's state of health is determined by the dimensions of functioning and disability and is moderated by how these dimensions interact with each other. The schematic representation of the model indicates that the dimensions of health (body functions and body structures; activities and participation) may influence one another. The model also indicates that contextual factors, such as environmental and personal factors, may influence the dimensions of health. Over the years there has been a modification in how the ICF is applied. Formally, the classification system clearly

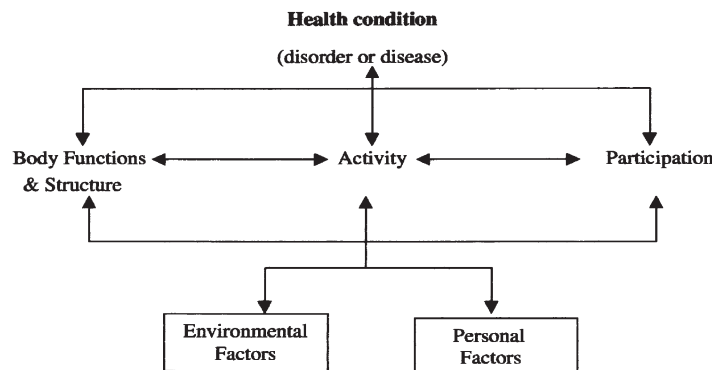


Figure 3–2. Illustration of the interactions of the concepts incorporated into the International Classification of Functioning, Disability, and Health (WHO, 2001).

Table 3–2. Definitions of Terms Used in the International Classification of Functioning, Disability, and Health (ICF: WHO, 2001)

Body Functions are physiological functions of body systems (including psychological functions).

Body Structures are anatomical parts of the body, such as organs, limbs and their components.

Impairments are problems in body function or structure such as a significant deviation or loss.

Activity is the execution of a task or action by an individual.

Participation is involvement in a life situation.

Activity Limitations are difficulties an individual may have in executing activities.

Participation Restrictions are problems an individual may experience in involvement in life situations.

Environmental Factors make up the physical, social, and attitudinal environment in which people live and conduct their lives.

delineates the domain of activity and activity limitations from the domain of participation and participation restrictions. In some cases, it may be difficult to determine whether a given behavior falls within the purview of activities or participation. Delineating what constitutes an activity from what constitutes participation in an activity can be an interesting but difficult and tedious academic exercise. More importantly, for rehabilitative purposes there is no real advantage of trying to identify behaviors as activities or participation. Typically, individuals with hearing loss experience difficulties when they are executing certain activities that involve hearing (e.g., discussion, remunerative employment, socializing, etc.).

The WHO (2001) has designed an extensive coding scheme to classify categories of body functions, body structures, activities and participation, and the environment. The coding scheme has a hierarchical structure with several levels, and was designed to provide a common language and framework to describe and quantify elements of the ICF across health disciplines. The broad categories of first-level classification (chapters) are listed in Table 3–3. The second level allows for a more detailed description of the chapters included at the first level of classification, and the third and sometimes fourth levels are even more specified than the second level. For example, the second-level category code “b230—Hearing functions,” is related to the first-level chapter “b2—Sensory functions and pain.” The third-level category codes related to b2 and b230 are “b2300—Sound

detection” and “b2304—Speech discrimination.” The category code d350 indicates an activity or participation in conversation, a second-level classification. A related third-level classification would be “d3501” sustaining a conversation.” Within the ICF framework a hearing aid is considered an assistive device that modifies the environment and thus would be coded as e1251, “assistive products and technology for communication.”

For the ICF domains of functions and structures, and for activity and participation, one may add a specific qualifying numeric code. Specifically, for these domains of functioning and disability there exist seven possible codes, ranging from *no problem* to *complete problem* (see Table 3–4 for a complete list of qualifiers). Qualifiers have also been developed to describe the impact of environmental factors. For this domain of the classification system, the qualifiers are considered either facilitators or barriers when executing tasks or actions or when engaging in life situations. The coding scheme used ranges from *complete facilitator* to *complete barrier* (see Table 3–5 for the complete list of qualifiers) that can be applied to descriptors in the domain of environment factors. It should be noted that at the present time, no categories exist to classify personal factors.

Beyond the elaborate coding scheme, conceptually, the ICF (WHO, 2001) differs from the original ICIDH classification system (WHO, 1980) in some fundamental ways. One difference between the conceptual frameworks of the ICIDH (WHO, 1980) and

Table 3–3. Aspects of Health Explicitly Covered in the ICF Classification System (WHO, 2001)

Body Functions	
Chapter 1	Mental functions
Chapter 2	Sensory functions and pain
Chapter 3	Voice and speech functions
Chapter 4	Functions of the cardiovascular, haematological, immunological, and respiratory systems
Chapter 5	Functions of the digestive, metabolic, and endocrine systems
Chapter 6	Genitourinary and reproductive functions
Chapter 7	Neuromusculoskeletal and movement-related functions
Chapter 8	Functions of the skin and related structures
Body Structures	
Chapter 1	Structures of the nervous system
Chapter 2	The eye, ear, and related structures
Chapter 3	Structures involved in voice and speech
Chapter 4	Structures of the cardiovascular, immunological, and respiratory systems
Chapter 5	Structures related to the digestive, metabolic, and endocrine systems
Chapter 6	Structures related to the genitourinary and reproductive systems
Chapter 7	Structures related to movement
Chapter 8	Skin and related structures
Activities and Participation	
Chapter 1	Learning and applying knowledge
Chapter 2	General tasks and demands
Chapter 3	Communication
Chapter 4	Mobility
Chapter 5	Self-care
Chapter 6	Domestic life
Chapter 7	Interpersonal interactions and relationships
Chapter 8	Major life areas
Chapter 9	Community, social, and civic life
Environmental Factors	
Chapter 1	Products and technology
Chapter 2	Natural environment and human-made changes to environment
Chapter 3	Support and relationships
Chapter 4	Attitudes
Chapter 5	Services, systems, and policies

Table 3-4. Codes Used to Qualify the Magnitude of a Health Condition

Qualifiers	Associated Descriptor	Example of Appropriate Qualifying Words	Example of Range of Performance in Percentages
.0	NO problem	None, absent, negligible	0–4%
.1	MILD problem	Slight, low	5–24%
.2	MODERATE problem	Medium, fair	25–49%
.3	SEVERE problem	High, extreme	50–95%
.4	COMPLETE problem	Total	96–100%
.8	Not specified		
.9	Not applicable		

Table 3-5. Codes Used to Qualify the Extent (or Nature) of the Environmental Factor

Barrier Codes	Associated Descriptor	Examples of Appropriate Qualifying Words	Facilitator Codes	Associated Descriptor
.0	NO barrier	None, absent, negligible	+0	NO facilitator
.1	MILD barrier	Slight, low	+1	MILD facilitator
.2	MODERATE barrier	Medium, fair	+2	MODERATE facilitator
.3	SEVERE barrier	High, extreme	+3	SEVERE facilitator
.4	COMPLETE barrier	Total	+4	COMPLETE facilitator
.8	Barrier, not specified		+8	Facilitator, not specified
.9	Not applicable		+9	Not applicable

the ICF (WHO, 2001) is that the latter allows activity limitations and participation restrictions to cause impairments at the level of body structures and body functions. There are experimental data to support this claim. For example, investigators have demonstrated differences in performance on certain listening activities such as sound discrimination and speech recognition tasks (Gatehouse, 1989; Hurley, 1993; Palmer, Nelson, & Lindley, 1998; Robinson & Gatehouse, 1995; 1996; Silman, Gelfand, & Silverman, 1984; Willott, 1996). Specifically, performance is poorer when the stimuli are presented in the ear in which a person with a bilateral hearing loss does not wear a hearing aid (hence, few or no listening activities are processed through that ear) than in the ear in which a person

uses amplification (i.e., the ear that is typically used for listening). In these examples, the data can be interpreted as illustrations of activity limitations (e.g., not listening to sound in the unaided ear) that led to an impoverished body function (e.g., poorer sound processing in that ear) and maybe even a disorder in body parts (e.g., perhaps some neural degeneration along the central auditory system).

Another significant difference between the ICIDH (WHO, 1980) and the ICF (WHO, 2001) classification schemes is the importance attributed to the contextual factors such as personal and environmental factors. Clinically, it is possible to illustrate the importance that personality factors may have on creating activity limitations and participation

restrictions. For example, two persons with similar hearing loss (i.e., impaired body structures and functions) may belong to different bowling teams. One person, Nancy, is an outgoing and assertive person. Whenever a team member says something she doesn't understand, Nancy politely asks the person to repeat the statement while speaking clearly and facing her ("because I have difficulty hearing"). Jack, on the other hand, is shy and has a passive personality. Whenever a member of his bowling team says something he doesn't understand, Jack usually just smiles and pretends he understands. In this example, differences in personality traits may account for the fact that Nancy experiences fewer activity limitations or participation restrictions than Jack during their respective bowling outings.

Similarly, both physical and social environmental factors may contribute to determining whether a disability is experienced. For example, due to his hearing loss, Tony may experience a participation restriction when he plays bocce ball with his friends in the afternoon, but not in the evening. The explanation is as follows. Within the neighborhood park, the bocce ball court is located just beside the waterslide playground for toddlers. During the day, because it is crowded with children, the waterslide playground is very noisy. The noise is so loud that Tony does not understand what his playing partners say to him. However, the waterslide playground closes at 5 PM every day. Tony does not experience any participation restriction when he plays bocce ball after 5 PM because the playing area is quiet and he has no difficulty communicating with his playing companions.

An example of how social environmental factors may influence whether a person with a hearing loss experiences a restriction while participating in a given activity may be as follows. Because of his hearing loss, Jay, who refuses to use hearing aids, often doesn't understand when people say things to him, even in relatively quiet environments. Typically, when he does not understand, Jay relies on his spouse as his "hearing aid" to repeat what the interlocutor said using louder, clearer speech. This strategy works fairly well when the couple is alone and watching a movie on the television in the living room. However, the couple cannot use the same strategy when they go to see a play or when they go to the movie theater. In those settings, it is socially unacceptable for Jay's spouse to repeat what was said to her partner.

Another benefit of the ICF (WHO, 2001) as a conceptual framework for rehabilitation is that it can be used to explain how, in certain situations, people with

normal hearing may experience an activity limitation or participation restriction due to a communication partner participating in the activity has a hearing impairment. For example, Héту (1996) reported that spouses of working men with hearing loss experienced activity limitations or participation restrictions. Specifically, the spouses reported that, because of their hearing loss, their husbands did not enjoy participating in many social gatherings, such as visiting friends, having dinner at the restaurant, or going to plays or movies. Thus, because of the unwillingness of the husbands to participate in those activities due to their hearing loss, the spouses also experienced a participation restriction. Hickson and Scarinci (2007) described this situation as a "third-party" disability. Their graphic representation of third-party disability is displayed in Figure 3-3. As can be seen from this diagram, in a given environment, the presence of a person with an impairment (e.g., a hearing loss) may influence the health condition of a person without any impairments.

ICF Core Sets for Hearing Loss

As mentioned previously, the ICF manual (WHO, 2001) offers an extensive coding scheme, involving more than 1,400 categories intended to describe health and health-related states. In describing overall health, the ICF attempts to cover the entire life spectrum. However, when the ICF is used in rehabilitation sciences, and especially when it is applied to certain specific health conditions (e.g., hearing loss), not all of the categories may be relevant. Specifically, whereas the complete list of categories may be relevant in providing a comprehensive description of one's overall, a more restricted list of categories may be called for in providing a description of specific health conditions (e.g., hearing loss). For example, as a first approximation, the category "maintaining a standing position (d4154)" is not likely to be related to classifying an activity limitation related to hearing loss. It would be tedious and irrelevant to apply and quantify all of the ICF categories for any single health condition. Thus, for reasons of practicality and efficacy, the WHO recommends that core sets be developed for specific health conditions. A core set consists of a listing of highly relevant ICF categories to a given health condition. In this regard, a core set can be considered a checklist that relates to a specific health condition. In designing a core set for a given health condition, the

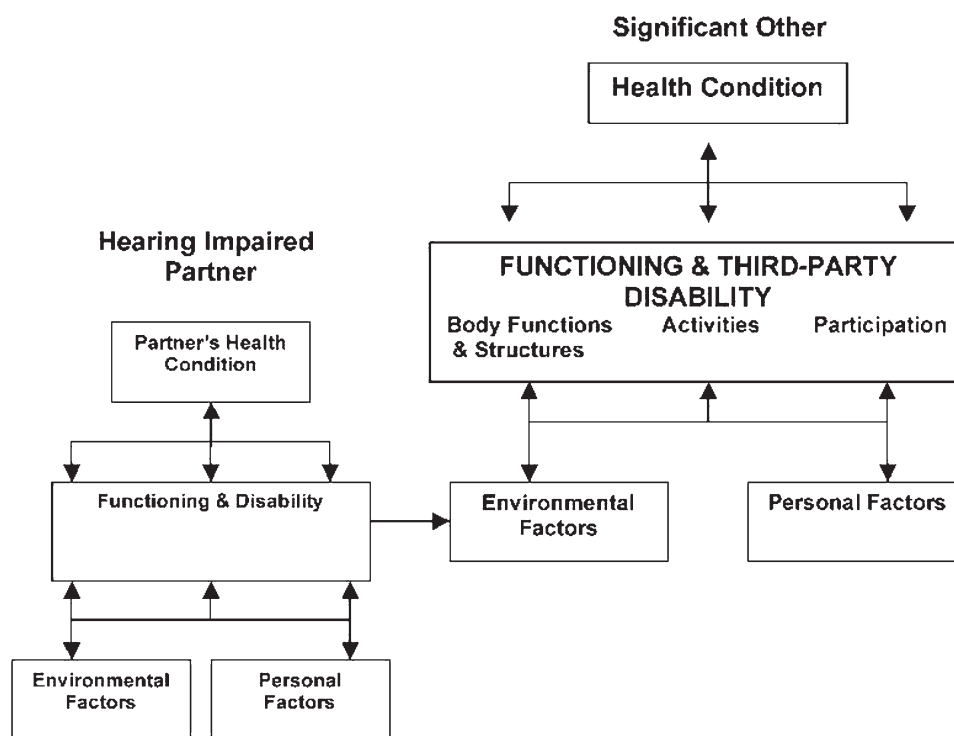


Figure 3-3. Model of the ICF (WHO, 2001) that incorporates third-party disability.
Source: Printed with permission from Worrall and Hickson, 2003.

goal is to include all of the ICF categories that apply to a given health condition while limiting the number of categories to as few as required to adequately characterize a health condition. For each health condition, two levels of core sets may be developed: a comprehensive core set and a brief core set. The list of categories included in a comprehensive core set are more extensive (typically between 70 and 150 categories) and capture all of the dimensions of the ICF relevant to a given health condition. Comprehensive core sets may be used in primary-care health facilities that treat many different health conditions. Or, they can be applied in cases where a multidisciplinary approach is used to provide health services to an individual. On the other hand, brief core sets are intended to capture only the most common and salient characteristics associated with a given health condition. Typically, fewer than 50 categories are included in a brief core set. Brief core sets may be used clinically to characterize the effects of a given health condition on a client and to outline the person's ability to function in society. Core sets may also be used in research, such as to describe the general functional characteristics of a group of participants in an experiment or as an outcome measure to quantify the effects of a specific treatment project.

The WHO has proposed a specific research protocol that must be applied to develop core sets for a given health condition. A detailed description of this protocol is beyond the scope of this chapter. Suffice it to say that the major steps of the protocol include a preparatory phase followed by two validation phases. The steps that must be completed to develop core sets for a given health condition have been described in several publications (Cieza et al., 2002; Danermark et al., 2010; Stucki, 2005). The initial components of the development protocol incorporate three different steps that are typically completed over a period of 2 years. At the present time, core sets have been developed for about 20 different health conditions, including osteoarthritis, rheumatoid arthritis, diabetes, and breast cancer.

In 2008, under the leadership of Professor Berth Danermark of the Swedish Institute for Disability Research, a group of international scholars in the domain of hearing loss undertook the challenge of developing core sets that would apply to adults with hearing loss. Consistent with the established core set development protocol, and following the completion of the preparatory phase of the process, in May 2012 a consensus conference was held to adopt the first version of the core sets for hearing loss.

The ICF categories included in the brief and comprehensive core set for hearing loss are listed in Tables 3–6 and 3–7. The core sets adopted at the consensus conference must now undergo a validation process that entails an international cross-sectional multicenter study among adults with hearing loss. The goal of that study is to examine the content valid-

ity and feasibility of the ICF Core Sets for Hearing Loss (Danermark et al., 2010).

Criticism of the ICF (WHO, 2001)

Generally, the ICF (WHO, 2001) has been widely accepted by professionals in rehabilitation sciences, including AR. The classification system now serves as the conceptual framework in most professional training programs as well as in many centers that provide rehabilitation services. However, some investigators have identified limitations of this classification system as a conceptual framework for AR (Noble, Tyler, Dunn, & Bhullar, 2008; Noble, 2009; Stephens & Danermark, 2005; Wade & Halligan, 2003). For example, according to Wade and Halligan (2003), issues related to a person's health are not appropriately accounted for in the classification system (Table 3–8).

Clinical Applications of the ICF

Using the ICF (WHO, 2001) as a conceptual framework for rehabilitation sciences, AR are considered procedures designed “to restore or optimize participation in activities considered limitative by persons who have a hearing impairment or by other individuals who partake in activities that include persons with a hearing impairment” (Gagné & Jennings, 2008, p. 390). Accepting the ICF as a conceptual framework for rehabilitation and agreeing with the aforementioned definition of AR has many ramifications for the way in which rehabilitation services are conceived, designed, organized, and dispensed.

Several authors have discussed how the ICF (WHO, 2001) can be applied in clinical settings (Gagné, 2000; Gagné & Jennings, 2008; Hickson & Scarinci, 2007; Hickson & Worrall, 1997; Smiley et al., 2005; Worrall & Hickson, 2003). For example, Gagné and Jennings (2008) described one approach to rehabilitation that is consistent with the ICF (WHO, 2001): a solution-centered problem-solving approach to rehabilitation. Similarly, Smiley et al. (2005) described how the ICF could be incorporated into rehabilitation programs designed for people with hearing loss. These authors used case presentations to illustrate one application of the ICF to AR. Hickson and her colleagues (Hickson & Scarinci, 2007; Laplante-Lévesque, Hickson, & Worrall, 2010; Worrall & Hickson, 2003) outlined how the ICF can be used by those who provide rehabilitation services to older adults with hearing loss.

Table 3–6. The Brief ICF Core Set for Hearing Loss

ICF Code	Title
b126	Temperament and personality functions
b140	Attention functions
b144	Memory functions
b152	Emotional functions
b210	Seeing functions
b230	Hearing functions
b240	Sensations associated with hearing and vestibular function
s110	Structure of brain
s240	Structure of external ear
s250	Structure of middle ear
s260	Structure of inner ear
d115	Listening
d240	Handling stress and other psychological demands
d310	Communicating with—receiving—spoken messages
d350	Conversation
d360	Using communication devices and techniques
d760	Family relationships
d820	School education
d850	Remunerative employment
d910	Community life
e125	Products and technology for communication
e250	Sound
e310	Immediate family
e355	Health professionals
e410	Individual attitudes of immediate family members
e460	Societal attitudes
e580	Health services, systems, and policies

Source: Obtained from B. Danermark, June 2012.

Table 3–7. The Comprehensive ICF Core Set for Hearing Loss

ICF Code	Title	ICF Code	Title
b117	Intellectual functions	d360	Using communication devices and techniques
b126	Temperament and personality functions	d440	Fine hand use
b130	Energy and drive functions	d470	Using transportation
b1300	Energy level	d475	Driving
b1301	Motivation	d620	Acquisition of goods and services
b140	Attention functions	d660	Assisting others
b144	Memory functions	d710	Basic interpersonal interactions
b152	Emotional functions	d720	Complex interpersonal interactions
b156	Perceptual functions	d730	Relating with strangers
b1560	Auditory perception	d740	Formal relationships
b1561	Visual perception	d750	Informal social relationships
b164	Higher-level cognitive functions	d760	Family relationships
b167	Mental functions of language	d770	Intimate relationships
b210	Seeing functions	d810	Informal education
b230	Hearing functions	d820	School education
b2300	Sound detection	d825	Vocational training
b2301	Sound discrimination	d830	Higher education
b2302	Localization of sound source	d840	Apprenticeship (work preparation)
b2304	Speech discrimination	d845	Acquiring, keeping and terminating a job
b235	Vestibular functions	d850	Remunerative employment
b240	Sensations associated with hearing and vestibular function	d855	Non-remunerative employment
b280	Sensation of pain	d860	Basic economic transactions
b310	Voice functions	d870	Economic self-sufficiency
b320	Articulation functions	d910	Community life
b330	Fluency and rhythm of speech functions	d920	Recreation and leisure
d110	Watching	d930	Religion and spirituality
d115	Listening	d940	Human rights
d140	Learning to read	d950	Political life and citizenship
d155	Acquiring skills	e115	Products and technology for personal use in daily living
d160	Focusing attention	e120	Products and technology for personal indoor and outdoor mobility and transportation
d175	Solving problems	e125	Products and technology for communication
d220	Undertaking multiple tasks	e130	Products and technology for education
d240	Handling stress and other psychological demands	e135	Products and technology for employment
d310	Communicating with—receiving—spoken messages	e140	Products and technology for culture, recreation, and sport
d315	Communicating with—receiving—nonverbal messages	e145	Products and technology for the practice of religion and spirituality
d330	Speaking	e150	Design, construction, and building products and technology of buildings for public use
d350	Conversation	e155	Design, construction, and building products and technology of buildings for private use
d3503	Conversing with one person		
d3504	Conversing with many people		
d355	Discussion		

Table 3–7. *continued*

ICF Code	Title	ICF Code	Title
e225	Climate	e455	Individual attitudes of health-related professionals
e240	Light	e460	Societal attitudes
e250	Sound	e465	Social norms, practices, and ideologies
e2500	Sound intensity	e515	Architecture and construction services, systems, and policies
e2501	Sound quality	e525	Housing services, systems, and policies
e310	Immediate family	e535	Communication services, systems, and policies
e315	Extended family	e540	Transportation services, systems, and policies
e320	Friends	e545	Civil protection services, systems, and policies
e325	Acquaintances, peers, colleagues, neighbors, and community members	e550	Legal services, systems, and policies
e330	People in positions of authority	e555	Associations and organizational services, systems, and policies
e335	People in subordinate positions	e560	Media services, systems, and policies
e340	Personal care providers and personal assistants	e570	Social security services, systems, and policies
e345	Strangers	e575	General social support services, systems, and policies
e350	Domesticated animals	e580	Health services, systems, and policies
e355	Health professionals	e585	Education and training services, systems, and policies
e360	Other professionals	e590	Labour and employment services, systems, and policies
e410	Individual attitudes of immediate family members	s110	Structure of brain
e415	Individual attitudes of extended family members	s240	Structure of external ear
e420	Individual attitudes of friends	s250	Structure of middle ear
e425	Individual attitudes of acquaintances, peers, colleagues, neighbors, and community members	s260	Structure of inner ear
e430	Individual attitudes of people in positions of authority	s710	Structure of head and neck region
e440	Individual attitudes of personal care providers and personal assistants		
e445	Individual attitudes of strangers		
e450	Individual attitudes of health professionals		

Source: Obtained from B. Danermark, June 2012.

Perhaps one of the most positive aspects of using the ICF (WHO, 2001) as a conceptual framework for AR is that it provides a functional description of the difficulties experienced by individuals with hearing loss (or someone with normal hearing who interacts with a person who has a hearing impairment: see Gagné & Jennings, 2008). All difficulties that a person might experience due to hearing loss can be described as an activity limitation or participation restriction.

Consequently, clients who seek rehabilitation services can be asked to, or be taught to, describe the effects that their hearing loss has on everyday activities. For example, Mrs. Kirk may report that the most important difficulties that she experiences due to her hearing loss are communicating with her friends at the bridge club and interacting with communication partners during social gatherings. Similarly, Mr. Hall might report having difficulty conversing when

Table 3–8. Some Limitations of the ICF Outlined by Wade and Halligan (2003)**According to Wade and Halligan (2003), the classification system:**

- Does not have neutral language (terminology) to describe body functions and body structures (i.e., positive terminology to counter the terms impairment and disabilities).
- Is not completely integrated with the International Statistical Classification of Diseases and related health problems (ICD-10, WHO, 1992) to form a comprehensive classification system of illness. In some respects the two classification systems may be redundant and in other respects there are links missing between the two classification systems.
- Does not take into account the element of time. That is, the ICF describes a person's health status at a particular point and time (i.e., the here and now) and does not capture the person's previous health status. According to these authors, in order to fully understand a person's health status, one must consider the individual over a period of time.
- Does not attribute sufficient weight (importance) to the person ('the individual') with the impairment. In particular, the ICF does not sufficiently consider how much a "person's choice" directly influences performance in activities. Specifically, the ICF does not take into account the notion of an individual's free will. That is, in most democratic and legal conceptions of human nature there is a pervasive and deep-seated notion of free will and individual responsibility. According to the authors, a person's free will will have an influence on whether or not an activity limitation /participation restriction occurs, as well as the magnitude of the limitations/restrictions.
- Does not implicitly take into account the role of other persons who may be involved in activity limitations and participation restrictions.
- Does not consider the phenomenon of quality of life. This is important because quality of life is a major determinant of an individual's well-being.

meeting clients one-on-one as well as in small groups, especially in his office and at restaurants. Involving clients in the process of identifying and describing the specific difficulties they experience in their everyday life ensures the activity limitations and participation restrictions targeted by the rehabilitation program will be relevant and will address difficulties that the clients deem important to resolve (for more information, see: Gagné & Jennings, 2008; Gagné, Jennings, & Southall, 2009).

Another advantage of describing difficulties in terms of activity limitations and participation restrictions is that it shifts the source of the problem—and the focus of the rehabilitation efforts—away from the person who has a hearing loss toward the problematic activity. That is, the focus of the rehabilitation program will not be the hearing loss or the person with the hearing loss per se. Rather, the focus of the rehabilitation program is the problematic activity. For example, the goal of the rehabilitation program may be to make it possible for Mrs. Kirk to participate actively

in conversations at social gatherings, or for Mr. Hall to minimize communication breakdowns when he negotiates with clients in his office and at restaurants. The solutions to those problems may involve the active participation of people other than Mrs. Kirk and Mr. Hall. Identifying the goals of rehabilitation in terms of activity limitations and participation restrictions may eliminate or reduce the guilt, shame (including the negative psychological impact), and responsibility that some people with hearing loss experience because they view themselves as singularly responsible for communication breakdowns. Perhaps other factors, such as the environment or other persons who take part in the activity, are also partially responsible for the difficulties experienced.

Another important advantage of describing the difficulties in terms of activity limitations and participation restrictions is that it allows the people who seek rehabilitation services to identify the difficulties that have a direct impact on their lives. For example, Ted, who has a hearing loss, may not be able to have

conversations with his friends when they are standing at a noisy street corner (a limitation/restriction that is very common among people with hearing loss). However, if the inability to participate in that activity is not important to him, when asked to report the most important limitations/restrictions experienced due to his hearing loss, Ted would likely not report this activity as being particularly relevant. On the other hand, Alice, who has a hearing loss, may place a lot of importance on an activity that is problematic even though it does not occur frequently. For example, she may report that she would like to resolve the difficulty she has conversing with her 6-year-old grandson when they play cards. For Alice, this participation restriction may be considered a high priority for rehabilitation even though the activity itself may occur only infrequently (e.g., once every 2 or 3 weeks).

Two important points about using the ICF to identify the goals of rehabilitation program are that: (a) they necessarily involve the person consulting in the process of identifying the goals of rehabilitation, as this person is best qualified to identify the most important activity limitations or participation restrictions they experience; and (b) the rehabilitation program, and more specifically the goals of the rehabilitation program, will be formulated in functional terms such as everyday activities specific to the client (Gagné & Jennings, 2008).

Clearly, defining the goal of the rehabilitation program is the most important component of any intervention program. If the goal of the program is unambiguously identified and described, it will be easier to select the intervention strategy that is most likely to overcome the targeted activity limitation or participation restriction. Moreover, once the intervention program is completed, it will be possible to evaluate the outcome of the intervention program. Essentially, the outcome of an intervention program will be assessed according to the answer to the following question: "Is the specific activity limitation/participation restriction targeted by the intervention program (the specific goal) still present, or has the problem been fully or partly solved?"

Using the ICF (WHO, 2001) as the conceptual framework for rehabilitation programs allows one to clearly distinguish the goals of an intervention program from the tools and the strategies selected to attain the intended goal. It is not rare for the goal of a rehabilitation program to be overshadowed by the application of the proposed treatment. For example, the rehabilitation audiologist may ascertain that wearing hearing aids would enable a client to under-

stand the television better. In this instance, the goal of the rehabilitation program is not to fit the client with hearing aids, but rather to make it possible for the client to understand the television. Using hearing aids constitutes the strategy, or the intervention tool, used to attain the specific goal. The outcome of the rehabilitation program will be favorable only if the client's ability to understand the television is improved. The goal is not directly related to whether or not the client uses hearing aids. The desired outcome (understanding the television) may not be attained even though the client regularly uses hearing aids for this activity (e.g., because of background noise in the room where the client watches the television). Similarly, improving visual or audiovisual speech perception skills using hearing assistance technologies (HATs) does not constitute a goal of rehabilitation programs. The goal of a rehabilitation program should not be that the client will use hearing aids for an average of 8 hours per day or that the client will purchase and install a luminous alerting system to the telephone. The goal of an intervention program should be stated in terms of the client's participation in specifically identified activities.

Another positive aspect of using the ICF (WHO, 2001) is that this conceptual framework for rehabilitation makes many options available to overcome activity limitations or participation restrictions. Recall that according to the ICF, an activity limitation or participation restriction occurs as a result of a complex interaction among several factors including: the person's impairment; the activity; the personal factors associated with the person with the impairment, as well as those of other people who are involved in the activity; and the physical and social environmental (Hickson & Scarinci, 2007). If all of these factors contribute to creating an activity limitation or participation restriction, they should also be considered when identifying possible solutions to overcome the limitations or restrictions. That is, the possible solutions to overcoming an activity limitation or participation restriction might be one or a combination of many of those factors.

As discussed above, with some minor modifications, the ICF (WHO, 2001) can also be used to account for some activity limitations or participation restrictions that a person without impairment might experience. For example, Pat, who has normal hearing, might experience a limitation or restriction when he plays bridge because Charlotte, his playing partner, who has a severe hearing loss, has difficulty understanding the bids made by the other three players. Or, Louise may explain that she no longer goes to the theater

(a participation restriction) because her husband, who has a severe hearing loss, will not accompany her. Or, Lou might have to forego his social dance class because his partner's hearing loss is such that she can no longer follow the beat of the music. In our view, it is legitimate and sometimes advantageous for an audiologist to provide rehabilitation services to someone with normal hearing who participates in activities that involve people with hearing loss. Sometimes the person referred to the audiologist will deny having problems due to hearing loss. In that case, the audiologist may ask the spouse with normal hearing if they experience any activity limitations or participation restrictions in their everyday activities; if so, the spouse with a hearing loss may be solicited as part of the solution to overcome the difficulties experienced when the couple participates in the targeted activities.

In summary, relative to a medical model of health, adopting the ICF (WHO, 2001) as a conceptual framework for AR services will have marked ramifications on how rehabilitation programs are conceived, described, provided, and evaluated. Most importantly, it will likely have a significant impact on the clients who seek rehabilitation programs. The rehabilitation programs will not focus on reducing the client's impairment. Rather, the rehabilitation programs will focus on reducing or eliminating the limitations or restrictions clients experience when they participate in certain activities they have identified as being important to them.

Implications of Applying the ICF to Research in Rehabilitation

In the introductory section of this chapter, it was suggested that one reason few studies have been able to demonstrate the effectiveness of AR services was that, in some cases, the conceptual framework underlying the research investigations was inappropriate. Gagné (2003) outlined four fundamental principles that should guide evaluative research in AR. A detailed discussion of these fundamental principles is beyond the scope of the present chapter. However, one principle relevant to this discussion was "treatment programs should be selected so that they meet the goals of audiological rehabilitation and be consistent with the conceptual framework from which these goals are defined" (Gagné, 2003, p. S105). Given that there is a wide consensus that the ICF (WHO, 2001) constitutes the most suitable conceptual framework for clinical services in rehabilitative sciences, including AR, it

is logical that the same framework be used to guide evaluative research in AR.

Gagné (2000) discussed some of the implications of applying the ICF (WHO, 2001) to treatment evaluation research. He explained that the most important ramification of the ICF is that the goal of the research should be consistent with and based on the conceptual framework that guides clinical services. From an ICF perspective, the goal of an evaluative research project must be to determine whether the treatment program under investigation eliminates or reduces activity limitations or participation restrictions rather than disorders, impairments, disabilities, or handicaps. This is not a trivial issue. This approach requires a major shift in how evaluative research should be conducted, because the framing of the research question will influence almost all aspects of the research project, including the choice of research paradigm, the data collected, the types of analyses conducted, and the interpretation given to the data obtained. For example, there is an important conceptual difference between an investigation that aims to determine whether a digital hearing aid with a specific algorithm improves speech perception in noise and an investigation that aims to determine whether a digital hearing aid with a specific algorithm will improve Mrs. Kirk's ability to converse with her friends at the bridge club. The first question is a generic question of hearing disability (i.e., speech perception in noise). The second question addresses the specific issue of Mrs. Kirk's participation restriction at a specific activity. It is likely that the most suitable research paradigm for a given research project will vary according to the questions posed. For example, a classical group research design conducted in the laboratory (e.g., presenting speech stimuli in noise under different experimental conditions, including one that involves the digital hearing aid under investigation) would constitute an appropriate paradigm to address the first question. However, that paradigm would not be appropriate to address the second question. For the second question, a single-subject experimental design based on data obtained from questionnaires administered to or interviews conducted with Mrs. Kirk would constitute an appropriate experimental paradigm. Both the classical group research design and the single-subject experimental design have positive aspects and negative aspects. They simply address different questions. The results obtained from the first study would make it possible to make a generalized statement concerning speech perception in noise with the hearing aid. However, it would not provide data that would help ascertain whether or not the hearing aid would resolve

the specific limitations or restrictions that Mrs. Kirk experiences at her bridge club because of the differences in noise, settings, tasks between the experimental conditions, and the actual activity limitation. On the other hand, the second experimental design would make it possible to address the specific objective set for Mrs. Kirk (i.e., does it help communication at the bridge club). However, the data obtained from that investigation would not make it possible to generalize the findings (i.e., whether or not the hearing aid is useful) to other situations nor to other individuals.

A discussion of the application of experimental designs that lend themselves to treatment-evaluative research that is consistent with treatment services based on the ICF (WHO, 2001) is beyond the scope of this chapter. However, there is a literature that addresses this issue (Barlow & Hersen, 1984; Cardillo & Smith, 1994a, 1994b; Gagné, 2003; Gagné et al., 1999; King, Leolane, & Verba, 1994; Kiresuk, Smith, & Cardillo, 1994; Kiresuk & Sherman, 1968; McReynolds & Thompson, 1986; Olswang, Thompson, Warren, & Minghetti, 1990; Smith & Cardillo, 1994). As this point relates to the present chapter, it should be noted that, to be useful and valid, treatment evaluative research must be consistent with the underlying conceptual framework used to provide clinical services. If it is not, the results of those investigations may not have practical clinical applications; or, even worse, the research will lead to findings and conclusions that are incorrect and invalid.

Summary

A major goal of the chapter was to point out that, whether it is officially acknowledged or not, there is always a conceptual framework underlying the rehabilitation services that audiologists provide. Another important goal was to describe some conceptual frameworks, namely the ICIDH (WHO, 1980) and particularly the ICF (WHO, 2001), that provide a valid and useful model for AR. Adopting the ICF (WHO, 2001) requires a major paradigm shift in the way audiologists conceive of rehabilitation and in the way rehabilitation services are organized and provided. However, there is little doubt that adopting the ICF (WHO, 2001) will result in the provision of better services to those who require them.

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4

THE SOCIOLOGICAL EFFECTS OF STIGMA

Applications to People With an Acquired Hearing Loss

Kenneth Southall, Jean-Pierre Gagné, and Mary Beth Jennings

Introduction

In many societies, negative stereotypes and prejudices are attributed to people who have hearing loss. The general population perceives individuals with hearing loss to be “old,” “cognitively diminished,” “poor communication partners,” and “uninteresting” (Blood, 1997; Doggett, Stein, & Gans, 1998; Erler & Garstecki, 2002; Franks & Beckmann, 1985; Kochkin, 1993, 2007b). Partly because of society’s view of deafness,¹ many people who acquire hearing loss as adults may be devalued by others. This phenomenon, known as *stigmatization*, is very prominent in many Western societies.

Stigmatization is a social construct, whereby society attaches a label to a devalued trait. As such, individual differences judged to be stigmatizing are defined by societies. Within this context, the term *society*, from a sociological perspective, is defined as a group of persons regarded as forming a single community, or any organized group of people joined together because of some interest in common (McKechnie, 1976).

The stigma related to hearing loss constitutes a major obstacle to audiologic rehabilitation. Due to the ubiquitous and harmful nature of stereotyping, a person who suspects they have hearing difficulties may conceal the effects from others and choose not to seek services from a hearing health care professional. Hence, the individual is depriving himself or herself of rehabilitation services that are potentially helpful, such as obtaining hearing aids and hearing assistance technologies (HATs), and learning to request and apply communication strategies that are known to be effective. These individuals deny themselves the opportunity to improve their communication skills and increase their level of participation in activities that are essential or considered important to them.

The first edition of this text (Gagné, Southall, & Jennings, 2009) discussed how stress and the coping process are important in understanding and determining the effects of stigmatization on psychological well-being and physical health. Specifically, we described hearing loss stigma using a framework called the stigma-induced identity threat model

¹The general population does not readily recognize degrees of hearing loss; mostly, people are thought to have normal hearing or no residual hearing. Hence, to members of the general population, people either have “normal hearing” or they are “deaf.”

(Case Study 4–1). Major and O’Brien (2005) described this model as being based on the transactional models of stress and coping (Lazarus & Folkman, 1984). The stigma-induced identity threat model is based on two premises: Experiencing stigma can induce stress, which manifests in a manner similar to any other stressful situation experienced by a person being stigmatized; and stigma threatens one’s identity. Individuals are thought to approach situations that are potentially stigmatizing by appraising collective representations, situational cues, and relevant personal characteristics. Based on this conceptualization of the stigma process, cognitive and emotional resources are expended when the various elements of a stigmatizing situation threaten one’s self-identity. This expenditure of energy and possible negative outcomes of stigmatization may lead to a decrease in overall quality of life and cause secondary health-related problems (Leary, Tambor, Terdal, & Downs, 1995). Readers who wish more detail on this model and its applications to the phenomenon of hearing loss stigma are referred to this team’s previous work (Gagné, Southall, et al., 2009; Southall, Gagné, & Jennings, 2009).

Prior to the early 1960s, research on health-related stigmas was dominated by psychocognitive frameworks of the stigma process. These frameworks were founded on the assumption that stigmatization was the result of symptom recognition. For example, a person with normal hearing may recognize that another person is experiencing numerous communi-

cation breakdowns, speaking loudly, or dominating conversations; infer that the person has a hearing loss; and consequently devalue this person. After the early 1960s, an alternative, sociological framework called *labeling theory* was proposed. According to this conceptualization of stigma, discrimination is the result of a social process of identification of difference, referred to as *labeling* (e.g., “She is deaf!”). The purpose of this chapter is to discuss the stigma associated with hearing loss using the sociological framework called *modified labeling theory*.

Labeling theory, developed by sociologists in the 1960s (e.g., Becker, 1963), holds that groups in positions of power have a tendency to assign negative labels on groups of individuals who have less influence and who demonstrate deviance from cultural norms. Based on symbolic interactionism (Blumer, 1969; Mead, 1934), labeling theory is an explanatory framework that accounts for the stigmatization of marginalized groups (Markowitz, 2005). According to this line of thinking, inherent meanings to social behaviors do not exist; rather, actions are interpreted through language and symbols. While previous conceptualizations of stigma referred to “marks,” “traits,” and “characteristics,” labeling theory focuses on the selection and attachment of labels of difference. An important presumption of labeling theory is that there are social groups in society that have power (i.e., a governing status) that dictate which characteristics are construed as deviant behavior. This governing group: (a) has the resources to repel suggestion of

Case Study 4–1

John has a hearing loss and has been fit with hearing aids. He uses the hearing aids at home without any hesitation and finds them to be beneficial. However, he doesn’t use his hearing aids in the workplace because he worries his coworkers will notice them and question his ability to do his job. His concern is based on his past experience in the workplace. Specifically, he has observed that many of his coworkers hold negative stereotypes toward older adults who use hearing aids (e.g., “they are senile and don’t understand anything”). He has also observed that he feels personally threatened in some situations at work, but not in others. John finds work-related meetings very stressful because he is afraid his coworkers will notice he has hearing loss because he doesn’t always understand

what is being discussed. During those meetings John tends to sweat under his armpits and develops red blotches in the neck area (i.e., John’s natural reactions to stressful situations). He always has a headache by the time the meeting ends. Recently, a new management position was advertised at work. Notwithstanding the fact that he feels he has the competencies required to take on a management position, John did not apply for a promotion. In his mind, that job would too stressful because it would make it even more likely that his coworkers would become aware of his hearing loss (i.e., the outcome of the stigma). Not only did he not apply for the promotion, but John is also contemplating early retirement. He finds going to work too stressful.

deviant status, and (b) acts in such a way to maintain their position of power. Conversely, it is assumed that the devalued group is less able to repel accusations of deviance.

Throughout the chapter, an attempt is made to discuss labeling theory within the context of adults who have an acquired hearing loss.² It is our view that a better understanding of the factors that influence the labeling and stereotyping of people who have an acquired hearing loss, as well as the effects and consequences of being stigmatized, will be helpful to hearing health care professionals. Moreover, this knowledge should make it possible to develop intervention programs that will cater to the specific rehabilitation needs of people with hearing loss who are vulnerable to being labeled by others and feeling the negative effects of social stigma.

The sections that follow provide definitions and a description of social stigma. Then, the pioneering work of Raymond Hétu (1996) is summarized; specifically, we describe the stigmatization and normalization processes experienced by people with hearing loss. That is followed by a presentation of the five requisite components of the modified labeling theory, proposed by Link and Phelan (2001). Finally, we discuss the clinical implications of stigma on seeking health care by hearing professionals, adherence to recommended treatments, and the types of intervention services that could be offered to individuals who may be affected by hearing loss stigma.

Definitions of Stigma

Individuals are stigmatized when they possess, or are thought to possess, an attribute or characteristic that conveys a social identity that is devalued in a particular social context (Crocker, Major, & Steele, 1998). Stigmatization is a complex and multifaceted phenomenon. In Erving Goffman's major treatise entitled *Notes on the Management of a Spoiled Identity*, stigma was defined as "an attribute that is deeply discrediting" that reduces the individual "from a whole and usual person to a tainted, discounted one" (Goffman, 1963, p. 3). It is generally agreed that stigmatization

may occur when two groups of people exist: *insiders* (people who have a stigmatizing trait), and *outsiders* (people who do not have this trait). When outsiders (i.e., the governing group) perceive a trait to be stigmatizing, they might devalue, prejudice, and discriminate against the insiders (Dovidio, Major, & Crocker, 2000; Heatherton, Kleck, Hebl, & Hull, 2000). This is sometimes referred to as *public stigma* when the general public reacts negatively to an individual or group of people based on a stigmatizing attribute (Watson, Corrigan, Larson, & Sells, 2007). *Self-stigma* occurs when an individual who possesses a stigmatizing trait directs their own prejudicial attitudes inward, and devalues themselves (Jones, Farina, Hastorf, Miller, & Scott, 1984; Major & O'Brien, 2005; Watson et al., 2007).

Goffman, a sociologist, proposed a taxonomy of stigmas including "abominations of the body" (e.g., amputated limbs), "blemishes of individual character" (e.g., ex-convicts), and "tribal identities" (e.g., some religious belief systems). According to this conceptualization of stigma, individuals who possess such traits or characteristics were considered "deviant" or "abnormal" relative to a reference group (i.e., the outsider group), were devalued as individuals, and discredited as members of their society. A variety of marks, attributes, and characteristics may be the source of stigmatization. For example, in many Western societies, people who are obese, short, HIV positive, diagnosed with a genetic syndrome such as trisomy 21, who smoke tobacco, belong to a given ethnic or a specific religious group or sect, or who speak with a foreign accent may be stigmatized, just to name a few.

Building on Goffman's work, Jones et al. (1984) proposed six key dimensions of stigmas, including concealability, course of the mark, disruptiveness, aesthetics, origin, and peril. Several of the dimensions identified by Jones et al. are particularly relevant to the stigma associated with hearing loss. *Concealability* refers to the extent to which a stigmatizing trait is apparent to others. The *course of the mark* refers to the possibility that the attribute may become more prominent over time. *Disruptiveness* is in reference to how the stigmatizing trait interferes with social interactions. *Aesthetics*, relates to physical—"observable"—traits, and the extent to which these traits

²In preparing the chapter, it became apparent that the factors that contribute to stigmatization differ according to several factors, including: the degree of hearing loss (e.g., individuals with a moderate to moderately severe hearing loss versus those with a profound hearing impairment); whether the hearing impairment was present at birth (or before the normal period for acquiring oral language) or whether it developed in adulthood; and the age range of the population of interest (children versus young adults versus older adults). A decision was made to focus the content of the present chapter to issues related to stigmatization among adults who acquire a hearing loss in adulthood.

are unattractive as perceived by others. *Origin* refers to the perceived responsibility of the individual in acquiring or creating the trait. Finally, *peril* relates to the perceived danger of the trait to others in the social setting. Of these dimensions of stigma, Crocker and colleagues (1998) proposed that concealability and origin are the most influential in giving rise to stigmatization.

Hearing Loss Stigma

In most Western societies, people who have hearing loss are stigmatized. Goffman (1963) described how the stereotypes associated with hearing loss (he used the term *deafness*) are frequently generalized to other personal characteristics that carry a negative connotation. Hearing loss is often misunderstood as an intellectual challenge or a deficiency in personality and character. It is noteworthy that both intellect and personality are characteristics that may potentially prompt devaluation from others. Stigmatization of persons who have congenital hearing loss can be traced back to Babylonian laws, pre-Christian laws, religious texts, and the writings of Aristotle and Saint Augustine: Persons who could not hear or speak were likened to animals who were not capable of having intelligence or faith in God (Roots, 1999).

Of the dimensions proposed by Jones et al. (1984), perhaps the most relevant issue to the discussion of hearing loss stigma is concealability. Goffman (1963) referred to people who have marks or traits that are immediately apparent as *the discredited*, whereas individuals whose mark or trait is not readily obvious to others were referred to as *the discreditable*. Quinn (2006, p. 84) defined a concealable stigma as “a stigmatized identity that is not immediately knowable in a social interaction.” Several aspects of the stigma process may differ depending on the concealability of the trait. For example, a person with a history of mental illness or a person who is HIV positive may decide to conceal this personal characteristic from some people with whom they interact. This option is not possible to someone who has a conspicuous attribute (e.g., skin color, visible physical deformity). Hearing loss is an impairment that an individual can sometimes conceal from others. This explains, at least in part, why persons with acquired hearing loss have consistently requested that hearing aid manufacturers make the smallest and most inconspicuous hearing aids possible.

Social stigma research has shown that, relative to conspicuous stigmatizing attributes, having an invis-

ible stigmatizing trait has both some advantages and disadvantages. For example, one advantage of having a conspicuous characteristic is that the individual can decide if, when, and to whom to reveal. Should she or he choose to conceal, one drawback is that there is always discomfort associated with the possibility of having this characteristic unexpectedly disclosed during a social interaction (Case Study 4-2).

There is an increasingly comprehensive body of literature that addresses the psychocognitive consequences of concealing a stigmatizing trait. Studies have shown that concealing a stigmatizing trait increases the cognitive load required to take part in social interactions (Lane & Wegner, 1995; Smart & Wegner, 1999,). In addition to exerting the cognitive effort normally required to participate in a conversation, a person who is attempting to pass as “normal” will expend additional cognitive resources to avoid divulging any information, signals, or cues that would betray efforts to conceal. This may be particularly stressful for a person who has a hearing loss, given that normal conversations (even those in which disclosing hearing loss is not an issue) invariably require more cognitive resources than are typi-

Case Study 4-2

At the request of her family, Jane had a hearing test. The results of the audiologic assessment revealed that she had a bilateral moderately severe sensorineural hearing loss. She was told that the hearing loss is most likely due to a normal aging process (i.e., presbycusis). Jane was miffed by the way her hearing results were communicated to her. In addition, she holds very negative stereotypes about hearing loss and aging. When she was a child, Jane’s grandmother had hearing loss and used hearing aids. Because her grandmother frequently misunderstood and often needed several repetitions before she understood, Jane and her sister thought of her as being old and senile. That image of hearing loss and aging has stayed with her. Because she doesn’t want to be viewed in the same way that she labeled her grandmother, Jane decided to conceal her hearing loss from everyone. In fact, she often avoids attending family social events because she is afraid her siblings will recognize that she has difficulty hearing and make fun of her.

cally required by persons with normal hearing who perform the same task (Fraser et. al., 2010).

Related to the previous point, the amount of stress associated with participating in a social interaction will vary according to the importance attributed to unwillingly disclosing the stigmatizing trait and the likelihood of that happening. For example, for someone who has an eating disorder that is not physically apparent to others, the stress associated with a conversation will be diminished when the topic of discussion does not center around food and eating, while the stress will likely be heightened when the discussion centers around those activities. For people with hearing loss, in most social situations there is a risk of communication breakdowns due to the hearing impairment. Thus, in every social interaction, there is risk of disclosure. This may explain why many people with hearing loss often avoid social interactions. Although from a social health perspective, this strategy may be considered “maladaptive,” it is nonetheless a “safe” strategy (Case Study 4–3).

The manner and the timing of disclosure will influence how others receive this information. For example, if a third party unwittingly discloses someone’s stigmatizing trait (e.g., when a person with hearing loss has difficulty following conversation with another person and their spouse apologizes

and tells the conversation partner of their spouse’s hearing loss), or if it is divulged only after a trusting relationship is formed, the person who possesses the trait may be seen as being dishonest or untrusting (for example, when a person with hearing loss tells a colleague, with whom they have worked closely for a period of time, about the presence of the hearing loss, the coworker may wonder if the person did not trust them enough to disclose the hearing loss earlier. If this is the case, then there may be other matters that may be of importance at work that the person with hearing loss may not divulge). If it is disclosed too soon after meeting someone, there is a danger that others will perceive this disclosure as an excuse for incompetence or social inaptitude. For example, it may seem odd to others if a person with hearing loss immediately discloses the presence of the hearing loss when meeting someone for the first time (Bairan et al., 2007; Corrigan & Matthews, 2003; Joachim & Acorn, 2000; Pachankis, 2007).

Studies have also shown that there is a direct relationship between the desire to conceal a stigma and the importance given to the stigmatizing trait. For example, Major and Gramzow (1999) demonstrated that the more women perceived abortion to be a stigmatizing attribute, the more they attempted to keep their own abortion secret from others. Extending this to hearing loss, one would presume that the greater the amount of negative stereotypes that someone attributes to having hearing loss, the more effort that person will expend to try to conceal the hearing impairment from others (Case Study 4–4).

A final point about concealable attributes is that the person who has the stigmatizing characteristic is more likely to know what family members, friends, and workmates think about that trait. That is, because they may not be aware that someone in their midst has a stigmatizing trait, others may express their attitudes about that specific stigmatized attributes more freely. This is different from a conspicuous stigma (e.g., obesity) because in the presence of a person with a visible stigma, others are likely to be more diplomatic (i.e., politically correct) about how they express their attitudes about the specific stigmatizing trait. This information may be of value for rehabilitation. Knowing the attitudes of significant others toward hearing loss should provide insights on how to proceed in disclosing the hearing loss to others. For example, a person with hearing loss may choose first to disclose the impairment to someone they trust and respect or to someone they know to be less prejudicial toward people with hearing loss. In addition, they may solicit that person’s help by using repair strategies during

Case Study 4–3

Since his retirement about 10 years ago Paul has played golf regularly with his two sons and his nephew. However, recently Paul noticed that he had more and more difficulty hearing his golf partners when they played. He often felt left out of the conversations or required multiple repetitions before he understood the topic of the discussion at hand. Paul does not want to admit to his golf partners that he has trouble hearing because he thinks they will associate that with the fact that he is getting old and cognitively challenged. In any case, Paul doesn’t think his hearing problems are that serious, and says, “It is just normal to hear less well as we get older.” Nonetheless, Paul is embarrassed because he feels he has become a burden to his golf partners. Recently, he has invoked excuses for not joining his sons and nephew for a round of golf. He would rather stay home than potentially embarrass himself in their presence.

conversations. This strategy will enable the person with hearing loss to improve his or her self-esteem and to gain confidence as a communication partner in a given setting.

Case Study 4-4

The results of a recent audiologic assessment revealed that Ruth has a hearing loss that is attributable to aging. Because she holds negative stereotypes associated with hearing loss in persons her age, Ruth has chosen not to reveal her hearing loss to her friends and family.

Ruth is an avid bridge player. At the bridge club, her perception is that most bridge players do not want to be paired with someone with hearing loss because they can't follow the hand and often make inexcusable mistakes because they don't hear well. Because she doesn't want to be seen as a hearing-impaired person, Ruth has chosen not to divulge her hearing difficulties at bridge. However, she finds that it is very stressful and effortful to keep this secret from her card-playing partners. Recently, she has noticed that she sometimes makes mistakes while playing bridge (e.g., inappropriate bids) because she is concentrating so much on not revealing her hearing loss to others.

Conceptualizations of the Stigma Process—Persons and Hearing Loss

To our knowledge, Héту (1996) was the first scholar to describe the issue of stigma associated with hearing loss from the perspective of individuals with an acquired hearing loss. In his seminal article "The Stigma Attached to Hearing Impairment," Héту (1996) proposed two conceptual models that he based on the generic literature available on stigma from the social sciences, and on the results of interviews with adult males with an acquired hearing loss (and their spouses). The first model described the *stigmatization process*. The second model described the *normalization process*. According to the stigmatization process model, people who are discredited (i.e., stigmatized) because of their hearing impairment experience shame. Shame, Héту suggested, is the emotion that accompanies threats to one's sense of social belonging. Shame is a social control mechanism that instills acceptable behavior and inhibits unacceptable behavior. According to Héту's stigmatization process, when people with hearing loss interact with people who have normal hearing, communication breakdowns and other "deviant" behaviors occur (Figure 4-1). These breakdowns prompt demeaning and discriminatory reactions from communication partners and result in the person who has a hearing loss feeling shame and guilt. As a consequence of the stress induced, and feelings of incompetency that develop, the person's self-esteem and social identity are dimin-

Stigmatization Process

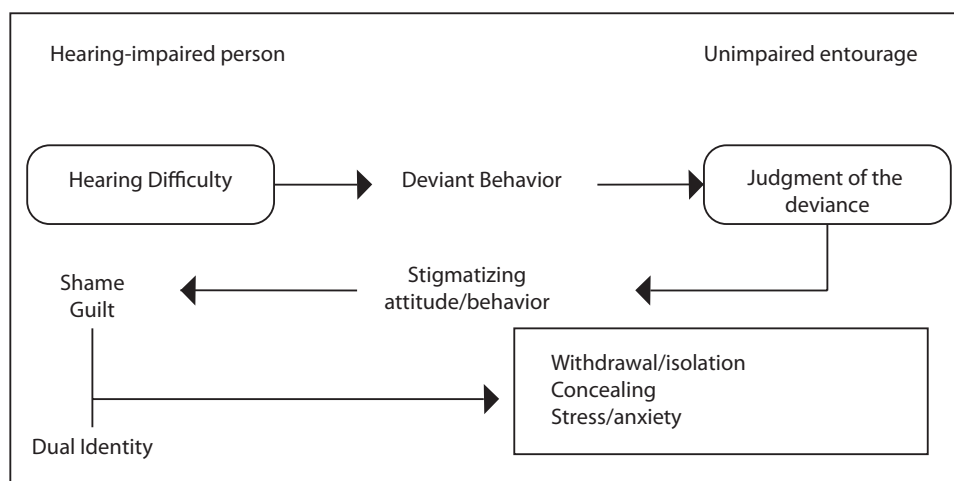


Figure 4-1. Héту's stigmatization process model (from Héту, 1996).

ished. The strategy often used to avoid stress and feelings of incompetency associated with unsatisfying social interactions is to conceal their hearing loss from their communication partners or to withdraw from social activities.

Hétu (1996) also proposed a two-step normalization process model to describe how people with hearing loss may overcome feelings of shame and guilt associated with hearing impairment, and how they might regain a more favorable social identity (Figure 4-2, upper panel). The first stage of the normalization process involves meeting and interacting with other people who also have a hearing loss (members of the *in-group*). Together, members of the in-group share experiences of hearing difficulties and the resultant unsatisfactory social interactions. It is proposed

that, through therapeutic activities, participants begin to realize that unsatisfactory social interactions are the result of the hearing loss, rather than other factors that may be unjustifiably attributed to them (e.g., being intellectually dull, an uninteresting communication partner, an unfriendly person, unwilling to communicate, or otherwise responsible for the communication breakdowns). And, through these activities, individuals who are stigmatized because of hearing loss begin to realize there are others who feel denigrated, diminished, or ashamed of themselves because of hearing difficulties. They realize they are not alone in these feelings. These realizations trigger the normalization process, and the restoration of a more positive social identity is initiated. Individuals with hearing loss become more willing to engage

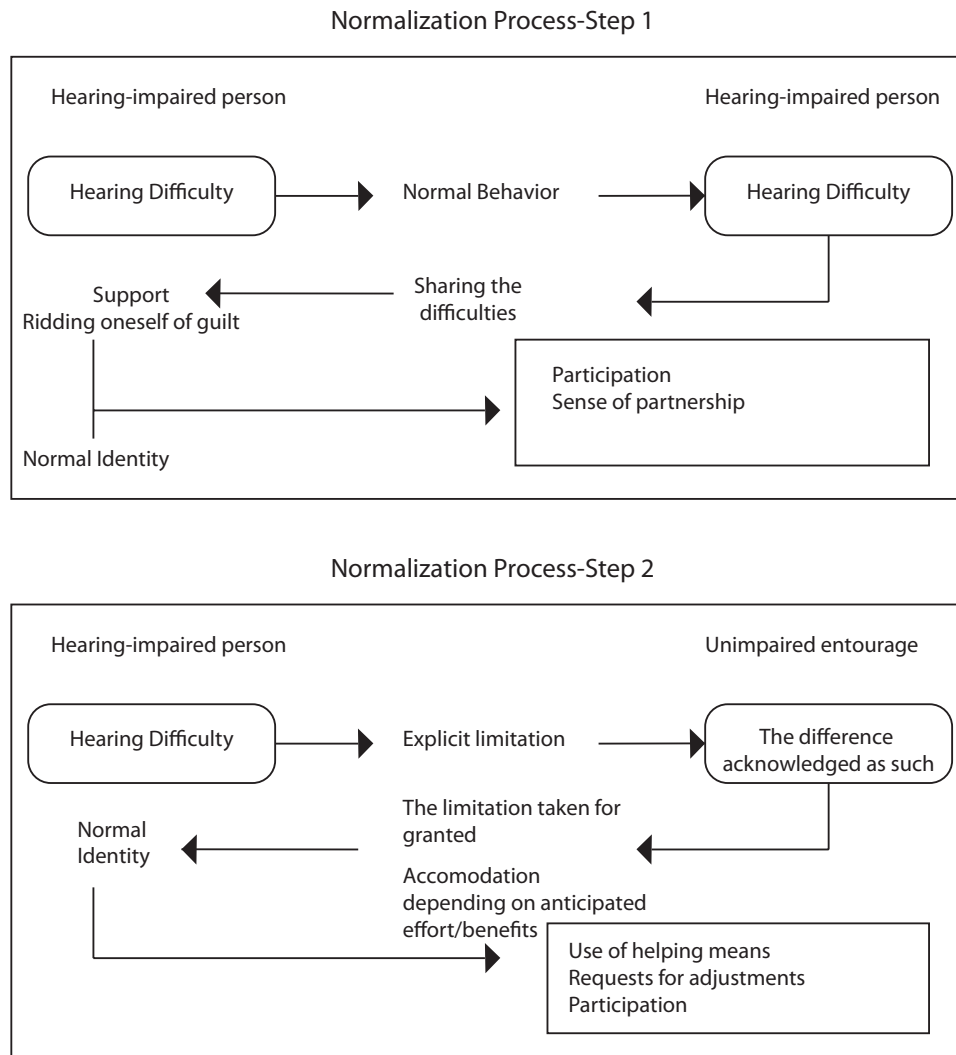


Figure 4-2. Hétu's two-step normalization process model (from Hétu, 1996).

in social activities and interactions. They may learn appropriate coping strategies and experience success in using these strategies when they interact in this favorable social environment (i.e., in the presence of others who have hearing loss). Through the normalization process, people with hearing loss start to feel better about themselves.

In the second stage of the normalization process (see Figure 4-2, lower panel) people with hearing loss are encouraged to interact with people in their entourage who do not have a hearing loss. They are taught and encouraged to inform their communication partners that they have a hearing loss and to solicit the use of communication strategies that will optimize the exchange of information. Under these circumstances, it is suggested that communication partners are more likely to acquiesce to the requests of the persons with hearing loss. It is likely that communication will become more efficacious and more satisfying for both communication partners. Participating in more satisfying verbal conversations (i.e., social interactions) restores a more favorable social identity for the person with hearing loss. As the process of restoration continues, the person with hearing loss will gain confidence in their ability to be a satisfying communication partner. Consequently, they are more likely to participate in valued activities that include people

who do not have a hearing impairment (members of the *outsider group*), and will likely regain a more positive image of themselves. Hétu's contribution of providing a conceptual model of stigmatization and guidelines for rehabilitative services that restore a favorable social identity contributed enormously to the domain of rehabilitative audiology as well as to the social integration of adults with an acquired hearing loss (Case Study 4-5).

Around the same time that Hétu published this article, important works were being published on the sociological aspects of stigmatization. The next section describes the modified labeling theory. In the authors' view, most of the concepts described in this model are applicable to the stigma associated with hearing loss. They believe that hearing health care professionals will benefit from having a better understanding of the self-stigmatizing process and its effects on people who have a hearing loss. Additionally, it is their belief that, largely due to Hétu's work, integrating aspects of this model into the domain of rehabilitative audiology will complement and extend the current level of knowledge that is available. Undoubtedly, a more comprehensive understanding of the self-stigmatizing process will lead to the development of more appropriate rehabilitation services for people who are stigmatized due to their hearing loss.

Case Study 4-5

Ruth, the avid bridge player from Case Study 4-4, agreed to attend a 5-week communication strategies Audiologic Rehabilitation program. Ruth found the group AR meetings very helpful. She enjoys the fact that everyone in the group shares their experiences, both pleasant and unpleasant, about hearing difficulties with the other group members. In this setting, the open discussions about hearing loss and its manifestations are very useful and nonthreatening. She has learned communication strategies that she applies in various settings.

Generally, she finds the other group members interesting. However, she is particularly impressed by two of the women in her group. They are intelligent, assertive, and confident in themselves. Both of them are actively involved in community activities and are quite open about the fact that they have hearing loss that makes it difficult to communicate in certain situations. Observing these women in the group setting has helped Ruth regain some confidence in herself and it served to boost her self-esteem.

Some group activities included role-playing activities. One of those activities consisted of learning how to inform others, without self-pity and shame, of one's hearing loss. Practicing this activity in the group provided Ruth with the skills and confidence required to eventually inform her bridge-playing partners of her hearing loss and to suggest communication strategies they can use to reduce the possibility of communication breakdowns. She realized her friends from the AR program did not make such a big deal about their hearing losses. Consequently, she found attending bridge club less stressful than it was when she was concealing her hearing problems from her friends.

Conceptualizations of the Stigma Process—Modified Labeling Theory

The modified labeling theory (Link et al., 1989) is a sociological perspective on the stigma process that has received general support among mental health theorists. According to this theory, an integral part of socialization is the learning of stereotypes about behaviors, objects, and attributes. The general public comes to understand that people who possess certain attributes are devalued in society (e.g., through reporting in the media), and the public tends to discriminate against them (e.g., distance themselves from the individuals who have the stigmatizing trait). Familiarization with these views is strengthened through daily social interactions (Link et al., 1989; Scheff, 1966).

According to this theory, certain stereotypes become personally relevant following confirmation that one possesses a stigmatizing attribute (e.g., diagnosis or entrance into treatment). At this time, some of the stereotypes learned earlier in life unexpectedly become applicable to oneself, and result in reduced self-esteem. Labeled individuals expect to be devalued and discriminated against, and anticipate these outcomes when they enter into social settings. To avoid negative outcomes, individuals use a variety of strategies, such as withdrawing from social events, keeping the stigmatizing trait a secret from others, or even educating others about the reality of the stigmatizing attribute. The modified labeling theory predicts that beliefs about societal attitudes and coping strategies employed enhance the anticipation of stigmatization and lead to negative outcomes, such as a restricted social network and negative impacts on work opportunities. Thus, for people who possess a stigmatizing attribute, the stigma process becomes a self-fulfilling prophecy (Case Study 4–6).

Link and Phelan (2001) proposed that stigma has five key components: labeling, stereotyping, separation, status loss, and discrimination, all of which take place in a context of power imbalance. The following section discusses each component as it relates to the stigma associated with acquired hearing loss.

Labeling

Labeling is the identification of distinguishing characteristics (i.e., differences), and the assignment of some social significance. In all societies, individual differences exist. Each individual has differences based on age, weight, IQ, etc. Yet, many differences are not ste-

Case Study 4–6

Paula is a 75-year-old widow who self-stigmatizes because of her hearing loss. She believes people with hearing loss are generally “getting on in years,” and “not particularly pleasant” to be around. Her adult daughter has noticed that Paula is lonely and becoming slightly depressed. She suggested that Paula get involved in volunteer work at her church. Recently, a call for volunteers was sent to church members; kitchen help was sought to prepare and serve a brunch for Easter Sunday. Paula is a good cook and would enjoy volunteering for that type of activity. However, due to the self-stigma associated with her hearing loss, including her self-ageism and her inability to enjoy being with others, Paula decided to pass on this opportunity. She thinks the members of the church’s social club will see her as a burden if she joined the group. She fears others will judge her as someone who can’t really contribute to the tasks that need to be done in the kitchen. Because of the poor perception she has of herself, Paula decided not to volunteer for this activity.

reotyped. For example, a person who has especially good hearing is not stereotyped in Western cultures, but someone who has hearing difficulties is stereotyped and may encounter discrimination. Thus, a process is in place whereby societies select differences that will be stereotyped. Societies categorize and assign labels based on broad criteria: this person is “old” or “young”; this person is “hearing” or “deaf.”

Based on the above description, two characteristics of labels are noteworthy. First, labels vary by place. Labels that are assigned in one society may not be assigned in another society. For example, being overweight may have a negative connotation in some societies, but it may be a valued personal trait in another society. A personal attribute that may be viewed as positive or neutral in one microsociety (e.g., hearing loss among members of the Hearing Loss Association of America), may be viewed negatively in another microsociety (e.g., hearing loss among members of a competitive bridge club).

Second, labels change over time. Within a given society, a trait may be viewed as positive or neutral at one point in time, and the source of stigmatization at another time. For example, only a few decades ago,

smoking tobacco did not carry much of a negative connotation among most middle-class Americans of European descent. People smoked in a variety of social settings. Nowadays, in many respects, smokers are generally viewed negatively. To various extents, people who smoke are devalued as individuals and discredited as members of society; they are stigmatized. The issue raised here is that stigmatizing attributes are defined by the collective perceptions and values that members of a society (outsider group members) hold at a given point in time. When this point is considered as it relates to hearing loss, this is reassuring. It is possible and hopeful that the negative and prejudicial attitudes currently associated with hearing loss in most Western countries may change over time. Perhaps as Baby Boomers get older and society's views become less conservative, the negative connotations associated with hearing loss will subside (Case Study 4–7).

In the studies that the present authors have conducted (Jennings, Southall, & Gagné, 2013; Southall, Gagné, & Jennings, 2010; Southall, Gagné, & Leroux, 2006; Southall, Jennings, & Gagné, 2011), participants have identified several concerning aspects of the labels assigned to people who have hearing difficulties. The language used to describe hearing deficits influence how people respond to people who have the condition.

Many people with hearing loss whom we have interviewed have objected to the label “deaf” because the term does not fit their conceptualization of their condition. Most people with hearing loss have some degree of residual hearing, a state that is not captured by the term *deaf*. One participant from the authors' workplace disclosure study commented, “People are sympathetic to the person that don't hear (people who are deaf) . . . but because you have a partial [hearing loss], you can hear partially you know. I think it's as big a frustration as not being able to hear.” Other labels many participants believed pejorative were the terms *hard of hearing* and *hearing impaired*. Individuals who the authors interviewed made comments such as, “I am a person who just happens to have hearing difficulties. It does not define who I am!” The authors came to understand that this frustration with language and labels encountered by participants extends into health care settings, including the offices of hearing health care professionals.

It is common, even among those who have hearing difficulties, to use the terms *hearing loss* and *deafness* interchangeably. For example, in the authors' investigation of factors that influence the use of HATs, one participant, who had a moderate to severe hear-

Case Study 4–7

Although there is not much experimental evidence, many audiologists have clinically observed that many Baby Boomers are less reticent to use hearing aids than adults of previous generations were. Several factors may account for the trend. One factor may be related to the current generation of hearing aids. First, many models of hearing aids are inconspicuous (e.g., behind-the-ear aids with a Receiver-In-Canal; hearing aids come in a multitude of colors). Also, technologically, hearing aids have become very performant and include advanced micro-processing components as well as sophisticated algorithms (e.g., noise reduction circuits, feedback elimination circuits, etc.). Also, connectivity options incorporated into many hearing aids (e.g., direct reception of telephone, television, tablets, stereo systems, and other signals) has made their use very convenient and congenial. Baby Boomers who are savvy users of cellular telephones, the internet, and computer technology appreciate benefits such as the versatility and quality of signal provided by hearing aids. For them, all forms of communication are important and they will not refrain from using hearing aids if they facilitate accessibility to the devices they use to communicate daily. Perhaps the social stigma associated with hearing loss and the use of hearing aids is decreasing. Experimental data that would shed light on this issue would be timely.

ing loss, commented, “My father was slightly deaf, but not a real—not as bad as I am” (Southall et al., 2006). From the same study, another participant said, “Both parents and my brother were deaf. And my brother is three years older than I am. And he has maybe got 50% loss” (Southall et al., 2006). This type of characterization of hearing difficulties is misleading to other people and may perpetuate labeling and stigmatization.

Stereotypes

Stereotyping refers to negative attributes affixed to an individual or group based on an oversimplification or biased frame of reference. According to Goffman

(1963), the association of the trait to the stereotype is the origin or root of the stigmatization process. Stereotyping is, in essence, categorizing, or making generalizations about things (e.g., people) that we encounter in social settings. We categorize because it is an efficient way to make sense of complex social settings, and to conserve exhaustible cognitive resources (Link & Phelan, 2001).

Many of the present-day stereotypes attributed to people with hearing loss date back to prejudicial attitudes held against people who were deaf in ancient Greek societies (Higgins, 1980). Like other cultural beliefs, prejudicial beliefs are transmitted from one generation to the next. Aristotle wrote that people who could not hear were speechless, and most likely senseless. At this time, the word used to express that someone is speechless was identical to the word used to express that someone is senseless. Goffman (1963) observed that this is a good example of how language used in social settings can influence social and self-stigma. Nowadays, should one encounter a person who is unable to communicate in a “normal” manner, one might revert to (i.e., rely upon) historically and culturally constructed stereotypes to understand what is perceived to be deviant behavior. As with some other chronic impairments (e.g., vision), the general public does not easily make the distinction between absolute and partial loss of function (Bunting, 1981). Thus, the general public may stigmatize people with hearing loss in the same manner that they would stigmatize people who are deaf. For example, the general public associates hearing loss with reduced intellectual capacity and mental health problems—two stereotypes commonly associated with people who are deaf (Waridel, 1993).

Many of the historically relevant stereotypes described above have emerged in the interviews we have conducted. For example, the following two quotations are taken from our exploration of factors that influence the use of HATs. Seemingly, still today, fears of stereotyping are a concern for some people who have a hearing loss. One participant from our workplace disclosure study was very embarrassed about what hearing aids symbolized to others in social settings, in particular because the most common stereotypes (described above) do not apply to her situation. She said, “It [reluctance to use a hearing aid] was a whole ego, vanity thing. When I first got my hearing aid, I was embarrassed. I hated it. I was very embarrassed by it. And at that point, I found myself having to explain it. Having to explain it right off, so that when they saw me, they wouldn’t think I was a freak. They would understand I wasn’t just a deaf person;

I was someone who got a virus and suddenly got deaf.” (Southall et al., 2011). Another participant said that one of her motivations to use hearing aids was because she wanted to be socially engaged, unlike the subject of a famous piece of art: “They say Mona Lisa was deaf. You know that? You know that story? . . . Because of that stare of hers. And if you have ever noticed that the people who cannot hear, they feel out of it” (Southall et al., 2006).

These two quotations demonstrate a certain level of self-stigma, deriving from attitudes likely developed before being informed that they had a hearing impairment. Héту, Riverin, Getty, Lalonde, and St-Cyr (1990) reported that this manifestation of cognitive dissonance (their general view of people with hearing loss versus the more positive view they have of themselves despite their hearing loss) was sometimes difficult to reconcile. This denigrates and contributes to a devalued self-identity (Case Study 4–8).

Many of the people who have participated in the authors’ studies have described strategies they have employed to avoid acknowledging the hearing loss to

Case Study 4–8

John was a reasonably well-adjusted manager in a noisy factory plant. At 50 years of age, he consulted an audiologist and learned that he has a moderately severe hearing loss, probably due to noise exposure. At work, he was regularly exposed to negative attitudes about people who have hearing loss, and he generally agreed with these views. Following his visit with the audiologist, John did everything possible to not divulge his hearing loss to his coworkers (superiors, peers, and subordinates). His self-assurance diminished considerably. Also, he found he was having difficulty concentrating on work-related issues because he was afraid his colleagues would realize he is hearing impaired. Recently at work his boss summoned him to a meeting because he had made several key mistakes. This only contributed to his devalued self-identity because John always considered himself very competent at his job. Presently, a new position that would be a natural promotion for John is advertised a work. However, John decided that he will not apply for the position because he does not feel he has the knowledge and competency for the advertised post.

other people in their home, work, and in many activities. Among these strategies are hiding their hearing aids, not telling employers and coworkers about hearing loss, and downplaying the severity of their hearing difficulties in consultations with health care professionals—all done, it could be argued, to avoid other people making the connection between the label *hearing loss* and the stereotypes associated with this condition. The modified labeling theory presupposes that people who have stigmatizing traits are aware of the relevant negative stereotypes, and are aware that they will be devalued in social settings (Crocker et al., 1998).

Fear of stereotyping has been repeatedly expressed by participants in the authors' studies. In many instances, participants agreed with the stereotypes, which led to diminished efforts to manage the condition. For example, a government worker described how stereotypes influenced her attitudes about hearing aids: "I thought those [hearing aids] are just for old people. Young people don't have hearing problems . . . In my mind only old people wore those big things" (Southall et al., 2010). In the authors' investigation of how stigma may impact help-seeking for hearing difficulties, one participant indicated, "If I admit that I have a hearing loss (i.e., the label), people will think that I'm old. They will think that I'm not as capable. It's still the stigma that ties hearing loss to a lack of mental agility [i.e., the stereotypes]" (Southall et al., 2010).

Yet, in our investigation of help-seeking behaviors, numerous participants described how it was important for them to overcome and better manage situations of stereotyping. A commonly expressed desire was to learn to manage their hearing loss while not allowing stereotypes to wear them down. One participant said, "It is hard enough having a hearing loss without feeling like it's only old people who have hearing loss" (Southall et al., 2010). From the authors' investigation of HATs, one participant said, "But if you don't use it then the other guy on the other side of the table knows that you have a problem . . . He thinks you are stupid. Oh sure. He thinks that you are dull" (Southall et al., 2006).

Separation

Separation refers to a dissociation or segregation between those who have the stigmatizing trait and those who do not. Separation may be created due to the dehumanizing and demeaning language frequently used to reference people with disabilities,

such as "She is impaired" "She is deaf." Labeling creates an "us" versus "them" divide between those who have stigmatizing traits and those who do not. Although this may not be an overtly expressed goal (to separate from the other), the stigma process results in both people who have a stigmatizing attribute, as well as those who do not have one, wishing a separation from the other group. Individuals who have been assigned a label expect to be devalued and discriminated against. Thus, for them, separation results in avoiding mistreatment. While those who do not have a stigmatizing trait may separate themselves to avoid stigma by association and avoid the awkwardness of communicating with someone who is stigmatized (Case Study 4–9). One person interviewed for the help-seeking study described how the challenges and frustrations of being associated with prejudicial attitudes in the workplace forced her into an extended depression in her home. She did not feel as though there was anyone who could relate to the problems she was having at work and in her personal life, and said

I really thought I was the only person like me. I didn't know that there were thousands and thousands of me's out there who are feeling the same isolation, the same sadness, the same frustration, the same anger, the same "what do I do about this?" type feeling. (Southall et al., 2010)

Another person interviewed for the investigation of workplace disclosure described how she wholly agreed with the stereotypes associated with hearing loss, and how those had a profound impact on her self-esteem. She said, "the psychosocial aspect is that I felt stupid. I felt very dumb. I didn't like answering things or talking to anyone because I would be talk-

Case Study 4–9

Mary, who does not have a hearing loss, is active in a local volunteer organization. Judy is another volunteer who has a hearing loss. During the social hour after the meetings, Judy often stays on her own due to difficulties following conversations. Mary avoids conversing with Judy because she is concerned that her association with Judy, who has observable hearing difficulties, may result in negative judgments being made about her.

ing out of context . . . I was a very withdrawn person” (Southall et al., 2011).

For individuals who have stigmatizing traits, separation also meant separating themselves from activities. This self-imposed separation was often the result of an expectation that participation in the activity would likely result in acknowledging the trait to others who may or may not have previously been aware. One person described her profound aversion to using the telephone at work as, “The things I did *not* to have to make a phone call is incredible. I could run to the other end of the building, I would fax. People would call and leave me messages. And my colleagues were really very good in helping me to save these messages. And then I would send off a fax or I would run after their office [and they would say] ‘well you could have just picked up a phone!’” (Southall et al., 2011). A different participant said, “I avoided the phone for so long that it (now!) produces anxiety when I pick up a phone. And I realize that I’m squeezing it so hard. This is not fun. I have a hammer lock on that instrument” (Southall et al., 2011).

Status Loss and Discrimination

Status loss refers to a reduced standing in social position. According to the modified labeling theory, status loss is the result of labeling, stereotyping, and separation. Status loss may arise in any milieu of a person’s life where a hierarchy exists. In his examination of the experiences of people who are Deaf (i.e., capital “D” Deaf community—those individuals who are born deaf), Higgins (1980) provided compelling evidence that derivations of ancient prejudices toward people who are Deaf continue to affect the present-day community of people who have hearing loss. In many ways, the stereotypes about people who are Deaf have affected the rights of people who have hearing loss. In North America, for example, in the 1900s, people who were Deaf endured a range of prejudicial and discriminatory policies and practices related to land ownership, the right to marry, and the right to vote.

Many participants have experienced status loss in the workplace. One particularly poignant interview conducted in the authors’ workplace disclosure study provides a good example of how having a stigmatizing trait can result in status loss. In this interview, a man who has held various jobs described how the mistreatment he received by coworkers robbed him of his coworker status. “Brian” explained that he lived in “no-man’s land,” a parallel existence in which he is secluded, because his workplace peers cannot

understand his personal situation (i.e., his hearing loss). He said,

[There is] the land of the hearing, and the land of the non-hearing. What I call no-man’s-land [is] in between. I feel like I am trapped there . . . “In” the world, but not “of” it. I know it sounds kind of silly, but . . . it’s a peculiar sensation . . . I don’t know how to describe it, but it exists. (Southall et al., 2011)

Status loss, however, was not restricted to workplace settings. One participant in the help-seeking study described how her relations with her grandchildren and her status as a grandmother had been drastically modified due a lack of understanding about her hearing difficulties, and this had created a separation. She said,

We did a lot of smiling and waving at my four grandchildren. But I wasn’t the kind of grandmother who would sit down and talk to them because I avoided having conversations with anybody. And particularly somebody with an accent or a child with their juvenile speech patterns. So I would just absent myself from interacting with them unless it was necessary-type interactions. (Southall, Gagné, & Jennings, 2010)

For many participants, separation only served to alienate further the person who has been assigned the label. Separation further constricted social networks.

Discrimination refers to behaving in such a way that this consistent with prejudicial attitudes. Discrimination against people with hearing loss is a very common in Western societies. For example, individuals who have hearing difficulties sometimes encounter difficulties finding paid work (Blanchfield, Feldman, Dunbar, & Gardner, 2001; El-Khiami, 1993; Parving & Christensen, 1993; Schroedel & Geyer, 2000). Employers may assign tasks that do not take full advantage of abilities (Danermark & Gellerstedt, 2004). Workers with hearing loss are sometimes denied promotions and/or career advancement (Royal National Institute for Deaf People, 2006). In the U.S., in households where one inhabitant had hearing loss, it was estimated that yearly household income was approximately U.S. \$12,000 lower per year depending on the severity of hearing loss (Kochkin, 2007a). An analysis of data compiled by the U.S. Equal Opportunity Commission (The U.S. Equal Employment Opportunity Commission, 2006) for the years 1992–2003 revealed more than 8,900 allegations of discrimination filed

by people with hearing loss, ranging from matters of discharge to absence of reasonable workplace accommodations (Bowe, McMahon, Chang, & Louvi, 2005).

Access to satisfactory telephone communications is another source of discrimination. One report found that federally regulated telephone communications for individuals who are deaf, deafened, or hard of hearing provide unsatisfactory service due to inadequate procedures to ensure equitable access, as well as a lack of proper personnel training, which hinders utilization (Canadian Human Rights Commission, 2006). The telephone has been the source of many interesting comments in the interviews we have conducted. One person who worked for a government agency described discrimination she encountered following a request for an inline telephone amplifier. She said

I remember that boss. I can still see his face as if it were yesterday. He was a tall son of a [. . .]. He said, "No. You can't fool around with Crown property. You can't do that. And if you can't do the job, we will just demote you and put you somewhere else." (Southall et al., 2011)

A person with hearing loss may choose, or be encouraged by their employers, to apply for disability leave or early retirement from their employment (Christensen & Gupta, 2017; Jennings & Shaw, 2008). For another participant in the help-seeking study, secondary health concerns resulted in her being granted disability leave. Yet this leave simply provided a good opportunity for her employer to distance himself from her. She reported that

My bosses were basically waiting to get me out. They started cutting my hours and my pay. I was becoming more trouble than it was worth to them. And so I was able to retire on disability. My audiologist and surgeon were able to write it up so that Social Security granted me going on disability. (Southall et al., 2011)

Power Imbalance

According to Link and Phelan (2001), the stigma process can only take place when there is an unfavorable balance of power between the individual who has the trait when compared to the others present in the situation who do not have the trait.

A power imbalance may manifest as one individual being aggressive or bullying others. One study

participant stated, "I can think of someone else I have to interact with who is as far away from warm and cuddly as can be . . . and I probably would take a little bit more time before I would [disclose his hearing loss to this person] simply because his attitude, I find aggressive. [Disclosing] would feel like I have a weakness. And I wouldn't want to give him the advantage of knowing I have it" (Southall et al., 2011).

Sometimes the power is on the side of the individual who has stigmatizing trait. In these instances, the favorable imbalance of power may allow the individual who has the trait to deflect or potentially extinguish stigmatizing situations. This was the case for one participant in our workplace disclosure study. Diane, a physician, alluded to her relative good fortune:

No one is going to accuse me of putting it on, or disbelieve I have problems. I am very fortunate, very senior. Whether I put lunch on the table today or tomorrow doesn't depend on someone firing me. I think this level of disability in another job would be very tough, if I was a waitress, or a casual worker. (Southall et al., 2011)

Clinical Implications

Based on the above discussion, several aspects of the modified labeling theory are relevant to the design and implementation of services for adults who have acquired hearing loss. Hearing health care professionals should be aware of the labeling process and how the labels and stereotypes associated with hearing loss may affect their clients' help-seeking behaviors.

First, according to the modified labeling theory, from a very young age children are aware of stereotypes, including those about hearing difficulties. Following the onset of a hearing loss, people may anticipate stereotyping and mistreatment in a given setting (e.g., a meeting) and act in a way to avoid stereotyping. Thus, one rehabilitation goal should be to teach clients and develop adaptive effective coping strategies (Hogan, 2001; Jennings, 1993; Trychin, 1986; Tye-Murray, 2002; Wayner & Abrahamson, 1996). Communication training programs that promote and teach the use of anticipatory strategies, that emphasize assertive behaviors and the use of effective conversational strategies should also serve to reduce an individual's stress level (Heydebrand, Mauze, Tye-Murray, Binzer, & Skinner, 2005). Obviously, if someone who has a hearing loss learns appropriate conversational skills, the likelihood of

communication breakdowns will decrease. If so, the stigma related stress associated with being an inadequate communication partner should also diminish. Similarly, the use of hearing aids when appropriate, as well as other HATs when called for, should reduce the probability of communication breakdowns and improve one's ability to participate in social activities. Hearing health care professionals should be aware that clients who have sought professional help in the past are not immune to the effects of labeling and stereotyping. It has been speculated that the benefits of mental health rehabilitation (treatment outcomes) are transient in nature, in part because stigma is an ever-present counterbalance (Link & Phelan, 2001). The same is likely true for people who have sought out audiologic rehabilitation. Over time, activity limitations or participation restrictions experienced by an individual with hearing loss are likely to change, and as a result the goals of a rehabilitation program are also likely to change. These suppositions are coherent with scheduling rehabilitation periodic coping strategy retraining (Gagné & Jennings, 2008; Gagné, Jennings, & Southall, 2009).

Second, fears and worries of future instances of stigmatization may result in separation, reduced self-esteem, and a lack of confidence. Intervention programs based on the principles of perceived self-efficacy (PSE; Bandura, 1977) have been promoted as an approach to the rehabilitation of people with hearing loss (Jennings, 2005; Smith & West, 2006). A priori, it would appear that the underlying principles of PSE are consistent with rehabilitation services designed to reduce perceived identity threat. Human functioning is viewed as the result of behavioral, cognitive, and other personal factors, as well as events in the environment. Each of these factors work together, but have different levels of influence depending on the specific situation, the specific environment, and the individual themselves (Bandura, 1986). PSE refers to a person's belief in their ability to organize and execute courses of action that are required to manage prospective situations (Bandura, 1995). The aim of a PSE-based rehabilitation program is to increase participants' levels of PSE using specifically designed approaches, including enactive mastery, vicarious experiences, social persuasion, and somatic and emotional states. Related to the model of stigma-induced identity threat, if the person has high levels of PSE, it is likely that the perceived demands of a relevant situation will not be appraised as exceeding their resources. On the other hand, for persons with low levels of PSE, it is likely that perceived demands of a relevant situation will be appraised as exceeding their resources, and a stigma-induced identity threat will

result. PSE level will affect volitional responses (i.e., problem-focused and emotion-focused coping strategies) as well as nonvolitional responses (i.e., arousal and increased blood pressure). For example, people with high PSE who have learned to identify their nonvolitional, physiologic responses are better able to interpret these responses as a "call to action." Thus, they are better prepared to use coping techniques to control the escalation of nonvolitional responses (Kramer, 2008; Kramer & Gussenhoven, 2014; Shaw, Jennings, & Kramer, 2013; Shaw, Jennings, Poost-Foroosh, Hodgins, & Kuchar, 2013).

Finally, providers should be aware that the stigma process is often accompanied by elevated levels of stress, and that this stress is associated with negative secondary health consequences. The challenge for the hearing health care professional is to persuade the person with hearing loss, whose self-esteem may be low and whose social identity may be threatened, to participate actively in a rehabilitation program that calls for overt behavior change, such as the use of amplification devices, behavioral coping strategies, and expressive communication strategies. In the authors' experience, it may be difficult to convince people who perceive themselves as being stigmatized to use any strategy that entails the disclosure of hearing loss. However, people are more likely to agree to disclose their hearing loss freely if the strategies are believed to be effective in reducing the level of stress they experience in a given situation. It is the authors' contention that it may be beneficial to adopt a cognitive-behavioral approach when designing rehabilitation programs that address social identity threats. The components of such an audiologic rehabilitation program are outlined in Table 4-1. Williams, Falkum, and Martinsen (2015) have also addressed this issue.

Conclusion

Anecdotal clinical reports and research investigations have clearly demonstrated that the stigma associated with hearing loss constitutes a significant obstacle to rehabilitation for many individuals who have a hearing loss. Until now, little was known about the social stigma associated with hearing loss from the perspective of the people who are being stigmatized. During the last decade, mainly in the social sciences, significant inroads have been made to broaden our understanding of the processes involved in creating a threatened social identity. The goal of the present chapter is to describe one sociological framework of the stigma process and to discuss its applications to

Table 4–1. Components of an Audiologic Rehabilitation Program Designed to Address Identity Threat

1. Describe and discuss the stigma-induced identity threat model to explain to the client the causes, consequences, and the potential costs of the stress related to identity threat.
2. Establish a hierarchy of situations in which identity threat occur.
3. Discuss, in parallel, the client's typical coping strategies and the impact and relative effectiveness of these strategies. Retain the strategies that are most appropriate or promising, and introduce new adaptive strategies.
4. Implement a problem solving approach to address a situation of stigma-inducing identity threat identified by the client.
5. Train and encourage the client to apply the selected coping strategies in a secure environment (initially, implementing the strategies are practiced during the therapy session).
6. Train and encourage the client to apply relaxation techniques in a secure environment.
7. Meet with the client to discuss the process of implementing and the consequences of applying the strategies (perhaps using the diagram of the model to identify the key elements).
8. Attempt a similar experience in a slightly more threatening situation/environment.
9. Increase the number of situations in which the client discloses his or her hearing loss and applies appropriate coping strategies, accompanied by reinforcing feedback rehabilitation sessions.

hearing loss. As a first approximation, the modified labeling theory appears to provide a framework that can be used to understand the manifestation of stigma among people who have hearing loss. Furthermore, the model offers a conceptual framework concerning the types of rehabilitative audiologic services that could be provided to people with hearing loss who perceive an identity threat. Our goal was to introduce the modified labeling theory to the field of audiologic rehabilitation. Hopefully the contents of the chapter will serve as an impetus for further discussions and research on the topic of stigma associated with hearing loss.

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5

PEER SUPPORT/ CONSUMER PERSPECTIVE

Samuel Trychin

Introduction

As a means of introduction, I am a licensed psychologist who has a moderate hearing loss in my left ear and a profound degree of hearing loss in my right ear, which was first diagnosed in 1953 while I was serving in the Air Force. I started college/university teaching in 1968 and have been doing that for more than 30 years. In 1976, I took time out to do full-time clinical work in psychiatric hospitals and have continued working part-time in community mental health services since then. I served as director of training from 1995 to 1998 on a large National Institute on Disability and Rehabilitation Research (NIDRR) hearing loss-related grant located at the School for Professional Psychology in San Diego, CA. I also served on state-level offices of vocational rehabilitation advisory councils for people who are deaf and hard of hearing in California and Pennsylvania. I currently conduct education and training programs for people who have hearing loss, their communication partners, and professionals who serve them. I also teach psychology classes at Penn State University and provide psychological services to Stairways Behavioral Health in Erie, PA.

Thirty-one years after receiving my first hearing aid at the Walter Reed Army Hospital in Washington, D.C., Rocky Stone, the founder and then executive director of Self Help for Hard of Hearing (SHHH; now Hearing Loss Association of America [HLAA]),

invited me to conduct a stress management group with his staff and volunteers at the Bethesda, MD, headquarters. It was 1984 and I began working with a group of people who had hearing loss and their communication partners. There were 10 of us in that first group: seven hard of hearing, one deaf, and two family members with no identified hearing problems. We elected to have a 2-hour meeting once a week for 8 weeks.

I had been wearing hearing aids for 31 years. During that time, I completed 12 years of university education, 8 years of college teaching, and 6 years of clinical practice, but I knew next to nothing about my own hearing loss and had not considered its effects on my biopsychosocial status. As a result, this group became my basic training in the dynamics of hearing loss.

The First Group

During each session, participants would take turns presenting a hearing loss-related problem or concern they had experienced either during the preceding week or previously. After each person presented an issue to be discussed, others, in turn, presented solutions they had used successfully or attempted unsuccessfully. Often, discussion would center on alternatives that had not been previously considered. The person who raised the problem or concern was given a homework assignment requiring the use of the suggested solutions

during the following week and was asked to report back on the results at the next session.

We discovered that most of the problems or concerns raised were commonly experienced among all of us and our communication partners. We learned the power of role-playing problem situations and their possible solutions in the group setting as a means of directly seeing the root causes of communication breakdowns and the essential features of what can be done to prevent or reduce them. Early on, we learned the necessity of practicing and polishing new communication behaviors in the safety of the peer group prior to attempting to apply them out in the real world. We were struck by how difficult it is to change long-standing communication behaviors that are over-learned and often have a strong emotional undercurrent. These seemingly easy-to-modify behaviors often require practice over extended time before they become habitual (Watson & Tharp, 2007).

The process of establishing techniques for running groups for people with hearing loss and their communication partners evolved over time, but many of the methods I currently employ were developed during this first group encounter. This group experience had a profound personal effect on me. I became aware, for the first time, of the social and emotional effects of my hearing loss. I began to connect hearing loss to my tendency to be a lone wolf professionally; that is, to always manage to be an independent practitioner while still working within a formal organization such as a hospital, clinic, or university. Also, until that group experience, I had always preferred working with patients one on one, most likely due to my ignorance of effective communication strategies that enable understanding in groups. Since that group experience in 1984 to 1985, I rarely work with people individually, always preferring a group format. I also began to relate my tendency to be somewhat on edge to a chronic, low-level anxiety that often accompanies a lessened ability to be connected with the physical and social environments. From that group experience, I learned the necessity of monitoring my own tension level and finding ways of reducing it when necessary.

At the end of the first 8 weeks, the group elected to continue for an additional eight sessions, followed again by another. That first group continued for 32 weeks, during which we produced several books and videotapes demonstrating what we found to be effective in dealing with hearing loss-related communication problems. During the same time the first group was meeting, another group composed mainly of SHHH chapter members asked for similar meetings in

Baltimore, MD. I went there as well and we followed the same format as the first group and continued weekly for 21 weeks. Over subsequent years, I have taken the information garnered from these early groups across the United States and to many locations in Canada. With each group, I learned a little more, but the basic information about people's experience with hearing loss gleaned from those first two groups remains the same no matter the geographic location or time, whether 1984 or 2012 (Trychin, 2003a, 2006).

Virtually all of this work has been done with adults, but in 2002 my wife, Janet, an audiologist, and I were asked to conduct a weekend-long program for adolescents in Portland, OR, who have hearing loss. Forty adolescents from various areas in Oregon, Washington, and Idaho participated in the program. The following year, we were invited back to Oregon and conducted another weekend program. Several other adolescent groups grew out of this program and what we found was a little surprising: Early, middle, and late adolescents experience the same basic problems, issues, and concerns as adults, and the same strategies for effectively coping with hearing loss apply similarly to adolescents as well as adults.

Importance of Including Communication Partners

Because hearing loss is a communication disorder, it affects both the individual who has it and those who are attempting to speak to him or her—the communication partners (CPs). The perspectives and experiences of both must be taken into consideration during assessment, treatment, and follow-up. Table 5–1 describes the importance of CPs in the life of people with hearing loss.

Certainly, more intimate CPs—those who communicate with the person who has hearing loss most frequently and meaningfully—are likely to be most strongly affected by communication breakdowns. They are also the ones who can be most helpful in establishing interpersonal and environmental conditions that increase the probability that the person with hearing loss will understand what is being said. However, the vast majority of people who have hearing loss find themselves in situations that demand communication with other people as well.

In the past, we have focused on the term *significant others* when referring to those people with

Table 5–1. Importance of Communication Partnerships (CPs) for People With Hearing Loss

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- Both the person speaking and the person listening contribute to communication breakdowns.
 - Both the person speaking and the person listening often need to change something in order to prevent or reduce communication breakdowns.
 - Communication partners will sometimes reveal important information that may not be in the awareness of the patient who has hearing loss or that the person with hearing loss is reluctant to reveal.
 - CPs can facilitate or inhibit the tendency of a person who has hearing loss to take effective action to deal with the hearing loss.
 - CPs share in the negative emotional experiences of their partners who have hearing loss and have a stake in helping the partner avoid those experiences as much as is possible.
 - Communication partners (CPs) provide a backup system ensuring that the person with hearing loss understands the information provided by professionals who are offering services.^a
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^aWe always attempt to minimize or eliminate any dependencies of the person who has hearing loss on his/her communication partner. We encourage people to manage their own communication situations and rarely, if ever, depend on a partner as an interpreter. Independence fosters increased self-confidence/self-efficacy and minimizes resentment on the part of CPs.

whom the person with hearing loss communicates. Usually, we think of significant others as a person's spouse, lover, child, parent, other family member, and/or close friend. Most people communicate with a variety of people, however, in the course of a day, week, month, or year. The term *communication partner* is preferable because it encompasses any and all of these other people as well.¹

The most important CP at any given time is that person with whom the individual with hearing loss is conversing. Mary, for example, may be in face-to-face communication with her physician only once a year, but at that moment, the physician is her most important CP. In addition, the physician needs to ensure Mary understands the information being presented. A comprehensive program whose goal is helping people prevent or reduce communication problems needs to incorporate the fact that there is a wide range of communication demands on a person, not only those that concern the intimate partners. Such a program needs to include information and training on what can be done to reduce problems in a wide variety of challenging communication situations.

It is unlikely that these other CPs will participate in a program designed to improve communication; however, the more intimate and frequent partners

may be more motivated to attend. The next section focuses on information, skills, and attitudes that have been found to be helpful for both the person with hearing loss and her/his intimate communication partner(s).

Strategies, Tactics, and Tools for PHLs and Their CPs

Overall, effective management of hearing loss requires specific attention to two different, but related, areas of human experience. First is learning to manage the external environments in which people find themselves, including the physical environment and the social environment. In the former are environmental stimuli that affect safety and pleasure, such as alarm signals and music. Support from the social environment is absolutely necessary for human survival at early ages and for continued health and well-being at later ages (Cacioppo et al., 2006; Hawkey & Cacioppo, 2003). It is most helpful for people who have hearing loss (PHLs) and their CPs to know about assistive alerting and listening devices to help stay in tune with the physical environment. It is necessary for

¹For more information on the importance of communication partnerships, the reader is referred to Chapter 18 of this text.

both PHLs and CPs to know what to do to maintain social connectedness as well.

The second area of human experience involves learning to manage the internal environment or learning to manage one's self. High levels of emotional arousal inhibit cognitive functioning, reducing the person's ability to use good judgment in problem-solving attempts (Childre & Rozman, 2005, 2006). When a person is angry or anxious in a communication situation, that anger or anxiety interferes with that person's finding ways to resolve the communication difficulty and often increases difficulty in understanding what is being said. It is important to address the effects of emotional responses because there will be times when people have done their best to improve understanding, but are unsuccessful because others will not cooperate or the environment cannot be altered.

Managing Communication Situations

We have found the following types of information to be critical in learning to manage communication situations: signs and symptoms of hearing loss, myths and misconceptions about hearing loss, problems frequently reported by PHL and CPs, and causes of and dysfunctional reactions to communication problems. Over time, we also found that this order of presentation of these topics (Table 5-2) works best and that it

helps to cover only one topic at a time, rather than causing confusion by attempting to include multiple issues all at once.

Identifying Communication Problems and Their Causes

Signs and Symptoms of Hearing Loss

Many people are reluctant to inform others about the fact that they have a hearing loss. Some people do not acquire hearing aids and others who have them do not wear their hearing aids in public settings out of fear of embarrassment, being seen as different, or being rejected. Social stigma concerning hearing loss certainly contributes to these concerns (Gagné, Southall, & Jennings, 2009). In our groups, members are asked to identify and list signs or cues that might indicate that a person has hearing loss, such as the cues noted in Table 5-3. Table 5-3 illustrates the behavioral signs of communications breakdowns.

The reason for this exercise is to indicate that if others are not informed about one's hearing loss, the signs and symptoms are still there and may be attributed to other, more negatively perceived characteristics (e.g., personality problems, rudeness, social ineptitude, dementia, uncooperativeness, etc.). I try to have members see that it is really in their own best interest to inform others that they have a hearing loss and that any difficulties that arise may be due to the

Table 5-2. Group Topics and Associated Activities

Behaviors Addressed in Group	Activities
Identify the signs and symptoms of hearing loss	List signs and cues that might indicate that a person has hearing loss (See Table 5-3.)
Myths and misconceptions about hearing loss	Listing and dispelling the myths (See Table 5-4)
Problems reported by people who have hearing loss	Group members list their problems (See Table 5-5)
Problems reported by communication partners	Communication partners discuss communication breakdowns (See Table 5-6)
Situational challenges	Group identifies problematic situations and develops compensatory strategies (See Table 5-7)
Causes of communication breakdowns	Group identifies causes of problems experienced (See Tables 5-8 and 5-9)
Reactions to communication breakdowns	Group identifies reactions they experience (See Table 5-10)

Table 5-3. Behavioral Signs of Communication Breakdowns

Frequently asking people to repeat	Inappropriate response to what is said
Difficulty understanding in groups	Puzzled expression when listening
Intently watching the speaker's mouth	Strained expression around eyes
Turning the head to one side to hear better	Avoids social situations; withdraws
Talks too loudly or very softly	Turns up TV or radio much too loud
Blames people for not speaking clearly	Defensive about communication problems

hearing loss and not to some undesirable personal characteristic.

Myths and Misconceptions About Hearing Loss

There are several misconceptions about hearing loss that impede taking effective action and making necessary accommodations. The misconceptions occur both among those with hearing loss and those in the general public. Providing accurate information opens the door for adopting realistic attitudes and taking effective action. Table 5-4 provides examples of misconceptions that are commonly addressed in groups.

Problems Reported by People Who Have Hearing Loss

Group members who have hearing loss are always asked to report any problems they experience due to hearing loss. This is a useful exercise for three reasons. First, it allows people to consider the full range of effects of hearing loss on their lives. Second, it may clarify other issues they might not previously have considered being related to their hearing loss (e.g., fatigue, irritation, or digestive problems). Third, it provides opportunity for CPs to increase their awareness of the impact of the effects of hearing loss and that these problems are not due to a personal weakness or deficit in their partner who has hearing loss.

There are three basic types of communication problems that are frequently raised in my groups. The first type of problem occurs when the person who has hearing loss is unaware that someone is speaking to him. Given this situation, there is no possibility that the message will be received, and even if the person at some point tunes in to the fact that he is being addressed, the speaker will most likely have to go back and repeat the earlier part of what was said. The second problem occurs when the per-

son who has hearing loss is aware of the fact that she does not understand what is being said to her. This type of problem is easy to correct if she knows how to identify the cause of the problem and offer a solution, e.g., "I need you to speak a little louder." Depending on who is talking and the nature of the situation, however, many people resort to bluffing rather than revealing that they do not understand. The third type of communication problem can produce the greatest difficulties and that occurs when the person who has hearing loss misunderstands what is being said.

The problems created by misunderstandings are often not revealed until a later time when it may be too late to make a correction. For example, the boss tells the employee to not exceed \$15,000 on the Smith contract and the employee who has hearing loss hears \$50,000 instead of \$15,000 and signs the contract for \$25,000—\$10,000 more than the boss had budgeted, thinking he has made a good deal. We often devote time during group sessions for members to report any misunderstandings they recently experienced. When shared with understanding peers and viewed more lightly, the emotional effects of the misunderstandings can be softened, especially when they can be reviewed with a dollop of good humor.

The following is a personal example of one of my own misunderstandings. During the time that President Clinton was running for office for his second term, I was listening to the radio and heard the announcer say, "Voters don't like politicians who take aspirin." I wondered what kind of pollster would go into people's medicine cabinets to determine how they would vote. Thinking that what I had heard was probably incorrect, I listened more carefully and discovered that he had said the word *action* and not *aspirin*. My brain had heard the beginning *a* and the ending *n* of the word, missed the middle part, and filled in the gap with a word from its lexicon. This process did not result from a conscious decision; it happened automatically and instantly.

Table 5–4. Common Examples of Myths and Misconceptions

Hearing aid myth: I call this one a myth because it seems to be so widely held among the general population and those who have hearing loss. Essentially, this misconception is that hearing aids and cochlear implants function like eyeglasses, returning most wearers to normal or close to normal hearing. Our group discussion of hearing aids or cochlear implants includes situations in which they are very useful and those in which they are less useful, pointing out that those people who are able to benefit from hearing aids and cochlear implants should have them even though they may not always be helpful.

Lip Reading myth: This is another generally held view that if you don't hear well you automatically become a great lip reader and that suffices to fill in what you missed due to difficulty hearing. Discussion focuses on personal and environmental constraints on information gained by lip reading. We always encourage people to take speechreading classes as another tool to add to their kit in accommodating hearing loss.

Sign language is for everyone: Learning sign language is another helpful tool for people to use if they have the time and energy to become proficient. Limitations are that friends, family, and coworkers are unlikely to learn sign language. However, even just finger spelling and a few basic signs can be helpful in some communication situations.

Professionals understand hearing loss: With rare exceptions professionals in most fields have not had education or training in the basics of hearing loss and its impact on a person's life. Even in medical school training, there is limited time devoted to the ear and that brief time is mostly focused on the structure of the ear and not its psychosocial functions. Teachers, medical personnel, and mental health providers, for example, all need to be educated about what they need to do in order to be understood. Many suffer from the belief that the hearing aid is always the solution.

People who are hard of hearing understand hearing loss: In my experience the vast majority of people who have hearing loss and their communication partners (including myself, as stated in the introduction) are unaware of the breadth of issues in their lives affected by hearing loss and have little knowledge about what can be done to accommodate the hearing loss.

Another example I frequently use during presentations is as follows. I was to do a week-long training on coping with hearing loss at the California School for the Deaf in Riverside, CA. Two people came down from Riverside, CA, to pick us up at the Los Angeles airport and guide us back to the training site. I was seated in the back seat next to the left-side window and one of the women who had driven down to get us was sitting next to me in the middle of the back seat. I noticed she was wearing hearing aids and decided to make some conversation with her. We drove past a large building and I asked her what its name was. She replied, "It's not very far." I said, "I don't think you understood my question; I asked the name of that big building we just passed." She responded (somewhat testily), "IT'S NOT VERY FAR!" Now, I was getting a little annoyed and said (somewhat louder than before), "We passed a large, tower-like structure about a mile ago, and I want to know its name!" She replied, "IT'S KNOTT'S BERRY FARM!" (which is a

local theme park). The main point of this example is that I had been doing this work with folks who have hearing loss for about 15 years at that time and would have bet \$1,000 that she had misunderstood and that I had heard correctly, but I was the one who had misunderstood. That experience really penetrated my consciousness and increased my awareness of the insidiousness of misunderstandings.

We find it very important to ensure that group members are aware of the possibility for, and results of, such misunderstandings because they occur frequently and in a variety of situations. A very recent example that is much more alarming than humorous occurred during a tour of the White House in Washington, D.C. A man who has hearing loss and his wife were attending the tour and standing with a group in the basement of White House waiting for the tour to begin. A uniformed security guard came into the room and told the tour guide to wait because, "We just found out Obama is on the first floor." The

husband had a panic attack, felt faint, and had to be seated on the floor. What he had heard was, “We just found out a bomb is on the first floor.”

It is important that both the people who have hearing loss and their frequent CPs know about misunderstandings and their effects. A good rule is: If something you hear seems strange, check it out. Another good rule is: Always repeat the important parts of a message, e.g., who, where, when, and numbers. In that way, if the person has misunderstood, it can be corrected on the spot. As illustrated in Table 5–5, the pervasive impact of hearing loss can affect all of a person’s life, from the time of arising to going to bed at night. For some, it even affects their sleep.

Problems Reported by Communication Partners

Having the CPs report the difficulties they experience related to the hearing loss highlights its impact on

others. Listening to someone else’s CP report difficulties that their own partner has been voicing for some time allows the person who has hearing loss to step back a moment and consider that the problem is valid and not simply due to some idiosyncratic complaint of their partner. It is truly eye opening for people who have hearing loss to see that CPs’ list of problems is as long as their own.

As reflected in Table 5–6, consideration of this list reveals the many ways in which hearing loss can negatively affect relationships, as well as the personal status of both the person with hearing loss and the CP.

Problems Reported by People Who Have Hearing Loss Related to Specific Situations

Along with the problems and concerns cited in the preceding list, people who have hearing loss are vulnerable to barriers to understanding that are often inherent in specific types of situations such as business

Table 5–5. Examples of Problem Situations Reported by People Who Have Hearing Loss

▪ Conversations in a moving car	▪ Family dinners at holidays
▪ Understanding conversations on the phone	▪ Several people talking
▪ Medical situations	▪ Outdoors-in wind, traffic noise, etc.
▪ Hearing alarm signals	▪ Voice from another room
▪ Can’t see speaker’s face	▪ Poor illumination
▪ People whispering	▪ Voice’s on TV or radio
▪ Person whose speech is not clear	▪ Misinterpretation of mistakes
▪ Unaware person is talking to me	▪ Movies, plays, lectures, classes

Table 5–6. Examples of Problem Situations for Communication Partners

▪ Difficulty remembering what I need to do	▪ Not knowing whether she/he understands
▪ Variability in his/her ability to understand	▪ Hard to get her/him to understand me
▪ TV or radio is much too loud	▪ Having to repeat a lot
▪ Having to interpret too frequently	▪ When we become frustrated or irritated
▪ When she/he doesn’t pay attention	▪ Becoming too dependent on me
▪ Being isolated from friends and family	▪ Loneliness
▪ Not doing things we enjoyed previously	▪ Having to repeat in pressure situations
▪ Not traveling or going new places	▪ Not talking as much as before
▪ When he/she is not understanding someone else	▪ Being asked to repeat embarrassing jokes or remarks

Note. Examples of communications partners referred to here are family, teachers, and friends.

meetings, church services, and classroom settings. Table 5-7 illustrates the unique barriers of one such situation—a work training site.

Employees or trainees who have hearing loss are at high risk for failure to miss important information in training environments. When appropriate accommodations are in place and instructors or trainers are using communication tactics that enable all trainees to understand what is being said, people who have hearing difficulty can reach or maintain their potential and be equally productive. It may be helpful to keep in mind that the major tasks for people who have adult onset hearing loss are continuing their education or training and maintaining their jobs or career, relationships, self-respect and confidence, and health. Appropriate accommodations during orientation and training sessions and on the job increase an employee's ability to accomplish these tasks and function up to her/his ability.

Causes of Communication Problems

It happens with some frequency that if you ask someone who has hearing loss about why they did not understand what was just said they respond with something like, "Because of my hearing loss." It can be fairly depressing if the person has bought expensive hearing aids and uses them regularly, but still has communication difficulties. Fortunately, that limited view of the cause of hearing loss-related problems is incorrect. There are many other causes of difficulty understanding what is being said and most of those can be prevented or reduced. There are things that the person speaking is either doing or not doing that contribute to communication breakdowns. The same

is true of the listener who has hearing loss. Additionally, there are features of the physical environment in which the message is delivered that contribute to communication problems (Trychin, 1994). Table 5-8 provides examples of the kinds of variables that can be causal factors creating communication difficulties. When the person who has hearing loss and his/her CP become aware of these causal factors, it is then possible to determine the source of a specific communication breakdown and suggest and implement a correction.

It is most important that people who have hearing loss and their CPs become aware of the physical environmental factors that can interfere with understanding what is being said and/or prevent hearing or locating the source of important environmental sounds. Environmental barriers to hearing well are often the major contributors to communication breakdowns and their effects can usually be prevented or reduced once their presence is known. Some of the environmental factors listed in Table 5-9, such as background noise, are obvious; others are more subtle and people need to consciously watch for them.

A major contributor to communication problems is the inability to pay attention. When it is difficult to understand what is being said, a normal response is to tune out and think about something else. Some people in our groups have practiced tuning out so frequently that they now have difficulty tuning in. Sometimes, the person who has hearing loss only attends to his/her frequent CP and becomes dependent upon that person for everything said by others.

A good example of this dependency occurred at Ghost Ranch in Abiquiu, NM, where for 12 years each summer I taught an intensive 7-day class for people who have hearing loss and their CPs. At the beginning of each session, there was a general orientation

Table 5-7. Examples of Problem Situations: Employees/Trainees Who Have Hearing Loss

▪ Trainer talks while facing the blackboard ▪ Trainer talks while walking around the room ▪ Trainer lectures while trainees take notes ▪ Air conditioning drones on during lecture ▪ Written outlines are not provided ▪ Teacher talks while trainees use computers or while lights are out during films or videos ▪ Trainer answers questions asked by other trainees without first repeating the questions	▪ Other trainees ask questions from behind ▪ Trainer uses ALD microphone as a pointer ▪ Uncaptioned TV, videotapes, or films shown ▪ Trainer has a heavy foreign accent ▪ Difficulty understanding PA announcements ▪ Unable to follow comments in work groups
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Table 5–8. Causes of Communication Breakdowns

A. Examples of <i>Speaker</i> Factors That Influence Understanding	
▪ Does not first get listener's attention	▪ Does not face listener
▪ Turns away while talking	▪ Talks too rapidly/too slowly
▪ Talks too softly/too loudly	▪ Does not enunciate clearly
▪ Uses too little/too much facial expression	▪ Uses too little/too much body language
▪ Has a foreign accent/regional dialect	▪ Has distracting mannerisms
▪ Has an untrimmed beard or mustache	▪ Has a boring delivery or message
B. Examples of <i>Listener</i> Factors That Influence Understanding	
▪ Severity of hearing loss	▪ Type of hearing loss
▪ Use of hearing aids	▪ Use of assistive listening devices
▪ Attention level	▪ Emotional state
▪ Fatigue level	▪ Distracting sensations
▪ Distracting thoughts	▪ Motivation to hear
▪ Expectations about ability to understand	▪ Unfamiliarity with terminology used

Table 5–9. Examples of *Environmental* Factors That Influence Understanding

▪ Background noise	▪ Poor lighting
▪ Poor room acoustics	▪ Too far from source of sound
▪ Visual or auditory distraction	▪ Objects interfering with seeing speaker
▪ Poor seating arrangements	▪ Inadequate ventilation
▪ Lack of assistive listening systems	▪ Lack of alerting systems
▪ Lack of visual aids: overheads, CART, etc.	▪ Illegible visual aids

meeting. I always put an induction loop system in an area of the auditorium where the general orientation session was conducted to allow people who have hearing aids to more easily understand speakers' remarks. One year, I was seated behind a married couple within the looped area and observed that after every statement made by a speaker, the husband, B., turned to his wife and asked, "What was that?" She then repeated what had been said. After watching this for a few minutes, I tapped B. on the shoulder to say that if he had a t-switch on his hearing aids he could hear directly what was being said. His response was, "Yes, I have them and they're on."

I was struck by the fact that B. could hear both me and his wife when his hearing aids were on the

telecoil program with the room microphone feature turned off, but could not understand what was coming directly through the telecoil. It was apparent that B. only took information from his wife, not from anything or anyone else in the environment. I separated B. and his wife during the first class session so that B. would not be able to rely solely on his wife for information. B.'s wife became visibly upset at being separated and I inquired about the cause of her appearing to be upset. She responded, "I don't think he's going to make it." I asked if she was concerned about his physical survival, and she stated that she did have that fear. We agreed to keep them separated for an hour or so to see how he would do. He was able to survive the separation so we continued keeping them

apart for the whole week. B.'s habit of depending on his wife was so strong, however, that it took two and a half days before he began to be able to function independently in the group. For the first 2 days, he made no active effort to understand what others in the group were saying. I repeatedly asked B. what had just been said and he responded with, "I have no idea!" in an irritated/angry tone of voice. I would then ask the speaker to repeat what had been said and ask B. to look at the person speaking. Then, I would again ask B. to repeat what had been said. Usually he was then able to understand correctly what had been said.

Other group members began to express irritation at this process, but on the third day B. began to learn that, with some effort and practicing paying attention to whomever was talking, he could follow what others were saying. From Thursday to Sunday, B. became an active member of the group and was mostly able to follow whatever was being discussed. At the end of the last day, his wife took me aside and said, "I think this experience has saved our marriage." She was caught in the bind of believing he was unable to "make it on his own," but hating the role of being a "walking hearing aid."

One moral of this story is that being a frequent interpreter is a real burden and CPs can quickly come to resent that role, resulting in damaged relationships. Another moral of this story is that some people who have hearing loss need training to recover their ability to pay attention to what is being said by whomever is speaking at the moment. There are habits, such as depending on others, that become so ingrained that special efforts are often required to establish new, more functional, habits in their place.

As discussed earlier, Tables 5-8 and 5-9 highlight multiple influences, often interacting, that render conversation difficult to understand. Experience indicates that when people are able to use this information to determine the cause(s) of a communication problem and find a way to correct or repair the situation, they are less likely to keep returning to their audiologist to have her fix the hearing aid so that it will work better, that is, eliminate communication difficulties.

Reactions to Communication Problems

Once a communication problem arises and is noticed, both the person speaking and the person listening will react to the communication breakdown in some way. Some reactions are productive—they resolve the problem. Other reactions are unproductive—they

fail to solve the problem, or make the situation worse (Trychin, 1995). It is most important for people who have hearing loss to identify their own unproductive reactions to communication breakdowns and replace them with reactions that are more productive.

Rather than becoming upset and withdrawing or blaming the other person, it would be more productive, for example, to identify the cause of the problem and offer a solution such as the following: "I was not paying attention when you said that and I'd appreciate your repeating it; I'd really like to know what you said" or "I'm unable to follow what you are saying unless I can see your face; can you please come around where I can see your face when you talk? That's much better, thanks."

People respond to a communication breakdown in several ways: physically, emotionally, cognitively, behaviorally, and interpersonally or socially. Table 5-10 illustrates these reactions to communication problems.

Physical Reactions

Physical complaints are a result of the body's activation as it attempts to prepare to defend against threat or danger. These built-in protective processes include changes in heart rhythm/rate; blood vessel constriction; blood shunting from digestive processes; respiration changes—shallow, rapid breathing; hormonal activation; increase of adrenalin, cortisol, etc.; neurotransmitter activation; and pupil dilation. These reactions occur below the level of conscious awareness and therefore are often not under conscious control. Acute, transient physiological reactions subside rather quickly and do not pose a risk to health, but prolonged or chronic physiological arousal is implicated as a causal factor in many different forms of disease. Chronic physiological arousal also produces fatigue. Fatigue affects ability to pay attention and inability to pay attention increases difficulty understanding what is being said: A cycle develops that needs to be interrupted.

Emotional Reactions

When emotional arousal is high or prolonged, the verbal areas of the brain shut down, rendering it difficult to use judgment to solve a problem, such as determining what might be the best solution for a communication breakdown. Additionally, negative emotional arousal produces discomfort that induces the sufferer to withdraw from the distressing situation that produced the discomfort (Kanfer & Gold-

Table 5–10. Frequently Reported Reactions to Communication Problems

A. Physical Reactions^a	
▪ Heart pounding	▪ Feeling out of breath
▪ Muscle tension/pain	▪ Headaches
▪ Stomach problems	▪ Sleep disturbances
▪ Jitteriness	▪ Restlessness/agitation
▪ Fatigue	
B. Emotional Reactions	
▪ Anger/irritation	▪ Fear
▪ Anxiety	▪ Depression/sadness
▪ Frustration	▪ Embarrassment
▪ Shame	▪ Guilt
▪ Loneliness	
C. Cognitive Reactions	
▪ Can't think straight—confused	▪ Hard to focus attention
▪ Easily distracted	▪ Difficulty making decisions
▪ Mind goes blank	▪ Decreased self esteem
▪ Decreased self-efficacy	▪ Reduced risk-taking
D. Behavioral/Social Reactions	
▪ Bluffing—pretending to understand	▪ Withdrawing—avoiding/escaping
▪ Blaming, e.g., “You mumble.”	▪ Demanding, e.g., “You’d better . . .”
▪ Dominating conversations	▪ Guilt tripping, “If you really loved me . . .”

^aNachtegaal et al. (2009).

stein, 1991). Unfortunately, some people who have hearing loss experience social anxiety resulting from repeated embarrassing communication difficulties and isolate themselves to avoid risking re-experiencing the anxious feelings.

Depending on the degree of self-isolation, the person may then experience loneliness, which is known to cause depression, damage health, and result in early death (Cacioppo, 2003; Hawkley & Cacioppo, 2003).

CPs also experience emotional arousal when communication difficulties arise. They report such feelings when experiencing difficulty being understood by the person who has hearing loss and when observing their intimate partner experiencing emotional reaction as a result of having difficulty understanding someone else. They feel empathy with the person who has hearing loss when seeing or hearing

that person show signs of being distressed (MacDonald & Jensen-Campbell, 2011). Additionally, mirror neurons in the CP's brain are activated by observing their partner's expressions of distress, which produces an empathetic response in the observer (Shiota & Kalat, 2012).

Cognitive Reactions

Being aware that a communication problem is occurring, looking for the probable cause(s) of the difficulty, and deciding on a course of action to repair the situation are cognitive activities requiring ability to focus attention, remember what to look for, and execute appropriate action. When cognitive processing is disrupted by emotional arousal or distractions,

there is a breakdown in problem-solving capacity that often results in dysfunctional reactivity (McCraty & Atkinson, 2003). The person who has hearing loss may resort to bluffing in an attempt to hide the problem, make an excuse to leave the situation, or become angry and verbally lash out at the person who is talking. None of these reactions are functional in terms of resolving the problem.

Behavioral Reactions

Because so many people do it, bluffing—pretending to understand what is being said—seems to naturally go with hearing loss. Bluffing is a major source of problems for people who have hearing loss and also produces problems for their CPs. Bluffing means pretending to understand when you know that you are not, in fact, understanding what is being said. You are aware that you are copping out, pretending something that is not true or, to be blunt, telling a lie. Bluffing contributes to lowering self-confidence and feeling incompetent. When the person speaking discovers the listener has been bluffing and not really understanding, the implication is that the listener was not interested in either what was being said or not interested in the person who is speaking. That does not bode well for maintaining a good relationship. One benefit of seeing people in a group setting is the opportunity to identify any bluffing as it occurs, call attention to it, and practice alternative behaviors that increase one's probability of understanding what is being said.

One difficulty encountered when dealing with bluffing and the other behavioral reactions is that they are most often habitual behaviors that have been overpracticed, sometimes for many years. Being habitual, they operate below the level of conscious awareness and require lots of time and many practice opportunities to bring them into conscious awareness to develop alternative behaviors to use in their place. When a group of peers become conscious of the risks associated with bluffing, they are able to identify it when it occurs and bring it to the bluffer's attention. Then, everyone in the group becomes better able to reduce that habit.

The same is true for habitually dominating conversations. Often people are unaware that they frequently talk excessively, butt in when someone else is speaking, or abruptly change the topic of conversation. Discussion of this behavior in a group setting can alert people to the potential for committing any of those social blunders and to the payoff for doing so. Increasing people's awareness of this type of problem

can motivate them to monitor their own social interactions and reduce the tendency to dominate conversations and risk losing friends.

Social Reactions to Communication Problems

The focus of this section is on what happens when people feel a loss of attachment to an important person or group. The loss of attachment may be due to an active rejection such as a divorce, being fired from a job, or someone saying, "I don't want to continue this conversation with you." Loss of attachment to others can also occur more passively by being or feeling ostracized, passed over, left out, or disregarded. People who have hearing loss are at risk of experiencing or imagining a loss of attachment to valued other people. We frequently see people who live or work in situations in which they are nominally included (i.e., physically present), but not included in conversations. So the fact of being in the presence of others does not mitigate feeling left out or rejected. Family members or coworkers can inadvertently cause feelings of being out of the loop by saying things like "it wasn't important," "I'll tell you later," or "never mind," when asked to repeat what they had said. The usual response to the "I'll tell you later" statement is that the person who has hearing loss feels as though he or she is not important enough to the speaker to repeat what was said. Also, it is difficult to feel like an equal member when others can know what has been said now, but the person who has hearing loss has to wait until later to know. It is these kinds of statements, usually made to save time and energy, but not meant to be hurtful, that result in people who have hearing loss feeling marginal in their own families, social gatherings, and work sites (MacDonald & Jensen-Campbell, 2011).

The reaction to the experience of some form of social loss is called *social pain* and is mediated by part of the same neural network as physical pain. Just as the built-in reaction to physical pain is withdrawal from the pain stimulus, the built-in reaction to social pain, when frequent or intense, is social withdrawal. One of the major benefits of participating in groups of peers who experience similar problems, issues, and concerns is the buffering of social pain. Research indicates that a strong social support system somewhere in one's life acts as a buffer against the negative effects of social pain, allowing the sufferer to endure effects that might otherwise produce reactions not in the sufferer's best interest, e.g., social isolation.

Mental Health Risks Associated With Hearing Loss

Risks to mental (and physical) health occur when reactions to situations are exaggerated and/or long-standing/chronic (Trychin, 2002). If people meet the requirements for being diagnosed with a mental health disorder, such as some forms of anxiety or major depression, they may need some medication along with the help provided by aural rehabilitation programs. If mental health disorder is suspected, a referral to a mental health provider may be in order for assessment and adjunctive treatment. Personally, I have not had to resort to such a referral in more than 24 years of doing this work, but it is something to consider. Otherwise, when the signs and symptoms of a mental health disturbance at a less than clinical level are present, the individual experiencing them probably needs more support and patience than others in the group who are less affected. Often it is very helpful just to raise these kinds of issues for consideration and discussion; the relief experienced by some members can be palpable. Some possible mental health risks are listed in Table 5–11.

Strategies and Tactics for Preventing and Reducing Communication Problems

Once people have a basic grasp of the salient issues related to the effects of hearing loss and are able to determine the causes of communication problems in specific situations, they are in a better position to begin to apply preventative and repair strategies and

tactics. Table 5–12 highlights guidelines for speakers and listeners by noting the effective communication behaviors derived from reports of success by people who have hearing loss and their CPs. One of the main virtues of working with people in a group setting is that problems—their causes, reactions to them, and solutions to reduce communication hassles—can be addressed on the spot, as they occur. As stated previously, when people begin to change communication behavior habits in the group setting, they are more comfortable trying them out at home and/or work. They then are able to bring information to the group about what they did and what happened as a result. This provides instruction for all group members about what is effective. If the result did not meet expectations, the group members can troubleshoot to see what might be done differently next time. The member is then urged to go back and try again using these new suggestions.

There are two essential components in a program that focuses on tactics and strategies for preventing or reducing communication problems. The first component is providing information about what to do in communication situations. The second component involves opportunity to practice how to do it effectively. It is through the practice with feedback on the adequacy of the performance that new, more functional communication behaviors become firmly established.

Other Effective Communication Behaviors

Inform Others About Your Hearing Loss

Experience conducting these groups indicates that many people are reluctant to inform others about the

Table 5–11. Mental Health Risks Associated With Hearing Loss

▪ Becoming chronically nervous/anxious	▪ Becoming chronically sad/depressed
▪ Feeling anger much of the time	▪ Loss of a sense of group identity
▪ Feeling marginal in one's own family	▪ Loneliness ^{a,b}
▪ Becoming distrustful of people	▪ Withdrawing from social contact
▪ Developing poor self-image	▪ Feeling incompetent
▪ Feeling unacceptable to others	▪ Feeling marginal socially
▪ Feeling loss of influence or control over the physical and/or social environments	

^aNachtegaal et al. (2009). ^bKnutson & Lansing (1990).

Table 5-12. Guidelines for Speakers and Listeners

A. Guidelines for Speakers	
▪ Pick the best spot to communicate	▪ Inform listener when changing the subject
▪ Get the listener's attention before talking	▪ Don't shout
▪ Be sure your face can be clearly seen	▪ Try to be patient, calm, and relaxed
▪ Do not have objects in your mouth	▪ Speak to, not about, the person who has hearing loss
▪ Speak slowly and clearly	▪ Use facial expressions and gestures
▪ Rephrase if you are not understood	▪ Ask for tips to improve communication
B. Guidelines for Listeners	
▪ Pick the best spot to communicate	▪ Do not bluff
▪ Anticipate difficult situation-plan ahead	▪ Inform the speaker about what you heard
▪ Inform others how best to talk to you	▪ Reinforce speaker's helpful communication
▪ Pay attention to the speaker	▪ Set realistic goals for understanding
▪ Look for visual cues of what is said	▪ Balance your needs with those of others
▪ Ask for key words in writing if needed	▪ Arrange for breaks if meetings are long

fact that they have a hearing loss. Part of this reluctance is due to not knowing what to say when doing so. This topic brings up an important point applicable to other parts of a rehabilitation program. The point is that people often need two things: information about what to do and to learn how to do it effectively. For example, I may know that I need to inform others about the fact that I have a hearing loss, but may bring it to their attention in ways that put them off rather than inducing their cooperation. For example, I may say or imply something like, "I can't hear, so you'd better speak up!" We find that people need opportunity to practice in the safe environment of a peer group before they are able to go out and competently do it in the real world.

Until participating in the group sessions in 1984 to 1985, I never considered telling other people about my hearing loss or talking about it in any way. I remember one situation in a graduate-level class I was taking when the professor mentioned to the class that my academic achievement was high in spite of my hearing loss. She meant well and it was a compliment, but I was quite embarrassed. In retrospect, I think I never really wanted to talk about my hearing loss. It was during those later group sessions that I first became aware of how essential it is to inform others about one's hearing loss and about what they need to do to be understood.

Inform Others About What to Do to Be Understood When Speaking to You

Informing people about accommodations you need so you can understand requires several things. First, the person with hearing loss must understand what strategies are effective. Second, the person has to know how to request the use of these strategies in a way that is likely to elicit cooperation. Third, the person needs permission to make such a request. Sometimes people are reluctant to instruct others about what to do to increase their ability to understand conversation because of concern that doing so would be impolite or socially unacceptable.

In one session, for example, a woman complained about attending a social event at a friend's house and being unable to follow the conversation occurring around the dinner table. She stated that there was a huge flower arrangement in the table that interfered with her ability to see some other people's faces and there was music playing that prevented her from hearing very well. When asked why she hadn't requested that the flowers be removed and the music be turned down, she stated that she couldn't do that because it would have been impolite. Group discussion then turned to what would be more impolite: making repeated mistakes, pretending to understand, and frequently asking people to repeat; or making the

request that could reduce the problem. After considering the options, she decided that it is more socially acceptable to be open and honest about one's needs than to be passive in the situation and do nothing to improve one's ability to participate.

Remind Others When They Forget

There is considerable annoyance, irritation, and anger among people who have hearing loss, produced by the faulty assumption that other people are unwilling to do what is necessary to help them understand or that others do not care enough about them to cooperate. For example, a woman in one of our groups came to a session appearing to be quite upset. I asked her what was going on and she said that she had just returned from visiting her daughter in Los Angeles and that it was obvious that her daughter doesn't "give a damn about me." She stated that the daughter had spoken to her from behind her back, from another room, while the TV was blaring, and so forth, even though "She knows I have a hearing loss." When asked how often she visits her daughter and how long she was with her in Los Angeles, she replied that she sees her once every 5 years and was there for a weekend visit. Her expectation that the daughter should remember her hearing loss and remember to make the difficult communication behavior changes she required was unrealistic.

People forget that we have a hearing loss. Even those people with whom we live forget, especially when they are preoccupied with doing other things while attempting to communicate. Ordinarily people are much more interested in *what* they are saying than in *how* they are saying it, and need to have frequent feedback about what they need to change. Also, people have been talking the way they talk for virtually their whole lives, and even if we need them to change those ingrained communication habits, it will not happen overnight. They, like us, need instruction, opportunity to practice, and feedback about the adequacy of the performance until the new habits replace the old ones.

Model the Communication Behavior You Need From Others

Much of what we do in situations is influenced by what we see others doing. In a restaurant, diners are all essentially engaging in the same behaviors: sitting, eating, talking. During a church service, the members of the congregation are all doing essentially the same thing: sitting, looking toward the front, and singing

on occasion. In a real sense we are all teachers—teaching others what we like or do not like, what we need, and what we desire. It makes sense then that if I need you to slow down when you talk, but I speak rapidly when I talk to you, that I am working against myself. People who have hearing loss need to pay attention to their own communication behaviors to ensure they are not demonstrating the wrong behaviors to others. There is also the issue of fairness that comes into play. If the person who has hearing loss requires the CP to come to where they are before beginning to talk, then the person who has hearing loss needs to go to where the CP is located when initiating conversation with that person.

Reinforce Others' Communication Efforts

When people go out of their way to help the person who has hearing loss understand what they are saying, it is important that this effort is recognized and appreciated. Granted that sometimes people are modifying their behaviors for their own benefit (i.e., to reduce their need to repeat), it is still helpful to the person who has hearing loss. Participants in our groups indicate that reinforcing communication behavior changes results in faster and more enduring communication behavior changes. A simple "Thank you for getting my attention before speaking" or "I really appreciate your turning that music down" stated on occasion is all that is usually required. We all need to know that our efforts are noticed and appreciated and that we are not being taken for granted.

What to Avoid Doing When Informing Others

One of the benefits of seeing people in a group setting is the opportunity to observe how they come across when interacting with others. This can happen during spontaneous group interactions, in response to questions asked, or when participants are role-playing scenes from their lives. Many times people project information about themselves that escape their conscious awareness. People transmit information verbally by the words they say and nonverbally by conscious gestures. They also transmit information about themselves subconsciously by facial expression, body language, and tone of voice, all of which are often below the level of awareness. It is important to recognize these factors because others will often react toward us based on their conscious and subconscious awareness of the signals we transmit to them. Several

things important to avoid when communicating information about our hearing loss and requesting communication behavior changes are listed in Table 5–13.

Another reason to refrain from communicating negative statements is that, if they are repeated often enough, the person with hearing loss may begin to act as if those statements or attitudes are really true, resulting in becoming incompetent, unacceptable, dependent, or victimized. Again, it is critical to pay attention to *what* we need to do, but just as important to pay attention to *how* we do it. In group sessions, role-playing helpful and unhelpful ways of informing others about one's hearing loss and/or what to do and not do when requesting communication behavior changes from other people is an excellent way of showing group participants the importance of considering how they present themselves (Trychin, 1994). People can appear to be apologetic, impolite, etc., not only by the words they say, but also by body language, tone of voice, and other, often subconscious, signals. Role-playing in front of the group and receiving feedback from peers brings all of this to the actor's conscious awareness and indicates the types of changes that would be helpful in conveying information and making requests (Trychin, 2003a, 2003b). It is very helpful to use the Wrong Way/Right Way method when role-playing.

Group Activity Exercises

Wrong Way/Right Way Method

In the present context, "wrong way" means not helpful in achieving goals, and "right way" means helpful in achieving goals. The Wrong Way/Right Way method is useful for bringing dysfunctional behaviors and attitudes to people's attention to facilitate helpful changes. It is also useful for providing opportunity for group participants to practice effective communication behaviors.

Steps for Setting Up Wrong Way/Right Way Scenarios

1. Provide a script and have two or more group members act out an interaction in which someone is doing something communicatively inappropriate or unhelpful.
2. Participants then identify what is wrong and discuss why it is wrong. Other participants then act out the same scene the right way by correcting

Table 5–13. Behaviors to Avoid When Communicating About Hearing Loss

Being apologetic as in, "I'm so sorry that it's such a drag to talk to me."

Putting yourself down as in, "I feel really stupid about misunderstanding that."

Being impolite as in, "Speak up!!!"

Expressing anger as in, "How many times have I told you to slow down, stupid?"

Guilt tripping as in, "If you really loved me you'd remember to speak up."

Being defensive as in, "There's nothing wrong with my hearing; you mumble!"

the situation using more effective communication behaviors. Following that, the group discusses why that version is better.

3. Do the same as in Steps 1 and 2 for three or four scripts.
4. Participants then bring up scenarios depicting their own personal problem experiences and have group members identify the cause(s) of a problem being presented and role-play ways of preventing or reducing that problem in the future.

Anticipating Difficulties and Planning Courses of Action

We also spend a lot of time focused on anticipating difficulties. When people are blindsided by unanticipated problems, they often react in ways that may not be in their best interest. Along with role-playing, another useful procedure for helping people learn to anticipate difficulties in upcoming events and plan strategies for preventing or reducing problems is The 15-Things Brainstorming Method (Trychin, 2003c). The 15-Things method is an excellent tool for expanding people's awareness of the possible options available to them should they decide to improve their situations. It also puts the onus on the group participants to develop solutions, rather than relying on professional advice.

The 15-Things Brainstorming Method for Use in Group Work

Steps for Setting Up 15-Things Brainstorming Method Scenarios:

1. Identify a problem experienced by a participant in the group.
2. Have the problem stated in objective, measurable terms. For example, "George does not go out in public since his hearing has worsened. I would like him to go out to a restaurant with me and another couple, understand most of what is being said, and enjoy himself."
3. Ask participants to identify at least 15 different things that could be done to facilitate doing this.
4. Write down every suggestion; criticisms or fault finding of suggestions is not allowed.
5. Have the person select whichever suggestions on the list seem best to him/her, try them, and report results in the next session.

Usually, participants come up with 25 to 30 suggestions for any particular problem, which indicates that even in situations that have been difficult there are many options available that can reduce the difficulty and make life easier. Once a group has gone through three or four different scenarios (e.g., restaurants, business meetings, family gatherings), participants realize that many of the solutions suggested that apply to one also apply to many other situations.

Helpful Ida Institute Tools

The Ida Institute is an organization in Copenhagen, Denmark, that has been funded to develop and disseminate information to audiologists and other hearing professionals regarding the broader psychodynamic issues related to hearing loss. It has provided a series of 3-day education and training seminars to audiologists from a variety of countries around the world. A number of clinical tools were developed at the Ida Institute. The attendees understand that they are to take the information home, apply it in their practice, and share their results with others in their region and around the world through the Ida Institute website (<http://www.IdaInstitute.com>).

The Ida tools are proving to be very effective in facilitating a more rapid and deeper understanding of many of the issues faced by people who have hearing loss and their CPs. These tools are an excellent resource for structuring group sessions, eliciting the concerns of group members, and suggesting helpful communication behavior changes. The Line and Box tools also provide excellent means of bringing

people's attention to the benefits of making certain communication behavior changes and the costs of refraining from making those changes.²

Self-Management

In the beginning of this chapter I briefly discussed the importance of people learning to manage themselves in distressing situations, such as when experiencing communication difficulties, and that such difficulties or rifts in social relations can produce distress. Human beings are prewired to feel secure when included in a social network and feel anxiety and other discomforting emotions when being or feeling excluded. Humans are dependent on other, older members of our species for quite a few years after birth and usually remain dependent on connection to others in some way for their whole lives. Even the recluse depends on electrical power and water availability, and their provision depends on someone else. Neurological studies indicate that the neural/hormonal networks in our brains are prewired to respond positively or negatively to signals of acceptance or rejection from other members of our species. Eye contact and smiles, along with pleasant vocalizations and touch, trigger the release of antiaging hormones and neurotransmitters associated with pleasure. A sense of social disconnection releases hormones and neurotransmitters associated with physiological and psychological threat or damage known as social pain. Research is clear that early relational trauma produced by abuse or neglect often has long-term psychological and physical health implications.

People who have hearing loss are at risk for experiencing social pain when communication breaks down and other people pull away or are perceived to be pulling away or disconnecting. There are mitigating factors such as genetic predisposition, amount and quality of other social support systems, and frequency and intensity of the sense of social disconnection. However, there are many people who have hearing loss, and sometimes their CPs as well, who show signs of social pain. The most obvious signal indicating social pain is the withdrawal from social situations. The withdrawal may take the form of being present among other people, but not participating in conversations. The most extreme form of social withdrawal is virtually complete self-isolation resulting from severe social anxiety.

²For information on the Ida tools, the reader is referred to Chapters 11 and 18 of this text.

People who have reached high levels of social pain reaction do sometimes come forward to seek help in our groups. There are also members who are at a lower level of emotional distress who benefit greatly by sharing experiences with peers who know and understand what they are going through. Therefore, discussion of emotional reactions to hearing loss-related problems and concerns is not only of value, but is also necessary. We find it useful to help people realize that emotional reactions to certain situations are prewired, normal, and valuable for survival. For example, embarrassment serves a useful social function. Embarrassment is a physical and psychological reaction to a relatively minor social error, or gaffe. Physical signs of embarrassment are head lowered, shoulders down, and blushing, among other indicators. Psychological aspects of embarrassment are discomfort and eagerness to rectify the situation through apology or some other way of acknowledging one's awareness of the mistake. Upon seeing the signs of embarrassment, the person with whom the mistake was made is usually moved to forgive the error or indicate how restitution can be achieved. When the signs of embarrassment are not present, the offended person usually regards the offender as being rude, uncaring, or boorish, and often responds with some form of anger or retribution. So, embarrassment serves the useful survival function of keeping the social unit intact.

Shame has similar psychological and physiological effects, but usually results from a more egregious social error or blunder and has more dramatic personal consequences for the offender. These effects include a strong desire to disappear from the face of the earth and more often result in some form of social withdrawal. A person who is ashamed feels that he or she is worthless and undeserving of any consideration from others and has no right to ask for any concessions or forgiveness. Everyone has probably felt embarrassed on occasion and perhaps ashamed of something they have done or failed to do in the past. These are normal reactions to events that happen along life's journey. But when these self-conscious emotions are experienced frequently, persist over time, or are very intense, resulting in social pain, the sufferer is more likely to resort to avoiding social situations whenever possible.

One of the major differences between physical and social pain is that the recall of physical pain some time following the painful event, e.g., tooth extraction, ordinarily produces much less intense pain than that actually experienced at the time of the event. Social

pain, on the other hand, can be remembered with the same, and sometimes greater, intensity at a later time and can also be elicited at the same intensity by anticipating a future recurrence of the pain at an upcoming event. The person with hearing loss who experiences shame due to a major misunderstanding at a business meeting may re-experience the same intensity of shame while remembering that event later that evening. The same or greater feelings of discomfort may arise upon thinking about an upcoming business meeting involving the same people as before.

One of the best antidotes for shame is the ability to talk about and, ultimately, find some humor in the embarrassing situation. Sharing such experiences among a group of peers who have similar experiences to report can deflate much of the emotional tension built up around past or current situations in which embarrassment or shame is experienced. Another benefit of airing these emotionally charged situations is that it can serve a preventative function by inoculating people against undue emotional reactions in the future when similar situations arise. Also, when people learn to prevent or reduce communication breakdowns, the frequency of occurrence of these distressing experiences is decreased.

One major purpose for considering and discussing these emotional issues is reducing the likelihood that participants will resort to self-isolation or otherwise curtail attending social functions. Both the people who have hearing loss and their CPs report loneliness as one of the problems they experience related to hearing loss. Loneliness has produced several public health concerns, including depression, increased illness, and earlier mortality. People who self-isolate are caught in the middle of a double-edged dilemma. If they return to social activities, they run the risk of experiencing real or imagined rejection, embarrassment, or shame, and suffer consequent social pain. If they stay isolated from these experiences, they run a great risk of developing loneliness-related physical and psychological problems. Again, one way out of the dilemma is learning to better manage communication situations to reduce the occurrence of situations that produce social pain.

A second helpful method is to learn how to better manage one's physiological reactions to distressing events. Teaching stress management procedures such as meditation, deep muscle relaxation, yoga, or Tai Chi is beyond the province of the kind of program being discussed here, but regularly taking 5 minutes during sessions to demonstrate and practice a deep breathing exercise can be very helpful. It

can also be useful to have knowledge of stress management programs available in one's geographic area to refer group members who would like to pursue those options.

Conducting Groups

The benefits of seeing people in group settings should already be apparent, but a few additional comments might prove useful (Trychin, 2012).³

Group Structure/Rules/Proceedings

A very important issue is that the participants in the group need to understand most, if not all, of what is being said during the sessions. Perhaps the second most important consideration is that the group leader or facilitator elicits rather than provides information. I find it very helpful to see how patients and their CPs function together, and the group setting provides this opportunity. During the sessions, participants' communication behavior is directly observable and suggestions for improving communication skills can be offered on the spot, as they occur.

Another reason that seeing patients in a group setting is beneficial is that behavior, particularly habitual behavior, is difficult to change. In regard to communication behaviors, people have been doing what they do for a long, long time, perhaps most of a lifetime. Simply giving a person advice about what to do to improve understanding in communication situations and then sending her or him on his way will not foster the behavior changes that are usually necessary. Practice in the group provides the time and opportunity to strengthen new behaviors until they become habitual. Appendix 5–A provides goals and suggestions for group audiologic rehabilitation (AR) services.

There are often many communication behavior changes necessary for both the person who has hearing loss and his or her CP to increase understanding and decrease misunderstandings. No one can make many changes simultaneously, so seeing people over a number of sessions allows opportunity to deal with behavior changes sequentially over a period of time.

Summary

We have worked with hundreds of people who have hearing loss and their CPs, who are usually close family members. In dealing with their hearing loss, several things stand out, some of which are quite surprising. In regard to people who live together as spouses, couples, or other adult family relationships, there appears to be a relatively small subgroup who discuss issues related to the hearing loss. Some people in this category, usually members of HLAA or another hearing loss-associated organization, focus on what can be done to minimize the negative effects of the hearing loss. A larger percentage of this subgroup also appears to discuss issues related to the hearing loss, but from a more negative and emotional perspective, e.g., blaming, criticizing, and showing irritation or frustration.

What is surprising to me is the larger subgroup of people who live together with hearing loss, often for many years, who have never discussed it in a meaningful way. They may have talked about getting a hearing evaluation and/or a hearing aid, but nothing is discussed on a more personal level, i.e., the effects of hearing loss on them personally. It is as though hearing loss is viewed as being inevitable and/or that nothing can be done to reduce the hassles it produces anyway, so just live with it. The saddest part of the lack of discussion and the negative reactions in regard to coping with hearing loss is that the hearing-related problems remain unsolved even though there is a lot that can be done to prevent or reduce them.

Overall, the benefits I see in participating in a group composed of people who have hearing loss and their CPs are as follows. People learn that they are not alone and that the problems they encounter are not due to some personal weakness or flaw. The people who have hearing loss learn that others who have a similar condition report many of the same problems and concerns. The CPs learn that other CPs share similar issues and dilemmas. The people who have hearing loss learn, often for the first time, about the hearing loss-related problems experienced by their CPs and vice versa. The partner may have been voicing similar complaints at home for years, but he or she was never really heard. Participating in a group can allow members to learn from each other. When a problem is raised and a solution sought, usually

³For additional information on the group process, the reader is referred to Chapter 17 of this text.

someone in the group has successfully dealt with it in some fashion. Hearing success stories from peers can have a greater impact than hearing the same solution from a professional, resulting in increased likelihood that the person will try out what is suggested because it has actually worked for someone they know.

Group sessions provide opportunity to practice the suggested behavior changes and receive feedback about the adequacy of the effort. Often, people are able to do the right thing, but the way they do it turns others off. For example, I may know that I need you to raise your voice to be understood, but defeat myself by saying something like, "Speak up, stupid!" I can convey the insult by tone of voice, facial expression and body language, as well as by the specific words I use. People may not be aware of the negative effects they have on others even if they use the right words, and group members pick up these more subtle signals and suggest other ways of making the request.

Another benefit of group participation is that it provides opportunity to go home and try out the suggested changes in real life and then come back the following week to report the results of the effort. If it was not completely successful, further suggestions can be offered and role-played, thus encouraging the person to go back and retry it again. In this way, a process of trial and error is instituted that ultimately results in success. People learn they are not helpless, but can take action to reduce communication hassles and manage emotional reactivity.

Group members can also provide the permission to do what is necessary to understand what is being said. Sometimes, a person knows what to do to resolve a difficult situation and is able to do it effectively, but does not take action due to fear of being impolite, as occurred when the woman mentioned previously failed to ask to have the flowers moved from the table and the music turned down. Finally, working in a group is beneficial for the audiologist as he/she is able to see and feel the fundamental issues experienced and reported by participants. The emotion is obvious, and feeling it imparts the depth of concern or hurt experienced by the people involved. The audiologist also benefits from the varied experiences reported by group members, including creative ways in which they have managed difficult communication situations. The audiologist is also in a better position to see the varied, sometimes subtle, psychosocial effects that hearing loss causes and, conversely, how these psychological effects can interfere with understanding what other people are saying.

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Appendix 5-A

GUIDELINES FOR GROUP AR

Goal #1:

Participants should understand most of what is being said in the group.

A first requirement for achieving this goal is establish ground rules for communicating. Guidelines or rules, along with clearly stated goals, establish and structure the way the sessions will be conducted and serve to prevent the group process from becoming chaotic, confusing, and unproductive. Some example of guidelines/rules for group sessions are as follows:

Only one person speaks at a time. This cuts out distracting background noise and allows the person with hearing loss to focus attention on whoever is speaking. It educates communication partners about the necessity for doing this.

The person speaking uses the microphone. This ensures that listeners will know who is talking and also amplifies the sounds coming from the speaker. It also reduces the effects of any background noise present, e.g., rustling of papers, feet shuffling, air conditioners.

Everyone gets an equal opportunity to speak. Some people are very reticent to talk in public or even small group settings. Others have a difficult time refraining from talking and can dominate group proceedings, resulting in frustrating, aggravating, or boring other members, and essentially reducing or destroying the effectiveness of the group process. One way of handling both those who are reluctant to speak and those who cannot stop talking is by asking discrete, nonthreatening questions, at least initially. For those who are reluctant to speak, asking these kinds of nonthreatening questions provides opportunity for them to get some practice talking in the group. If someone chooses not to answer a question, he/she can simply pass the microphone to the next person. That is perfectly acceptable and he/she has at least had the opportunity to speak. It is helpful to discuss this issue with the group at the outset of the first session, informing everyone of the reason for, at least initially, asking discrete questions.

Inform the person speaking as soon as you are aware that you are not understanding what is being said. This guideline does several things. First, it provides permission for group members to stop the proceedings when they are aware that they do not understand something being said. In fact, throughout the group process something important is happening. Participants are being given permission to apply effective communication skills when necessary. Some people know what they should be doing to improve communication when a difficulty arises, but are reluctant to do so out of fear that doing so may be impolite or socially unacceptable. Second, informing others immediately when not understanding teaches the effective communication tactic of stopping someone immediately upon awareness of not understanding, instead of waiting until the person speaking has gone on for five minutes and then saying, "What?" Third, many people who have hearing loss have developed a long-standing habit of pretending to understand when they know they do not understand: They bluff. This guideline allows for stopping the proceedings when it appears that a participant is bluffing and asking the person to repeat what had just been said. If the person had in fact been bluffing, nicely ask him/her to request a repeat from the person who had been talking. This is a way of bringing a long-standing, and often unconscious, habit of bluffing to conscious awareness and increasing awareness of the negative social consequences resulting from bluffing.

Stay with the topic under discussion; do not change the subject. People who have hearing loss have great difficulty when someone changes the subject without informing them of the change. In a group situation, the topic of conversation can easily change from person to person and result in confusion, if not bewilderment. Some people tend to go beyond the question asked and bring in irrelevant information or questions that are on their minds at the moment. If, for example, the group leader says, "We'll go around the group and ask each person to tell us his or her name. Usually, someone will respond by saying something like, "My name is John Smith and I've had a hearing loss for twenty years. I think it's work related, my father also had hearing loss, etc." In such a circumstance, the

group leader should say something like, “Please only answer the question asked; we’ll come back for the other information in a few minutes.” With just a little practice, participants learn the advantages of adhering to the topic under discussion.

We treat each other with respect and kindness. This guideline is helpful in alerting people to another important communication issue—that *how* one says something is as important as *what* one says. An important message to be conveyed throughout the group process is that making polite requests for communication behavior changes is vastly more effective in achieving results than is demanding or displaying negative emotion. If, for example, a group member conveys irritation or anger through tone of voice or use of threatening words, it is useful to ask the person if she or he might be able to say that in a way that others might want to hear it. It might be useful for a person displaying anger to take a break for a few minutes in order to gather their thoughts and calm down. Often people are unaware that they are coming across to others as being angry and the recognition of that can be helpful in changing their style of communicating.

Goal #2:

A major focus of the group is to increase awareness of the variety of hearing loss-related communication problems, the causes of those problems, and their solutions.

Achieving this goal is accomplished by keeping the focus of discussion on issues related to hearing loss. One tactic used by people when discussion of a topic becomes emotionally charged or otherwise difficult is changing the subject. When people who have hearing loss become lost during a discussion or feel anxious about not being sure they are understanding, a frequent tactic used is changing the subject so they will then know what is being discussed.

Goal #3:

The proceedings during group sessions should be mainly driven by the problems, issue, and concerns of the group members and *not* by those of the group leader/facilitator.

A very good method for achieving goal of the group driving the sessions (Trychin, 1994) is having the group facilitator ask a specific question and ask each person in the group, in turn, to give one answer. For example, the facilitator may ask, “What is a problem you experienced this week related to your hearing loss?” Then, getting one response from each person (or a pass), going around the group and listing each response in writing on a whiteboard, blackboard, overhead transparency, or any other medium so that the participants can see the list. It is important to vary the location of the person who is first asked a question so that the same participant is not always the last one to answer. For example, if ten people are in the group and seated like this



It would be good to ask person 1 the first question and proceed left to right, then ask person 10 the second question and proceed right to left, then ask person 5 the next question and proceed left to right, and then person 6 the next question and proceed right to left, etc. Once the group’s responses to a question have been exhausted, the leader/facilitator can then provide any additional information related to that question obtained from preceding groups or the audiology literature. This kind of approach ensures that the sessions are driven by the information, needs, and desires of the group members and should ensure the relevance of the proceedings.

Part II

BUILDING THE AUDIOLOGIC REHABILITATION PLAN

6

CLINICAL UTILITY OF SELF-ASSESSMENT

Kathleen M. Cienkowski

Introduction

Quantifying and improving health care services, specifically hearing rehabilitation services, has taken on increased importance, most notably with the recent passage of the Over the counter (OTC) Hearing Aid Act of 2017 (Abrams & Chisholm, 2000; Cox et al., 2000; Gatehouse, 2003; Humes et al., 2017). A database search in PubMed using the terms *outcome assessment*, *self-assessment*, *hearing aids*, *audiologic rehabilitation*, and *aural rehabilitation* resulted in an average of 204 articles published per year in the last 5 years in which an outcome assessment was either the primary focus of the article or included in the experimental design. That is an increase from an average of 153 articles per year over the 5 years prior. Audiologists offer diverse services for the management of hearing loss (ASHA, 2018; Gatehouse, 2003). These services include the dispensing of devices, management of the acoustic environment, and rehabilitative support and counseling. In order to navigate successfully through these choices, clinician selections need to conform to client needs as identified through self-assessment. Similarly, measures of quality of life (QoL) have received increased recognition in health care service (Beck, 2000; Emanuel et al., 2007; Fitzpatrick, 1999; Institute of Medicine, 2015). The use of QoL assessments has become standard practice in measuring health care outcomes and may be used to allocate health care-

related funds (Fitzpatrick et al., 1992; Higginson & Carr, 2001). Both the American Academy of Audiology (AAA) and the American Speech-Language-Hearing Association (ASHA) have recommended the use of self-assessment as part of their best practice guidelines (AAA, 2004; ASHA, 2006). Despite this, many audiologists remain unsure about how to implement self-assessment into their daily clinical practice.

What Is Self-Assessment?

The terms *self-assessment*, *subjective assessment*, *subjective questionnaires*, and *outcome assessment* are often used interchangeably in the literature. This approach can be misleading, however, and does not always accurately convey the clinical utility of these measures. Outcome assessment in a health care context measures the effectiveness of a treatment by attaining a specified goal post-treatment (Frattali, 1998; Institute of Medicine, 2015). For example, a hearing loss may be remediated by dispensing a hearing aid to the client. Outcome assessment, in this case, can be accomplished in a variety of ways, including the use of behavioral measures such as speech in noise testing, real ear measures, questionnaires, or informal client reports. Questionnaires are the actual survey instruments that may be used for self-assessment. Subjective assessment is a measurement of a quality

or condition based solely on the opinion of the examiner or client. For example, a client may report that he experiences severe tinnitus, yet the tinnitus is not audible to the audiologist. The term self-assessment reflects a client's own self-evaluation of his/her perceptions and performance. This is a reflective process in which the individual considers his/her own strengths, weakness, and overall performance (Harvey, 2004). This type of measure can be administered after a treatment as an outcome assessment, before a treatment to develop rehabilitation goals, or during a treatment to monitor treatment progress. The term *self-assessment* will be used throughout the remainder of this chapter as it best reflects the diversity of clinical application.

Why Is Self-Assessment Important?

The routine use of self-assessment as part of comprehensive audiologic rehabilitation (AR) is a philosophically based decision. From the client's perspective, AR should not be an afterthought to a diagnostic evaluation, but rather in keeping with the primary purpose of the client seeking our services. A client does not come to the audiologist because of a hearing loss, but rather because he/she is experiencing communication difficulties. The remediation of those difficulties clearly goes beyond simply measuring an audiogram. Planning for AR services begins the moment the client

walks through the door. Client-centered care necessitates that the individual's experiences be part of that process. Self-assessment is a quick and easy way to document these client needs and expectations.

Self-assessment measures can be used in a variety of ways. Table 6-1 includes applications for both disease-specific and generic self-assessment measures. Disease-specific assessments evaluate a particular disease or condition. For example, the Abbreviated Profile of Hearing Aid Benefit (APHAB) can be used to evaluate auditory disability before and after the provision of hearing aids (Cox & Alexander, 1995). The Hearing Handicap Inventory for the Elderly (HHIE) can be used to assess a client's perceived handicap as a result of hearing loss (Ventry & Weinstein, 1982). Disease-specific instruments have been shown to be sensitive to pre- versus post-treatment measures in audiology (Abrams & Chisolm, 2014). Generic measures, in contrast, allow for comparisons to be made across diseases or conditions. General QoL measures fall into this latter category. Saunders and colleagues (2005) noted that generic measures afford audiologists several advantages over disease-specific measures, including the consideration of hearing rehabilitation in a broader health care context; the application of cost-benefit analysis; which may affect resource allocation; a better understanding of underlying issues that may affect rehabilitation efforts; and, finally, a comparison across sensory or rehabilitation aids, as is the case when a client wears both hearing aids and eyeglasses.

Table 6-1. Applications of Self-Assessment Measures

Use	Disease Specific	Generic Quality of Life
Screening and monitoring of psychosocial problems	x	x
Identify perceived health problems	x	x
Medical compliance/or audit	x	x
Treatment outcome measure	x	x
Research evaluation	x	x
Clinical trials	x	x
Cost-utility analysis	x	x
Prioritizing aural rehabilitation goals	x	

Note. Adapted from Fitzpatrick, 1992.

The literature on the application of generic measures in audiology has shown equivocal results. Bess (2000) reported that commonly used generic measures such as the Sickness Impact Profile (SIP) (Gilson et al., 1975) are often insensitive to the impact of hearing loss. Abrams et al. (2005) also noted that generic measures are not sensitive to the effects of treatment for hearing loss. In contrast, Saunders and Jutai (2004) found that subscales from the Psychosocial Impact of Assistive Devices Scale (PIADS) (Jutai & Day, 2002) were more strongly correlated with daily hearing aid use than subscales of disease-specific instruments were, namely the Satisfaction with Amplification in Daily Life (SADL) (Cox & Alexander, 1999), the Expected Consequences of Hearing Aid Ownership (ECHO) (Cox & Alexander, 2000), and the APHAB. In addition, Brodie and Ray (2018) noted that hearing loss is associated with reduced QoL as measured with generic measures such as the SF-36. Despite these mixed results, proponents argue that the use of generic measures keeps the treatment and evaluation focus on the client's needs, rather than on the disease (Higginson & Carr, 2001). Arguably, both types of instruments have merit depending upon the information desired (Saunders et al., 2005).

An Opportunity to Remain Client-Focused

Clinicians may argue that self-assessments are not needed: "I can find out my client's needs with a thorough case history" or "I can simply ask how my client is doing." For many years this was the reason given for clinicians' failure to adopt the use of self-assessment tools (Nemes, 2003). Fortunately, clinicians are recognizing the need for formal verification of benefit and satisfaction using a self-assessment tool, at least when dispensing hearing aids. In *The Hearing Review Dispenser Survey of 2002* (Strom, 2002), only 19% of the more than 600 respondents reported administering a self-assessment tool for hearing aid

clients "always or nearly always." Nearly half of the sample (43%) reported at that time that they "seldom or never" used such measures. In *The Hearing Review Dispenser Survey of 2006* (Strom, 2006), 77% of dispensing audiologists and 76% of hearing instrument specialists reported using self-assessment tools. Most hearing instrument specialists reported that a survey developed in their own office was the instrument of choice, while more than half of dispensing audiologists reported using a published survey such as the Client Oriented Scale of Improvement (COSI) (Dillion et al., 1997) or the APHAB (Cox & Alexander, 1995). This represents a substantial increase in the reported usage of self-assessment tools. Although it is unclear how audiologists incorporate this information into their clinical practice (Erdman, 1993, 2009) with recent survey data suggesting that many audiologists are using the information to validate treatment outcomes (ASHA, 2012).

It is important to note that self-assessment is a multidimensional process. The psychosocial impact of hearing loss and communication difficulties cannot be easily predicted from the audiogram or other diagnostic measures (Newman et al., 1990; Saunders & Cienkowski, 1996; Weinstein & Ventry, 1983). Furthermore, help-seeking behaviors are more strongly correlated with self-reported hearing loss than with audiometric thresholds (Duijvestijn et al., 2003; Laplante-Lévesque, Hickson, & Worrall, 2011a; Swan & Gatehouse, 1990). It is imperative to use a formal assessment because clinician impressions and client impressions may not always be in agreement. A clinician's initial impressions of a client's communication problems or needs may artificially constrain future assessments and influence outcomes (Schneider et al., 2005). Social psychologists refer to this as an *anchoring effect* when a clinician's first impression may bias future interactions and interpretations (Leary & Miller, 1986). When new or different information comes to light, it may be difficult for the clinician to change course or revise a rehabilitation plan (Friedlander & Stockman, 1983). See an example of this in the case study below.

Case Study 6–1. Mrs. X., the Finicky Hearing Aid User

Mrs. X. was a 75-year-old woman with a long-standing sensorineural hearing loss. Her hearing loss was identified at age 50 and she has worn four sets of hearing aids since that time. She has a family history of hearing loss as well as a history of noise exposure. She was an

accomplished violinist and actively plays in a community orchestra. Her general health was good, with the exception of high blood pressure, which was controlled by prescription medication. She was widowed and lived alone in a single-family home. She had two adult children who resided in neighboring towns and one granddaughter away at college. Her most recent audiologic evaluation revealed a stable, moderately severe to severe sensorineural hearing loss bilaterally. Word recognition was fair in both ears when recorded monosyllabic words were presented at elevated conversational levels. Tympanometry revealed Type A tympanograms bilaterally with absent ipsilateral acoustic reflexes. Otoscopic results were unremarkable and ear canals were clear bilaterally. Mrs. X.'s current hearing aids were approximately 7 years old. Electroacoustic evaluation found the aids to be weak when compared to manufacturer specifications and it was recommended that Mrs. X. purchase a new set of hearing aids rather than repair her current aids, given the age of the devices and the outdated technology. Mrs. X. agreed. The clinician selected and ordered new receiver in the canal hearing aids with micromolds, took ear impressions for the micromolds, and scheduled a follow-up appointment.

Mrs. X. returned to the clinic to pick up her new hearing aids. The audiologist programmed the aids with three stored functions. The first program was "listening in quiet" with prescriptive gain set to match a standard NAL-NL2 target. The second program was "listening in noise." Based on the prescribed gain from the first program, the second program also incorporated adaptive directionality and multiband noise reduction. The third program was set for telephone use with auto telecoil activated. The audiologist confirmed a match of the first program to the prescriptive gain target within 4 dB using real ear measures. The micromolds were modified slightly for comfort. Mrs. X. was counseled on the care and use of her new devices and advised that she had a 30-day trial period with the new aids. She signed up for a 3-week "Living With Hearing Loss" class.

A few days later Mrs. X. returned to the clinic. "Something is wrong with these new hearing aids," she stated. "They are not loud enough." The audiologist confirmed by electroacoustic evaluation that the aids were in working order. She reprogrammed the aids, increasing the gain by 5 dB. Mrs. X. reported the aids "do sound louder" and continued with her trial period. She returned two additional times during the first week to have the aids tweaked. "I think my old aids sounded better," she stated. The following week the "Living With Hearing Loss" class began. Mrs. X. was excited. Although she was an experienced hearing aid user, she was not happy with her new devices and felt there was a lot she could still learn about managing her hearing loss. As part of the class, each participant was asked to complete a COSI and a HHIE. Ranked from most important to least important on the COSI, Mrs. X. identified listening to music, conversations with her children on the phone, and watching television as the three areas in which she would like to receive benefit from her hearing aids. She scored a 36 overall on the HHIE, indicating that she experienced a moderate hearing handicap, even while wearing her hearing aids. Realizing that her trial period was nearly finished, Mrs. X. returned to the clinic to return her hearing aids. After receiving the self-assessments, the audiologist reassessed Mrs. X.'s fitting. It was clear from the information Mrs. X. provided that her new fitting was not satisfactory. It was also apparent that listening to music was the situation that Mrs. X. felt was most important. Although her new aids provided adequate audibility for everyday listening, the aids were not ideal for nonspeech stimuli. The audiologist returned the aids and ordered a different pair that was known to provide a wider dynamic range for input signals. Mrs. X. returned to the clinic and was fit with the new aids. Two weeks into her new trial period Mrs. X. reported much improved sound quality when listening to music. On a HHIE administered 1 month post-fitting, Mrs. X. scored a 10, indicating a significant reduction in her hearing handicap. Consider how the use of self-assessment altered the outcome of this routine hearing aid fitting. How might self-assessment be integrated into the clinical process earlier to facilitate the aural rehabilitation?

Incorporating Self-Assessment Into Clinical Practice

Clinicians may be overwhelmed by the number of self-assessment tools at their disposal. There is a plethora of instruments currently available to practitioners. The selection of a self-assessment tool begins with the identification of the underlying theoretical framework for the assessment to determine what question(s) the clinician wants to ask. All self-assessments are inherently designed to provide the clinician and the client with feedback on the AR process. But it is important that reflection or self-assessment results in action. A framework is a way of linking the goals of AR with change and the activities leading to perceived change and sustainable development (Lusthaus et al., 2002). The World Health Organization's International Classification of Functioning, Disability, and Health (WHO-ICF) (World Health Organization, 2001) outlines a

conceptual classification system that describes the consequences of health conditions, including hearing loss, and has been applied to AR (Bruyère et al., 2005; Gagné et al., 2014). Within this model, sensorineural hearing loss may be viewed as a chronic health condition and, although treatment plans may result in successful management of the communication difficulties associated with this loss, the hearing loss itself remains unchanged (Gagné et al., 2014). Notably, the ICF also takes into account individual factors such as personality, culture, social situation, and environmental factors that are critical components in the management of hearing loss.

Gagné and colleagues (Gagné et al., 2014; Gagné & Jennings, 2008) advocate the use of the ICF (Figure 6-1) as a conceptual framework. They note that, within that framework, AR is defined as an intervention procedure that attempts to restore participation in activities that was limited or restricted by the

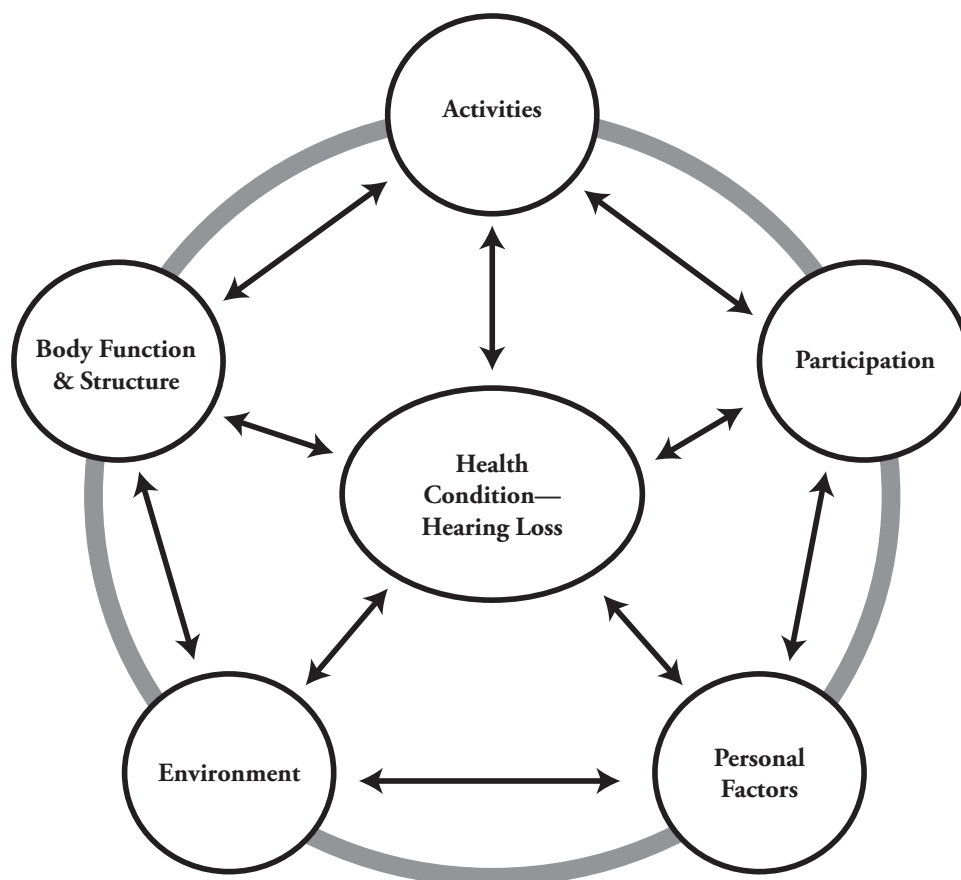


Figure 6-1. Model of disability and function for hearing loss based on World Health Organization's International Classification of Functioning, Disability, and Health (WHO-ICF, 2001).

hearing loss. This functional description of difficulties may affect both the individual with hearing loss and his/her significant others or communication partners. Clients are directed to describe their communication problems in a context that is meaningful to them. This, in turn, will direct the AR in a way that is relevant to the client's everyday life. For example, instead of the clinician asking the client, "Do you have difficulty hearing in noisy places?" through self-assessment the client might state, "I have difficulty hearing my wife and children when we go to Sunday brunch at a restaurant." The latter example provides a specific goal to be addressed and assessed post-intervention. With this in mind, the following section describes specific questions or areas that audiologists may wish to assess in daily practice. Table 6-2 provides an overview of these areas, along with examples of self-assessment tools that might be beneficial in addressing these topics. Again, these are examples only. There are many self-assessment tools that address each of these areas.

Identifying and Prioritizing Problems

One challenge in the management of hearing loss is that hearing loss affects individuals in different ways. Reduced audibility results in diminished access to auditory information (Humes & Dubno, 2010; Van-Tasell, 1993). This is factored into threshold-based prescriptive fitting of amplification devices (Dillon, 2012). For the use of devices to be successful, however, the clinician must also consider the nature of the input signal to the device; that is, the speech or nonspeech signal and the environmental conditions in which the signal is heard. Environmental factors such as distance, noise, and reverberation should be considered (Van Tasell, 1993). Furthermore, the amplification device needs to provide benefit in those situations in which the client is having difficulties. For example, a directional microphone can improve signal-to-noise ratio, but is only effective in the near field or close distance (Dillon, 2012). If a client's complaint is listening to a lecture in background noise when seated at the rear of an auditorium, a directional microphone will provide limited benefit. Management of this communication difficulty may be better addressed with preferential seating or hearing-assistive technology. A self-assessment tool can be useful to identify those situations in which the client is experiencing difficulties. For example, the Hearing Performance Inventory (HPI) developed by Giolas and colleagues (1979) examines client performance in everyday listening situations. The inventory looks at speech understand-

ing, social communication, personal communication, work communication, and reactions to unsuccessful communication (Bentler & Kramer, 2000). The original version contains 158 items and takes approximately 30 to 40 minutes to administer. The revised version contains 90 items (Lamb, Owens, & Schubert, 1983) and takes approximately 20 minutes to administer if occupational items are omitted (Huch & Hosford-Dunn, 2000). The Communication Profile for the Hearing Impaired (CPHI) (Demorest & Erdman, 1987) is another example of a comprehensive tool that assesses communication problems in a variety of situations. Areas addressed in the 145-item questionnaire include communication performance, communication environment, communication strategies, personal adjustment, and denial. The CPHI incorporates evaluation of activity limitations, participation restrictions, and environmental factors; as such, it is a comprehensive tool. The drawback to the CPHI, similar to that of the HPI, is that it takes approximately 30 to 40 minutes to administer (McCarthy & Alpinier, 2000). As an alternative, the Screening Test for Hearing Problems (STHP), developed by Demorest, Wark, and Erdman (2011), is based on items from the CPHI but includes only 20 items, and thus is more time efficient. The COSI (Dillon, James, & Ginis, 1997), in contrast, is an open-ended self-assessment tool. The client can identify up to five situations in which communication or listening is a problem and prioritize them, after consultation with the audiologist. The client rates those situations with regard to the degree of change provided by the amplification device and also rates his/her final ability with the device. The COSI is highly individualized and easily incorporated into the rehabilitation plan. The drawback is that clients may find it difficult to identify specific situations and prioritize them easily.

Facilitating Communication

Good rapport between the client and clinician is critical for successful rehabilitation. Each individual is unique and has his/her own individual characteristics, so there is no one assured way to establish rapport. The ability to make a connection with the client may influence future interactions and rehabilitation outcomes (Thomas & Hersen, 2009). It has been shown that client-centered communication is significantly correlated with a person's willingness to disclose information. Clinician interaction styles that

Table 6–2. Incorporating Self-Assessment Into Clinical Practice

Target Area	Examples of Self-Assessment Tools
Identifying and prioritizing problems Hearing loss can create a wide range of communication problems. Self-assessment can help clients and clinicians decide which areas are most important. It can help focus the client on identifying specific areas of concerns rather than more general problems such as “difficulty hearing in noise.”	Communication Profile of Hearing Impairment (CPHI: Demorest & Erdman, 1987) Client Oriented Scale of Improvement (COSI: Dillon, James, & Ginis, 1997) Telegram (Thibodeau, 2004)
Facilitating communication With a clearer understanding of the clients’ needs and help-seeking concerns, the clinician may utilize time more effectively and efficiently. Amount of time spent counseling has correlated with increased benefit from amplification and reduced hearing aid return rates.	Attitudes Toward Loss of Hearing Questionnaire (ALHQ: Saunders & Cienkowski, 1996) Communication Profile of Hearing Impairment (CPHI: Demorest & Erdman, 1987) Expected Consequences of Hearing Aid Ownership (ECHO: Cox & Alexander, 2000)
Screening for hidden problems Hearing loss can often go undetected. Older individuals in particular may not attribute communication problems to hearing loss because of the gradual onset of age-related changes in hearing. Similarly, communications demands are highly individual. Advocates propose that screening for hearing loss through self-assessment can be a cost-effective way to identify problems that may otherwise go unreported.	Screening Test for Hearing Problems (STHP: Demorest, Wark, & Erdman, 2011) Hearing Handicap Inventory for the Elderly/Adults Screening Version (HHIE-S: Ventry & Weinstein, 1983)
Facilitating shared clinical decision-making Shared decision-making is a process where by clinician and clients work together to evaluate probable treatments and outcomes, arriving at a mutually agreed upon plan. It has been shown that using self-assessment measures can assist clients and clinicians develop a shared AR plan.	Client Oriented Scale of Improvement (COSI: Dillon, James, & Ginis, 1997) Glasgow Hearing Aid Benefit Profile (GHABP: Gatehouse, 1999) Speech, Spatial and Qualities of Hearing Scale (SSQ: Gatehouse & Noble, 2004)
Monitoring change or responses to treatment Change can be measured in a variety of ways. The inability to measure improvements that are seen as relevant to the client, however, may affect adherence to the AR plan. For example, behavioral test measures are often insensitive to changes in amplification strategies. Self-assessment tools may be able to measure subtle differences that are meaningful for clients.	International Outcome Inventory for Hearing Aids (IOI-A: Cox, Alexander, & Beyer, 2003) Dynamic Assessment of Hearing Aids (DAHA: Cienkowski et al., 2006) Effectiveness of Auditory Rehabilitation (EAR: Yueh et al., 2005) World Health Organization Disability Assessment Schedule: (WHO-DAS II: World Health Organization, 1999)
Impact of hearing loss on significant others Evidence has shown that spouses of individuals with hearing loss report increased burden, stress, and decreased participation in social activities even though they do not have hearing loss themselves (Hetu, 1988). An evaluation of the family unit including family dynamics, communication mode(s), and compensatory strategies may increase adherence to the rehabilitation plan.	Self-Assessment of Communication (SAC: Schow & Nerbonne, 1982) Significant Other Assessment of Communication (SOAC: Schow & Nerbonne, 1982)

Source: Adapted from Higginson & Carr, 2009.

are focused on client's needs or concerns result in more effective communication between the clinician and client (Berrios-Rivera et al., 2006). Without trust between the client and clinician, a barrier is erected that may reduce adherence to the rehabilitation plan and limit positive health care outcomes.

In a study reported by Wasson et al. (1999), it was found that the use of self-assessments facilitated communication between physicians and elderly patients living in the community. The research conducted over a period of 2 years examined perceived quality of care for 1,651 patients aged 70 and older being seen in community clinics for general geriatric health care. Patients in the intervention group completed self-assessments on their health status that allowed physicians to provide individually tailored educational materials and support to the patients to facilitate the management of chronic health conditions. Patients in the "usual treatment" group received no individualized materials. Patients in the "intervention" group reported receiving better medical treatment, having a better understanding of their health-related concerns, and receiving more support for their functional difficulties when compared to the usual treatment group. The authors concluded that self-assessments can enhance client/clinician communication and, along with individualized education, can improve the quality of care for older adults with chronic health conditions.

Screening for Hidden Problems

Screening tools are a quick, inexpensive way to identify individuals at risk for communication problems associated with hearing loss. Since hearing loss due to age is typified by a gradual onset, adults may compensate for incremental changes in hearing sensitivity and may be unaware of communication difficulties. Demorest and colleagues (2011) have advocated for the use of self-assessment tools for the screening of hearing loss that may contribute to increased communication difficulties and a decline in psychosocial well-being. It has also been noted that self-assessment of hearing loss may be more convenient in a busy clinical setting (Kamil et al., 2015). Weinstein and Ventry (1983) were among the first to recommend the use of self-assessment tools as a complement to audiometric screenings and to tap into the social and emotional impact of hearing loss. The screening versions of the Hearing Handicap Inventory for the Elderly (HHIE-S) (Ventry & Weinstein, 1983) and the Hearing Handicap

Inventory for Adults (HHIA-S) (Newman, Weinstein, Jacobson, & Hug, 1990) are among the most widely used self-assessment screening tools for hearing loss. Numerous investigations have shown the HHIE-S to be a useful tool to screen for hearing loss among older adults (Lichtenstein, Bess, & Logan, 1988a, 1988b; McBride, Mulrow, Aguilar, & Tuley, 1994; Mulrow, Tuley, & Aguilar, 1990; Weinstein, 1986). Demorest and colleagues (2011) reported on the development of the STHP. Based on data from the CPHI, the researchers developed a 20-item self-assessment tool with two subscales: the Communication Scale and the Adjustment Scale. Items are scored on a pass/fail basis. According to the authors, an individual must pass both subscales to pass the test. A failure on either subscale constitutes a failure on the test and warrants further evaluation.

Facilitating Shared Clinical Decision Making

Self-assessments can help guide the clinician and client to listening situations that hold the greatest importance for the client. Incorporating what the client values, along with what the clinician views as best practice, is known as *shared decision-making* (Charles, Gafni, & Whelan, 1997; Frosch & Kaplan, 1999; Laplante-Lévesque et al., 2010). The audiologist brings expertise about hearing loss, amplification, and aural rehabilitation. The client brings expertise about his/her life experience. Both experts have valuable information that is necessary for the management of the hearing loss to be successful. AR plans may include a variety of treatment options including hearing aids or other sensory devices, assistive devices, group rehabilitation, or no treatment at all. When more than one treatment is viable, shared decision-making gives weight to the client's values when a choice is to be made. Laplante-Lévesque and colleagues (2011b) reported that, for adults with hearing loss, the decision to seek AR services using shared decision-making could be predicted from seven factors: (a) application for subsidized hearing services; (b) degree of hearing impairment; (c) communication self-efficacy; (d) external locus of control; (e) perceived hearing disability; (f) perceived communication program effectiveness; and (g) perceived suitability of individual communication program. They noted that clinicians can use self-assessments as a guide when interviewing clients, especially when

trying to uncover information related to the predictive factors. Shared decision making is an element of family-centered care, as discussed in several other chapters in this book.

Monitoring Change or Response to Treatment

Monitoring change or response to treatment is one of the most common reasons self-assessment tools are used by audiologists. Consider the scenario in which a client is identified with a hearing loss and is subsequently fitted with a hearing aid. The audiologist may ask a client to complete a pre-/post-fitting questionnaire to measure benefit or satisfaction derived from using the device. Commonly used tools include the APHAB, SADL, or International Outcome Inventory for Hearing Aids (IOI-HA) (Cox et al., 2003). The APHAB and its predecessor, the Profile of Hearing Aid Benefit (PHAB) (Cox et al., 1991), were designed to assess the impact of hearing loss across a variety of common listening situations. The inventory includes four subscales: ease of communication; listening in background noise; listening in reverberant environments; and aversiveness to loud sounds. The client rates his/her abilities on a Likert scale in unaided and aided conditions. A global score can be computed, as can scores for the subscales. Normative data are available for comparison. The APHAB has been researched extensively and judged to be a valid and reliable instrument (Cox & Alexander, 1995; Johnson et al., 2010). Further, it has been utilized with users of cochlear implants and other implantable devices, as well as hearing aids, making it attractive for practices with a diverse amplification clientele (deWolf et al., 2010; Litovsky et al., 2006; Plyer et al., 2008). The SADL, also developed by Cox and colleagues (1999), evaluates client satisfaction with amplification. Cox defines satisfaction as a multidimensional concept that includes physical, social, psychological, and financial changes associated with the use of amplification (Cox, 2003). Measures of satisfaction may include benefit derived from device use as well as other elements. For example, it is quite possible for a client to receive benefit from wearing a hearing aid while listening in certain environments, but yet he/she may still not be satisfied with the outcome of the device fitting because of unmet needs or issues related to his/her personality that affect satisfaction (Cox et al., 2007). The IOI-HA is a brief, seven-item question-

naire. The original intent of the survey was to develop an instrument that would serve as an addendum to existing outcome measures (Cox et al., 2003). Because of its brevity, however, it appeals to the clinical audience. The seven items are scored on a 5-point scale and address benefit, satisfaction, and use. The IOI-HA has been translated into 23 languages and has normative data available for the English version. The use of self-assessment tools, as noted above, offers a unique insight into the benefit and satisfaction derived from amplification use that might not otherwise be documented from objective assessment of devices.

Impact of Hearing Loss on Significant Others

The communication difficulties associated with hearing loss affect both the individual with hearing loss and his/her communication partners. Early work by Hétu and colleagues (1988) showed that spouses of individuals with hearing loss report increased burden, stress, and a decreased participation in social activities even though they do not have hearing loss themselves. This work is supported by recent studies that have found that communication partners may be affected throughout the identification and treatment of hearing loss of a significant other (Manchaiah et al., 2013; Wallhagen, 2010). Further, studies investigating the perception of hearing-related QoL for individuals with hearing loss and their significant others often report discrepancies between the partners (Chmiel & Jerger, 1993; Newman & Weinstein, 1986; Preminger, 2002). Individuals with hearing loss may be unaware of the compensatory strategies employed by significant others to manage communication breakdowns (Preminger & Meeks, 2010). A case can be made that plans are more likely to be successful when significant others are included in the rehabilitation decision-making process. Preminger and colleagues (2010) have shown that participation of significant others in AR groups results in better congruence between communication partners, reduction in stress and negative affect by persons with hearing loss, and improved marital communication.

There are not many tools available to assess the impact of hearing loss on communication partners. Schow and Nerbonne developed the Self-Assessment of Communication (SAC) (Schow & Nerbonne, 1982) and the Significant Other Assessment of Communication (SOAC) (Schow & Nerbonne, 1982). The SAC is a

short, 10-item inventory that assesses communication difficulties in a variety of situations. It includes questions related to disability, handicap, and perceptions of others' attitudes toward hearing loss. The SOAC is a complementary questionnaire that contains the same 10 items with pronoun changes to reflect the communication partner's point of view. Scarinci et al. (2009) developed a self-assessment inventory to measure third-party hearing loss-related QoL in spouses over 50 years of age called the Significant Other Scale for Hearing Disability (SOS-HEAR). SOS-HEAR is based on five themes identified in a qualitative analysis study of spouses (Scarinci et al., 2008). The SOS-HEAR measures the effects of hearing loss on communication, everyday activities, emotions, the spousal relationship, and social factors. In addition, Preminger and colleagues (2012) devised the Hearing Impairment Impact-Significant Other Profile (HII-SOP). The HII-SOP allows the activities limitations and participation restrictions experienced by partners of individuals with hearing loss to be quantified. The instruments described above may be useful in gaining a greater understanding of the impact of hearing loss beyond just the individual with hearing loss to immediate communication partners and aids in documenting third-party disability.

Barriers to Implementation

Incorporating self-assessments into clinical practice may mean the clinician needs to change current practice. For some, change is not easy, especially when practice procedures have been well established over time. Clinicians may fear adverse judgment from colleagues; the culture of the workplace becomes important to successful implementation (Higginson & Carr, 2001). Barriers to the use of self-assessment measures in everyday clinical practice include concerns about cost, time, feasibility, and clinical relevance. For an instrument to be clinically useful, it must be valid, reliable, and relevant. It must also be simple, quick, and easy to complete and score (Noble, 1998; Saunders & Cienkowski, 1996; Saunders et al., 2005).

Lack of or limited evaluation time is given as a reason for the failure of clinicians to adopt self-assessment measures (Nemes, 2003). Many self-assessment measures were developed in research settings where time constraints are not the same as in the clinical setting. As noted earlier, there are instruments that can take 30 minutes or more to administer. The majority of tools have been developed with the clinician

in mind, however, and take fewer than 10 to 15 minutes to complete. In addition, some instruments have short forms that take far less time to complete. The clinician does not always need to be present for the client to complete the questionnaire. Many tools can be completed by the client at home and returned by mail or uploaded via an electronic patient portal. Furthermore, a number of tools (e.g., APHAB, DAHA, SAC/SOAC) can be completed on a computer, which facilitates scoring and comparison to normative data. Treatment timelines are also important when considering the best time to administer a self-assessment tool. In a clinical setting when the self-assessment is used as an outcome measure, it may be desirable to administer the assessment within the device trial period. If the tool is to be used to develop an AR plan or serve as a baseline for post-fitting testing, then administration at an initial visit might be desirable. This information is only useful, however, if it is predictive of outcome or results (Cox, 2003). Therefore, the psychometric properties of the instrument are important to consider. What is the test/retest reliability of the instrument? Has it been validated, and for which population? Is normative data available to which results can be compared? Fortunately, audiologists have a number of self-assessment tools with known psychometric properties at their disposal, so this need not be a concern. It is, however, imperative that the clinician be a wise consumer and review the literature on a given instrument prior to its use to be familiar with its strengths, weaknesses, and proper application.

Research supports the conclusion that personality may also affect the self-assessment reports. In a series of studies, Cox and colleagues (1999, 2003) reported that the stability of self-reported outcome assessments were influenced by the personality trait of neuroticism as measured by the NEO-FFI Personality Inventory (Costa & McCrae, 1992). They found that subjects who scored highest on neuroticism reported more benefit from amplification than those subjects with lower scores for neuroticism, as measured at intervals of 3 weeks and 6 months post-fitting. Benefit scores for those subjects with lower neuroticism scores tended to be more stable across measurements taken at the different time intervals. The authors concluded that personality does influence self-assessment measures. It is to be expected that the self-assessed outcome will vary over time, especially in the first few months of device use, but that certain domains of outcome may remain stable. The influence of personality has been supported by other investigators (Garstecki & Erler, 1998; Gatehouse, 1994). It

should be noted, however, that there is not extensive literature addressing this topic; it is likely a complex issue that warrants further investigation.

Final Remarks

There is an increased interest in the use of self-assessments in hearing health care. An individualized approach to the management of hearing loss is central to client-centered care. The use of self-assessment in clinical audiology is a complex and multidimensional process. The incorporation of such measures is not without challenges to clinicians, including changes to existing clinical protocols and time management. These challenges are not new to audiologists. As an evidence-based profession, it is incumbent upon the clinician to maintain and update their clinical practice continually as new evidence becomes available. In our current health care climate, it is imperative that clinicians not only provide what they know to be the best management possible, but also support the efficacy of their work with appropriate documentation measures. Self-assessment reflects the perception that hearing-related needs are unique to individuals and identifies communication challenges in a way that is not possible through other means. Self-assessment will allow audiologists to communicate better with their clients and incorporate their needs directly and effectively into AR.

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7

MEASURING HEALTH-RELATED QUALITY OF LIFE IN AUDIOLOGIC REHABILITATION

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As advances in health care science and technology yield cures for diseases and extend life, society is turning its focus from length of life to quality of life (QoL). The advances in medicine come at a great economic cost, however; indeed, the Centers for Medicare & Medicaid Services (CMS) project national health expenditures to increase at an average rate of 5.5% from 2017 to 2026, reaching a total of \$5.7 trillion in 2026 or 19.7% of this nation's gross domestic product (GDP) (CMS, 2017). As we continue to spend an increasing percentage of our nation's financial resources on health care, can we demonstrate that our expenditures have translated into a better quality of life for our citizens? This question has emerged as an important health care policy and research issue. For example, the United States Federal Drug Administration (FDA) published guidance encouraging pharmaceutical companies to include health-related quality of life (HRQoL) outcome measures as part of its clinical trials when investigating new drugs. FDA recommends patient-reported outcomes (PROs) for the adequate labeling of medical claims or concepts that are known best by the patient (United States FDA, 2009). In 15 guidance documents published by the FDA in which PROs are recommended for use in clinical trials, three specifically recommend the use of HRQoL measures (Marquis et al., 2011). In addition, as part of its value-based payment system, the CMS is

encouraging any quality measure development plan to include a quality of life PRO performance measure (Centers for Medicare & Medicaid Services, 2016).

Within the field of audiology, it is important to determine whether advances in hearing-related technology (e.g., cochlear implants, middle ear implants, bone-anchored and conventional hearing aids) as well as other nontechnological features of audiologic rehabilitation (AR) (e.g., individual counseling, group sessions, computer-based auditory and speechreading training) result in associated improvements in the quality of life of our patients. Determining the impact of intervention on the HRQoL of individuals with hearing impairment assumes great importance when we consider that the number of people with disabling hearing loss worldwide is expected to grow from 466 million in 2018 to 933 million by 2050 (World Health Organization, 2018); indeed, results from the most recent Global Burden of Disease (GBD) study suggests hearing loss is ranked among the top 10 diseases in years lived with disability (YLD) in 193 countries and territories (GBD, 2017). This chapter reviews those concepts associated with the measurement of HRQoL with an emphasis on techniques appropriate for the assessment of AR outcomes.

In our discussion of QoL and its relationship to AR, we are utilizing the definition proposed in the Preferred Practice Patterns of Audiology (Section 15.0),

Audiologic Rehabilitation for Adults published by the American Speech-Language-Hearing Association (ASHA, 2006) and, for simplicity, we are focusing on the adult population. ASHA (2006) defines AR as a “facilitative process that provides intervention to address the impairments, activity limitations, participation restrictions, and possible environmental and personal factors that may affect communication, functional health, and well-being of persons with hearing impairment or by others who participate with them in those activities.” The clinical processes involved in AR include, but are not limited to, counseling; selection and fitting of hearing instruments and other hearing-assistive technologies; individual and group training; and other forms of follow-up care. One of the expected outcomes of AR is the enhancement of well-being and QoL for those individuals with hearing impairment, as well as for their family members and/or caregivers. The challenge to the clinician is measuring the extent to which an improvement in HRQoL has been achieved. In order to make such a measurement, we first need to determine what we mean by HRQoL.

Defining Health-Related Quality of Life

QoL, in its simplest form, can be thought of as “how good or bad you feel your life to be” (Bradley et al., 1999). Of course, many factors can influence QoL. According to the National Institutes of Health (NIH), factors influencing QoL involve the cultural, psychological, interpersonal, spiritual, financial, political, temporal, and philosophical domains, in addition to one’s health status (NIH, 1993). Although health status may or may not affect other aspects of QoL, nonhealth domains also can influence the impact of a disease or disorder and a person’s response to treatment. As early as 1948, the World Health Organization (WHO) defined health as a “state of complete physical, mental and social well-being and not merely the absence of infirmity and disease” (WHO, 1948, p. 100). More recently, the WHO Quality of Life Group (WHOQOL) offered a definition of QoL as “individuals’ perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns” (WHOQOL Group, 1995, p. 1405). It should also be noted that the Healthy People 2020 initiative (U.S. Department of Health and Human Services, 2010) recognizes the importance of measuring and improving HRQoL among United States citizens and extends the definition of HRQoL to include

measures of well-being; specifically, those aspects of a person’s life associated with positive emotions and life satisfaction.

Although numerous definitions of HRQoL can be found, there is general agreement that HRQoL is a multidimensional concept that focuses on the impact of diseases and disorders, as well as their treatments, on the well-being of individuals (Fairclough, 2002). It’s important to note that not only the immediate impact of disease or disorder must be considered but also the length of time an individual must deal with the consequences of the health condition. A definition of HRQoL that considers quantity as well as quality of life is suggested by Patrick (1993, p. 82): “Health-related quality of life is the value assigned to duration of life as modified by the impairments, functional states, perceptions, and social opportunities influenced by disease, injury, treatment or policy.”

Measuring HRQoL

HRQoL assessment involves examining the extent to which one’s usual or expected physical, emotional, and social well-being is affected by a health condition and its treatments. HRQoL cannot be measured objectively, but rather reflects the subjective assessment of an individual’s self-perception of their health status and well-being. As illustrated in Figure 7–1, there are two general types of HRQoL measures: health status, and health decision or preference assessment measures. Health status assessment is based on well-established psychometric principles and involves the posing of a question or questions about a patient’s well-being, with scores being derived from their responses. Patient preference assessment is derived from econometrics and is influenced by the concept of *utility*, or the making of decisions under known or assumed probabilities of outcomes (i.e., under risk or uncertainty). Scores can reflect trade-offs between quality and quantity of life.

As shown in Figure 7–1, both health status and preference-based HRQoL assessments can be classified as being either disease-specific or generic (NIH, 1993). Disease-specific measurement focuses on the effects of a single disease or disorder (e.g., heart disease, depression, hearing loss) and its treatments (e.g., surgery, medication, hearing aids). Disease-specific instruments serve an important function for the clinician because they are responsive to intervention (i.e., pre- versus post-treatment). Although audiologists can use a variety of disease-specific instruments to measure the impact of hearing loss and validate

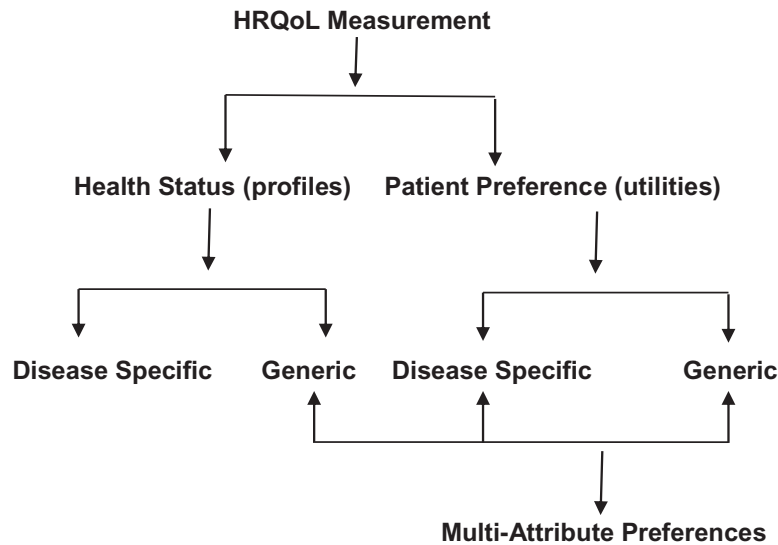


Figure 7-1. Schematic of types of health-related quality of life measurements.

the benefits of AR, the results do not allow for comparison with other diseases and disorders and their treatments. For example, with a disease-specific instrument, we cannot compare the impact of depression to the impact of hearing loss on HRQoL. Likewise, the effects of hearing aids on the HRQoL of an adult with hearing impairment cannot be compared to the effects of surgery for a patient with heart disease or medication for a patient suffering from depression. The ability to compare the effects of different diseases and disorders and their treatments is taking on increasing importance as society struggles with the question of which health conditions and interventions are worthy of third-party reimbursement and public funding for research.

To determine if depression is worse than hearing loss or whether we should be spending more money to find a cure for hearing loss versus a cure for atherosclerosis, generic HRQoL measures are required. Generic measures do not focus on any particular disorder or treatment, but rather on the self-perceived overall health of the individual. Although generic measures have demonstrated negative impacts of hearing loss on HRQoL (e.g., Bess, Lichtenstein, & Logan, 1991; Chia et al., 2007; Mulrow et al., 1990), most generic measures are not sensitive to the effects of treatment for hearing loss (Abrams, Chisolm, McArdle, & Wilson, 2005; Arnoldner et al., 2014; Bess, 2000). The reason is simple: Whereas most generic profiles include items related to major health domains—such as physical, social, and mental health—generic measurement instruments, with few exceptions, do not

include questions that specifically address problems with communication or hearing.

As noted above, HRQoL also can be assessed by measuring the preference an individual or community expresses for a particular health state. These measures are referred to as *utilities* and can be considered a cardinal measure of the strength of one's preference, expressed as a value ranging from 0.0 (least desirable health state) to 1.0 (most desirable health state). The techniques associated with utilities emerged from von Neumann and Morgenstern's (1944) decision-making theory and are increasingly being used by health care researchers to compare HRQoL across and within conditions and interventions.

What follows is a brief description of measuring instruments that have been used within the adult population to assess the impact of hearing loss and audiology interventions for hearing loss on self-perceived HRQoL. The measures are categorized according to the model presented in Figure 7-1.

Health Status Instruments

Disease-Specific Profiles

Many self-reported measures are available to the field of audiology. To determine if a given measure truly assesses the HRQoL construct, it is important to consider its content rather than simply the name of the instrument. Thus, although the term *handicap* may

be used rather than *HRQoL*, it is clear from a review of the items in the Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982) that the instrument was designed to assess the impact of a hearing loss on an individual's emotional and social well-being, which are key components in the definition of *HRQoL*. Indeed, the HHIE is perhaps the most widely known disease-specific *HRQoL* instrument used in the assessment of the effects of AR. The 25-item HHIE assesses the self-perceived psychosocial effects of a hearing loss on the older individual. Thirteen of the HHIE items measure the emotional impact of the hearing loss and 12 items measure the effect of hearing loss on social/situational functioning. The question, "Does a hearing problem cause you to feel embarrassed when meeting new people?" is an example of an emotional-domain item; on the other hand, the question, "Does a hearing problem cause you difficulty when listening to radio or TV?" is a social/situational domain item. For each item, the respondent answers *yes* (4 points), *sometimes* (2 points), or *no* (0 points). The scores for each item response are summed to provide a total score (ranging from 0 to 100), as well as emotional (0 to 52) and social/situational (0 to 48) subscale scores. The higher the score for each of these values, the greater the self-perception of hearing handicap. The questionnaire can be administered pre- and post-AR to assess the change in *HRQoL*. When administered in a face-to-face format, a reduction in score of 18.7 points, post-intervention, is needed for the clinician to conclude that real benefit has been attained. If a paper-and-pencil format is used, however, test-retest reliability diminishes and the 95% confidence interval for a true change in score becomes 36 points (Weinstein, Spitzer, & Ventry, 1986). Variations of the HHIE include a 10-item screening version, the HHIE-S (Ventry & Weinstein, 1983); a 25-item version for younger, working adults, the Hearing Handicap Inventory for Adults (HHIA and its shortened version, HHIA-S) (Newman, Weinstein, Jacobson, & Hug, 1991); and a Spanish version (Lopez-Vazquez, Orozco, Jimenez, & Berruecos, 2002).

Another example of a disease-specific profile that targets the impact of hearing loss and subsequent audiology intervention on *HRQoL* is the Hearing Handicap Questionnaire (HHQ) (Gatehouse & Noble, 2004). The HHQ consists of 12 questions scored on a 5-point Likert scale ranging from *never* (0 points) to *almost always* (5 points). Items address such *QoL* issues as avoiding social situations, feelings of being cut off from things, loss of self-confidence, and feelings of self-consciousness.

While some instruments assess only *HRQoL*, other audiology-specific instruments include one or more items to assess *HRQoL*, along with constructs such as impairment, activity limitations, and participation restrictions. A prime example is the International Outcomes Inventory-Hearing Aids (IOI-HA) (Cox & Alexander, 2002). One of the seven items on the inventory asks individuals how much the use of hearing aids has changed "enjoyment of life." Individuals respond on a 5-point Likert scale, with 1 indicating *worse* and 5 indicating *very much better*. Originally designed as an assessment of a variety of outcome domains for hearing aid intervention (i.e., number of hours per day of hearing aid use; improvement in hearing-related activities; residual activity limitations; satisfaction; residual participation restrictions; impact on others; quality of life), variations of the IOI-HA have been developed for assessing other aspects of AR intervention and the perception of intervention effectiveness by significant others (Noble, 2002).

The Audiological Disabilities Presence Index (ADPI) (Joore et al., 2002) was developed to measure *HRQoL* and differs from other disease-specific instruments by inclusion of a preference-based item. The ADPI was adapted from the Amsterdam Inventory for Auditory Disability and Handicap scale (AIADH) (Kramer, Kapteyn, Festen, & Tobi, 1995). Although the AIADH is a 30-item questionnaire designed to assess the impact of hearing loss in daily life, the ADPI includes only five items. The questions in both the AIADH and ADPI assess sound distinction, intelligibility in quiet, intelligibility in noise, auditory localization, and sound detection. For the ADPI, each item has three possible answers: no problems, moderate problems, and severe problems. In the case of ADPI, higher numeric scores indicate fewer problems. In addition to the five descriptive questions, the ADPI includes a preference measure, a horizontal visual analog scale (VAS) of 10 centimeters in length, with 0 equal to *deaf* and 100 equal to *perfect hearing* (Joore et al., 2002). The inclusion of the hearing-specific VAS is an important innovation, as the scores can be compared to those obtained with the generic visual analog scales described below.

A disease-specific profile designed specifically to measure self-assessed *HRQoL* among cochlear implant recipients is the Nijmegen Cochlear Implant Questionnaire (NCIQ) (Hinderink, Krabbe, & van den Broek, 2000). The 60 items on this questionnaire are answered on a 5-response Likert scale ranging from *never* to *always*. The items are categorized into three general domains (physical functioning, psychological functioning, and social functioning) and six sub-

domains (sound perception basic, sound perception advanced, speech production, self-esteem, activity, and social interaction). The NCIQ has been found to be a sensitive and reliable measure of self-perceived changes in HRQoL resulting from cochlear implants (Damen, Beynon, Krabbe, Mulder, & Mylanus, 2007; Loeffler et al., 2010; Hinderink et al., 2000).

Generic Profiles

As previously noted, although generic instruments can be sensitive to the effects of hearing loss in adults, they tend to be relatively insensitive to the effects of AR. Nonetheless, several generic measures have been used to study the impact of hearing loss and audiology interventions on self-perceived HRQoL.

The Sickness Impact Profile (SIP) (Bergner, Bobbitt, Carter, & Gilson, 1981) is a 136-item, self-administered questionnaire designed to assess the impact of a disease or disorder on an individual's HRQoL across 12 activities of daily living categories: emotional behavior, body and movement, social behavior, sleep and rest, home management, mobility, work, recreation, ambulation, alertness behavior, communication, and eating. In addition to providing 12 subscale scores, scores can be combined to provide a physical score, a psychosocial score, and an overall or total score. The higher the SIP scores, the greater the negative impact is on HRQoL. Bess and colleagues (1989) demonstrated that SIP scores increased as hearing loss progressed in older adults. In addition, Crandall (1998) demonstrated statistically significant improvements in SIP total, physical, and psychosocial scores as a function of hearing aid intervention in 20 older adults. Although Crandall's data suggest that SIP might be a useful generic instrument for AR, the SIP's length makes it time consuming and difficult to administer, thus limiting its clinical utility.

At only 36 self-recorded items, the MOS SF-36 Health Survey (SF-36) (Ware & Sherbourne, 1992) is much shorter and easier to administer than the SIP; it is popular among HRQoL researchers. The SF-36 measures several areas related to overall general health-related well-being, including physical functioning, role limitations due to physical health problems, bodily pain, general health perceptions, vitality, social functioning, role limitations due to emotional problems, and mental health. In addition to providing eight subscale scores, the SF-36 yields two component scale scores: a Physical Component Summary (PCS) scale score and a Mental Component Summary

(MCS) scale score. For the subscale and component scales, scores are standardized to a mean of 50 and a standard deviation of 10, with higher scores indicating more positive health status. Variations of the original SF-36 have been developed to include a screening version (SF-12), a health survey for children (SF-10), and a health survey designed for use in very large population studies (SF-8) (Ware, Kosinski, Dewey, & Gandek, 2001).

Interestingly, in the aforementioned study by Crandall (1998), hearing aid intervention did not have a significant effect on the eight SF-36 subscale scores. Crandall's finding, however, may be due to a lack of statistical power, since there were only 20 participants. Support for a conclusion that the Crandall study was underpowered, comes from Abrams et al. (2002) and Hyams et al. (2018) with larger numbers of participants ($n = 105$ and $n = 100$, respectively). Abrams et al. (2002) demonstrated significant improvements for the SF-36 MCS scale score and Hyams et al. demonstrated significant improvements on the SF-36 general health subscale score as a function of hearing aid intervention. As generic instruments tend to include few, if any, questions directly related to hearing and communication, it is likely that any treatment effects will be relatively small and the demonstration of statistical significance will require a relatively large number of subjects.

A generic health-status instrument that does include questions about communication is the WHO's Disability Assessment Schedule 2.0 (WHODAS 2.0) (Üstün et al., 2010). The WHODAS 2.0 is a 36-item questionnaire that assesses self-perceived functioning over an individual's most recent 30 days in six different life domains: communication (i.e., understanding and communicating with the world), mobility (i.e., moving and getting around), self-care (i.e., attending to one's hygiene, dressing, eating, and staying alone), interpersonal (i.e., getting along with people), life activities (i.e., domestic responsibilities, leisure, and work), and participation in society (i.e., joining in community activities). These six subscale scores map directly onto activity and participation domains of the WHO's International Classification of Functioning, Disability, and Health (WHO-ICF) (WHO, 2001). In addition to providing six subscale scores, the WHODAS 2.0 provides a total score. Scoring for the WHODAS 2.0 is accomplished by averaging responses and then normalizing scores into a standard scale, with higher scores indicating better HRQoL. Chisolm and colleagues (2005) evaluated the psychometric properties of the WHODAS 2.0 for the measurement of functional health status in adults with acquired hearing

loss. The results indicated that the communication and participation domains, as well as the total score, could be considered psychometrically sound assessments of HRQoL in an adult hearing-impaired population in convergent validity, internal consistency, and test-retest stability.

Health Decision and Preference Instruments

The value added by a given treatment is often of significant interest to patients, clinicians, insurers, and policy makers. With rising health care costs and shrinking reimbursement rates, applied health economics are helpful in demonstrating improvements as a result of the intervention, which can in turn influence resource allocations. HRQoL measures are commonly used in health economics to evaluate and compare intervention gains in relation to a monetary value or cost. Two methods are commonly used in AR: cost-effectiveness analysis (CEA) and cost-benefit analysis (CBA). CEA relates monetary costs with effectiveness in health outcomes (e.g., reduced mortality, pain reduction, activities of daily living) and is measured in monetary units (e.g., U.S. dollars) per quality-adjusted life year (QALY). A QALY is a generic measurement of the quality and the quantity of life lived with a disease, is used to assess the health-related value of an expenditure in medical intervention, and is equal to 1 year in perfect health. Cost-utility analysis (CUA) is a subtype of CEA that applies measures of utility, or health preferences under uncertainty. While both are theoretically similar, CUA relies on a cardinal measure of utility, or a health preference index (i.e., 0.0 = death; 1.0 = perfect health) whereas CEA compares observed effects of different treatments. CBA, on the other hand, compares costs with outcomes or benefits, both measured

in monetary terms. Table 7–1 summarizes the characteristics for each of these decision- and preference-based techniques.

Cost-Effectiveness Analysis

CEA and CUA are best suited for assessing technical efficiency in achieving a predetermined goal within a given budget and cannot be performed on one intervention alone, but rather require comparisons of two or more interventions for the same disorder (e.g., HA versus HA + AR, or pre-HA versus post-HA). CEAs are performed under the assumption that each patient and health care system has a specific monetary amount they are willing to pay (i.e., willingness to pay [WTP] or cost-effectiveness threshold) per QALY. WTP thresholds are specific to a given health care system and are estimated to be \$50,000/QALY in North America, compared to £20,000 to £30,000/QALY in the United Kingdom and €30,000/QALY in Spain (Shiroiwa, Igarashi, Fukuda, & Ikeda, 2013; Bobinac, van Exel, Rutten, & Brouwer, 2010; Pérez-Martín, Artaso, & Díez, 2017). CEA can often provide information on allocative efficiency via an incremental cost-efficiency analysis (ICER), which demonstrates the amount of extra benefit that is obtained as a result of extra cost (National Information Center on Health Services Research and Health Care Technology [NICHSR], 2016; Pérez-Martin et al., 2017; Bond et al., 2009). A treatment is deemed to be cost-effective if the results for the treatments of interest are below the WTP threshold and can identify the most cost-effective intervention among the two options. CEA employs probabilistic decision models in which data are simulated through a flowchart of probabilities for items such as test sensitivity, specificity, and disease prevalence, and provide simulated data for outcomes.

Table 7–1. Summary of Health Decision and Preference Tools

Technique	Measured Benefits	Unit of Measurement
CEA	Comparative health-related effectiveness of treatments in relation to monetary cost	Cost/QALY
CUA	Health preferences in relation to monetary cost	Cost/QALY
CBA	Health and nonhealth outcomes or benefits in relation to cost	Monetary cost (e.g., US\$, €)

Commonly utilized CEA modeling tools include decision trees, Markov processes, and stochastic trees. A detailed description of these tools is beyond the scope of this chapter; however, for an introduction to these methods, the reader is directed to Chapman and Sonnenberg's text (2000). The basic formula for cost-effectiveness analysis is expressed in a ratio: Cost Benefit Ratio = (Cost A – Cost B) / (Effect A – Effect B).

CEA has historically been the most common method of economic evaluation in AR. Hearing aids have been found to be a cost-effective treatment for adults aged 50 to 80 when compared to nonuse of hearing aids at an estimated \$13,615/QALY for men and \$9,702/QALY for women, both of which fall below a WTP threshold of \$20,000/QALY (Chao & Chen, 2008). Favorable findings for the cost-effectiveness of hearing aids were reported for fittings in the Netherlands with an estimate of €15,807/QALY, which was below the €16,000 threshold for consideration of insurance coverage and reimbursement in the Netherlands (Joore, van der Stel, Peters, Boas, & Anteunis, 2003). Newman and Sandridge (1998) explored the cost-effectiveness of various hearing aids that were available at the time. Unilateral and bilateral cochlear implantation have been found to be cost-effective solutions for hearing loss (Bond et al., 2009; Pérez-Martín et al., 2017). Unilateral cochlear implantation in adults and children was found to be cost-effective for adults and children with severe to profound hearing loss in the United Kingdom when compared to no intervention and hearing aids (Bond et al., 2009). Bilateral cochlear implantation, both simultaneous and sequential, was found to be cost-effective when compared to unilateral implantation (Pérez-Martín et al., 2017). Other applications of CEA in audiology include tinnitus management (Maes et al., 2014), magnetic resonance imaging studies for asymmetric hearing loss (Hojjat et al., 2016), adult hearing screening (Baltussen & Smith, 2009; Rob et al., 2009), and newborn hearing screening (Chiou et al., 2017).

Cost-Utility Analysis

Generic Utilities

As previously described, CUA is a type of CEA that measures utility. A utility is defined as a cardinal measure of the strength of one's preference and ranges from 0.0 (death) to 1.0 (perfect health). A cardinal utility is an economic concept that assumes peo-

ple are able to assign a meaningful number (i.e., utility) to their satisfaction in one situation as compared to their satisfaction in an alternative situation. Utilities are assumed to be similar to, for example, temperature or distance—that is, the incremental units are reasonably constant and objective. Although it is generally accepted that, in practice, the measurement of utilities is only a rough approximation of value, utility assessment has taken on an important role in health care economics. Utilities are particularly useful in making comparisons across diseases and interventions and are often used to calculate the cost of treatment outcomes in terms of improved HRQoL. Such economic evaluations are known as CUAs, the result of which is expressed as a cost per QALY gained (Drummond, Sculpher, Torrance, O'Brien, & Stoddart, 2005). The basic formula for CUA is expressed in a ratio (NICHSR, 2017): Cost-Utility Ratio = (Cost A – Cost B) / (Utile A – Utile B).

The most common direct methods used to measure utility include the standard gamble (SG), time tradeoff (TTO), and visual analog scale (VAS) techniques (Bennett & Torrance, 1996). Each is described briefly below.

Standard Gamble

The classic method of measuring health preference is the SG approach. In a SG, a person is offered a choice between two alternatives: living with Health State B with certainty (which, in a clinical population, is presumably a patient's present health state) or gambling on Treatment X. The result of Treatment X is either perfect health (Health State A) or immediate death (Health State C). The probabilities of achieving Health State A or Health State C as a result of Treatment X are manipulated until a person is indifferent between his present Health State B and Treatment X. An assumption of the technique is that the higher the probability of death a person is willing to consider, the poorer the health state (or HRQoL) perceived in remaining with Health State B. Utility is calculated as 1.0 (perfect health) minus p (the probability of death associated with Treatment X) when a person is unable to choose between "A" and "B").

Time Trade-Off

Another approach commonly used to measure health state preferences is the TTO technique. In TTO, a person is offered a choice between living a normal life span in the present health state or a shortened life

span in perfect health. The number of years spent in perfect health is manipulated until a person is indifferent to the shorter period of perfect health and the longer period in the less desirable state. An individual who is willing to trade years of life for a shorter life in perfect health (or perfect hearing) is assumed to be revealing a great deal about his or her perceived HRQoL as imposed by a disease or a disorder.

Visual Analog Scale

The use of a VAS was briefly described earlier in the introduction of the ADPI (Joore et al., 2002). Measuring utility with a VAS or a “feeling thermometer” has the advantage of simplicity. A person is simply required to rate his or her perceived health state on a scale marked with 0 at one end, representing death or some least desirable state, and 100 at the other end, representing perfect health or a most desirable state.

Disease-Specific Utilities

Disease-specific utilities are a relatively new approach to measuring HRQoL (Drummond et al., 2005). They combine the benefits of a generic utility measurement—the ability to generate a single preference rating—with the advantage of a disease-specific measure that is more sensitive to the disorder or intervention under study. The techniques used to measure utilities for specific diseases are no different from those previously described (SG, TTO, VAS) with the exception that the choices or “anchors” describe the extremes of a specific disorder, rather than perfect health or death. For example, instead of asking the patient to choose between living a normal life span in his or her present health state or a shortened life span in perfect health, the TTO may be worded in such a way as to require the patient to choose between living a normal life span with his or her current hearing loss or living a shortened life span with perfect hearing. Examples of disease-specific utility studies include those examining the HRQoL impact of prostate cancer (Saigal, Gornbein, Reid, & Litwin, 2002), schizophrenia (Lerner, Rupnow, & Elnitsky, 2005), and hearing aid use (Kenworthy, 2002; Piccirillo, Merritt, Valente, Littenberg, & Nease, 1997).

In our own work, we have developed a software program, Utility Measures for Audiological Applications (UMAA) and have obtained preliminary disease-specific utilities for tinnitus, benign paroxysmal

positional vertigo, and hearing loss (Abrams, Roberts, Lister, & Hnath Chisolm, 2006). Figures 7–2, 7–3, and 7–4 demonstrate the UMAA screenshots for hearing-specific utility assessments using SG, TTO, and VAS techniques, respectively. Significant correlations were obtained between the utilities as measured with all three techniques and responses to the HHIE from 48 adults with hearing loss (Condill, 2006), as well as between utilities measured with all three techniques and responses to the Dizziness Handicap Inventory (DHI) (Jacobson & Newman, 1990) from 52 patients with benign paroxysmal positional vertigo (BPPV) (Roberts et al., 2009).

As described by Drummond et al. (2005), the use of disease-specific and generic utilities in a two-stage approach may allow for more appropriate resource allocation within disorders (because disease-specific utilities will be sensitive to the effects of different interventions within the same disorder) and across disorders (because generic utilities provide data for making CUA comparisons across different disorders). An alternative to performing a two-stage approach is using a multi-attribute health status classification system that evaluates both health profiles and utilities.

Multi-Attribute Preferences

Direct measurement of utilities for health outcomes using the SG, TTO, and VAS approaches can be a time-consuming and complex task. To overcome some of the difficulties with direct assessment of utilities in clinical populations, researchers have developed prescored health status classification systems. Within each system, several dimensions or attributes of health are assessed with a number of levels of functioning defined for each dimension. For example, the Health Utilities Index Mark 3 (HUI3) (Feeny, Furlong, Boyle, & Torrance, 1995; Furlong et al., 1998) measures six health attributes: sensation, mobility, emotion, cognition, self-care, and pain. The sensation attribute is subdivided into vision, hearing, and speech. Each of these subattributes contains a list of level descriptors where Level 1 (single attribute utility factor = 1.0) describes the highest health state or preference and Level 6 (single attribute utility factor = 0.0) describes the lowest health state or preference. With regard to hearing, for example, Level 1 is associated with the statement “Able to hear what is said in a group conversation with at least three other people without a hearing aid” whereas the Level 6 statement indicates that the person is “Unable to hear at all.” In

Time Trade-off

Exit

Now think about the following two alternatives.

A You continue to live with your current hearing for 24 years, followed by death by natural causes. Your condition neither improves nor worsens.

or

B Your hearing returns to normal immediately and you live with normal hearing for 24 years, followed by death by natural causes.

Which alternative would you choose?

Select A

Select B

Cannot Choose

Figure 7-2. Screenshot from the Utility Measures for Audiological Applications (UMAA) showing an example of the time tradeoff technique for hearing.

Standard Gamble

Exit

Now think about the following:

A You can live with your current hearing for the rest of your life. Your condition neither improves nor worsens.

or

B You can take a magic pill with a 50% chance of your hearing returning to normal AND a 50% chance of total deafness.

Which would you choose?

Select A

Select B

Cannot Choose

Figure 7-3. Screenshot from the Utility Measures for Audiological Applications (UMAA) showing an example of the standard gamble technique for hearing.

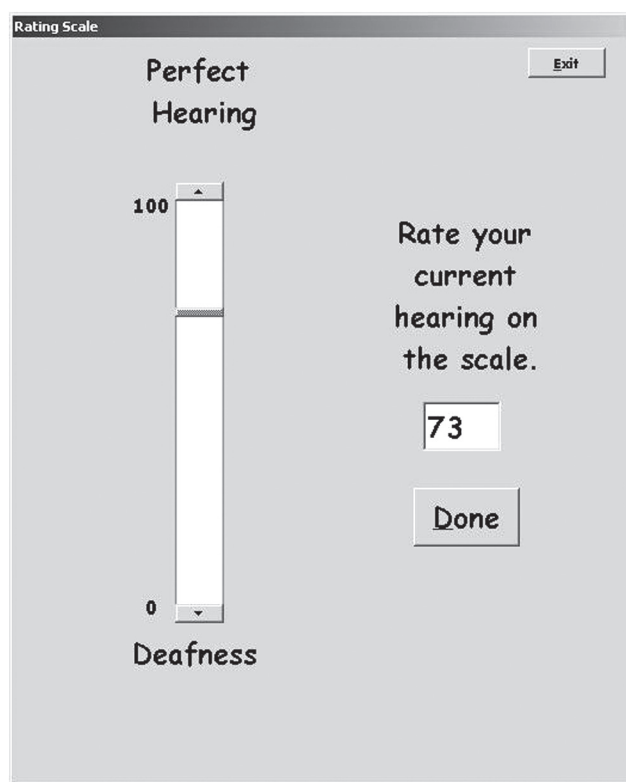


Figure 7-4. Screenshot from the Utility Measures for Audiological Applications (UMAA) showing an example of the visual analog scale for hearing.

the HUI3, the scores of the six attributes are weighted and combined in a complex formula to create a single utility estimate. The multiplicative formulas used to determine the utility estimates for the HUI3 and for a related instrument, the Health Utilities Index Mark II (HUI2), are based on the direct assessments of utility using SG and VAS techniques among Canadian adults.

Another commonly used multi-attribute instrument is the EuroQol Five-Dimension (EQ-5D) (EuroQoL Group, 1990). With only five questions, the EQ-5D provides a simple descriptive profile and a single index value. Each of the five questions relates to a different attribute of the respondent's self-perceived health status regarding mobility, self-care, usual activities, pain or discomfort, and anxiety or depression. For each attribute, there are three levels: no problems, some problems, and major problems. Taken together, the questions can classify a person into one of 243 different health states. Initial preferences for the health states were based on a population study in the United Kingdom utilizing the TTO technique. The EQ-5D was originally designed to complement other instruments, such as the aforementioned generic profiles,

the SIP and SF-36, but is now increasingly used as a stand-alone measure.

In addition to the HUI3 and the EQ-5D, other commonly used multi-attribute preference instruments include the Quality of Well-Being scale (QWB) (Kaplan & Anderson, 1988) and the Short-Form Six-Dimension (SF-6D) (Brazier, Roberts, & Deverill, 2002). The QWB is a preference-weighted measure that combines three scales of functioning (mobility, physical activity, and social activity) with a measure of symptoms and problems. Scoring functions for the QWB were obtained from a sample of the general public in San Diego, CA. The SF-6D (Brazier et al., 2002) is based on the generic QoL measure, the SF-36, and was developed as means of converting SF-36 results into utilities. The SF-6D consists of six attributes: physical functioning, role limitations, social functioning, pain, mental health, and vitality. Four to six levels of functioning are associated with each attribute. The scoring model for the SF-6D was developed using SG utilities obtained from the general population in the United Kingdom. The SF-6D can only be used in combination with the SF-36 or the SF-12.

Cost-Benefit Analysis

Compared to CEAs and CUAs, the use of CBAs is relatively limited in the AR literature. CBA involves a weighing or ranking of the benefits and costs (i.e., advantages and disadvantages; gains and losses) (McIntosh, 2010, pp. 1–18). CBA is unique in that both benefits and costs are measured monetarily, rendering information on health and nonhealth attributes in monetary units. This method is commonly used because results are more easily understood by policy makers. It can be said that ranking of alternatives can indirectly measure the value of a given health care good or service; benefits to well-being and satisfaction of consumer preferences inherently have economic value. The most common CBA methods are: direct (stated preferences, hypothesized behavior) or indirect (revealed preferences, actual behavior). Because the indirect or revealed preferences method would be strongly tied to a disease state, this method is scarcely seen in health care. Direct methods such as WTP, willingness to accept, and stated preference discrete choice experiments (SPDCEs) are used instead. For an introduction to applications of CBA in health care, the reader is directed to the text of McIntosh and colleagues (2010).

Examples of CBA in AR include a WTP experiment by Palmer and colleagues in 1995 in which vari-

ous sound quality judgements were valued. It was found that for each 1% increase in the rating of sound quality, value increased by \$6.75. In our own work, a WTP approach was investigated with respect to hearing aid benefit as measured by the Abbreviated Profile of Hearing Aid Benefit (APHAB) (Chisolm & Abrams, 2001). For every one-unit increase on the Ease of Communication, Background Noise, and Reverberation subscales of the APHAB, hearing aid value (WTP) increased by \$16.32, \$16.88, and \$13.78, respectively. Average WTP was \$981.71 per hearing aid. In a later study using WTP, participants estimated their WTP for several cosmetic and signal processing features; the most highly valued was feedback management, averaging \$318.42 (Abrams, Chisolm, & Block, 2004). Grutters and colleagues (2008) applied a SPDCE approach using WTP and willingness to accept (i.e., compensation to forgo the good) for determining the hypothetical value in creating a new organization for hearing aid provision in the Netherlands but concluded that more research and methodological consideration was needed before applying this method to a CBA in hearing health care. Later work from the same group found that both open-ended questions (i.e., fill in the blank) and payment scales (i.e., choose one) were valid measures of WTP among hearing aid users in the Netherlands (Grutters, Anteunis, Chenault, & Joore, 2009). However, the open-ended question was found to be more strongly associated with actual payment, making it the superior method between the two.

Does Audiologic Rehabilitation Improve HRQoL?

Given that the definition utilized in this chapter for AR includes counseling, selection and fitting of hearing instruments and other hearing-assistive technologies, individual and group training, and other forms of follow-up care, evidence related to hearing technologies and other forms of training are briefly reviewed.

Hearing Aids and HRQoL

In an extensive systematic review of the literature, in which the data from individual studies were combined in meta-analyses, Chisolm et al. (2007) concluded that hearing aids do, in fact, improve adults' HRQoL by reducing the psychological, social, and emotional effects of sensorineural hearing loss as measured by some of the generic and disease-specific HRQoL pro-

files described above (e.g., MOS SF-36, HHIE). An important finding from the meta-analyses relates to the differences in effect size estimates obtained for the generic and disease-specific instruments. Effect sizes are considered measures of practical significance and, unlike measures of statistical significance, are independent of sample size (Cohen, 1988). In interpreting effect sizes, medium to large effect sizes are often considered indicative of robust clinical changes, whereas small effect sizes, although not to be ignored, are suggestive of a more subtle clinical change. The effect sizes calculated for the generic outcome measures by Chisolm et al. (2007) were small, whereas those calculated for disease-specific outcome measures were medium to large. These results highlight, as previously discussed by Bess (2000), the difficulty of demonstrating the positive effects of hearing aid intervention utilizing generic HRQoL instruments. The findings of Chisolm et al. (2007) were supported in a recent Cochrane Collaboration systematic review (Ferguson et al., 2017). An analysis of five randomized controlled trials (RCTs) involving 825 participants with mild to moderate hearing loss led the authors to conclude that hearing aids are effective at improving both disease-specific and generic HRQoL.

The importance of including questions about hearing and/or communication in generic instruments used in assessing changes in HRQoL as a function of hearing aid intervention was highlighted in a study by McArdle and colleagues (2005). In this large, multisite, randomized, controlled trial examining the outcomes of hearing aid intervention in adults, the researchers found statistically significant HRQoL benefit from hearing aid use measured by both generic (WHODAS 2.0) and disease-specific (HHIE) HRQoL instruments. As expected, a large effect size was found for the HHIE. Although the effect size reported for the total score on the WHODAS 2.0 was small, the effect size associated with the communication subscale of the WHODAS 2.0 was large.

Further support for the importance of including questions regarding hearing and/or communication in generic HRQoL instruments used to assess the outcomes of AR comes from studies examining changes in health state utility from the use of hearing aids. For example, Barton, Bankart, Davis, and Summerfield (2004) failed to demonstrate a statistically significant improvement as a result of hearing aid fitting in 609 adults using the EQ-5D or the SF-6D. Significant improvements, however, were demonstrated on the HUI3. These findings highlight the importance of considering the descriptive system used to derive generic HRQoL utilities. Indeed, the HUI3 is

the only multi-attribute preference instrument that includes specific questions about hearing. Similarly, Grutters et al. (2007) reported finding statistically significant changes on the HUI3, but not the EQ-5D, in 70 adults fit with hearing aids. As a result, Grutters and colleagues concluded that the HUI3 should be the instrument of first choice when assessing utilities in populations with hearing impairment.

Despite the limitations associated with attributing generic HRQoL improvement with hearing aid use, Gopinath and colleagues (2012) were able to demonstrate a reversal of SF-36 scores (signifying improvement in self-perceived HRQoL) among participants in the Australian Blue Mountains Eye and Hearing Studies. Specifically, 829 participants for whom baseline self-perceived HHIE-S (Ventry & Weinstein, 1983) and HRQoL (SF-36) data were available were followed up with 10 years later. The study findings suggested that self-perceived hearing handicap at the baseline was associated with poorer HRQoL (as measured by the SF-36) 10 years later, whereas hearing aid use was associated with improvements in the SF-36. Several domains were specifically identified as having improved relative to baseline measures: role limitation due to physical problems, vitality, and mental health.

Cochlear Implants and HRQoL

Perhaps the area where HRQoL measures have been most extensively and successfully used with respect to AR has been associated with cochlear implant research (e.g., Cheng et al., 2000; Francis, Nelson, Yeagle, Cheng, & Niparko, 2002; Summerfield, Marshall, & Archbold, 1997; Wyatt, Niparko, Rothman, & deLissovoy, 1996). An interesting example of a study that incorporated a CUA to demonstrate the cost effectiveness of cochlear implants was that of Evans, Seeger, and Lehnhardt (1995). They used utilities to compare the cost per QALY gained among cochlear implant users (\$15,590) against previously published costs per QALY gained for recipients of coronary angioplasties (\$11,490), implantable defibrillators (\$29,220), and knee replacements (\$49,700).

In an investigation of the impact of cochlear implants on self-perceived HRQoL, Chung and colleagues (2012) compared pre- versus post-implantation SF-36 scores among 283 adult unilateral cochlear implant recipients. On average, the participants' scores improved on five of the eight SF-36 domains: vitality, physical role functioning, mental health, emo-

tional role functioning, and social functioning. The three domains in which improvement could not be demonstrated were physical functioning, bodily pain, and general health perception. By contrast, in an earlier study, Mo, Lindbaek, and Harris (2005) demonstrated improvement in only one SF-36 domain score, general health perception, following 15 months of unilateral cochlear implantation among a group of 27 adults. The difference in findings between these two studies highlights the need for adequate sample sizes in studies examining intervention for hearing loss using generic HRQoL outcome measures.

Auditory Training and HRQoL

The issue remains, however, as to whether or not non-instrument-based intervention (e.g., auditory training or counseling-based AR) provides any HRQoL benefits separate from those obtained through the use of hearing aids or cochlear implants alone. Few studies have addressed this question with regard to auditory training. For example, in a recent systematic review of auditory training in adults, Sweetow and Palmer (2005) found only two studies that incorporated what has been identified in this chapter as a HRQoL instrument. Both studies were conducted by Kricos and colleagues (1992, 1996) using the HHIE. Whereas Kricos and colleagues (1992) found a significant change in HHIE scores as a function of auditory training, Kricos and Holmes (1996) did not. In a study examining the efficacy of a home-based, interactive adaptive computer program, Sweetow and Sabes (2006) demonstrated statistically significant improvements on the HHIE or HHIA following 4 weeks of auditory training for 30 minutes a day, 5 days per week.

Group AR and HRQoL

The HHIE/A has also been used in several studies examining the effects of post-hearing aid fitting group AR. In a systematic review of group AR programs, Hawkins (2005) concluded that there was good evidence for the efficacy of the programs, with two studies that used the HHIE showing significant changes as a function of AR treatment. One of the studies demonstrating positive benefits was conducted by Abrams, Hnath Chisolm, Guerreiro, and Ritterman (1992). Their study demonstrated significant improvement in HHIE scores among a group of adults who received hearing aids plus counseling-based AR compared to

two other groups who received either hearing aids alone or no treatment. Positive results from group AR were also reported by Preminger (2003), who found greater improvements, as measured by the HHIE and the HHIA, among participants in a group counseling-based AR program who attended the sessions with a significant other as compared to those participants who did not. In another study by Preminger and colleagues (2008), participants in a speech perception training group demonstrated improvements in self-perceived HRQoL, as measured by the emotional subscale of the HHIE/A, even though they failed to exhibit any significant improvement on measures of speech perception. These findings suggest that the dynamics associated with AR group interaction, including the informal sharing of experiencing and coping with hearing loss, can have a positive impact on HRQoL. Similarly, Moser and colleagues (2017) observed a direct relationship between self-perceived social support and HRQoL among elderly adults with age-related hearing loss. The authors concluded that the HRQoL of older individuals with hearing loss might be improved by actively engaging family members and communication partners in the rehabilitation process.

Additional evidence supporting the importance of post-hearing aid fitting AR programs for improving disease-specific HRQoL outcomes is provided through a randomized trial by Kramer and colleagues (2005). Kramer and colleagues examined the short- and long-term effects of an at-home, videotape- or DVD-based education program for older adults with hearing loss and their significant others. The outcome measure used in this study was the IOI-HA. Recall that one of the items on the IOI-HA was designed to assess QoL by asking about enjoyment in life. Immediately post-intervention, results revealed that those hearing-impaired individuals who participated in the home education program had statistically significant, higher scores on the HRQoL item than those who did not participate. Interestingly, when long-term outcomes were assessed, the QoL scores for the control group (no home education program) decreased, while remaining relatively stable for the home-based education participants.

Although there is good evidence that post-hearing aid fitting AR improves HRQoL when measured with disease-specific instruments, few studies have examined the effects of post-hearing aid fitting AR using generic measures. One of the first studies to utilize a generic HRQoL instrument to assess the outcomes of group AR was conducted by Abrams,

Hnath Chisolm, and McArdle (2002). They administered the MOS-SF36V (a slightly modified version of the MOS SF-36 designed for the veteran population) to veterans receiving treatment with hearing aids alone and those receiving treatment with hearing aids plus participation in a post-hearing aids fitting group AR program. Overall, intervention resulted in a statistically significant improvement on the Mental Component Summary (MCS) scale, with a mean change of approximately 2 points. Although those who participated in the group AR showed a mean change in the MCS scale of 3 points and those who received hearing aids alone had a mean change of only about 1.4 points, there was not a statistically significant difference in the pre- versus post-intervention scores between the two groups. Despite the lack of a statistically significant differential treatment effect, Abrams and colleagues (2002) argued that the change in scores needed to be considered relative to the cost of treatments. When a cost-effectiveness analysis was conducted, it was found that hearing aids alone intervention cost \$60 per QALY gained, whereas the addition of group AR reduced costs to only \$31.91 per QALY gained.

Although it is encouraging that Abrams et al. (2002) concluded group AR was cost-effective based on the MCS scores from the SF-36, it is important to consider findings of a randomized, controlled trial reported by Hickson, Worrall, and Scarinci (2007). In this study, participants were randomly assigned to a placebo, control social interaction group, or to an AR group focused on communication training. Interestingly, the control social group demonstrated a significant change on the MCS scores of the SF-36, whereas the AR group participants did not. These results suggested that it might simply have been the social interaction and not the communication-specific training that influenced changes in self-perceived HRQoL as measured by the SF-36. Given that the SF-36 does not contain any specific questions associated with hearing and communication, but does contain questions regarding social functioning, this may be a plausible conclusion. Indeed, in a study that examined the effects of group AR content on HRQoL, Preminger and Yoo (2010) compared changes in self-perceived HRQoL as measured by both disease-specific (HHIE/HHIA) and generic (WHO-DAS 2.0) instruments among hearing aid recipients randomized into one of three AR groups: a communication strategies training group, a communication strategies training plus psychosocial exercise group, and an information lecture plus psychosocial exercise group. Better HRQoL

outcomes were observed on selected subscales for both the HHIE/A and WHO-DAS 2.0 for the two groups that included psychosocial exercise.¹

Conclusions

In this chapter, we reviewed the principal concepts associated with HRQoL, including its definitions, measurements, and AR-related research. The measurement of HRQoL will remain a critical determinant of the impact of disease and injury and the extent to which interventions are successful. The potential of HRQoL measures to allow comparisons within and across diseases and interventions makes QoL assessment a powerful tool among clinicians, researchers, third-party payers, and health care policy makers. Although the research findings appear to be mixed, evidence is emerging that AR, including both sensory aids and non-device-based interventions, has a positive impact on the self-perceived HRQoL among those with hearing loss. More research in this area is clearly needed. The hearing health care environment is undergoing significant changes characterized by big-box stores, direct-to-consumer internet sales, and the legalization of over-the-counter hearing aids. There is a critical need to examine the efficacy of AR through careful and thoughtful research design, to include a HRQoL measurement component, if the profession of audiology hopes to substantiate claims that what we do makes a difference in the quality of life for the patients to whom we provide care.

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8

HEARING HELP-SEEKING BEHAVIOR

Gabrielle Saunders

My hope is that after reading this chapter, you will know more about some of the barriers and facilitators of help seeking for hearing loss. I also hope you will recognize how health psychology and health behavior theories can assist our understanding of help seeking, and how these theories can be used to develop behavior change interventions to promote help seeking for hearing loss at both the individual and public health levels.

Introduction

It is often noted that people wait between 7 and 10 years between first suspecting they have a hearing loss and seeking help for their difficulties (Davis et al., 2007; Fischer et al., 2011; Simpson et al., 2018). This seems to be the case regardless of whether or not there is a direct financial cost to the individual (Saunders et al., 2016a). Those in the field of audiology tend to assume the underuse of hearing health care is associated with issues specific to hearing loss and hearing aids. For example, underuse has been attributed to hearing aids being stigmatizing, too expensive, and not working well enough to warrant their use (Barnett et al., 2017; McKee, 2018). However, help seeking and intervention uptake for other chronic medical conditions are similarly low, even in developed countries and in countries in which services are provided free of charge. To illustrate, up to 86% of women with

chronic pain associated with dysmenorrhea do not seek help from health care professionals despite there being evidence-based management interventions (Chen et al., 2018). The data are similar for pain arising from other conditions (Thortensson et al., 2009). Only 20% of individuals over age 65 years with correctable vision impairment use rehabilitation services (Gresset & Baumgarten, 2002), only about one-third of older adults with self-reported vision loss have had an eye test in the previous 12 months (Iliffe et al., 2013), and on average, less than 35% of individuals with symptoms of urinary incontinence or lower urinary tract symptoms seek medical help (Koch 2006; Suen et al., 2017).

In a similar vein, the reasons given for seeking (or not seeking) help for symptoms of hearing loss are the same or similar to those given for seeking (or not seeking) help for other chronic medical conditions. These reasons include but are not limited to symptom severity, social stigma, and perceived support from others. For example, individuals who perceive their symptoms to be more severe or disabling are more likely to seek help for hearing impairment (Claesen & Pryce, 2012; Gussekloo et al., 2003), as well as for depression (Lexis et al., 2010), urinary incontinence (Burgio et al., 1994), and chronic pain (Thorstensson et al., 2009). Likewise, more perceived social stigma is associated with lower likelihood of acquiring hearing aids/hearing assistive technologies (van den Brink et al., 1996; Wallhagen, 2010), as well as with lower rates of help seeking and intervention for depression

(Y-Garcia et al., 2012) and chronic pain (Slade et al., 2009). Finally, more perceived social support from friends and family is associated with greater likelihood of seeking and acquiring hearing aids (Laplante-Lévesque et al., 2013), as well as with better hypertension self-management (Flynn et al., 2013) and adherence to diabetic therapy (Miller & DiMatteo, 2013).

The implication of these parallels between help seeking for hearing loss and help seeking for other conditions is that we should look beyond audiology to the field of health psychology to understand hearing help-seeking behaviors and health behavior change, especially because there are many models that explain the underlying motivators, barriers, and facilitators of health behaviors that can be used to develop behavior change interventions. Indeed, such models have been used as the basis for developing smoking cessation programs (Ben Natan et al., 2010), understanding dietary choices (Sun et al., 2006), changing the use of hearing protection (Griest et al., 2007; Saunders et al., 2014), and creating diabetes management programs (Jennings et al., 2014). While no health behavior model can perfectly describe why and how health behavior changes occur, models are valuable because they provide insight into the interaction between various factors that influence health behaviors to facilitate or impede behavior change. Furthermore, models provide a basis from which to develop behavior change interventions, and data show that behavior change interventions with a theoretical basis tend to be more effective than interventions developed without a theoretical basis (National Cancer Institute, 2005; Noar & Zimmerman, 2005).

Recently, audiology researchers have examined how health behavior theories can be applied to understand help seeking for hearing loss. This work is promising in that models are identifying barriers and facilitators to hearing help seeking. To illustrate this, some of the models and the associated work on hearing help seeking are described below.

Health Behavior Models for Help Seeking

Transtheoretical Model

The Transtheoretical Model (TTM) of behavior change was developed in the 1980s by Prochaska and colleagues (Prochaska & DiClemente, 1983; Prochaska & Velicer, 1997). The premise of the model is that an individual's readiness for change determines whether he/she will engage in, and maintain, a health behavior. It is proposed that individuals move from being unready for change (precontemplation) to a stage at which change is being considered (contemplation), leading them to a time during which they prepare for change (preparation), until they reach the point at which behavior change occurs (action). That behavior may continue (maintenance), or it may stop (relapse). If the latter, the cycle begins again (Figure 8–1). The model acknowledges that progression is not necessarily continuous, in that individuals have the potential to progress and regress between stages. Movement between stages is affected by a process of decisional balance, which is the weighing of the perceived pros

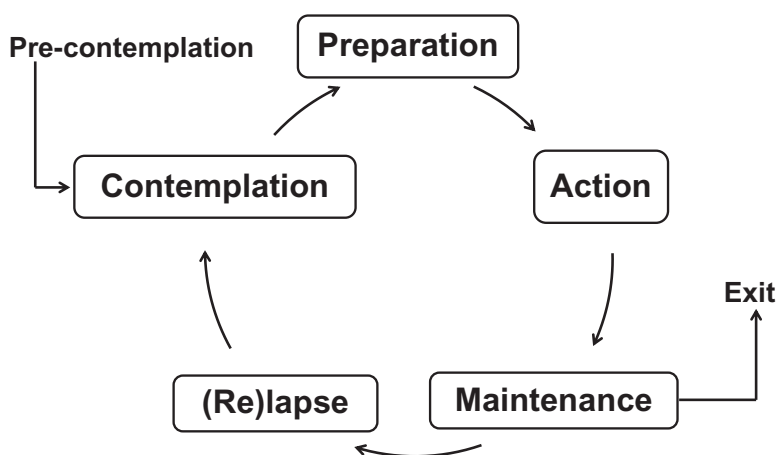


Figure 8–1. Transtheoretical Model.

and cons of taking up the new behavior, and by the individual's self-efficacy for conducting the new behavior, such that to progress from one stage to the next, the individual must perceive more pros than cons and believe in their capability to be successful. This model is not without its critics. Some have questioned whether stages of change should be described as categories or a continuum (Littell & Girvin, 2002), which in turn has raised questions as to whether the

stages can be accurately measured (Bridle et al., 2005). Further, this model assumes a rational approach to decision making and largely ignores social and emotional influences. Nonetheless the constructs of the model can well describe hearing health behaviors (Table 8–1), and several studies have applied this model to hearing help seeking, as follows. (*Editors' note:* For more information on the Transtheoretical Model, the reader is referred to Chapter 12 of this text.)

Table 8–1. Transtheoretical Stages of Change Model: Definition of Constructs and Influences on Help Seeking for Hearing

Definition	Associated Behaviors	Application to Hearing Help Seeking	Comment
Precontemplation			
Individual is not ready to change and does not intend to do so in the foreseeable future.	Individual does not acknowledge a problem, is unaware if a problem, or does not want to think about the problem.	Individual may be in denial about hearing loss or may be unaware that they have hearing problems.	Individual may be in this stage because they are uninformed or underinformed about the consequences of untreated hearing loss or may simply be unaware they have hearing loss.
Contemplation			
Individual is aware of a problem and is weighing the pros and cons of behavior change.	Individual acknowledges a problem and begins to see value in taking action.	Individual begins to seek out information about hearing health care and/or talk to others about hearing loss and hearing aids.	The process of decisional balance can lead to ambivalence, which others may view as procrastination.
Preparation			
Individual is preparing to make behavior changes and may have formed a plan of action.	Individual gathers information needed for behavior change and shares their plans with others.	Individual looks for an audiologist and obtains concrete information about hearing loss and hearing aids.	Self-efficacy for help seeking knowledge may play a large role at this time.
Action			
Behavior change occurs.	Individual changes behavior.	Individual schedules an audiology appointment.	In the context of help seeking for hearing loss and behavior change, scheduling an audiology appointment is defined as action.
Maintenance			
Individual follows through with behavior change.	Individual maintains planned behavior.	Individual attends audiology appointment.	

In 2002, Milstein and Weinstein applied the TTM to understand the help seeking behaviors of individuals who failed a free hearing screening that took place at a senior center. Those who failed the screening were asked to identify with one of four statements representing the stages of change, ranging from “I do not think I have a hearing problem and therefore nothing should be done about it” (precontemplation) to “I know I have a hearing problem, and I am here to take action to solve it now” (action). If the TTM applies to hearing help seeking, one would expect the majority of individuals in this study to have agreed with the statement “I do not think I have a hearing problem and therefore nothing should be done about it.” This was indeed the case, with 76% of the 147 participants being in the precontemplation or contemplation stages of change. In the second part of the study, the 135 individuals who failed the hearing screening were given a written recommendation to have an audiological evaluation. Two weeks later, they were contacted by the investigators to find out whether they had had an audiology evaluation, and the reason(s) for their decision. It turned out that there were no relationships between the stage of change at baseline and the individual’s help seeking behavior. However, given the short 2-week time frame, and the fact that almost all participants were in the precontemplation or contemplation stages of change, this is perhaps unsurprising. The reasons given by those who had had a hearing test said they did it to improve their quality of life, to find out their hearing status, because they were aware of hearing difficulties, and/or because it was recommended at the screening. The reasons for not getting a hearing test included having other health problems, having had no time to get a hearing test, and/or not noticing any hearing difficulties. Take note of these factors because they are relevant to some of the other health behavior models described below.

Laplante-Lévesque and colleagues used the TTM to examine hearing health behaviors in two studies. They examined the stage of change of 153 adults with acquired hearing loss who were seeking help for the first time (Laplante-Lévesque et al., 2013) and the stage of change of adults who failed an online hearing screening (Laplante-Lévesque et al., 2015) using a version of the University of Rhode Island Change Assessment (URICA; McConaughy et al., 1983) adapted for hearing loss. Based on the TTM, one would predict that the first-time help seekers would be at a more advanced stage of change than those who failed an online hearing screening. Indeed, this was the case: almost 80% of the first-time help seekers were in the

action stage of change, while in contrast, just 2.7% of the adults who had failed a hearing screening but had not sought help were in the action stage. The majority of the latter group was in the preparation stage (50%), with an additional 38% being in the contemplation stage. Laplante-Lévesque et al. (2013) also assessed whether the TTM had predictive validity. Using logistic regression analyses, they determined that 67% of the first-time help seekers who were in the action stage had taken up an intervention 6 months later, as compared with 33% of those who were in the preparation stage. Similar results were obtained by Saunders et al. (2016a), who found that 77.5% of their 182 first-time help seekers were in the action stage, with just 3.7% in the precontemplation stage and 16% in the contemplation stage. Furthermore, 6 months later, less than 15% of the individuals who were in the precontemplation stage at baseline had acquired hearing aids, while almost 80% of those in the action stage had acquired hearing aids (Saunders et al., 2016b).

Health Belief Model

The Health Belief Model (HBM) was developed in the 1950s by social psychologists in an attempt to understand why people were not taking up free health screenings to detect asymptomatic diseases. It was revised by Rosenstock (1966) to understand patients’ responses to illness symptoms and compliance with prescribed health care regimens and, later again, by Janz and Becker (1974). The current version of the model consists of six constructs that, together, aim to explain and predict health-related behaviors. The model postulates that the likelihood an individual will engage in a health behavior is determined by the extent to which he/she perceives a threat from a health condition. The perceived threat is affected by the interplay between the individual’s perception about the positive consequences of adopting the behavior (perceived benefits) and the negatives of doing so (perceived barriers), the individual’s assessment of the risk of having or acquiring the condition (perceived susceptibility), the seriousness of the consequences of the condition (perceived severity), the individual’s confidence in his/her ability to successfully adopt a behavior and of a successful outcome from the behavior (perceived self-efficacy), and by external influences (cues to action), such as symptoms, media communications, and information from a health care provider. Additionally, the HBM assumes that demographic and psychosocial variables such as

age and personality influence the beliefs and attitudes associated with each construct. Figure 8–2 illustrates the way in which the constructs are thought to interact, and Table 8–2 describes how each construct can be applied in the context of hearing help seeking. As with the TTM, critics have pointed out that the model relies on rational decision making with little acknowledgment of the role of automatic motivational processes such as emotions and habits that affect behavior.

The HBM has been examined relatively extensively in the context of hearing help seeking. In 1996, van den Brink and colleagues developed an attitude questionnaire that assessed perceived severity, perceived benefits, perceived barriers, and cues to action (which van den Brink et al. refer to as “perceived social norm”). They determined that individuals who did not seek help for their hearing loss had significantly lower perceived severity and perceived benefits scores and had received fewer cues to action than those who had sought help. In a similar vein, Saunders et al. (2013) developed the Hearing Behaviors Questionnaire (HBQ) to assess all six constructs of the HBM. The HBQ was completed by 223 individuals, of whom 127 reported they had a hearing loss, 55 reported normal hearing, and 51 were unsure about their hearing status. Seventy-five had recently had their hearing tested and 44 owned hearing aids. Using logistic regression analysis, the HBQ correctly classified 78% of the non-help seekers as non-help seekers and 68% of the help seekers as help seekers, as well as 85% of the hearing aid owners as hearing aid owners and 77% of nonowners as nonowners.

Meyer et al. (2014) used a variety of measures to assess the constructs of the HBM among individu-

als who had or had not sought help for their hearing. They determined the help seekers saw more potential benefit from hearing aids, reported more activity limitations and had higher perceived ability to manage hearing aids than the non-help seekers—reflecting the HBM constructs of perceived benefits, perceived severity, and perceived self-efficacy, respectively. As with the participants of the Milstein and Weinstein (2002) study, the non-help seekers reported barriers (e.g., lack of time, cost), negative cues to action (e.g., others had negative experiences with hearing aids and/or significant others less supportive of hearing aids) and low perceived benefit (e.g., hearing not bad enough) as reasons for not seeking help.

Schulz et al. (2016, 2017) recently conducted a study to determine whether adding a seventh construct, “perceived burden on communication partners,” would enhance the predictive capability of the HBM for hearing help seeking. Using the HBQ of Saunders et al. (2013), and data from 413 older adults, they determined that the help seekers ($n = 272$) had significantly lower perceived benefits and cues to action scores, and significantly higher barriers scores, than non-help seekers. They also determine that adding the “perceived burden on communication partners” construct improved the fit of the model from correctly classifying 66.7% of individuals to correctly classifying 72% of individuals. (*Editors’ note:* For further discussion of the role of family and friends on hearing help seeking and on AR outcomes, the reader is referred to Chapter 16 of this text.)

It is also worth noting that other studies investigating help seeking indirectly illustrate the applicability of the HBM. For instance, when individuals

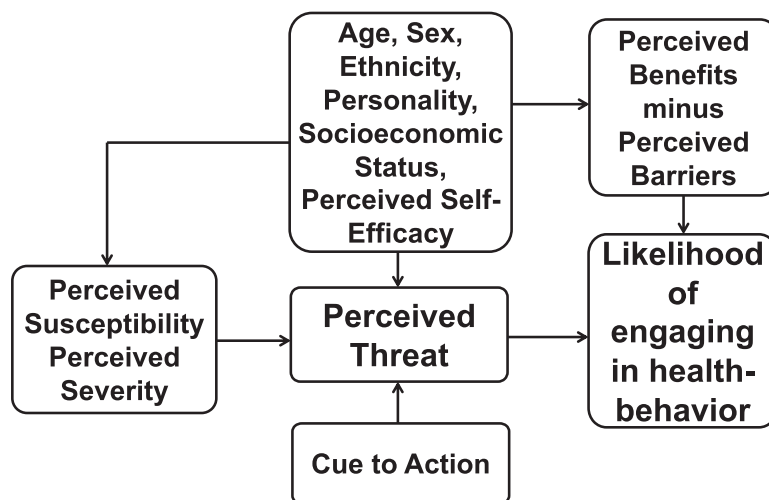


Figure 8–2. The Health Belief Model.

Table 8–2. Health Belief Model: Definition of Constructs and Influences on Help Seeking for Hearing

Definition	Attitude Likely to Lead to Positive Help-Seeking Behaviors	Application to Hearing Help-Seeking
Perceived Susceptibility		
Individual's perception of the risk of acquiring a condition	Feeling vulnerable to hearing loss	An individual who feels they are at risk of losing their hearing is likely to attribute hearing difficulties to their having a hearing loss, and thus would be likely to seek help for hearing loss.
Perceived Severity		
Individual's perception of how serious the consequences of a condition would be if it was acquired	Believing that a hearing loss would have negative consequences	An individual who considers hearing difficulties problematic would be motivated to address them.
Perceived Benefits		
Individual's perception of the positive outcomes of adopting a health-seeking behavior	Believing that hearing well is important, and that seeking help would have positive outcomes	An individual who considers hearing well to be important will be motivated to seek help if they have hearing difficulties.
Perceived Barriers		
Individual's perception of the negative outcome of adopting of a health behavior	Believing that it is not costly (socially, emotionally, financially, etc.) to get help for hearing loss	An individual who thinks seeking help for hearing loss is costly in some way is not motivated to seek help. Ultimately, the perceived benefits of seeking help must outweigh the perceived barriers to doing so.
Perceived Self-Efficacy		
Individual's perception of his/her ability to be successful if a health-seeking behavior is adopted	Believing one has the knowledge and abilities to acquire help for hearing loss and that there is an effective solution to the problem	An individual who believes they have the necessary ability to seek help for hearing loss will be more likely to do so than an individual who believes otherwise.
Cues to Action		
External influences on a health behavior (e.g., information from a health care provider, input from others, media, etc.)	Receiving prompts from others to get help for hearing loss	Individuals who have received cues to action are aware of the need for, and possible outcomes of, seeking help for hearing loss.

were queried about why they had not sought help for symptoms of hearing loss, their responses reflected constructs of the HBM: "My hearing isn't bad enough" (perceived severity), "Hearing aids won't help me" (perceived benefits), "Hearing aids are too expensive" (perceived barrier), "I've got other things to worry about" (perceived barriers), "Hearing aids

are embarrassing/stigmatizing" (perceived barriers), "I don't know where to go to get help" (self-efficacy), and "Others have had poor experiences with hearing aids" (cues to action). Likewise, reasons for seeking help included "hearing aids would improve quality of life/help maintain social participation" (perceived benefits), "I can't cope without help" (perceived

severity), “Hearing aids will decrease the burden on my friends/family” (cue to action), “Other people want me to get help for my hearing” (cue to action), or “Someday I will find a solution to my hearing loss” (self-efficacy) (e.g., Milstein & Weinstein 2002, Meyer et al., 2011; Rolfe & Gardner, 2016; Laplante-Lévesque et al., 2012; Fischer et al., 2011; Lockey et al., 2010; Southall et al., 2006, 2010; Schulz et al., 2017).

Self Determination Theory

Self-Determination Theory (SDT) (Deci & Ryan, 1985) proposes that individuals have three universal innate psychological needs—a desire to master the environment (referred to as *competence*), a desire to feel connected to others (referred to as *relatedness*), and the desire to have a sense of free will (known as *autonomy*)—and that motivation influences the extent to which these three needs are met. The model posits that people are more likely to engage in an activity when they have a sense of free will or of being in control of their actions, feel competent and capable, and feel connected with others and that their motivation for engaging in an activity is on a continuum from being intrinsic or autonomous (i.e., being self-motivated or motivated from within) to being extrinsic or introjected (being motivated by others to gain reward or avoid punishment). The model is illustrated in Figure 8–3. In the context of hearing help seeking, the model would suggest that individuals who value hearing well and thus are self-motivated to address hearing loss to achieve the personal goal

of hearing better, will be more likely to seek help and will have better outcomes than those who are seeking help to avoid family nagging or to assuage guilt. SDT has been applied to hearing help seeking and hearing aid adoption by Ridgway and colleagues (2015, 2016, 2017). They showed that autonomous motivation was positively associated with hearing aid satisfaction, and that individuals with higher autonomous motivation wanted hearing aids more, reported more hearing difficulties, and were younger than those with lower autonomous motivation. Conversely, those with higher extrinsic motivation were more often prompted by others to get a hearing test and had a greater desire for hearing aids than those with lower controlled motivation. Further, a logistic regression analysis showed that autonomous motivation, along with perceived difficulties and degree of hearing loss, were significant predictors of hearing aid adoption. These variables correctly classified 68.3% of the hearing aid adopters and 84% of the nonadopters.

Theory of Planned Behavior

The Theory of Planned Behavior (TPB) (Ajzen, 1991, 2002) focuses on intentions rather than motivations. It assumes that health behaviors are under volitional control (i.e., the individual chooses whether or not to engage in a behavior), and that a person’s intention to perform a behavior is the best predictor of whether or not they will do so. The person’s intention is influenced by factors that include attitudes toward the behavior, how they perceive others would

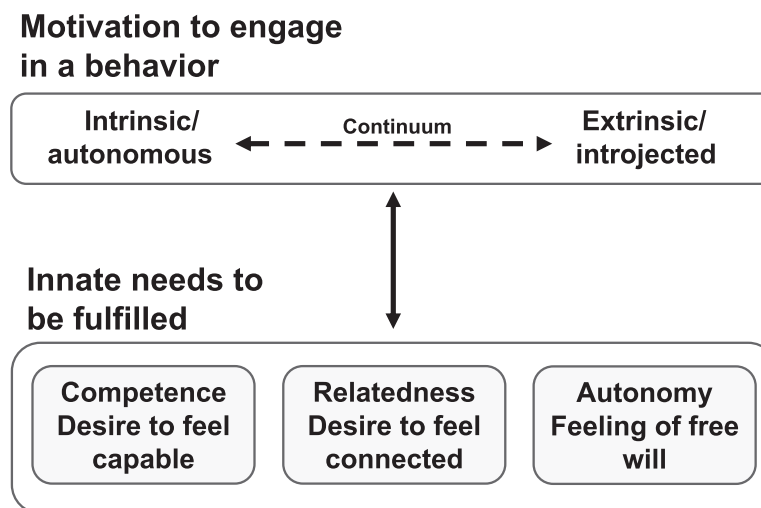


Figure 8–3. Self-Determination Theory.

expect them to act (subjective norms), and how well they think they can perform the behavior (perceived behavioral control). In turn, the person's attitude is influenced by what they think the consequences of performing the behavior might be (behavioral beliefs), while their subjective norms are influenced by their beliefs about how others would like them to behave (normative beliefs), and their perceived behavioral control is influenced by the control they have over the behavior and their confidence about doing it. Figure 8-4 illustrates the model.

As applied to hearing help seeking, it can be hypothesized that an individual who has a positive attitude toward hearing health and expects a positive outcome from seeking help, who perceives that others want them to seek help, who feels confident in their ability to seek help, and feels they can make their own decisions about doing so will have more intention to seek help than if the converse is true. Of course, for the intention to translate into behavior, the individual must also be capable of completing the behavior—a construct referred to as *actual behavioral control*.

Meister et al. (2014) developed a questionnaire to assess the constructs of the TPB as applied to hearing health behaviors and compared the responses of individuals at different stages in the help-seeking process. They reported that as people moved from the pre-help seeking stage to using hearing aids, the expectations of others (subjective norms) played a lesser role, while perceived behavioral control (ability to do the behavior) and intentions to do the behavior became greater. Arnold et al. (2019) also developed a TPB questionnaire. It differed from the questionnaire of Meister

et al. because it focused specifically on help seeking for hearing loss. Arnold et al. studied 407 older adults all of whom had failed a community hearing screening. Of the 407 who completed the questionnaire, 362 were successfully contacted 6 months after the hearing screening. Of these, 98 had had a hearing test, as recommended at the time of screening. The participants who had sought help had significantly higher intentions and perceived behavioral control, and more positive attitudes than those who did not seek help ($n = 264$), reflecting stronger intentions, feelings of being in control of decisions around help seeking, and more positive feelings toward having a hearing test than the non-help seekers.

Behavior Change Interventions

Theory-Based Clinical Interventions

From the above, at least two things should be clear. The first is that these models encompass many overlapping constructs, even though the construct is labeled differently model by model. For example, the construct of needing self-confidence in one's ability to complete a health behavior is present in all four models; the influence of others as a determinant of health behaviors is seen in the HBM, SDT, and TPB; and the process of weighing the pro and cons of taking up a behavior is seen in the TTM, HBM, and TPB.

The second observation is that the constructs are clearly applicable to help seeking for hearing loss;

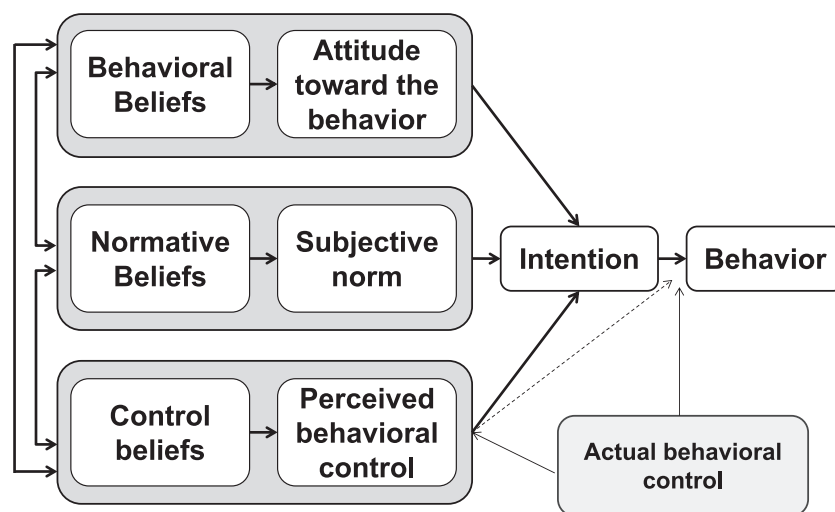


Figure 8-4. Theory of Planned Behavior.

thus, it makes sense to use them for direction and guidance when developing behavior change interventions. For example, if using the TTM as the theoretical basis of a behavior change intervention, the content of the intervention and how it would be delivered should presumably differ by stage. For individuals in the precontemplation stage, the focus would be on increasing awareness about the impacts of hearing loss and the role hearing plays in psychosocial well-being, while those in the preparation stage would need specific information about local audiology services. At the action stage, it would be important to facilitate appointment scheduling and clinic access, and at the maintenance stage, the focus should be on providing follow-up rehabilitation, support, and counseling. As a result, at each stage of change, the individual's self-efficacy would increase, and their decisional balance process would be more likely to result in progress through the stages of change. In a similar manner, the channel through which information and education is provided would, by necessity, differ. At the precontemplation and contemplation stages, the audiologist is unlikely to be the provider. Instead, the information would need to come from a different source such as a general health practitioner, the internet, or a public health campaign. On the other hand, at the action and maintenance stages, the responsibility for providing the intervention falls on the audiologist.

If one were to apply the framework of the HBM to behavior change strategies, an intervention could include education to ensure individuals have an accurate perception about the risk of hearing loss from age, noise exposure, etc. to address perceived susceptibility; access to simulations of hearing loss to address perceived severity; opportunities to experiment with hearing-assistive technologies to address perceived benefits; correction of misinformation about hearing aids to address (some) perceived barriers; information about how, where, and when to get a hearing test, and options for auditory rehabilitation to address perceived self-efficacy; and the use of primary care physicians and public health messages to promote awareness about hearing and hearing loss to address cues to action. Once again, some of this information would, by necessity, be provided via the internet or other publicly accessible fora, while other information would be provided in person by the audiologist.

Saunders et al. (2017) developed and evaluated a theoretically based intervention to promote help seeking for hearing loss that was based on the constructs of the three theoretical models—the TTM, HBM, and SDT—using data from Saunders et al. (2016a)

that showed that individuals who were less ready to seek help for their hearing loss (as assessed using the adapted URICA) perceived fewer benefits in obtaining hearing aids, had lower self-efficacy for doing so, and had encountered fewer cues to action than individuals who were more ready to seek help. In the same study, Saunders et al. showed that individuals with high URICA action scores (i.e., those making change) perceived more negative effects from their hearing loss (high perceived severity), had higher self-efficacy for seeking help, and had encountered more cues to action than individuals with lower action scores. The intervention, therefore, focused on increasing perceived benefits, perceived severity, perceived self-efficacy, and cues to stimulate action, as follows. While in the waiting room of a health care provider's office, the recipient of the intervention is provided with nine emotionally evocative photographic images that depict stigmatization, old age, isolation, peacefulness, stress, embarrassment, exhaustion, frustration, and happiness. These particular emotions were selected because they tend to be the triggers for hearing help seeking and/or intervention uptake (Barker et al., 2017; Claesen & Pryce, 2012; Preminger & Laplante-Lévesque, 2014). The recipient of the intervention is asked to view the images and reflect on whether they can identify with any of them when thinking about their hearing. During the face-to-face clinical interaction, the health care professional asks the recipient of the intervention which photographs (if any) he/she identified with and why. If the recipient reports a negative emotional impact of hearing loss (e.g., "I get mad when I can't hear my wife speaking" or "I feel lonely when I sit at dinner with my family"), the health care professional suggests the recipient has a full audiological evaluation and provides a list of local audiologists. The theoretical rationale behind the intervention is fourfold. First, by encouraging the recipient to consider the negative emotional effects of hearing difficulties, the perceived benefits for seeking help will increase. Second, by providing information about local audiology services, perceived self-efficacy for seeking help will increase; third, by having a health professional provide the intervention, there will be a cue to action. Finally, the use of emotionally evocative photographs will lead the recipient to the realization that hearing loss has negative effects, thus increasing autonomous motivation to seek help. The intervention was evaluated with 87 individuals who reported noticing hearing difficulties and yet had not sought help (Saunders et al., 2017). They were randomly assigned to either receive the intervention or no intervention. Six months later, participants were

contacted to see whether they had had their hearing tested within the prior 6 months. Twenty-nine percent (12 out of 41) of participants in the intervention group and 15.2% (7 out of 46) of participants in the control group had done so. Analyses showed that the odds ratio of having had a hearing test were 2.3 times greater for those who received the intervention than for those who did not, suggesting the intervention is somewhat effective at promoting help seeking for hearing difficulties.

Population-Based Approaches to Promoting Help Seeking for Hearing Loss

A wider-reaching approach to changing help seeking behavior would be to address the issue at a population level. Such an approach could be considered to be critical because of the 7- to 10-year time delay between hearing aid candidacy and hearing aid uptake (Simpson et al., 2018), because of the enormous economic and societal costs of untreated hearing loss, and because hearing tends to be undervalued at the individual level. To elaborate, a 2017 World Health Organization Report (World Health Organization, 2017) estimated the global burden of untreated hearing loss to range between \$750 billion and \$790 billion associated with health care, education, loss of productivity due to unemployment and premature retirement, and societal costs resulting from social isolation, communication difficulties, and stigma. That hearing is undervalued is illustrated by the knowledge that while primary care is an important entryway to almost all health care services, screening for hearing loss in primary care is uncommon—overall rates are only around 23% (Abrams & Khim, 2015; Wallhagen, 2010). Additionally, it is not uncommon for older patients to report that their primary care practitioners do not view hearing loss as a health-related issue or as a priority during office visits (Chou et al., 2011). Perhaps more alarmingly, even when hearing loss is raised as a concern, practitioners often minimize or negate it, and view hearing aids as stigmatizing or ineffective (Wallhagen, 2010). Further, older adults prioritize managing other health conditions over managing hearing loss (Barnett et al., 2017), and often dismiss hearing loss as a normal part of aging (Saunders et al., 2018).

Unfortunately, despite the negative consequences of untreated age-related hearing loss, a public health approach to rehabilitation for the aging population has remained underprioritized for a long time (Davis

et al., 2016; Reavis et al., 2017). This is unfortunate because public health approaches aim to provide the maximum benefit for the largest number of people (Davis et al., 2016). On a positive note, there are indications that priorities about hearing loss are changing around the world. In the U.S., the Institute of Medicine and the National Research Council hosted a “Hearing Loss and Healthy Aging” workshop that focused on determining how age-related hearing loss can be addressed as a public health issue (Institute of Medicine and National Research Council, 2014). This workshop led to a cascade of outcomes, including a President’s Council of Advisors in Science and Technology (2015) report recommending, among other things, that the FDA approve a class of hearing aids for over-the-counter (OTC) sale and, in turn, the OTC Act of 2017 (Warren & Grassley, 2017). In the U.K., the National Institute for Health and Care Excellence (NICE) recently released guidelines that aim to improve the quality of life for adults with hearing loss by advising health care staff on assessing hearing difficulties, managing earwax, and referring people for audiological or specialist assessment and management (NICE, 2018). Finally, a World Health Organization report of 2017 highlighted not only the monetary costs of untreated hearing loss but also the societal impacts and the cost-effectiveness of interventions. These reports are raising awareness about hearing loss in the population. This will hopefully change negative perceptions about hearing loss, increase the value individuals place on hearing well and, in turn, result in more older adults seeking help for hearing difficulties.

Public health approaches to interventions include legislation and taxation, mass media campaigns, social marketing, community programs, and point-of-sale promotions. These approaches have been successful with other conditions (NICE, 2007) and can also be applied to hearing help seeking as follows:

- Legislation and funding to mandate annual hearing screening for older adults could increase access to hearing health care in a similar manner to that which has been successfully implemented at the federal level for newborn hearing screening (Public Law No. 115-71). This law, in existence since 2005 and recently reauthorized until 2022, has resulted in more than 97% of newborns being screened for hearing loss within 1 month of birth; of those who did not pass the hearing screening, more than 71% were diagnosed as either having or not

having a hearing loss before 3 months of age (Centers for Disease Control and Prevention, 2017). A law mandating annual or biennial hearing screening for older adults could substantially raise awareness about hearing loss among patients and could result in providers taking age-related hearing loss more seriously.

- Mass media campaigns and social marketing can be used to enhance awareness about the negative ramifications of untreated hearing loss. While no particularly large campaigns have as yet been implemented, organizations such as the American Association of Retired Persons (AARP), Age U.K., and Canadian Association for Retired Persons (CARP), provide educational content on their websites about hearing loss and hearing health care. Possible marketing campaigns could be used to educate older adults about the negative effects of untreated hearing loss and about possible solutions and interventions, thus increasing perceived benefits, decreasing barriers, and raising self-efficacy for seeking help.
- Community programs can be used to address barriers to access and decrease health care disparities among various underserved populations. One such program is the Oyendo Bien (Hearing Well) program of Marrone and colleagues (Ingram et al., 2016; Marrone et al., 2017; Sánchez et al., 2017), for which community health workers (*promotoras*) are trained to work with members of their local Spanish-speaking community to identify signs of hearing loss and facilitate educational and peer support groups for individuals with hearing loss and family members. The program has shown positive changes in self-efficacy and stigma following participation in the outreach program. Other programs with similar goals exist, such as the “Walk and Talk for Your Life program,” located out of the University of British Columbia, designed to increase socialization and healthy lifestyles among seniors (Jutras et al., 2015), and the Baltimore HEARS Pilot Study (Nieman et al., 2017), which provides hearing screening, subsidized hearing devices, and communication education and counseling to community-dwelling individuals living in subsidized, independent housing to

low- and middle-income older adults in Baltimore, MD. To date, community programs have been implemented by independent groups and are relatively small. Large-scale government funding of such programs has the potential to have major impacts on promoting hearing help seeking.

Summary

This chapter summarized research illustrating the need to change hearing health behaviors so that more older adults with hearing loss seek auditory rehabilitation. You have learned why there is value in applying health psychology to understanding hearing health behaviors and the rationale behind several health behavior theories (i.e., Transtheoretical Model, Health Belief Model, Self-Determination Theory, and Theory of Planned Behavior). Perhaps more importantly, you have seen examples of how these theories can be used to understand and change hearing help seeking behaviors at the individual and public health levels. It is hopefully clear that we need to find a way to effectively change the status quo of a 7- to 10-year time delay between hearing aid candidacy and hearing aid uptake, and that applying knowledge and techniques from the fields of health psychology and public health has enormous potential to positively affect this.

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9

AUDIOLOGIC ASSESSMENT FOR IMPLANTABLE TECHNOLOGIES

Jaclyn B. Spitzer, Dean M. Mancuso, and Rachel A. Schepeler

Introduction

The subspecialty of implantable devices is in constant evolution. The changes that occur affect which devices are offered to potential candidates and evaluation methods to determine which treatment is appropriate. Our field is continually faced with a diversity of implantable hearing devices (IHDs) (de Souza, Roland, & Tucci, 2017) in varying stages of approval by the Food and Drug Administration (FDA), clinical trials, and off-label applications. In this chapter, “IHD” will refer to the full range of implantable and semi-implantable hearing devices, ranging from bone-anchored to middle ear implants to cochlear and brain stem implants. Audiologists must determine how to evaluate potential candidates for these devices and what methods to apply in an ongoing assessment of the devices’ safety and efficacy.

The first, and most obvious, question to be asked is “What is the purpose of assessment?” This seemingly simple question has several answers. The most direct objective in performing assessments for IHDs is to identify candidates that meet selection criteria. In this way, we can undertake proper counseling regarding the range of appropriate options. Based on finding that a candidate does, in fact, meet the selection criteria for a class of IHDs, we still have an ethical obligation (Berg, Herb, & Hurst, 2005; Hyde & Power, 2006) to provide information about competing tech-

nology and communication methods, including non-surgical options (Spitzer, Mancuso, & Cillum, 2013).

Alternative rehabilitation and educational paths are overtly discussed with parents of infants and children being evaluated for cochlear implants (CIs), for example, but such discussion must also take place when the candidate is an adult. It is widely acknowledged that we have this obligation while we are engaged in a clinical trial, but the requirement to present alternatives continues beyond an IHD’s approval by the FDA for clinical use. Even when the audiologist is an advocate for treatment with a specific IHD, accepting that that class of device is the current standard of care, there is still an obligation to present information about the possible benefits and limitations of alternatives.

The second objective of assessment for IHDs is to identify noncandidates and to counsel them and their families regarding alternatives. As the rehabilitation specialist on an implant team, the audiologist should formulate possible alternative recommendations, guided by knowledge of competing technology, counseling options, and rehabilitative planning. Periodic reexamination, review of needs, changes in selection criteria (Davids et al., 2007; Gifford et al., 2010), and emerging technologies, such as those for acoustic hearing preservation (Incerti, Ching, & Cowan, 2013) may eventually move a current noncandidate into the range of qualifying for an IHD when followed longitudinally. In this vein, Bibee, Gende, and Shohet

(2017) argue that the currently applied criteria for CI result in underreferral of persons who could potentially benefit, and that “traditional” CI candidates are underreferred, as are persons who meet expanded guidelines.

Another important purpose in carrying out a comprehensive evaluation for IHDs is to prepare the candidate and the implant team for postimplantation needs. With thorough assessment, rehabilitative planning or support services can be made available in anticipation of the implantee’s needs or limitations. For example, some studies have shown that vertigo or dizziness is not only a risk factor following cochlear implantation (Spitzer et al., 2015), but many times also presents as a preoperative complaint in implant candidates (Rah et al., 2016). For these individuals, vestibular rehabilitation may be indicated in an extended recovery period, and alerting the candidate to this possibility is essential to acceptance.

Assessment in the Historical Context

Assessment has been at the heart of demonstrating to governmental agencies (especially the FDA), the professional community, the Deaf community, and the general public that IHDs are safe and provide substantial benefit that outweighs the risks. Beginning in the 1970s, clinical trials with CIs received considerable media attention. Early work with the single-channel CI was criticized as having excessive reliance on anecdotal reporting of benefit. Thus, later studies, such as the randomized trial by the Department of Veterans Affairs (Cohen et al., 1991), needed to be more fastidious in producing tight pre- and postdesigns to document safety and efficacy.

Multichannel CI devices have undergone a stepwise progression in hardware and software design. With each iteration, the clinical trials have had rigorous, structured baseline testing with contemporary hearing aids and postoperative examination using reliable measures in controlled test environments. Many of the procedures that were used in the clinical trials of the 1980s and 1990s are still used in the post-FDA approval clinical milieu. As new IHDs are developed, with differing target patient populations and divergent capabilities, it is reasonable to project that the clinical protocols in use today will require modification to demonstrate that advanced features make a significant difference in patient outcomes. The CI battery for adults was revised nearly a decade ago to address the observation that many CI users demonstrated ceiling effects on the previous generation of

tests (Gifford et al., 2010; New Minimum Speech Test Battery, 2011). Further change can be anticipated relating to all types of IHDs. For other IHDs, there is also an increasing interest in speech perception measures in noise (Snapp et al., 2010).

Selection Criteria: Influence on Assessment Methods

In designing an evaluation protocol, documentation of preintervention function is needed to serve as a baseline. The assessment methods should be clearly related to determining whether the selection criteria are met. The dimensions to be assessed should address areas known to be affected by the intervention with an IHD, as well as parameters that may have an indirect impact. The bone-anchored hearing implant or osseointegrated device (also known as a bone-anchored cochlear stimulator, implantable prosthetic device, or bone-anchored implant; formerly known as a bone-anchored hearing aid) will be used as an illustration of how the evaluation components need to address selection criteria and form a relevant baseline against which comparisons can be made effectively. For the purposes of this chapter, we will refer to this category of implanted devices as osseointegrated devices (OIDs). It should also be noted OIDs include both percutaneous and transcutaneous stimulation, as well as a new category of nonimplantable devices that deliver a bone-conducted signal across the skin.

As seen in Table 9–1, outlining the evaluation protocol for OIDs, multiple dimensions of patient characteristics and performance are examined. The medical criteria relate to factors intrinsic to the patient, such as age or diagnosis (e.g., congenital aural atresia), or historical information, such as prior surgery for cholesteatoma, in addition to physical findings. The otologist will initiate the requests for computed axial tomography (CAT or CT scan) or magnetic resonance imaging (MRI), dependent on the underlying pathology and device option being explored, to visualize the anatomy to determine if a condition exists that precludes implantation or that may alter the surgical approach.

At various points during the evaluation protocol, the otologist may seek input or clearance from the primary care physician or specialists, especially in regard to the candidate’s ability to undergo surgery and the impact of health conditions on the healing process and implications for choice of transcutaneous or percutaneous devices. For example, a patient

Table 9–1. What Constitutes a Basic OID Selection Protocol?

Category of Assessment	Methods/Tests	Considerations
Medical/surgical evaluation	• History • Physical examination • Radiologic workup	• Age over 5 years (applies to U.S.) • Anatomic appropriateness for IHD • Determining contraindications to surgery, often in consultation with primary care physician and/or specialists
Unaided evaluation under headphones/insert earphones/soundfield	• Basic comprehensive audiologic examination • Tympanometry and acoustic reflexes	• Where there are limitations in administration level by air conduction testing, consider administration of speech stimuli via bone oscillator • Importance of using recorded stimuli to ensure ability to make pre- and post- comparisons
Aided evaluation in soundfield	• Warble tone thresholds • Spondee thresholds • Speech recognition in quiet • Speech recognition noise with the location of noise source chosen based on side to be implanted • Localization evaluation, if equipment is available	• Using personal amplification • Using competitive technology • Conventional hearing aid, if appropriate • OID with headband for conductive or mixed HL, especially atresia • CROS or BICROS for SSD application
Nonaudiologic criteria	• Direct observation of manipulation of device and batteries • Handicap assessment • Interview and counseling by both otologist and audiologist and/or other team members, with conference on impressions • Interview with OID user • Psychological screening if necessary • Visual information access • Communication effectiveness, as evaluated by speech-language pathologist or auditory-verbal therapist	• Adequate manual dexterity or support system for management of device • Assistance with visual inspection of the abutment and surrounding skin interface • Support system and availability determined through interview of patient and/or significant other • Examine motivation for surgery and for adjustment to novel devices or stimulation • Realistic expectations • Provides realistic feedback about life with an OID • May be able to address unrealistic expectations or barriers to proceeding with intervention • May require consultation by optometry or ophthalmology • Ability to communicate preoperatively, including use of visual information via lipreading and gestures • Utilization of repair strategies

with diabetes may pose a problem in terms of healing of incisions, or a percutaneous OID may prevent the healing of the skin around the abutment. For a user of a transcutaneous device, pressure between internal and external components should be monitored periodically to avoid necrosis (Chen, Mancuso, & Lalwani, 2017). In such a condition, the physician's monitoring of the patient may be more vigilant than in a less complex case.

Audiologically, the scenarios that are appropriate for OID candidates may entail bilateral conductive or mixed hearing loss, unilateral conductive hearing loss, or unilateral profound sensorineural hearing loss, also called "single-sided deafness (SSD)" (Wazen et al., 2003). Attention is directed to the levels of bone-conducted responses, which cannot exceed 45 dB Hearing Level (HL) for the standard head-level OID devices and 55 dB HL for power models. In cases of bone conduction responses up to 65 dB HL, a body processor is on the market that can accommodate that level of conductive component. In terms of performance with a hearing aid or OID on a headband, there are no explicit candidacy requirements for the extent of aided threshold improvement or speech recognition performance. OID is an example, albeit a rarity in the field of IHDs, of a situation in which we can evaluate the candidate with a simulator (usually a power model OID on a headband) to obtain preoperative data. This also affords the patient an opportunity to listen to a sound that simulates the postoperative outcome; while there may be some underestimation of the gain to be obtained, the candidate nonetheless can formulate an impression of the sound quality to be achieved with an OID.

The audiologic procedures of interest for OIDs are to demonstrate improved auditory sensitivity, speech recognition in quiet and noise, and localization ability. Thus, the preoperative baseline can include conditions that pose communicative difficulty for the candidate's level and configuration of hearing loss. It is also generally accepted in evaluation of IHDs, particularly with middle ear implants and CIs, that evaluation should be accomplished after the candidate has had sufficient experience with competitive, less invasive technology to permit the assessment to reflect maximal performance. Thus, a trial with a Contralateral Routing of Signal (CROS) hearing aid or equivalent, such as a power aid fitting to promote hearing crossover, may precede an evaluation of a candidate for the SSD application of an OID or CI. The audiologist must ensure the trial is with a device that truly is competitive technology, and neither outdated as some candidates' older personal hearing

aids might be, nor suboptimal as in an underpowered in-the-canal hearing aid for a severe to profound hearing loss.

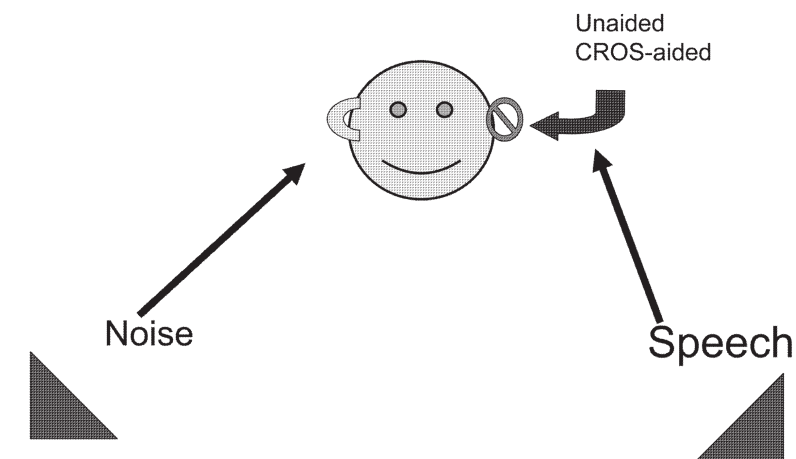
It is worthwhile to explore responses to noise in differing juxtapositions to the listener. As in Figure 9-1A, unaided speech recognition testing or measurement with an adaptive procedure such as the Hearing-in-Noise Test (HINT) (Nilsson, Soli, & Sullivan, 1994), the BKB Sentences (Bench, Kowal, & Bamford, 1979; Etymotic Research, 2005), AzBio Sentences (Spahr & Dorman, 2004), or the QuickSIN™ (Etymotic Research, 2001) can document for both the SSD patient and the clinician the disadvantage experienced in conditions when speech originates on the poorer side and noise is on the better side (Bosman et al., 2011). The results of this condition are then compared to those in the aided condition, as in Figure 9-1B, when the IHD is in place. Such testing can provide reinforcing data to the potential IHD user as an illustration of the advantage to be gained (see Case Study 9-1, below). In addition to audiometric measures, Ortega et al. (2011) have demonstrated cross-validation of perceived improvement in noise using questionnaires.

The audiologist also has a significant role in determining whether the candidate can contend with the IHD's practical application. Verification that the patient has adequate manual dexterity or a support system for management of the day-to-day maintenance of the device can be accomplished by watching the patient handle a demonstration device and batteries. In the case of OIDs, the audiologist and/or otologist must determine how the patient will be able to care for the abutment site and surrounding skin interface, or whether the patient will need assistance with visual inspection of the area.

Psychological Dimension and Assessment of Handicap

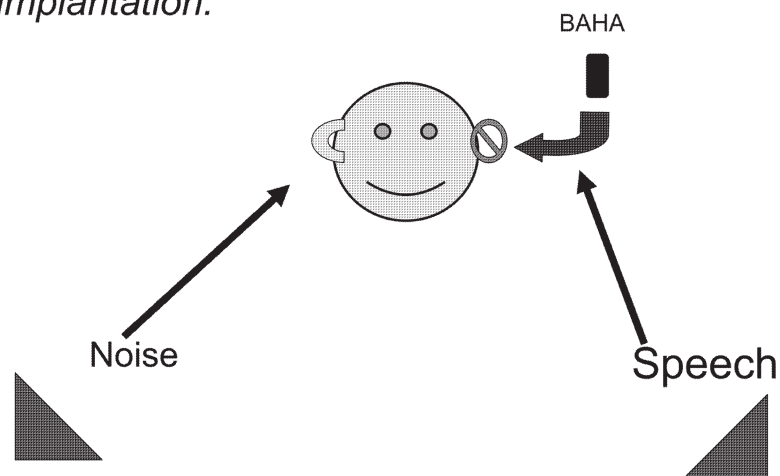
Although involvement of a psychologist is highly variable in current implant team settings, some form of psychological screening or full evaluation is generally acknowledged as necessary to examine such intrinsic factors as psychological status and acceptance of implantation whenever questions arise. The general understanding is that a candidate must have realistic expectations about outcomes from the IHD. The method for determining that someone has realistic expectations is also highly variable, and may be based on a team consensus. Extrinsic factors that may have an impact on the candidate's success with

Listening in noise may be one of the key challenges for the unilateral hearer.



A

Speech perception testing in noise may reveal the advantage offered by implantation.



B

Figure 9-1. A. Speech recognition performance: unaided. **B.** Speech recognition performance: aided. *Note.* Figures 9-1A and 9-1B illustrate the use of speech in noise testing. In comparing speech recognition performance in the unaided (A) and aided (B) conditions with noise on the side of the better hearing ear, it is possible to demonstrate the advantage offered in the Single-Sided Deafness (SSD) scenario to a potential BAHA user. This arrangement of noise and speech may be useful in demonstrating benefit to users of other IHDs.

an IHD should also be addressed, including family support, social network, vocational demands, and avocational interests. In some teams, the latter issues

are addressed by a social worker, but these social dimensions often arise in counseling sessions with the audiologist as well. When psychologists are involved

in the initial assessment, it is reasonable for them to follow the implantee over time to assess the impact of implantation (Knutson et al., 1998). Individuals with hearing impairment (and their spouses) often experience greater incidence of social isolation, depression, suspiciousness, and loneliness than the general population (Knutson, Johnson, & Murray, 2006), and may experience greater psychological burden that could make the auditory rehabilitation process more difficult following implantation (Brüggemann et al., 2017).

Adjustment to hearing loss and its impact on social, emotional, and vocational aspects can also be assessed formally using self-assessment and hearing handicap scales. Selecting the appropriate scale for the level of hearing loss is critical. Thus, using the Hearing Handicap Inventory for Adults (HHIA) (Newman, Weinstein, Jacobson, & Hug, 1990, 1991) for adults under 60 years old, or the Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982; Weinstein & Ventry, 1983) for adults over 60 years old, would be reasonable for mild to moderate levels of hearing loss. Other scales, such as the Screening Test for Hearing Problems (Demorest et al., 2011), should also be considered depending on the depth of information sought. When the loss is severe to profound, however, a scale should be selected that is aimed at the communicative challenges associated with that level of loss, such as the Performance Inventory for Severe and Profound Loss (PIPSL) (Owens & Raggio, 1988). Specialized scales have been developed to assess the impact of cochlear implantation (e.g., the Nijmegen Cochlear Implantation Questionnaire [NCIQ], Hinderink, Krabbe, & van den Broek, 2000). In the setting of SSD, a scale that provides insight into problems with localization and speech in noise is suggested (e.g., Speech Spatial Qualities [SSQ], Gatehouse & Noble, 2004; Noble et al., 2013). When selecting a handicap scale, it is important to keep test-retest reliability in mind, since high reliability is required (Weinstein, Spitzer, & Ventry, 1986) if the measurement is to be repeated postintervention and periodically thereafter; in this way, changes measured can be attributed to the impact of intervention, rather than variability inherent to the handicap scale. (*Editors' note:* For further information about use of handicap scales, the reader is referred to Chapter 6 in this text.)

Reviewing the handicap scale responses with the candidate and communication partner(s) is a suitable method to launch discussions of adjustment to loss, coping mechanisms, need for changes in or implementation of repair strategies, and how the postop-

erative function may be significantly improved on an auditory basis but still require use of strategies to maximize successful communication. These discussions in a family context can reveal the level of support available to the potential implantee and what obstacles must be addressed in the family milieu.

Other Nonauditory Factors

It is important to determine if there are any visual limitations that may impinge on the ability to access visual communication information. Sometimes, visual deficits become apparent during meetings with other team members. It is often in the formal assessment of baseline communication abilities, as carried out by the speech-language pathologist, audiologist, or auditory-verbal therapist, that limitations in accessing visual communication are discovered. When such visual deficits are identified, it is necessary to obtain consultation with an optometrist to improve or correct refraction or with an ophthalmologist to address any remediable ocular problem. Although good vision is not a prerequisite for IHDs, and in fact persons with low vision or blindness have been implanted with OID or CIs (Daneshi & Hassanzadeh, 2007; Saeed, Ramsden & Axon, 1998; Damen et al., 2006), maximization of communication skills through every available modality should be part of the preoperative phase.

Current Audiologic Methods

There is no universally accepted test battery for IHDs. If a device is in clinical trials, its assessment must follow a rigorous protocol to document safety and efficacy. After FDA approval in the United States, considerable discretion is applied, with implant centers seeking an economical balance (Cheng & Niparko, 1999) between scientific rigor and clinical feasibility. The pressure to streamline assessment is a reality that each practitioner must face.

A test battery should be device oriented so that assessment reflects areas in which a particular IHD will or may provide benefit. For example, Gifford et al. (2010) described a revision of a cochlear implant battery, which was adapted in 2011. The new battery added elements to challenge the prospective (and eventual implant) candidate in competing noise or speech babble. A brief summary of the current audiologic benefits of IHDs appears in Table 9-2.

Table 9–2. Sample Audiologic Benefits With Current Generation of Implantable Hearing Devices (IHDs)

Device	Application	Sample Audiologic Benefit	Representative Citation Documenting Benefit
OID	Bilateral conductive or mixed HL ^a	Improved threshold sensitivity	Tjellstrom et al. (1981); Wazen, Caruso, & Tjellstrom (1998)
		Speech recognition in quiet	Spitzer, Ghossaini, & Wazen (2002); Hol et al. (2005)
	Unilateral conductive or mixed HL	Improved threshold sensitivity	Wazen et al. (2001); Snik, Mylanus, & Cremers (2002)
		Speech recognition in quiet	Yuen et al. (2009)
	Single-sided deafness	Improved speech recognition in noise	Wazen et al. (2003); Snapp et al. (2010)
Middle ear implantable device (IMEHD) ^b	Mild to severe sloping SNHL ^c	Improved threshold sensitivity	Luetje et al. (2002); Uziel et al. (2003); Jenkins et al. (2004)
	Conductive or mixed HL ^d	Improved speech recognition in quiet	Luetje et al. (2002) Re: Conductive HL: Venail et al. (2007); Bassim & Fayad (2010)
		Reduction in occlusion effect	Luetje et al. (2002)
Cochlear implant	Severe to profound SNHL; auditory neuropathy; SSD	Improved threshold sensitivity	Connell & Balkany (2006); Re: Auditory neuropathy: Shallop et al. (2001); Peterson et al. (2003) Re: SSD: Arndt et al. (2011); van Zon et al. (2015)
		Improved speech recognition in quiet	Balkany et al. (2007)
		Improved speech recognition in noise	
		Reduced hearing handicap	Spitzer, Kessler, & Bromberg (1992); Zheng et al. (2011)
Auditory brainstem implant	In patients with neurofibromatosis II, neural HL or HL affecting eighth nerve and extracanalicular space; in cases of failed cochlear implantation ⁴	Improved threshold sensitivity	Otto et al. (2002) Re: Failed cochlear implant: Colletti et al. (2004)
		Support to lipreading	Schwartz et al. (2003)
		Limited open-set speech recognition	Colletti et al. (2009) Otto et al. (2002)

^aHL (hearing loss).^bThe only FDA-approved devices at the time of writing are the Envoy Medical Esteem, Med-El Vibrant Soundbridge, Otologics MET, Otologics Carina, Ototronix Maxum, and Soundtec (inactive).^cSNHL (sensorineural hearing loss).^dNot an FDA-approved application at the time of writing.

Preoperative Assessment

Threshold and speech recognition improvement are documented in a straightforward manner, using measures with which audiologists are very familiar; however, the standard battery of tests should be “sharpened” by consistent use of recorded speech stimuli to control test-retest variability influences and routinely obtaining the maximum speech recognition scores (phonetically balanced maximum performance or PB_{max}). It is also worthwhile to remember that the preoperative examination is a medicolegal document, which substantiates the need for intervention, conformance with accepted selection criteria or, where there are deviations from such criteria, the rationales or aspects of performance that motivated a surgical treatment.

In the selection protocol, there is a focus on speech perception performance that (1) establishes preoperative function; and (2) is used to determine conformance with published guidelines. For example, the current criteria for adult candidacy for a traditional CI include speech recognition for sentence material no better than 50% in the ear to be implanted, and no better than 60% in the opposite ear. Gifford et al. (2010) suggested that these criteria be modified to no better than 40% correct on a monosyllabic test

in the ear to be implanted. In the case of implantable middle ear hearing devices (IMEHDs), this criterion is often stated as a wide range, such as 20% to 80% correct for monosyllables, and is sometimes not stated explicitly (Magnan, Manrique, Dillier, Snik, & Hausler, 2005).

Hybrid CIs entail the application of electrical stimulation and acoustic stimulation in the same ear. This is one form of electroacoustic stimulation (EAS). Hybrid criteria rely heavily on performance in noise as the potential candidate, someone who has residual hearing in the low to mid frequencies, is likely to perform better on speech recognition tasks in quiet than those being assessed for traditional CI devices (Woodson et al., 2010). The addition of electrical stimulation is intended to supplement acoustic cues. Thus, a hybrid candidate, having residual hearing, risks losing that function upon introduction of an electrode into the cochlea. Determination of candidacy for EAS devices is heavily reliant on demonstrating a speech perception breakdown in various noise conditions.

The following case study demonstrates that a single test protocol, in this case to determine candidacy for an OID, may not be an appropriate approach. Both flexibility and creativity must be employed to demonstrate the potential benefit of an IHD to a given patient.

Case Study 9–1. Applying an Evaluation Protocol

Patient S. M., a 20-year-old male, was referred to our center for evaluation of candidacy for OID implantation.

Clinical History

The patient’s mother reported that he was born full-term via forceps delivery. The patient presented with left congenital aural atresia and stenosis. The right pinna appeared normal; however, the patient’s mother reported an unspecified abnormality of the ossicular chain in the right ear. The patient took synthroid for hypothyroidism. He was also a longtime wearer of a digital behind-the-ear-style hearing aid in the right ear.

Summary of Audiologic Findings

A Type A tympanogram was obtained in the right ear (–75 daPa peak pressure). Ipsilateral acoustic reflexes were absent in the right ear at equipment limits. Immittance testing could not be performed for the left ear due to the aforementioned stenosis. As seen in Figure 9–2, the results showed a moderate to profound mixed hearing loss, bilaterally. Due to the masking dilemma, it was not possible with currently available equipment to determine the exact, ear-specific bone-conduction thresholds for all test frequencies in each ear. However, unmasked bone conduction thresholds reveal that at least one ear has sufficient bone conduction

thresholds to consider OID implantation. Additionally, though the masked air-conduction thresholds indicate a profound impairment, the unmasked thresholds were used in judging candidacy, as this reflected his hearing status in the real world without the presence of masking noise. The speech recognition thresholds corroborated pure-tone averages bilaterally.

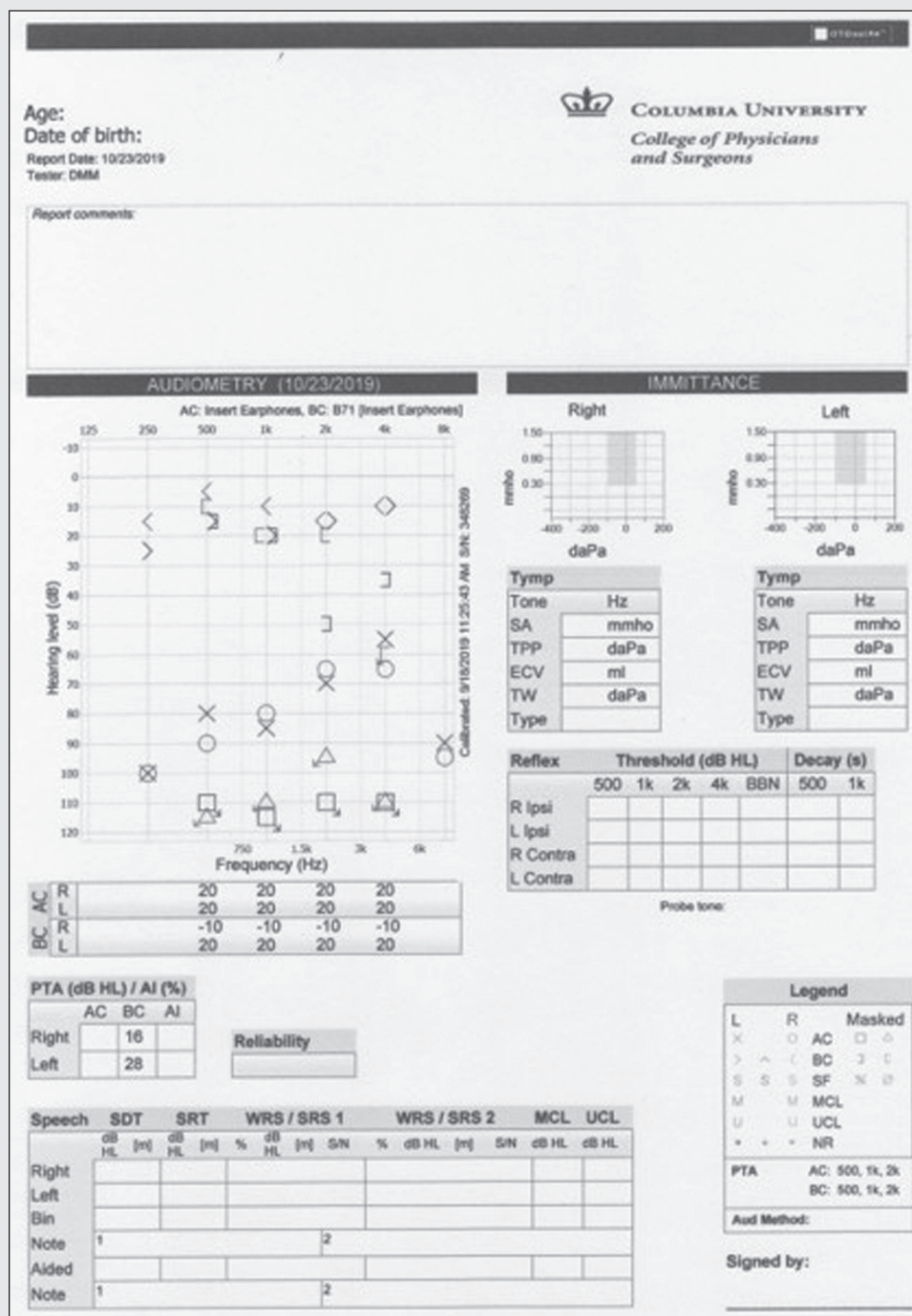


Figure 9-2. Case Study 1: Audiologic findings for S. M., the 20-year-old described in the case study.

BAHA Evaluation

Aided soundfield evaluation was completed. For the majority of the evaluation, the patient wore a Baha™ demo OID on the left side. The left side was chosen because the patient was unable to wear a conventional hearing aid on that side due to the aforementioned stenosis/aural atresia. Pure-tone thresholds were obtained, along with speech testing using a 50-word CNC list and HINT sentences. All speech testing was performed using recorded materials, and stimuli were presented at 70 dB HL, with the patient seated in front of the loudspeaker at 90 degrees azimuth. Aided hearing was within normal limits for pure tones from 500 Hz to 2000 Hz. The CNC phonemes correct score was 89%; the HINT sentences score was 98% in quiet, 0% in noise with the Baha™ in “omni” mode, and 74% in noise (+10 dB S/N ratio) with the directional microphone engaged. The patient also used his personal high-end digital power BTE during the evaluation, as it was likely he would continue to do so regardless of whether or not he ultimately elected to pursue OID implantation. The HINT score in quiet for the hearing aid only was 98% phonemes correct, whereas the HINT in noise (+10 dB S/N ratio) with the hearing aid in directional mode was 18%. The combined performance of the Baha™ and the hearing aid in noise (also +10 dB S/N ratio) was 73% of words correctly identified. The patient also expressed a favorable opinion of the sound quality of the Baha™.

Summary of Candidacy Findings

Results of this evaluation indicated that the patient was an excellent OID candidate. That is, this patient was unable to use a conventional aid due to the stenosis on the left side. Audiologists measured substantial improvement in speech perception in noise over the hearing aid alone, and the patient’s reaction to the device was positive. It was likely that the patient would continue to use a hearing aid on the side opposite the Baha™. Speculatively, this may yield improved sound localization relative to using the Baha™ only. Additionally, due to the presence of bilateral conductive pathology, the patient was also a candidate for future evaluation to consider bilateral Baha™ implantation.

Treatment Plan

Appropriate manufacturer’s literature was provided to the patient, and he was referred back to his otologist to further discuss the medical and surgical considerations associated with OID implantation.

Discussion

This case illustrates that the clinician cannot be content with the utilization of a cookie-cutter approach when evaluating OID candidacy. Frequently, as in this case, the patient’s individual needs and issues will provide the impetus to modify the typical evaluation protocol. In this case, assessment of the patient’s performance using his hearing aid was added to the usual protocol, and the inability to determine ear-specific masked bone-conduction thresholds for the majority of frequencies tested was considered but not found to be an impediment to the patient’s candidacy. Other possible scenarios that may require modification of the “standard” protocol include patients with SSD, patients with congenital versus acquired hearing loss, duration of hearing loss, the patient’s performance in noise, the patient’s listening environments and lifestyle issues, the patient’s ability to perform the whole test battery (i.e., younger children who may have reduced attention span). During the OID evaluation, as with virtually any other audiological test battery, flexibility and insight on the part of the clinician are paramount to successful outcome.

In the past, assessment using nonspeech stimuli was common, especially in the evaluation of persons with profound hearing loss. The Minimal Auditory Capability (MAC) battery (Owens, Kessler, Raggio, & Schubert, 1985) included subtests with identification of environmental sounds and discrimination of voice versus noise. As we observed greater benefits to cochlear implantation than initially anticipated, the focus shifted to assessing the perception of speech stimuli. The thinking is that such measures provide too low a target for postimplant comparisons with the current, successful generation of devices. Because success from cochlear implantation has continued to exceed expectations, candidacy has relaxed to include both individuals with higher levels of auditory function preoperatively and prelinguistically impaired adult candidates (Waltzman & Cohen, 1999). There is a need for test materials to reflect that same diversity, especially if one purpose of the preoperative assessment is to have it serve as a baseline against which to compare postoperative function. Potential implantees with limited auditory experience are expected to perform poorly pre- and postoperatively on open-set monosyllabic or sentence material despite some improvements that can be documented. To demonstrate that implantation improved some dimensions of auditory perception, tasks in addition to those focusing on open-set speech recognition, such as those included in the MAC battery, should be considered. Indeed, Shafiro et al. (2008) showed that environmental sound perception abilities correlated strongly with speech perception abilities, suggesting that there may be a role in the CI assessment for environmental sound testing. Since that time, additional modifications have been suggested, such as the inclusion of a nonlinguistic measure of spectral resolution (Drennan et al., 2014; Gifford et al., 2014) and objective and subjective measures of complex listening abilities. For individuals performing at a higher level preoperatively, more complex materials, such as music, might need to be considered. Finally, for candidates who do not speak English, nonlinguistic measures such as spectrally rippled noise can provide another means to assess auditory function.

Auditory Electrophysiology

Auditory electrophysiologic tests have an essential role in preoperative assessment of hearing when the candidate is young or otherwise difficult to test behaviorally due to comorbid conditions. Auditory Brainstem Response (ABR) and/or Auditory Steady State Response (ASSR) are commonly used to estimate the configuration, type, and degree of hear-

ing loss (Spraggs, Burton, & Graham, 1994) and, in combination with otoacoustic emissions (OAEs), to identify the profile referred to as auditory neuropathy/auditory dyssynchrony spectrum disorder (Shallop, Peterson, Facer, & Fabry, 2001). In an unselected sample of people with congenital hearing loss, Cross, Stephens, Francis, Hourihan, and Reardon (1999) demonstrated that a functional overlay or nonorganic hearing loss may be encountered. For adults who can reliably perform behavioral audiometric testing, the electrophysiological tests are not commonly included in the preoperative assessment. However, in the interest of differential diagnosis and/or ruling out a functional component in an adult IHD candidate, ABR/ASSR may contribute to confirm hearing loss degree.

For cochlear implantation, promontory electrical stimulation via a transtympanic electrode was historically evaluated as a means for distinguishing between sensory and neural etiologies for individuals with profound hearing loss (House & Brackmann, 1974). As candidacy has relaxed to include individuals with residual hearing, and with high resolution imaging available to assess the physical structure of the auditory system, there is less concern about implanting an ear that will not be stimuable. Even so, outcomes with implantation remain variable, and promontory stimulation has continued to be explored as a means for estimating residual neural function and predicting outcomes. Although there is some indication that preoperative measures of auditory sensation with promontory stimulation are correlated with postoperative speech-perception scores (Kileny et al., 1991), this procedure is not used widely for preoperative assessment.

Vestibular

Electro- or videonystagmography (E/VNG), vestibular-evoked myogenic potentials (VEMPs), the head impulse test (HIT), and the Dizziness Handicap Inventory (DHI) (Jacobson & Newman, 1990) are examples of commonly used vestibular assessments used by some cochlear implant teams to document pre- and postoperative vestibular function (Abouzaïd et al., 2017; Brey et al., 1995). Given the time-intensive nature of the audiological portion of the preoperative assessment, and that vestibular function does not have strong predictive value for CI speech perception performance outcomes (Chen, Shipp, Al-Abidi, Ng, & Nedzelski, 2001), vestibular testing is often eliminated from the protocol. However, not only do candidates sometimes report vestibular symptoms preoperatively, but also balance problems sometimes accompany insertion of the IHD electrode as it entails invasion of the cochlea or round window (RW) or

placement of a vibrating prosthesis in the RW (see Abouzayd et al., 2017 for review of cochlear implantation; see Klein, Nardelli, & Stafinski, 2013 regarding dizziness in Vibrant Soundbridge). The frequency of postoperative symptoms reported in the literature ranges from 6% to 80% and the timing can vary from immediate to delayed. Patients over age 60 years appear more at risk (Abouzayd et al., 2017). This suggests that if vestibular assessment is eliminated from the general IHD protocol, there may be value in retaining the test for older candidates or those with a history of dizziness to have both baseline information on those candidates and insight into the mechanism for postimplantation recovery of function.

Self-Reporting in IHD Evaluation

A variety of self-reported measures have been used to evaluate candidates for IHDs. In CI candidates, assessment has employed questionnaires in hearing (Spitzer, Kessler, & Bromberg, 1992; Vermeire et al., 2006), tinnitus (Tyler, 1995), dizziness (Enticott et al., 2006; Jacobson & Newman, 1990), handicap scales, balance confidence (Enticott et al., 2006), quality of life assessment (Faber & Grøntved, 2000; Spitzer et al., 1992; Vermeire et al., 2006; Wanscher, Faber, & Grøntved, 2006), and employment and job satisfaction (Fazel & Gray, 2007). (*Editors' note:* For a review of self-reports of the impact of hearing impairment and tinnitus, the reader is referred to Chapter 6 in this text.) As is true in other applications, self-reported measures may provide a starting point for counseling based on responses to the scales or questionnaires.

In addition to handicap and quality of life assessments, other scales have been used extensively with OID and to a lesser extent with middle ear implants, such as the Med-El Vibrant Soundbridge and Otologics MET. These other scales include the Abbreviated Profile of Hearing Aid Benefit (APHAB) (Cox & Alexander, 1995), Glasgow Hearing Aid Benefit Profile (Kemper & Holmes, 2004), and a sound quality rating (Gabrielson, Hagerman, Bech-Kristensen, & Lundberg, 1990). Table 9-3 summarizes the outline of audiologic methods for evaluation of IHDs in adults and illustrates the linkage with various dimensions of anticipated benefit.

Implantee Contact

Many IHD teams routinely set up a meeting or contact between a current IHD user and an IHD candidate. When a prospective implantee has the opportunity to meet with a current user, there are information and emotional exchanges that cannot be conveyed by the

professionals on the team. Such a meeting can promote realistic expectations and reduce fears about implantation. If an appropriate experienced user is not available from within the evaluating clinic, it is possible to provide a contact from an external pool. The CI manufacturers, recognizing the potential benefit to persons undergoing evaluation, have facilitated meetings with CI users through associations of implantees, such as the Bionic Ear Association (Advanced Bionics) and online resources, such as the Nucleus Forum, the Cochlear™ Community (a website for CI and Baha™ users by Cochlear Americas), and Hearing Companions (Med-El Corporation). There are parallel opportunities to meet previous implantees with bone-conduction stimulators and middle ear implants as well. These resources allow users to express their viewpoints and provide opinions, in much the same way as self-help organizations do.

The Special Case of Single-Sided Deafness (SSD)

The handicap imposed by Single-Sided Deafness (SSD) represents a unique problem for the implant team. SSD has been recognized in the audiologic otologic communities as imposing communicative, localization, speech-in-noise, and psychologic functional challenges for the person with this hearing loss picture (Spitzer, Korres, & Lalwani, 2015). Additionally, suppression of tinnitus after CI has also been reported (Amoodi et al., 2011; Arts et al., 2012). While the use of OID for improvement of communication, especially in noise, is well documented, cochlear implantation remains controversial due to technical issues in fitting and, though generally favorable, some mixed outcome reports. The situation is reflected in the results of a survey of otolaryngologists' practices by the American Neurotological Society (Carlsen et al., 2018) in which CIs were provided to adults with asymmetrical hearing (in which at least one ear was better than the established performance criteria cutoff, representing 61% of respondents) and in single-sided deafness (37, 46% of respondents), illustrating a trend toward off-label implantation in SSD.

For example, Sladen et al. (2014) presented the preliminary findings of an SSD study at two CI centers. Their data demonstrated improvements in speech recognition for monosyllables and sentences in quiet in pre- and postimplantation comparisons. There was also a trend toward improved speech recognition in noise, but it did not reach statistical significance. Among those who reported preoperative

Table 9–3. Outline of Audiologic Methods for Evaluation of Implantable Hearing Devices (IHDs) in Adults: Linkage With Dimensions of Anticipated Benefit

Test Phase	Dimension to Be Assessed	Relevant Devices	Tests in Current Usage
Preoperative baseline	Description of residual hearing sensitivity, nature of loss, speech recognition ability	All	Comprehensive audiologic evaluation under headphones/insert phones: ▪ pure-tone air- and bone-conduction thresholds ▪ Speech recognition for NU-6 words
		OID	Speech recognition via bone-conduction oscillator if air-conduction headphones cannot deliver sufficient sensation level
	Performance with competitive technology in soundfield	All	Warble tone thresholds Speech recognition: ▪ NU-6 words ▪ Consonant-nucleus-consonant (CNC) monosyllables ▪ Hearing-in-Noise Test (HINT) sentences in quiet ▪ HINT sentences in S/N = +10 dB ▪ HINT sentences in S/N = +5 Db ▪ AzBio sentences ▪ BKB sentences ▪ Handicap measurement ▪ Satisfaction/quality of life questionnaires ▪ For congenitally impaired candidates: ▪ Minimal Auditory Capabilities battery ▪ Monosyllable-Trochee-Spondee test
		OID; CI for SSD	HINT paradigm to document dB advantage in noise (Nilsson, Soli, & Sullivan, 1994); Quick SIN (Snapp & Telischi, 2008; Snapp et al., 2010)
		IMEHDs	Speech Perception-in-Noise (SPIN) sentences
		Severe to profound hearing loss; CI	ENG/VNG VEMP HIT DHI
Perioperative	Integrity of electrode	Cochlear implant (CI)	Electrical impedance test
	Confirmation of neural excitation		▪ Electrically evoked Compound Action Potential (ECAP; AKA Neural Response Telemetry, Neural Response Imaging, Auditory Response Telemetry) ▪ Electrically evoked Auditory Brainstem Response (ABR)
	Estimate of programming	CI	▪ Electrically evoked Compound Action Potential (ECAP; AKA Neural Response Telemetry, Neural Response Imaging, Auditory Response Telemetry) ▪ Electrically evoked Stapedial Reflex Thresholds (ESRTs) ▪ Electrically evoked Auditory Brainstem Response (ABR)

continues

Table 9–3. *continued*

Test Phase	Dimension to Be Assessed	Relevant Devices	Tests in Current Usage
Perioperative <i>continued</i>	Confirmation of placement of electrode array/paddle	Auditory brainstem implant (ABI)	Electrically evoked Auditory Brainstem Response (ABR)
Postoperative, 3 months, 6 months, 12 months, annually thereafter	Description of residual hearing sensitivity, nature of loss, speech recognition ability ^a	All	Comprehensive audiologic evaluation under headphones/insert phones: ▪ pure-tone air- and bone-conduction thresholds ▪ speech recognition for NU-6 words
	Performance with implanted technology in sound field	All	Warble tone thresholds Speech recognition: ▪ NU-6 words ▪ Consonant-nucleus-consonant (CNC) monosyllables ▪ Hearing-in-Noise Test (HINT) sentences in quiet ▪ HINT sentences in S/N = +10 dB ▪ HINT sentences in S/N = +5 dB ▪ AzBio sentences ▪ BKB sentences ▪ Handicap measurement ▪ Satisfaction/quality of life questionnaires
	Electroacoustic assessment	CI plus hearing aid	Comparable to above
	Localization	Bilateral or SSD applications of OID, CIs, EAS	Requires soundfield array

^aPostoperatively and at intervals required for confirmation of impact of surgery and standard of care.

tinnitus, 12 out of 13 reported improvement in annoyance and the 13th indicated no change.

As illustrated in the following case study, the

options for a patient with SSD should be based on a comprehensive evaluation and patient preferences, among other historical and physical parameters.

Case Study 9–2. Illustrating SSD Evaluation

Patient W. H., a 45-year-old female, was referred for evaluation of candidacy for OID implantation as treatment for SSD of her left ear.

Clinical History

The patient reported that she suffered a total hearing loss in her left ear s/p CPA tumor surgery 4 years prior. She has never tried amplification, and was not interested in trying a CROS hearing aid system.

Summary of Audiologic Findings

A Type A tympanogram was obtained in the right ear (−15 daPa peak pressure) and the left ear (−27 daPa peak pressure). Ipsilateral acoustic reflexes were present in the right ear and absent in the left ear. Contralateral acoustic reflexes were present stimulus right, and absent stimulus left. As shown in Figure 9–3, the results indicate a profound SNHL in the left ear and normal hearing in the right. The speech recognition thresholds corroborated pure-tone averages bilaterally.

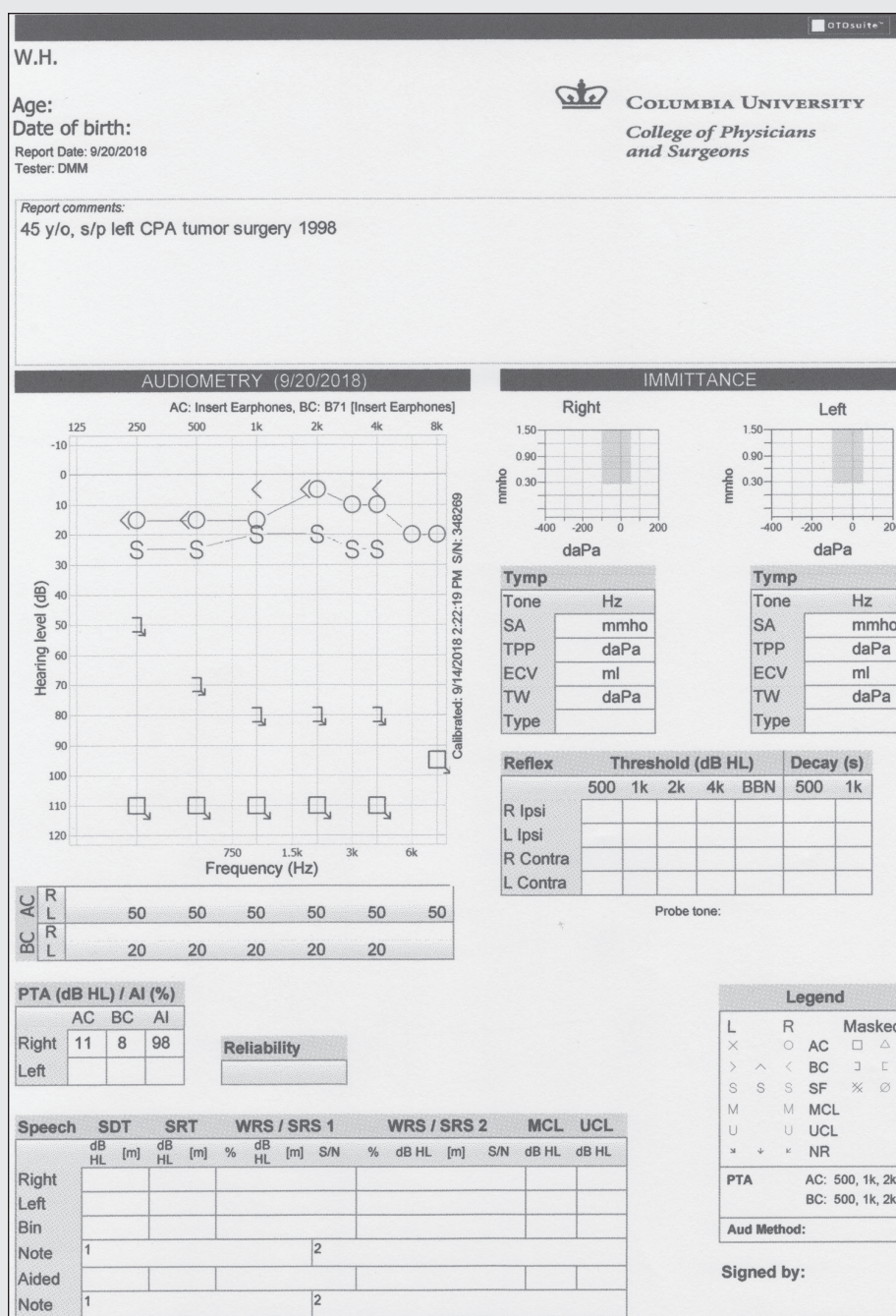


Figure 9–3. Case Study 2: Audiologic findings for W. H., the 45-year-old described in the case study.

BAHA SSD Evaluation

Aided soundfield evaluation was completed. For the evaluation, the patient wore a Baha5™ on the left side. The device was held in place with a test band supplied by the manufacturer. The right ear was occluded with an E-A-R earplug, and, additionally, muffled with a Sennheiser HDA-300 circumaural headphone. Masking was not used in the right ear as this would interfere with the patient's ability to hear via the intended interaural crossover on that side. Aided pure-tone thresholds were obtained, along with speech testing using a 50-word CNC list, AZ-BIO sentences, and Bamford-Kowal-Bench Sentences in Noise (BKB-SIN). All speech testing was performed using recorded stimuli presented at 55 dB HL in soundfield, with the patient seated in front of the loudspeaker at 90 degrees azimuth. Aided hearing was within normal limits for pure tones from 250 to 4000 Hz. The CNC phonemes correct score was 93%. The AZ-BIO sentences score was 97% with the Baha5™ (+5 dB S/N ratio) with the D-mic engaged and the digital noise reduction activated. BKB-SIN testing revealed a 4.5 dB SNR loss, which is nearly comparable to normally hearing individuals. The patient expressed a favorable opinion of the subjective sound quality of the Baha5™.

Summary of Candidacy Findings

Results of this evaluation indicated that the patient was an excellent candidate for OID implantation. She was able to demonstrate substantial benefit for speech perception in noise, and her reaction to the sound quality of the device was positive.

Treatment Plan

The patient was given applicable manufacturer's literature and was directed to various online resources, including the manufacturer's website and Baha™ user's groups. She was referred back to her otologist to further discuss the medical and surgical considerations associated with implantation.

Discussion

While other treatments exist (e.g., CROS hearing aid, cochlear implantation in some cases), this case illustrates that OIDs such as the Baha 5™ can be successfully used to treat patients suffering from SSD.

Perioperative Testing

Although the role of the audiologist in perioperative testing for OIDs and IMEHDs is still evolving, there are well-established and expanding roles for cochlear and auditory brain stem implantation.

Cochlear Implants

The most familiar perioperative measurements for cochlear implantation include electrode impedance; electrically evoked compound action potentials (ECAP) from cranial nerve VIII (AKA neural response telemetry [NRT]) (King, Polak, Hodges, Payne, &

Telischi, 2006; Potts, Skinner, Gotter, Strube, & Brenner, 2007); neural response imaging (NRI) (Caner, Olgun, Gultekin, & Balaban, 2007); auditory response telemetry (ART) (Alvarez et al., 2010); electrically evoked stapedial reflexes (ESR) (Battmer, Laszig, & Lehnhardt, 1990; Buckler, Dawson, & Overstreet, 2003); and electrically evoked auditory brain stem responses (EABR). Testing the impedance of the just-inserted electrode array provides information about device integrity and the local environment (e.g., high impedance or open circuits can be observed in instances of extracochlear electrode position, but also when air bubbles are present on the electrode surface. Extremely low impedance in cases of short circuits suggests abnor-

mal coupling across electrodes and a decrease in the number of independent stimulation sites). The measurements involving the neural pathway (e.g., ECAP, ESR, EABR) confirm a stimuable peripheral auditory system. In some instances, threshold measurements permit a gross estimate of the stimulation levels used for programming. Very rarely is an EABR performed in the operating room for adults.

For difficult-to-test individuals, the information obtained during surgery is at times invaluable for planning how to proceed with programming. For cooperative adults, except in rare instances of device failure at the time of surgery, the benefit gleaned from this standard set of measurements is more limited compared to the time and expense associated with performing these measurements in person. As such, many clinics have begun moving away from in-person intraoperative monitoring as a standard practice for adults. Remote monitoring makes performing these procedures more time efficient and cost effective for the audiologist and thus feasible to perform as a standard of care for all patients, including adults (Shapiro et al., 2008).

Expanding Roles for Perioperative Measurements. One challenge associated with cochlear implantation is optimal placement of the array with the least amount of trauma. In addition to the potential structural damage to the cochlea, another risk is that of the tip folding back on itself (Briggs et al., 2011; Zuniga et al., 2017; McJunkin et al., 2018). Imaging can be used perioperatively to provide the surgeon an estimate of electrode position following insertion. If films are not available, audiologists can measure the ECAP within a spread of excitation paradigm or evaluate the pattern of an impedance matrix for evidence of tip foldover (Grolman et al., 2008; Zuniga et al., 2017). If a tip foldover is identified at the time of surgery, the surgeon has the opportunity to reposition the electrode array, which is the ideal solution. If the electrode cannot be repositioned, the information gleaned at the time of surgery can be useful as it allows the audiologist to plan which electrodes might need to be deactivated in the program.

As cochlear implant candidacy has continued to expand to include individuals with more residual hearing, the role of perioperative assessment has also expanded to include physiological measures that can provide real-time feedback to the surgeon during electrode insertion (Campbell et al., 2016; Harris et al., 2017a, 2017b). In implant recipients with residual hearing, acoustically evoked electrocochleography can be recorded from the cochlear implant electrodes,

thus providing a way to monitor cochlear function. Although perioperative electrocochleography (ECoG) is not performed globally as a standard of care, these measurements are being explored by research centers as a way to refine the surgical approach, minimize trauma, and ultimately improve postoperative outcomes.

Auditory Brain Stem Implant

Nowhere is perioperative testing more important than to aid in the placement of an auditory brain stem implant (ABI) at the cochlear nucleus. The ECAP (arising from the cochlear nucleus) or the EABR is used to confirm stimulation of the auditory system (Waring, 1996; Mandala et al., 2014). Monitoring of neighboring cranial nerves is also performed to assist in placement, with the goal of minimizing the risk of nonauditory percepts and maximizing auditory percepts.

Value of Pre- and Post-Testing

Although we may be treating a candidate with the IHD that is currently regarded as the standard of care, as CIs are considered for profound SNHL, continued evaluation in a time-structured protocol is important. Again, using OID as an example, follow-up testing is necessary to: (1) document benefit of actual implant performance, which may exceed that with the headband simulation; (2) establish a baseline of implanted function against which future assessments will be compared; (3) provide feedback to the IHD user about performance; and (4) follow up annually, ideally, to detect alterations in the underlying condition and the impact of OID use.

Recognizing that upgrades in performance are to be anticipated with both hardware and software improvements in IHDs, we must establish a data reference that permits critical examination of advances in the field and alters our expectations of performance. The materials must be of sufficient difficulty that a large portion of implantees do not achieve very high scores, thus avoiding ceiling effects (Gifford et al., 2008). For example, we can review outcomes recently reported regarding the Nucleus Freedom implant and processor. For this device, Balkany et al. (2007) reported mean scores for CNC words of 57%, HINT sentences in quiet of 78%, and HINT sentences in noise of 64%. In a similar vein, Gifford et al. (2010) assessed a variety of implant device users and reported mean scores of 67% for the CNC words. Thus, our expectations increase for performance in both quiet and noise.

Assessment of Localization and IHDs

Evaluation of localization with OID has focused on postimplantation performance (Bosman, Snik, van der Pouw, & Mylanus, 2001; Wazen, Ghossaini, Spitzer, & Kuller, 2005). Bosman et al. (2001) demonstrated that bilateral BAHAs resulted in localization ability. In addition, Snik, Mylanus, and Cremers (2002) reported that, at least for patients with adventitious unilateral conductive hearing loss, OID permitted localization, whereas the performance of congenitally impaired persons was erratic or poor.

Wazen et al. (2005) pursued responses on questionnaires by several SSD patients that they believed their localization ability had improved while using OID. The study was undertaken to evaluate the impact of using an OID on localization performance and to quantify the deficit in localization encountered by persons with profound SNHL on one side and normal hearing in the better ear. The method entailed presentation of narrow bands at 500 Hz and 3000 Hz via speakers separated by 45 degrees of arc. Randomized trials of five presentations per speaker from each of eight speakers yielded 40 stimuli per frequency band. The performance of normal hearing controls was compared to SSD subjects. An unaided trial preceded an OID-aided trial. Accuracy did not exceed chance for most SSD subjects unaided and accuracy did not improve when wearing the OID.

A number of authors have demonstrated that bilateral cochlear implantation improves the implantee's ability to determine the direction of a sound source (Grantham, Ashmead, Ricketts, Labadie, & Haynes, 2007; Neuman, Haravon, Sislian, & Waltzman, 2007). Grantham et al. demonstrated that subjects with CIs were able to localize sounds only in the bilateral implant condition. Participants were performing with excellent accuracy at 5 months poststimulation, and showed the ability to learn to apply cues so that poor performers improved in accuracy with additional listening experience. Mosnier et al. (2009) reported that the ability to localize sound was most apparent at 12 months postimplantation. Preece (2010) reviewed the literature and concluded that "the majority of adults with bilateral implants are much better at horizontal localization with two implants compared with one, but their abilities do not reach those of normal hearing listeners" (p. 37). In this vein, Zheng, Koehnke, and Besing (2017) demonstrated that bilateral CI users had significantly poorer performance on localization tasks when challenged by the combined effects of noise and reverberation.

At this time, counseling for bilateral implantation, for CI or OID, acknowledges that audiologists anticipate improvement in localization and speech perception in noise but lack methods that can determine that such abilities are likely for a particular candidate. Based on both OID and CI experience, the person with an adventitious loss is more likely to develop good or possibly normal accuracy in localization tasks, and experience with bilateral stimulation will likely result in further improvements in this ability.

Developing New Procedures

We are at a significant juncture in assessment of potential IHD candidates and in documentation of benefit postintervention. Several criticisms may be leveled at current tests and methods, including ceiling effects, lack of real-world challenges, and a need for assessments of sound quality, especially as it affects music perception.

Ceiling effects are reflected in reports of exceptionally high performance by each successive generation of IHDs. For example, Staller, Arndt, and Brimacombe (2000) reported speech recognition performance in open-set sentences of 79.6% in quiet, in users of the Nucleus 24 Contour electrode and ear-level speech processor. Other investigators have reported similar outcomes for the other two FDA-approved cochlear implants (Hamzavi, Franz, Baumgartner, & Gstottner, 2001; Shapiro, Green, Bromberg, Gomoilin, & Waltzman, 2000; Valimaa & Sorri, 2000). The range of scores includes high performers who achieve 100% or very high scores early in their CI experience, implying that audiologists cannot accurately assess continued growth in perception as a consequence of experience and/or training. Thus, there is a need to develop materials that will present greater challenges in CI assessment. Real-world challenges might include introduction of highly variable speech babble competition, simulation of reverberation or reverberation plus noise, and elevated level of linguistic sophistication. Such ecologically valid stimuli might be applicable for the "star" performers who find the current materials and methods insufficiently difficult. In addition, when an implantee complains that speech comprehension is poor in large or echo-y venues, testing in reverberation is feasible without special equipment (Spitzer et al., 2015). Given our diverse population, there is also a glaring lack of non-English standardized materials to be used in the United States.

Dorman and Gifford (2017) presented their combined data on testing in adverse listening conditions, with ramifications for both test procedures as well as the impact of a variety of instrument-related features. Their evaluation entailed complex listening environments with multiple, diffuse noise sources (simulating restaurant and cocktail party environments). They examined the impact of bimodal fittings (i.e., electroacoustic), beamforming and remote microphones, hearing preservation (contralaterally to the CI or ipsilaterally), and bilateral CIs. They also evaluated the additive effect of audiovisual information over auditory-only conditions. Their findings support the use of remote microphones in noise, with beamforming and visual supplementation as important additional factors supporting speech understanding in complex environments. Further, bilateral CIs and utilization of residual hearing also resulted in better performance in noise when compared to other conditions. The conditions employed in this report differ from standard techniques seen in many clinics and should be considered for expansion of and IHD battery to augment the challenges in pre- and postevaluation.

When asked about listening habits and enjoyment of music, 52% of a sample of postlinguistically deaf adults stated that they enjoyed listening to music postimplantation, although music was less enjoyable than it was preimplantation (Lassaletta et al., 2007). This finding reinforces earlier reports by Gfeller and coworkers (Gfeller et al., 2000a; Gfeller, Witt, Woodward, Mehr, & Knutson, 2000b), in which CI users indicated an overall decrease in musical enjoyment compared to the remembered preimplant condition. These findings have added impetus to development of processing strategies to increase naturalness of sound and improve music appreciation. Recently, investigators have examined whether reengineering the music itself to reduce its complexity is an effective means to mitigate the limitations of CI signal processing (Kohlberg et al., 2015). Reduction in the number of musical instruments playing simultaneously (Kohlberg et al., 2016), reverberation time (Certo et al., 2014), and reduction of harmonic series (Nemer et al., 2016) are among the parameters of the musical listening experience being investigated. (*Note:* For additional information on music perception and CIs, see Plant [2014] in the second edition of this text.)

Despite the recognition of the importance of music enjoyment to IHD users (Buckler, Dawson, Lisbona, & Zimmerman-Phillips, 2006; Plant, 2009), including CI recipients, the application of new tests of musical appreciation (Brockmeier et al., 2011; Fitzger-

ald et al., 2006; Spitzer & Mancuso, 2006; Spitzer, Mancuso, & Cheng, 2008) and perceptual skills (Nimmons et al., 2008) has been limited. Questionnaires regarding musical appreciation have not yet been broadly utilized (Looi & She, 2010; Mirza et al., 2003). Studies with these measures in CI recipients using advanced processing strategies, incorporating fine structure of signals, continue.

Later latency auditory evoked potentials have been explored as a means to document alterations in processing following intervention with cochlear implantation (Bauer, Sharma, Martin, & Dorman, 2006; Burdo, Razza, di Berardino, & Tognola, 2006; Eggermont & Ponton, 2003; Ponton, Eggermont, Waring, & Masuda, 1996; Sharma, Dorman, & Spahr, 2002a, 2002b). For instance, the latency for N₁-P₂ components has been observed to decrease postimplantation in both children and adults, although the effect is more dramatic in children (Burdo et al., 2006). These changes provide physiological evidence of neuroplasticity that might underlie behavioral changes. Although the use of late potentials is not common in clinical practice, the measure continues to be explored as a way to evaluate an adult IHD user's adjustment to his/her device and to understand the wide variety of outcomes encountered in clinical practice (Purdy & Kelly, 2016).

Another promising area for evaluation of the brain's accommodation to sensory deprivation is functional imaging, including functional magnetic resonance imaging (fMRI) and activated positron emission tomography (activated PET). Although fMRI may be an unsuitable evaluation post-some IHDs' implantation, as a preoperative measure it may provide insight into how the auditory cortex is functioning in relation to various nonauditory stimulations. Activated PET has been used in a variety of studies (Catalán-Ahumada et al., 1993; Ito et al., 2004; Lee et al., 2001; Lee et al., 2003; Roland, Tobey, & Devous, 2001) as an index of the impact of CI use. PET offers the possibility of studying the impact of reintroduction of stimulation after sensory deprivation and of observing brain changes in the auditory cortex and association areas. For example, Ito et al. (2004) evaluated eight postlingually deafened CI users within the first 2 months post-CI initial stimulation. An abnormal pattern of activation was obtained in comparison to a normal listener control group. In the CI subjects, when stimulated by a 1000-Hz tone burst, strong and broad activation was obtained in the ipsilateral primary auditory cortex, with a weaker activation in the contralateral primary auditory cortex; in the control group, activation was in the contralateral

primary auditory cortex only. When stimulated by word stimuli, activation was observed in the superior frontomedian cortex (supplementary motor area and cingulate gyri) in the CI users, with no such finding in the normal listeners. Activation was also reduced in the immediate association areas. Evidence of plasticity was seen in activity in the periphery of association cortex. Thus, Ito et al. demonstrated both abnormal activation patterns and plasticity by mobilization of association cortex during the early phases of adjustment to electrical stimulation.

Tobey et al. (2005) described the impact of drug therapy in conjunction with 2 months of audiologic rehabilitation (AR) on cerebral blood flow. In their line of research, Single Photon Emission Computed Tomography (SPECT) findings in auditory primary and association areas were diminished in CI users compared to normal listeners. The authors evaluated the impact of pharmacological enhancement of rehabilitation using either amphetamine or placebo during intensive auditory training. An enhancement effect was documented in pharmacologically assisted therapy. These intriguing findings provoke the questions of how to maximize behavioral training methods in IHD users. An exciting new technique, functional near infrared spectroscopy (fNIRS) (Chenet et al., 2016; Lawler et al., 2015; Saliba et al., 2016) makes use of hemodynamic changes related to neuronal activation, as a response to sensory stimulation and has been applied to evaluation of brain response to CI stimulation. As fNIRS is entirely noninvasive and entails no magnetic equipment, it may be a promising method for evaluation of CI users of all ages (see Anderson, Lazard, & Hartley, 2017, for a review).

Conclusion

As is clear from this chapter, the clinician's creativity is an element that cannot be undervalued in the development of protocols and test materials for assessment of candidates for IHDs, and measurement of the impact of intervention. While it is important to adhere to a protocol to describe a possible candidate's auditory function and challenges, it is also vital to recognize when a standardized approach fails to provide sufficient insight. We do need to document that a candidate meets current selection criteria, but often the variations in an individual patient's need and lifestyle, as well as clinical profile, necessitate flexibility in assessment.

The continued evolution in technology must be accompanied by evolution in assessment to capture the dramatic changes experienced by the hearing-impaired patients making use of continued enhancements in implantable devices. The growing demand for improved performance in noise or with nonspeech signals, such as music, will require that evaluation routinely include these aspects in the battery of measures for both candidacy and postoperative function.

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10

HEARING REHABILITATION RECONSIDERED THROUGH THE LENS OF SOCIAL RELATIONSHIPS

Gurjit Singh and Stefan Launer

Introduction

Literature and cinema are replete with examples of the lone wolf: a character who prefers solitude over company and who remains unavailable in both presence and emotion. Despite the romantic characteristics of the lone wolf, humans are a pack species who form social collectives to survive and prosper. Moreover, it is exactly those individuals who remain socially engaged who tend to thrive as they age. The academic recognition of the role of others for personal well-being was brought to wider attention in a landmark review of epidemiological studies of social isolation (House, Landis, & Umberson, 1988). This work made the fascinating observation that social integration, social isolation, and social support are significant risk factors for both morbidity (the state of being diseased or unhealthy) and mortality (the incidence of death in a population). Further still, the finding that stunned researchers and media alike was not only the breadth of health conditions influenced by social factors, but also the magnitude of risk associated with impaired social relationships. Poor social relationships are as strong a risk factor for morbidity and mortality as other, more well-known risk factors such as smoking, obesity, sedentary lifestyles, and high blood pressure,

an observation confirmed over decades of research (Berkman & Glass, 2000; Berkman et al., 2004; Berkman & Syme, 1979; Cacioppo & Hawkley, 2003; Holt-Lunstad, Smith, & Layton, 2010; Seeman, 1996; Smith & Christakis, 2008; Uchino, 2006; Waldron, Hughes, & Brooks, 1996).

For hundreds of millions of people globally, disabling hearing loss compromises the ability to participate fully in society. It is estimated that in 2018, there were 466 million people with disabling hearing loss globally, with the number projected to increase to 630 million in 12 years. Hearing loss is considered to be the third-most common chronic disease in human (Vos et al., 2017). Hearing loss is further estimated to influence negatively the global economy by 750 billion international dollars each year (World Health Organization, 2018). Left untreated, hearing loss is associated with negative consequences in economic, physical, cognitive, psychosocial, emotional, and behavioral domains of one's life (Dalton et al., 2003; Lin, 2011).

Although there are potentially many reasons why an individual wishes to address a loss of hearing (e.g., environmental awareness, communication, enjoyment of music, etc.), it is typically the case that the motivation for better hearing is rooted in improving communication, which by definition refers to a

social relationship. This observation further suggests that hearing can be considered a “social sense.” Given the critical role of social relationships for mortality, morbidity, and quality of life (García, Banegas, Pérez-Regadera, Cabrera, & Rodríguez-Artalejo, 2005), this chapter is designed to provide a high-level overview of research on hearing loss and hearing rehabilitation through the lens of social relationships. Specifically, we set out to:

1. Define terms and describe concepts commonly used in hearing research to investigate social relationships;
2. review measures commonly used in hearing research that assess social relationships, as well as provide a framework to assist readers to select measures for either clinical or research purposes;
3. review research investigating the effect of hearing loss on social relationships, with a particular emphasis on specific domains of social relationships;
4. review research investigating the effects of hearing rehabilitation on social relationships;
5. provide descriptions of concepts relevant to social influence, as well as an overview of how the behaviors of others may influence the thoughts, attitudes, and behaviors of a person with hearing loss;
6. review research investigating how social relationships influence help seeking and hearing rehabilitation uptake; and
7. provide several recommendations for improving clinical practice.

Part I: Definition of Terms

Over the past four decades, there have been important developments in our understanding of the role of social relationships in audiology. However, a review of the literature observes that the use of terminology is not used consistently within hearing research. How one researcher defines and operationalizes social isolation is often different than when investigated by others. For example, social isolation has been measured by the use of composite measures (i.e., the combination of questions inquiring about marital status, size of friendship networks, etc.), by the use of formal questionnaires coupled with measures of social belonging (Ramage-Morin, 2016), availability of social support (Mick, Kawachi, & Lin, 2014), and by the use of a single question (i.e., “Do you often feel lonely?”; Dawes et al., 2015). While such approaches may be

appropriate given a particular research goal, a consequence is that comparisons between studies become more challenging. Hence, in Part I of this chapter, we set out to provide clarification by first defining commonly used terms.

Social Isolation, Social Integration, and Social Ties

Researchers have used a number of terms to describe effects on interpersonal or social well-being, including *social isolation*, *social ties*, and *social integration*. These terms are loosely defined, used interchangeably, and depending on the study, operationalized differently (Valtorta, Kanaan, Gilbody, & Hanratty, 2016). Of the terms just described, *social isolation* is likely most commonly used in the audiology literature. Cacioppo and Cacioppo (2014) distinguish between *objective social isolation*, or the extent to which an individual is socially isolated, and *subjective social isolation*, or the extent to which a person feels socially isolated. In scientific communication, it is often necessary to make use of umbrella terms such as *social isolation* that encompass subdomains of the phenomena. For the remainder of the chapter, we use the term *social relationships* as an umbrella term that encompasses domains of interpersonal well-being (described below). It should also be mentioned that there can be conceptual overlap between/among various terms.

Loneliness

Throughout the academic literature, researchers conflate the terms *social isolation* and *loneliness* (Valtorta et al., 2016). Whereas social isolation tends to be used as a “catch-all” term, and whereas objective social isolation can be empirically quantified using markers such as living alone, having few social network ties, and/or having infrequent social contact, loneliness is the perception of social isolation or the subjective experience of being lonely. Loneliness can stem from a mismatch between desired and actual social network size, depth, or frequency of contact.

Loneliness is a complex emotion, but critically it is dependent on cognitive appraisals of experience. Theoretical descriptions of loneliness suggest an affective component as well as a social-cognitive component (Weiss, 1973). Affective loneliness represents the emotional response arising from a lack of social bonding especially with specific others, and results in felt emotions such as sadness, fear, and emptiness. Social-

cognitive loneliness stems from evaluations of the self, such as when an individual perceives a relationship as unsatisfactory or a social network as unavailable, as well as from social comparisons such as that which occurs when individuals recognize a lack of bonding as compared to others with well-formed social bonds. Experiences of social-cognitive loneliness result in feelings such as exclusion, marginality, and boredom (Peplau & Perlman, 1982; Weiss, 1973).

Social Support

In contrast to measures of relationship quality that focus on the number of significant others within a social network, social support refers to the availability of members within one's social network that can provide supportive behaviors in times of need (Cohen 2004; House, 1981). Social support describes the functional content of relationships, and can be categorized into four types of supportive behaviors:

1. *Emotional support* involves the provision of empathy, love, trust, and caring.
2. *Instrumental support* involves the provision of tangible aid and services that directly assists a person in need.
3. *Informational support* involves the provision of advice, suggestions, and information to address issues and problems.
4. *Appraisal support* involves the provision of feedback and affirmation that is useful for evaluative purposes.

Notably, there is limited research investigating social support *per se*, because of the methodological challenges associated with measuring supportive behaviors within a social network. Instead, researchers use measures, typically questionnaires, that assess perceived social support; that is, the belief that social support is available. Interestingly, there is evidence to support the idea that perceived social support is a better predictor of health outcomes than actual levels of social support (McDowell & Serovich, 2007).

Social Network

A social network refers to the interpersonal linkages that surround an individual—people who may provide care, social support, companionship, and influence, and who act as points of social comparison. Social network analysis (Scott, 1991) is a method of

investigating social structures through consideration of both nodes (e.g., individuals, groups, etc.) and the ties (i.e., relationships or interactions) that bind them. In contrast to other measures of social relationships, analyses of social networks focus on objective characteristics of ties (e.g., relationship roles, frequency of contact, etc.) rather than appraisals of relationship quality, such as perceived availability of others and social support provided by one's network. Social networks can be assessed along multiple dimensions including size (the number of people in the network), density and complexity (the extent to which network members know and interact with each other), reciprocity (the extent to which supports and obligations are equal within a network or between members), homogeneity (the extent to which network members are similar along factors of interest such as age, ethnicity, social class, etc), proximity (a measure of geographic distance between a person of interest and their network), durability (a measure of the duration of relationships within a social network), and accessibility (the ease with which one can access their social network) (Wellman, 1981).

Part II: Measures of Social Relationships

Given the importance of social relationships to the human experience, hearing health care practitioners and researchers are often interested in better understanding how hearing loss and hearing rehabilitation affects social relationships. This can be accomplished using various methods, ranging from discussions with patients and their families to the use of standardized measurement tools. In audiology, there are a number of questionnaires designed to assess aspects of social relationships, and a recent systematic review identified 54 such instruments available in epidemiology research (Valtorta et al., 2016). Further complicating matters, clinicians and researchers are at times interested in understanding social relationships in their totality, and at other times interested in specific domains of social relationships, such as whether an individual is experiencing loneliness or reduced social activity. Accordingly, one challenge facing clinicians and researchers alike concerns how one might select a specific measurement tool to assess a social phenomenon of interest. Part II of this chapter attempts to serve as a guide for clinicians and researchers to assist in the selection of measurement tools when assessing social relationships.

Commentary on Measures

Provided in Table 10–1 is a categorization scheme of measures commonly used to assess social relationships in audiology practice and research. Notably absent are measures assessing handicap experienced by the significant others of persons with hearing loss, as these are covered in greater detail in Chapter 6 (Scarinci, 2019). The classification scheme presented here is based on Valtorta et al. (2016) but modified to be better suited for audiology. In this classification scheme, the measures reviewed are assessed along the following four dimensions of social relationships:

1. *Social Activity*: Assesses participation in activities that involve interaction with others or mem-

bership in various relational roles (e.g., spouse, parent, friend, etc.). These items are designed to gauge degree of involvement for a specified list of activities, size, and range of social relationships;

2. *Availability of Social Support*: Assesses the extent to which social relationships are perceived to be available to an individual;
3. *Adequacy of Social Support*: Assesses the extent to which social relationships satisfy interpersonal needs, either in quality or quantity;
4. *Feelings Related to Social Relationships*: Assesses affective experiences arising from interpersonal relationships or interactions.

The assessment tools are categorized along two additional dimensions. First, we report whether the

Table 10–1. Categorization of Measures Commonly Used in Hearing Research to Assess Social Relationships

	Social Activity	Availability of Social Support	Adequacy of Social Support	Feelings Related to Social Relationships	Hearing Specific	Unidimensional Measure
HHIE	Yes	—	—	Yes	Yes	No
CPHI	Yes	Yes	Yes	Yes	Yes	No
SPaRQ	Yes	—	—	Yes	Yes	Yes
EMO-CHeQ	No	—	—	Yes	Yes	No
SF-36	Yes	—	—	—	No	Yes
SF-6D	Yes	—	—	—	No	Yes
SELF	Yes	—	—	—	No	No
SNI	Yes	—	—	—	No	Yes
UCLA Loneliness Scale	—	—	Yes	Yes	No	Yes
de Jong Gierveld Loneliness Scale	No	—	Yes	Yes	No	Yes
FSSQ	—	—	Yes	—	No	Yes
MOS	—	Yes	—	—	No	Yes

Note. HHIE: Hearing Handicap Inventory for the Elderly; CPHI: Communication Profile for the Hearing-Impaired; SPaRQ: Social Participation Restrictions Questionnaire; EMO-CHeQ: The Emotional Communication in Hearing Questionnaire; SF-36: Short-Form 36 Health Survey Questionnaire; SF-6D: Short-Form 6 Dimensions; SELF: Self-Evaluation of Life Function Scale; SNI: Social Network Index; FSSQ: Duke-UNC Functional Social Support Questionnaire; MOS: Medical Outcomes Study Social Support Survey.

tool is “hearing specific” or not. This column denotes whether the self-reported measure was explicitly developed for the purpose of assessing experiences of hearing or was designed to provide a more global assessment of experience. Second, we examine whether the tool is a “unidimensional measure” or not. This column denotes whether the tool or subscales of the tool provides a score quantifying a particular domain or domains of social relationships (i.e., domain-specific) or not (i.e., domain-general). For example, the de Jong Gierveld Loneliness Scale is considered a unidimensional measure of social relationships because it contains a subscale assessing adequacy of social support as well as a subscale assessing feelings related to social relationships. In contrast, the HHIE is not considered a unidimensional measure of social relationships because its subscales reflect both interpersonal and noninterpersonal aspects of communication. (For additional information on the use of self-reported inventories, the reader is referred to Chapter 6 of this text.)

Hearing Handicap Inventory for the Elderly (HHIE)

The HHIE (Ventry & Weinstein, 1982) is a 25-item self-reported measure of situational and emotional effects of hearing loss, with higher scores indicating more hearing handicap. Respondents are asked whether they experience hearing or hearing-related difficulties and respond with a “yes,” “sometimes,” or “no.” The 12-item situational subscale asks participants to rate both hearing difficulty in different listening situations (e.g., when listening to TV or radio, shopping, etc.) as well as limits imposed on interpersonal activities (e.g., attending religious services, visiting family, etc.). The emotional consequences subscale assesses feelings and experiences in a variety of nonsocial and social situations. Three caveats regarding the measure are worth noting. First, about half of the items on the situational subscale inquire about hearing in difficult listening situations, and the other half of the items on the situational subscale inquire about limits on social activities. Second, slightly less than half of the items on the emotional subscale assess emotions experienced in social situations. Hence, the subscales in and of themselves, while excellent in many respects, are not ideal if the goal of the clinicians or researchers is to use a measure whose subscales solely assess impact of hearing loss and rehabilitation on social relationships. Third, it is worth mentioning that since about half of the items from the HHIE assess social consequences associated with hearing loss, item-by-item analysis

could enable one to understand effects of hearing loss on activities and emotions related to social relationships from effects of hearing loss on emotion that are not related to social relationships.

Communication Profile for the Hearing-Impaired (CPHI)

The CPHI (Demorest & Erdman, 1987) is a 145-item self-reported questionnaire designed to assess experiences of communication across 25 subscales. Of most relevance for this chapter are the subscales assessing: “Attitudes of others,” which measures the respondent’s views of communication partner’s attitudes (e.g., “People treat me as if I’m stupid because I can’t understand what they say”); “behaviors of others,” which measures the respondent’s views of communication partner’s behaviors (e.g., “Members of my family leave me out of conversations or discussions”); “verbal strategies,” which measures the respondent’s use of verbal behaviors with others that compensate for hearing loss (e.g., “When I don’t understand what someone has said, I explain that I have a hearing loss”); “maladaptive behaviors,” which measures the respondent’s use of undesirable interpersonal coping behaviors (e.g., “I tend to avoid social situations where I think I’ll have problems hearing”); and a subscale assessing “withdrawal,” which measures feelings of withdrawal and isolation (e.g., “Because of my hearing loss, I keep to myself”). The CPHI is a communication-focused questionnaire that includes components assessing social activity participation and perceived availability of, perceived adequacy of, feelings related to, and avoidance behaviors designed to discourage social relationships.

Social Participation Restrictions Questionnaire (SPaRQ)

The SPaRQ (Heffernan, Coulson, & Ferguson, 2018; Heffernan, Maidment, Barry, & Ferguson, 2019) is a 19-item hearing-specific, patient-reported outcome measure that could be used in research or clinical practice to assess efficacy of auditory rehabilitation interventions on participation restrictions in adults with hearing loss. The SPaRQ contains two subscales: a nine-item *social behaviors* subscale that measures difficulties in performing behaviors in a social context due to a hearing loss, and a 10-item *social perceptions* subscale that measures thoughts and feelings experienced in social contexts due to hearing loss. Thus, the SPaRQ assesses perceived involvement in and feelings related to social relationships.

The Emotional Communication in Hearing Questionnaire (EMO-CHeQ)

The EMO-CHeQ (Singh, Liskovoi, Launer, & Russo, 2018) is a 16-item questionnaire assessing hearing difficulties when listening to voices which contain emotion information. Difficulties are assessed across four subdomains: (a) characteristics of encountered talkers (e.g., voices on television), (b) communication in challenging listening situations (e.g., noisy environments), (c) speech production (e.g., the ability to convey emotion in a subtle manner using one's own voice), and (d) the associated impact of such deficits on socioemotional well-being (e.g., Item 16: My difficulty identifying emotions expressed in speech negatively affects my relationships with friends and family). Thus, subdomain (d) is seemingly most relevant when assessing the social impact of vocal emotion hearing difficulties.

Short-Form 36 Health Survey Questionnaire (SF-36)

The SF-36 (Brazier et al., 1992; Brazier, 1993) is a 36-item patient-reported survey of general health covering eight dimensions of health (vitality, physical functioning, bodily pain, general health perceptions, physical role functioning, emotional role functioning, mental health, and social role functioning). The social role functioning subscale contains two items. The first social item asks if "Emotional problems interfered with your normal social activities with family, friends, neighbors, or groups?" with five response options ranging from *"not at all"* to *"very severe"*. The second social item asks "During the past 4 weeks, how much of the time has your physical health or emotional problems interfered with your social activities (like visiting with friends, relatives, etc.)?" with five response options ranging from *"all of the time"* to *"none of the time"*. Thus, the social functioning questions of the SF-36 assess social activity participation.

Short-Form 6 Dimensions (SF-6D)

The SF-6D (Brazier, Roberts, & Deverill, 2002) is a 6-item patient-reported survey of general health derived from the SF-36. The SF-6D assesses physical functioning, role limitations, pain perception, vitality, mental health, and social functioning. All dimensions are assessed with a single item, with social functioning assessed by the item "Your health limits your social activities ____." The blank is completed with

one of five possible response options ranging from *"none of the time"* to *"all of the time"*. Thus, the social functioning question of the SF-6D assesses social activity participation.

Self-Evaluation of Life Function (SELF) Scale

The SELF Scale (Linn & Linn, 1984) is a 54-item comprehensive self-reported measure of health for older adults that assesses six areas of functioning: physical disability, symptoms of aging, depression, self-esteem, personal control, and social satisfaction. A total of four items assess social satisfaction. (These items are: Item 18: "How often do you see your friends and relatives?"; Item 19: "How often have you made telephone calls over the past month?"; Item 20: "How often have you worked on a hobby or some activity of interest over the past month?"; Item 21: "How often have you attended meetings at associations, church, organizations, get-togethers, or clubs over the past month?") For items 18 to 21, there are four response options that range from *"often/several times a day"* to *"rarely or never/not at all"*. Two of the items from the social satisfaction subscale of the SELF do not ask questions regarding interpersonal relationships. Thus, the social satisfaction subscale of the SELF mostly assesses social activity participation.

Social Network Index (SNI)

The SNI (Cohen, 1991) is a 12-item measure of social network diversity whereby respondents report membership in various social roles (e.g., marital status, parent, employee, student, etc.) and degree of interpersonal contact every 2 weeks. One point is assigned for each type of relationship for which respondents indicate that they speak (in person or on the phone) to someone in that relationship at least once every 2 weeks. Thus, the SNI assesses social network breadth and social activity participation.

UCLA Loneliness Scale

The UCLA Loneliness Scale (Russell, Peplau, & Ferguson, 1978) is a 20-item self-reported questionnaire designed to measure subjective feelings of loneliness. Respondents read statements (e.g., "I have nobody to talk to") and report their experience with one of four response options ("often," "sometimes," "rarely," or "never"). The UCLA Loneliness Scale assesses both perceived adequacy of social relationships and feelings relating to social relationships.

de Jong Gierveld Loneliness Scale

The de Jong Gierveld Loneliness Scale (de Jong & Tilburg, 2006) is an 11-item scale whereby respondents read statements (e.g., Item 7: “There are many people that I can count on completely” and Item 10: “Often I feel rejected”) and report experiences with one of three response options (“yes,” “more or less,” or “no”). The scale assesses both perceived adequacy of social relationships and feelings related to social relationships.

Duke-UNC Functional Social Support Questionnaire (FSSQ)

The FSSQ (Broadhead et al., 1988) is an eight-item self-reported questionnaire that is commonly used in research investigating the effects of perceived social support (e.g., Item 1: “I have people who care what happens to me”; Item 2: “I get love and affection”; Item 7: “I get useful advice about important things in life”). Responses are made on a 5-point Likert scale with higher values corresponding to greater perceived social support (i.e., 1 = *Much less than I would like to* to 5 = *as much as I would like*). The FSSQ can be analyzed to yield scores on two subscales, confidant support and affective support. Confidant support refers to having someone with whom to discuss important and personal matters, whereas affective support refers to being cared for emotionally. Thus, the FSSQ assesses perceived adequacy of social relationships.

Medical Outcomes Study (MOS) Social Support Survey

The MOS Social Support Survey (Sherbourne & Stewart, 1991) is a 19-item self-reported questionnaire that investigates four dimensions of perceived availability of social support: (1) emotional/informational support (availability of expressions of positive affect, empathetic understanding, encouragement of expressions of feelings, offering of advice, feedback, or guidance); (2) tangible support (availability of material aid or behavioral assistance); (3) affectionate support (availability of expressions of love and affection); and (4) positive social interactions (availability of others to do fun things with). Responses are made on a 5-point Likert scale with higher values corresponding to greater perceived social support (i.e., 1 = *none of the time* to 5 = *all of the time*). Thus, the MOS Social Support survey assesses perceived availability of social relationships.

Part III: Social Relationships

One of the long-standing beliefs in audiology, held by many researchers, clinicians, patients, and significant others, concerns the relationship between hearing loss and effects on social relationships, whereby poorer hearing abilities leads afflicted individuals, and often their significant others, to withdraw from others and social situations, and to experience social disconnectedness. Indeed, there is considerable evidence that severity of hearing loss and associated communication difficulties can negatively affect our social world (Barker, Leighton, & Ferguson, 2017; Dalton et al., 2003; Héту, Jones, & Getty, 1993; Pronk et al., 2011; Kramer, Kapteyn, Kuik, & Deeg, 2002; Vas, Akeroyd, & Hall, 2017; Wallhagen, Strawbridge, Shema, & Kaplan, 2004; Weinstein & Ventry, 1982). As discussed previously in the chapter, there are different domains of social relationships that can be affected. Next, we discuss the findings from various approaches adopted by researchers to understand the social consequences associated with hearing loss and hearing rehabilitation.

Before moving on to this review, we draw attention to an important caveat. Numerous studies describe social effects of hearing loss and rehabilitation, but in many cases, it is unclear whether the measure or item used in a study reflects interpersonal or noninterpersonal aspects of experience. This is understandable because items or questionnaires are often designed to understand experiences of hearing that are both social and nonsocial. For example, consider Item 14 from the social/vocational subscale of the modified Denver Scale of Communication Function: “I do not enjoy my job as much as I did before I began to lose my hearing” (Schow & Nerbonne, 1980). This item could be construed to reflect difficulties with interpersonal communication at work, difficulties with interpersonal bonding and connectedness, auditory challenges that do not involve others such as the ability to localize sounds, or some other difficulty that is not social. Hence, in this chapter, we attempt to limit our discussion and review of the literature by focusing on research studies that use measures that more concretely assess social relationships.

A number of qualitative studies have examined the impact of hearing loss on everyday social relationships. It is often observed that individuals with hearing impairment report engaging in various interpersonal coping behaviors such as the avoidance of social activities (Hallberg & Carlsson, 1991; Heffernan,

Coulson, Henshaw, Barry, & Ferguson, 2016; Lockett, Jennings, & Shaw, 2010), less active participation while attending social activities (Hétu, Riverin, Lalonde, Getty, & St-Cyr, 1988), or active communication management during social interactions (i.e., controlling conversational topics, altering communication patterns, explicit requests for repetition, optimizing body position relative to others) (Hallberg & Carlsson, 1991; Heffernan et al., 2016; Hétu et al., 1988; Lind, Hickson, & Erber, 2006).

Hearing Loss and Involvement in Social Activities

In the quantitative literature investigating the effects of hearing loss on social relationships, researchers have used a myriad of research designs and questionnaires. Overall, there appears to be good evidence that there are significant effects of hearing loss on social activity (Andrade, Pereira, & Silva, 2017; Dalton et al., 2003; Hallam & Brooks, 1996; Helvik, Jacobsen, & Hallberg, 2006).

In classic work first describing the HHIE, perhaps the most popular measure of handicap in audiology, Ventry and Weinstein (1982) assessed 100 adults aged 65 years or older with varying degrees of audiometric hearing loss from normal to severe. On the situational subscale (of which about half the items assess social activity limitations imposed by hearing loss), it was observed that greater handicap is systematically observed with poorer hearing, thus suggesting that activity limitations are associated with hearing loss. Complementing this work, a related finding was observed by the National Council on the Aging (1999) who obtained responses from 2,304 individuals with varying degrees of hearing loss. When asked whether respondents participate regularly in social activities, 37% of those with milder hearing losses reported regular activity, whereas this number drops to 32% for those with more severe hearing losses.

Longitudinal evidence has also observed that hearing-impaired respondents report poorer social activity (indexed by the HHIE-S, a shorter version of the HHIE; Ventry & Weinstein, 1983) compared with normal hearing peers (Andrade, Pereira, & Silva, 2017). Using data from the Epidemiology of Hearing Loss Study in Beaver Dam, WI (Dalton et al., 2003), it was found that relative to individuals with normal audiometric thresholds, hearing loss severity was associated with poorer scores on the HHIE-S. Importantly, the study included a measure of social activity

other than the HHIE-S, and the authors also observed poorer social relationships on this measure of involvement in social relationships (SF-36).

Notably, there are important limitations regarding the effect of hearing loss on social activity participation. To date, there is a dearth of evidence demonstrating the effect of hearing loss on actual involvement in social relationships. Instead, the literature relies on proxy measures of social activity, namely self-reported questionnaires. Moving forward, it is anticipated that novel research methodologies using smartphones (e.g., ecological momentary assessment; Kimball, Singh, John, & Jenstad, 2018; Stone & Shiffman, 1994; Timmer, Hickson, & Launer, 2018) will address methodological limitations imposed by self-reported paper and pencil questionnaires, and will be better able to track individuals close in time to events to study social interactions.

Hearing Loss and Loneliness

Recall that loneliness represents a negative affective state resulting from an unfulfilled desire to have friends, deprivation of affective bonding, and a perception of the gap between actual and desired levels of bonding. Both cross-sectional and longitudinal research methodologies, and multiple measures of loneliness have been used to investigate whether hearing loss is associated with loneliness.

Data from several cross-sectional datasets all suggest an association between loneliness and hearing loss. Using data from the first wave of the Canadian Longitudinal Study of Aging, Mick, Parfyonov, Wittich, Phillips, and Pichora-Fuller (2018) investigated whether self-reported hearing loss was associated with experiences of loneliness in 21,241 adults aged 45 to 59 years. They measured loneliness with the following single item: "In the past week, how often did you feel lonely?" and found that self-reported hearing loss was significantly associated with loneliness. Similarly, Dawes et al. (2015) assessed loneliness (labelled as "social isolation" in their paper) using the single item "Do you often feel lonely" (yes/no) using data from 164,770 individuals aged 50 to 69 years of age from the U.K. BioBank dataset. It was observed that social isolation was associated with poorer hearing. Finally, Sung, Li, Blake, Betz, and Lin (2016) assessed experiences of loneliness in 145 hearing-impaired adults aged 50 to 94 years using the UCLA Loneliness Scale and found that greater hearing loss was associated with higher levels of loneliness.

The effect of hearing loss on loneliness and social disconnectedness has also been investigated longitudinally. In the Alameda County longitudinal study (Strawbridge et al., 2000), the effect of self-reported hearing difficulty was assessed on three questions of social well-being measured 1 year later: (1) "Not feeling close to others" (response options of "true" or "false"), (2) "Feeling left out even in a group" (response options of "true" or "false"), and (3) "Feeling lonely or remote" (response options of "never," "sometimes," or "often"). Compared with individuals who reported no hearing difficulty in a conversation, individuals who reported "a little" hearing difficulty indicated both feeling left out of groups and greater rates of loneliness, but still feeling close to others. The group with self-reported moderate (or worse) hearing difficulty scored significantly poorer on all three social measures compared to those reporting no impairment.

Finally, Pronk et al. (2011) assessed how well hearing loss predicts loneliness four years later than initial measurement using data from the Longitudinal Aging Study Amsterdam. Hearing loss was measured in two ways, self-reported and performance on a speech-in-noise test. Loneliness was assessed using the de Jong Gierveld Loneliness Scale, which assesses both perceived adequacy of social relationships and the affective experience of loneliness. They found that whereas poorer performance on the speech-in-noise test predicted the emotional experience of loneliness years later for men but not women, self-reported hearing loss predicted social-cognitive appraisals of loneliness (i.e., perceived adequacy of social relationships) for individuals, but only if they had a cohabitating partner. Thus, Pronk et al. (2011) highlights two important considerations. First, negative effects of hearing loss on perceptions of affective bonding are somewhat dependent on how one measures hearing loss. This suggests that differences between hearing assessment methodologies are relevant, although it is unclear what differences account for the differential pattern of findings. Whereas self-reported hearing loss taps into hearing loss constructs such as subjective evaluations of communication in everyday environments, speech-reception threshold testing evaluates function of speech discrimination in noisy backgrounds. Second, the findings from Pronk et al. underscore the importance of individual differences when assessing the potential impact of hearing loss on social relationships. The identification of subgroups at greater risk of experiencing loneliness associated with hearing loss draws attention to the need

to consider lifestyle factors and their potential role in moderating or exacerbating patients' well-being (see also Pronk, Deeg, & Kramer, 2013; Pronk et al., 2014). Altogether, the results from these studies suggest that there is good evidence that hearing loss is associated with increased feelings of loneliness and appraisals of social disconnectedness.

Hearing Loss and Social Network Size

Relatively few studies of hearing loss have attempted to characterize social network size, although there are some exceptions. Kramer et al. (2002) assessed loneliness in 3,107 adults (aged 55 to 85 years) participating in the first wave of the Longitudinal Aging Study Amsterdam. Hearing loss and social network size were assessed using self-reporting. They found that compared with peers reporting normal hearing, hearing-impaired adults reported smaller social networks. This finding stands in contrast to other work which observed that self-reported hearing loss was not associated with social network size (assessed using the SNI; Mick et al., 2018). Hence, there is mixed evidence to support the idea that hearing impairment leads to fewer individuals in one's social network.

Hearing Loss and Perceived Social Support

There is limited work evaluating consequences of hearing loss on the perception of the availability and adequacy of social support. To date, this relationship has been investigated in two population-based cross-sectional studies. Mick et al. (2018) found that self-reported hearing loss was associated with poorer appraisals of social support across all four subscales of the MOS Social Support survey (emotional/information, tangible, affectionate, and positive social interaction support). In contrast, Mick, Kawachi, and Lin (2014) assessed social support using one item asking about availability of emotional social support and one item asking about availability of financial support. They found lower perceived social support on both items, but only for women aged 60 to 69 years—not in women aged 70 to 84 years or men between 60 and 84 years of age. Because both studies employed cross-sectional designs, it is unclear whether there is a causal relationship between hearing loss and perceptions of social support.

Interpersonal Relationships

Up to now, we have discussed the effects of hearing loss on domains of social relationships (e.g., social activity, loneliness, etc.). In this next section, we focus on the effects of hearing loss on specific interpersonal relationships (e.g., spouses, adult children, etc.). Given the central role of communication in interpersonal relationships, it is perhaps not surprising that hearing loss can have a significant effect on relationship quality and associated outcomes. In the literature investigating the role of hearing loss on adult interpersonal relationships, most research has been conducted in either the context of spousal relationships or relationships between older persons with hearing loss and their adult children, although there are some exceptions, which we also describe.

Spousal Relationships

In the current context, spousal relationships refer to intimate relationships regardless of legal status, and includes married, unmarried, heterosexual, or homosexual relationships. Although there is research to suggest that hearing loss does not appear to cause marital breakdown, there are a number of consequences for relationships when one partner experiences impaired hearing (Hétu et al., 1993; Morgan-Jones, 2001; Scarinci, Worrall, & Hickson, 2008; for a review, see Barker et al., 2017). In longitudinal research of 418 older married couples, Wallhagen et al. (2004) found that for persons with self-reported hearing loss, their spouses reported lower mental, physical, and social well-being when measured after a 5-year interval. However, other research with 8,607 couples observed that audiometrically measured hearing loss was not associated with poorer mental health in spouses (Ask, Krog, & Tambs, 2010). Based on analyses of the data from Wallhagen et al. (2004) and Ask, Krog, and Tambs (2010), Bainbridge and Wallhagen (2014) suggest that the impact of a husband's hearing loss on a spouse may be greater than the impact of a wife's hearing loss on a spouse.

Other research observes poorer outcomes on relationship measures assessing interpersonal tension, irritation, frustration (Hétu et al., 1993; Oyer & Paolucci, 1970), increased blame resulting from violations of social rules regarding communication (Hétu & Getty, 1991), increased effort to maintain adequate communication (e.g., having to repeat oneself, facilitating communication for the partner with impaired hearing, always having to answer the phone; Hétu & Getty, 1991), restrictions on social life (e.g., avoiding

noisy social situations, being isolated as a couple in social situations; Hétu & Getty, 1991), and poorer intimacy (Hétu et al., 1993; Jones, Kyle, & Wood, 1987; Wallhagen et al., 2004).

Importantly, several factors can buffer relationship strain when one partner in a dyad experiences hearing loss. For the better-hearing partner, it is key to consider the process by which she/he explains behaviors of the poorer-hearing partner. To the extent that better-hearing partners assign causes of behavior of poorer hearing partners to internal characteristics (e.g., "my partner withdraws from social situations because of a lack of interest or care for others"), relationship quality suffers. To the extent that better hearing partners assign causes of behavior of poorer hearing partners to external or situational factors (e.g., "my partner withdraws from social situations because of difficulties communicating in noisy environments"), relationship quality is maintained (Anderson & Noble, 2005; Piercy & Piercy, 2002). For the person with hearing loss, relationship strain can be buffered through the use of technological (i.e., hearing instruments and assistive listening devices) and nontechnological (i.e., communication tips such as the need to communicate in acoustically favorable environments) aids, and through a typically gradual process of a shared acceptance of the hearing loss (Yorgason, Piercy, & Piercy, 2007).

Intergenerational Relationships of Older Adults

Because older adults with hearing impairments experience an "invisible disability" (Wright, 1983), significant others with good hearing may not consider loss of function to be disability related. As a result, the behaviors of older adults are susceptible to being misunderstood, and stemming from this misunderstanding, experience a number of social consequences, many of which are similar to those experienced within spousal relationships (e.g., poorer relationship quality and decreased social activity). The person with hearing loss may experience increased anxiety, fear, and distrust of others. Better-hearing significant others may experience increased resentment (Schow, Christensen, Hutchinson, & Nerbonne, 1978).

Relationships in Other Contexts

Perhaps understudied is the effect of hearing impairment on relationship quality in contexts other than romantic and intimate interpersonal relationships. There is, however, evidence to suggest that hearing

loss may influence social relationships in employment settings and in interactions with medical professionals (Punch, Hyde, & Power, 2007). Compared with normal hearing peers, persons with hearing loss are more likely to experience unemployment (Hogan, O'Loughlin, Davis, & Kendig, 2009; Parving & Christensen, 1993), earn less income (Jung & Bhattacharyya, 2012), are less likely to be employed in highly skilled jobs and overrepresented in low-paying jobs (Hogan et al., 2009), experience more psychological distress (Järvelin, Mäki-torkko, Sorri, & Rantakallio, 1997), and take more stress-related work leave (Kramer, Kapteyn, & Houtgast, 2006). Importantly, it is unclear to what extent the poorer outcomes experienced by persons with hearing loss are mediated by social relationships with coworkers or by nonsocial mechanisms. (For further discussion of the impact of hearing loss on vocational issues, the reader is directed to Chapter 25 of this text.)

In medical contexts, there is evidence to suggest that hearing loss may lead to provider-patient communication breakdown, potentially leading to medical errors. Based on interviews with 100 hearing-impaired older adults, 43% reported mishearing either a physician or nurse during a hospital visit or medical appointment (Cudmore, Henn, O'Tuathaigh, & Smith, 2017). This finding is further supported from an analysis of the responses of 122,556 participants from the Medical Expenditure Panel Survey Household Component, a nationally representative survey of the U.S. civilian noninstitutionalized population. It was found that after controlling for sex, age, hearing aid use, and self-reported visual impairment, poorer self-reported hearing loss was independently associated with lower ratings of physician-patient communication effectiveness and health care provision quality (Mick, Foley, & Lin, 2014). Collectively, these results suggest that individuals with hearing loss may have greater difficulty understanding and engaging in discussions with medical personnel.

In a systematic review of the literature investigating physician-patient communication with older patients, it was observed that across 67 papers on the topic, only 16 studies mentioned the role of hearing loss on communication (Cohen et al., 2017). This suggests that, even among highly trained health professionals, consideration of hearing loss on efficacy of communication during health visits may be lacking. This is particularly troubling as medical appointments can take place in noisy environments where discussions often include unfamiliar medical terminology, concepts, and complex instructions regarding treatment and care.

Part IV: Hearing Rehabilitation and Social Relationships

Given the body of evidence observing impaired social relationships that follow hearing loss, it is important to consider the effect of hearing rehabilitation interventions on interpersonal relationship outcomes. As discussed throughout the textbook, there are a variety of possible interventions to address hearing loss including the provision of hearing instruments, counseling about hearing loss and strategies to improve communication, group AR classes, and auditory training. Next, we discuss the details regarding the effect of hearing rehabilitation on social relationships.

Hearing Aids and Social Relationships

In addition to investigating the effects of hearing loss on loneliness, Pronk et al. (2011) also investigated, in a longitudinal fashion, how hearing aid use affects ratings of loneliness (using the de Jong Gierveld Loneliness Scale) measured years later. Importantly, they found that for persons with hearing loss, nonusers of hearing aids reported adverse effects on subscales assessing emotional and cognitive appraisals of loneliness, but that no such adverse effects were observed for users of hearing aids. These findings support the notion that hearing aids reduce feelings of loneliness and appraisals of social disconnectedness. Notably, Dawes et al. (2015) observed that, counterintuitively, hearing aid use was weakly associated with more loneliness.

Cochrane is a nonprofit, nongovernmental organization formed to organize medical research findings and facilitate systematic reviews, with the goal of supporting evidence-based decision making by patients, health professionals, and policy makers. A Cochrane review of the literature was conducted by Ferguson et al. (2017), investigating the effects of hearing aids for older adults aged 69 to 83 years with mild to moderate hearing loss. In summarizing the literature, they concluded that hearing aids are associated with significant benefit regarding involvement in social activities and feelings relating to social relationships. Notably, the effects considered by Ferguson et al. were based on assessments of social relationships when measured by hearing-related measures (i.e., HHIE) of social relationships.

Next is a detailed discussion of effects of hearing aids on domain-specific (i.e., hearing-related) measures of social relationships, followed by a review

of studies that employ domain-general measures of social relationships. Several laboratory-based studies have investigated hearing aids and social relationships using the HHIE, and a reduction in handicap post-hearing aid fitting is typically observed on the situational subscale of the HHIE (Abrams, Hnath-Chisolm, Guerreiro, & Ritterman, 1992; Malinoff & Weinstein, 1989; Newman & Sandridge, 1998). For example, in a randomized control trial of 194 older adult veterans, Mulrow et al. (1990) observed significant improvement in social relationships as assessed by the situational subscale of the HHIE, but interestingly, failed to observe improvements in social relationships when assessed by a more general (i.e., not hearing-specific) measure assessing involvement in, perceived availability of, and feelings relating to social relationships (i.e., the SELF questionnaire). Similarly, Stark and Hickson (2004) failed to observe improvements in social function using the SF-36 in their study of 93 first-time hearing aid wearers.

The failure to observe improvements in social relationships on domain general measures by some researchers stands in contrast to other research investigating social relationships post-hearing aid fitting. Embedded within a larger study investigating acceptability, benefit, and costs of early screening, Davis et al. (2007) assessed the effect of hearing aid provision on social function using the 6F-SD. Three months post-fitting, it was observed that hearing aid provision resulted in significant and substantial improvement to the level of social activity reported by participants. Similarly, in a sample of 80 moderately hearing-impaired adults, Joore, Brunenberg, Chenault, and Anteunis (2003) found that first-time hearing aid provision resulted in more social activity when assessed by the SF-36 and on a single item measuring frequency of visiting others. Dissimilarities in findings between those studies observing significant effects on domain general measures of social relationships and those that do not may be due to several methodological differences between studies such as selection of measures, the point in time post-fitting when outcomes are assessed, and sample size employed to observe potential differences to name a few.

In addition to methodological differences between studies, analyses of data from the National Health and Nutrition Examination Survey (Mick et al., 2014) and the Longitudinal Aging Study Amsterdam (Pronk et al., 2013) may offer additional clues as to why there are differences in the literature regarding the effect of hearing aids on domain general measures of social relationships. Specifically, both studies draw

attention to the fact that negative effects of hearing loss on interpersonal relationships are not universally observed across populations. Instead, what both studies highlight is that social effects of hearing loss are at times present in some subgroups and not others (e.g., depending on sex, age, socioeconomic status, etc.). Because research investigating efficacy of hearing aids is typically conducted on relatively small samples, this raises the possibility that differences in sampling (i.e., a sampling bias) may account for the failure to observe effects of hearing aids on domain-general measures of social relationships in some studies but not others.

Interestingly, to date, there is minimal “objective” research investigating the effect of hearing aids on social relationships. One exciting exception is research investigating social interaction patterns that are indexed by coding body position and interpersonal movement such as orientation behaviors (e.g., turning one’s head toward a talker, leaning toward a talker, etc.). Typically in this work, social interactions are videotaped and later coded by trained raters. Preliminary work using the methodological approach suggests increased social interaction is observed when hearing-impaired listeners use hearing aids equipped with adaptive beamforming technologies than when using fixed directional beamformers (Paluch, Latzel, & Meis, 2015; Schulte, Meis, Krüger, Latzel, & Appleton-Huber, 2018).

Aural Rehabilitation and Social Relationships

In a systematic review designed to examine the effectiveness of aural rehabilitation programs, Hawkins (2005) concluded that there is good evidence that participation provides short-term reduction of perceived hearing handicap and improved perceptions of quality of life. Importantly, Hawkins’ review was not designed to assess effects of aural rehabilitation on social relationships, and we next review the literature with this goal in mind.

Chisholm, Abrams, and McArdle (2004) examined subjective benefits from providing hearing aids alone or hearing aids plus a 4-week counselling-based group audiologic rehabilitation program for 106 veterans with hearing loss. Benefit was assessed using the CPHI at multiple time points following the intervention including 1 year post-intervention. They found that, whereas hearing aid provision resulted in less maladaptive social behavior (e.g., less avoid-

ance of social situations, less avoidance of strangers, etc.), hearing aids plus counselling resulted in more use of interpersonal verbal strategies to enhance communication (e.g., asking others to repeat themselves, reminding others of their hearing loss, etc.). In another study, Preminger and Yoo (2010) investigated self-reported benefit on the HHIE/HHIA (the Hearing Handicap Inventory for Adults) (Newman, Weinstein, Jacobson, & Hug, 1991) following different types of group aural rehabilitation classes for 52 older experienced users of hearing aids. The aural rehabilitation course consisted of six classes, each lasting between 60 and 90 minutes; outcomes were assessed both at the conclusion of the course as well as 6 months later. Significant improvement in social relationships was observed for aural rehabilitation classes at both testing intervals compared to baseline scores on the HHIE/HHIA. The findings suggest that group aural rehabilitation classes result in improved evaluations of social relationships (see also Preminger, 2003).

In contrast, there are several investigations that have failed to observe that social relationships improve as a result of attending group aural rehabilitation classes. For example, Abrams et al. (1992) considered whether hearing aids plus aural rehabilitation classes decreased handicap on the HHIE beyond that provided by hearing aids alone, and found no additional benefit from aural rehabilitation classes on the situational subscale of the HHIE. A similar result was also observed in Abrams, Chisholm, and McArdle (2002) when assessed using the SF-36. Altogether, the results from studies investigating the effect of aural rehabilitation classes on social relationships seem to suggest improved social relationships when assessed using hearing-specific measures, but the failure to observe benefit for social relationships using domain-general measures of social relationships.

Part V: Social Influence

Because we exist in social collectives, it is important to consider the ways in which the behavior or presence of others influences the thoughts, attitudes, and behaviors of the person with hearing loss. There is a long (i.e., over a century) tradition of academic research on social influence; however, it is only relatively recently that such forces have been considered within the context of hearing research. What follows is a basic description of some social influence forces

relevant to audiology, followed by a brief review of research on social influence conducted in audiology.

Compliance refers to the act of acquiescence to the requests of others. Such requests can be explicit (e.g., "Please wear your hearing aid") or implicit (e.g., "Communication is much easier when you wear your hearing aids"). In both cases, the person with hearing loss recognizes that they are being urged to behave in a goal-directed fashion. One interpersonal factor that can influence compliance behaviors is emotion, and emotion may be relevant either when embedded in requests for compliance or to alleviate post-behavior experiences of emotion (e.g., guilt; Boster et al., 1999).

Obedience, in contrast, refers to acquiescence behaviors done in accordance with the directives, advice, and opinions of authority figures. Positions of authority can be based on one's expertise (e.g., audiologist) and/or relative position within a social hierarchy (e.g., parent). Within the context of obedience, a distinction is made between soft influences, which are based on internal factors within the agent of influence (e.g., credibility), and harsh influences, which are based on factors external to the agent of influence (e.g., an existing social structure; Koslowsky, Schwarzwald, & Ashuri, 2001).

Advice is another mechanism by which others can influence behavior is through the mitigation of decision uncertainty. As discussed earlier in the chapter, significant others can act as sources of informational social support. In doing so, others minimize perceptions of risk and uncertainty (Kiecker & Hartman, 1994). Significant others can also act as interpersonal sources of information, support decision-making processes, and increase confidence that wise decision making takes place (Furse, Punj, & Stewart, 1984; Kiecker & Hartman, 1993; 1994).

Decision-making is typically assumed to be a deliberate and conscious process, affected by *normative social influence*. For decades, standard models of decision-making characterized individuals as rational deliberators who objectively weigh the pros and cons of decisions and behaviors. However, there is a great deal of consensus among psychologists that witnessing others' behavior exerts a powerful effect on one's own behavior, including lying (Asch, 1956) and failing to respond to imminent threats (Latané & Darley, 1970).

Social norms are the customary rules that govern behavior, and they can operate at the level of small groups (e.g., friends taking turns providing snacks for an evening of cards) to societies (e.g., shaking hands when meeting someone). Social norms influence

behavior from the human desire for affiliation and acceptance with others (Bicchieri, 2006). Interestingly, normative social influence operates both at conscious (Maxwell, 2002) and subconscious levels of awareness (Epley & Gilovich, 1999). Factors that can affect the relationships between social norms and acquiescence include reciprocity (e.g., obligations to repay others for what we have received from them; Cialdini & Goldstein, 2002) and culture (i.e., behavioral expectations within a society or subgroup; Reno, Cialdini, & Kallgren, 1993).

An important consideration about the influence of social norms on behavior concerns the role of *stigma*. Whereas social norms communicate behaviors which are acceptable and expected by society, stigma refers to social disapproval (Goffman, 1963). Social stigmas represent a means to exert control or communicate disapproval on those who do not adhere to socially defined norms. Despite the lack of experimental work investigating stigma and hearing, there is broad consensus in audiology that stigma is a relevant consideration that inhibits hearing rehabilitation uptake (Southall, Gagné, & Jennings, 2010; Wallhagen, 2010; for a review, see David & Werner, 2016). Notably, it is unclear whether stigma inhibits hearing rehabilitation because of stigma related to hearing loss and hearing aids (Southall, Gagné, & Jennings, 2009) or stigma related to age (Chasteen, Pichora-Fuller, Dupuis, Smith, & Singh, 2015), or both. (For more information on the psychological effects of social stigma, the reader is referred to Chapter 4.)

Theories of *social comparison* (Festinger, 1954) state that individuals can determine personal and social worth based on how they measure up against others. Typically, social comparisons are conducted with relevant and similar others within one's social network so as to provide judges with meaningful self-evaluations. Importantly, social comparisons can serve other purposes. For example, a desire to improve oneself can foster *upward social comparisons* (i.e., with individuals we believe are better than ourselves), and motivations to improve one's mood can foster *downward social comparisons* (i.e., with individuals we believe to be worse off than ourselves).

Part VI: The Role of Others in Help Seeking and Hearing Aid Uptake

Based on informal personal communication with hundreds of family members of persons with hearing loss,

it became quickly apparent that family members and significant others are often well aware that they have a critical role in raising awareness of hearing loss and motivating loved ones to address communication difficulties. Conducted mostly using qualitative research methods, it has been observed that significant others can potentially foster a greater understanding of the communication and psychosocial consequences of a hearing loss (Lockey et al., 2010; Manchaiah, Stephens, & Lunner, 2013; Schulz et al., 2017), encourage and motivate hard-of-hearing individuals to seek help (Duijvestijn et al., 2003; Lockey et al., 2010; Mahoney, Stephens, & Cadge, 1996; Manchaiah et al., 2013; van den Brink, Wit, Kempen, & van Heuvelen, 1996), and help patients adjust to the psychological distress (e.g., depression, anxiety, and paranoia) associated with hearing loss (Frankel & Turner, 1983). Interestingly, there is also evidence suggesting that negative attitudes of significant others towards hearing aids can delay help seeking or reject treatment options (Brooks 1989; Kochkin 2007).

The influence of social context has also been explored in research using quantitative research methodologies (Singh & Launer, 2016). In this work, the role of social influence on hearing aid adoption was investigated, specifically, whether hearing aid adoption is associated with significant other attendance at audiology appointments. The sample consisted of 60,964 individuals, 33,933 of whom attended the appointment alone and 27,031 of whom attended the appointment with a significant other. Hearing aid adoption was significantly greater when patients attended audiology appointments with a significant other (63.8%) than when attending appointments alone (50.6%). Notably, it was further observed that the association between hearing aid adoption and attendance by a significant other was hearing dependent. For patients with mild hearing losses, hearing aid adoption increased by 96% when patients attended appointments with a significant other than when attending appointments alone. Typically, uptake of hearing rehabilitation is low for individuals with mild hearing loss, but the mere presence of a significant other in this study was associated with hearing aid uptake typically observed with individuals with moderate hearing losses.

Interestingly, others influence experiences of help seeking even if they are not physically present. In a qualitative study of seven women with presbycusis, Carson (2005) set out to better understand factors that motivated help seeking. One of the major themes identified in the study was that of the role of social

comparisons. Specifically, it was found that one of the cues that prompted individuals to seek help was self-evaluation of communication difficulties in relation to others in their social network.

The Role of Others in Hearing Rehabilitation Outcomes

Not only do significant others influence help seeking and hearing aid rehabilitation uptake, but they also influence hearing rehabilitation outcomes. Erber (1996) suggested that successful use of hearing aids is likely dependent on support from communication partners, and subsequent quantitative research supports this view. Noteworthy correlations that link positive audiologic outcomes with social relationships have been observed in three studies. Hickson et al. (2014) investigated factors that differentiate successful from unsuccessful users of hearing aids and found that perceived availability of social support was the factor that best determined success for users of hearing aids. Furthermore, in two studies, the perceived availability of social support was found to be the best predictor of hearing aid satisfaction, outperforming other known predictors of hearing aid satisfaction such as reported benefit from hearing aids, personality, and style of hearing aid (Singh, Lau, & Pichora-Fuller, 2015). Finally, Garstecki and Erler (1998) compared groups of older adults with hearing loss who either accepted or rejected advice from a hearing health care professional to obtain and use amplification, and found that perceived availability of social support was positively associated with hearing aid use. Altogether, the results from these studies suggest that hearing rehabilitation outcomes improve with the perceived availability of social support. It should be noted that to date there is a lack of experimental evidence linking the availability of social support and successful use of hearing instruments.

Several studies have including significant others in aural rehabilitation classes (Hétu & Getty 1991; Hickson, Worrall, Yiu, & Barnett, 1996), but only one study was designed to assess benefits arising from social support (Preminger, 2003). The latter study compared perceived hearing handicap for persons with hearing loss who attended rehabilitation sessions either with or without a significant other. They found that attendance with a significant other resulted in larger reductions in hearing handicap scores than if the classes were attended alone.

Clinical Implications

In this chapter, we have reviewed effects of hearing loss on social relationships, effects of hearing rehabilitation on social relationships, and the effects of social relationships on hearing loss and hearing rehabilitation. Next, we use this information to outline implications for improving clinical practice.

- In light of the increased risk of impaired social relationships experienced by hearing impaired populations, we argue that, to develop a more complete understanding of a particular patient, it is important to inquire about the impacts of hearing loss on their social relationships. This can be achieved using methods ranging from discussions of the topic with the patient and their significant others, the use of formal assessment tools assessing social impacts of hearing loss described earlier in the chapter, and/or novel methodological approaches such as ecological momentary assessment (EMA).
- When assessing social impacts of hearing loss with patients and their families, a deeper understanding of issues can potentially be obtained by inquiring about different domains of social function. Such domains include social activity participation, experiences of loneliness, interpersonal affective perception, perceptions regarding the availability of social support, and effects of hearing loss on interpersonal relationships. Consideration of different risk factors (e.g., age, sex, living arrangements, etc.) can inform these inquiries.
- Persons with hearing loss and their significant other should be made aware that communication in a variety of interpersonal contexts (including interactions with health care professionals) may be vulnerable to effects of hearing loss.
- In light of evidence suggesting that significant others have a critical role in raising awareness about hearing loss and associated effects on patient well-being, and evidence suggesting improved hearing rehabilitation uptake and clinical outcomes when significant others participate in audiological care, it is recommended

that hearing care professionals structure appointments and design interventions that involve significant others (Singh et al., 2016).

Conclusion

Hearing loss is highly prevalent globally, and is associated with negative consequences in economic, physical, cognitive, psychosocial, emotional, and behavioral domains of one's life. Moreover, evidence in hearing research suggests that hearing loss has a deleterious effect on social relationships. Decades of research has established that social relationship factors have a profound effect on well-being, morbidity, and mortality, thus reinforcing the need to address hearing loss and mitigate effects of hearing loss on social relationships. Evidence suggests that hearing rehabilitation has a positive effect on measures assessing social function.

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Part III

THE REHABILITATIVE TOOLBOX: THERAPEUTIC MANAGEMENT

11

BIOPSYCHOSOCIAL APPROACHES TO AUDIOLOGIC COUNSELING

Patient-, Person-, Family-, and Relationship-Centered Care

Sue Ann Erdman

Introduction

The ideal outcome of rehabilitation is successful management of a chronic disease or disability by the individual with the condition. To say that audiologists manage hearing impairment, or that they manage patients with hearing impairment, in reality misses the mark. The audiologist's responsibility is to establish a therapeutic relationship and facilitate adjustment to hearing impairment by engaging patients in the actual management of their hearing problems. Although the semantic differences may seem minor, they are, in fact, focal to the rehabilitation process. An acute health problem typically warrants specific, even immediate medical intervention that is intended to resolve the problem. A chronic condition, the onset of which may be gradual and subtle, persists; indeed, it may be permanent. Incurring a chronic condition, even one as insidious as hearing impairment, can affect one's self-perception. It detracts from one's being "the picture of health" or "whole." Living with a chronic condition requires significant adaptations to minimize the impact on one's quality of life; there

are no quick cures for enabling an individual to do so. Our goal is not merely to test our patients' hearing and fit them with electroacoustically appropriate hearing aids. Our goal is to ensure that our patients are able to cope with the challenges hearing impairment imposes on their daily lives and their relationships. They must adjust to their loss of hearing and to the use of whatever technologies are most appropriate, be it assistive devices, cochlear implants, or hearing aids. They must adjust to the ways in which their self-image is altered. They must make environmental and situational adaptations and develop the skills necessary to manage the communication and adjustment issues that they experience. The outcome of audiologic rehabilitation (AR) will not be successful unless or until they are able to make these adjustments.

This chapter is an examination of the need to shift audiologic practice from the traditional biomedical model of service delivery to a model that better meets the needs of our patients and their families by addressing the impact hearing impairment has on communication, psychosocial functioning and adjustment, and the quality of life. By definition, such a model focuses on the individual and what it

means to live with hearing impairment. The need for this paradigm shift has been echoed repeatedly in audiology (Erdman, 1993a, 2009a; Erdman, Wark, & Montano, 1994; Gagné, 1998; Hickson, 2012; Ross, 1987, 1997). The shift mirrors widespread recommendations and changes occurring throughout health care that are designed to enhance treatment efficacy and outcomes by implementing service delivery models that are whole person rather than disease or impairment centered. These models have evolved, directly or indirectly, from Engel's (1977) biopsychosocial model and include such approaches as patient- and person-centered care, family-centered care, and relationship-centered care. The commonalities among these approaches are many (Beach et al., 2006; Mastro, Flynn, & Preuster, 2014; Smith, Fortin, Dwamena, & Frankel, 2013) and the differences more often relate to their specific goals (Eklund et al., 2018) or to areas of emphasis. Eklund and colleagues suggest that the goal of patient-centered approaches is related to functioning whereas the goal of person-centered approaches is meaningfulness. This is consistent with the emphasis on spirituality that Miles and Mezzich (2011a, 2011b) include in their descriptions of person-centered care. Family-centered care is often advocated for situations involving pediatric and intensive-care patients in which parents' and caregivers' involvement is of vital importance. In audiology practice, the impact of hearing problems on spouses and other family members clearly warrants a family-centered approach; the person with hearing impairment incurs communication and adjustment difficulties, as do the family members. The fallout in terms of the impact of hearing problems on marital relationships and family dynamics constitutes yet a third dimension that counseling must often address. Considering the importance of the therapeutic relationship, relationship-centered care is also highly relevant (Kirkegaard & Ring, 2017). As Beach and colleagues (2006) indicate, relationship-centered care enhances patient-centered care, the relationship between practitioners and the patients and families they serve, the relationship among members of the health care team, and the practitioner's relationship with themselves by enhancing self-awareness, resilience, and resourcefulness. In this chapter, and throughout this entire book, the focus is on the valuable contributions that these biopsychosocial approaches are making to practice in rehabilitative audiology.

In its landmark publication, *Crossing the Quality Chasm: A New Health System for the 21st Century*, the Institute of Medicine identifies quality health care as that which is safe, effective, patient centered, timely,

efficient, and equitable (IOM, 2001). Specifically, patient-centered care is "respectful of and responsive to individual patient preferences, needs, and values and ensures that patient values guide all clinical decisions" (p. 6) and "encompasses qualities of compassion and empathy" (p. 48). Passage of the 2010 Patient Protection and Affordable Care Act and creation of the Patient-Centered Outcomes Research Institute (PCORI) are further evidence of the major shift toward a more patient-centered focus throughout health care. This evolution in health care is prompting changes that place increasing emphasis on patient satisfaction and treatment outcomes.

As we realign our thinking toward an approach wherein the individual's experience of hearing impairment as a chronic, often life-altering condition is the focus of intervention, we must ask ourselves what this paradigm shift entails and what is required to implement it. The answer, simply stated, is counseling. Counseling enhances benefit from and satisfaction with all aspects of intervention. It influences adherence to treatment recommendations and the realization of treatment goals. It is the means by which we facilitate patients' realization of the confidence and skills they need to manage their hearing problems effectively. It empowers. It instills hope and the belief that one can cope. It enables patients to eliminate or reduce the limitations and restrictions that they experience secondary to hearing impairment. It establishes the audiologist as a credible and valued ally in minimizing the effects of hearing impairment on communication, psychosocial functioning, and quality of life. At times counseling may involve empathizing, reassuring, validating, explaining, or motivating. At other times, it may simply entail listening, understanding, and appreciating. More than any other factor and regardless of the form it takes, the effectiveness of our counseling influences the opinion patients have of our competence, our professionalism, and our commitment to their well-being.

Although counseling often is viewed as a specific clinical activity, it can be appreciated more fully when considered in its more complex nature as it is throughout this chapter—as a process and as the context in which all of audiology services are provided. Appreciation of this broader conceptualization of counseling is evident in efforts of the World Health Organization's International Classification of Functioning, Disability, and Health (WHO-ICF, 2001) (Danermark et al., 2010; Gagné, Jennings, & Southall, 2009; Hickson & Scarinci, 2007). These inroads to a biopsychosocial, patient-centered approach in audiology are indicative of clinicians' growing awareness

of the need to engage patients in the management of their hearing problems to maximize the benefits of audiology intervention. They are also indicative of the need to acknowledge, accept, and embrace counseling, and our role and responsibilities in it, because ultimately, it is our most important professional skill.

Health Care: Crisis and Change

... nothing will change unless or until those who control resources have the wisdom to venture off the beaten path of exclusive reliance on biomedicine as the only approach to health care.

—George L. Engel (1977a, p. 135)

The need to view counseling in audiology from a different perspective is understood best when examined in the context of changes occurring throughout health care. Historically, research and development, and the allocations of resources in health care, have been disproportionately focused on technological advances. At the same time, there have been glaring shortages in preventive and primary medical care, and woeful inadequacies in how chronic illness and disability are addressed, particularly in patient education and counseling. These shortages and inadequacies have led to unacceptably high rates of nonadherence to treatment recommendations creating a vicious cycle that includes:

- treatment failures;
- spurious results in outcome research;
- deteriorations in health, functioning, and quality of life;
- unnecessary medical tests and hospital readmissions;
- increased demands on limited resources; and
- escalations in health care costs.

In short, a health care system that has not focused adequately on chronic illness and disability has been drained by those very problems.

It is impossible for practitioners and patients to avoid reminders of the challenges and changes that health care is facing. The financial and political implications of these issues continue to be the subject of endless debate. What is abundantly clear, however, is the fact that chronic disease and disability cannot be addressed adequately by a system that is structured to address acute conditions. At the same time, the aging population and advances in acute care are contributing to an ever-increasing number of individuals with

one or more chronic health conditions. The financial and human costs of these exigent circumstances, particularly the multiple comorbidities associated with increased longevity, have long demanded new approaches to health care. Collaborative/interprofessional health care teams, accountable care organizations, patient-centered medical homes, and increased emphasis on healthy lifestyles and prevention are but a few of the changes stemming from the obvious need to focus on chronic health conditions. Growing emphasis on person and patient-centered care offers hope that education and counseling to promote self-management of chronic disease and disabilities will become more prevalent throughout health care, particularly when patient satisfaction and the benefits of such interventions become more evident.

Why has health care been so ill prepared to address chronic illness and disability? The answer is literally centuries old. Medical science and practice have long been anchored in the biomedical model, a mechanistic product of *dualism* and *reductionism*. Dualism, in philosophy of mind, refers to the belief that mind and body are separate and independent entities. The church adamantly upheld this view through the 16th and 17th centuries because the mind was perceived to be soul-related and, therefore, strictly in God's purview. The church, however, did permit study and dissection of the body. As a result, for centuries, the focus of medical science and practice in Western medicine has been on the human body. From the perspective of reductionism, all matter and phenomena are reduced to their parts, and those parts, in turn, are reduced to their parts. Hence, molecular biology has emerged as the science underlying modern medicine. Simply stated, the biomedical model has entailed assuming a detached perspective, finding the broken part, and fixing it.

Fueled by the Industrial Revolution, medicine made, and has continued to make, rapid advances. Not surprisingly, high-tech, cutting-edge medical care has come to involve increasingly sophisticated treatment, repair, or removal of diseased, impaired, or malfunctioning parts. The biomedical model has been instrumental in eradicating many heretofore incurable diseases, in developing previously unthinkable surgical procedures, and in the discovery of miraculous drugs. Within this model, however, physicians have been trained to disregard patients' subjective accounts of illness; to be detached, objective observers; and to use only "reliable and valid" technical procedures and laboratory measurements. From this perspective, only observable, objective phenomena have been deemed relevant. Patients have become

“cases.” Symptoms and variables that do not fit the biomedical model have been ignored because they do not constitute medical issues. Consequently, the biomedical model is responsible for the inadequacies and failures in treating chronic conditions in which subjective, psychosocial, and behavioral phenomena figure prominently. It is not surprising, therefore, that the biomedical model has not been particularly successful in preventing or managing hypertension, asthma, chronic obstructive pulmonary disease (COPD), diabetes mellitus, chronic pain, addictions, or eating disorders—all of which are influenced by cognition, affect, behavior, and environment. Whereas disease and impairment have been focal, the experiences of illness and disability have been discounted. Consequently, the biomedical model has failed to provide viable mechanisms with which to address issues such as disablement, coping skills, nonadherence to treatment recommendations, lifestyle, health beliefs, hypochondriasis, illness behaviors, defense mechanisms, or any of the myriad societal factors that impinge upon health such as illiteracy, poverty, and other factors that limit access to care. Not surprisingly, psychiatry, which could provide valuable insights into many of these issues, has often been viewed disparagingly by other medical specialties that question whether psychiatry actually constitutes medicine. The enormous impact that professionals in the rehabilitation fields and health psychology could have in rectifying this situation also has been limited by the health care system’s structural and financial adherence to the biomedical model. Increased focus on prevention, health maintenance, and patient-centered care is a welcome sign that health care delivery is being reoriented to address these increasingly prevalent and complex, chronic health issues. Obstacles to such changes, including those as insidious as institutional resistance, highlight the need for creative strategies and systemwide incentives to implement policies and procedures that promote a paradigmatic shift toward a biopsychosocial approach that is patient/person and relationship centered.

The Biopsychosocial Model in Health Care Today

... the ongoing process of “becoming biopsychosocial” develops mental suppleness, diagnostic agility, thoughtful approaches to therapeutics, and a holistic vision; it provides a focus for finding meaning in clinical practice and pathways to strong relationships with patients

—Epstein and Borrell-Carrió (2005, p. 430)

George L. Engel (1913–1999), an internist and psychiatrist whose career at the University of Rochester Medical Center spanned five decades, argued passionately that health care has long outgrown the utility of the biomedical model. He faulted the scientific limitations of reductionism and dualism in understanding health and illness, and deplored the objectification and the disempowerment of the patient inherent in the biomedical approach. Engel also viewed the model as inherently flawed because it cannot account for individual differences in perceived “illness,” the subjective experience of disease. As Engel (1977a) repeatedly pointed out, illness cannot be understood without knowing why this particular patient is presenting in this particular way at this particular time. The issue of individual differences is critical when it comes to patients’ experience of disability; that is, the subjective experience of impairment and its associated limitations and restrictions. Research in audiology has demonstrated the effects of individual differences in a number of ways. For example, studies spanning a half-century have consistently shown that although audiometric measures and self-reported hearing problems are correlated, one cannot predict a patient’s communication and adjustment difficulties or quality of life from the audiogram (Brainerd & Frankel, 1985; Erdman & Demorest, 1998b; Gopinath et al., 2012; Hallberg, Hallberg, & Kramer, 2007; Hannula et al., 2011; Hawes & Niswander, 1985; Helvik et al., 2006; Hideki, Kyoko, & Eiji, 2004; High, Fairbanks, & Glorig, 1964; Hornsby & Kipp, 2016; Kielenen & Nerbonne, 1990; Kramer, Kapteyn, Festen, & Tobi, 1996; Rowland, Dirks, Dubno, & Bell, 1985; Speaks, Jerger, & Trammel, 1970; Weinstein & Ventry, 1983a, 1983b). Furthermore, the decision to seek audiologic services is correlated more strongly with self-reports of hearing problems than it is with the audiogram (Brink, Wit, Kempen, & Heuvelen, 1996; Laplante-Levesque, Hickson, & Worrall, 2012; Swan & Gatehouse, 1990). Indeed, the individual variability in difficulties experienced secondary to hearing impairment is one of audiology’s most well documented phenomena.

Engel’s alternative to the biomedical model, the biopsychosocial model is based on general system theory (von Bertalanffy, 1969) rather than dualism and reductionism and is widely viewed as one of the most influential developments of 20th-century medicine. As just one indicator of this impact, a recent check of Google Scholar reveals that Engel’s seminal article, *The Need for a New Medical Model: A Challenge for Biomedicine* (1977a), has been cited well over 13,000 times. Borrell-Carrió, Suchman, and Epstein (2004) succinctly describe Engel’s contribution as having broadened

“... the scope of the clinician’s gaze” (p. 581). In Engel’s mind, he was simply presenting a scientific model that includes what is left out of the biomedical model: the patient as a person. In the biopsychosocial model, biological, psychological, and social systems are integral and interactive elements of health and illness that are implicated in the diagnosis and, importantly, in the prognosis. The patient’s “humanness” influences and is influenced by his or her biological and social systems. It was clear to Engel that effective mechanisms for managing chronic health conditions will not emerge until the fixated, narrow focus on one specific aspect of care—be it the disease or the impairment—is replaced by an all-encompassing recognition of the complexity inherent in human experience. A giant in the field of rehabilitation, Franklin Shontz (1977), eloquently summarized this point:

Of all the factors that affect the total life situation of a person with a disability, the disability itself is only one, and often its influence is relatively minor. This implies, in the final analysis, that the understanding of psychological reaction to physical disability requires the understanding of individual human beings in all of their complexity (p. 210).

The key aspects of the biopsychosocial approach as proposed by Engel (1977a, 1977b, 1980) include the following:

- a view of the patient as a whole; a person whose fundamental nature is at once biological, psychological, and social;
- a triadic process of observation, introspection, and dialogue through which the patient’s subjective experiences become scientific data;
- a clinical interview in which the patient’s narrative is allowed to unfold without interruptions and with minimal prompting or interrogation;
- a practitioner-patient relationship that fosters shared and complementary communication and responsibilities;
- a mutual understanding of the patient’s narrative that ensures inclusion of his or her perceptions and experiences in the assessment and diagnostic process;
- patient engagement in the treatment process and plans intended to alleviate or resolve perceived illness or disability; and
- system theory rather than reductionism as the approach to analyzing and understanding health and illness.

As shown in Figure 11–1, systems theory facilitates an understanding of illness and disease through a hierarchical ordering of systems along a continuum ranging from the molecular level to the societal level. This framework allows both intra- and intersystem study. The person, who is the focus, can be viewed as the end of the organism/biologic continuum and the beginning of the societal/social continuum. These systems can also be represented as a nested, concentric progression from the molecule at the center to the biosphere as the outermost layer. When viewed as a nested continuum, it is evident that each system

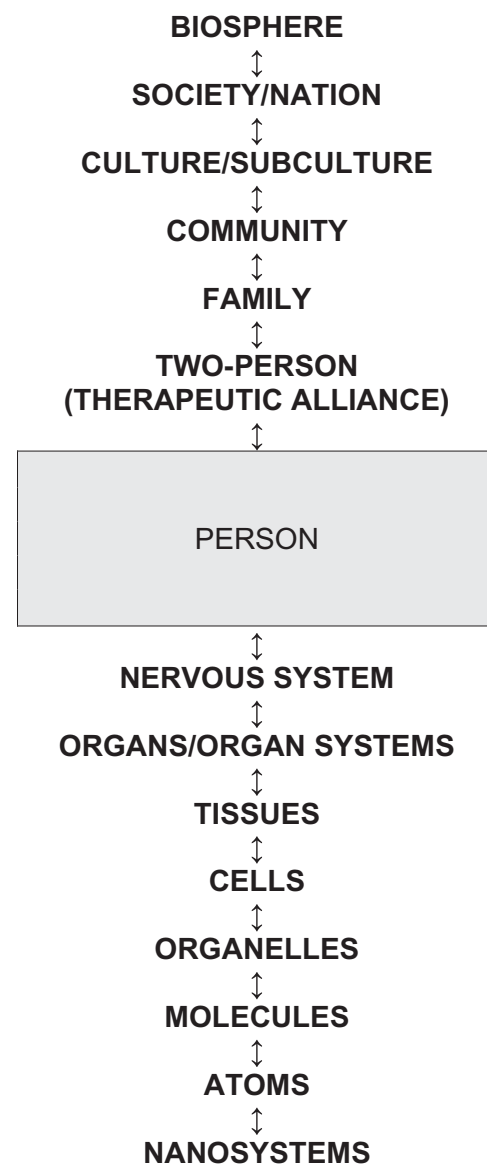


Figure 11–1. Hierarchy of levels of natural systems organization. *Source:* Adapted from Engel (1980).

is both a whole and a part of a larger system. How a person idiosyncratically experiences the effects of these internal and external systems governs the way in which those internal and external systems are affected in return.

The interdependencies and interactions among multiple systems that are implicated in pain, for example, cannot be examined or studied within the biomedical framework. Research has demonstrated that pain-related fear, hypervigilance, and catastrophizing play a significant role in the transition of acute and subacute pain to chronic pain (Crombez, Vlaeyen, Heuts, & Lysens, 1999; Goubert, Crombez, & van Damme, 2004; Peters, Vlaeyen, & Weber, 2005; Vlaeyen, Crombez, & Goubert, 2007; Vlaeyen & Morley, 2005). Indeed, comprehensive reviews confirm that emotions are primary factors in the conceptualization, assessment, and treatment of persistent pain (Gatchel, 2004; Gatchel & Turk, 2008; Lumley et al., 2011). Patients' appraisals of their pain (i.e., the meaning they ascribe to their pain) and their capacity to manage fears of reinjury or exacerbation of pain contribute to the development of chronic pain (Turk & Okifuji, 2002; Vlaeyen, Crombez, & Linton, 2016). The more traditional biomedical approach to managing pain, prescription painkillers, has been blamed for contributing to addiction problems, which has added even more credence to the biopsychosocial approach to pain management (Bonnie, Schumacher, Clark, & Kesselheim, 2019; Gourlay, Heit, & Almahrezi, 2005; Passik & Kirsh, 2008). The benefits of applying a biopsychosocial approach to pain management following injury were made abundantly clear in a Colorado experiment in which the state mandated a "biopsychosocial law" to control worker's compensation costs (Bruns, Mueller, & Warren, 2012). Briefly, the goal was to provide a biopsychosocial intervention that included rehabilitation and psychological intervention for the treatment of work-related injuries, especially when there were signs that the injury could become disabling. Colorado achieved a marked reduction in the inflation rate of medical and disability costs. Other states are incorporating aspects of this plan and it has become a fascinating case study for health care reform.

The increasing presence and influence of the biopsychosocial model can be seen throughout medicine and the allied health professions including those in the social and behavioral services (Wade & Halligan, 2017). Wampold, Ahn, and Coleman (2001) maintain that there is no empirical support for the biomedical model in the counseling professions because the

person must be understood in relation to his or her particular circumstances. Sensitivity to individual and cultural differences is contingent upon a model that espouses a person in context approach. Suls and Rothman (2004) emphasize the particular relevance of the biopsychosocial model for health psychology and advocate emphasis on interprofessional collaboration and on efforts to expand research promoting the multilevel, multisystem, and multivariate nature of health. Anchin (2008), in a thought-provoking treatise on the pursuit of a unifying paradigm for psychotherapy, suggests that the biopsychosocial model "creates an organizing framework for systematically interrelating accumulating knowledge about biological, psychological, and social subsystems in relation to personality, psychopathology, and psychotherapy" (p. 328). Meyer and Melchert (2011) echo this point, stressing the importance of adhering to the biopsychosocial framework for mental health intake assessments to guide not only diagnosis but also treatment planning, implementation, and monitoring. On an even more fundamental level, it is impossible not to appreciate the relevance of the biopsychosocial framework and system theory in view of continuing developments in the neurosciences. Cognitive therapy, meditation, and even repeated practice of a skill are implicated in neuroplasticity and other observable changes in neuroanatomy (Garland & Howard, 2009). Ongoing and, at times, controversial research continues to delineate the possible role of mirror neurons in empathy, with potential implications for understanding social skills deficits in disorders such as autism and schizophrenia (Iacoboni, 2009; Pineda, 2010).

At about the same time that Engel's biopsychosocial approach was being introduced, renowned psychologist Albert Bandura, whose work also has important implications for audiologic counseling, proposed the principle of reciprocal determinism (1978). According to this principle, psychological functioning includes reciprocal interaction of behavioral, personal inner experiential (e.g., cognitions, perceptions, and affect), and environmental variables as shown in Figure 11-2. Bandura's and Engel's models both feature bidirectional effects and influences from and on the external world; that is, the environment or society. Reciprocal determinism and the biopsychosocial approach have been transformative developments in our understanding of human health and functioning.

As opposed to a rigid, testable model, the biopsychosocial model is best viewed as an overarching conceptualization within which clinical and research endeavors in health care can be couched. Just as the

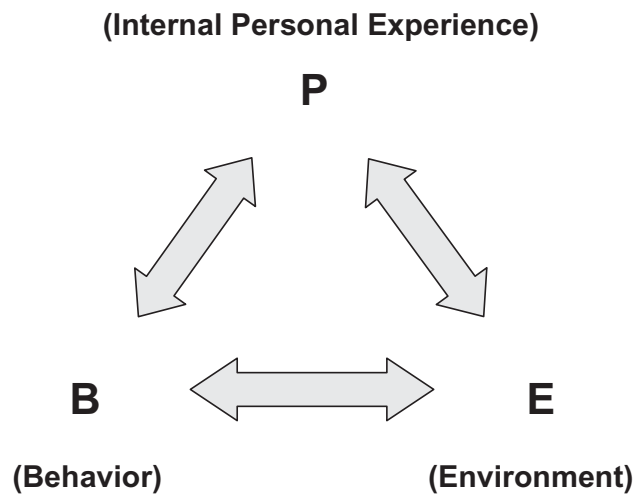


Figure 11-2. Principle of reciprocal determinism. *Source:* Adapted from Bandura (1978).

term *biomedical model* implies a particular approach to health care, a biopsychosocial approach should be considered a different approach within which knowledge gleaned from the social and behavioral sciences is as meaningful and important as that derived from the biologic and physical sciences. The biopsychosocial model evolved as a function of Engel's own life experiences (see Engel, 1996). Nonetheless, Engel insisted that the model simply reflects good practice; others insist his practice epitomized the good doctor (Smith & Strain, 2002). One cannot fully appreciate the changes occurring throughout health care, particularly the development of patient-, person-, family-, and relationship-centered approaches to care, without understanding the biopsychosocial model. The confluence of these movements and the degree of concordance in their focus on the patient's experience and on interpersonal interactions are guiding the transformation health care is undergoing today. Skeptics, who initially thought the biopsychosocial model was too complicated, have come to value the more complex ways in which disease and health can be examined. As a result, the biopsychosocial model is making inroads throughout health care that have triggered an unprecedented outgrowth of discoveries and innovations, which, in turn, are having a profound impact on the provision and outcome of treatment. As Smith and colleagues (2013) stress, patient-centered methods that embody the biopsychosocial approach are making health care more scientific and, at the same time, more humanistic, the ultimate goal of Engel's biopsychosocial model.

Patient-Centered Care: Hearing the Patient's Story

The practice of listening to patients' stories and responding sensitively to them provides an opportunity for more clinically effective, as well as more empathic, care.

—Garden (2010, p. 210)

Engel's emphasis on the patient's narrative is but one facet of the biopsychosocial approach that has had considerable influence on clinical routines. Engel believed passionately that the narrative, the patient's experience, and the systems relevant to that experience should be heard and that patients' stories are "naturally" disclosed. Seaburn (2005), who worked with Engel, has found that attentively listening to patients' stories allows the elements of the biopsychosocial approach to emerge on their own. The key in allowing this to occur lies in truly listening. Many clinicians find it difficult to refrain from interrupting. In one study, the mean length of time before physicians interrupted patients was a mere 18 seconds (Beckman & Frankel, 1984); another study found a mean length of 23 seconds (Marvel, Epstein, Flowers, & Beckman, 1999). Some practitioners resist the narrative approach because they believe that a standardized intake form saves time. Studies have shown, however, that once one has developed skill with the biopsychosocial interview, consultations run less than a minute longer than the more traditional intake interview (Margalit, Glick, Benbassat, & Cohen, 2004; Stewart, Brown, & Weston, 1989). On the plus side, the investment of this additional time reportedly results in greater patient satisfaction (Charlton, Dearing, Berry, & Johnson, 2008; Epstein et al., 2003) as well as in the need for fewer referrals, prescriptions, and diagnostic tests (Margalit et al., 2004; Margalit & El-Ad, 2008; Soler & Okkes, 2012), all of which have implications for long-term cost containment. Regular intake forms tap standard symptoms; an individual's narrative is more likely to reveal pertinent details and underlying factors—particularly affective, cognitive, and behavioral variables—that are implicated in his or her condition and situation. Insufficient specific information can impede or derail diagnoses, interfere with the patient's ability to feel understood and, ultimately, have a negative impact on outcome. The IOM (2001) affirms that communication lapses and relationship obstacles can contribute to preventable harm and even death.

The patient's narrative is the key to establishing open communication and an empathic, trusting,

therapeutic relationship between the practitioner and patient. The importance ascribed to these stories can be found in the increasing attention being given to narrative health care (Brown, Thorton, & Stewart, 2012; Charon, 2006; Charon et al., 2016; Engel, Zarconi, Pethtel, & Missimi, 2008; Meza & Passerman, 2011). The invaluable content and the multiple functions of patient narratives, as highlighted by Greenhalgh and Hurwitz (1999) (see Sidebar 11–1) underscore the critical nature of effective listening, reflective practices, and interpretive skills for health care providers. Indeed, interpersonal and communication skills are now requisite competencies for medical school graduates (Accreditation Council for Graduate Medical Education, 2012). Reflective writing and narrative

study are being stressed in an ever-increasing number of medical and allied-health professions' academic programs to help students appreciate the meaning and significance of illness and disability in patients' lives. In 2000, Rita Charon, MD, established the Program in Narrative Medicine at Columbia University, which now offers a master's degree. While many programs have incorporated standardized patients into their training experiences, other programs such as the University of Michigan's Family Centered Experience (Kumagai, 2008; Kumagai, White, & Schigelone, 2005; White, Perlman, Fantone, & Kumagai, 2010) match students with actual patients and emphasize both patient and student narratives to explore the experience of illness and, importantly, the complexities of patient-practitioner interactions.

The emphasis on the patient's story and perspective, and on the subsequent importance of practitioner's communication and interpersonal skills, has also resulted in the development of numerous approaches to conducting clinical interviews. The following are among those that have had particularly widespread use.

Sidebar 11–1. Why Study Narrative?

For diagnostic purposes, narratives:

- are the phenomenal form in which patients experience disease and disability,
- encourage empathy,
- promote understanding between clinician and patient,
- allow for the construction of meaning, and
- supply useful analytical clues and categories.

In therapeutic processes, narratives:

- encourage a holistic approach to management,
- are intrinsically therapeutic or palliative, and
- often suggest or precipitate additional therapeutic options.

In educating patients and health care professionals, narratives:

- are often memorable,
- are grounded in experience, and
- encourage reflection.

For research purposes, narratives:

- help set a patient-centered agenda and
- help generate hypotheses.

Source: Adapted from Greenhalgh & Hurwitz, 1999.

- Calgary-Cambridge Observation Guides (Kurtz & Silverman, 1996; Kurtz, Silverman, Benson, & Draper, 2003; Silverman, 2007; Simmenroth-Nayda, Heinemann, Nolte, Fischer, & Himmel, 2014)
- Smith's Patient-Centered Interviewing Method (Fortin, Dwamena, Frankel, Lepisto, & Smith, 2019; Smith, 1996)
- Four Habits Model (Frankel & Stein, 1999; Gulbrandsen et al., 2008; Krupat, Frankel, Stein, & Irish, 2006; Lundebj, Gulbrandsen, & Finset, 2015; Stein, Frankel, & Krupat, 2005)
- Kalamazoo Essential Elements Checklist (Duffy et al., 2004; Joyce, Steenbergh, & Scher, 2010; Makoul, 2001)
- Patient-Centered Clinical Method (Stewart et al., 2013)

These approaches vary widely in length, detail, specific areas of content, and the extent to which they are amenable to teaching and assessing clinicians' communication skills (Duffy et al., 2004; Hullman & Dailey, 2008; Schirmer et al., 2005). All, however, clearly have a biopsychosocial scope and emphasize the need to (a) elicit the patient's perspective, (b) provide an empathic response, and (c) engage the patient in the decision-making process. As an example, the basic elements of the Four Habits Model are shown in Table 11–1. Each of the four habits consists of specific

Table 11–1. Basic Elements of the Four Habits Model

HABIT	SKILL
Invest in the beginning	▪ Create rapport quickly ▪ Elicit the patient's concerns ▪ Plan the visit with the patient
Elicit the patient's perspective	▪ Ask for patient's/family's ideas about problem ▪ Elicit patient's specific requests ▪ Explore impact on patient's life/work/family/activities
Demonstrate empathy	▪ Be open to patient's emotions ▪ Make an empathic statement ▪ Convey empathy nonverbally
Invest in the end	▪ Deliver diagnostic information ▪ Provide information/education/counseling ▪ Involve patient in decision making ▪ Complete the visit

Source: Adapted from Frankel and Stein (1999).

communication tasks or skills that can be observed and assessed. With its companion Four Habits Coding Scheme (Krupat, Frankel, Stein, & Irish, 2006), the Four Habits Model has been used for many years throughout the Kaiser Permanente system and elsewhere to train and evaluate students and practicing clinicians (Gulbrandsen et al., 2008; Stein, Frankel, & Krupat, 2005; Jensen et al., 2011). Richard M. Frankel, Robert C. Smith, and their colleagues have long argued passionately for many of the changes occurring throughout health care (Fortin, Dwamena, Frankel, Lepisto, & Smith, 2019; Frankel & Quill, 2005; Frankel, Quill, & McDaniel, 2009; Smith, 1996, 2002; Smith, Dwamena, Grover, Coffey, & Frankel, 2011). They have figured prominently in promoting the patient- and relationship-centered approaches, which, in their view, represent health care's incorporation of the biopsychosocial model.

Patient-centered care has become somewhat of a generic term for many of the transformations occurring in health care; however, the need to define it systematically has been addressed repeatedly. Mead and Bower (2000) cite five dimensions of patient-centered care:

- a biopsychosocial perspective,
- the “patient as person,”
- sharing power and responsibility,
- the therapeutic alliance, and
- the “practitioner as person.”

The IOM (2001) defined patient-centered care as “respectful of and responsive to individual patient preferences, needs, and values, and ensuring that patient values guide all clinical decisions” (p. 6). Stewart (2001), in response to concerns that perhaps not all patients prefer a patient-centered approach, or need all aspects of such an approach, clarified that patient-centered clinical practice is “a holistic concept in which components unite and interact in a unique way in each practitioner-patient encounter” (p. 445). Since those descriptions, there have been continuing efforts to refine and expand the definition of patient-centered care. Epstein, Fiscella, Lesser, and Stange (2010), for example, point out six measurable aspects of patient-centered care:

- fostering a healing relationship,
- exchanging information,
- responding to emotion,
- managing uncertainty,
- making decisions, and
- enabling self-management.

Epstein and colleagues advocate a national health care policy that “helps health professionals acquire and maintain skills related to patient-centered care. It also should encourage organizations to move toward a culture of patient-centeredness. In turn, these organizations should support patients in self-management and shared decision-making” (p. 1494).

Efforts to operationalize person-centered care as a standardized clinical method have also been ongoing for a number of years at the University of Western Ontario (Brown, Stewart, McCracken, McWhinney, & Levenstein, 1986; Levenstein, McCracken, McWhinney, Stewart, & Brown, 1986; McWhinney, 1986; Stewart, Brown, Levenstein, McCracken, & McWhinney, 1986; Stewart et al., 2013). Drawing from Engel's model, Rogerian person-centered counseling (Rogers, 1951), Balint group experiences (Balint, 1969, 2000) and other approaches, Stewart and colleagues (2013) developed and implemented a clinical process composed of six interactive components:

- exploring the disease and the illness experience,
- understanding the whole person,
- finding common ground,
- incorporating prevention and health promotion,
- enhancing the patient-provider relationship, and
- being realistic.

Hudon et al. (2012) suggest that validating the illness experience, acknowledging and capitalizing on patients' expertise, and adding hope are other dimensions that should be included in patient-centered care, particularly when chronic conditions are involved.

The relationship-centered approach has also received considerable attention (Beach, Inui, & Relationship-Centered Care Research Network, 2006; Frankel, Eddins-Folensbee, & Inui, 2011; Frankel & Quill, 2005; Nundy & Oswald, 2014; Roter, 2000; Schoenthaler, Hassan, & Fiscella, 2019; Suchman, 2005; Tresolini & Pew-Fetzer, 1994; Williams, Frankel, Campbell, & Deci, 2000). According to Beach and colleagues (2006), relationship-centered care is founded upon four principles:

- Relationships in health care should include the personhood of all participants.
- Affect and emotion are important components of these relationships.
- All health care relationships occur in the context of reciprocal influence.
- Forming and maintaining genuine relationships in health care is morally valuable.

Relationship-centered care is viewed, to some extent, as an expansion of patient-centered care, but it is also a direct outgrowth of the biopsychosocial model (Smith,

2002; Suchman, 2006). Some proponents of a patient-centered approach suggest that the relationship-centered approach stresses the interaction rather than the goal of care (Brown, Stewart, Weston, & Freeman, 2003; Churchill, 1997). In their report on health professions education and relationship-centered care, however, Tresolini and the Pew-Fetzer Task Force (1994) emphasized the fact that the therapeutic relationship is among the most salient variables in health care:

This relationship is a medium for the exchange of all forms of information, feelings, and concerns, a factor in the success of therapeutic regimens, and an essential ingredient in the satisfaction of both patient and practitioner. For patients, the relationship with their provider is frequently the most therapeutic aspect of the health care encounter. (p. 9)

Without discounting any aspects of the patient-centered approach, Roter (2000) concurred with the Pew-Fetzer report acknowledging that mutuality and reciprocity, as hallmarks of the relationship-centered approach, are optimal throughout health care. Moreover, she contends, the emphasis on mutuality and interaction avoids the potential implications of paternalism and consumerism sometimes associated with the traditional biomedical model and the patient-centered approach respectively. The quality of practitioner-patient relationships reportedly influences patient satisfaction, adherence to treatment recommendations including medication use and behavioral change, and continued health care utilization (Williams et al., 2000). In the field of physical therapy, studies indicate that the therapeutic relationship is positively associated with: (a) treatment adherence in patients with brain injury, (b) depressive symptoms in patients with cardiac conditions and those with brain injury, (c) treatment satisfaction among patients with musculoskeletal problems, and (d) physical function in geriatric patients and those with chronic low back pain (Hall, Ferreira, Maher, Latimer, & Ferreira, 2010).

Nowhere has the significance of the therapeutic relationship in treatment outcome been more clearly established than in the realm of counseling (Elvins & Green, 2008; Gelso, 2018; Gelso, Kiviglian, & Horvath, 2006; Norcross, 2011; Norcross & Wampold, 2011). Within the practitioner-patient relationship, practitioner variables have more of an effect on treatment outcome than do patient variables (Baldwin, Wampold, & Imel, 2007; del Re, Flückiger, Horvath, Symonds, & Wampold, 2012; Holdsworth, Bowen, Brown, & Howat,

2014; Zuroff, Kelly, Leybman, Blatt, & Wampold, 2010), which places the responsibility for establishing effective therapeutic relationships squarely on the practitioner's shoulders. This fact has innumerable implications for clinical training of those in counseling roles and for the continued professional development of current clinicians.

Proponents of relationship-centered care contend that it enhances patient-centered care as a result of the emphasis given to mindful practice and clinicians' self-awareness (Dobie, 2007; Frankel, Eddins-Folensbee, & Inui, 2011; Frankel & Quill, 2005; Krasner et al., 2009; Smith, Dwamena, & Fortin, 2005). As Dobie (2007) observes, relationship-centered care "orients us to the importance of self-awareness and mutuality, which enrich us beyond our clinical practice" (p. 432). Of added significance, the relationship-centered approach encompasses relationships beyond the practitioner-patient dyad. Involvement of family members is clearly envisioned as critical, particularly those in caretaker roles or in positions to provide autonomy support (Williams et al., 2000). Additionally, clinician-clinician relationships are becoming increasingly important as the emphasis on interprofessional practice, collaboration, and teamwork becomes more widespread (D'Amour, Ferrada-Videla, San Martin Rodriguez, & Beaulieu, 2005; Dix, Steggle, Baptiste, & Risdon, 2008; Gaboury, Lapierre, Boon, & Moher, 2011; Gittell, Godfrey, & Thistlethwaite, 2013; Nester, 2016).

The biopsychosocial model, along with patient-, person-, family-, and relationship-centered care approaches, represent somewhat philosophical, theoretical albeit observable, even measurable ideals that are being promoted throughout health care. At the same time, related but more pragmatic changes are unfolding that will have a direct impact on the provision of health care services. These other changes, also aimed at improving health care, focus specifically on quality management and quality processes. The variable that will constitute the foundation of this wave of change is the patient-reported outcome (PRO). In short, PROs are becoming the mechanism for determining the effectiveness and the efficiency of treatments and providers. The IOM (2001) and the Patient Protection and Affordable Care Act (ACA) (2010) both call for patient-centered care and emphasize the importance of engaging patients in shared decision making to determine treatment goals and strategies and enhance care. Inherent in this process is the monitoring of progress and outcomes (Lavalley et al., 2016). The ACA mandated creation of the PCORI, which is testimony to the central role PROs now have in health care analyses. This broad-based utilization

of PROs is necessary to ensure that care of individuals translates to populationwide improvements in health care quality and to long-term cost containment (Bobbit, Cate, Beardsley, Azocar, & McCulloch, 2012). The most commonly utilized PRO is patient satisfaction with provider care and treatment. Inasmuch as the literature indicates that the biopsychosocial approach, as manifested in patient-centered care is associated with patient satisfaction (Cvengros, Christensen, Cunningham, Hillis, & Kaboli, 2009; Margalit et al., 2004; Sherbourne, Hays, Ordway, DiMatteo, & Kravitz, 1992; Stewart et al., 2000) and, importantly, with cost containment (Bertakis & Azari, 2011; Bruns, Mueller, & Warren, 2012; Stewart et al., 2000; Stewart, Ryan, & Bodea, 2011), the rationale for implementing these approaches in all of health care is abundantly clear. Interestingly, Sherbourne and colleagues (1992) found that although patient satisfaction with the interpersonal aspects of care is predictive of adherence to treatment recommendations, satisfaction with the technical aspects of care is not. As mentioned earlier, the costs associated with nonadherence to treatment recommendations are staggering; hence, efforts to improve adherence are critical. Clearly, excellence in technical skills, albeit important, is not sufficient to ensure patient satisfaction, adherence to treatment recommendations, or reductions in health care costs. The fact that significant advances in technology have not given rise to comparable advances in the management of chronic health problems that call for patient self-management is testimony to this fact. Given the role that the interpersonal aspects of care have in patient satisfaction and long-term adherence to treatment regimens, it behooves all clinicians to maximize their interpersonal skills by incorporating those critical aspects of person- and relationship-centered care; in short, by adhering to a biopsychosocial rather than a biomedical approach to service provision.

The Need for a Paradigm Shift in Audiology

It is not that audiology ever explicitly abandoned A/R . . . We just do not do it very much or manage to get it rewarded very well. . . If we want to keep in touch with our roots, and have a future as an independent, self-supervised profession, evaluating and managing the communicative and psychosocial impact of hearing loss has to be our core purpose, our reason for being.

—Mark Ross (1997, p. 14)

Despite its origins as a rehabilitation endeavor, audiology has become entrenched in the biomedical model. To consider how we can, with the rest of health care, undo the constraints imposed by the biomedical model and adopt a more patient-centered approach, it is important to recognize how firmly ingrained the biomedical model has become throughout health care and how its dogmatic effects have been resistant to change. Hewa and Hetherington (1995) suggest that the biomedical model's survival can be attributed to the fact that adoption of a broader model could erode the authority of medical practitioners. Resistance to the addition of social and behavioral sciences into medical school curricula has also been viewed as a sign of vested interest in perpetuating a professional hierarchy that precludes a team approach to the management of illness and disability (Carr, Emory, Errichetti, Johnson, & Reyes, 2007; Dobie, 2007). Additionally, the underlying presumption is still a mechanistic approach; that is, to fix whatever is broken. The health care industry has produced equipment, devices, and pharmaceuticals to diagnose and treat virtually every imaginable ailment. Moreover, some believe that if there is a problem the biomedical model has not yet solved, it will certainly be able to do so in the future. This mechanistic approach tends to be more prevalent in some medical specialties than in others. Family practice, pediatrics, rehabilitation medicine, and psychiatry, for example, are more patient oriented than are pathology, surgery, or anesthesiology.

Biomedical thinking is found in many aspects of audiology. Diagnostics rule. From the days of the tuning fork, our armamentarium of diagnostic technologies has come to include computerized and automated audiometers, tympanometers, ABR equipment, ENG equipment, OAE systems, real ear measurement and hearing aid test systems, computer administered self-assessments, and video-otoscopy and microscopy. Treatment-wise, the list is equally high-tech: remote-controlled hearing aids, in-the-ear hearing aids, behind-the-ear hearing aids, and CROS and BICROS hearing aids; various bone-conduction implantable devices with FM, infrared, Bluetooth, and wireless technologies; telehealth; video- and computer-administered training; and an ever-increasing array of assistive technologies. A true assembly line of technologies awaits the patient who comes to see an audiologist. The phrase "high tech, low touch" epitomizes much of audiologic practice.

The importance of patients' perceptions of their hearing difficulties is not something to which audiologists have been oblivious. During the 1980s and

1990s, as the patient's perspective became a matter of increasing interest, research in the area of self-reported measures to assess patients' perceptions of their hearing problems was particularly active. Some of the instruments were developed for screening purposes; others were designed to provide a global perspective of the individual's experience of communication problems and associated adjustment difficulties. Still others were focused on specifics such as hearing aid use or tinnitus. As noted earlier, a wide range of studies has demonstrated that although correlations between audiometric and self-reported measures are statistically significant, they are not very strong. The following conclusions (Erdman, 1994) can be drawn from these findings:

- The communication and adjustment difficulties evidenced in self-assessment are related to hearing impairment.
- This relationship, however, does not permit the audiogram to predict the difficulties experienced secondary to hearing impairment.
- The variability in the experience of hearing impairment indicates that other factors determine the extent to which hearing impairment does or does not become a disabling condition.
- Assessment of hearing impairment is an inadequate means of assessing an individual's rehabilitative needs.

In short, individual differences account for the variability in the experience of hearing impairment. Nonetheless, despite everything we know about individual differences in adjustment to hearing impairment and despite how often we see virtually identical audiograms from individuals with vastly different subjective accounts of hearing problems, we continue to concentrate our time, effort, and energy on fitting the hearing aid to the audiogram. As Kooser (2014) poignantly observes, all too often, audiologists never talk to patients about the actual communication and adjustment difficulties that prompted them to seek audiologic assistance.

The clinicians and researchers who have developed self-assessment tools for use in audiologic practice anticipated that audiologists would use the results to identify relevant issues for each patient, to guide the development of intervention goals, to counsel patients, to monitor their progress, to assess treatment outcomes, and to evaluate the effectiveness of treatment programs (Erdman, 1993b). Unfortunately,

many audiologists say they simply do not know what to do with the results of self-reported measures. Although the results of these instruments point to areas in which counseling would be of value, the vast majority of audiologists do not take advantage of this readily available information. To the extent that self-assessment instruments are used at all, there is a strong preference for short instruments and ones that are computer administered, neither of which facilitates development of a practitioner-patient relationship founded on rapport and mutual goals. Ostensibly, this is due to time constraints. Interestingly, the amount of time required to complete an audiogram, even with a difficult patient, is never questioned. If complications develop during the course of an ABR, there is always time to complete it or to schedule an additional evaluation. The same commitment to assessing the actual impact that hearing impairment has on an individual's daily living and quality of life simply does not exist. The time factor, in actuality, is a rationalization. The patient's experience of hearing impairment simply does not fit into the biomedical approach to which audiology practice adheres. Because the communicative, psychosocial, and behavioral implications of hearing impairment are not integral to the audiologic evaluation, this information is all too often avoided and ignored.

There is evidence of a biopsychosocial approach to understanding the implications of hearing impairment in the body of international audiologic literature. Audiologists and psychologists in Australia, Canada, Denmark, England, New Zealand, Norway, Scotland, and Sweden have made innumerable contributions to the literature on hearing impairment as it relates to coping, well-being, family, marriage, occupation, psychological adjustment, and stigma. This invaluable body of work is noteworthy in many respects. Much of the work stems from systematic dedicated research that has advanced our understanding of the experience of hearing impairment and produced ongoing learning threads for the audiology community and beyond. A second point worth noting is the extent to which this work consistently draws on knowledge from other disciplines. It reflects an awareness of what the social and behavioral sciences can offer to a more holistic understanding of the wide-ranging effects of hearing impairment on the human condition.

The work of the late Raymond Héту and his colleagues, for example, provides a systematic examination of occupational hearing loss (OHL). A qualitative analysis of hearing handicap among individuals with OHL (Héту, Riverin, Getty, Lalande, & St-Cyr, 1988) and an investigation of the psychosocial disadvan-

tages experienced by family members of individuals with OHL (Héту, Lalande, & Getty, 1987) were followed by a study of workers' reluctance to admit to hearing problems (Héту, Riverin, Getty, Lalande, & St-Cyr, 1990). These investigations led to the development of a rehabilitation program for individuals with OHL (Getty & Héту, 1991; Héту & Getty, 1991), an examination of how workers overcome difficulties in the workplace (Héту & Getty, 1993), and an evaluation of coworkers' attitudes toward individuals with OHL (Héту, Getty, Beaudry, & Philipert, 1994; Héту, Getty, & Waridel, 1994). The culmination of Héту's research is his seminal article on the stigma related to hearing impairment (Héту, 1996). This critical work has been continued by Gagné, Southall, and Jennings (2009).

The work by Héту and colleagues has also raised awareness of the effects of hearing impairment on significant others. A survey conducted by Héту, Lalande, and Getty (1987) indicates that family members experience several consequences from the hearing impairment of workers' with OHL and from their noise exposure. Their conclusions indicate that spontaneous coping efforts do not achieve effective resolution of communication difficulties; moreover, immediate family members do not automatically seek mutually acceptable solutions to hearing problems. This highlights the importance of including family members in audiologic counseling. Anderson and Noble (2005) investigated couples' attributions regarding the behaviors associated with hearing difficulties. Their findings suggest a relationship between such attributions and marital satisfaction. Scarinci, Worrall, and Hickson (2008) conducted a qualitative analysis of interviews with older couples to (a) describe the experience of living with a person who has hearing impairment, (b) describe the effect of hearing impairment on their communication and relationship, and (c) identify the coping strategies adopted by their spouses with hearing impairment. Their analysis indicates that the range of effects hearing impairment has on spouses' lives is very broad. The results also emphasize the spouses' continual need to adapt to their partners' hearing problems. Another important variable influencing the impact on spouses is the acceptance of the hearing impairment by the hearing-impaired partner. Specifically, the greater the extent to which the individual appears to have accepted hearing impairment, the less effect it has on the spouse. Aging and retirement issues also emerged as a consistent theme. Hearing problems became more evident upon retirement when more time was spent together. Spouses perceived hearing loss as an inevitable part

of aging about which nothing can be done. Spouses also felt that age had a limiting effect on the extent to which their partners could adapt or learn coping skills. The authors have continued this line of investigation (Meyer, Scarinci, Ryan, & Hickson, 2015; Scarinci, Meyer, Ekberg, & Hickson, 2013; Scarinci, Hickson, & Worrall, 2011; Scarinci, Worrall, & Hickson, 2009, 2012), elucidating the factors associated with third-party disability as experienced by spouses of individuals with hearing impairment and clarifying the ways in which rehabilitative endeavors can address the needs of spouses and couples in resolving communication and adjustment difficulties (Scarinci, Meyer, Ekberg, & Hickson, 2013).

In the U.S., research related to spouses has frequently focused on the dyadic communication difficulties related to disparate perceptions of hearing-related problems and behaviors. Correlations between patients' self-assessments and their spouses' corresponding measures using the Denver Scale (McCarthy & Alpinier, 1983), the Hearing Handicap Inventory for the Elderly (HHIE; Newman & Weinstein, 1986), and the Communication Profile for the Hearing Impaired (CPHI; Erdman, 1995) consistently indicate discrepancies in couples' perceptions of the communication difficulties experienced as a result of one partner's hearing impairment. Mean scores are similar; correlations, however, albeit significant, are not strong. In short, this supports clinical observations that couples frequently have different perspectives on the problems experienced in relation to hearing impairment. This has important implications for couple or family counseling. It is not surprising that couples do not come to agreement on solutions when perceptions of the problem differ markedly. Common complaints such as "My wife mumbles all the time" and "He can hear just fine when he wants to" illustrate the lack of shared understanding of the problems hearing impairment creates. The spousal assessments of their partner's hearing problems are not intended to corroborate the patient's difficulties. Rather, they provide valuable insight into the nature of communication difficulties that develop subsequent to those caused by the hearing impairment. For example, when spouses come to believe that their partners simply "do not care enough to listen," the nature of the communication problems becomes significantly more complicated.

Of course, many audiologists in the clinical arena do focus on how hearing impairment affects their patients, and they provide counseling to ensure that patients adjust to their hearing problems and manage them successfully. Many more audiologists would do so if institutional resistance did not exist on several

different levels. Professional training in audiology typically is not approached from a biopsychosocial perspective. Physical sciences dominate the curriculum with less emphasis on social and behavioral sciences. This mirrors the situation that existed in medical schools, although in those programs, change has been mandated; as stated above, communication and interpersonal skills are now considered critical competency areas (ACGME, 2012). In practicum settings for audiology students, the emphasis is on quickly and accurately administering the battery of audiologic measures and determining a diagnosis from those findings. Service delivery is based on a top-down biomedical model in which the clinician provides the diagnosis and recommends the treatment. In view of the fact that dramatic changes are occurring throughout health care, audiology must address the need to modify its service delivery approach. In many respects, this will entail a return to our roots.

The Military Experience

This program has changed a lot of people since Day 1.

—Army officer, one of a 12-member,
9-day, AR group (1989)

Audiology's origins can be found in the AR programs that were implemented during World War II to care for military personnel who incurred deafness or hearing impairment secondary to their military service (Gaeth, 1979; Morrissett, 1957). At the time, the AR programs actually approximated a biopsychosocial approach. The programs were situated at three medical facilities: Borden General Hospital in Chickasha, OK; Hoff General Hospital in Santa Barbara, CA; and Deshon General Hospital in Butler, PA. Army Captain Raymond Carhart, came from Northwestern University to Deshon, where he served as an acoustic physicist; he later returned to Northwestern and started the first academic program in audiology. Otolaryngologic and psychiatric evaluations were conducted on an in-patient basis as part of the admission process. Following the in-patient medical exams, the soldiers resided in convalescent centers for the duration of the 3-month AR program. Once accepted into the program, a Red Cross caseworker was assigned to each patient to handle any concerns he might have and to facilitate contact and communication with family members. Lipreading teachers, speech pathologists, and social workers were involved in the daily training. Dr. Walter Hughson, an otologist serving as a

civilian consultant, urged the inclusion of evaluations of mental ability and educational achievement to ensure that the patients could be expected to benefit from the program and, subsequently, to realize successful vocational rehabilitation. Hughson praised the realistic approach to the total problem of hearing loss with its recognition of the psychologic implications. According to Morrissett (1957), who became chief of the otolaryngology branch in the Surgical Consultants Division in the Army Surgeon General's Office in 1944, the programs gave specific attention to:

- social and psychiatric treatment to overcome possible trauma associated with combat related hearing loss and to rule out "hysterical" deafness,
- medical and surgical treatment as necessary,
- hearing aid fittings for those who were anticipated to accrue benefit from them, and
- "education for social living" (p. 449).

Education for social living included a program of recreational and educational activities, training in lip-reading and speech correction, and "indoctrination of the soldier's family" (Morrissett, 1957). Absent the technologic advances in diagnostic audiology and hearing aids, the emphasis on rehabilitation focused directly on the patient and his communication and adjustment difficulties.

A special effort was always made to make clear to each patient precisely what he could expect from the rehabilitation course. It was explained unequivocally that a . . . person could not expect to regain his full hearing but that he could be taught methods of compensating for his disability. It was emphasized that nothing could be accomplished without his own full cooperation; and it was only in the exceptional case that such cooperation was not given. (p. 469)

In 1945, Truex (1957) conducted a follow-up survey of patients from the AR program at Deshon General Hospital at least 6 months after their hospital stay. Of the 261 respondents, only two did not consider the time spent in the program worthwhile. Of even greater significance is the fact that 91.6% indicated that they were using their hearing aids. Among those who were not wearing their hearing aids, the primary reasons given included headaches, difficulty obtaining batteries, the hearing aid was in need of repair, they did fine without aid, the aid was too noisy or too cumbersome, and symptoms of external otitis.

In terms of daily use, 23% indicated they wore their hearing aid 5 to 8 hours per day, 21% reported 9 to 12 hours of hearing aid use, and 22% said they wore their hearing aids more than 12 hours per day. Of those who only used their hearing aids 1 to 4 hours per day, the vast majority indicated they did not wear their hearing aids at work in factories or on farms, and so forth. Not surprisingly, the results of this study sparked an interest in establishing similar programs in the civilian sector after the war.

Following the war, the military AR programs were consolidated and based at Walter Reed General Hospital, later designated Walter Reed Army Medical Center. With the move to Walter Reed, audiologists rather than otolaryngologists managed the AR program. The vast majority of patients presented with bilateral high-frequency sensorineural hearing impairment secondary to long-term noise exposure. Referrals for psychiatric evaluations were essentially nonexistent. Although changes in the program were implemented over time, an AR program was offered at Walter Reed for more than 50 years. Northern, Cilix, Roth, and Johnson (1969) conducted a study of patients who had attended a 3-week AR program. The results indicated that 93.5% of the respondents wore their hearing aids "always" or "often." Only 4.5% said they "never" wore their hearing aids, and 2% indicated they seldom wore them. When asked if they would refer a military colleague with hearing loss to the program, only one of 199 patients indicated that they would not.

By the 1980s, the AR program had been shortened to 9 days. Spouses were frequently included and the number of female patients had increased. Better hearing aid technology and milder hearing impairments facilitated acceptance of and adjustment to the use of amplification. As the degree of patients' average hearing loss became milder, less emphasis was placed on speechreading and auditory training. Instead, emphasis shifted toward enhancing patients' awareness of the benefits of hearing aid use and on the resolution of specific problems experienced secondary to hearing impairment. Recorded exercises in varied listening environments were used to compare unaided and aided performance. These exercises were effective in demonstrating the benefits of hearing aid use in terms of speech intelligibility and even more so in terms of listening ease and the associated decrease in stress and anxiety. Patients were constantly amazed to realize how much they had been missing and how much more relaxed and confident they were when listening in the aided condition. Many also observed that listening was actually enjoyable and no longer

fatiguing. The course of the program represented an opportunity for patients to adjust to the use of amplification while working closely with program audiologists. Adjustment difficulties, specific fitting problems, and perceived benefit and satisfaction were closely monitored throughout the program to ensure acceptance of and adjustment to the use of amplification. Patients were afforded the opportunity to try alternate fittings and to evaluate monaural versus binaural amplification when appropriate. Their active engagement in this evaluation and decision-making process undoubtedly was implicated in the program's success. Results of the long-term follow-up of patients who participated in the National Institutes on Deafness and Other Communication Disorders/Veterans Affairs (NIDCD/VA) Hearing Aid Clinical Trial also suggest that patients' active engagement in the selection of an optimal hearing aid fitting has positive effects on treatment outcome (Kricos, Erdman, Bratt, & Williams, 2007).

The AR program at Walter Reed featured individual and group counseling sessions that included assertiveness training, stress management techniques, and problem solving based on results of patients' self-assessment inventories and on analysis of the patients' written reflections on specific communication and adjustment difficulties. The group format was invaluable. As one patient stated in a review of the program,

I really didn't think I needed to come to this course. But so many things were helpful, especially the feelings others expressed as to their personal problems with hearing loss and the suggestions everyone gave each other. I'm a bit of a loner, but I felt really comfortable being with others who are experiencing the same problems I've been having.

This statement exemplifies just some of the therapeutic factors that commonly occur in group counseling: universality, group cohesion, imparting information and altruism (Erdman, 2009).

Routine follow-up of the AR patients through the 1980s consistently revealed hearing aid use rates comparable to those reported by Truex (1957) and by Northern and colleagues (1969). Erdman and Demorest (1987) compared pre- and post-program CPHI (Demorest & Erdman, 1986, 1987) results for former AR patients ($N = 70$) 8 months to a year following their enrollment. Hearing aid use rates in the 90% range were reported. Moreover, significant improvements in scores were demonstrated in all areas of communication performance, attitudes of others, maladapt-

tive strategies, and verbal strategies, and all but one scale (displacement of responsibility) in the personal adjustment area (see Erdman & Demorest, 1998a, 1998b, for a review of the current CPHI scales and Erdman, 2006, for CPHI interpretation guidelines). The results indicate broad-based treatment effects for new hearing aid users attending the counseling-based AR intervention.

Throughout the counseling and rehabilitation professions there is widespread evidence that group counseling is as effective, if not more effective, than individual counseling (Berg, Landreth, & Fall, 2018; Burlingame, Fuhrman, & Mosier, 2003; Fuhrman & Burlingame, 1990; Holmes & Kivlighan, 2000; McRoberts, Burlingame, & Hoag, 1998; Payne & Marcus, 2008). This is especially true for individuals with problems or conditions that can be isolating, stigmatizing, chronic, or recurring (Huebner, 2004; Yalom & Leszcz, 2005). The interactive nature of group counseling provides unique and powerful mechanisms of change that are evident in virtually all counseling groups regardless of the focus or purpose of the group, the theoretical approach, or age of its members. These therapeutic factors are also readily apparent in AR groups (Erdman, 2009). Excellent reviews of these mechanisms of change can be found in Corsini and Rosenberg (1955), Kivlighan and Goldfine (1991), and of course, in Irvin D. Yalom's classic, *The Theory and Practice of Group Psychotherapy*, now in its fifth edition (Yalom & Leszcz, 2005).

John Hersey Pratt, a physician whose practice included treatment of tuberculosis patients, appears to have serendipitously discovered the therapeutic value of the group approach when, to save himself time, he organized group meetings for tuberculosis patients who could not afford treatment in a sanatorium (Pratt, 1906, 1912). After recognizing the healing qualities that stemmed from patients becoming concerned about and helping one another, he adopted the group approach with other patient populations as well (1922). Sabin (1990), who reviewed Pratt's work, hails the comprehensive, empathic, biopsychosocial nature of Pratt's group approach and emphasizes its relevance for reforms throughout health care. Given the overall effectiveness and the economic benefits of group approaches, it is not surprising to see their use now being encouraged throughout health care, particularly for patients with chronic health conditions.

Following a review of the literature on group counseling programs in AR, Hawkins (2005) concluded that in addition to a decrease in perceived hearing problems, there are benefits in hearing aid use, the use of communication strategies, quality of

life, and personal adjustment. He recommends that future research include heterogeneous populations of individuals with hearing impairment, randomized controlled trials, adequate numbers of subjects, multiple outcome measures, evaluation of the effects of including significant others, and assessment of short- and long-term benefits.

It is encouraging to see recent developments in the implementation of group approaches to AR (Habanec & Kelly-Campbell, 2015; Hickson, Worrall, & Scarinci, 2007; Jennings, 2009; Marrone & Harris, 2012; Tucker, Compton, Mankoff, & Rulison, 2011) and the enthusiasm these approaches are generating among patients and clinicians alike. Audiologists who have worked with groups recognize how instrumental they are in promoting adjustment to hearing impairment and to hearing aid or cochlear implant use. The rationale for a group approach in AR does not only relate to its clinical benefits; the group approach is also optimal in terms of resource allocation. More patients, therefore, can be seen by fewer clinicians in less time.

For more than half a century, the military AR programs achieved successful hearing aid outcomes at a rate that has yet to be replicated anywhere in the clinical arena. Ironically, the influence of the biomedical model on audiology triggered an end to this remarkable program just as efforts to improve treatment outcomes for chronic conditions are leading to an integration of psychosocial interventions throughout health care. Patients' belief in the program and support for it remained solid throughout its life span. Unfortunately, as audiologists who worked in the program moved on or retired, replacements became difficult to find. Failure to emphasize the human element in audiology training programs has led to fewer audiologists who are counseling oriented. Unlike the convalescent centers that housed the AR programs during the war, military hospitals are acute care facilities, and it became increasingly difficult to justify intervention for a chronic condition, albeit one that was service related. The lack of standard referral and funding procedures for outpatient rehabilitative care, in conjunction with scarce housing options, resulted in a recurring need to justify the AR program. The paucity of audiologists committed to rehabilitation and the combination of logistical factors ultimately led to discontinuation of the program.

The image of audiology as a microcosm of the health care system makes it easier to address how the limitations and inadequacies of the biomedical model have affected our practice. Addressing these issues and the ways in which a biopsychosocial approach

can transform our practice is a necessity on all levels ranging from the individual clinician, to clinical supervisors, to clinic directors, to the hearing aid industry, to faculty members and deans, to the governance of professional organizations, to third-party payers, and government agencies. A comparable range of stakeholders participated in the Changing Health Care Landscape Summit (American Speech-Language-Hearing Association, 2012), a conference to address issues related to the ongoing changes in health care, including:

- how do we create new, timely, cost-effective models to dynamically update clinical best practices, and
- what changes are needed in how care is delivered and/or documented that will help assure payers, employers, and consumers of the value of our services?

Not surprisingly, but encouragingly, following 3 days of deliberation among audiologists from academic, clinical, and research facilities and representing American Academy of Audiology (AAA), Academy of Doctors of Audiology (ADA), Academy of Rehabilitative Audiology (ARA), and the American Speech-Language-Hearing Association (ASHA), a dominant and consensual theme that emerged is the need to infuse a patient-centered approach to all aspects of audiology care with, importantly, this emphasis beginning at the point of academic entry into the profession. In essence, there is widespread agreement that, especially in view of impending changes throughout health care, a fundamental change is necessary that reflects a shift in perspective from the hearing impairment to the person with hearing impairment—a simple thought with myriad implications.

Just as some have disputed the need for a biopsychosocial approach in medicine, some may debate the need to shift audiology's focus. As one colleague, a basic scientist at heart, has argued, "Fixing what's wrong, fixes everything." Still others will argue, "The biopsychosocial approach and all it entails will only make things more complicated." Certainly, expanding the focus of practice to facilitate diagnosis and patient engagement entails looking at more factors, but it also means we are more likely to discover the variables that ultimately can facilitate adjustment and enhance outcomes.

The benefits of the biomedical model in medicine are mirrored in audiology. The diagnostic technologies available to us today were unfathomable in

the early years of the profession. These advances and others throughout medicine have contributed to an unprecedented increase in our understanding of the anatomy and physiology of the auditory system down to its molecular bases. From body-worn hearing aids to implantable devices, there is no question that innovations in amplification technology have improved the quality and sophistication of products available for the remediation of hearing impairment. Nonetheless, there are signs of limitations to a biomedical focus in audiology. The emphasis on technology and procedures has detracted from our original focus on the individual. To what extent has progress in technology translated into improvements in the lives of those with hearing impairment? Are those with hearing problems more accepting of hearing aids today? Has satisfaction with hearing aids increased significantly? Are those who wear hearing aids experiencing fewer residual problems than hearing aid users in the past? How are we addressing the problems of patients who are not achieving sufficient benefit or are simply not adjusting to hearing aid use? How are we addressing the needs of family members who are unavoidably affected by their loved one's hearing impairment? Why have we not seen improvements on the human level that correspond to improvements on the technological level? How successful have we been in addressing hearing impairment as a chronic condition? These questions are not a matter of self-recrimination; they are representative of the crisis that has precipitated the major changes health care is now undergoing. Moreover, they are questions we are now forced to address for ethical as well as pragmatic reasons. The classic biomedical response to suggestions that patient variables are to blame for treatment failures has been, "That is not our problem." To the contrary, when advances in biomedical thinking do not result in comparable advances in health and well-being, it is everyone's problem. It is a failure of the system. Professionally, morally, and ethically, it is our responsibility to respond accordingly. Audiologists are faced with a multitude of situations in which the biomedical approach can pose limitations. Dissatisfaction with hearing aids and the number of hearing aids residing in dresser drawers, residual communication difficulties, and nonadherence to treatment regimens are all limitations linked to the inadequacies of the biomedical approach in audiology practice. It has not adequately served patients' specific communication needs, nor has it adequately addressed the impact of hearing impairment on patients' personal adjustment and on the members of their families. The evidence for improved patient satisfaction and

broad-based treatment outcomes from biopsychosocial, patient-centered approaches is overwhelmingly positive, particularly for those with chronic health conditions (Bodenheimer, Wagner, & Grumbach, 2002; Carr, Emory, Errichetti, Johnson, & Reyes, 2007; Earp, French, & Gilkey, 2007; Falvo & Holland, 2018; Fuertes, Boylan, & Fontanella, 2009; Gilbert et al., 2011; Stewart et al., 2013). Such a shift in audiology will represent an actual return to our roots (Ross, 1997). In the early AR programs, the purpose was to restore the person rather than to restore the hearing. Shifting our perspective allows us to fundamentally realign our philosophical underpinnings to more accurately reflect audiology's identity as a rehabilitative profession. This shift in paradigm is a matter of priorities and a matter of focus; it is a matter of understanding what living with a hearing impairment means to each patient. The patient-centered focus, as Engel (2005) describes it, involves "learning how to embed the illness (disability) in the patient's life" (p. 377). It is by incorporating this philosophical reorientation into our clinical perspective and practice that the role of counseling emerges front and center, making patient-centered and relationship-centered care a reality in audiology practice. Heaton (1992) raised this issue in a thought-provoking discussion of quality care:

We need to see what can be changed so that it is easier for us to live with ourselves, so that the public can readily perceive how we contribute to the welfare of clients, and so that, at the same time, we do not compromise our ethical standards. For if we are not comfortable with the way in which we offer service, if we are unable to believe in ourselves, then how can we expect others to believe in us? (p. 272)

In their discussion of service delivery models, Erdman, Wark, and Montano (1994) concur with Heaton's observation and state:

The minimal emphasis on rehabilitation perpetuates the perception of audiology as a diagnostic profession with adherence to a medical model of service delivery. Failure to modify service delivery to better meet the needs of individuals with hearing impairment, when we know change is needed and when we know we can do better, is unethical. To assume the position that change is not possible because of time constraints or reimbursement constraints when we know change is indicated is indefensible. (p. 55)

Incorporating Change by Shifting Our Perspective

Clinicians and patients should maximize the therapeutic effects of communication by explicitly orienting communication to achieve intermediate outcomes (e.g., trust, mutual understanding, adherence, social support, self-efficacy) associated with improved health.

—Street, Makoul, Arora, and Epstein (2009)

To maximize our patients' ability to manage their hearing problems and increase their satisfaction with audiology care, emphasis rests on our ability to provide effective counseling. Incorporating counseling into our view of audiology practice facilitates a biopsychosocial, whole person approach as manifested in patient/person, family, and relationship-centered care. This is abundantly clear when recognizing the triadic nature counseling has in rehabilitative intervention as depicted in Figure 11-3. As mentioned in the introduction, counseling in rehabilitation can be viewed on three different therapeutic levels. The broadest, most comprehensive of these is counseling

as the therapeutic context in which all of audiology intervention takes place. The therapeutic context, which can also be viewed as the philosophical context, is at once, biopsychosocial, patient- and person-centered, and empathic. Within the therapeutic context, the direction and goals of the counseling process, and the audiologist's role and responsibilities in it, are established for each and every patient. It is within this context that the therapeutic process emerges, evolves, and functions. The process is interactive, i.e., relationship-centered, facilitating development of the therapeutic alliance, a shared understanding of the patient's problems, and opportunities to enhance patients' self-efficacy (Bandura, 1977, 2004; Smith & West, 2006)—which is the key to successful management of a chronic condition—and engagement of the patient in the identification of acceptable strategies for self-management. As the counseling process unfolds, individualized therapeutic activities are incorporated to promote the patient's capability to manage his or her chronic condition successfully. These may consist of basic counseling interventions such as a hearing aid orientation or may involve more specialized techniques such as motivational

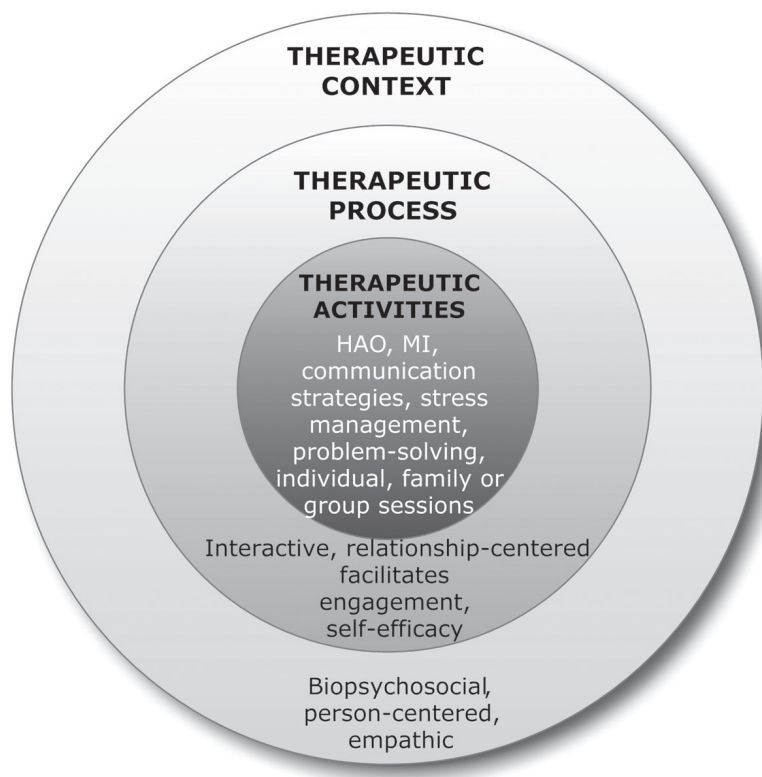


Figure 11-3. Triadic nature of counseling in audiology rehabilitation.

interviewing, assertiveness training, stress management, brief-focused problem-solving approaches, or ongoing individual, family, or group counseling sessions. The key to comprehending the actual scope of counseling in audiologic practice and hence, our responsibilities in it, is to recognize the presence and nature of this triadic phenomenon. It is essential that audiologists develop the skills necessary to (a) communicate empathy effectively, (b) engage patients in the identification and self-management of their hearing problems, (c) promote self-efficacy in patients, and (d) ensure adherence to the recommended management strategies.

The depiction of counseling as a dichotomy consisting of informational counseling and adjustment counseling has hampered effective integration of counseling into audiologic practice. Wylde (1987) and Erdman (1993a, 2000) have stressed the erroneous nature of the information-adjustment dichotomy: It ignores the relationship between cognition and affect and is, therefore, flawed and unsustainable. Information has very powerful effects on adjustment; this is the very principle on which cognitive approaches to counseling are based. Furthermore, neuroscience corroborates the extent to which cognition and affect are interconnected, if not interdependent phenomena. Functional brain imaging depicts the two separate but interacting processes mediated by separate but interacting brain systems (Greenberg, 2008).

On a practical level, consider a patient who incorrectly believes his recent diagnosis of hearing impairment portends eventual deafness. Consequently, he may experience feelings of despair, fear, anxiety, and sadness. The audiologist who voices understanding of his concern and reassures him that eventual deafness is highly unlikely, particularly given the etiology of his hearing problem, can resolve his emotional distress immediately. Correction of an erroneous belief through cognitive intervention facilitates a change in affective state, i.e., emotional adjustment. Affect, cognition, and behavior are interconnected phenomena, a concept covered in entry-level courses in psychology. It is important to couch our understanding of counseling theory and its methodologies in established psychological conceptualizations to ensure our clinical practice and our research are consistent with scientific knowledge. There is no rationale or justification for doing otherwise. Understanding the experience of hearing impairment, adjustment to hearing impairment, and the sources of individual differences inherent in each requires that we do so. As Luterman (2006) has pointed out, individuals who are troubled by hearing problems are not emotionally disturbed, but they may be understandably emotionally upset.

It is safe to say that most people who experience hearing impairment sometimes feel stressed, irritated, annoyed, sad, lonely, discouraged, and left out. To the extent that the emotions a patient reports are related to his or her hearing problems, audiologists should be prepared to listen, to empathize, to validate the patient's experience and feelings. And, importantly, it is essential to recognize that such reactions are, in fact, perfectly normal. Our role in counseling is to help our patients improve their hearing, adopt effective coping skills, and find resolutions to the communication difficulties they have experienced so they do not feel as stressed, irritated, annoyed, sad, lonely, discouraged, and left out. Because we understand why our patients are having the problems they are having better than any other professionals do and because we have the knowledge and expertise to help them resolve those problems, it is our professional responsibility to do so by counseling them effectively. The Preferred Practice Patterns for counseling in audiology (American Speech-Language-Hearing Association, 2006), as shown in Table 11-2, are an excellent guide to the scope of counseling services that audiologists should be prepared to provide in their clinical practice.

Implications for Academia and Professional Development

The role of counseling in audiology and the critical need for audiologists to have skills in this area are issues that have been apparent for some time and our professional organizations have emphasized their importance (e.g., AAA, 2004; American Speech-Language-Hearing Association, 2001, 2006, 2018; CCC-ASLP, 2018). The paucity of social and behavioral sciences in graduate programs in audiology is troubling. Coursework is largely rooted in the physical sciences (e.g., anatomy and physiology, hearing disorders, genetics, neuroscience, diagnostic procedures, acoustics, electroacoustics, instrumentation, evaluation of amplification systems, etc.). It is ironic that, historically, the study of communication sciences and disorders has been viewed as a social and behavioral science. There have been repeated calls to increase the emphasis on counseling in the curriculum and practicum experiences in academic programs in audiology (Crandell, 1997; Crandell & Weiner, 2002; Culpepper, Mendel, & McCarthy, 1994; Erdman, 1993, 2000; McCarthy, Culpepper, & Lucks, 1986; Meibos, Muñoz, & Twohig, 2019; Whicker, Muñoz, Butcher, Schultz, & Twohig, 2017). Although a survey conducted by English and Weist (2005) indicates that the number of academic programs requiring a counseling course has

Table 11-2. Preferred Practice Patterns in Audiology: Counseling**Expected Outcome(s)**

- Counseling enhances patients' and their families' understanding of, acceptance of, and adjustment to auditory, vestibular, or related disorders.
- Counseling enhances acceptance of and adjustment to hearing aids and hearing assistive technology systems designed to maximize communication skills.
- Counseling engages patients in the management of their communication problems and enhances the physical and psychosocial well-being and quality of life for individuals with hearing impairment and other auditory disorders.
- Counseling increases awareness of the need for prevention of further damage to auditory, vestibular, or related systems.
- Counseling enhances compliance with treatment recommendations.
- Counseling enhances benefit from and satisfaction with treatment.

Clinical Indications

- Counseling is indicated for all patients and their family members/caregivers as an integral part of audiology services.

Clinical Process

- Counseling goals are established based on assessment of patients' needs.
- Counseling goals and approaches are modified to facilitate patients' motivation, progress, and engagement in the management of auditory and nonauditory effects of hearing impairment and other auditory, vestibular, or related disorders.
- Counseling is individualized for each patient using culturally and linguistically appropriate language.
- Counseling approaches may be cognitive, affective, behavioral, or eclectic in nature based on the patient's specific needs and target goals.
- Counseling for patients and their families/caregivers may focus on one or more of the following:
 - evaluation procedures
 - diagnosis and results of evaluation
 - treatment options
 - communication problems experienced secondary to hearing disorders
 - effects of hearing and balance disorders on psychosocial and behavioral adjustment including interpersonal relationships, social activities, education, and occupational options and performance
 - affective/emotional reactions to auditory, vestibular, or other related disorders
 - development of problem-solving skills and compensatory behaviors
 - development and coordination of self-help and support groups
- Counseling should include referral to, and consultation with, appropriate professionals and nonprofessionals, as appropriate.

Source: Reprinted with permission from the American Speech-Language-Hearing Association, 2006.

increased since the transition from a master's degree to the AuD, a review by Phillips and Mendel (2008) suggests that there is still continued variability in the extent to which counseling courses in the social and behavioral sciences are offered. Moreover, according to Phillips and Mendel, the majority of graduates do not feel fully prepared to conduct counseling with

their patients regarding their communication problems. The variability in course provision is disconcerting. An informal survey conducted by ASHA's Special Interest Group on Aural Rehabilitation and its Instrumentation (SIG 7) (Erdman, Gagné, Mauzé, Wark, & Jennings, 2010) found that some audiology programs offer their own counseling course; others

refer students to an elective course in another department. Still other programs feature one or two courses that cover self-awareness, loss and grieving, health psychology, rehabilitation theory, psychosocial adjustment to disability, and psychosocial adjustment to hearing loss. Nonetheless, the total number of credit hours that could be subsumed under the social and behavioral sciences is minute compared to the biological and physical sciences. Meibos, Muñoz, and Twohig (2019) conducted a survey of counseling audiologists on the basis of which they compiled a list of knowledge, skills, and attitudes that are critical in providing audiologic counseling. In conjunction with the most recent certification requirements (ASHA, 2018) and Preferred Practice Patterns (2006), these areas of knowledge and skill provide relevant and appropriate content to include in audiology curriculum and practicum. Faculty members with strong interests in rehabilitative audiology try to incorporate counseling as a fundamental ingredient throughout the audiology curriculum. This approach is consistent with a scenario in which counseling is a part of the very fabric of audiology. It is also consistent with demands throughout health care to infuse social and behavioral sciences into the education of health care professionals and into the health care process. Along these lines, the inclusion of interprofessional education and practice (IPE and IPP, respectively) to develop health care skills is increasingly emphasized in allied health programs and in professional development throughout health care, which offers alternative options for counseling experiences (Cavanaugh & Konrad, 2012; Congdon, 2016; Koch, Gitchel, & Higgins, 2009; Sheldon et al., 2012). The inclusion of AR programs and assessments of communication and related adjustment difficulties as a routine part of comprehensive audiologic evaluations in academic clinic programs provides opportunities for students to obtain supervised counseling experience. Such experience would increase the ease with which they will be able to assume counseling responsibilities in their own clinical practice. Furthermore, it would ensure that counseling is perceived as an integral aspect of audiologists' professional identity.

Given the limited counseling experience students have during their academic training, it is not surprising that many audiologists question their qualifications to provide counseling. Two important points warrant consideration. First, one's qualifications to provide audiologic counseling consist of a recognized degree in the profession, the requisite certification or licensure, and the necessary knowledge and skills. Clinicians who believe they need

to expand their knowledge and skills in counseling should enroll in a counseling course, pursue continuing education offerings in this area, or seek mentoring from a respected colleague who has demonstrated counseling skill. Secondly, counseling is included in audiologists' scope of practice and American Speech-Language-Hearing Association certification standards (American Speech-Language-Hearing Association, 2018). The Preferred Practice Patterns (2006) describe a wide range of relevant counseling responsibilities in audiology; they appear in Table 11-2, shown earlier. This is further evidence that audiologists are expected to provide counseling to help patients and their families resolve the communication problems and adjustment difficulties they experience because of hearing impairment. The lack of confidence and uncertainty regarding qualifications and the associated intimidation some audiologists feel just hearing the words "counselor" and "counseling" stem, in large part, from the fact that we are not trained to view ourselves as counselors. To instill counseling into audiologists' conceptualization of their role and responsibilities, it must be viewed as a focal aspect of their professional identity (Erdman & Wark, 2007). Hence, faculty members and clinical supervisors in graduate programs should consistently affirm the following:

- Audiologists are the professionals most qualified to counsel those who experience communication and adjustment difficulties related to their hearing problems.
- Counseling is fundamental to rehabilitation.
- Effective counseling enhances treatment benefit, satisfaction, and outcomes.
- Effective counseling skills can be learned and developed.

This shift in perspective is necessary if our next generation of clinicians is to include counseling as a key element of their professional identity and be prepared to provide care that is based on biopsychosocial, patient-centered, and relationship-centered principles.

Calls for the inclusion of social and behavioral sciences in medical school curricula were essentially ignored for some 30 years. In 2002, the IOM convened a committee to (a) review the status of behavioral and social sciences in medical school curricula, (b) determine a priority list for such curricula, and (c) identify barriers to incorporating new curricula into medical education (Carr, Emory, Errichetti, Johnson, & Reyes, 2007). Recommended priorities in subject matter include interpersonal and communication skills, mind-body interactions, patient behavior, physician

role and behavior, physician patient interactions, social and cultural issues in health care, and health policy and economics. Not surprisingly, these priorities reflect growing acceptance of the biopsychosocial model and emphasis on patient- and relationship-centered care in clinical practice.

Similar changes can expand the curriculum and practicum experiences in audiology to augment our focus on the psychosocial implications of hearing impairment. Moreover, communication and counseling skills need to be taught and evaluated systematically. To ensure current practitioners, and faculty members in particular, are informed and capable of infusing the necessary psychosocial and behavioral elements into curriculum and practicum offerings throughout health care, continuing education and professional development activities in these areas are proliferating (Branch et al., 2009; Frankel, Eddins-Folensbee, & Inui, 2011; Jayatilleke & Mackie, 2013; Stein, Frankel, & Krupat, 2005). Professional development activities are frequently available from a number of sources that emphasize counseling and biopsychosocial/patient-centered approaches to care and their applicability to audiology practice. The Academy of Rehabilitative Audiology, American Speech-Language-Hearing Association's SIG 7 (Aural Rehabilitation and its Instrumentation), and the Hearing Rehabilitation Foundation are among the organizations that hold conferences and provide publications and other materials to aid in the provision of patient-centered interventions for individuals with hearing problems and members of their families. The efforts of these organizations, in conjunction with works emanating from the WHO on the ICF (Danermark et al., 2010; Gagné, Jennings, & Southall, 2009; Granberg et al., 2014; Grenness, Meyer, Scarinci, Ekberg, & Hickson, 2016; Scarinci, Worrall, & Hickson, 2009), are indicative of professionwide awareness of the relevance and importance of focusing on the individual and not just the diagnosis of hearing impairment in our practice.

Hearing Impairment: Implications for Mental Health and Well-Being

One doctor, who is now becoming hard of hearing, recently admitted to me that he now realizes he was wrong to keep dismissing my complaints of how much I was missing.

—Woman who sought counseling from a mental health professional

Long-standing misperceptions regarding mental health among individuals with hearing impairment may also contribute to the doubts some audiologists have about their qualifications to counsel. For hundreds of years, individuals with hearing impairment were institutionalized along with individuals manifesting every imaginable mental health problem because their behavior and strange vocalizations (unexplainable at the time) were viewed as aberrant. Even when hearing impairment became a diagnosable condition, the "suspiciousness" attributed to people with hearing problems led to a belief that they were paranoid.

This lack of insight into hearing problems, unfortunately, remains highly pervasive. Stika (1997) conducted a survey of members of the Self Help for the Hard of Hearing (SHHH) organization (now the Hearing Loss Association of America (HLAA) to obtain information regarding the use of mental health services by those who have hearing problems. The results show that, in general, respondents sought mental health services for many of the same reasons the general population does. Many, however, also indicated that they sought counseling for issues directly or indirectly related to their hearing problems. Their comments revealed a disconcerting lack of knowledge and understanding of hearing impairment among mental health counselors. Stika quotes the following comments from those who responded to the survey:

"She had no understanding of hearing loss. I had to spend part of my session—that I was paying for—educating her."

"She did not understand the stress and responsibility I feel to make the communication process successful. At times, she would say things like, 'You're taking communication too seriously. Everyone misses part of what is being said anyway.'"

"I understand well in one-on-one situations so they were unable to comprehend the problems I said I had. They were somewhat incredulous and disbelieving. They would argue as if they could convince me that my hearing loss didn't exist!"

"The psychiatrist could not differentiate between neurotic defenses and defensive behavior caused by the frustration of not hearing well, especially at parties, professional meetings, etc. One doctor, who is now becoming hard of

hearing, recently admitted to me that he now realizes he was wrong to keep dismissing my complaints of how much I was missing."

Although many of the respondents had received satisfactory counseling services, mental health professionals' lack of awareness of the difficulties associated with hearing loss is disconcerting. Some of the counselors appeared to be unaware of the special communicative needs of their patients (e.g., the importance of visual cues, good lighting, and a well-modulated voice). There was limited understanding of the effects of hearing loss and background noise on speech comprehension. Some of the mental health professionals were not cognizant of the vastly different needs of those with congenital versus acquired deafness and hearing impairment. A lack of insight into the psychosocial effects of hearing impairment in terms of personal adjustment, relationships, family life, and occupation was especially difficult to comprehend. Most alarming, however, is the apparent tendency for mental health professionals to discount or refute patients' complaints of hearing difficulties when such problems are not apparent in the one-on-one counseling encounter. This introduces the very real possibility of potentially damaging misdiagnoses. Moreover, it is the antithesis of an empathic understanding of the patient's problems and precludes development of a therapeutic relationship based on trust and mutual understanding. It is discouraging to see continuing evidence of what Meadow-Orlans (1985) refers to as the "persistent myth" regarding the mental health of people who have hearing problems. Stika stresses the need for change in meeting the mental health needs of those with hearing problems. Hallam, Ashton, Sherbourne, and Gailey (2006) echo this need for counseling services from professionals who are cognizant of the issues facing those with hearing impairment. Their investigation of individuals with acquired profound hearing impairment suggests that although this is not a frequent clinical phenomenon, the counseling needs of these individuals pose a critical need for expanded service in this area.

Many of the individuals from Stika's (1997) survey were seeking assistance for problems related to their hearing difficulties. The irony is, of course, that audiologists are exceptionally well qualified to be of assistance with many of these problems. A number of respondents stated that they realized later how much hearing impairment was implicated in their decision to seek counseling in the first place. The insidious nature of hearing impairment manifests itself in many different ways. Audiologic counseling can be

and should be an option for patients with adjustment problems related to their hearing loss. For patients in genuine need of psychologic or psychiatric intervention, audiologists should facilitate a team approach to ensure that the patient's hearing and mental health needs are both adequately addressed. This is a serious unmet need and mechanisms for addressing it are sorely needed.

Thomas (1984) disputes the idea that suspiciousness and paranoid psychoses are more common among those with hearing impairment and maintains that hearing impairment does not affect basic personality structure. His findings are concordant with rehabilitation psychology, which holds that disabilities are not associated with type or degree of personality traits, or with adjustment. Nonetheless, he stresses that people can experience significant psychological distress secondary to hearing impairment. Thomas' work represents a marked departure from some of the psychoanalytic thought that dominated the "psychology of hearing loss" for a number of years. Knapp (1953) suggested the possibility that "unconscious attitudes centered in the ear may actually alter its physiology" (p. 107) and cites a review with evidence that emotional factors may serve as an activating agent in otosclerosis. Knapp contended that the ear has a definite role in psychic life. Many of those trained in Freudian theory believed that the sexual symbolism associated with the ear and with hearing allows an understanding of the psychological reactions seen in patients with hearing loss (Rousey, 1971). Psychoanalytic writings such as *The Madonna's Conception Through the Ear* (Jones, 1951) reveal some intriguing, if not improbable, conceptualizations. The intensity and duration of psychoanalysis preclude its utility in most rehabilitation situations; however, knowledge of psychoanalytic theory is invaluable in many areas of rehabilitation counseling. For example, ego defenses first described by Freud are often implicated in adjustment to disability. Descriptions of such defenses are included in Table 11-3. Counselors can gain a better understanding of patients' coping behaviors by understanding the role such defenses can play in the adjustment process. In addition to the importance of defense mechanisms in rehabilitation, other psychoanalytic applications pointed out by Livneh and Siller (2004) include the effect of disability on self-perception and body image; the study of reactions to loss, trauma, and disability (e.g., anger, denial, depression, mourning); and the implications of attitudes toward people with disability.

The relationship between hearing impairment and psychological state has been the subject of con-

Table 11-3. Examples of Psychoanalytic Ego Defenses Manifested in Reactions to Disability

Defense Mechanism	Definition	Example
Repression	Forcing intrapsychic conflicts, painful experiences, and disturbing memories out of conscious awareness	Person with a visible congenital disability repressing feelings of shame and embarrassment triggered by early life reactions of others
Projection	Externalizing unconscious forbidden ideas, needs, and impulses and attributing them to other people or environmental conditions	Blaming others for onset of disability or attributing lack of progress in rehabilitation to staff incompetence rather than own lack of effort
Rationalization	Using after the fact, false reasons to offset negative emotional or consequences	Person who gradually loses hearing and attributes lack of participation to boredom, lack of interest, or fatigue
Sublimation	Adopting useful, socially acceptable behaviors to express forbidden and socially unacceptable wishes and impulses	Anger and desire to retaliate against an uncaring society channeled into artistic endeavors
Reaction formation	Substituting and expressing responses and feelings that are exact opposites of those that are deemed verboten or unacceptable	Replacing initial feelings of aversion and rejection toward a child born with a severe disfigurement with overly demonstrative affection and protectiveness
Regression	Reverting to childlike behaviors first exhibited during an earlier developmental stage	A recently disabled person whose temper tantrums are activated when needs are not immediately gratified or who daydreams rather than pursue treatment
Compensation	Seeking to excel in functionally related activities or behaviors to make up for disability-generated loss	Person who lost sight at an early age and has achieved success as a musician
Denial	Resolution of emotional conflict and reduction of anxiety by refusing to perceive, accept, or acknowledge threatening aspects of external reality	Failure to perceive or acknowledge effects of hearing impairment on job performance
Displacement	Shifting energy towards a less intimidating or more accessible object or person to reduce anxiety	Blaming others for not speaking clearly rather than admit to having a hearing problem

Source: Adapted from Cubbage and Thomas (1989); Livneh and Siller (2004); and Livneh and Cook (2005).

siderable debate over the years. A number of studies have been reported in which investigators specifically examined the effects of hearing impairment on psychological well-being. Scherer and Frisina (1998) found that even marginal hearing impairment has an impact on the well-being of older adults. The authors recommend identifying the personal and interpersonal concerns of these individuals to inform treatment planning and achieve a higher rate of adherence to rehabilitation efforts. Consistent with the literature

relating audiologic and self-reported data, Helvik and colleagues (2006) found that well-being (defined as health-related quality of life) among adults with acquired hearing impairment is associated with activity limitations and participation restrictions, but not with degree of hearing impairment. Their results indicate that the well-being of older adults is less likely to be affected in comparison to younger adults. Erdman and Demorest (1998b) report similar results with the possible exception of poorer adjustment among the

oldest adults. Gopinath, Schneider, and colleagues (2012) report that perceived hearing handicap rather than audiometric configuration constitutes a risk to quality of life for older adults. A survey of cochlear implant users in the U. S. and in Sweden indicates that their psychological well-being is influenced by age, perceived social support, participation restrictions in social life, and the attitudes of other people (Hallberg, Ringdahl, Holmes, & Carver, 2005).

The etiology of hearing impairment can also have an impact on quality of life and psychological state. The multiple symptoms associated with Ménière's disease, for example, precipitate avoidance of daily activities (Hägnebo et al., 1997) and negatively affect overall quality of life (Söderman, Bagger-Sjöbäck, Bergenius, & Langius, 2002). Furthermore, perceived intrusiveness of vertigo and tinnitus is associated with symptoms of depression among those with Ménière's (Arroll, Dancey, Attree, Smith, & James, 2012). Given the multiple symptoms associated with Ménière's disease, the importance of a patient-centered approach that includes counseling to facilitate a wide range of coping skills is readily apparent. This is also especially important when working with patients with sudden onset of hearing impairment and with those whose diagnosis may portend a progressive hearing impairment as in otosclerosis (Eriksson-Mangold, Erlandsen, & Jansson, 1996; Luntz, Yehudai, & Most, 2009).

The literature pertaining to hearing impairment and depression reveals disparate findings. Thomas (1984) refuted the notion that hearing impairment results in some specific psychologic disturbance or that it alters personality. He was also the first to suggest that psychological disturbance may be the independent rather than the dependent variable. Lee and Gomez-Marin (1997) conducted a study of Puerto Ricans, Cuban-American, and Mexicans with hearing impairment and found no increased risk for depression. Yueh, Collins, and Souza (2007) examined more than 100 veterans and concluded that depressive symptoms may influence responses on self-reported measures of hearing difficulties. Correlations between scales from the personal adjustment section of the CPHI and selected psychological measures similarly suggest that psychological distress exacerbates adjustment to hearing impairment and not vice versa; moreover, mean scores for measures of anxiety and depression for patients with hearing impairment fell well within the normative range (Erdman, 2001, 2007).

Results of some studies have been interpreted to suggest that hearing impairment plays a role in the onset of depression. Results of a survey by the

National Council on the Aging (NCOA, 1999), for example, have been interpreted as evidence that hearing loss is associated with depression among older adults and that such depression is attenuated by hearing aid use. Respondents were simply asked if they had felt sad, blue, or depressed for 2 weeks or longer during the past years. Although hearing problems can certainly be "depressing," and an individual may feel "depressed" about hearing problems, such feelings do not constitute a diagnosis of depression. Depression is a complex disorder that has psychological, physiological, and behavioral symptoms that can manifest in different ways. More recent research has cautiously reported evidence of depressive symptoms (Garnefski & Kraaij, 2012; Gopinath et al., 2012; Saito et al., 2010). Poor research design, disparate measures and populations, and investigators' lack of understanding of the effects of hearing impairment or conversely, the effects of psychological variables, have contributed to equivocal findings regarding the relationship between adjustment and hearing impairment. Advanced age, comorbidities, and other confounding variables must also be carefully controlled. A study by Mahapatra (1974), for example, involved testing patients while they were hospitalized and about to undergo surgery for otosclerosis. Not surprisingly, the results indicated high levels of psychological distress. It behooves clinicians and researchers to remember that correlation does not equal causation and to exercise caution when interpreting results of studies investigating the relationship between hearing impairment and psychological status.

On the positive side, there is ample evidence that rehabilitative intervention can mitigate the emotional distress associated with hearing difficulties and precipitate improvements in quality of health (Acar, Yurekli, Babademez, Karabulut, & Karasen, 2011; Jennings, 2009; Preminger & Yoo, 2010). Further investigation is warranted to elucidate the emotional impact of hearing impairment as well as the role that preexisting or unrelated emotional distress may have on adjustment to hearing impairment. This adds emphasis to the importance of viewing each individual in the context of his or her own life situation. Of equal importance is the need to develop and implement programs that address the psychosocial consequences of hearing impairment. Preminger and Yoo's (2010) determination that discussion of psychosocial issues in rehabilitation results in greater improvement than does focusing on communication strategies, reinforces widespread recommendations that rehabilitation focus on the emotional aspects of hearing difficulties

for the person and for significant others (Danermark, 1998; Hallberg, Hallberg, & Kramer, 2007).

Although some controversy remains over the relationship between hearing impairment and mental health, particularly with respect to depression, the literature allows one to draw the following conclusions:

- Hearing impairment does not affect personality, nor does it “cause” depression.
- Hearing impairment can be depressing, upsetting, stressful.
- The best predictor of depression is a prior episode.
- Those with hearing impairment and with a predisposition for depression may be more likely to have a recurrence.
- Depression can be detrimental to adjusting to hearing impairment.
- Depression may cause patients to increase their complaints about problems.
- Intervention in the form of hearing aids (HAs) and cochlear implants (CIs) can often alleviate stress, loneliness, and feelings of sadness.

There is overwhelming agreement and evidence that individual differences play a major role in adjustment to hearing impairment; hence, the experience of hearing impairment is highly idiosyncratic. Based on statistical and clinical analyses of CPHI results over a period of 10 years (Erdman, 2006), it is possible to conclude the following:

- There are predictable “normative” patterns of adjustment to hearing impairment.
- Individual patterns of adjustment to hearing impairment are unpredictable.
- Individuals’ adjustment patterns should be identified and assessed to determine the nature and extent of variables impinging upon communication and psychosocial adjustment.
- AR programs should address the normative needs of clinical populations and be flexible enough to meet the needs of individuals.
- Clinicians should be prepared to address the psychosocial and behavioral variables associated with adjustment to hearing impairment.

These conclusions can serve as guidelines for defining the scope of audiologists’ role and responsibilities in counseling individuals with hearing impairment and

the members of their families. In essence, we must be prepared to meet the individual needs of each patient; to do so, counseling will play a central and indispensable role.

What Is Counseling?

It is a process, a thing-in-itself, an experience, a relationship, a dynamic.

—Carl Rogers (1951)

The chronic nature of hearing impairment indicates the need for a biopsychosocial, patient-centered approach in audiology that promotes long-term self-management. Shifting the focus to our patients and attending to the actual reasons they seek audiologic intervention yields a foundation upon which a patient-centered approach that is biopsychosocially rooted can be based and from which relationship-centered counseling can evolve. Perceived obstacles to developing and implementing effective counseling in audiology can be overcome. The most important first step is to admit that the biomedical model places unacceptable limitations on the provision of audiologic care. That simple acknowledgement, a change in mindset, will significantly enhance our ability to overcome remaining obstacles because the justification for them no longer exists. Undoubtedly, systemic difficulties will need to be confronted and resolved. Awareness of the limitations of the current model, however, is a critical move in the right direction.

Professionals’ effectiveness in health and rehabilitation involves many factors. Berven, Thomas, and Chan (2004) identify several key ingredients:

- establishing a therapeutic relationship with patients,
- communicating with patients and their families in facilitative ways,
- obtaining information from individuals in a thorough, attentive manner,
- helping patients identify their problems and explain their needs,
- conceptualizing problems in ways that will facilitate an appropriate treatment plan, and
- facilitating adherence to plans that patients have decided to pursue.

These responsibilities mirror key elements of the biopsychosocial approach and the principles upon which patient- and relationship-centered care are based; they

are considered focal areas throughout rehabilitation (Chan, Berven, & Thomas, 2015; Dell Orto & Powers, 2007; Martz & Livneh, 2007; Parker & Patterson, 2012) and patient education (Falvo, 2010; O'Donahue & Levensky, 2006). Additionally, each of these responsibilities is among the commonalities that exist across a wide range of counseling approaches (Corey, 2017; Norcross & Goldfried, 2019).

The terms *counseling* and *psychotherapy* are, more often than not, used interchangeably in the literature. The titles and content of current textbooks reflect the fact that the conceptual foundations, theories, and methodologies for counseling and psychotherapy are not just similar, they are the same. Consider, for example, *The Theory and Practice of Counseling and Psychotherapy* (Corey, 2017) and *Ethics in Psychotherapy and Counseling: A Practical Guide* (Pope & Vasquez, 2010). Carl Rogers summed it up well in his book *Counseling and Psychotherapy: Newer Concepts in Practice* (1942):

Most frequently, [these processes] are termed counseling, a word in increasingly common use, particularly in educational circles. Or such contacts, with their curative and remedial aim, may be classed as psychotherapy, the term most frequently used by social workers, psychologists, and psychiatrists in clinics. These terms will be used more or less interchangeably in these chapters, and will be so used because they all seem to refer to the same basic method—a series of direct contacts with the individual, which aims to offer him assistance in changing his attitudes and behavior. . . . It is also plain that the most intensive and successful counseling is indistinguishable from intensive and successful psychotherapy. Consequently, both terms will be employed, as they are in common use by workers in the field. (pp. 3–4)

Interestingly, Rogers also noted that a “great many professionals” provide these services and that, regardless of how such individuals might refer to themselves, their approach was the relevant consideration. Consistent with its inception in psychiatry and a traditional focus on pathology, psychotherapy tends to connote long-term treatment of psychological and personality disorders. Counseling, on the other hand, which evolved from education, suggests a problem-focused and adjustment-oriented learning process. Within the profession, there appears to be universal agreement that the latter more appropriately describes the process involved in AR.

The impact of Rogers’ theories on rehabilitation counseling has been enormous. His concepts regard-

ing ideal versus real self-discrepancies (i.e., differences in how people wish to view themselves and how they actually view themselves) are central to understanding why people react differently to physical disabilities. It is not the disability, but the significance one attaches to it that determines acceptance and adjustment. Self-acceptance is instrumental in adjustment to disabilities including hearing impairment. Self-acceptance, as measured by the CPHI, is predictive of overall psychological adjustment to hearing impairment, perceptions of others’ attitudes and behavior, and less-frequent use of maladaptive compensatory strategies (Erdman & Demorest, 1986).

The number of different approaches in counseling has grown exponentially during the past several decades. Implications of the major approaches (humanistic/affective, cognitive, and behavioral) and recent integrative movements for audiology counseling have been described previously (Erdman 1993a, 2000). There are specific counseling variables and concepts, however, that are fundamental to a patient-centered approach. Consistently, these variables are evident in the therapeutic context, therapeutic process, and therapeutic activities involved in the triadic nature of rehabilitative counseling, as shown in Figure 11–3. They include:

- patient narrative;
- empathy;
- verification/clarification to establish an accurate, mutual understanding;
- counselor characteristics;
- practitioner-patient relationship and therapeutic alliance;
- validation and reassurance;
- promotion of self-efficacy; and
- engagement of patient in managing his or her condition.

Erdman (2014) described a model of patient-centered care in which the patient’s narrative and clinician’s empathy provide a foundation for a therapeutic relationship that engages patients in the effective management of their hearing problems to facilitate adherence and optimize outcomes (Figure 11–4). Operationalizing patient-centered care in such a process promotes a systematic whole-person approach with important implications for clinical purposes (Erdman, Wark, & Montano, 2019) as well as research applications because it is replicable and the key elements are measurable and observable. The model served as the control condition in a randomized controlled trial of tinnitus retraining therapy

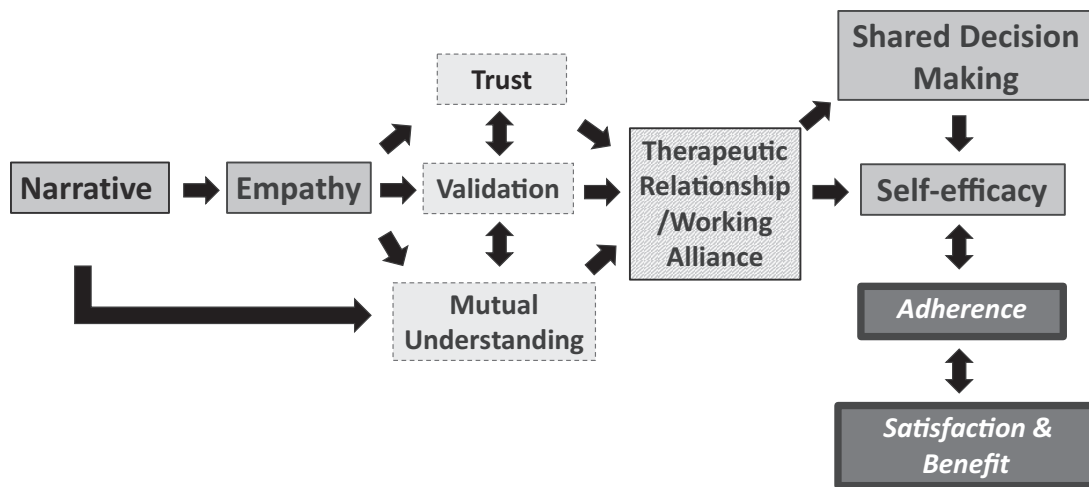


Figure 11-4. Patient-centered care as a process. *Source:* Adapted from Erdman (2014).

(Erdman, Scherer, Sierra-Irizarry, & Formby, 2019; Scherer et al., 2014) and as the research protocol in an investigation comparing face-to-face versus remote applications of audiology rehabilitation for adult cochlear implant recipients (Robinson, Hughes, & Erdman, 2015).

Audiologic Counseling: The Therapeutic Context—Biopsychosocial, Patient-Centered, and Empathic

Sometimes your patients have to hit you over the head before you take any notice of them.

—Michael Balint, MD

The health care system is undergoing seismic change; it is evident that the old way of doing things had serious inadequacies and will not survive. The biomedical model in audiology forces a band-aid solution, in the form of a hearing aid, on a problem that, more often than not, requires intervention with broad-based psychosocial and behavioral elements.

Audiologists provide services for one of the most common chronic conditions in our society. Audiology is a health care profession, but it is not a medical profession. It is a rehabilitative profession. To optimize our success as a profession, rehabilitation is an identity we must embrace fully. Diagnostic audiology is an essential component of audiology rehabilitation but it is just one component. It represents the “bio” portion of the patient-centered care model. Adhering to the patient-centered care model brings counseling to the forefront as the therapeutic

context in which comprehensive, patient-centered care is empathically provided. As illustrated earlier in Figure 11-3, this care involves an interactive and facilitative relationship-centered process that promotes patients’ self-efficacy and engages them in the effective management of their hearing-related problems through activities that enable them to adapt the problem-solving skills and coping strategies needed for the communicative and psychosocial difficulties they experience. The effectiveness of this three-tiered biopsychosocial approach ultimately determines the extent to which treatment outcome is successful.

The first critical phase in the counseling approach is to establish a therapeutic relationship, which evolves as a function of trust, and mutual validation and understanding. It is then possible to engage patients in shared decision-making regarding treatment goals and strategies, promote their belief that they can adhere to these recommended strategies to meet their goals, and ultimately, to ensure that the mutually agreed upon counseling goals are met and maintained. Specific counseling activities can include ongoing exchanges that take place with the clinicians, patients, and their family members; dedicated sessions such as hearing aid orientations; cognitive, affective, and/or behavioral interventions such as assertiveness training, motivational interviewing, and/or stress management targeted at specific communication and adjustment issues; and family and group sessions. The characteristics of counseling as the therapeutic context of rehabilitative audiology extend to all aspects of the counseling process and counseling activities. This is achieved by ensuring that

all aspects of intervention are patient centered. Examining the biopsychosocial, patient-centered, empathic, interactive, and facilitative elements of audiologic counseling reveals a promising framework for rehabilitation planning, evaluation, and research as well as professional education.

The IOM (2001) identified a continuous healing relationship as the most critical ingredient in improving the quality of patient care. It is precisely in this vein that patient-centered care is enhanced by relationship-based care. For well over 50 years, research has shown that the quality of the practitioner-patient relationship is a major, if not *the* major, contributor to treatment success (Castonguay, Constantino, & Grosse Holtforth, 2006; Elvins & Green, 2008; Fierman, 1997; Gelso & Carter, 1994; Gelso & Hayes, 1998; Horvath, 2006; Norcross & Wampold, 2011, 2019; Rogers, 1957, 1958; Tresolini & Pew-Fetzer, 1994; Truax et al., 1966; Truax & Mitchell, 1971). Rogers (1975) consistently maintained that empathy from the clinician is the most powerful ingredient in solidifying the therapeutic relationship and, therefore, the therapeutic process. Others from a wide range of professions have supported his contention over the years (Elliott, Bohart, Watson, & Greenberg, 2011; Frankel, 2009; Hojat, 2016; King, 2011; Norfolk, Birdi, & Walsh, 2007; Riess, 2015; Schrooten & de Jong, 2017; Switankowsky, 2004).

Empathy consists of (a) a cognitive component that represents an accurate perspective of the individual's experience, (b) an affective component that consists of appropriate emotional reactivity, and (c) an accurate attunement to and reaction to the cognitive and affective points raised in the patient's narrative. The latter, which can be viewed as a behavioral component, can include verbal and nonverbal elements that are critical in communicating an understanding of the patient's story.

Squire's (1990) model of an empathic practitioner-patient relationship illustrates how the affective and cognitive components of empathy facilitate engagement of the patient in the management of his or her health condition, resulting in adherence to a treatment regimen to achieve the desired health benefit. Squire's model is shown in Figure 11-5, adapted to illustrate the empathic relationship process in audiologic care. Briefly, in the model, the audiologist strives to understand, both cognitively and affectively, the patient's experience of hearing impairment as revealed in the patient's story. The practitioner must then articulate an empathic understanding of this experience to the patient. As stated above, this understanding is based

on (a) an accurate perspective of the patient's experience and (b) appropriate emotional reactivity. In doing so, the accuracy of the empathy conveys important information to the patient; it provides the patient with a better understanding of hearing impairment and the effects it has on communication. The affective component of empathy allows the burden of the experience to be shared instilling a sense of confidence that the problem is one with which it will be possible to cope. The audiologist's cognitive and affective communications (indicated by the solid arrows leading to the Phase 1 level) allow the person to know he or she is understood. The affective component of empathic understanding, which is supportive and reassuring, combined with the cognitive exchange, facilitates the development of a trusting relationship between the audiologist and the patient. Phase 1 of the empathic understanding model represents development of this trust and rapport—the critical foundation for engaging the patient in the treatment process.

The bidirectional arrows between Phase 1 and Phase 2 are indicative of a working, therapeutic alliance. Increased understanding of hearing impairment permits the patient to volunteer additional information to describe problems more accurately, thereby further informing the audiologist's understanding of the patient's experience. The parallel affective-motivational process reduces the level of overall distress and enhances satisfaction with treatment thus far.

Successful cognitive and affective interactions during Phase 2 ultimately facilitate psychosocial and behavioral adjustment including the requisite knowledge and skills to manage the condition and the self-acceptance, self-efficacy, and motivation to do so as indicated at the Phase 3 level. A successful Phase 3 in the cognitive and affective areas results in an increased likelihood of adherence to the agreed-upon treatment regimen. Adherence enhances treatment outcome in terms of overall adjustment, patient satisfaction, and the benefits derived from treatment. These, in turn, solidify adherence (as indicated by the bidirectional arrows between adherence and outcome). Hence, the model includes the patient's story; development of the practitioner-patient relationship; engagement of the patient in the rehabilitative process; interaction of the working/therapeutic alliance; facilitation of the patient's management of hearing impairment through increased motivation, hope, and self-efficacy; and adherence to the agreed-upon treatment regimen.

The empathic understanding model is a reasonable approach to providing audiologic counseling

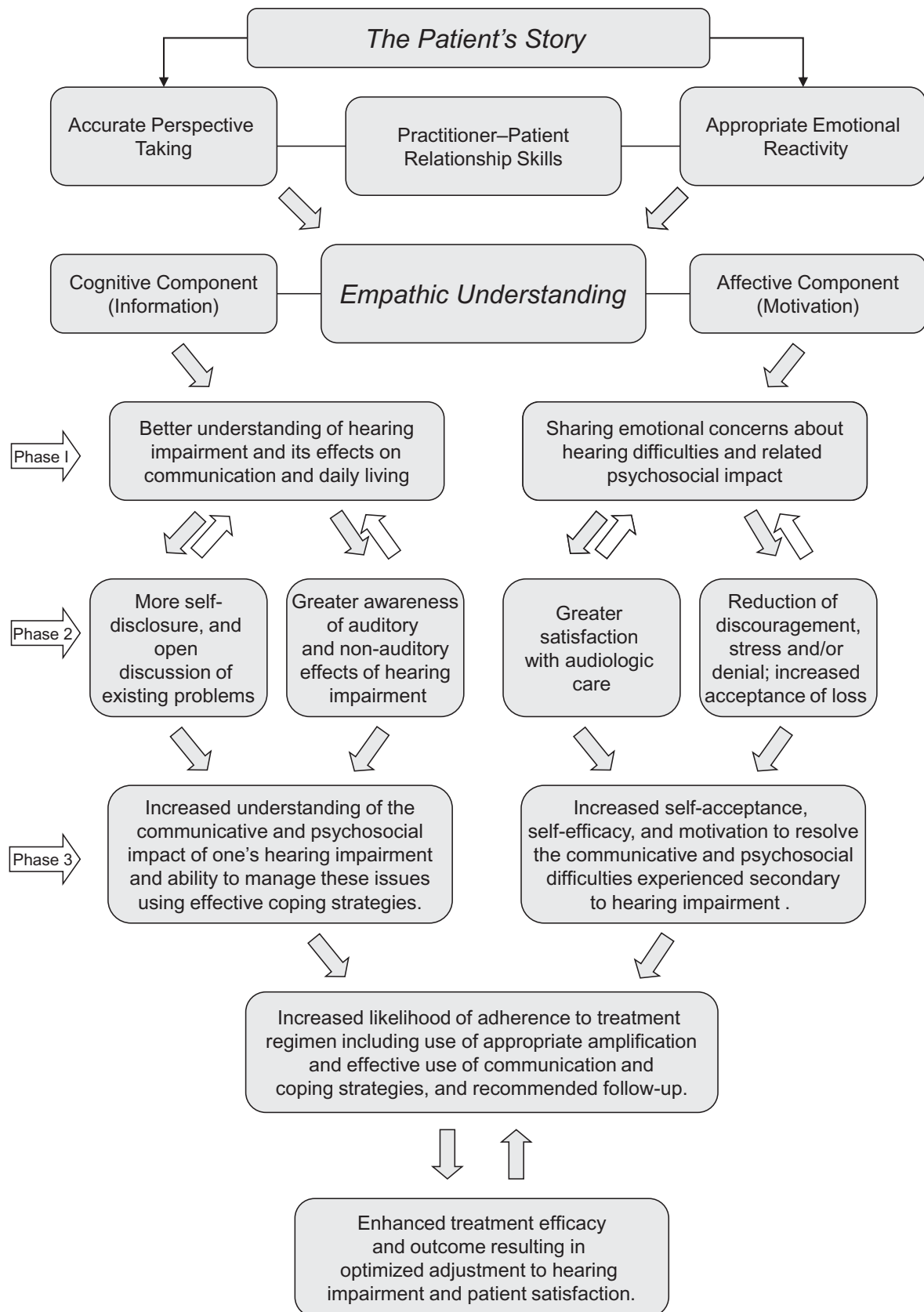


Figure 11-5. The role of empathic understanding in AR counseling. *Source:* Based on Squier (1990) and Erdman (1993).

that is biopsychosocial, interactive, facilitative, and conducive to adherence to one's treatment regimen. Evidence suggests that experienced clinicians become less empathic and that less-complicated conditions elicit less empathy from practitioners (Maxfield, Delzell, & Chumley, 2011). Additionally, differences in age, gender, or cultural background can affect one's ability to empathize effectively because such differences can make it more difficult to relate to one another (Riess, 2017; Sue, Sue, Neville, & Smith, 2019). The repetitive nature of audiologic evaluations can sometimes leave audiologists inured to the uniqueness of each patient. It becomes easy to forget that to each patient, hearing impairment is a significant and probably unwelcome development in life. Eliciting narratives allows us to see our patients as the unique, idiosyncratic individuals that they are, protecting us from potential burnout or tedium, and affording them our full range of empathy and professional expertise.

It is clear that the patient's narrative provides the foundation for establishing an empathic rapport with patients. When Engel (2005) responded to concerns that the biopsychosocial model was difficult to implement, his primary response was, ". . . the key is the interview . . . it is the basic scientific tool of the clinician—the door that opens to everything else. Successful application of the biopsychosocial model is 100% dependent upon the clinician's facility with interviewing" (p. 378).

When hearing impairment and the audiogram are the focus of an audiologic encounter, the patient's experience of hearing loss receives little to no attention. Unfortunately, this makes it virtually impossible to initiate a practitioner-patient relationship that is based on an understanding of the patient's concerns. Without this relationship, without a shared understanding of why the patient is presenting, the basis for engaging the patient in the rehabilitation process does not exist. Hence, our first responsibility is to hear each patient's story. Each story is the spark that activates biopsychosocial intervention. Eliciting that story is the clinician's first responsibility in each new encounter; ultimately, it may be the determining factor in the patient's outcome. Engel was fascinated by Margaret Mead's (1975) use of video recordings to study human behavior as this supported his contention that behavior, feelings, transactions, and relationships can be scientifically studied through observation, introspection, and dialogue (Engel, 1977a, 1977b, 1987, 1997). He advocated allowing patients to talk about themselves, their families, and symptoms and observing their verbal and nonverbal communication. He considered all of this scientific data.

In contrast, what usually occurs during the audiology patient's first visit is an interview focused on the etiology and progression of the hearing impairment. Obviously, information regarding the onset of hearing impairment is vital in establishing a diagnosis. Additionally, the nature and cause of a patient's hearing impairment undoubtedly can play a part in how the loss is experienced. Nonetheless, questions that are focused on auditory disorder and impairment are specific in nature and elicit responses that are similarly limited in scope. In fact, many answers are simply two- or three-word phrases or yes/no answers. They do not reveal the essence of the patient's experience.

The quotes included in Table 11-4 are "sound bites" from actual patient stories that are presented to illustrate the potency of hearing loss narratives. Even these brief snippets are compelling, poignant, warm, funny, sad, and, at times, joyful. They are distinctly human elements that provide the critical, initial link between the audiologist and the patient. Listening to patients' stories, clinicians come to share their patients' experience of hearing impairment in ways simply not possible through an audiogram, case history, or even self-reported measures. The clinician who listens intently to the father as he laughingly describes his little girl grabbing his face to make sure he hears everything she has to say cannot help but connect with him on a human level. In that connection, a patient recognizes that he is understood. The groundwork for empathy, trust, and rapport has been established.

Beginning clinicians often wonder how to elicit stories from patients that reveal such distinctively human elements. The fact is, every patient we see has a story. Moreover, the reason that patient is seeing you is to tell you that story. Open-ended statements such as "What brings you here today?" or "Tell me about the hearing problems you have been having" are appropriate ways to encourage patients to tell you about their experiences. Nonverbal factors are also important. Ideally, you are in a private area, seated face to face, maintaining eye contact, and listening while your demeanor communicates your genuine interest in what a patient is telling you. You are not walking toward the test booth with the patient following you or thumbing through a chart without looking at the patient. The challenge is not so much how to get them to reveal information as it is how to stay out of their way so they can do so. After asking the initial question, simply allow them to answer. They may not respond immediately. Giving someone room to answer is actually giving them time to answer. It is

Table 11–4. Excerpts from Patients’ Stories

“My wife’s small talk sometimes goes unanswered. I regret the difficulties she must endure. I know I must be a burden to her at times. It is more for her benefit than mine that I’m thinking about getting hearing aids.”

“When someone has to repeat for me, I feel embarrassed and inadequate. I scold myself—you should be listening more closely, concentrating more! I get angry at myself. When I hear something incorrectly and respond inaccurately, then I am really embarrassed! I feel like my credibility is slipping away. In class, I try to cover up by saying, ‘Speak up and talk clearly so the whole class can hear your question.’ Inside though I don’t think I’m really fooling anybody. Everyone must see through my sham. I feel so unsure of myself as a professional.”

“I get annoyed when I don’t hear someone and they yell loud enough for God and everyone to hear, ‘What are you, deaf?’ I tell them what they can do!”

“At parties, I have a terrible time. I move closer to the speaker and they back away. I ask them to speak up, but after a few words, their voices drop again. Not only am I a social loser, I’m a bore! When I’m in the backseat of our carpool, I have a terrible time understanding the conversation up front. So, I lean forward and ask them to speak louder. When they don’t, which is most of the time, I sit back and drop out of the conversation. There I am again, the social outcast.”

“My 5-year-old son has it figured out. He says, ‘Daddy, come in here away from the TV so you can hear me. I got something to tell you!’”

“I work on a psychiatry ward and listen to patients eight hours a day. Asking patients to repeat is not only frustrating for them and for me, it also alienates me from them. This is disastrous in a practitioner/patient relationship and causes extreme anxiety on my part.”

“My family has stopped trying to communicate with me anymore when we’re watching TV in the evening. I feel like I am losing touch with my family!”

“At a crowded cocktail party, when I’m talking to women with high pitched, or soft voices (small talk in which I may or may not be interested) I can’t understand a thing they say. I get tired of faking it, kill my drink, and excuse myself to get a refill. It gets to the point where it’s not worth the effort.”

“A woman and I were talking soft and low like lovers do. I asked her to repeat something but she said, ‘Never mind’ and the mood abruptly changed. It made me wonder how many ‘yesses’ I’ve missed.”

“At a party where the music is already 3 times louder than I care for, three people are all trying to talk to me at once. I pick out the one who is talking the loudest and listen to him. Usually, I still can’t understand what he’s saying. So, if he smiles, I smile. If he frowns, I frown. When he is done talking I end up walking away wondering what the devil he was trying to tell me.”

“I never purposely try to ignore anyone, but often it probably seems that way. Some may see me as unapproachable or think I’m rude.”

“I don’t think I will ever accept my hearing loss. I’ll always believe this is just temporary.”

“I cannot tell you what a difference my husband’s hearing aids have made in our life. He wasn’t listening to me. I was sure he didn’t care enough to listen anymore. He just ignored me. I was so afraid there was someone else.”

“I get very unhappy with myself when I don’t hear something someone in my family tries to tell me. It’s important to them. It must give them the impression I don’t care which isn’t true. If I ask someone to repeat what they said, they’re put off. I know I’ve hurt my children’s feelings. I have some fence-mending to do.”

continues

Table 11-4. continued

"My four-year-old daughter makes sure I hear her. She stands right in front of me or climbs on my lap. Sometimes she even grabs my face and looks straight at me! How can I miss what she has to say?" "At work I attend and/or conduct meetings, participate in several counseling sessions, and communicate on the phone or in person all day. At the end of the day, I'm drained from the mental strain of concentrating and trying to compensate. After wearing a hearing aid for only two days, I can already see a marked difference in how I feel by 5 o'clock. I regret that I didn't pursue correcting my hearing loss earlier in my career." A month later she added, "I can stay up an hour to an hour and a half later than I did before. It's great getting to watch the late news with my husband." "My teenage daughter gets away with murder. She'll tell my wife that she asked me if she could do something or have something and I said OK. When I say, 'You never asked me that!' She responds with, 'I did so! You said it was OK! It's not my fault if you didn't really hear what I said!'" "I've been accused of being off in my own little world." "I tried to go to college but the ambient noise in a classroom (papers rustling, background chatter, etc.) made listening to the professor difficult to impossible and not only made me anxious but also made him angry because he had to keep repeating. I stopped taking classes to remedy this particular situation." "Telephone use at work is critical for task accomplishment. If I can't get this part down I may as well fold up my tent and say sayonara." "My husband's hearing loss has affected us all. We love him and want him back."
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always tempting to ask another question if the patient does not respond immediately. Allow patients time to gather their thoughts and to formulate an answer. To refrain from interrupting the patient's thoughts, mentally count to 10, slowly. Literally, do so. If necessary, count to 10 again. It is exceedingly rare for a patient not to respond within that amount of time. If on a rare occasion that should occur, simply rephrase your question and again, wait. Allow the patient to see that you are genuinely interested in what he or she has been experiencing and that your question is a sincere and important one. Listen, appreciate, and provide an empathic response. If clarification is needed, ask them to explain in more detail. Verify that you correctly understand. Ask them if there is anything else they would like to tell you about their hearing problems.

The goal is to understand the patient's story—to appreciate that story and to come to a shared understanding of what the experience of hearing impairment means to the individual (DiLollo, 2014; DiLollo & DiLollo, 2014). The goal in audiologic counseling is to bring about change, to the extent possible, in the patient's story as it reflects the difficulties and distress in experiencing hearing impairment. Our role is a facilitative one; we engage patients in the treatment

process and we enable patients to make decisions that are both meaningful and acceptable to them. The narrative story is an increasingly common way of facilitating understanding. DasGupta and Charon (2004) recommend the use of narrative to teach empathy to medical students. The increasing use of narratives to promote understanding of patients' experiences and of practitioner-patient relationships is having a humanizing effect throughout health care.

Counselor Characteristics

... in the mid '70s ... people spoke of "bedside manner," which was regarded as a personal attribute that you were either born with or not. If you were, so much the better for you (and for your patients) and if you weren't, oh well. . .
—Suchman (2003, p. 677)

Effective counseling involves broad-based professional knowledge and skills combined with a set of personal aptitudes and characteristics. Typically, the knowledge and skills are acquired through academic programs and clinical training, whereas the personal aptitudes and characteristics often appear to be exten-

sions of one's personality. Nonetheless, many of the behaviors associated with these aptitudes and personality characteristics can be learned and honed. Although the practitioner-patient relationship is one of the most carefully examined variables in health care, audiologists' relationship skills have not been the subject of research. Moreover, our counseling skills are not often the focus of clinical training and evaluation. Work by English and colleagues (2007) is a notable exception.

Counselor characteristics are an essential part of the therapeutic relationship. The therapeutic relationship consists of an interpersonal relationship and the working alliance—essentially a measure of how well the practitioner and patient “work” together (Horvath, del Re, Flückiger, & Symonds, 2011). The latter involves the critical functions of engagement in goal and treatment planning. In his person-centered therapy, Carl Rogers posited that empathy, congruence, positive regard, and “unconditionality” are the necessary and sufficient conditions to effect therapeutic change. Moreover, he claimed that untrained persons who demonstrate an interest in others and a desire to help, and who present with these necessary and sufficient conditions, can be as effective as trained counselors in the helping process. Since that time, others (Corey & Corey, 2016; Egan & Reese, 2019) have stressed the veracity of Rogers' observation.

In Table 11–5, characteristics of effective counselors are categorized in terms of traits related primarily to personality, character strengths, and the core professional characteristics. Chief among these, of course, is empathy. Rogers maintained that empathy,

unconditional positive regard, and congruence enable patients to be more positive, realistic, self-directed, and open in their experiencing. Rogers (1951) defined empathy as:

... the counselor's ability to assume, in so far as he is able, the internal frame of reference of the client, to perceive the world as the client sees it, to perceive the client himself as he is seen by himself, to lay aside all perceptions from the external frame of reference while doing so, and to communicate something of this empathic understanding to the client. (p. 29)

Empathy is a prerequisite for the development of rapport, i.e., an alliance based on trust, cooperation, and a shared understanding of the patient's perspective. As in the model discussed earlier, it is generally concurred that empathy has an emotive (affective) component and an intellectual (cognitive) component. Norfolk, Birdi, and Walsh (2007) suggest that an element of motivation, possibly fueled by openness and curiosity, must exist for empathy to occur. The motivation essentially consists of wanting to understand the patient. Another key element of their vision of empathy is the ability to understand another's experience and perspective.

Given the critical role empathy has in terms of patient satisfaction and treatment outcomes, strategies to teach and develop it have received considerable attention throughout health care (Batt-Rawden, Chisolm, Anton, & Flickinger, 2013; Bayne, 2011; Beckman et al., 2012; Coulehan et al., 2001; Koch, Gitchel,

Table 11–5. Characteristics of Effective Counselors

Personality	Character Strengths	Professional
Patient	Self-aware	Empathy
Warm personality	Culturally aware	Warmth and caring
Good listener	Ability to analyze own feelings	Openness
Perceptive and sensitive	Ability to serve as a model	Positive regard and respect
Like people	Altruistic	Concreteness and specificity
Non-threatening demeanor	Strong sense of ethics	Communication competence
Sense of humor	Responsible	Intentionality
Desire to help		
Positive attitude		
Problem solver		

& Higgins, 2009; Krasner et al., 2009; Lim, Moriarty, & Huthwaite, 2011; Riess, Bailey, Dunn, & Phillips, 2012; Stepien & Baernstein, 2006). The Association of American Medical School's (AAMC, 2005) stated goal of preparing physicians who are compassionate and empathetic coupled with evidence that empathy actually declines during professional training (Bellini & Shea, 2005; Chen, Lew, Hershman, & Orlander, 2007; Maxfield, Delzell, & Chumley, 2011), highlights the need to focus on empathy throughout training for careers in health care. Among the interventions that have been implemented to increase student and current practitioners' empathy are reflective writing, theatrical performances, and literature courses based on illness narratives, mindfulness practices, and experiential learning that has included admitting students to the hospital with a fake diagnosis.

Pinkerton (2008) added to the list of important counselor characteristics in a wonderfully thought-provoking article entitled, *"I Apologize for Being Late": The Courteous Psychotherapist*. He emphasizes that above and beyond the obvious importance of responding courteously to the patients we serve, such behavior has positive effects on the therapeutic relationship. Inadvertent rudeness can have deleterious effects as in the case of the therapist who dozed off during a patient's session. Suffice it to say, the patient became irate and left, never to return. Pinkerton also discusses the possibility that patients may feel insignificant or diminished by a practitioner's chronic tardiness. Taking telephone calls, checking the time, and mispronouncing patients' names are other potential pitfalls that clinicians should guard against. Novice and long-standing clinicians alike would benefit from periodically reminding themselves of the points Pinkerton raises.

It would be ideal if all clinicians were able to demonstrate the necessary characteristics and skills at all times with all patients. For most of us, it remains an ideal. Our professional responsibility, however, is to strive to achieve that ideal. Increasing emphasis is being placed on "self-monitoring" or "mindful practice" as a means of enhancing the likelihood that we convey empathy, warmth, and genuineness; that we eliminate bias and narrow thinking in our clinical judgments; and that we demonstrate attentiveness to our patients and to our clinical decision-making.

Epstein (1999, 2003a, 2003b, 2017) stresses the need for self-awareness in practice and describes four key elements to mindful practice: attentive observation of oneself, the patient, and the problem; critical curiosity; "beginner's mind;" and presence. The self-monitoring that Epstein advocates facilitates clear thinking, multitasking, and personal growth. Those

who practice yoga or who are familiar with ancient Buddhist thought will recognize the Eastern philosophy principles in mindfulness approaches. On a very simple level, being mindful consists of clearing one's mind, being fully present in the moment, and adapting a stance of open, nonjudgmental curiosity. Remaining mindful in our interactions with our patients enables us to convey empathy, instill trust, and cultivate active patients who engage in the rehabilitative process. Beckman and colleagues (2012) found that a 52-hour mindful communication program for practicing physicians improved their self-awareness as well as their ability to be attentive, to listen deeply to their patients, to respond to patients more effectively, and to develop adaptive reserve. As Dobkin (2011) points out, when listening with an open, curious mind, we can hear our patients' stories—who they are, what their problems are, and why they are seeking our help. Without this information, how do we know what they need? Moreover, how can we possibly ensure that they will be satisfied with and benefit from the help we can provide?

Engaging Patients in the Rehabilitative Process

Clinicians are responsible for providing patients with the knowledge, self-efficacy, hope, and motivation to manage chronic health problems. Activating patients and making them partners in the management of their conditions develops from a working alliance that is built on trust. Effective counseling involves using a variety of techniques to engage patient in self-management of chronic conditions. Long-term adherence to a recommended course of treatment or behavior is maximized if the patient is involved in deciding what that course of action will be. It should be feasible and appropriate for their lifestyle. Central to successful self-management is Bandura's principle of self-efficacy (1977, 1982, 1997). Self-efficacy plays a major role in behavior change because it represents the perception that one is capable of performing a specific task or accomplishing a particular goal. To promote patients' self-efficacy beliefs, it is necessary to:

- show or persuade individuals that they have the capability to accomplish the desired behavior,
- enable individuals to master specific skills, and
- build on existing and ideally related skills that the individual has already demonstrated.

In learning new behaviors, a sense of mastery is required that is achieved through practice. In audiology counseling this may focus on adjusting to a new hearing aid, developing new communication skills, or becoming more assertive in difficult communication situations. Smith and West (2006) contributed an excellent tutorial on the applicability of self-efficacy principles to AR.

Successful engagement of patients in the management of their hearing problems may also involve instilling hope. Kent and La Grow (2007) conducted an investigation of the role of hope in the relationship between disability characteristics and adjustment to hearing impairment. Hope was found to have a substantial role, suggesting that it can be regarded as a principal factor in facilitating adjustment to hearing loss. The instillation of hope is a potent mechanism of change in group counseling (Yalom & Leszcz, 2005). Audiologists can be instrumental in initiating the instillation of hope by conveying the belief that AR will help resolve the communication and adjustment difficulties patients experience secondary to their hearing impairment. As such, it motivates patients to participate in the AR process. Meeting others who have benefited from AR in the past or interacting with other group members who are making progress in resolving similar issues, can promote additional hope, which is optimized when the patient begins to experience success in resolving his or her own hearing problems (Erdman, 2009).

Zubialde, Eubank, and Zink (2007) stress the need for a relationship-centered approach that engages patients in ongoing learning, growth, and management of their health. Self-management programs have been developed in health education programs to address the management of chronic health conditions (Holman & Lorig, 2000, 2004; Lorig & Holman, 2003). The programs promote the development of six skills: problem solving, decision making, utilizing resources, forming a practitioner-patient relationship, action planning, and self-tailoring. These skills are applied to three primary self-management tasks: medical management, role management, and emotional management. For individuals with chronic illness or disability, self-management (i.e., management of the condition and the effects it has on daily living) becomes a lifelong responsibility. Stanford University's Patient Education Research Center has developed self-management programs for patients with arthritis, HIV-AIDS, and back pain. It also has a program for individuals with varied chronic diseases, the Chronic Disease Self-Management Program (CDSMP), which has been replicated with diverse populations and is now licensed to organizations throughout the U.S.

and in more than 20 other countries. Additionally, there are now CDSMP web-based programs for specific health conditions including heart disease, arthritis, and diabetes.

Given the extent to which the needs of individuals with chronic disease and disability have been unmet, the self-management programs have been a valuable and successful health care option for many. Holman and Lorig have consistently maintained that there are ways to meet the needs of those with chronic condition despite the health care system's failure to do so. Self-management education is one option, and a strength inherent in these programs is the group approach. Another option involves group visits; groups of patients periodically meet with their practitioner. Comparable to group hearing aid orientations or hearing aid follow-up visits, this concept could be extended to address a range of issues including communication difficulties. The third option includes remote management, an area in which audiology practice continues to expand exponentially (Saunders, 2019). The internet is permitting contact between practitioners and patients without involving appointment time and travel for patients. Individuals in remote, underserved areas can now be evaluated and counseled. Although clinicians have had concerns regarding the impact of not having face-to-face contact with patients, such concerns pale in comparison to the problems endured by those who do not have access to care. The flexible options afforded by advances in technology should be embraced for the good they can do, but they must also be evaluated and monitored as thoroughly and frequently as any of the clinical services we provide.

Another approach that has been suggested for engaging patients in management of their hearing problems is motivational interviewing (Miller & Rollnick, 2012; Rollnick, Miller, & Butler, 2008), which includes concepts from social psychology, attribution theory, and cognitive dissonance. Motivational interviewing is based on the stages of change approach (Prochaska & DiClemente, 1982; Prochaska et al., 1994). Unlike Rogers' client-centered approach to counseling, motivational interviewing is a more directive approach that focuses on removing barriers to change. The goal of motivational interviewing is to promote choice as well as responsibility in implementing change. Four guidelines in motivational interviewing include: (a) express empathy, (b) "roll" with resistance even when the patient is defensive and argumentative, (c) explore discrepancies between present behavior and behavioral goals, and (d) support self-efficacy.

Motivational interviewing has been beneficial primarily in treatment programs involving cessation

of addictive behaviors, but it has not been as effective in promoting adherence to medication regimens for diabetes (Rosenbek Minet et al., 2012) or for osteoporosis (Solomon et al., 2012), leading to questions about its ability to promote adherence (Berkowitz & Johansen, 2012). Randomized clinical trials are indicated to evaluate its appropriateness for specific behavioral changes given that results to date have been equivocal. To the extent that motivational interviewing is focused primarily on specific behavior changes, it may be more appropriate for issues related to the acceptance and use of hearing aids and other technologies than for more complex issues related to adjustment problems and communication difficulties.

Like other health care professions, audiology needs to be proactive and creative in developing ways in which to meet the needs of our patients. New interventions and group approaches, and a recent upsurge in audiologists' interest in what it means for our patients to live with hearing impairment are positive signs that the changes health care is currently undergoing will lead to a more comprehensive, patient-centered approach to audiologic practice. Acknowledging that counseling constitutes not only a therapeutic process or specialized clinical activities but also our therapeutic context will enable us to practice relationship-based, patient-centered care and enhance patients' satisfaction with the interpersonal and technical aspects of audiologic services.

Summary

Modern health care is the product of the biomedical model. The result is a health care system that has achieved remarkable success in acute care but that has been woefully inadequate in managing chronic illness and disability. The biopsychosocial model espouses approaches in which patients' subjective experiences of illness and disability and the impact their conditions have on their lives become integral aspects of intervention.

The biopsychosocial approach as manifested in patient-, person-, family- and relationship-centered care places counseling at the forefront of intervention as the essence of audiology's therapeutic context, therapeutic process, and therapeutic activities. The emphasis is shifted from the hearing impairment and the audiogram to the patient, the subjective experience of hearing impairment, and the impact on all who are affected by the limitations the impairment imposes. By adhering to biopsychosocial approaches

to care and establishing empathic, facilitative relationships with patients and their families, audiologists will be able to promote patients' self-efficacy and engage them in a mutual decision-making process regarding the optimal management of their hearing problems. The outcome of successful audiologic care will be patients and families who are able to manage hearing-related communication and psychosocial problems effectively and who are satisfied with the audiologic care we have provided.

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12

IMPROVING PATIENT ADHERENCE

Perceived Importance, Comfort Rankings, and Decisional Scales

John Greer Clark

Introduction

Nearly one-third of the more than 40 million American adults with reported hearing difficulties have failed to seek the help they need (Mahboubi, Lin, & Bhattacharyya, 2017). When they do make appointments, it appears that the hearing loss exploration and evaluation process is not always as comprehensive as it should be, and that the consultation could be structured better to ensure greater adherence to the audiologist's recommendations.

Audiologists make recommendations for their patients based on a comprehensive case history and accurate diagnostics. These recommendations are greatly enhanced when time is taken to assess the impact of background noise on speech perception and to explore the patient's perceived impact of the presenting complaint. Both of these are requisite to the most efficacious recommendations and are key components of best-practice guidelines (American Academy of Audiology [AAA], 2006). In spite of this, only 29% of surveyed audiologists directly assess the prevalent complaint of difficulty hearing in noise with more than half of the patients they see, and they reportedly explore hearing loss impact as perceived by the patient on a routine basis only 15% of the time

(Clark, Huff, & Earl, 2017). Yet even when best-practice guidelines are followed and careful explanations of test results and how these relate to the challenges associated with diminished hearing are provided to patients, professional recommendations are not always accepted. This lack of adherence to recommendations is not a unique challenge to audiology, having been evidenced for many years within other areas of health care as well (Doughty, Baker, Jacoby, & Lavaud, 2003; Hammar et.al., 1972; Prochaska & DiClemente, 1984a; Zolnierek & DiMatteo, 2009).

The term *compliance* implies one who obeys, obliges, or yields to another, especially in a submissive manner. As such it suffers the same dependent connotations as do the words *client* and *patient*. *Client*, deriving from the Latin *cliens*, has at its root one who leans on another for protection, while the word *patient* (*patiens*) suggests one who passively receives care (Martin, 1996). The more contemporary word, *adherence*, implies that patients may be active participants in their health care decisions. If our goal is greater adherence to recommendations, then it is inherent that we fully embrace the precepts of person-centered care and engage our patients as active participants in the hearing loss treatment process (Clark & English, 2019).

It is clear to both professionals and their patients that hearing aids do not fully restore hearing. Indeed,

even the most advanced hearing aid technologies can restore only approximately half of the degree of sustained hearing loss (Dillon, 2012; Lybarger, 1944). As such, many patients, even after the most successful hearing aid fittings, continue with significant restrictions to their active participation in various areas of their lives. For these individuals, augmentative training can be invaluable (see Chapters 13 and 16). Hearing rehabilitation training through a variety of venues has been demonstrated to be beneficial when implemented (Abrahamson, 2000; Kricos & McCarthy, 2007). A primary means of coping with a handicap, however, is often to try to appear no different than others (Hetu, 1996), which may create a desire to hide one's hearing loss. The active use of perceptually visible hearing technologies or the advocated use of communication management and repair strategies often runs counter to this desire.

Patient motivation toward the acceptance of and action on professional recommendations has been successfully enhanced in a variety of health-related areas, including substance abuse, smoking cessation, changes in diet, medication adherence, and the establishment of exercise regimens (Tonnesen, 2012). Similarly, motivational enhancement techniques have proven successful in hearing loss management (Ferguson, Maidment, Russell, Gregory, & Nicholson, 2016). The purpose of this chapter is, therefore, to present the concept of motivational engagement as a means to help patients improve adherence to their audiologist's recommendations of needed hearing technologies and communication management strategies.

A Readiness for Hearing Loss Help

As hearing aid technology has improved over the years, the bedrock of audiologic rehabilitation for most with hearing loss has become the proper selection, fitting, and adjustment to personal acoustic amplification. Certainly, as noted throughout this text, amplification alone often falls short of fully addressing all of the daily communication challenges the adult with hearing loss confronts in the home, at work, and within the larger milieu of society. Yet it remains that without the procurement of appropriate amplification when needed, little progress toward hearing rehabilitation can be made.

Historically, hearing aids once were of only marginal benefit for those with sensorineural hearing loss. Many potential consumers of hearing health services (and their physicians) remember the larger sized hearing aids of the past that provided limited

benefit. This, coupled with the stigma of hearing loss, that—while arguably decreasing—still remains (David, Zoiner, & Werner, 2018; Gagné, Southall, & Jennings, 2011; Kochkin, 2012), can often place a direct obstacle in the path of audiologists when recommendations for improved hearing include the fitting of a visible prosthesis or the use of communication management strategies that necessitate drawing attention to the hearing loss itself.

The marketing of many commercial products panders to the desire for an appearance of maintained vitality if not Juan Ponce de Leon's dream of eternal youth. Given human nature to avoid unwanted life changes that require a necessary alteration in one's self-concept with aging (Clark & English, 2019), it is not surprising that hearing aid market penetration remains at approximately only 30% (Abrams & Kim, 2015). Those who do seek hearing assistance, inevitably come to the audiologist's office within varying stages of readiness to accept recommendations.

Stages of Readiness

Nearly 40 years ago, Goldstein and Stevens (1981) described four levels of patient motivation that serve to build perspective on the spectrum of readiness for change so often encountered in the clinic. The first two levels of motivation represent the majority of patients seen for hearing consultations and include those with relatively high motivation for change. Specifically, Type I individuals anticipate treatment with a positive outlook and are the most likely to accept hearing care recommendations. Type II individuals hold similarly positive attitudes and the desire for improvement, although their particular case may present an additional challenge such as a coexisting condition that may limit treatment options, as in the case of significant dexterity or tactile issues, severe visual limitations, or cognitive decline. Still, the challenge in working with Type II individuals is not overcoming a lack of readiness to receive help but rather designing the most appropriate treatment plan to serve the patient best.

As can be surmised, the remaining Goldstein and Stevens patient readiness levels account for those individuals with more negative outlooks on hearing management. Whereas Type III patients are generally open to some level of treatment planning exploration, those classified as Type IV frequently display open rejection of amplification and audiologic rehabilitation, and possibly the existence of any real need for either.

Motivational engagement is most appropriate for patient types III and IV. If these patients come

to the audiology office, reluctantly on their own or at the behest of another, the audiologist becomes responsible for coaching these new patients toward self-reflection on their current perceptions of hearing care, their willingness and readiness for change, and their confidence and comfort in being able to follow any recommendations that bring about that change. Without this form of motivational engagement, these patients will likely leave the audiologist's office without the care and assistance they truly need. Such a scenario presents a failure for the patient and the family, as the hearing loss communication difficulties that prompted the appointment have not been adequately addressed. This scenario also presents a failure for the audiologist, who has lost the opportunity to provide solutions that he or she knows would prove beneficial. If, on the other hand, a patient is not truly motivated to use hearing aids, yet is persuaded to obtain

hearing aids by family members or the audiologist, it is possible that the patient will fail to accept fully and adapt to the purchased devices, yielding suboptimal outcomes.

But, using hearing aids, which often can be quite visibly discreet, is different from openly sharing your communication needs with others. As such, even the Type I and Type II patients may need coaching toward use of communication management strategies. The professional's instruction on the use of strategies is frequently insufficient by itself.

The Change Circle

An alternate view of readiness for change was presented by Prochaska and DiClemente (1984b) (Figure 12-1). This visual representation of the various

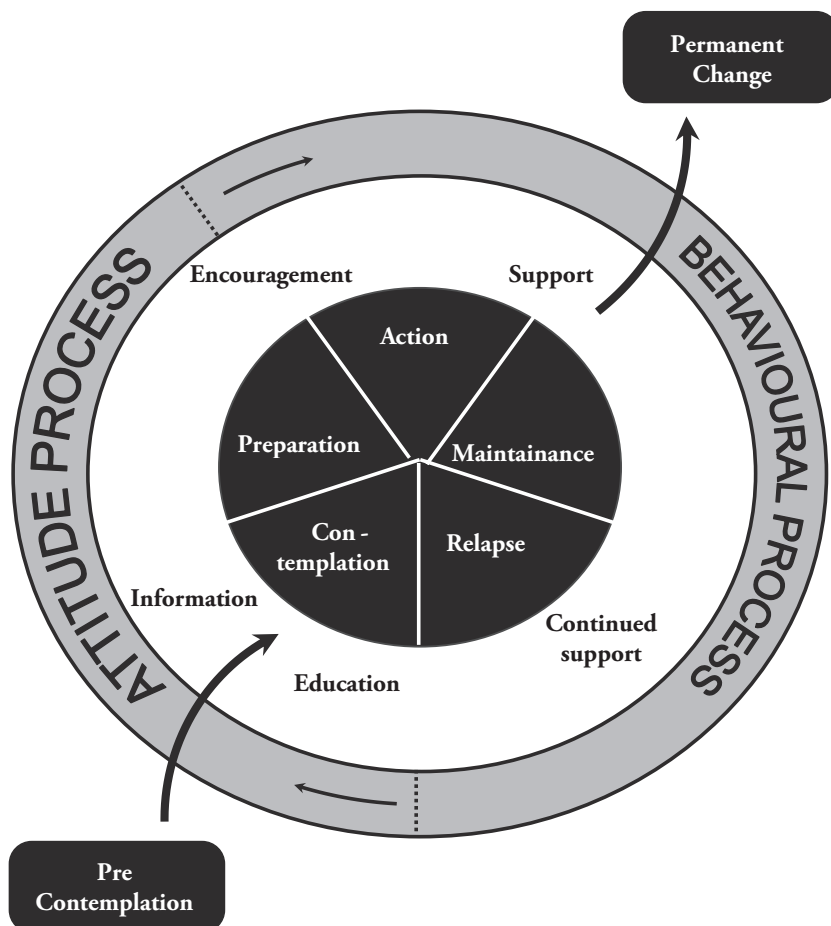


Figure 12-1. The Change Circle used to envision the cycle of stages experienced when considering audiologic treatment options. All stages require some form of clinician support. *Source:* Adapted from Prochaska and DiClemente, 1984b, with permission from the Ida Institute, idainstitute.com

stages of readiness for change provides a framework for understanding the patient's incoming perspectives. Self-assessment measures are particularly useful in facilitating focused discussion with our patients and providing clues for identification of a patient's specific stage of readiness for change. (*Editors' note:* For further information regarding self-assessment, the reader is referred to Chapter 6 in this book.) A patient's specific location within the change circle reflects his or her readiness to work with hearing management recommendations and may reveal a need for further information or support prior to reaching a point of action.

As audiologists, we are privy to only a small slice of each patient's hearing loss journey. Much transpires before patients arrive at the audiologist's door, and these patients frequently leave their last post-fitting follow-up appointment with a life of residual hearing challenges ahead of them. As we look at the change circle, we see that many begin their hearing loss journey in a precontemplative stage in which they may not fully recognize the negative impacts of hearing loss that others in their familial, vocational, or social lives more readily see. It is also possible that these patients are aware of decreased communication effectiveness but do not fully attribute the noted changes to deficiencies on their end but rather blame communication difficulties on the speaking habits of others. Gradually, as communication becomes increasingly

frustrating for either the person with the hearing loss or those with whom he or she interacts, there may be movement into a more contemplative stage, which may hold considerable ambivalence toward change.

Prochaska and DiClemente's circle of behavioral change notes a transitional "preparation" phase between contemplation and action in which one further prepares for what is beginning to be viewed as an inevitable change. While normal ambivalence toward change may continue, the patient is frequently not fully confident in how to proceed. This is an information-gathering stage, often beginning with closer attention to marketing, searches on the internet, or discussions with friends who have already obtained hearing aids, and often eventually culminating in seeking advice from a professional. It can be a mistake for the audiologist to assume that patients in this transitional preparation stage are ready to purchase hearing aids. It may be that, for patients in the transitional preparation phase, motivational engagement is most beneficial as an aid to transition into action.

As discussed in Chapter 6, audiology has a wide array of self-assessment measures that can shed light on a patient's perceptions of hearing loss impact. The value of these measures in hearing loss management is second only to the results of routine audiometrics, and their use has been advocated as part of clinical best practice for years (AAA, 2006). Given the value that these measures bring to the rehabilitation pro-

Case Study 12-1

Dr. Reeves has completed the hearing evaluation for his 10:30 patient, Mr. Rodrigues, and is explaining his test results. As he explains the findings of Mr. Rodrigues' high-frequency hearing loss and how this can impact speech understanding especially when listening in noise, Mr. Rodrigues nods attentively. But when he says, "Actually, the degree of hearing loss we have here can benefit greatly with today's hearing aids, especially with the newer microphone technology that helps to block out some of the background noise," he senses a resistance. Mr. Rodrigues leans back in his chair, crosses his arms and says, "I'm not sure I need hearing aids yet. I just wanted to see where things are."

►Dr. Reeves is finding he is losing this patient and he isn't sure why. He is discussing the hearing loss and its impact based on what he knows of the effects of diminished hearing on the reception of speech in different environments. But he has not explored the impact of the hearing loss from his patient's perspective. What he does not know is that Mr. Rodrigues rarely finds himself in noisy environments. His primary concern is that his grandson is becoming more distant as he senses his grandfather's frustrations when he can't hear the boy's softer, higher-pitched voice. Dr. Reeves is not talking to his patient's concerns or needs as he never learned what these were. And he is failing to meet Mr. Rodrigues where he is: squarely in the transitional preparation stage and not quite ready for action.

cess, it is surprising that only 15% of surveyed audiologists use these measures routinely and that 57% report they use them seldom if ever at all (Clark, Huff, & Earl, 2017). If Dr. Reeves, in the earlier case study, had more fully explored Mr. Rodrigues' perceptions of hearing loss, he not only would have been able to relate his test findings more directly to his patient's true concerns, he also would have been well poised to help Mr. Rodrigues explore the issues that would bolster his own internal motivation to pursue treatment.

Most people with hearing loss will eventually reach a point in which they are ready to take positive action toward the help they need before they make an appointment with the audiologist. Those who may still be in the transitional preparation stage and not ready to act on improving their hearing when they arrive at the audiologist's office may reach a readiness for action while in consultation with the audiologist. Either way, once in the action stage, these patients may still be experiencing the normal ambivalence we all face when we recognize the positive side of the action we have decided to take, but have not fully reconciled ourselves with the negatives of that action. At this stage, patients frequently need more than information alone, but also may need to explore actively their attitudes toward hearing loss and audiologic rehabilitation to bolster their decision to move forward. A common clinical mistake is to counter ambivalence with more information. Instead, the audiologist's skills in motivational engagement are required to explore the fears and concerns a patient still may have even at this stage of behavioral change.

The change circle concludes with two final stages. In the maintenance stage, some degree of ambivalence may persist but generally patients are pleased with the decision to be fit with hearing aids. The audiologist must continue to provide attentive follow-up and support for patients into the maintenance stage after a sufficient readiness for change has been reached. This stage is designed to maintain the patient's willingness to work through adjustment and acceptance of amplification and audiologic rehabilitation. The audiologist's role becomes one of support and encouragement to maintain the successes that have been reached. At this stage, it is critical to establish a plan to address the residual hearing difficulties that are often present following successful hearing aid fittings and to provide continued routine patient follow-up.

The long-term goal of audiologic rehabilitation is to provide sufficient aftercare for patients so that they never enter the relapse stage. However, some patients do discontinue hearing aid use, which may suggest that the perceived benefit was insufficient to sus-

tain the change. Support in the earlier stages—along with a well-implemented follow-up program and the provision of a more comprehensive treatment that includes communication strategy training—helps to avoid potential relapse. Should patients lose their positive attitude about treatment, however, the cycle continues via continuation of support as the patient prepares to reenter the action stage.

The circle of behavioral change is a visual reminder that our counseling and presentation of treatment options are to be customized to meet each patient in his or her current state. Thus, when used in conjunction with self-assessment measures, trained listening skills, and open discussion, this cyclical representation provides the audiologist a context in which motivational engagement can then increase success. Yet the success of any well-orchestrated and competently facilitated journey through the stages of behavioral change is highly dependent on trust.

Building Trust

Listening is a precursor to trust, and trust is a precursor to success. Without trust, acceptance of the information we provide to patients is compromised, attempts toward motivational engagement are jeopardized, and subsequent adherence to professional recommendations is threatened. Effective listening is a necessary foundation for trust and is facilitated by a concentrated effort to avoid known trust barriers (Table 12-1). Possibly one of the greatest of these barriers is communication mismatch, which occurs when clinicians respond to the content of patient's statements but fail to recognize and acknowledge the emotions that may lie beneath the spoken words.

Table 12-1. Barriers to the Establishment of Trust

- Poor listening skills
- Clinician responses that fail to acknowledge unexpressed emotion (communication mismatch)
- Multitasking during clinical exchanges
- Assuming an understanding of patient needs based upon previous patients with similar hearing profiles
- Basing clinical treatment decisions predominantly on audiometric findings without learning about patient concerns and desires.

Surveys have indicated that patients can perceive audiologists as insensitive or indifferent to their plight (Glass & Elliot, 1992; Martin, Abadie, & Descouzis, 1989). While this is of course far from true, it strongly suggests that audiologists may not be adequately displaying the listening behaviors requisite for perceived empathy and established trust. Discussion of counseling principles that enhance listening and fortify trust is beyond the scope of this chapter. Readers interested in learning more about effective patient counseling are encouraged to read counseling texts for audiologists that address these issues (Clark & English, 2019; Luterman, 2016).

Patient Education

Within several stages building toward behavioral change, those with hearing loss will be seeking information and clarification. Of all the counseling that patients need in the course of audiologic rehabilitation, patient education (AKA content counseling) is an area in which most audiologists feel comfortable. Unfortunately, the delivery of further information is often poorly timed. The confirmation of a hearing loss may release a flood of patient questions and concerns ranging from potential progression, etiology, risk factors for future generations, implications for anticipated quality of life, and others. It is frequently at these most emotionally charged times that audiologists provide critical information on the implications of test findings or options for treatment (Clark & Brueggeman, 2009). However, neuroscientific research reveals that heightened emotional states directly impede cortical processing of new information (Cahill, Babinsky, Markowitsch, & McGaugh, 1995; Canli, Zhao, Brewer, Gabrieli, & Cahill, 2000; Richardson, Strange, & Dolan, 2004).

To aid comprehension, patients should be given the opportunity to express their thoughts and concerns prior to the delivery of information. Questions or invitations for dialogue that will serve to facilitate comprehension might include: "Tell me what you have been told before about your hearing loss (or about hearing aids)" or "Before I share details on your test results, or any of my recommendations, what questions do you have for me now that we have confirmed there is a hearing loss?"

Patients from any of the Goldstein and Stevens readiness stages or at any placement on Prochaska and DiClemente's change circle may benefit from these or similar invitations to field concerns prior to the clinician's information dissemination.

There is a significant body of literature across health care settings that reveals poor retention of information delivered by health professionals (Anderson, Dodman, Kopelman, & Fleming, 1979; Kessels, 2003; McGuire, 1996). While the emotional state of the patient plays a large role in reduced information processing, the sheer volume of information delivered in a single appointment can be overwhelming for many (Tirone & Stanford, 1992; cited in Lesner, Thomas-Frank, & Klinger, 2001). Therefore, audiologists must strive to support patients openly in the face of unanticipated and unwanted diagnostic discoveries, acknowledging the importance of processing the news. Movement away from a monologue-driven transfer of information is a solid first step to ensure greater comprehension (Clark & English, 2019).

A Neglected Need: Motivational Engagement

Even with the clear establishment of trust and well-delivered information, there will always remain some patients who find the transition to action difficult. When requisite patient motivation for positive action is lacking, audiologists may counter with further information on the hearing loss and the technological benefits of amplification or find themselves defaulting to other external motivators including celebrity endorsements, financial incentives, or third-party testimonials. However, life lessons have repeatedly shown that motivation that arises from within is both more successful and sustainable than any external motivator.

When amplification is indeed warranted, patient apprehension including cost and maintenance demands, cosmetic factors, stigmatic impact, and acquaintance with others who did not succeed with hearing aids can, for some patients, hold greater sway over the internal decision process than potential benefit that may be afforded through successful hearing loss treatment. Directly addressing the possibility of a patient's negative perceptions of hearing aids and hearing loss can result in greater adherence to recommendations and a greater opportunity to provide hearing assistance to patients and their families. Toward this end, motivational engagement helps build the requisite internal motivation to accept and act upon recommendations (Rollnick, Miller, & Butler, 2008).

The need for an internal motivation to take action to address hearing difficulties has been addressed in the audiologic literature (Beck & Harvey, 2009; Beck, Harvey, & Schum, 2007; Harvey, 2003). The practice of motivational engagement helps audiologists guide

patients in (a) reflection on the impact of hearing loss; (b) their own willingness to make the necessary changes to address this impact; and (c) their perceived abilities to make needed changes (Clark, 2010; Clark & English, 2019; Clark, Maatman, & Gailey, 2012; Ferguson, Maidment, Russell, Gregory, & Nicholson, 2016). As audiologists help patients find an internal motivation for successful hearing loss treatment, there are few better insights than those that may be obtained through self-assessment measures.

For patients who arrive ready to accept treatment recommendations (Goldstein and Stevens' Types I and II, or Prochaska and DiClemente's "action" stage), it is not external motivators that induce action, but rather the effect of these external motivators on preexisting internal motivation to accept treatment options, including amplification use. For more reluctant patients, whose internal motivation is low, trying to entice action through external motivators is generally a weak approach, as it ignores the patient's true concerns.

The clinician's interactions and experiences with patients from opposite ends of the readiness spectrum may be quite different. For those with relatively low internal motivation, the audiologist's recommendations likely come as unwelcome news, often eliciting a number of defense mechanisms. But, for patients who are actively seeking an improvement of their hearing ability, little or no resistance may be encountered. Though the latter patient is often ideal, audiologists must be prepared to coach the full spectrum of readiness for change.

For those patients clearly not ready to proceed with treatment, audiologists frequently will acknowledge the lack of current motivation, provide what information they can, and offer a follow-up appointment to see if either hearing levels or impact recognition have changed. However, this approach should be a last resort after directly addressing motivation. When we fail to address lacking motivation, we have failed these patients and those who interact with them. These patients return to their lives with the same communication limitations they arrived at our offices with, destined to perpetuate the same failures and frustrations they have been contending with for some time.

Engaging the Patient Toward Change

While detailed descriptions of audiologic test results and their relation to speech understanding are informative (if a patient is in a state to process the infor-

mation effectively), when readiness is not previously existent, this information delivery cannot instill the same level of motivation as does a personal exploration of the negative impact of hearing loss and the perceived barriers to treatment. Building motivation to accept audiological treatment recommendations requires patients' acknowledgment and understanding of the influence of hearing loss within the context of their daily life activities. While a first step in more effectively working with our patients should be to increase our consistent use of self-assessment tools, a close second would be to engage the patient's primary communication partner in the consultation process, a resource that continues to remain underutilized in the clinic (Clark, Huff, & Earl, 2017; Stika, Ross, & Cuevas, 2002). Together, these may be keys to jump-starting internal motivation. Motivational engagement, when combined with a horizontal approach to patient management in recognizing the valuable contributions of both patients and communication partners to the rehabilitation process, facilitates a guided patient encounter that is as individualized as our treatment recommendations. (*Editors' note:* For further information on communication partners, the reader is referred to Chapter 19 of this text.)

In audiology, denial of hearing loss, unwillingness to make a life-altering change, opposition to diagnostic findings, and indecision to proceed demonstrate how uncomfortable change can be (Clark, 1999). As such, it is paramount that we accept each patient's feelings and attitudes of the moment (Rogers, 1951). Motivational engagement allows for that acceptance while helping patients to challenge themselves to view their perceptions differently, thereby moving them in a positive direction.

The need to engage our patients and enhance their motivation through guided self-realization is obvious; the exact method of choice, perhaps less so. In practical health care, the chosen protocol must be time efficient, especially when utilized within a busy facility and when discussion of any recommendations is not the sole planned activity during a given appointment. Unless explored, the audiologist rarely knows the patient's readiness for change and motivational level. Clinicians quickly discover that when the patients' readiness level is relatively high, the stages of motivational engagement move rapidly and take an insignificant amount of time. The up-front time investment is a bit greater when it is discovered that patient readiness and motivation are low. However, the net amount of clinical time saved is high when clinical time is not spent on recommendations that the patient likely will not pursue or may ultimately reject after attempting to follow.

Drawing the Line: Assessing Perceptions

Using a clean and simple scaled line, patients can visually rank their perceived importance for change and their belief in their own ability to make a needed change (Figure 12–2). Earlier responses to self-assessment measures may reveal specific difficulties resulting from untreated hearing loss, such as frustration when talking with others (e.g., Hearing Handicap Inventory for the Elderly, Ventry & Weinstein, 1982) or others expressing concern or annoyance with hearing abilities or being less social due to difficult communication (e.g., Self-Assessment of Communication, Schow & Nerbonne,

1982). The paramount factor is that the importance-ranking line links to the patient's own story, providing a self-reported internal motivator capable of compelling him or her to make a significant change.

Where Do We Go Next?

The paths to follow in motivational engagement differ at this point depending on the patient's answer. We ultimately will want to know two things: (a) how important change is, and (b) how strongly the patient believes he or she can make needed changes. (*Note: For further information regarding self-efficacy, the reader is referred to Chapter 13 of this text.*)

Case Study 12–2

Audiologist: It sounds like you're experiencing quite a few frustrations with your daughter when she comes to visit on the weekends. She seems to think it might be related to your hearing, but you feel that she talks too softly and mumbles. Am I listening to you accurately?

Patient: Yeah, I always tell her to speak up, and she never does. I try to hear her, but it's like she doesn't care if I understand or not. I feel like she's wasting her time coming over to talk sometimes because we argue so much over this.

Audiologist: I definitely agree with you that the way people talk can have a significant impact on our ability to understand what they're saying, and it would be helpful to address that with your daughter at some point. But also, as we discussed, the test results do show that you have some hearing loss. There are clearly several factors contributing to the frustration you and your daughter are experiencing, and certainly your hearing loss is enough to be a part of this.

Patient: Sure, it's probably a combination of things, but I'm sure it's not all me.

Audiologist: I agree. And we may need to tackle things from more than one front. But just so I have a better understanding on how to help you, can you tell me, say on a scale of zero to 10 with zero being not important at all and 10 being extremely important, how important is it to you, in light of the frustrations you have described, to improve the communication between you and your daughter.



Figure 12–2. The graduated line used to help patients visually represent their self-ranked reflection on the importance of a specific change or perception of personal ability to achieve such a change. Zero represents “not important” or “highly unlikely to be able to change” with increases in these scales up to 10, representing that a specific change is “very important” or that the individual is “highly likely to be able to change.” *Source:* Modeled after Rollnick et al., 2008.

Our paths are as follows:

1. **A high importance ranking followed by a high self-efficacy ranking.** This is the outcome we might expect for the majority of patients we see who will arrive in clinic at the Goldstein and Stevens Readiness Levels of 1 or 2 or at the action stage on the Prochaska and DiClemente change circle. Uncovering which patients are ready to move forward takes no significant clinical time and allows the audiologist to proceed with amplification discussion, order placement and subsequent fitting with a strong assurance of success.
2. **A low importance ranking,** which will require exploration before proceeding to the self-efficacy ranking. These patients will be fewer in number and represent those on the Goldstein and Stevens Readiness Level of 3 or 4, or the contemplation or preparation stages on the Prochaska and DiClemente change circle. This quick ranking ensures the audiologist does not proceed with a patient who is not onboard with the recommendations about to be made. The next step with these patients is to aid them in an introspective cost-benefit analysis of action versus inaction, as discussed later.
3. **A high importance ranking followed by a low self-efficacy ranking.** A patient may initially have a high importance ranking for change, or a low importance ranking that rises to a ranking of 7 or above after an introspective cost-benefit analysis as in Path 2, yet have a low self-efficacy ranking for actual implementation of a rehabilitation plan. It is not uncommon for some patients to see the benefits of using hearing aids, but have reservations toward openly revealing the presence of hearing loss through use of amplification, or fear that they may not be able to effectively handle the technology. The value of a self-efficacy ranking is that it allows the audiologist to explore concerns that patients have before these concerns become a true impediment to success. An interweaving of the clinician's person-centered counseling skills and implementation of cognitive and behavioral counseling principles are valuable when working with patients who may have a lower self-efficacy (Clark & English, 2019).

When the importance ranking is low (less than 7) as in Path 1, further discussion of the importance-ranking is warranted. In that event, an appropriate follow-up question might be: "What would it take to bring your ranking from a 4 to a 7?"

Attention to patient answers to this question can maximize use of clinic time and provide guidance for discussion. In addition, effective use of reflective silence during counseling could further facilitate productive discussions by providing an opportunity for patients to reflect and expand on statements (Clark & English, 2019; Luterman, 2016). If, however, no clear factors arise, the next move is to use a decisional box (Figure 12–3, as discussed later). In contrast, a ranking of a 7 or higher on the importance-ranking line indicates high prioritization for making a change.

When introducing the self-efficacy ranking, we are looking to see how confident the patient is in, or comfortable with, addressing hearing loss, which may involve amplification in addition to other avenues of audiologic rehabilitation. This ranking seeks the patient's level of confidence in being able to adopt a significant life change. Introduction of the self-efficacy ranking may look something like this:

"Let's take a look at this line again. This time, I would like to know how strongly you believe in your ability to follow my recommendations, which may include hearing aids, so that we can improve the frustrations you experience with your daughter? Here, zero is not strongly at all, and 10 is very strongly."

A strong ranking reflecting a sense of high self-efficacy for change is expected from most patients at or nearing the action stage on the change circle. Further discussion of the line is warranted, however, when a lower response indicates relatively low confidence in one's ability to change successfully. As stated earlier, it is not uncommon to see patients who demonstrate a high priority for lessening frustrations on the importance-ranking line paired with decreased confidence in their ability to successfully make desired changes. It is helpful to first focus on the positive at this juncture. For example, the audiologist might ask the patient with a low self-efficacy ranking, "Why do you give yourself a 4 on this scale rather than a 0?"

Reflection on the personal strengths that one might bring to a perceived difficult task can begin to bolster confidence. It is equally important to uncover the factors that might be undermining the patient's confidence by asking directly, "Why do you feel your ability to be successful in improving the communication between you and your daughter is so low?"

The concerns expressed may stem from a previous failure with hearing aids, anxiety over an ability to maneuver complicated technology, or fear of making one's hearing loss "public." Regardless of our view of

the concern, we cannot disregard the fact that, for the patient, these may be substantial stumbling blocks. It is vital that we acknowledge the patient's hesitations as legitimate apprehensions, regardless of our personal assessment of their validity. Our appreciation of patients' efforts to express their concerns does not automatically equate to validation of those worries. In being direct with our counseling, our question allows worries to be brought directly into the open for discussion and inspection, thereby increasing the likelihood of success. A cognitive-behavioral counseling approach can be used to strong advantage in addressing what may sometimes be poorly founded concerns (Clark & English, 2019).

Stuck? Think Inside the Box

When a patient has not yet reached a readiness stage requisite for action, a hearing consultation can quickly come to a stall. The decisional box (Figure 12-3) is held in reserve for those patients who give a low ranking to the importance of change. Many patients who benefit from a guided cost-benefit analysis ultimately come from Goldstein and Stevens Readiness Levels 3 or 4 or the contemplative and preparation stages of the Prochaska and DiClemente change circle. This 2×2 matrix provides an organizational framework

Benefits of Status Quo	Costs of Status Quo
Potential Costs of Change	Potential Benefits of Change

Figure 12-3. The cost-benefit analysis box used to help patients explore and consider their self-reflected advantages and disadvantages of inaction versus action. *Source:* Modeled after Janis and Mann, 1977.

for the perceived benefits and drawbacks of following the audiologist's recommendations. In most cases, following completion of the decisional box, the pros of action will outweigh the cons of inaction, including those that may be more difficult to eliminate.

A common mistake when using the decisional box occurs when we follow our natural inclination to offer examples of various pros and cons to initiate the discussion. This follows health care professionals' familiarity with information delivery but decreases the value of this motivational tool. Pausing to allow patients a moment to process, reflect, and offer their own personal thoughts can be a rewarding strategy for both parties. By allowing the individual to populate personal concerns and thoughts for the matrix, we tap into internal motivators, the power of which cannot be matched through external sources. We can offer nothing that can compel patients toward action like the strength of their own words that support a rationale for change.

When the patient's importance ranking is low, introduction of the box may look something like this:

"So, based on your ranking, you don't seem to feel it's important to improve the communication between you and your daughter. That could be the case; but, from our discussion about the frustrations you're feeling, it seems like something needs to change. Let's briefly explore this box which helps us compare the pros and cons of change. Using this first box (top left quadrant), what advantages do you see for your life if you don't do anything to help your hearing and communication issues?"

This exercise ultimately promotes decision making by revealing the cost-benefit trade-off of action versus inaction (Janis & Mann, 1977). The audiologist guides the patient through the matrix from left to right, then top to bottom. The top left quadrant likely will contain patient-reported pros of inaction that reflect actual concerns such as financial strain, social stigma, or perceived ability to use technology. By acknowledging these factors as valid concerns instead of countering their existence, we facilitate the patient's decision-making process and maintain open dialogue. The top right quadrant details the cost of inaction and highlights the many frustrations that brought the patient into the clinic in the first place. Discussion of answers on previous self-assessment measures or case history forms can offer additional patient-reported drawbacks, such as family discord, decreased social

involvement, and reduced enjoyment or quality of life. Already, the patient may start to be aware of the influential impact of untreated hearing loss on communication success. The value of having a supportive communication partner present during the consultation is clearly apparent.

The third and fourth (lower) quadrants will generally mirror responses from the first two boxes, being that they are the possible benefits and disadvantages of action, respectively. Any treatment recommendations made should target the cons of inaction while drawing attention to the potential benefits to be gained through action. Most often, the benefits of action carry more weight than the benefits of inaction just as the costs of inaction dominates the benefits of the status quo. Taken together as a visual representation, the total box helps lead patients to realize the necessity to address their hearing and communication needs. A subsequent reassessment of the importance-ranking line most often reflects both a new sense of dedication to making necessary life changes and an accompanying forward movement within the circle.

Time Well Spent

When contemplating a redirection of clinical approach to incorporate a style of motivational engagement, audiologists frequently wonder if they have enough time to do justice to this approach. The greater question may be, if we fail to move a patient forward on his or her rehabilitative journey, in what ways have we failed ourselves, our individual practices, the patient, and the patient's family? In reality, the time invested to build needed motivation when it is lacking is not only time well spent, but rarely is it a significantly greater amount of time than when we do not address motivation.

When greater patient motivation is needed and we fail to address this need directly with our patients, a patient may decline our recommendations (a failure for the patient and the audiology practice and a significant loss of invested time). Or, a patient may purchase hearing aids but return them within the allotted adjustment period (again, a failure for the patient and the audiology practice and a loss of invested time). Or, a patient may purchase hearing aids but later discontinue their use (a failure for the patient).

Starting with a simple inquiry of one's perceived importance to make a change toward success takes no time and if high the audiologist can quickly attain

a ranking of self-efficacy. If self-efficacy is high, no clinical time has been expended. If self-efficacy is low, the ensuing dialogue will greatly bolster the patient's success and decrease the chances of either hearing aids that are returned for credit, or purchased hearing aids that are later "worn in the dresser drawer." There is no need for use of the decisional box for this patient. Very little time was invested.

If the importance for change is low, bring in the decisional box. This exercise takes a little longer (5 minutes or so), but the payoff is well worth the added time. After looking at the pros and cons displayed in the decisional box, reassess the importance ranking. If this is still low, set a future appointment to see how things have evolved (along with instruction on things to look for in their communication exchanges, such as, "Watch to see how many times in a day you failed to hear something correctly; how many times in a day others seemed to be frustrated because of your misunderstanding; how many times you declined to participate in an activity because you were afraid you may not hear sufficiently," etc.). However, following use of the decisional box, most patients will find their internal motivation has increased sufficiently to take the forward-moving action that is required to improve communication success.

Use With Tweens and Teens

The use of motivational engagement in clinical practice goes beyond helping adult patients to find the requisite internal motivation to use recommended amplification. Preteen and teenaged patients who reject once successfully used amplification can be engaged in dialogue through importance rankings, self-efficacy rankings, and decisional boxes just as our adult patients can. Our first step is to ensure that we are not aligned within the enemy camp (Clark, 2013).

These young patients have often placed parents and educators in the enemy camp as those who just do not understand them. The need to fit in with one's peers is powerful. External motivators such as reminders that not wearing hearing aids will adversely affect academic performance may have little impact given a young person's desire for peer acceptance. Similarly, rewards for consistent use of hearing aids will often fail to move one toward better decisions. A more powerful approach may be to begin with an acknowledgment of our young patients' decisions with a display of understanding: "If I were 12 years old and

in middle school, I wouldn't want to wear hearing aids either."

Such a statement of understanding and acceptance of position opens a window for discussion. Just like for adults, use of a self-assessment in teens (e.g., Self-Assessment of Communication—Adolescents, Elkayam & English, 2003) can provide a springboard for importance rankings and can help one to consider the pros and cons underlying decisions. This approach will take young patients through an introspective exercise often unfamiliar at their age and help to uncover personal motivators to consider when thinking of abandoning hearing aid use.

Beyond the Technological Fix

While the use of hearing aids can reduce the family discord, anger, frustration, social isolation, and other negative sequelae associated with hearing loss, these negatives often continue to some degree even with amplification (Kochkin & Rogin, 2000). It is known that significant residual hearing deficits can remain with hearing aid use (Dillon, 2012) and that communication enhancement training can improve communication success with hearing loss (see Chapters 16 and 18).

While not all patients may need or significantly benefit from aural rehabilitation or communication training, Taylor (2012) suggests that those with poor performance on speech-in-noise testing (e.g., a signal-to-noise ratio loss greater than 10 dB on the QuickSIN) and who also have lower tolerance for background

noise (e.g., an Acceptable Noise Level score greater than 10 dB) are less likely to be successful with amplification alone.

Training in improved communication strategies for patients and life partners, recognition of barriers to successful communication, and improved management of the communication settings, all can heighten communication success. Yet, to be successful, these most often require a disclosure of one's hearing deficit—an action that is not easy for many patients to take.

Adults with hearing loss continue to be influenced by the stigma associated with hearing aid use (David, Zoiner, & Werner, 2018; Chapter 8). As one mechanism for coping with disability is to avoid drawing attention to one's difference (Hetu, 1996), we must question how successful the provision of communication strategies and other means to enhance communication may be when these are provided without an exploration of self-efficacy and comfort of use. The motivational engagement practices of importance rankings, self-efficacy and comfort rankings, and decisional boxes should be used routinely when discussing communication management designed to enhance hearing aid benefit.

It is unfortunate that most audiologists do not provide or recommend communication training, clear speech instruction, computer-based AR training, or even communication management handouts (Clark, Huff, & Earl, 2017). While group audiologic rehabilitation services (see Chapter 18) provide valuable benefits for many with hearing loss, even one-on-one discussions of useful strategies can be beneficial if provided in a meaningful fashion.

Case Study 12–3

Last week, Dr. Robbins fit Mr. Billings with his first set of hearing aids. Now that he has been successfully using his new hearing aids for a week and appears to be adjusting well, Dr. Robbins broaches a communication strategy that she believes will be helpful to Mr. Billings.

Dr. Robbins: Mr. Billings, all of the testing we did the other week is in agreement with your original complaint of significant difficulty hearing in background noise. How are the hearing aids working for you for that?

Mr. Billings: Well, like I said, I'm doing a whole lot better with my hearing aids than without, but noise can still be a problem. Better, I'd say. But I go out for lunch once a month with a group of five other guys I've known for years. We went this Saturday and I had a lot of trouble... some let their voices trail off, some get to talking too fast. You know how it is.

Dr. Robbins: Yes, I know. Let me give you a suggestion that I know has helped for others. Next time, what do you think if you started by reminding the group that you have a hearing loss and saying, “I really want to hear as much as I can, but I know it can be tiresome if I interrupt the flow of the conversation all the time. So if you see me go like this (pumps hand with palm up) that’s me asking you to speak up a bit, and if I do this (pumps hand with palm down) that means slow down some.

(Mr. Billings nods in agreement while Dr. Robbins is speaking. Dr. Robbins ends by saying, “Do you think that would help?”)

Mr. Billings: I don’t know. I suppose so.

Dr. Robbins: Well, let me ask you, on a scale from 0 to 10 like we’ve done before, how comfortable would you be doing that.

Mr. Billings, with a chuckle: That ranking thing again, huh? I don’t know. Honestly, probably a 3.

►Without a quick ranking, Dr. Robbins would never know how her suggestion sat with her patient. The low ranking provided a window to explore how Mr. Billings felt his group would react to such an out of the ordinary request. It is often our unfounded fears that hold us back and an exploration of these through a cognitive counseling approach can be quite beneficial (Clark & English, 2019).

Summary

The practice of motivational engagement can help patients find the requisite internal motivation for greater adherence with professional recommendations. Combined with self-assessment measures, effective and sensitive timing, an interactive clinician-patient dynamic, and an integration of a primary communication partner into the process, motivational engagement requires few materials and little time commitment while positively engaging persons with low motivation to embrace hearing management solutions. While useful when employed in the hearing aid consultation process, this same approach is beneficial when introducing communication strategies or encouraging continued use of amplification.

Without question, change is difficult, even for audiologists who are accustomed to a comfortable pattern of connecting with patients. Audiologists must be willing to consider change in their own clinical approaches in the same way they expect patients to make necessary changes in attitude and behavior to follow through with hearing care recommendations. To be the most effective in providing successful treatment for those we help, we must guide and coach our patients to explore the impact of hearing loss in their

own lives, helping them recognize for themselves that the benefits of change are valuable and indeed worth more than the potentially uncomfortable cost of that same change. Providing effective support for patients with low motivation can be a challenge, yet one that is greatly facilitated by understanding creative resources.

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13

THE ROLE OF SELF-EFFICACY IN THE AUDIOLOGIC REHABILITATION PROCESS

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Disclaimer: The contents of this manuscript do not represent the views of the Department of Veterans Affairs or the United States Government.

Introduction

Modern-day audiology rehabilitation (AR), as highlighted in this textbook, emphasizes a patient-centered care approach to minimize any activity limitations and participation restrictions arising from hearing impairment. In the patient-centered care approach to AR, the clinician collaborates with the patient with hearing loss to better understand their unique needs, goals, and preferences (Grenness et al., 2014a, 2014b; Institute of Medicine, 2001). These needs and goals often include those related to speech understanding in specific situations, but may also include those related to psychosocial adjustment, vocation, or other unique needs associated with hearing loss. Because the needs and goals of the patient can vary widely, there may be several appropriate AR interventions including but not limited to amplification or hearing-assistive technology, auditory training, communication strategies training, assertiveness training, psychosocial adjustment counseling, or a combination of multiple

interventions that could be incorporated in a tailored AR approach. Regardless of which AR intervention(s) the patient chooses, the patient likely needs to acquire new knowledge and requisite skills, and implement behavioral changes associated with managing their hearing loss in their daily life. Self-efficacy plays a key role in implementing and maintaining health behaviors (Schwarzer & Fuchs, 1996), including those associated with managing hearing loss or tinnitus (Smith & West, 2006a; Meyer et al., 2014). This chapter will briefly review the concept of self-efficacy, the role that self-efficacy plays in the AR process, how to measure self-efficacy, and how clinicians can help to increase self-efficacy in their patients with hearing loss.

Self-Efficacy Theory

Overview

Self-efficacy refers to the beliefs (or confidence) individuals have in their abilities to execute courses of actions required to produce a specific task or behavior within a given context (Bandura, 1977, 1986, 1997; Stajkovic & Luthans, 2002). The concept of self-efficacy plays a central role in a larger theory known

as *social cognitive theory* that explains human functioning (Bandura, 1977, 1986). Social cognitive theory describes human functioning as a triadic interaction among personal factors (cognition, affect, and biological events), environmental factors, and behavior (Bandura, 1986). Each component of the triad influences the other components, and a person can exert control over each component to achieve a desired outcome. For example, individuals are not solely products of their environment, but have the capability to exert control over their environment through their personal factors (e.g., self-regulation, forethought, etc.) and behavior. Individuals' beliefs about themselves influence their environment and are critical to how they behave. Self-efficacy beliefs are core to social cognitive theory and thus human functioning. Individuals need to believe that they can produce a desired outcome or accomplishment; otherwise, there is little motivation to do so.

It cannot be overemphasized that self-efficacy beliefs are domain specific and pertinent to a particular behavior or task. With each task or behavior, certain knowledge and requisite skills must be organized and executed to achieve the desired outcome successfully. Consider the knowledge and skills associated with successful computer use versus public speaking. They are quite different. Therefore, an individual may be highly confident in their abilities to use a computer (i.e., computer self-efficacy) while at the same time completely lack confidence in their public speaking abilities (i.e., public speaking self-efficacy). In the context of AR behaviors, for example, a patient may be highly confident in their hearing aid use skills (i.e., hearing aid self-efficacy) but at the same time have low confidence in their ability to execute assertiveness in demanding communication situations (i.e., assertiveness self-efficacy) with unfamiliar communication partners.

Self-efficacy beliefs are thought to influence individual functioning in several ways including but not limited to the choices people make about the tasks or behaviors they pursue; the amount of effort they will put forth to succeed; how long they will persevere when tasks or behaviors become difficult or failure is experienced; and their ability to mobilize motivation to set goals and commit to them (Bandura, 1992, 1997, 1998; Pajares, 2002; Stajkovic & Luthans, 2002). In the context of AR, these influences should be carefully considered, as patients have to make decisions regarding which AR intervention option(s) they pursue and the effort involved with each option. Individuals tend to choose to engage in tasks that they believe they can confidently execute to achieve the

desired outcomes. Patients may lack the knowledge about the requisite skills and behaviors involved in a particular AR intervention, which highlights the importance of carefully reviewing the details associated with a desirable outcome and the support the clinician can provide as the patient implements new skills and behaviors. The good news is that self-efficacy is a key and modifiable mediator involved with behavioral change, including those associated with adopting and maintaining health behaviors (Bandura, 1997, 1998; Schwarzer & Fuchs, 1996; Shortridge-Baggett, 2001). Data from the larger health literature clearly show that patients who have high self-efficacy for managing a chronic condition have higher treatment outcomes, better adherence to treatment plans, better resilience when adversity is encountered, and a better outlook on their health in general (Bandura, 1992, 1997, 1998, 2004; Holden, 1991; Marks et al., 2005; Schwarzer & Fuchs, 1996). Later in this chapter, techniques clinicians can use to increase self-efficacy in their patients will be reviewed.

Role of Self-Efficacy in Audiologic Rehabilitation Domains

Help-Seeking

Self-efficacy beliefs influence the choices people make regarding whether or not to adopt a new behavior. In the context of AR, a primary treatment option is the adoption of amplification options such as hearing aids, cochlear implants, or other assistive technologies. Previous research has posited that individuals with hearing loss may not seek amplification options if they have low self-efficacy regarding their abilities to manage a hearing aid (Kricos, 2000; Smith & West, 2006a; Weinstein, 2000). Meyer and colleagues (2014) conducted a study and confirmed that individuals who had high self-efficacy for handling a hearing aid were more likely to seek professional help and adopt hearing aids. Laplante-Lévesque et al. (2011) examined the factors that predicted choice for an AR intervention (hearing aids, individual or group communication programs, or no intervention) adopted by a group of 139 adult participants. Communication self-efficacy, or how confidently an individual can plan and execute actions to successfully manage a communication situation, was one predictor variable used in the study. The study showed that participants who had higher levels of communication self-efficacy were less likely to choose hearing aid adoption, sug-

gesting these patients with hearing loss may rely on other resources for communicating. Ferguson et al. (2016a) showed that a group of first-time hearing aid users who participated in motivational engagement during a prefitting assessment had higher self-efficacy for following health care recommendations than those participants who engaged in a standard of care approach without motivational engagement. Finally, the role of self-efficacy has been examined in the context of multifaceted health behavior change models for which self-efficacy is a factor. Saunders and colleagues (2013, 2016) showed that self-efficacy plays a small role in where the patient is in their hearing health care help-seeking journey, whereas Pronk and colleagues failed to show that self-efficacy was a significant predictor in help-seeking behaviors of older patients with hearing loss. Overall, however, the results of these studies demonstrate that self-efficacy plays a role in help-seeking behaviors in patients with hearing loss. Therefore, counseling that incorporates self-efficacy (among other factors) might help patients promote help-seeking behaviors or take action regarding the treatment of their hearing loss (Saunders et al., 2016).

Associations Among Self-Efficacy and Other Measures Related to AR

The relations between self-efficacy and other measures associated with hearing have been evaluated in an effort to understand better its role in the context of AR domains, including audiometric data and self-reported measures. For example, Smith and West (2006b) evaluated hearing aid self-efficacy as a function of degree of hearing loss. They found that hearing aid users with mild hearing loss had higher aided listening self-efficacy than those with greater degrees of hearing loss. Kelly-Campbell and McMillan (2015) made a similar finding. Confidence levels in following conversations with a variety of auditory communication skills were found to be significantly correlated with degree of unaided hearing loss, word recognition abilities in quiet and in noise, and self-perceived handicap (Sweetow & Sabes, 2010). Consistent with other research, Sweetow and Sabes showed that poorer performance on behavioral and self-reported measures is associated with lower levels of confidence in following conversations in various listening conditions. Jennings et al. (2013) found that although communication self-efficacy was not associated with degree of hearing loss or age, it was associated with hearing attitudes and handicap. Those who reported

that they were more affected by their hearing loss and who had more negative attitudes towards hearing loss also reported less confidence in managing various listening situations. Taken together, these results suggest that listeners with more significant degrees of hearing loss, greater speech perception difficulties, or more negative perceptions regarding their hearing need additional counseling to improve their abilities to confidently engage in challenging listening conditions whether they wear hearing aids or not.

Kelly-Campbell and McMillan (2015) explored the relation between hearing aid self-efficacy and hearing aid satisfaction in a group of 47 new and experienced users who completed the outcome measures 12 weeks following their hearing aid fitting. Their data showed that, in general, users who exhibited high hearing aid self-efficacy were more satisfied with their hearing aids. Ferguson et al. (2016b) showed that hearing aid self-efficacy, when measured prior to the hearing aid fitting, is predictive of hearing aid satisfaction, but not other domains of hearing aid outcomes used in their study (e.g., data logging, Glasgow Hearing Aid Benefit Profile [Gatehouse, 1999]), whereas Dullard and Cienkowski (2014) showed that hearing aid self-efficacy was correlated with hours of use.

Johnson and colleagues (2018) examined outcomes in domains related to benefit, satisfaction, and hearing aid self-efficacy for advanced digital hearing aids in listeners with mild hearing ($n = 153$) who underwent best practice hearing aid fitting procedures. They found their sample had positive outcomes in all domains in general and was equivalent to published data. They did note, however, that some users lacked confidence in their advanced hearing aid skills (e.g., troubleshooting) and in using their hearing aids in difficult listening situations. These were the same users who had moderate levels of satisfaction for some items related to negative features of hearing aids (e.g., picked up sounds that prevented them from hearing the target). This finding, as the authors indicated, suggests that close follow-up and counseling are crucial for patients to optimize hearing aid use in challenging situations (i.e., noise) confidently.

Smith and Fagelson (2011) examined associations between self-efficacy for managing chronic tinnitus and self-reported tinnitus ratings. They found that patients with tinnitus who had high confidence in their abilities to manage their tinnitus tended to report being significantly less aware of their tinnitus while awake and less distressed and handicapped by their tinnitus compared to those who had lower confidence in their abilities to manage their tinnitus.

In addition, tinnitus self-efficacy was found to be significantly lower in patients who had both tinnitus and post-traumatic stress disorder (PTSD) than those patients with tinnitus who did not have PTSD (Fagelson & Smith, 2016). These data suggest that self-efficacy plays a role in tinnitus management and even more so in patients who exhibit additional comorbidities. Although AR interventions tend to place emphasis on communication, patients may also present with goals associated with reducing the effects of tinnitus and thus it is not uncommon that tinnitus is part of an AR approach in patients with hearing loss and is the focus of Chapter 22.

Changes in Self-Efficacy Through AR Intervention

Consistent with the self-efficacy concept, individuals' confidence in their abilities should improve with education to increase knowledge, practice, or training to teach new skills, and experience with using those skills in a variety of situations. Studies that have evaluated changes in self-efficacy levels through an AR intervention will be illustrated in this section.

West and Smith (2007) developed a hearing aid self-efficacy measure called the Measure of Audiologic Rehabilitation Self-Efficacy for Hearing Aids (MARS-HA). To demonstrate criterion validity of the measure, or to validate that experience with hearing aids leads to greater confidence in hearing aid skills, they reported on data examining hearing aid self-efficacy levels in 29 first-time hearing aid users preintervention and 1 month postintervention. Each participant received a standard hearing aid orientation that also highlighted the skills assessed on the measure (basic and advanced handling, adjustment, and aided listening). A subsample of these participants received enhanced counseling that incorporated self-efficacy enhancing techniques during the orientation (this will be described later in this chapter, but a detailed description can be found in Smith & West, 2006a, and Smith, 2014). The data showed a 30% increase in hearing aid self-efficacy levels when comparing prefitting self-efficacy levels to 1 month postfitting self-efficacy levels. These data support that training and experience lead to increased self-efficacy.

Sweetow and Sabes (2010) carried out a similar study in 51 patients receiving hearing aids for the first time. They showed that, on their measure of self-efficacy, the Communication Confidence Profile (CCP), nearly all of the patients reported improvements in self-efficacy following 2 to 4 weeks of hear-

ing aid use. Their measure focused on self-efficacy for a variety of auditory communication skills such as understanding conversation when talking with one to two individuals while at home or understanding an unfamiliar talker in a noisy environment.

In a somewhat different approach, McMullan and colleagues (2018) evaluated whether or not enhanced education materials would affect self-efficacy levels. In their study, they developed an enhanced hearing aid user guide (HAUG) by improving its readability and suitability. In addition, they made a video to accompany the revised HAUG. The use of education materials in written and video form is a technique suggested to increase hearing aid self-efficacy (Smith & West 2006a, Smith, 2014). They randomized 16 adults to a group that used the original HAUG (OG group) and 15 adults to the experimental group, which used the revised HAUG and video (RG group). Minimal counseling was provided by the examiner and both groups were allowed to review their assigned materials and practice their skills with a hearing aid without a time limit. Both groups completed a practical skills test along with the MARS-HA. The results revealed that the RG group had significantly higher self-efficacy for hearing aid skills compared to the control group (OG). These findings support that the use of enhanced written materials and videos results in higher hearing aid confidence in adults with no prior hearing aid experience.

In summary, self-efficacy plays a role in all aspects of human behavior, including those behaviors in the context of audiologic rehabilitation. Emerging data in our field show that self-efficacy is an important factor in hearing health care help-seeking behaviors. An individual's self-efficacy level for hearing aids, if low, may prevent them from seeking help regarding hearing aid adoption or from advancing through the steps in the patient journey. On the other hand, if communication self-efficacy is high, individuals may not pursue amplification because they are confident they can manage in their current situation. Taken together, these findings highlight the importance of examining self-efficacy in the target domain or context of interest. In addition, self-efficacy levels have been shown to be associated with attitudes and perceptions about hearing loss and also with hearing aid outcomes. These associations are good in the sense that they are modifiable through additional intervention or counseling. In addition, there are strategies that will be reviewed later to increase self-efficacy levels of patients in the context of AR. Finally, through AR intervention and experience, AR self-efficacy levels have been shown to increase. In the next sections, how an individual

formulates their self-efficacy beliefs will be discussed, followed by a review on how clinicians can use targeted techniques to help patients increase their self-efficacy levels.

Making Self-Efficacy Judgments

Individuals make judgments or appraisals about their self-efficacy levels by interpreting information from four sources. These sources of information include the following: (a) mastery experience, (b) vicarious experience, (c) verbal persuasion, and (d) physiologic and affective states (Bandura, 1986, 1977). For the purposes of illustrating these sources of information in action, we will consider some skills and behaviors associated with successful hearing aid use.

Mastery Experience

The strongest source of information from which individuals make self-efficacy judgments is through mastery experience or enactive attainment. This refers to the judgments made by acting out the skill or skill set associated with achieving the behavior of interest or judgments based on past experiences. Simply stated, if the skill was performed successfully, then self-efficacy often is judged as being high for that skill, whereas failures to perform the skill successfully often result in low self-efficacy judgments (Bandura, 1997).

The ability to adjust manual programs on a hearing aid, for example, often is needed to optimize hearing aid use in different listening situations. If a patient successfully identifies when to use a given program and navigates to the desired program correctly for a given listening situation, then they will likely judge their self-efficacy via mastery experience as high for this skill. In the future, they likely will attempt to use a manual program when the need arises and persevere with its use if difficulties are encountered. In contrast, an individual who cannot determine or navigate to the proper manual program for a given listening situation will likely judge their ability as being low, thus forming a low self-efficacy belief via lack of mastery experience. In this latter case of low self-efficacy, the patient is at risk for avoiding future attempts of using manual programs with their hearing aid or may simply give up after a feeble attempt.

Vicarious Experience

The old adage, “if they can do it, then I can do it,” represents judgments based on vicarious experience.

By observing others, or models, perform tasks or behaviors, individuals form beliefs about their own abilities. If the model successfully performs the skills needed to achieve a behavior, then a person will likely judge their own self-efficacy beliefs about their abilities as high. On the other hand, if a person observes a model struggling with achieving a behavior, then they will likely have low self-efficacy beliefs in their own abilities for accomplishing that behavior (Bandura, 1997).

It is not uncommon for older patients, particularly first-time hearing aid users, to struggle with inserting their batteries into their hearing aids. If the model successfully identifies and opens the battery door, removes the tab on the battery, orients the battery correctly, and inserts the battery into the door without dropping it, then the patient may judge their ability for battery insertion as high via vicarious experience. They may think “if they can do it, then so can I” and then be motivated to attempt battery insertion on their own and more eager to try again if failures occur along the way. If the model’s ability for their battery insertion skills is less than ideal, then the patient may judge their own abilities for the same skills as low. When the time comes for the patient to try out their skills for battery insertion, they may find that they face uncertainty in their ability.

The relationship between the patient and the model also is an important factor. Judgments based on vicarious experience are more effective when the model is viewed as a peer compared to when the model is viewed as a significant upward or downward social comparison. For example, new golfers are more likely to judge their self-efficacy levels more realistically when observing fellow new golfers who are similar in social comparison rather than observing a golf professional.

Verbal Persuasion

Individuals make judgments about their self-efficacy levels based on the feedback of others. Most commonly, verbal persuasion comes in the form of feedback one expresses about the abilities of the person, which can be very influential, particularly if the source is considered credible and when the feedback immediately follows the behavior. Positive verbal persuasion tends to boost the self-efficacy beliefs of the person, whereas negative verbal persuasion can undermine self-efficacy judgments (Bandura, 1997). Furthermore, feedback is more effective if it is specifically related to the ability or effort put into the skill rather than general in nature. For example, telling a

new hearing aid patient “You can do it” may not be as effective as “You were able to change the programs in your hearing aid correctly. You will be able to change the volume too.” The latter feedback is more specific and more likely to be motivating compared to the former statement that is more general in nature. The patient can provide verbal persuasion via self-talk, or say to himself that he believes he can do the task. Other sources such as viewing videos or reading materials about the subject matter may also serve as verbal persuasion (Smith & West, 2006a), especially if they are provided in an enhanced format (McMullan et al., 2018).

Physiologic and Affective States

The final source of information regarding self-efficacy judgments is how individuals feel emotionally and physically when performing behaviors. For example, if individuals are stressed or frustrated, or have sweaty palms or increased heart rate when performing skills, then they likely will interpret those states and judge their self-efficacy as low. Individuals with high self-efficacy for their abilities will likely have positive physiologic and affective states when performing behaviors.

It is not uncommon for first-time hearing aid users to have uncertainty about inserting hearing aids into their ears correctly. They may drop the hearing aid or have repeated unsuccessful insertion attempts. For these patients, they may express feelings of frustration or develop anxiety during these unsuccessful attempts. In this instance, their self-efficacy beliefs related to hearing aid insertion are likely to be judged as low. On the other hand, a patient who is able to insert the hearing aid into their ear with ease will have neutral or positive physiologic and affective states and judge their ability for this skill as high.

Techniques to Enhance Self-Efficacy

There are several techniques that clinicians can use to help their patients increase self-efficacy. These techniques can specifically target self-efficacy for a given source of information that was described previously. In other words, there are different techniques available for a clinician to use with the patient depending on whether or not the clinician wants to increase self-efficacy through sources of information related to mastery experience, vicarious experience, verbal persuasion, or physiologic and affective states.

Regardless of the AR intervention being targeted, the clinician can incorporate self-efficacy-enhancing techniques.

Recall that mastery experience is the strongest source of information by which individuals make self-efficacy judgments. Clinicians thus may focus on techniques to increase self-efficacy for mastery experience, although tapping into all four sources of information is most effective (Bandura, 1997; Maddux & Lewis, 1995). Extensive practice is a primary way to increase self-efficacy with mastery experience. The clinician can offer the patient several opportunities to practice the skills they are learning during the session and encourage them to practice at home to reinforce what has been learned. To ensure the task is successful while the patient is in the clinic, simpler skills can be introduced before moving to more advanced skills. Role play can be used to help the patient learn to execute the skill or behavior. Goals can be set for at-home practice and future follow up for skills or behaviors that posed some challenge or uncertainty.

Clinicians can use modeling as a means to increase self-efficacy through vicarious experience; however, they may not be viewed as a peer. The patient’s communication partner may serve as a better model and can learn and practice the skills along with the patient. The use of a family-centered AR approach can be helpful in this area. A group setting for the AR intervention may be a venue to elicit vicarious experiences. In addition, videoing can be used so that the patient can serve as their own model (i.e., self-modeling). By watching oneself perform the skills, the patient can identify what is being done correctly or incorrectly and use that information to reinforce or alter performance for that skill or behavior. If video-taping is unreasonable for the clinical setting, the patient can watch an instructional film of the skill or behavior in action. Ideally that instructional video would include star performers who would be considered peers by the patient. Cognitive rehearsal, which is a mental visualization of oneself executing the skills or behavior, can be used to increase self-efficacy for vicarious experience.

Clinicians can increase self-efficacy through verbal persuasion. One way to do this is to provide realistic feedback immediately following the patient performing the skill. This feedback should focus on the patient’s capabilities or effort and be specific in nature. If the communication partner is involved in the AR, then the clinician can encourage the partner to provide positive feedback about the patient’s capabilities and efforts. Verbal persuasion also can

come in the form of educational materials. The more knowledge a patient has about the AR skill or behavior of interest, the higher their self-efficacy may be for that skill.

Increasing self-efficacy for the source of information related to physiologic and affective states is accomplished primarily by reducing any negative emotional and/or physical stressors. For example, the clinician should provide a calm environment with ample time to practice skills so as to not rush the patient. If time is limited, then the skills associated with the AR behavior can be broken down into subsets and taught over multiple sessions. Another technique is to take a break from the skill that may be causing heightened negative states and work on a different, possibly easier, skill. The clinician can inquire about why the patient is having these negative states when performing the skill and counsel them accordingly.

The above are examples of techniques that can be used to increase self-efficacy for a given source of information. There can be some overlap in the techniques across these areas. For example, the communication partner can be included in the AR and used as

a source of vicarious experience and also encouraged to provide supportive feedback to increase self-efficacy through verbal persuasion. In short, these self-efficacy enhancing techniques add to the armamentarium available to clinicians engaged in AR services and are summarized in Table 13–1 (also see Smith, 2014). Smith and West (2006a) provide a comprehensive tutorial on self-efficacy and a detailed example of an interaction between an audiologist and patient during a hearing aid orientation session in which the audiologist is utilizing these self-efficacy-enhancing techniques.

Measuring Self-Efficacy for AR Behaviors

Given that self-efficacy beliefs are one's perceptions, measuring one's self-efficacy levels is done formally through the use of questionnaires. Consistent with the domain-specific nature of self-efficacy, an important characteristic of these measures is that the questionnaire targets the context of interest. Therefore, an all-purpose self-efficacy measure is unsuitable. Rather,

Table 13–1. Summary of Techniques to Increase Self-Efficacy

Mastery Experience	Vicarious Experience	Verbal Persuasion	Physiologic and Affective States
Practice	Clinician model	Specific, positive feedback	Calm environment, free from distractions
Repetition	Peer model	Immediate feedback	Extra time to practice skills
Role-play	Communication partner model	Self-talk	Multiple appointments
Build on skills based on difficulty level	Cognitive rehearsal	Encouragement from communication partner	Schedule breaks
Set realistic goals	Instructional videos with peer models	Patient education materials	Reassuring feedback
Give “homework” exercises to practice skills in real-world environment	Self-modeling via video recording		Counsel accordingly to reduce anxiety/stress

Note. This table summarizes the techniques that clinicians can use to increase self-efficacy based on the four sources of information.

Source: Adapted from Smith and West (2006a).

self-efficacy questionnaires in AR need to focus on the skills and behaviors that the clinician is addressing during the particular AR intervention. Furthermore, the items should target skills over which the patient can exhibit some control that will foster behavior change. Bandura (2006), who pioneered the self-efficacy theory, also recommends that self-efficacy questionnaires have certain characteristics so self-efficacy is being measured accurately. First, the items should be worded to inquire about current abilities (e.g., “I can . . .”) rather than intentions (e.g., “I will be able to . . .”). Second, each item should inquire about one skill or task at a time as individuals may have different levels of self-efficacy for different skills or tasks. Third, items should have gradations of challenge so that the measure contains skills that vary degree of difficulty. Fourth, the response scale should be positive and unidirectional (i.e., no negative values); it is best if there is a 0 to 100 response scale in 10-unit intervals (i.e., 0, 10, 20, . . . 100). Figure 13–1 illustrates an example of a response scale on self-efficacy measures. As seen in Figure 13–1, 0% represents complete uncertainty in the respondent’s ability to accomplish the task in question, 50% represents moderate certainty, and 100% represents complete certainty. Finally, Bandura recommends that self-efficacy measures contain practice items not related to the domain of interest so that the respondent can become familiar with the scale. For example, the following two practice items have been used in self-efficacy questionnaires developed within the authors’ laboratory:

- 1. I can lift a 10-pound object with ease.
- 2. I easily can tell the weight difference between a 19-pound object and a 20-pound object (West & Smith, 2007; Smith, Pichora-Fuller, Watts, & La More, 2011; Smith & Fagelson, 2011).

Self-efficacy questionnaires that could be used to assess various areas of AR behaviors are described

below. Five questionnaires focus on hearing-, communication-, or hearing aid-related self-efficacy and three focus on tinnitus self-efficacy.

Hearing/Communication/Hearing Aid Self-Efficacy Questionnaires

Communication Confidence Profile (CCP)

Sweetow and Sabes (2010) developed the CCP, which measures a patient’s confidence, in either unaided or aided conditions, on auditory communication skills. The CCP contains 12 items and uses a 5-point Likert scale where 5 represents *extremely confident* and 1 represents *not at all confident*. The CCP contains two subscales that inquire about a patient’s confidence in their (a) abilities to hear in easy and difficult listening situations, and (b) abilities to improve hearing skills through the use of a device or strategy. Responses are summed with higher scores reflecting higher confidence in auditory communication skills. Scores of >50 points indicate confidence, 40 to 50 points indicate the patient is cautiously certain, 30 to 39 points reflect tentativeness, and scores <29 indicate the patient is insecure.

Ida Institute’s Line Tool

As part of the motivational toolkit available from the Ida Institute (2009), there is a question on the line tool that focuses on self-efficacy. The question is “How much do you believe in your ability to use . . .” The clinician completes the end of the question with the AR intervention being targeted. The intervention could be device related (e.g., hearing aids, remote microphone) or not (e.g., communication strategies, assertiveness). The patient responds to this question using a 0 to10, unmarked visual analog scale.

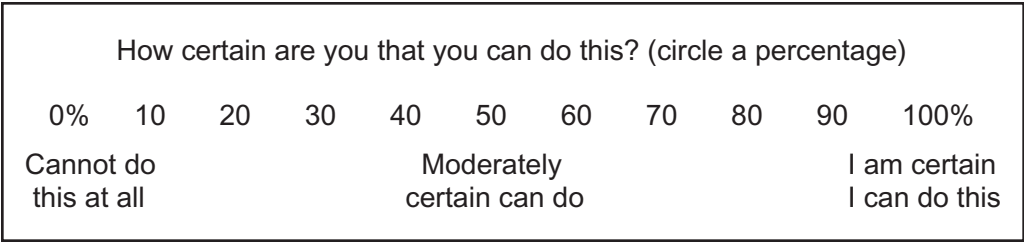


Figure 13–1. An example response scale on a self-efficacy questionnaire.

Listening Self-Efficacy Questionnaire (LSEQ)

The aim of the LSEQ is to quantify listening self-efficacy, which is defined as “the beliefs (i.e., confidence) listeners have in their capabilities to plan and execute the actions needed to understand speech in a variety of listening situations” (Smith et al., 2011, p. 418). The LSEQ is an 18-item measure, containing three subscales, that assesses self-efficacy for understanding speech in the following listening situations: (a) *dialogue in quiet*, which focuses on 1:1 conversations in quiet settings; (b) *directed listening*, or situations where the listener focuses on a single sound source (e.g., lecture, doctor in medical appointment, TV, etc.); and (c) *complex listening*, or challenging listening conditions such as in a noisy background or communicating with someone who is whispering. Self-efficacy levels for understanding speech in various listening situations are judged by the respondent using a 0% (*cannot do at all*) to 100% (*certain can do*) scale in 10-unit intervals. Mean listening self-efficacy scores are calculated for the overall LSEQ and for each subscale. Higher scores indicate higher self-efficacy levels. It is recommended that self-efficacy levels should be $\geq 80\%$.

Measure of Audiologic Rehabilitation Self-Efficacy for Hearing Aids (MARS-HA)

The MARS-HA was developed to quantify hearing aid self-efficacy, or “beliefs individuals have in their ability to perform the skills needed to be successful hearing aid users” (Smith & West, 2006b, p. 325; West & Smith, 2007). This 24-item questionnaire has four subscales. The *basic handling* subscale quantifies self-efficacy for basic handling skills related to insertion and removal of hearing aids and batteries, care and cleaning, use of controls, and the ability to tell the right hearing aid from the left hearing aid. The *advanced handling/knowledge* subscale assesses self-efficacy levels for knowledge about hearing aid components (e.g., microphone, battery door, etc.), battery size, as well as abilities related to feedback management and troubleshooting. The *adjustment* subscale focuses on the acclimatization abilities of the person related to sound quality, occlusion and own voice effects, and the physical fit of the hearing aids. Finally, the *aided listening* subscale assesses speech understanding abilities across a wide variety of listening situations. The respondents judge their self-efficacy levels for various hearing aid abilities using a 0% to 100% (*cannot do at all* to *certain can do*), 10-unit interval

response scale. Mean hearing aid self-efficacy scores are calculated for the overall MARS-HA and for each subscale. Higher scores indicate higher self-efficacy levels. It is recommended that self-efficacy levels should be $\geq 80\%$.

Self-Efficacy for Situational Communication Management Questionnaire (SESMQ)

Jennings et al. (2013) developed the SESMQ to assess the listener’s confidence for managing 20 different communication situations on a 0 (*not confident at all*) to 10 (*very confident*) scale. An interesting feature of this questionnaire is that the listener also rates each item on their hearing ability in that situation using a 0 (*not well at all*) to 10 (*very well*) scale. Therefore, a combined hearing score and a self-efficacy score is calculated by summing the ratings across the 20 items (possible range: 0 to 200). Higher values for the hearing and self-efficacy scores indicate better self-perceived hearing ability and greater self-efficacy for managing specific communication situations, respectively.

Tinnitus Self-Efficacy Questionnaires

Although interventions related to tinnitus management are not directly related to improving communication per se, they are often included as a part of an AR program (*Editor’s note*: For further information on tinnitus management, the reader is referred to Chapter 22 in this text). Three available questionnaires that measure tinnitus self-efficacy, or the beliefs individuals have in their abilities to manage their tinnitus (Smith & Fagelson, 2011), will be reviewed.

Self-Efficacy for Managing Reactions to Tinnitus (SMRT)

Henry and colleagues developed the SMRT measure to use in their series of clinical trials examining the skills taught in the Progressive Tinnitus Management (PTM) intervention program. The SMRT focuses on assessing self-efficacy levels for managing tinnitus reactions. A manuscript describing the psychometric properties of the measure has yet to be published, but preliminary psychometric data from a clinical trial are available (Henry et al., 2017). In its current form, the SMRT contains 17 items, the first six for which were adapted from a measure of managing chronic disease and 11 additional items were added that focused on

the skills targeted with the PTM. A 10-point Likert response scale is used where 1 represents *not at all confident* and 10 represents *totally confident*. Indexed total scores are calculated and range from 0 to 100 with higher scores indicating higher self-efficacy for managing reactions to tinnitus. This measure and the final scoring scheme may be modified after the psychometric properties are formally established.

Self-Efficacy for Tinnitus Management Questionnaire (SETMQ)

Smith and Fagelson (2011) developed the SETMQ to quantify tinnitus self-efficacy. The measure has 40 items across five subscales related to the following areas of tinnitus management: (a) routine management, (b) emotional response to tinnitus, (c) internal thoughts and interaction with others, (d) tinnitus concepts, and (e) use of assistive devices. Self-efficacy levels on this measure are judged using a 0% (*cannot do at all*) to 100% (*certain can do*), 10-unit interval, response scale and mean scores are calculated for each subscale and for the overall scale. Desirable levels of tinnitus self-efficacy on this measure are $\geq 70\%$.

Tinnitus Self-Efficacy Questionnaire (TSEQ)

The TSEQ is a seven-item measure that assesses self-efficacy in coping to areas affected by tinnitus sufferers and include relaxation, sleep, distraction, distress in quiet environments, distress in noisy places, depression, and concentration (Aug, Kavanagh, & Wilson, 1991). Each item is judged on a 0 (*can't do it*) to 100 (*extremely confident*), 10-unit interval scale. A nice feature of this measure is that for each item, the respondent rates several gradations of challenge indicating how confident he is in his abilities to do a certain task ranging from easier to harder. In the item regarding sleep, for example, the respondent will rate their confidence in their ability to fall asleep in 2 hours, 1 hour, 30 minutes, and 15 minutes. Clinicians can use this information to set goals and to focus skills in which the patient has low self-efficacy.

Using Self-Efficacy Measures

When selecting a self-efficacy questionnaire to administer, the clinician should consider the content of the measure to ensure it is appropriate for the target skill set and behavior of interest. Likewise, the clinician should take care when administering the question-

naire that the patient is indeed judging their current (i.e., right now) beliefs in their ability for accomplishing the task described in each item and not making appraisals regarding their intentions or future outcome expectancies about the task. To accomplish this, the clinician should encourage the patient to read the instructions carefully and allow the opportunity to ask questions. In addition, the practice items available on some questionnaires will give the clinician some assurances that the patient is appropriately judging their current self-efficacy beliefs with the given the response scale.

The self-efficacy questionnaires described above (see Table 13–2 for a summary of the questionnaires) can be administered to patients for several clinical purposes. Patient self-efficacy for given AR skills may facilitate the clinician in identifying areas of uncertainty for the patient so that a more tailored AR approach can be developed. The MARS-HA, for example, can be administered to new or experienced users (Smith & West, 2006b, West & Smith, 2007). If the self-efficacy level of a subscale or even an item is moderate or low, the clinician can work on skills needed to raise their self-efficacy to a high level (i.e., 80% or higher) (e.g., West & Smith, 2007). The MARS-HA, however, has been reported to be problematic for use in patients with no hearing aid experience as their knowledge of hearing aid use is limited (Ferguson et al., 2016b); however, West and Smith (2007) did not find this issue as the questionnaire instructions address how to respond to the questions if the patient has no prior hearing aid experience. In addition, self-efficacy questionnaires can be used to identify individuals with moderate to high levels of self-efficacy as they may be more likely to pursue certain AR interventions (Jennings, 2005; Meyer et al., 2014). Self-efficacy questionnaires also can be administered to monitor changes pre or post intervention (e.g., hearing aid orientation, counseling, etc.) or to monitor self-efficacy levels over time to ensure they are stable. In addition, these questionnaires can be administered to compare self-efficacy levels between groups of listeners or different types of AR interventions, especially those interventions that incorporate self-efficacy enhancing techniques (McMullan et al., 2018). The key point is to select the measure that focuses on the target skills and behaviors of interest and be administered in the appropriate point in time along the patient journey. Below are two hypothetical cases that represent how the MARS-HA and LSEQ can be used by the clinician to help steer the AR process.

Table 13-2. Summary of Available Self-Efficacy Questionnaires Related to Audiologic Rehabilitation

Questionnaire	Reference	Self-Efficacy Content Area	No. Items	No. Subscales	Subscales	Scoring Procedure
Communication Confidence Profile	Sweetow & Sabes, 2010	Communication	12	2	Hear under various conditions Improve hearing skills	Sum
Ida Institute Line Tool	Ida Institute	Customizable to specific intervention	1	N/A	N/A	0–10 Visual Analogue Scale
Listening Self-Efficacy Questionnaire (LSEQ)	Smith et al., 2011	Listening and understanding speech	18	3	1:1 listening in quiet Directed listening Complex listening	Average
Measure of Audiologic Rehabilitation Self-Efficacy for Hearing Aids (MARS-HA)	West & Smith, 2007	Hearing aids	24	4	Aided listening Basic handling Advanced handling Adjustment	Average
Self-Efficacy for Managing Reactions to Tinnitus (SMART)	Henry et al., 2016	Tinnitus	17	Undetermined	Undetermined	0–100
Self-Efficacy for Situational Communication Management Questionnaire (SESMQ)	Jennings, 2005	Communication management	20	1 ^a	Hearing score Self-efficacy score	Sum
Self-Efficacy for Tinnitus Management Questionnaire (SETMQ)	Smith & Fagelson, 2011	Tinnitus	40	5	Routine management Emotional response Internal thoughts and interaction with others Tinnitus concepts Devices	Average
Tinnitus Self-Efficacy Questionnaire (TSEQ)	Aug et al., 1991	Tinnitus	7	1	N/A	

^aOne subscale, but two scores are calculated.

Hypothetical Case Study 13–1. Hearing Aid Self-Efficacy

Background

Mr. Jones is a 76-year-old patient with a moderately severe, high-frequency sensorineural hearing loss. He is a new hearing aid user of behind-the-ear hearing aids (about 3 months of experience) that are programmed with automatic settings along with a manual program for speech-in-noise environments that he accesses via a program button. He also has a manual, toggle volume control. He came in to see the audiologist with the complaint that his “hearing aids don’t work in background noise and the volume control doesn’t work right.” Real-ear verification is performed and the hearing aids are functioning well, but it is noted that they are in need of cleaning and that the batteries are weak. The audiologist inquires about the condition of the hearing aids and the patient cannot explain how to clean them satisfactorily. The audiologist asks him how he wears his hearing aids in background noise and how he adjusts the volume control and realizes he is not utilizing those features appropriately. He reports being very discouraged about wearing his new hearing aids. He seems uncertain about how to care for his hearing aids and how to use them in noisy listening situations. The audiologist determines that the hearing aids are functioning properly and programmed appropriately for the patient’s listening goals. The clinician then decides to administer the MARS-HA to measure the confidence that Mr. Jones has in his hearing aid abilities. Table 13–3 illustrates one possible scenario of responses on the MARS-HA and how a clinician might use those responses to guide AR recommendations and improve the patient’s hearing aid self-efficacy to a high level (i.e., ≥80%).

Table 13–3. Responses on the MARS-HA for a Hypothetical Case

Subscale	Mean Score	Self-Efficacy Level	Individual MARS-HA Items <80%
Basic Handling	76%	Moderately-high	Battery insertion (50%) Battery removal (70%) Operate controls (50%) Clean/care (70%)
Advanced Handling	66%	Moderate	Identify components (60%) Feedback management (50%) Troubleshooting (40%)
Adjustment	87%	High	N/A
Aided Listening	59%	Moderate	Phone (60%) TV (70%) 1:1 in noise (40%) Small group in noise (20%) Announcement over PAS ^a (40%) Car (50%)

^aPAS = public address system.

Impression

Overall, Mr. Jones has an overall MARS-HA aggregate score of 69%, suggesting that he is moderately certain in his abilities to perform various hearing aid skills. He reports being highly confident in having adjusted to wearing hearing aids and is moderately confident in the remaining areas of hearing aid self-efficacy. Recall that our goal is that patients have high levels of self-efficacy (i.e., 80% or higher). A deeper look into the individual items that are <80% suggest that he may be having trouble with certain skills (self-efficacy levels for these items are listed in the far-right column in the Table 13–3). In particular, the audiologist notices that he is reporting lower than desirable self-efficacy levels for items related to battery insertion/removal and care/cleaning. Further, the patient reported having trouble identifying the hearing aid controls and being able to use them, which may contribute to his difficulty hearing in noise (i.e., accessing his noise program when desired) and complaints of the volume control not working correctly.

Plan

Based on this information at today's appointment, the audiologist decides to implement a hearing aid orientation session using the self-efficacy-enhancing techniques for skills in which the patient reports being less than highly confident (i.e., <80%). The audiologist gives him a cleaning tool with a magnet to help him with battery insertion and removal and provides several attempts to practice identifying and operating his hearing aid controls. The clinician plays background noise through the real-ear verification system and role plays using his hearing aids in noisy situations while encouraging the use of cognitive rehearsal as he practices manipulating his hearing aid controls in this simulated listening condition. Realistic feedback about his abilities is provided. At the conclusion of the appointment, the patient is given enhanced educational materials (McMullan et al., 2018). The patient is offered enrollment in a communication strategies training program in which he can learn facilitative strategies for listening in background noise and other situations in which he is not highly confident.

Hypothetical Case Study 13–2. Listening Self-Efficacy**Background**

Mrs. White, age 65 years, is an experienced hearing aid user and is new to the clinic. She recently moved to the area with her husband. In her previous hometown, Mrs. White was very active in her community despite her hearing loss. She reports that she is trying to get involved in her new community, but is having difficulty doing so because she cannot understand speech in many situations. She came to see the audiologist because she wants to make sure her hearing is stable and her hearing aids are working properly. Hearing testing is performed that shows her hearing is stable compared to her last hearing test by her former audiologist. Real-ear verification demonstrates that her hearing aids are indeed working and fitted properly. The audiologist inquires further about the difficulty she is encountering and the patient reports that she feels uncomfortable in many different listening situations because she is new to the area and the individuals with whom she is interacting are unfamiliar. She does not want to ask strangers to repeat themselves as that may give them a bad first impression. To gain a better understanding of her confidence to understand speech in various listening situations, the audiologist decides to give her the LSEQ (Aided Version). Table 13–4 illustrates one possible scenario of hypothetical responses on the LSEQ and how a clinician

Table 13-4. Responses on the LSEQ for a Hypothetical Case

Subscale	Mean Score	Self-Efficacy Level	Individual LSEQ Items <80%
One-on-One in Quiet	95%	High	N/A
Directed Listening	80%	High	TV (70%) Announcement over PAS ^a in noise (50%)
Complex Listening	43%	Low	1:1 another part of house (50%) 1:1 while doing dishes and facing away (40%) Whisper (40%) 1:1 in continuous noise (50%) 1:1 in a group (40%) Group conversation in noise (30%) Cell phone in noise (40%)

^aPAS = public address system.

might use those responses to guide AR recommendations and improve the patient's listening self-efficacy.

Impression

Overall, Mrs. White has moderate levels of listening self-efficacy while using her hearing aids. She is highly confident in her abilities to listen to and understand speech when in one-on-one conversations in quiet places. She also is highly confident in most situations in which she is focusing on one source, except for when listening to TV and listening to an announcement over a public address system while in noise. In more complex listening situations, however, she has low levels of listening self-efficacy. The audiologist inquires further about which situations are important to her. She states that she is really interested in hearing better in situations where people are not facing her; one-on-one conversations in noise and in groups; and listening to group conversations while in a noisy background as these are the situations in which she finds herself in when participating in community activities.

Plan

Based on the patient's low self-efficacy for communicating in these situations and her report of being uncomfortable asking people for help when she does not understand them, the audiologist decides to offer her enrollment in a group AR program that focuses on (a) communication strategies training, (b) conversational repair, and (c) assertiveness training. Of course, the audiologist will incorporate self-efficacy-enhancing techniques as part of this group AR program. After Mrs. White completes the AR program, the clinician plans to administer the LSEQ again to determine whether or not her self-efficacy levels have increased. If not, it could be that incorporating hearing aid accessories, such as a remote microphone, might help in her identified difficult listening situations.

Conclusion

Planning and executing the many skills needed to master an AR behavior (e.g., hearing aid use, conversation repair, speechreading, and so forth) can be overwhelming for many patients. This is particularly true when the set of skills needed for a given behavior is new to them. Some patients may lack confidence, or self-efficacy, in their abilities for AR skills and behaviors. Such patients may put forth insufficient effort to achieve the behavior, give up when challenged, have setbacks, be less compliant with treatment recommendations, and be at risk for poorer outcomes compared to patients with high self-efficacy.

The self-efficacy theory has been applied successfully in managing chronic health conditions. A patient's self-efficacy for a given AR behavior can be measured using available standardized questionnaires. The information gleaned from the questionnaires can be useful for clinicians to identify the skills in which the patient has low self-efficacy. Clinicians can then draw upon the source of information for which individuals make self-efficacy appraisals and use specific techniques to help patients increase their self-efficacy for the skills in which they are less confident in performing. The versatility of the self-efficacy theory is that it can be incorporated into any AR treatment. The information presented in this chapter provides clinicians with an alternate approach to enhance their AR service-delivery in hopes to improve patient confidence in their AR behaviors and thus patient outcomes.

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14

AUDIOVISUAL SPEECH PERCEPTION AND SPEECH PERCEPTION TRAINING

Nancy Tye-Murray

“Look at me so I can hear you,” said a colleague’s mother one day. Although this remark caused the author’s colleague and her siblings to chuckle, her mother was absolutely right. She had a hearing loss and truly needed to watch her daughter’s lip movements to recognize the words that she could only partially hear. In a study performed at Washington University in St. Louis (Sommers, Tye-Murray, & Spehar, 2005), the author’s research team found that a group of older adults with age-appropriate hearing loss recognized only about 30% of the words in an open-set word test that was presented in noise. However, when the researchers let them see the talker, so they could lipread as well as listen, performance jumped to 80% correct! These findings highlight two important points about unimodal and multimodal speech perception, the former being the deleterious effects of hearing loss on auditory speech perception and how these effects are exacerbated by aging and background noise, and the latter being the power of the visual speech signal to augment the corresponding auditory signal (Erber, 1975; Sumbly & Pollack, 1954). These findings have an important implication for audiologic rehabilitation, namely, to the extent that intervention can maximize adults’ ability to utilize the visual and auditory speech signals, the greater will be the intervention plan’s success in enhancing their everyday face-to-face conversations.

This chapter will start with a consideration of vision-only speech perception, a topic that is typically given only cursory treatment in the field of audiology. The chapter will then consider audiovisual speech perception, with particular emphasis on how unimodal performance affects multimodal performance, which will set the stage for the final topics: speech perception training and how to integrate auditory training into clinical practice.

Vision-Only Speech Perception

Vision-only speech perception, sometimes called lipreading or called speechreading, refers to utilizing the information available on a talker’s face to recognize the spoken message, where the conversational partner attends to the talker’s jaw, tongue, and lip movements. This chapter, will use the term *lipreading* to refer to vision-only speech perception and the term *speechreading* to refer to audiovisual speech perception for the sake of clarity. In a broader definition, lipreading and speechreading may also encompass occasions where someone attends to facial expressions and supplementary hand and body gestures. For example, a talker who is scowling and pointing to her shoulder as she says, “I bumped into the refrigerator in the dark

last night” will impart an entirely different meaning than a talker who is grinning and disentangling a refrigerator magnet from her hair.

In addition to imbuing a spoken message with nuance and subtle semantic meaning, the visual speech signal conveys basic information, such as when a talker is about to speak, and the rhythm and cadence of speech. It also conveys phonemic information, such as indicants of place of production (e.g., /p/ versus /f/ versus /g/) and manner (e.g., /l/ versus /d/). Interestingly, studies with fMRI have shown that silent lipreading activates core regions of the temporal auditory cortex (Calvert, Bullmore, Brammer, Campbell, Williams, McGuire, Woodruff, Iverson, & David, 1997; Pekkola, Ojanen, Autti, Jääskeläinen, Möttönen, Tarkainen, & Sams, 2005), just as if an individual were listening to auditory speech. The visual speech signal also intensifies the auditory cortex’s response when added to the auditory speech signal, as during speechreading (Okada, Venezia, Matchin, Saberi, & Hickok, 2013). Such cortical activation is “speech-specific” and does not happen when a person is watching simple mouth movements such as the mouth performing a chewing behavior.

Lipreading Performance

The visual speech signal is very difficult to recognize when it is presented apart from the auditory speech signal. For example, on a test of isolated word recognition, a group of young adults recognized only about 35% of the words correctly when they could see but not hear the talker, where the talker’s head and upper head were clearly displayed on the monitor. Performance was even worse for sentences—participants recognized only about 16% of the words (Sommers, Tye-Murray, & Spehar, 2005). With this said, lipreading shows remarkable variability across people. For example, in this study, performance ranged from 5% correct to 80% words correct, and on the same stimuli! This variability has baffled scientists for decades and there is no consensus as to what variables predict performance, although an individual’s general intelligence, degree of education, sex, duration of acquired hearing loss, and gaze behavior appear not to be predictive of who is and is not a talented lipreader (Hygge, Rönnberg, Larsby, & Arlinger, 1992; Rönnberg, 1995; Tye-Murray, Sommers, & Spehar, 2007a). Certain cognitive skills, such as spatial working memory and processing speech, are somewhat predictive (Feld & Sommers, 2009), with better skills being associated with better lipreading ability, but

even so, these cognitive skills tend to account for little of the variance found in group studies. Adults who have prelingual hearing loss—that is, hearing loss acquired before the acquisition of spoken language—also tend to be better lipreaders (Pimperton, Ralph-Lewis, & MacSweeney, 2017). The most reliable predictor—and of particular relevance to the topic of audiology rehabilitation for adults—is age, which appears not only to be the most reliable predictor of lipreading performance but also the most robust predictor ever considered in the literature.

The Effects of Age on Lipreading Performance

It is a sad twist of fate that just as hearing sensitivity and word discrimination begin to decline because of aging, so too does lipreading ability (Sommers et al., 2005; Tye-Murray, Sommers, & Spehar, 2007b; Tye-Murray, Sommers, & Spehar, 2008). For example, Figure 14–1 shows results from the author’s study in which the team presented words in a closed-set matrix sentence test (called the “build-a-sentence test”) to a group of adults who ranged in age from 22 to 92 years of age (Tye-Murray, Spehar, Myerson, Hale, & Sommers, 2016). Matrix-style tests are useful for testing vision-only speech perception because they avoid the “floor effects” often associated with open-set tests of sentences or words. In the build-a-sentence test, the target words for each test sentence are selected randomly without replacement from a closed set of 36 nouns and placed in one of possible sentence contexts (e.g., “The *boys* and the *dog* watched the *mouse*” or “The *snail* watched the *girls* and the *whale*”). A response screen for one version of the build-a-sentence test appears in Figure 14–2.

As Figure 14–1 reveals, lipreading performance decreased across the lifespan, falling on average by 30% words correct from young adulthood to old age. The reasons for this decline are not well understood, although changes in peripheral visual acuity can be ruled out since participants had to have visual acuity within normal limits to be included in the study (with corrective lenses, when appropriate).

Factors That Influence Lipreading Performance

As indicated in Table 14–1, a number of factors can affect lipreading performance, including the talker, the message, and the environment (see Tye-Murray,

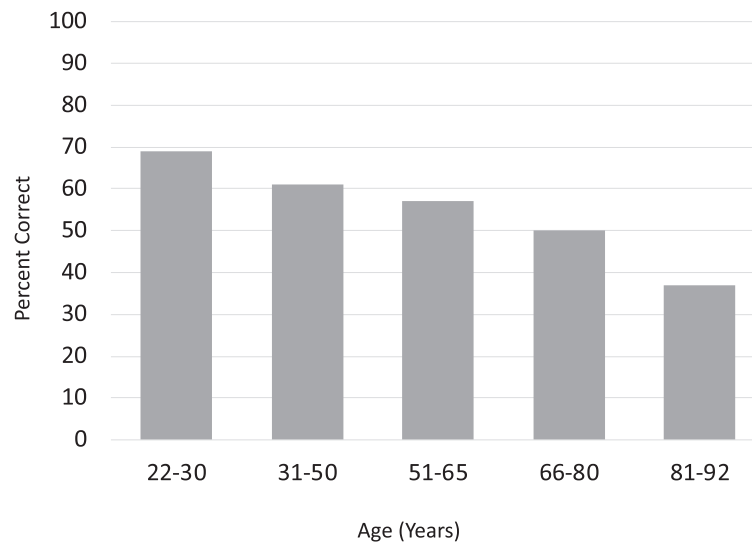


Figure 14-1. Lipreading performance declines across the adult life span. These are average scores from groups of adults varying age collected with a Spitzer test presented in a vision-only condition.

Please choose one of these types of sentences:

The ____ and the ____ watched the ____ and the ____.
 The ____ and the ____ watched the ____.
 The ____ watched the ____ and the ____.
 The ____ watched the ____.

Please choose all words from this list:

bear	cat	deer	fawn	geese	men	saint	team	whale
bird	cook	dog	fish	girls	mice	seal	toad	wife
boys	cop	dove	fox	goat	mole	snail	tribe	wolf
bug	cow	duck	frog	guest	moose	son	troop	worm

Figure 14-2. Example of a response screen for a build-a-sentence test.

Table 14-1. Factors That Can Affect Speechreading Performance

Talker	Message	Environment
- Facial expressions - Body gestures - Familiarity to the lipreader - Unfamiliar accent - Use of intonation - Use of clear speech (Gagne, Rochette, & Charest, 2002)	- Length - Syntactic complexity - Semantic complexity - Linguistic and situational context cues (Spehar, Goebel, & Tye-Murray, 2015) - Visemes^a and homophenes^b (Erber, 1974; Lesner et al., 1987)	- Viewing angle - Distance - Background noise, including the type of background noise (e.g., babble versus white noise; Myerson et al., 2016) - Room acoustics - Visual distractions

^aA viseme is a group of speech sounds that appear identical on the face (e.g., /p/, m/, /b/).

^bHomophenes are words that appear identical on the face (e.g., *fad* and *van*).

2020, for a review). For example, if a talker speaks very rapidly with minimal mouth movement, that talker will be extremely difficult to lipread. A word like *basketball*, which has few words in the English language that look similar on the mouth, will be easier to lipread than a word such as *can*, which not only had many visually similar words (e.g., *car*, *guard*, *hen*, etc.) but also is not produced with very visible articulatory movements (i.e., speaking the word does not entail lip movement and the initial consonant is produced at the back of the palate). Background noise can diminish lipreading, apart from how it adversely affects audiovisual and auditory-only speech recognition. For example, young and old adults can lipread speech more accurately in a quiet test condition as compared to a condition that has white noise presented simultaneously with the visual speech signal. Performance declines even more if the background noise is multi-talker babble (Myerson et al., 2016).

Many of these factors can be externally controlled, and might be targeted during family-centered care in an adult audiologic rehabilitation plan (Lind et al., 2019). For example, during a discussion of “where and when communication goes well and where it is a problem” (p. 35), a family member might note that the patient never seems to respond when he or she talks to the person while water runs from the kitchen faucet, which in turn could lead to a discussion about the effects of noise on speechreading and listening. For a family member who “talks a mile a minute,” training for the person in the art of speaking with clear speech (Picheny, Durlach, & Braida, 1985) might be included in an audiologic rehabilitation plan. Patients themselves can learn to optimize their ability to utilize the visual speech signal; for example, by seeking favorable seating during a lecture or podium presentation.

Audiovisual Speech Perception

The contribution of the visual signal to face-to-face communication is best demonstrated during speech perception in noisy environments. Compared to listening alone, when a conversational partner can both see and hear the talker, speech perception is substantially more accurate (Tye-Murray, Sommers, & Spehar, 2007b). Conversational partners can also tolerate more unfavorable signal-to-noise ratios without a loss in accuracy when speechreading as compared to listening (Grant & Seitz, 2000; MacLeod & Summerfield, 1987; Sumby & Pollack, 1954).

Being able to see the talker has greater benefits than simply improving accuracy. An important

advantage of speechreading over listening is a reduction in perceptual effort (Gosselin & Gagné, 2011). As the name implies, perceptual effort is the effort that someone expends to perceive speech, sometimes at the expense of depleting other cognitive resources such as working memory and attention. It is perhaps for this reason that audiovisual presentations as compared to auditory-only presentations also lead to better comprehension of short stories and dense philosophical texts (Arnold & Hill, 2001; Reisberg, McLean, & Goldfield, 1987).

Assessment of Audiovisual Speech Perception Performance

Sadly, routine audiologic examinations do not regularly include assessment of audiovisual speech perception in either quiet or with a noisy background, even though most everyday conversations occur in a face-to-face format and people are more likely to engage in speechreading than in listening alone.

Scores of performance in an audiovisual condition could serve as a powerful counseling tool, for both the patient and family members. For example, knowing how well a patient can speechread could set the stage for developing realistic expectations about the kind of benefit a hearing aid might provide or for communications strategies training, which might include a discussion about the importance of making sure the patient can see the talker’s face. Tests are available for this purpose. The test formats may include any of the following:

- Matrix style sentences such as that shown in Figure 14–2;
- Word lists that have been constructed so as to avoid floor effects and that include words like *basketball* (Tye-Murray & Geers, 2002);
- Topically related sentences that the patient learns beforehand, such as *summertime* (Boothroyd, Hanin, & Hnath-Chisolm, 1985);
- Sentences spoken by a variety of talkers (Tyler, Preece, & Tye-Murray, 1986);
- Sentences that either are supported by a picture illustration (i.e., the patient sees a picture of a flower in a window sill and then is asked to lipread the sentence, “The flowers were placed on the window ledge.”) or that require a “gist response” (i.e., the patient sees a sentence spoken and then selects a corresponding illustration from a choice of four) (Tye-Murray, Hale, Spehar, Myerson, & Sommers, 2014).

Perhaps one of the easiest and most effective ways an audiologic practice can distinguish itself in today's competitive market is to incorporate comprehensive speech perception assessment into their diagnostic battery, which includes the ecologically valid measure of audiovisual speech perception.

The Audiovisual Speech Advantage and Analyzing Test Results

The advantage of seeing and hearing the talker, as opposed to only hearing the talker, is often referred to as the *audiovisual speech advantage*, and might also be discussed using two other terms, *visual enhancement* and *auditory enhancement*. The term *visual enhancement* is used when we talk about how the visual signal enhances the ability, typically of someone who has hearing loss, to recognize speech. In this case, we might measure how well a person with a new hearing aid performs on a speech test in an auditory-only condition, without the visual signal, and then repeat the testing in an audiovisual condition, presenting both the talker's auditory signal and visual speaking image. The boost in the patient's performance when the visual signal is added is the visual enhancement, and indicates how the patient's use of the visual signal enhances his or her listening-alone performance.

The term *auditory enhancement* is used when we talk about how the auditory signal enhances listening performance. For example, in the early days of cochlear implants, a primary goal of providing a patient with a cochlear implant was simply to improve audiovisual speech perception—auditory-only word recognition was a dream for the future (since realized) (Tyler, 1993). In this case, how well a person with a new cochlear implant performed on a speech test in a vision-only condition was measured without the auditory signal, and then the test was repeated in an audiovisual condition. The boost in performance that occurred with the auditory signal was added was the amount of auditory enhancement, and was considered to be an indicant of the extent to which the electrical stimulation provided by the cochlear implant was improving face-to-face speech perception.

To compute either visual enhancement or auditory enhancement, patients must be tested in an audiovisual condition and either an auditory-only or vision-only condition, respectively. Comparable test stimuli must be used in each condition so any gains (or losses) are due to better audiovisual speech perception as opposed to differences in test list difficulty. For this reason, the type of matrix test presented in Figure 14-2 is often used because the test items are the

same across test conditions and learning effects with repeated exposure to test items is not a problem, as would be the case with open-set test lists. Computations might be either a simple difference score, computed by subtracting the percent words correct in the unimodal condition from the multimodal condition, or a normalized ratio, whereby you first determine how much room there is available for improvement in the unimodal condition and then referencing the amount of improvement gained in the multimodal condition to that amount available. For example, if you are interested in measuring visual enhancement, then you would use the formula:

$$AV\% \text{ correct} - A\% \text{ correct} / 100\% - A\% \text{ correct},$$

Where AV% correct refers to the percentage of words identified correctly in an audiovisual condition and A% correct refers to the percentage of words identified correctly in an auditory-only condition (and of course, 100% is perfection).

Imagine the power of knowing a patient's visual enhancement score when counseling family members. For example, you might say to a patient's wife, "When your husband can only hear you, he recognizes 50% of what you have to say. However, if you are sure to get his attention before you begin to speak, and if you make sure that he can see your face clearly, he'll be able to recognize 90%." Most likely, a light bulb will go off in the wife's head, as she realizes the potential for reducing both the occurrence of communication breakdowns and the need for using expressive repair strategies.

A patient's auditory enhancement score can also be useful, especially as an outcome measure of hearing aid benefit. For example, you might be able to demonstrate to a patient that without amplification, his or her ability to recognize speech during everyday communication situations is 50% but when he or she is wearing the new hearing aids, performance improves by 25%.

Theoretical Underpinnings of the Audiovisual Speech Advantage

A great deal of research has been directed towards understanding the audiovisual speech advantage (Blamey, Cowan, Alcantara, Whitford, & Clark, 1989; Braida, 1991; Grant, Walden, & Seitz, 1998; Massaro, 1996; Tye-Murray et al., 2008) and there is no consensus as to why seeing and hearing the talker is "super-additive," meaning that a patient will perform significantly better in an audiovisual condition than

would be predicted by adding together the patient's percent correct scores obtained in an auditory-only test condition and a vision-only condition. For example (and similar to the point made at the onset of this chapter), one patient may score 30% in auditory-only condition and score 16% in a vision condition, and yet, in an audiovisual condition, performance is not 46%, which one might predict if performance was simply additive, but rather, 70% words correct. Researchers also have attempted to understand the wide variability of visual enhancement found within groups of research participants, such as that reported by Sommers et al. (2005).

Some researchers posit a distinct stage of audiovisual integration in the speech perception process, and the integration stage enhances accuracy and also accounts for why some people are better integrators than are others (Grant et al., 1998). Conversely, others have suggested that a distinct stage does not exist, but rather, because visual and auditory processing occur simultaneously, it is the complementary nature of the auditory and visual speech signals that account for the audiovisual advantage (Tye-Murray et al., 2016). For example, cues about place of articulation are difficult to hear in the presence of hearing loss whereas they are easy to see on the face (e.g., lip closure for /p/); cues about voicing are relatively easy to hear whereas they are not visible on the face (e.g., /p/ versus /b/). Aligned with this latter viewpoint is the notion of overlapping auditory and visual lexical neighborhoods (Tye-Murray et al., 2008), which is a kind of "process of elimination." For example, if the test word is *carnation*, the visual signal eliminates word candidates that do not match what appears on the face (e.g., the word cannot be the single-syllable word *can*) and the auditory signal eliminates candidates that do not match what is heard (e.g., if the listener hears __-/naI/-__ /n/ the word cannot be the single syllable word *shell*, which has no nasal phonemes). In this alternative conceptualization, variability in visual enhancement can be attributed to differences in unimodal performance, which appears to be the case.

The Predictive Power of Unimodal Performance

For the purpose of this chapter, an important and relatively recent finding pertaining to audiovisual speech perception is the power of unimodal performance to predict speechreading performance. Because it has important implications for speech perception training, it is the topic that we will consider next as well

as for theoretical understanding of the audiovisual speech advantage.

Using a build-a-sentence test, Tye-Murray et al. (2016) measured vision-only word perception (lip-reading) in conditions where the visual signal was blurred to varying degrees and found no indication that a distinct integration ability exists in adults ranging in age from 22 to 92 years. The researchers also measured audiovisual word perception in the same blur conditions, and where enough background noise was presented so that if the participant had been listening in an auditory-only condition, performance would have been 30% words correct. Multiple regression analyses of the data from each blur condition revealed that once performance in the auditory-only and visual-only conditions were controlled for statistically, age was not a significant predictor of audiovisual speech perception; that is, performance in the auditory-only and vision-only conditions completely accounted for age-related differences in audiovisual speech perception. This outcome has been repeated using very different experimental methods: one using an open-set word test (Myerson, Spehar, Tye-Murray, Hale, & Sommers, submitted) and one using variations in speech rate (Sommers, Spehar, Tye-Murray, Myerson, & Hale, In press). This finding has important practical implications because although sensory and perceptual abilities typically decline with age, these changes are often correctable using hearing aids or corrective lenses. Moreover, the findings suggest that if we can improve a patient's unimodal speech perception, once appropriate hearing aids and eyeglasses have been provided, then not only might unimodal performance improve, so too might the patient's multimodal (i.e., audiovisual) performance.

Speech Perception Training

In today's world, speech perception training for adults typically focuses on developing patients' auditory speech recognition and is conducted without presenting the corresponding visual speech signal. If you were to trace speech perception training back to the heyday of its popularity, which would be the turn of the last century, speechreading training was more commonplace, most likely because patients did not have as much potential to develop their auditory skills because there were no sophisticated hearing aids and no cochlear implants available. Historically, speech perception training has been classified as either analytic, which focuses patients' attention on discriminating phonetic distinctions, or synthetic,

which focuses patients' attention on recognizing the gist of a message. More recently, speech perception training also focuses patients' attention on developing those perceptual cognitive skills that are necessary to decode spoken language. These skills include working memory, which is the ability to hold in memory the beginning of the message until the end of the message so processing of meaning can occur; attention, which is the ability to attend to a single talker in the presence of background noise; and processing speed, which is the ability to recognize words quickly, as speech may have an average rate of anywhere from 100 to 200 words per minute.

Two Examples of Computerized Auditory Training Programs

Many commercially available programs are computer based and are available either on CDs or on the internet, two examples being Listening and Communication Enhancement (LACE) (Sweetow & Sabes, 2006) and customized learning: Exercises for Aural Rehabilitation (cLEAR) (Tye-Murray, 2016). Both of these programs provide only auditory training and not speechreading training. In a typical training scenario, patients will train for about a half-hour at a time, give or take a few minutes either way, and two or more times a week over the course of several weeks or months. The spacing of training sessions is not critical, as research has shown that massed training (i.e., a set amount of a training condensed into a short amount of time) is equally effective as spaced training (i.e., the same amount of training spaced out over a longer amount of time) (Tye-Murray, Spehar, Barcroft, & Sommers, 2017).

LACE provides analytic training and training in the use of repair strategies, providing a variety of tasks that entail degraded speech, cognitive skill development, and use of communication strategies. Its main objective is to enhance listening and communication skills by increasing confidence and by providing auditory training activities. LACE does not attempt to make training gamelike but rather presents rote activities that are more like workbook exercises. For example, a patient might hear the sentence, "The point of the steel pen was bent and twisted." The sentence would then appear orthographically as follows, with the target words underlined: "The point of the steel pen was bent and twisted." The patient's task is then to use the computer cursor to indicate, "How many of the underlined words did you understand correctly?"

cLEAR provides both analytic and synthetic speech training, and also places an emphasis on cog-

nitive skill development. All training is "gamified." For example, one game plays like a slot machine (TreasEAR Island), one game plays like the card game Concentration (EARonaut), and a brand-new game plays like a card game of five-card draw (pokeEAR). The theoretical foundation for cLEAR comes from cognitive psychology, second-language learning, and auditory neuroscience (Barcroft, Sommers, & Tye-Murray, 2007; Barcroft et al., 2011; Tye-Murray, 2020). One of the most robust findings from research in cognitive psychology and second-language learning concerns the importance of meaning-based training. Training that uses meaningful stimuli as opposed to nonsense syllables or simple verbatim repetition is much more effective, which is why the cLEAR training stimuli are all meaningful words and sentences presented in a meaningful context. From neuroscience, we know that training that is engaging and entertaining is much more likely to lead to perceptual learning. This is because pleasurable activities lead to the release of dopamine in the brain, which leads to neural plasticity, which in turn potentiates perceptual learning, even in older adults (Whitton et al., 2017). The cLEAR games have been incorporated into an APP called "Amptify," which provides an ongoing daily hearing wellness journey for people who have hearing loss, with a special emphasis for those who have new hearing aids or a new cochlear implant (Amptify.com).

Speechreading Training Programs

Speechreading training tends to be much more short term and less experiential than auditory training, and might simply consist of a handout that is distributed during counseling. The handout might include a definition of speechreading and the importance of seeing the talker's face to enhance one's ability to recognize speech. If patients are participating in a communication strategies training class or in a hearing orientation group session, they might engage in a discussion about rules to follow when speechreading, perhaps with the guidance of a handout. "Rules" might include such simple suggestions as "Watch the talker's face," "Make ambient lighting work for you and not against you," and "If your conversational partner is engaging in a behavior that makes it difficult to speechread, say something!" (Tye-Murray, 2020, p. 145).

Efficacy of Speech Perception Training

Most current research about the effectiveness of speech perception training has concerned auditory training.

The research that has focused on speechreading training has revealed modest improvements incurred as a result of training (Walden, Prosek, Montgomery, Scherr, & Jones, 1977; Rishiq, Rao, Koerner, & Abrams, 2016). Lipreading, or vision-only speech perception, appears to be an innate skill that is seemingly impervious to training (Summerfield, 1992).

Because the term *auditory training* has been used to describe an array of training approaches that differ in terms of focus and content (e.g., LACE and cLEAR), it is difficult to make a blanket statement about the effectiveness of auditory training because, by so doing, we would be lumping together the proverbial “apples and oranges.” For example, Moore, Rosenberg, and Coleman (2005) provided phoneme-based, nonsense syllable training (e.g., *ama* and *afa*), with the assumption that by learning the “building blocks” of language, patients would learn to recognize better words and sentences. Humes et al. (2009) provided training in the most frequently used words of the English language, assuming that if patients could recognize the words that occurred routinely, overall sentence recognition would improve. Boothroyd (2008) presented topically related sentences, with the assumption that patients could learn to maximize contextual cues.

With this as a caveat, overall, the research suggests that auditory training can enhance adults’ auditory-only speech perception (Barcroft et al., 2011; Burk & Humes, 2007; Burk, Humes, Amos, & Stauser, 2006; Sweetow & Sabes, 2007; Tye-Murray et al., 2017), although at least two have shown minimal effects (Stacy et al., 2010; Saunders et al., 2016). Some studies have used outcome measures other than measures of speech perception. For example, studies have shown that auditory training can increase confidence to engage in conversation (Tye-Murray et al., 2012), reduce perceptual effort (Sommers et al., 2015), and reduce the return rates of hearing aids (Martin, 2007).

Implementing Auditory Training Into Clinical Practice

Auditory training is not routinely offered to adult patients. For example, only about 20% of patients who receive hearing aids receive additional hearing aids and only about 5% of them receive formal “re-training opportunities” (Kochkin, 2005). Audiologists generate 80% of their practice revenue from the sale of hearing aids (Taylor, 2018) and many have a “retail bent.” However, this state of affairs may change as

the ground is shifting with new governmental rules and regulations, and the ready availability of personal sound amplification products (PSAPS) and over-the-counter hearing aids. Increasingly, audiologists are looking for ways to distinguish themselves from the competition, and they can do so by integrating auditory training (and, as noted earlier, audiovisual assessment) into their clinical practice. Two important issues when considering how to integrate auditory training into clinical practice are how to determine candidacy and how to engage patients into the process.

Candidacy

While not all patients are candidates for auditory training, many of them are. As a general rule of thumb, patients who have undergone a recent change in hearing status, such as receipt of a new hearing aid or a cochlear implant, are prime candidates for auditory training. Perhaps not so obviously, so too are patients who are not yet ready for hearing aids—and there are many of those, as roughly only 20% of adults who might benefit from hearing aid actually obtain them (National Institute of Deafness and Other Communication Disorders, 2016). Auditory training can help patients maximally use the residual hearing and, importantly, draw attention to what they are not hearing. Through training, they may come to realize the magnitude of their listening problems, especially if their training program provides feedback about performance on the activities, as most computerized programs do in this day and age. “Wow!” a patient might realize, “I had no idea I was missing this much.” A third group of patients who stand to benefit from auditory training are those patients who are currently using hearing aids but who are still experiencing listening challenges. Many auditory training programs have activities that explicitly address the listening challenges that people with hearing loss are likely to experience, such as listening in the presence of background noise or listening to women’s and children’s high-pitched voices.

An important consideration in determining candidacy is the patient’s motivation to participate in an auditory training program and, if training is going to be computer-based, the patient’s wherewithal in working with a computer. A quick and easy way to gauge the former is to ask the question, “On a scale of 1 to 7, where 1 means *not at all* and 7 means *very much*, how interested are you in improving your listening success?” A score of 4 or higher bodes well for successful participation in a program. To gauge the latter,

the patient can answer this question, “Do you routinely use the internet at home for either social media, shopping, or news updates?” An answer of “Yes” also bodes well for successful participation.

Engaging Patients in the Auditory Training Process

When integrating auditory training into a clinical practice, it is important to gain buy-in from the patient. In so doing, it is important to describe to the patient what is auditory training, why is it helpful, and how will it be provided.

Jessen (2018) begins to educate the patient about the hearing health care journey on the first visit. She developed a written guide that serves as a template, where the audiologist sits with the patient at the onset of treatment and reviews the journey, which in her plan will be composed of the following components:

- completion of intake forms;
- determination of the top three communication and hearing goals;
- diagnosis of the degree of hearing loss and determination of word recognition ability;
- receipt of educational materials about communication strategies and consumer resources; and
- treatment options, including hearing aids and auditory training.

Clinicians who use the cLEAR program rely on a three-piece educational package. The first piece, given to the patient on the first visit, is a simple postcard that graphically portrays the three steps of a hearing health care journey: diagnosis (“Find out why conversations are becoming difficult”), hearing aids (“Obtain and learn to use the hearing devices recommended for your needs”), and listening therapy (“Exercise and develop your listening skills”). The purpose of this piece is to plant the seed for auditory training, so when the time comes to implement the training program, the patient is not taken by surprise but has expected it all along. The second piece of the educational package, provided during either the hearing aid assessment or the hearing aid fitting, is a simple brochure that describes auditory training. This is described in simple language, and might even be likened to the kind of physical therapy that one receives after receiving a knee implant, but in this case, it is listening therapy after receipt of a hearing aid: “Hear-

ing and listening are not the same thing. While hearing aids help improve audibility, they can’t help you develop the cognitive skills needed to understand conversation easily.” The brochure also describes the benefits of auditory training: “Users typically experience increased confidence in conversations and find it takes less effort to listen to and understand speech.” Finally, on the follow-up visit, the patient receives a “get started” piece, which is a simple handout about how to use the auditory training program.

The important points to remember when developing a means to integrate auditory training into routine clinical practice is that patients must be educated that this kind of listening therapy is part of the hearing health care journey, and that this education must occur in a way that is time efficient and economically feasible for a hearing health care professional to implement within the context of a busy clinical setting.

The Importance of Joining the Patient on the Hearing Health Care Journey

A relatively unheralded advantage of providing computer-based auditory training to patients is that it provides a mechanism by which the hearing health care professional can provide support and encouragement to the patient during the hearing health care journey. Many programs provide a means for an audiologist to follow their patients’ progress and training times via performance charts that are stored on the internet and an integrated messaging system whereby they can provide oversight and encouragement to their patients. Patients appreciate this. For example, in a study of 93 adults with hearing loss, after completing a computer-based auditory training program, participants were asked, “What did you like best about the program?” (Tye-Murray et al., 2012). One of the top answers was, “Regular contact with a hearing health care professional,” even though that contact consisted of less than 5 minutes per visit.

Summary

During everyday conversations, most of us engage with our conversational partners in a face-to-face fashion, which is why understanding audiovisual speech perception and devising ways to incorporate speechreading assessment into routine clinical practice is so important when developing an aural rehabilitation plan.

This chapter first considered lipreading and how, even though it is something that is very difficult to do, with *difficulty* defined as being able to understand speech accurately using only the visual speech signal. Nonetheless, it can greatly enhance a patient's ability to recognize speech in an audiovisual condition. Because most conversations occur face to face, there is good reason to include audiovisual speech assessment in routine clinical practice, and we considered a test format, tests, and formulae for doing so.

Research about audiovisual speech perception indicates that how well a patient can recognize speech in an auditory-only and a vision-only condition reliably predicts how well he or she can speechread. The implication of this finding is that if we can improve a patient's unimodal performance, then not only might we improve the patient's listening performance, but we may also enhance the patient's speechreading performance.

One way to affect unimodal speech recognition is to provide speech perception training. Although formal speechreading training rarely happens, several auditory training programs are available for this purpose, two of which were reviewed in this chapter. It is not enough merely to suggest that patients engage in auditory training; rather, it must be systematically integrated into clinical practice in such a way that it becomes a natural part of the patient's hearing health care journey.

In summary, the take-home messages from this chapter are twofold: (a) assessment of patients' speechreading performance in noise (and, if possible, in quiet) should be standard element in clinical practice and results can be considered both an outcome measure and a counseling resource, and (b) auditory training is listening therapy for the auditory system and is a service that will benefit patients, and not of least importance, it can be a very effective way for audiologists to distinguish the hearing health care they provide from that of their competitors

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15

KNOWLEDGE IS POWER

Interactive Multimedia to Improve Outcomes in the Digital Age

Melanie Ferguson

Introduction

The power of knowledge, a phrase commonly attributed to Sir Francis Bacon in 1597, has more recently been recognized as a key component of patient-centered care (Grenness, Hickson, Laplante-Levesque, & Davidson, 2014) and self-management of hearing loss (Barker, Mackenzie, Elliott, Jones, & de Lusignan, 2016). However, knowledge and understanding of hearing aids, communication, and the negative consequences of hearing loss by people with hearing loss, their family and friends, nonaudiologic health care professionals, and the general public are generally poor (Desjardins & Doherty, 2009; Greengross, 2014). This can lead to ineffective communication and social engagement between the person with hearing loss and those whom they communicate with (Barker, Leighton, & Ferguson, 2017; Kamil & Lin, 2015). The psychosocial consequences arising from this can result in social isolation, withdrawal and loneliness, stigma, and depression, leading to reduced quality of life (Barker et al., 2017; Heffernan, Coulson, Henshaw, Barry, & Ferguson, 2016; Strawbridge, Wallhagen, Shema, & Kaplan, 2000).

Hearing aids are the main clinical intervention for managing hearing loss. High-quality evidence from a Cochrane systematic review has shown that hearing aids are effective at improving listening, participation in everyday life and health-related quality of life (Ferguson et al., 2017). Despite this, hearing aid uptake and adherence is often far from ideal (Davis, Smith, Ferguson, Stephens, & Gianopoulos, 2007). Only one in three adults who would benefit from hearing aids has them, and nonuse of hearing aids ranges from 3% to 24% (Ferguson et al., 2017). There are a number of reasons for non- or suboptimal hearing aid use, including physical factors and comfort, hearing aid handling and maintenance, psychosocial and situational factors, and attitude of health care professionals (McCormack & Fortnum, 2013). Furthermore, expectations about the benefits of hearing aids may be set too high (Ferguson, Woolley, & Munro, 2016). Many of these reasons can be addressed by better knowledge and understanding of these issues.

In his seminal paper on adult aural rehabilitation, Arthur Boothroyd included instruction on hearing instrumentation as one of the four cornerstones of aural rehabilitation (Boothroyd, 2007). Hearing aids are complex, as are consequences of hearing loss.

Therefore, for a person to become an effective and successful user of hearing aids, they need to have the knowledge to understand how hearing aids work within everyday communication situations, often involving other people. Crucially, as will be discussed later, Boothroyd highlighted that "... there is a difference between 'instruction' and 'telling' . . ." (p. 65). The information offered to hearing aid users needs to be understood, assimilated, and acted upon within the context of their communication and participation in everyday life. At that time, there was relatively little research on the effectiveness of instruction, often known as hearing aid orientation (Reese & Hnath-Chisolm, 2005). However, over the last decade a number of interventions have been developed to help improve hearing aid knowledge and communication using a range of delivery systems. These methods include group work (Hickson, Worrall, & Scarinci, 2007), and remote delivery using videotapes (Kramer, Allesie, Dondorp, Zekveld, & Kapteyn, 2005), DVDs, and online materials (Ferguson, Brandreth, Leighton, Brassington, & Wharrad, 2016; Thorén, Öberg, Wänström, Andersson, & Lunner, 2014).

In the U.K., recent national guidelines on hearing loss recommend that good quality information should be provided for both those with hearing loss and their families (National Institute for Health and Care Excellence, 2018). However, clinical provision of information to promote knowledge on hearing loss, hearing aids and communication is often delivered verbally or through written materials. This often comes in the form of passive telling rather than active instruction, which is more likely to promote learning. In education and other health care fields, remote digital educational interventions have been shown to improve knowledge acquisition, satisfaction, and adherence.

To address knowledge limitations of hearing aid users and promote greater hearing aid use and better communication, a series of multimedia, interactive educational videos (or reusable learning objects, RLOs) have been developed. These RLOs, known collectively as C2Hear, provide education on important practical and psychosocial aspects of hearing loss, hearing aids, and communication. In addition to the RLOs, C2Hear includes a short introduction, patient testimonials, and a multiple-choice quiz after each RLO. Originally developed for remote delivery via DVD for television, personal computer, and the internet, C2Hear has been freely available online via YouTube since 2015 (<http://www.youtube.com/C2HearOnline>). More recently in 2019, a standalone, freely available "go-to" website (<http://c2hearonline.com>) was launched that contains a range of resources

including the RLOs, demo and waiting room clips, and journal and newsletter articles. This chapter focuses on the C2Hear research program and will discuss the (a) theoretical underpinning, including self-management of hearing loss; (b) development of C2Hear; (c) evaluation of C2Hear when used with patients, partners, and professionals; and (d) current developments to increase individualization and interactivity.

Theoretical Underpinning to Support Development of Educational Interventions

Development and evaluation of an intervention based on theory is more likely to lead to an effective intervention than using an empirical or pragmatic approach (Medical Research Council, 2006). A theoretical underpinning can provide an understanding of how an intervention might affect change in terms of what can be expected and achieved. Here, we focus on two theories that have informed the development of the C2Hear RLOs and their further development: (a) the constructivist learning theory, incorporating pedagogical participatory design principles; and (b) the health behavior change theory, specifically the COM-B model.

The *constructivist learning theory* posits that learning is an active process of constructing and conceptualizing knowledge whereby the learner constructs an internal representation of the learning materials. By actively interacting with learning materials, particularly in the form of relevant and specific activities, the learner is more engaged and motivated to learn than if the learning environment was passive. Incorporating past experiences and prior knowledge, the learning can result in a unique version of knowledge for the individual, resulting in more effective knowledge acquisition. The use of interactive videos delivered within an e-learning environment has been shown to provide better learning performance and greater satisfaction than more passive and traditional learning environments (Zhang et al., 2006).

The pedagogical design principles for C2Hear were based on the principles of learning theory, and used an e-learning format known as the RLO (Windle & Wharrad, 2010). RLOs are bite-sized chunks of interactive multimedia e-learning that focus on a specific learning goal. They are used widely in the educational field (e.g., universities), and have also been used in health care education. The theoretical framework underpinning the pedagogical design of the RLOs is the International Machine Standard (IMS) Learning

Design (Koper, 2003). Activities and self-assessments are built into the RLOs and are aligned with a specific learning goal (Biggs, 2003). These activities are important because they help users to actively engage in the process of learning, and feedback from self-assessments helps users to determine whether they have successfully achieved the learning goal.

RLOs comprise four components: (a) presentation of the concept or procedure to support the learning goal; (b) an activity that enables the learner to engage with the content; (c) self-assessment to test for mastery of the content; and (d) links to other resources to reinforce the learning. The visualization process that forms the computerized animations or videos is an important feature as this enhances learning and enables a more effective way to demonstrate complex concepts compared with learning from static text (Thatcher, 2006). Key to the effectiveness of RLOs is a sense of control and ownership of the learning process, along with the ability to reuse the resources on multiple occasions (Windle, McCormick, Dandrea, & Wharrad, 2010).

A weakness of many e-health and educational interventions is that there is limited or no involvement of end users (van Velsen, Wentzel, & van Gemert-Pijnen, 2013). The use of participatory and codesign approaches are increasingly being used to ensure that both end users and key stakeholders are at the core of the design and development of such interventions. This coproduction ensures educational materials are aligned to the needs of the end user, and improve usability and satisfaction (Bruno & Muzzupappa, 2010; Latif, Carter, Rychwalska-Brown, Wharrad, & Manning, 2017). Coproduction of RLOs typically involves end users and stakeholders in a design workshop that provides a forum for discussion and engagement with others to create the RLO content. This leads to high-quality and relevant materials, specifically designed to meet the needs of the end user. Behavior change is also a desirable outcome from an educational intervention (Yardley et al., 2016). The creative workshops allow personal stories and anecdotes to be captured during the storyboarding process, which can provide triggers for behavior change. A participatory and codesign approach initiated through workshops is a key feature of RLO development methodology.

Health behavior change is relatively new to audiology, whereas the field of health psychology has worked with behavior change for more than four decades. In recent years, audiology has focused on some of the more popular models that aim to understand, predict and promote health-related behavior

(Coulson, Ferguson, Henshaw, & Heffernan, 2016). Examples include the Health Belief Model (Rosenstock, 1974), the Transtheoretical Model (Prochaska & DiClemente, 1983), and the Theory of Planned Behavior (Ajzen, 1991).

The use of these models to provide theoretical underpinnings to hearing research has been a positive development. However, within the field of health psychology there is a well-developed body of literature to suggest that these models do not, and cannot, reliably explain the variability in health behaviors (Coulson et al., 2016). More recently, a new approach has been developed to address these limitations that has a psychological model of human behavior at its core, known as the COM-B system of health behavior change (Michie, van Stralen, & West, 2011). (*Editors' note:* For additional information on health behavior change, the reader is referred to Chapter 8 of this text.)

The Behavior Change Wheel is an overarching framework specifically developed to characterize behavior change interventions and link them to analysis of the target behavior (Michie et al., 2011) (see link <http://www.behaviourchangewheel.com/about-wheel>). The hub of the Behavior Change Wheel is the COM-B system that has three core components predicting behavior via capability, opportunity, and motivation. *Capability* is the individual's psychological and physical capacity to engage in a behavior, including having the necessary knowledge and skills. *Opportunity* considers factors that are external to the individual that prompt or make the behavior possible. *Motivation* considers brain processes that energize and direct behavior, including habitual processes, emotional responses, and analytical decision making.

The Theoretical Domains Framework (TDF) (Cane, O'Connor, & Michie, 2012) enables theoretical constructs relating to behavior change to be mapped directly onto the COM-B system. The Behavior Change Technique (BCT) Taxonomy (Michie et al., 2013) enables users to specify the smallest components of interventions that can bring about behavior change. This provides a common language by which to develop, define, and report behavior change interventions in terms of their "active ingredients." Components of the Behavior Change Wheel, namely the COM-B system, TDF, and the BCT Taxonomy, are now being used in audiological rehabilitation research to improve understanding of the underpinning mechanisms of health behavior change, and to theoretically inform intervention development and assessment (Barker, Atkins, & de Lusignan, 2016; Barker, Mackenzie, & de Lusignan, 2016; Maidment & Ferguson, 2017; van Leeuwen et al., 2018).

The C2Hear interventions described here are theoretically based. In the early development stages of the C2Hear RLOs, both learning theory and participatory/codesign principles were core to the development. Further developments of C2Hear into shorter, more individualized interactive RLOs developed for delivery through a mobile-enhanced platform (mRLOs) were theoretically designed using the COM-B model.

Self-Management of Hearing Loss

Self-management of hearing loss as a long-term chronic condition is also coming to the forefront in hearing research. More generally, there is a growing awareness that self-management of long-term conditions (LTCs) can enhance the efficiency of health and social care. Self-management is defined by the U.S. Institute of Medicine as “the tasks that individuals must undertake to live with one or more chronic conditions. These tasks include having the confidence to deal with medical management, role management and emotional management of their conditions” (Corrigan, Greiner, & Adams, 2004, p. 57). Self-management focuses on behaviors relating to a specific health condition. Those who take an active role in their self-management, and who are appropriately motivated to participate actively in their care, are more likely to adopt better health behaviors. This in turn leads to better patient outcomes (Mosen et al., 2006). (*Editors’ note:* For additional information on self-efficacy, the reader is referred to Chapter 13 of this text.)

A meta-review of a range of LTCs has highlighted five key components of self-management (Taylor et al., 2014). These are:

1. provision of education about the LTC;
2. psychological strategies to support adjustments to life with a LTC;
3. strategies to support adjustment to life with a LTC as well as adherence to treatments;
4. practical support tailored to the individual with the LTC; and
5. social support.

In terms of interventions to support self-management of LTCs, Pearce et al. (2016) developed a 14-item taxonomy of self-management support that classifies components of interventions. The taxonomy includes four specific elements: mode of delivery; personnel who deliver the support; target of the intervention;

and intensity, frequency, and duration of the intervention. Such a taxonomy would help underpin the development and evaluation of interventions for self-management of hearing loss.

More recently, there has been a growing awareness about self-management of hearing loss. A Cochrane systematic review has examined the use of interventions to promote hearing aid use (Barker et al., 2016b), based on the generic Chronic Care Model (CCM) used to describe self-management of LTCs (Bodenheimer, Wagner, & Grumbach, 2002). The review included 37 randomized controlled trials (RCTs), which included a range of interventions associated with adult aural rehabilitation (e.g., education, auditory training, counseling), with the primary outcome being hours of hearing aid use. The quality of the evidence was rated as very low to low. Therefore, the results of the numerous meta-analyses conducted, aligned to the components of the CCM, need to be considered within the context of the low-quality evidence. Overall, the review concluded that any improvements in outcomes had small effect sizes, and only some people showed improvements in certain outcomes. Although there were improvements in hearing “handicap” and verbal communication in the short term, there were no improvements in either the short-, medium-, and long-term outcomes for hours of hearing aid use. However, the review did highlight that one large RCT showed that the interactive multimedia videos (C2Hear) encouraged more people to wear their hearing aids (Ferguson, Brandreth, Brassington, & Wharrad, 2016). Furthermore, of all the 37 studies reported in the review, this study was shown to have the lowest risk of bias in terms of how it was conducted, therefore indicating a high-quality study.

Much of the relatively limited evidence looking at self-management relating to hearing loss is limited to adherence to interventions (e.g., hearing aid use), with relatively little focus on the more psychosocial aspects such as coping (Convery, Keidser, Hickson, & Meyer, 2019). A recent study by Convery et al. (2019) identified self-management of hearing loss as a multifactorial construct that includes three factors:

1. actions (e.g., shared decision-making, taking action);
2. psychosocial behaviors (e.g., coping); and
3. knowledge.

Independent variables associated with these factors and overall self-management of hearing loss were hearing aid self-efficacy, health literacy, hearing health care experience, and problem-solving skills.

Convery et al. (2019) propose that although fixed factors such as age, gender, and health literacy may influence the self-management intervention, modifiable factors such as self-efficacy may be targeted to improve self-management. These findings, which bring together self-efficacy and behavior change as constructs underpinning self-management of hearing loss, fit well with the outcomes from the C2Hear research, discussed in a later section.

Development of C2Hear

The C2Hear RLOs were coproduced using a participatory approach involving hearing aid users and hearing health care professionals, who were core to the development process and were involved at all stages of design (Ferguson, Leighton, Brandreth, & Wharrad, 2018). At the time that C2Hear was being developed in 2011, there was relatively little in the scientific literature to describe what good-quality information for first-time hearing aid users looked like. To address this, a two-stage approach was taken.

Delphi Review

The first stage used a Delphi review method to reach a consensus amongst hearing health care professionals on the informational needs of first-time hearing aid recipients. This is an iterative process that refines opinions on a specific subject until a group of experts reaches an accepted degree of consensus. A Delphi review has four core characteristics: an expert panel, a number of iterations and controlled feedback, statistical feedback, and anonymity of responses (Diamond et al., 2014). The expert panel comprised 33 U.K. hearing health care professionals including National Health Service (NHS) audiologists, hearing therapists, hearing researchers, hearing charity representatives, hearing aid company representatives, and independent hearing aid dispensers.

An anonymized three-round Delphi review was conducted by email. In round 1, 10 open-ended questions sought to establish the ideal information for first-time hearing aid users and reasons for non-use of hearing aids. Thematic analysis of responses resulted in a bank of 67 statements on information needs. These statements formed the basis of rounds 2 and 3, and participants scored the statements on a 5-point Likert scale (*strongly agree* to *strongly disagree*). In round 3, the statements were presented alongside

the summary statistics from round 2. Consensus on a statement was reached when $\geq 90\%$ of the expert panel “agreed” or “strongly agreed” with the statement. To provide general topic areas for the RLOs, the panel was asked to rank order the 15 proposed topics that arose from round 1.

At the end of round 3, 100% agreement was reached for 21 statements (31.3%), and between 90% and 99% agreement for a further 21 statements (31.3%). There was less than 50% agreement for nine statements (13.4%). The final top 10 ranked topics were: hearing aid insertion; hearing aid controls; hearing aid maintenance; getting used to hearing aids; communication tactics; hearing aid benefits and limitations; information for communication partners; listening in different situations; expectations of hearing aids; and telephones and assistive listening devices.

Workshops

The second stage involved hearing aid users and audiologists in the development of the RLO content through the coproduction of A0 (841 × 1189 mm) storyboards (Figure 15–1). This provided an opportunity for participants to conceptualize their personal experiences of hearing aids and hearing loss. A total of 32 hearing aid users and 11 audiologists participated in three 1-day workshops. The top 10 topics of ideal information from the Delphi review were discussed by the participants, and their views and perspectives of these topics were captured on the storyboards that provided visual representations of their personal thoughts and experiences around these topics. Each group developed two or three storyboards, and a total of 23 storyboards were generated, with at least two storyboards per topic.

Development and Production of the RLOs

The Delphi statements that reached consensus ($\geq 90\%$ agreement) and key points from the storyboards were mapped and integrated onto the relevant RLO title derived from the top 10 topics. Written specifications for the RLOs were developed, which included key pedagogical components: learning goals; detailed description of visual imagery and sounds, such as illustrations, video clips, static and dynamic animations, and photographs; transcripts of the text to accompany the media, which formed subtitles; and a two- or three-question multiple-choice quiz with

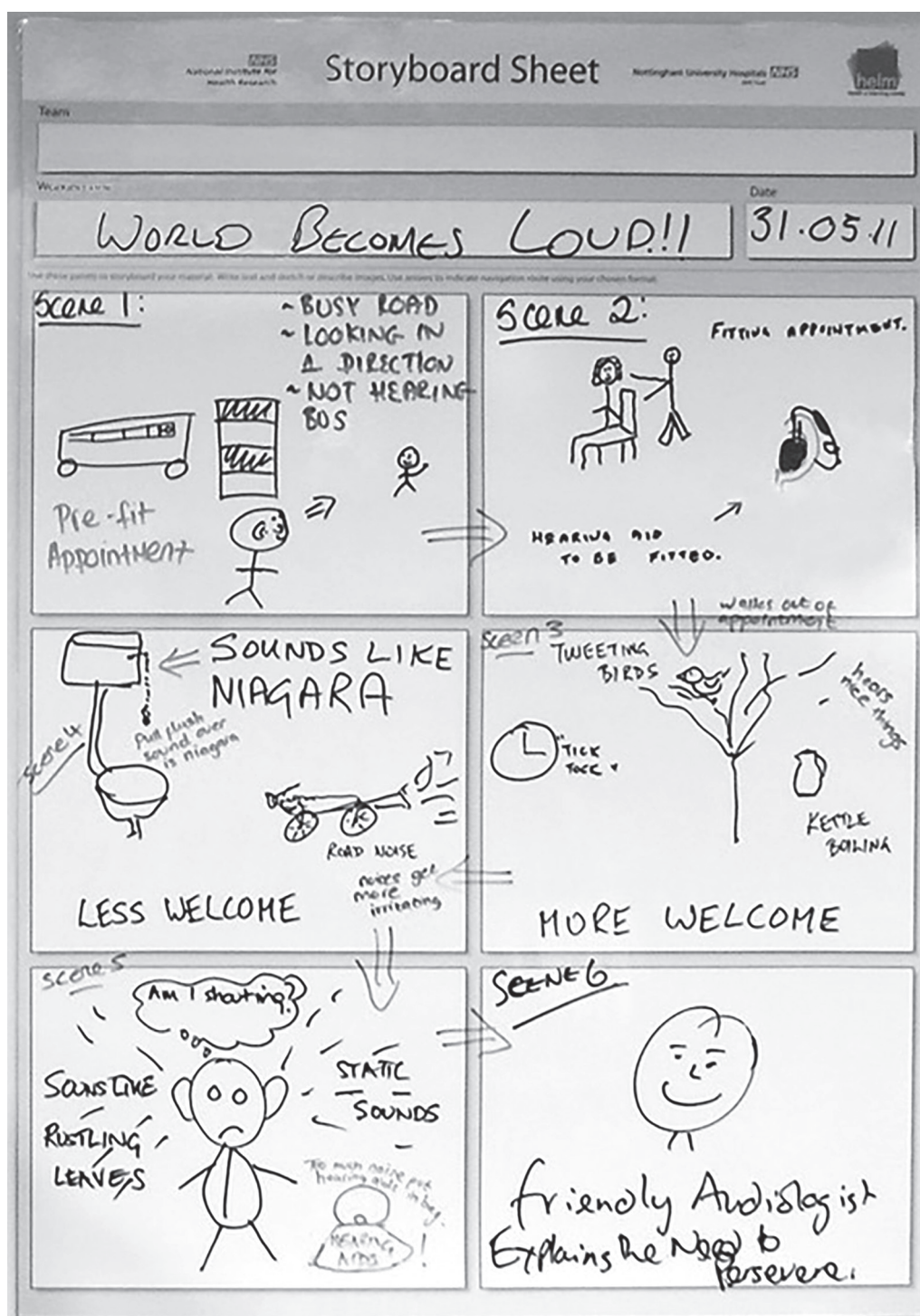


Figure 15-1. Example of a storyboard developed by hearing aid users.

additional feedback. The specifications were iteratively developed and peer reviewed with input from the project-specific patient and public involvement (PPI) panel that consisted of hearing aid users. This

ensured that the RLOs were meaningful and relevant, and clearly explained the concepts in a language that could be easily understood by the future end user (i.e., first-time hearing aid users). In addition, there

was also iterative peer review from a panel of audiologists to ensure the RLOs had clinical validity. Each RLO specification was reviewed two to three times.

The written specifications were then developed into the multimedia RLOs with subtitles. At the time the RLOs were developed in 2011 to 2012, the use of PCs and internet in the typical first-time hearing aid user age group (70 to 74 years) in Nottingham, U.K., was quite low, at 36% and 17%, respectively (Henshaw, Clark, Kang, & Ferguson, 2012). Although the interactivity potential for RLOs delivered online was greater, in order to maximize accessibility for the typical first-time hearing aid user population, a DVD delivery platform was chosen. This meant the RLOs could be delivered through DVD for TV or computer, as well as online through the internet via a custom-made, password-protected portal.

Initially, there were a total of seven RLOs plus an introductory RLO, some of which were developed for both slim open-fit tubes and custom earmolds (e.g., How to Insert Hearing Aids). The duration of the RLOs ranged from 4 minutes and 34 seconds to 11 minutes and 35 seconds (Mean = 9 min, 25s) (Ferguson, Brandreth, Brassington, & Wharrad, 2015). Participants had the freedom to choose which RLO to watch, and when, and for as many times as they liked, with the option to fast forward, rewind, and pause. Additional interactivity was achieved through the interactive multiple-choice quiz. The aim of this participatory approach to the development of the RLOs was to ensure that as many hearing aid users would find them relevant and useful. However, the ultimate aim of the research was that the C2Hear RLOs would provide benefits to hearing aid users.

Evaluation of C2Hear in Patients, Partners and Professionals

First-Time Hearing Aid Users

Following development of the RLOs, their effectiveness was evaluated in a registered RCT of 203 first-time adult hearing aid users (ISRCTN11486888) (Ferguson, Brandreth, Brassington, & Wharrad, 2016). The RCT aimed to (a) establish the accessibility, uptake, acceptability, and adherence of the RLOs, and (b) assess the benefits and cost-effectiveness of the RLOs. Patients were prospectively recruited and remotely randomized by the Nottingham Clinical Trials Unit into either the intervention group (RLO+, $n = 103$) that received the RLOs, or the waitlist control group (RLO-, $n =$

100) that received standard care. Those allocated to the RLO+ group were able to choose their preferred RLO delivery method. The most common delivery format was DVD for TV (50.6%), followed by internet delivery (32.9%) and DVD for TV (15.2%). One person used a DVD with autoplay.

The primary outcome measure was hearing aid use from the Glasgow Hearing Aid Benefit Profile (GHABP) (Gatehouse, 1999), and the study was powered to show a 12.5% difference between the RLO+ and RLO- groups. Allowing for a 15% attrition rate, 200 patients needed to be recruited. The RLO+ group received the RLOs at the hearing aid fitting appointment, and both groups were seen for a follow-up evaluation at 6 to 7 weeks postfitting (Mean = 6.8 weeks). The research audiologists were blinded as to the group allocation, and at the end of the follow-up session the RLO- control group were offered the RLOs.

Secondary outcome measures included: GHABP part 2, International Outcome Inventory for Hearing Aids (IOI-HA) (Cox & Alexander, 2002), Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982), Satisfaction with Amplification in Daily Life (SADL) (Cox & Alexander, 1999), Practical Hearing Aid Skills Test (PHAST) (Desjardins & Doherty, 2009), Hearing Health Care Intervention Readiness (HHCIR) (Weinstein, 2012), Hearing Aid and Communication Knowledge (HACK) (Ferguson et al., 2015), and the Hospital Anxiety and Depression Scale (HADS) (Zigmond & Snaith, 1983). In addition, a video diary was used to record use and usefulness of the RLOs. Patient feedback on the RLOs was obtained based on 20 statements, such as "I found the videos enjoyable," using a 5-point Likert scale (*strongly agree* to *strongly disagree*). To gain a deeper understanding of the benefits of the RLOs, three focus groups were held that included both the research participants ($n = 20$) and communication partners ($n = 5$).

Uptake and Adherence Was High; Feedback Was Positive

Uptake and adherence of the RLOs was high (78.4% and 97.4%, respectively). The average number of views was 13 per participant, and around half (49.2%) watched the RLOs more than once, with 22% watching at least three times. The multiple use of the RLOs suggests that they were used to self-manage their hearing loss, and this was confirmed in the focus groups and open-ended questions from the feedback questionnaire. The RLOs were rated as highly useful on average (Mean = 8.9, where 0 = not useful, 10 = extremely useful), and 78.4% said they would recommend the

RLOs to other people. Patient feedback was favorable. For content, the vast majority (92%) agreed the illustrations and videos aided their understanding. For activity and engagement, 91% agreed the quiz gave a clear message, and 88% would refer back to the RLOs if they had a problem. Finally, participants reported they preferred the RLOs to written information (82.9%), and felt more confident in discussing hearing aid and communication with others (81.3%).

This positive feedback was reflected in the responses from the open-ended feedback questions and the focus groups. Participants reported that the RLOs:

- provided useful and helpful advice;
- were easy to understand and reliable;
- reinforced their knowledge and confidence;
- provided reassurance and helped perseverance with hearing aids;
- were a helpful resource to refer back to if needed; and
- helped with coping in difficult situations.

Focus groups confirmed that the RLOs were generally very well received, with key themes being repeated watching, sharing RLOs, providing reassurance, and helping people to remember, reflected in the following statements:

“... I found that I had missed something, so I went back and looked again.”

“I have passed my DVD onto an old couple who both have hearing aids... I kept telling her, play that DVD and you will know why...”

“The DVD explained how we have to relearn to rehear things. That was not an aspect that I was aware of, to re-educate your brain to interpret what you hear.”

Not all the comments were positive, however. Some people reported that the RLOs were either too long or could have delivered the key messages in a more concise way. Others thought the RLOs were too simple. A surprising finding was that some participants thought that the hearing aid users who appeared in the RLOs and were the age of typical first-time hearing aid users were sometimes perceived to be too old. Even though they were broadly the same age as those participating in the study, this was seen as a negative and off-putting. Future development

of interventions that depict hearing aid users should take into account these views to minimize negative perceptions.

RLOs Benefited Patients

There were significant improvements in outcome measures for the RLO+ group compared to the control RLO- groups for:

- knowledge of hearing aids (HACK, $p < .001$), with a large clinical effect size ($d = 0.95$);
- practical handling skills (PHASI, $p < .001$); and
- hearing aid use (GHABP) in those who did not wear their hearing aids all the time ($<70\%$) ($p = .03$), with a large clinical effect size ($d = 0.83$).

There was no difference in overall hearing aid use for the whole sample (median RLO+ = 100%, RLO- = 96.7%, $p = 0.48$), in part because many participants reported 100% use, indicating ceiling effects. For other outcomes, there were no group differences. Finally, a health economic analysis showed that the RLOs in DVD format were highly cost-effective.

To summarize, the results from this RCT of the C2Hear educational intervention showed key benefits of knowledge improvement, hearing aid handling skills, and hearing aid use, as well as providing reassurance and increasing confidence with hearing aids and communication. RLO uptake and adherence was high, and C2Hear was rated as highly useful.

Through the mixed-methods approach using both quantitative and qualitative methods, the study demonstrated that C2Hear was effective in providing a range of benefits for first-time hearing aid users. However, it was also relevant to find out why this intervention was effective, and to establish what these underlying effects were. To do this, the Medical Research Council process evaluation framework was used that aims to examine causal mechanisms and contextual factors (Moore et al., 2015). Causal mechanisms, also known as mechanisms of impact, refer to *how* an intervention works, highlighting where future improvements can be made. Contextual factors are those that are external to the intervention, which can strengthen or impede its effects, and can be used to inform *why* an intervention does or does not work. Based on these principles, General Linear Model (GLM) analyses were conducted to gain some insights about how and why C2Hear provides benefits to hearing aid users.

The causal mechanism of impact (the *how?*), which was improved knowledge of hearing aids, led to greater hearing aid use, benefit, and satisfaction, and reduced emotional consequences of hearing loss (i.e., anxiety). For contextual factors (the *why?*), hearing aid self-efficacy (derived from the HHCIR) was the factor that led to improved hearing aid handling skills and knowledge, greater hearing aid use, reduced emotional consequences of hearing loss (i.e., anxiety), and patient activation (Maidment, Brassington, Wharrad, & Ferguson, in press.). The causal mechanism of improved knowledge that arose from using C2Hear is one of the three factors that Convery et al. (2019) identified as contributing to self-management of hearing loss. This suggests that C2Hear can be employed by hearing aid users in the self-management of their hearing loss, and as a result improve their outcomes. Furthermore, the contextual factor of self-efficacy meshes well with the conclusion from Convery et al. (2019) that self-efficacy is a behavior that can be targeted to improve self-management of hearing loss. Using C2Hear to improve self-efficacy was specifically examined in a later study.

After the RCT was completed, further feedback obtained from research participants was reviewed and the RLOs were revised. The end result was nine RLOs plus an introductory RLO. Changes included

patient testimonials being removed from the main body of the RLOs, removing aspects that were considered negative such as the use of black-and-white photos that compounded the older age of people who appeared in the RLOs, and the total time was reduced from 1 hour to 45 minutes (Figure 15–2).

Early Delivery of C2Hear

The RCT provided C2Hear at the fitting appointment, but increasing patient's knowledge prior to receiving hearing aids has obvious advantages, such as enhancing self-efficacy for hearing aids and readiness to take action. To investigate this, an RCT of 47 first-time hearing aid users was conducted where C2Hear was delivered earlier in the patient journey, at the hearing assessment. First-time hearing aid users received either C2Hear ($n = 24$, intervention group) or the standard clinic booklet ($n = 23$, control group) (Gomez & Ferguson, 2019). Outcomes were assessed 4 weeks later, prior to the fitting of the hearing aids. The primary outcome measure was the Measure of Audiologic Rehabilitation for Self-Efficacy for Hearing Aids (MARS-HA) (West & Smith, 2007). Other outcome measures were the Ida Institute motivation line tools that measure readiness and self-efficacy (Ferguson, Maidment, Russell, Gregory,



Figure 15–2. C2Hear RLOs made publicly available online following further patient input.

& Nicholson, 2016), and the HACK that measured knowledge of hearing aids and communication.

Adherence with C2Hear and the booklet in both groups was high (>85%). As seen with the previous RCT, knowledge of hearing aids and communication was significantly greater for the C2Hear group compared to the booklet (control) group ($p < .001$). Results were similar to the previous RCT, but with an even larger clinical effect size ($d = 1.38$). Self-efficacy for hearing aids improved for all the subscales of the MARS-HA between assessment and fitting appointment in the C2Hear group (global score, $p < .001$, $d = 0.83$). Strikingly, there was no significant pre- or post-change in self-efficacy for any of the MARS-HA subscales for the control booklet group. There was also a significant between-group improvement in the Ida self-efficacy measure when looking at those who were not at ceiling at the outset ($p = .002$, $d = 1.4$). There was a borderline significant between-group difference in readiness ($p = .053$). These results support the use of C2Hear at the assessment appointment to improve knowledge and self-efficacy in first-time hearing aid users, both of which are integral to self-management of hearing loss.

Early delivery of C2Hear at the hearing assessment appointment improves knowledge of hearing aids and communication, and effectively primes patients so they have greater self-efficacy and are better prepared to use hearing aids when they receive them. Further research is needed to identify what the longer-term effects of early delivery are.

Involving Others in Knowledge Exchange: Partners

As interpersonal communication is a two-way process, hearing loss affects not only the person who has the hearing loss, but also those with whom they communicate (Barker et al., 2017). Those who communicate regularly with people who have hearing loss, such as their family and friends (e.g., communication partners, or CPs), are an intrinsic component in managing hearing loss and improving communication. For example, involving CPs in the rehabilitation process results in greater inclusivity and has been shown to be highly beneficial (Carson, 2016; Knudsen, Öberg, Nielsen, Naylor, & Kramer, 2010; Scarinci, Worrall, & Hickson, 2008; Schulz et al., 2017; Stark & Hickson, 2004). Furthermore, many CPs experience some degree of third-party disability (Kamil & Lin, 2015; Scarinci, Worrall, & Hickson, 2012). This may

be alleviated through shared understanding of the challenges associated with their partners' hearing loss (including its management) and communication, leading to more aligned coping strategies and better outcomes for both parties (Barker et al., 2017).

Focus groups held with family members following the initial development of the RLOs indicated that they would benefit from RLOs that were specifically tailored to them (Ferguson, Brandreth, Brassington, & Wharrad, 2016). To address this, one of the original RLOs, Communication Tactics, was redesigned with wording that was more generic for "others," such as family members and the general public, rather than for hearing aid users. In addition, the RLO was developed for a mobile technology platform (mRLO) to increase the capacity of the user for greater interaction (Ferguson, Maidment, Henshaw, & Gomez, 2019). The RLO was divided into 10 separate 'pages' rather than being one continuous RLO, and included activities, such as identifying where best to sit in a restaurant, and identifying behaviors that either enhanced or reduced communication ability. (*Editors' note: For additional information on communication partners, the reader is referred to Chapter 19 of this text.*)

Quality, usability, relevance, and impact of the repurposed mRLO were examined for individuals (Hearing aid users or CPs) and for joint working (hearing aid users and CPs) using mixed methods (Ferguson, Maidment, Henshaw, & Gomez, 2019); quantitatively via the User-Mobile Application Rating Scale (uMARS) (Stoyanov, Hides, Kavanagh, & Wilson, 2016), and qualitatively using video-recorded Think Aloud techniques (Olmsted-Hawala, Murphy, Hawala, & Ashenfelter, 2010). The Think Aloud recordings were analyzed using inductive thematic analysis using Behavioural Observation Research Interactive Software (BORIS) video coding software (Friard & Gamba, 2016), and generated codes were interpreted according to the COM-B system of health behavior (Michie et al., 2011).

Nine dyads comprising hearing aid users and their CPs participated in the research, where CPs were primarily the spouse, with the exception of one daughter. Both hearing aid users and CPs preferred the enhanced mRLO over that used in the original RCT. Quantitative results from the uMARS (range 0 to 5, where 5 is the highest score) showed users rated the mRLO as moderately high quality (overall: hearing aid users = 3.7, CPs = 3.9), slightly higher than the score on the original RLO by hearing aid users (Mean = 3.6). Although there was relatively little difference in the perceived quality of the mRLO and the origi-

nal RLO, both the hearing aid users and CPs rated engagement with, and aesthetics of, the mRLO more highly (engagement, hearing aid users = 3.8, CPs = 3.8; aesthetics, hearing aid users = 3.9, CPs = 3.3) than the original RLO rated by hearing aid users (engagement = 3.1; aesthetics = 3.0). Therefore, it seems that the revised mRLO did indeed address some of the negative aspects reported about the original RLOs. Interestingly, the mean perceived impact of the mRLO to change behavior was higher for the CPs (Mean = 4.1) than the hearing aid users (Mean = 3.5), suggesting that the CPs valued the benefits arising from the mRLO.

Comments from the Think Aloud analysis provided insights into why the participants preferred the mRLO to the original RLO.

“It’s more interactive . . . ”

“This one [mRLO] is more, softer and perhaps persuasive, rather than telling you what you are like, it’s persuading you how to get better.”

“It’s split up into smaller chunks, nice little bits of information, rather than one big long video.”

The Think Aloud analysis of the videos of the dyads indicated that the greatest impact was for joint working conditions (i.e., when the hearing aid user and their CP were working together with the mRLO). Key results were that the mRLO:

- was aligned to CPs’ and hearing aid users’ experiences;
- increased CPs’ knowledge of hearing loss, hearing aids and communication;
- provoked novel discussions about communication difficulties; and
- prompted CPs to change their behaviors to aid communication.

The mRLO enabled users to become jointly aware of inhibiting factors that prevented optimal communication, such as overly high expectations of the CP in what hearing aids can do, diffusion of responsibility where at least one person in a pair does not engage with the communication process, and misconceptions as to why the person with hearing loss does not appear to hear (e.g., perceived to be ignoring the other). The most severe inhibitor was mindlessness, where the CP forgets how to cope with their partner’s hearing loss. The mRLO also highlighted

facilitating factors, which could help counteract these issues. These included promotion of positive intentions and behavior change, and an increase in mindfulness of the communication difficulties that can occur. This improved awareness and mindfulness of behaviors that inhibited communication resulted in joint responsibility for improving communication between CPs and PHL, and encouraged users (primarily CPs) to adopt facilitative behaviors to improve communication.

“Now, having seen this video, that’s what we shall look for [a quiet restaurant table with good lighting].”—CP

In summary, the greatest impact from the Communication Tactics mRLO was shown for joint working, where hearing aid users and their CPs had the opportunity to watch and discuss the issues that arose from the mRLO that were specific to them. This suggests greater aligned coping shown previously to result in better outcomes for both the hearing aid user and their CP (Barker et al., 2017).

Involving Others in Knowledge Exchange: Professionals

In addition to CPs, nonaudiologic health and social care professionals also interact with people who have hearing loss and would benefit from good quality information and support for hearing loss, hearing aids, and communication. A number of preliminary studies have been conducted to evaluate the benefits of using the RLOs to train nonaudiologic professionals on hearing-related matters to improve their knowledge base. Two studies examined whether C2Hear (either in DVD or online format) would improve hearing aid and communication knowledge in residential home care workers. In one study, care home workers were allocated dedicated time to watch the original C2Hear RLOs, and the HACK and PHAST were completed 2 weeks later (Rocks & Ferguson, 2013). There was a highly significant increase in knowledge of hearing aids and communication ($p < .001$), and a significant improvement in hearing aid handling skills ($p < .001$), with large effect sizes (3.6 and 3.4, respectively) (Figure 15–3). In another study, semistructured interviews indicated that the carehome staff found that C2Hear reinforced some of the information they already knew (e.g., correct insertion of hearing aids) (Finegold, 2014). Additionally, other information was

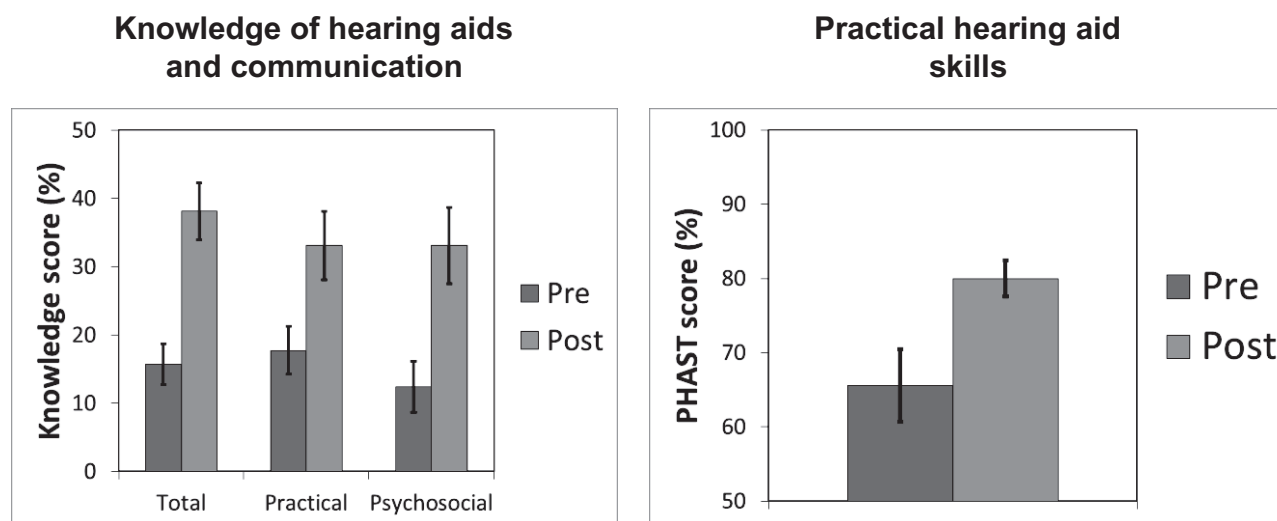


Figure 15-3. Knowledge of hearing aids and communication, and practical hearing aid handling skills scores for residential care home workers, before and after using C2Hear. Error bars are 95% confidence intervals.

highlighted that the care workers were not previously aware of (e.g., troubleshooting when hearing aids did not work, communication tactics). Typical comments following the use of C2Hear were:

“This should be made mandatory at induction with annual refresher sessions.”

“Awareness of hearing aids has definitely gone up with all of us.”

Similar results were seen in another mixed methods study (McShea, Corrigan, Greatrex, & Ferguson, 2016) of support staff in a residential home for deaf-blind residents. Knowledge and confidence in hearing aids were significantly greater in those staff who had used C2Hear compared to those who had not ($p < .001$). Staff reported that the C2Hear training was very useful, and improved their efficiency and effectiveness when dealing with those with hearing loss. For the specific impact of the training, key themes arising from semistructured interviews were the workers’ increased awareness of hearing aids and how to address problems that arose with them, alongside feeling more encouraged and empowered to engage in opportunities to help residents with their hearing aids and communication. The workers said:

“Hearing aids often drop off the radar. We shouldn’t be waiting for a crisis.”

“This project has changed the things I’m doing and saying on a daily basis.”

A study of 95 nursing students who watched C2Hear showed highly significant improvements in knowledge of hearing aids and communication ($p < .001$), as well as improved confidence to communicate with adults in their care who had hearing loss ($p < .001$) (Hussein, 2017). Importantly, the nursing students reported that this would change their future practice as nurses dealing with older adults.

Aligned with the views of CPs, one of the recurrent themes identified across all groups in these studies was the desire for RLOs to be tailored for their own workplace situations. Future research aims to expand the mRLO concept to flexible, individualized tools for CPs and a range of health and social care professionals to meet their educational and training requirements around hearing loss, hearing aids, and communication.

Finally, a study looked at the views that audiologists had C2Hear (Hajat, 2016). Twenty audiologists were asked to rate the statement “I am confident that the information given is remembered and acted upon by my hearing aid patients” on a 5-point Likert scale (*strongly agree* to *strongly disagree*). The audiologists were then asked to watch the C2Hear RLOs across a week and rate the statement again, but this time to consider that C2Hear was to be offered as part of standard care. There was a significant pre/post improvement in confidence ($p < .001$) (Figure 15-4), suggesting that audiologists recognized the value of C2Hear for their own clinic clientele. The majority (85%) stated that they would consider using C2Hear in their own clinics in future.

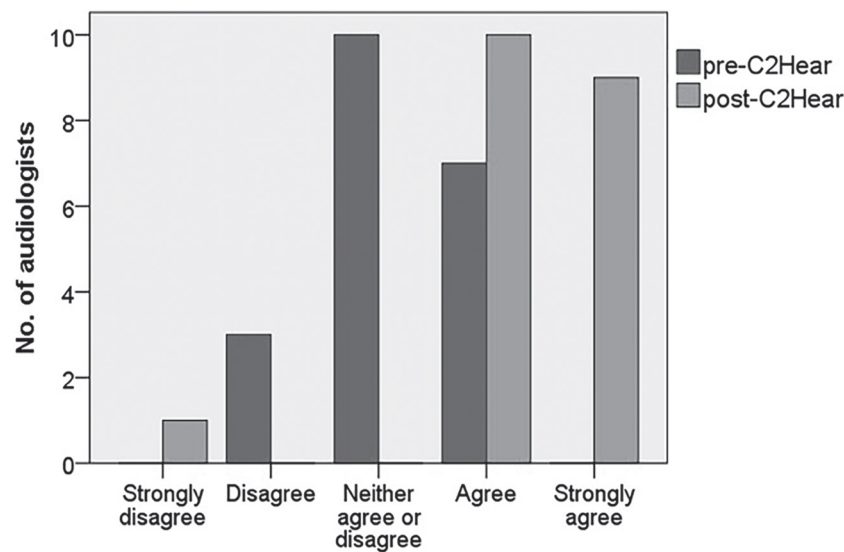


Figure 15-4. The response of audiologists to the statement “I am confident that the information given is remembered and acted upon by my hearing aid patients,” before and after using C2Hear.

Individualization: Tailoring Information to Meet Individual Needs

The functionality of mobile technologies (e.g., smartphones, tablets) provides a platform that has been shown in other health care domains to increase user-interactivity to enhance learning potential (Bennett & Glasgow, 2009; Zhang, Zhou, Briggs, & Nunamaker, 2006). Furthermore, mobile technologies can enable tailoring of an intervention to meet individual needs, so that patients can better manage their condition (Murray, Burns, See, Lai, & Nazareth, 2005). Over recent years, there has been a year-over-year increase in smartphone ownership in 55+-year-olds (2012 = 19%, 2017 = 71%) (Deloitte, 2017), and the proportion of 55- to 64-year-olds accessing the internet on the go via a smartphone has also increased exponentially (2012 = 29%, 2017 = 64%) (Office for National Statistics, 2017). (*Editors' note:* For additional information on e-health, the reader is referred to Chapter 23 of this text.)

The previously described C2Hear research with patients used RLOs that were one-size-fits-all. Their duration was between 4 and 11 minutes long (Mean = 9.4 minutes), which 5 to 6 years ago was generally acceptable. However, today there is an expectation that videos, particularly those that appear online and are accessed via smartphones, will be short and concise. Although the main C2Hear RCT resulted in posi-

tive outcomes, a criticism by some participants was that the RLOs were too long. Furthermore, their development for a DVD platform, while appropriate for the intended first-time hearing aid user population at that time, resulted in lack of specific activities (other than the quiz), with limited levels of interactivity.

The increasing digital literacy in the older adults suggests the time is right to develop and evaluate mobile-enabled RLOs (or mRLOs) to enable greater individualization and interactivity. To this end, the C2Hear RLOs have been redeveloped from their one-size-fits-all approach into a theoretically-driven individualized, interactive intervention for use with mobile technologies, known as m2Hear (Ferguson, Maidment, Henshaw, & Gomez, 2019).

The development of m2Hear was informed by the COM-B system (Michie et al., 2014). Specifically, the TDF can be used to identify the constructs, or “active ingredients” that are necessary to bring about behavior change, thereby informing the design and implementation of an intervention. Prior to this redevelopment, the C2Hear RLOs were analyzed using the TDF to establish which aspects were present in each RLO. The results are shown in Figure 15-5. The radar plots show the three components of the COM (capability, opportunity, and motivation), and the TDF domains. The black segments show the proportion of time that the RLO focused on each particular TDF domain. The active ingredients differ for different RLOs. The Hearing Aid Insertion RLO that

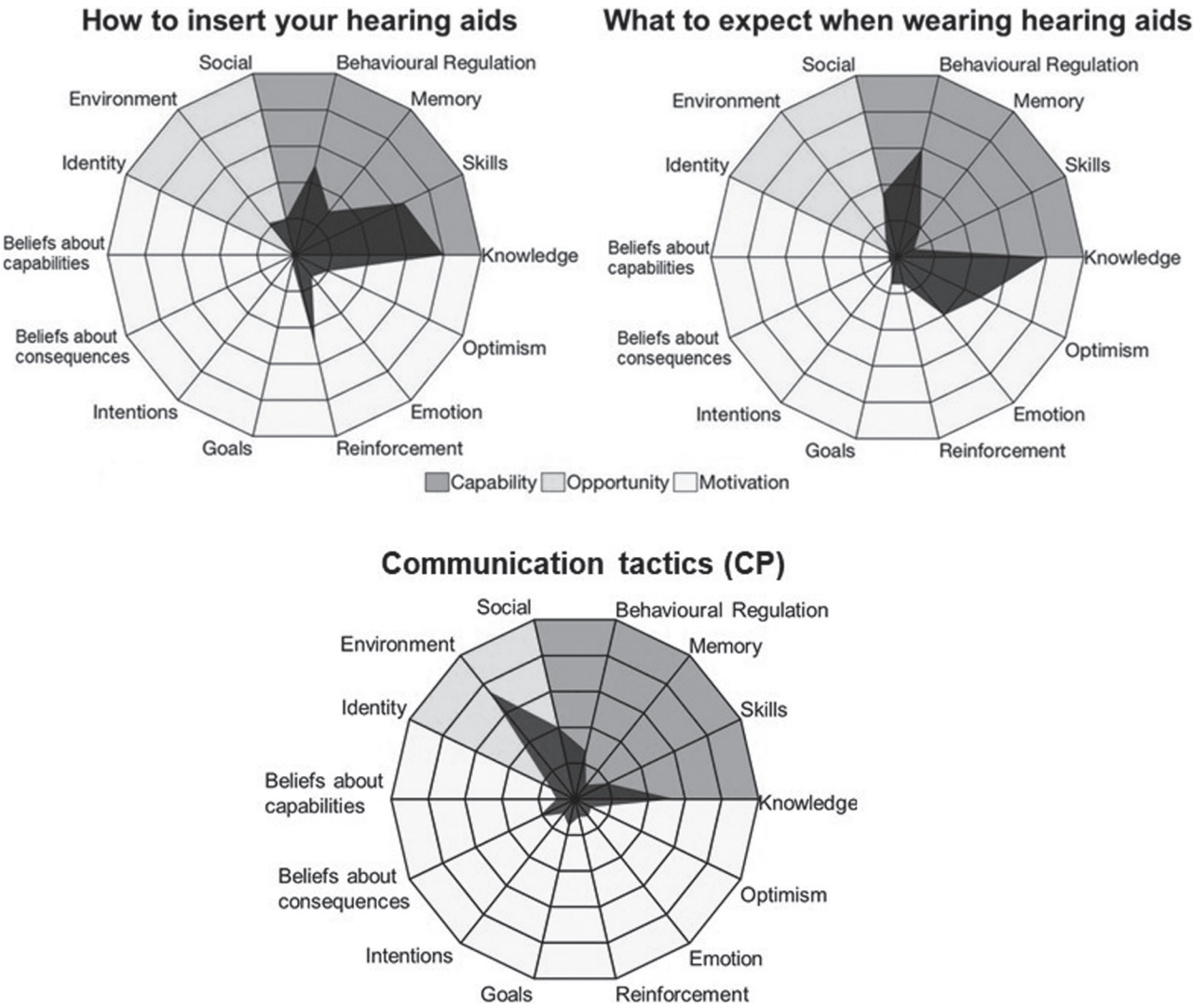


Figure 15-5. Radar plots demonstrating the “active ingredients” of three RLOs identified by the Theoretical Domains Framework and COM-B model. “How to insert your hearing aids” and “What to expect when wearing hearing aids” were the original C2Hear RLOs developed for hearing aid users. The “Communication tactics” was developed for communication partners.

discusses mainly practical issues focuses mostly on capability, particularly knowledge and skills. There was also behavioral regulation and reinforcement (motivation), which related to the feedback offered by the RLO; for example, indicating when the hearing aid was correctly or incorrectly inserted. The Expectations of Hearing Aids RLO that is more focused on psychosocial aspects of hearing aids still has a significant amount on knowledge, as this is a key function of all the RLOs. However, there was a greater reference to motivational aspects, in particular the TDF domains for social, emotion, and optimism. The CP mRLO showed a different pattern altogether. Again, there is some knowledge, but the overriding TDF domain was the environmental aspect (e.g., where to sit in restaurants) with some behavior regulation, both of which were highlighted in the CP Think Aloud study, reflecting the main purpose of the mRLO.

Having developed a better understanding of the underlying behavior change aspects of each RLO, all the C2Hear RLOs were repurposed by dividing them into short learning segments that were approximately 30 to 60 seconds in duration. This process was theoretically grounded, whereby each segment was classified according to the TDF, linking to a specific determinant of behavior. The end result was a series of 42 short (1 to 2 minute) mRLOs, each identified by a set of domains that are necessary to bring about the target behavior (e.g., hearing aid use).

To complement this theoretical approach, an ecological method was adopted to provide labels for each m2Hear RLO to embed the end-user perspective. Fifteen hearing aid users were asked to take part in a Think Aloud analysis. They were asked to view and simultaneously talk about the mRLOs in terms of what was important, relevant and valuable to them for their hearing and communication experiences and needs. All sessions were designed to encourage participants to reflect upon and describe in their own words their views on the mRLOs with minimal guidance from the researcher, who simply guided the task. The Think Aloud sessions were audio recorded and transcribed verbatim. These data were then used to label each mRLO with a specific question. For example, "What can I expect when wearing hearing aids for the first time?" and "What can I change to help me improve conversations?" This provides greater individualization, which enables hearing aid users to tap into the specific information that they need more readily.

In addition, the majority of participants commented that the information that users would need to know immediately following fitting (e.g., how to change the hearing aid battery) should be separated

from that needed after hearing aids had been used for some time (e.g., how to clean and maintain hearing aids). Therefore, the mRLOs were clustered into the following five areas: (a) using hearing aids; (b) getting used to hearing aids; (c) looking after hearing aids; (d) communication with others; and (e) using the telephone and other devices. Together with the questions generated for each mRLO, these five superordinate areas act as an interface to enable individualized tailoring of information. Finally, the mobile platform provided opportunities for greater interactivity and a series of activities were also included, such as where to sit in a restaurant and how to clean earmolds/open-fit tubes. The multiple-choice quizzes remained.

Following the development of the mRLOs, the m2Hear online platform was constructed using background programming to suit any mobile device (e.g., smartphone, tablet, laptop). The development of the platform was iterative, informing any content or usability modifications. As with the original RLOs, this phase employed a user-centered design, where a panel of hearing aid users from the Think Aloud evaluation ($n = 4$), as well as members of the project-specific patient and public involvement panel, formatively reviewed the intervention both in the lab and independently from home. These sessions were designed to assess user perceptions and interactions with the platform, as well as identify any potential problems.

Following 10 to 12 weeks of independent use, m2Hear was assessed for accessibility, usability, acceptability, and adherence, using both patient-reported outcome measures ($n = 50$ participants) and semistructured interviews ($n = 15$). Preliminary findings indicate that patients like the short and concise nature of the mRLOs, as they provide key take-home messages that users can apply in their everyday lives. In addition, participants have reported that they particularly appreciate the opportunity to return and revisit the mRLOs whenever they are needed, and find the questions valuable in identifying which mRLOs to watch. Additional interactivities that we have incorporated into the platform, such as where to sit in a restaurant, are also reported as highly useful, often acting as an *aide-memoire* to the information provided in the mRLOs.

From Research Into Practice

The ultimate dissemination for this research is to make the RLOs widely available to whoever wants to use them. To maximize accessibility, C2Hear Online

was launched in November 2015 on YouTube (<http://www.youtube.com/C2HearOnline>) and is now freely available. To date, there have been more than 200,000 unique views from more than 50 countries around the world (38% from North America). C2Hear Online is used in audiology clinics across the United Kingdom, and is included in national U.K. guidelines (BSA, 2016; National Institute for Health and Care Excellence, 2018). There are continuing developments to increase ease of use for patients and audiologists, such as the stand alone website (<http://c2hearonline.com>) that compensates for some of the disadvantages of YouTube. Finally, C2Hear has also been translated into American English.

Summary

C2Hear has evolved from its early development, with patients and audiologists at its core, to an individualized, interactive mHealth intervention for use with mobile technologies. These developments are not only for hearing aid users but also include others who would benefit from better knowledge of hearing loss, hearing aids, and communication, such as communication partners, health and social care professionals, and the wider public. Benefits extend to more than just improved knowledge, and include greater hearing aid practical handling skills and hearing aid use, self-efficacy, confidence, and aligned coping. This research extends beyond the lab, as C2Hear now has a global reach. The longer-term goal is to develop a self-management system that incorporates latest developments in technology, whilst taking a patient-centered approach that builds on the latest in health behavior psychology.

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16

FAMILY-CENTERED CARE IN ADULT AUDIOLOGIC REHABILITATION

Nerina Scarinci

Introduction to Family-Centered Care in Adult Audiology Rehabilitation

Family-centered care (FCC) has long been promoted as the gold standard of delivering health care services in pediatric audiology practice (Moeller, Carr, Seaver, Stredler-Brown, & Holzinger, 2013). In recent years, however, there has been growing recognition of the potential application of FCC in adult audiology rehabilitation. FCC is an extension of patient-centered care (PCC), a model of care that has recently received increasing attention in the adult audiology rehabilitation literature (Grenness, Hickson, Laplante-Lévesque, & Davidson, 2014a, 2014b; Laplante-Lévesque, Hickson, & Worrall, 2014). PCC focuses on the importance of the therapeutic relationship (Rogers, 1965) and the active engagement of patients in their own health care (Mead & Bower, 2000) through services that are sensitive to, and respectful of, the needs and preferences of patients (Committee on Quality of Health Care in America, 2001). Nevertheless, the definitions and descriptions of PCC do not explicitly highlight the central role of family in the health care alliance, and it is for this reason that FCC is the focus of this chapter and is the approach preferred for working with adults with hearing impairment and their families.

FCC emerged from family systems theory in psychology, which acknowledges the importance of viewing an individual's behavior within the context of their family network (Goldenberg & Goldenberg, 2013). Specifically, in family systems theory, family relationships and interactions between family members are viewed as important factors in psychological health, with studies demonstrating the significant impact of family relationships on biological systems, behaviors, and well-being (Martire & Schulz, 2007). Therefore, family systems theory proposes that interventions that involve both parties will improve the quality of family relationships and subsequently improve health outcomes. Research in the field of psychology has confirmed this hypothesis, with interventions involving family members being found to be more effective than interventions with the patient alone (Emmelkamp & DeLange, 1983; Mehta, 1990). Acitelli and Badr (2005) proposed that when older couples take a relationship perspective when faced with a disability, they are able to acknowledge the impact of the disability on their relationship, rather than just seeing it as something affecting only one person. Therefore, they are more likely to work together in adapting to and coping with the disability. As such, in FCC, the focus of the service is on the entire family, rather than only the person with the health condition,

and therefore the entire family is considered to be the “client,” or recipient of services (Hamilton, Roach, & Riley, 2003).

Although FCC has primarily been used within pediatrics, it is seen as applicable to all patient groups in any health care setting (Hughes, Bamford, & May, 2008), including adults with hearing impairment and their family members (Scarinci, Meyer, Ekberg, & Hickson, 2013). This chapter provides an overview of the principles and practices of FCC, the rationale for its use in adult audiologic rehabilitation, and the outcomes of FCC in adult audiologic rehabilitation. It is important to note that although the term *family* is used most frequently in FCC, the definition of family should be considered more broadly, with family defined as “two or more individuals who depend on one another for emotional, physical, and economic support” (Hanson, 2005, p. 7). This is especially important given other terms that are used frequently in the literature, including *significant others* and *communication partners*. For example, although the majority of literature on family members’ involvement in audiologic rehabilitation has focused on spouses of older adults with hearing impairment, Preminger, Montano, and Tjørnhøj-Thomsen (2015) found that adult children are also affected by hearing impairment and, therefore, could also be involved in audiologic rehabilitation programs.

The focus on family in this chapter aligns with other chapters in this book that highlight the importance of including family members or communication partners in the audiologic rehabilitation process, including Chapter 19 (Incorporating Communication Partners Into the AR Process) and Chapter 20 (Communication Partnership Therapy in Audiologic Rehabilitation), which expand upon the specific role of family members in adult audiologic rehabilitation.

Principles and Practices of Family-Centered Care

Although definitions and descriptions of FCC vary, one of the most recent conceptualizations came from a systematic review of the definitions used in the education, social work, and health literature (Epley, Summers, & Turnbull, 2010). This systematic review built on an earlier review of FCC definitions in the literature (Allen & Petr, 1996). The five elements of FCC identified by Epley et al. (2010) were: (a) the family as the unit of attention; (b) family choice; (c) family strengths; (d) family-professional relationship; and (e) individualized family services. Each of these

concepts will now be described within the context of adult audiologic rehabilitation.

Family as the Unit of Attention

When applying a family-centered approach to adult audiologic rehabilitation, the audiologist should consider the needs of not only the individual with hearing impairment, but also those of their family member. When viewing the entire family as the unit of attention, both the patient and their family become the focus throughout the rehabilitation journey, including assessment, planning, and intervention (Allen & Petr, 1996). This shift in focus away from just the patient with hearing impairment highlights the importance of acknowledging the needs of the family as well and considering the family unit as the client in rehabilitation, with the aim of improving family quality of life and well-being, and measuring the outcomes of this rehabilitation (Epley et al., 2010).

Family Choice

The second component of FCC identified by Allen and Petr (1996), and confirmed by Epley et al. (2010), is family choice. This means organizing and providing services according to the wishes and choices of the entire family (Epley et al., 2010). First and foremost, choice means giving the patient the option of defining their family. As discussed earlier, while family in adult audiologic rehabilitation has been traditionally associated with spouses or adult children, it is important to note that, alternatively, a neighbor, good friend, or colleague may be defined as family by the individual with hearing impairment. Thus, when providing FCC, it is important to take the time to ask the patient who they consider their family to be. Family choice ensures that both the patient and family are well placed to take a central role in the direction of their health care and an active role in decision making. It allows both the patient and family member to be involved in the main activities of care, including information sharing, setting goals, and making decisions about service delivery and intervention options (Epley et al., 2010; Park et al., 2018).

Family Strengths

In FCC, the clinician is encouraged to move away from a deficit model and take a strengths-based approach to audiologic rehabilitation. This means in the assessment, treatment, and counseling for

hearing impairment, the audiologist acknowledges, incorporates, and builds upon the strengths of both the patient and the family (Allen & Petr, 1996). Importantly, this requires family members to become partners in the treatment process and to contribute their strengths, abilities, and resources (Allen & Petr, 1996). This strengths-based approach plays a critical role in empowering all involved to manage the hearing impairment and resultant communication difficulties (Epley et al., 2010), including the provision of education and training (Park et al., 2018).

Family-Professional Relationship

Just like in PCC, building an effective therapeutic relationship is crucial to the delivery of FCC; however, this therapeutic relationship extends beyond the individual with hearing impairment. Some of the key concepts behind the development of the family-professional relationship are equality, mutuality, and teamwork (Epley et al., 2010). Importantly, in the development of an effective family-professional relationship, audiologists need to be cognizant of cultural responsiveness and respectful and sensitive to family needs and perspectives with honesty and active listening (Epley et al., 2010). Developing a trusting therapeutic relationship has been specifically emphasized in audiologic rehabilitation given the identified trust issues that patients have reported in audiologic service provision (Grenness et al., 2014b).

Individualized Family Services

Finally, when providing family-centered services, it is essential that audiologists provide services in assessment, goal setting, and intervention that match the unique needs and resources of each family (Allen & Petr, 1996). These individualized family services include services from both audiologists and other patients and families, such as support groups (Epley et al., 2010).

The Rationale for Family-Centered Care

Now that the common principles and practices of FCC have been proposed, it is important to discuss how and why it can be applied to adult audiologic rehabilitation. The rationale for and implementation of FCC in this population is twofold: first, family members are also affected by hearing impairment, and could therefore also benefit from rehabilitation; second, the

inclusion of family members in adult audiologic rehabilitation equips them to support the rehabilitation journey of the individual with hearing impairment.

Addressing the Impact of Hearing Impairment on Family Members

Hearing impairment affects communication in not only the person with the loss but all those involved in the communication interaction. The impact of hearing impairment on others is commonly referred to as a third-party disability (World Health Organization [WHO], 2001). A third-party disability is defined as the disability experienced by family members as a result of their significant other's health condition (WHO, 2001).

Scarinci, Worrall, and Hickson (2008, 2009a, 2009b, 2012) initially explored third-party disability in spouses of older adults with hearing impairment through a series of mixed methods studies, and found that spouses experience a range of effects on their everyday lives as a result of their partners' hearing impairment (Scarinci et al., 2008), with the majority of spouses reporting some degree of third-party hearing disability, and communication difficulties between the couple being the central source of stress (Scarinci et al., 2012). More recently, there have been a number of reviews that have specifically examined the impact of hearing impairment on communication partners, highlighting a broad range of impacts (Kamil & Lin, 2015; Vas, Akeroyd, & Hall, 2017). Vas et al. (2017) summarized these impacts across the domains of auditory (i.e., listening, communication, and speaking), social (i.e., relationships, isolation, social life, occupation, and interventions), and self (i.e., effort and fatigue, emotions, identity, and stigma).

Auditory Impacts

As a result of hearing impairment, family members have reported a range of auditory impacts, including tension over the need to repeat (Stephens, France, & Lormore, 1995; Scarinci et al., 2008), and coping with the loud volume of the television (Kelly & Atcherson, 2011; Lormore & Stephens, 1994; Scarinci et al., 2008). Family members have also reported third-party disability as a result of having to raise the volume of their voice when communicating (Govender, Maistry, Soomar, & Paken, 2014; Scarinci et al., 2008, 2012).

Social Impacts

Relationship difficulties for family members can range from reduced relationship satisfaction (Anderson

& Noble, 2005; Hallam, Ashton, Sherbourne, & Gai-ley, 2008; Scarinci et al., 2008; Scarinci et al., 2009a) and marriage difficulties (Knutson, Johnson, & Murray, 2006; Preminger & Meeks, 2010a; Scarinci et al., 2009b) to reduced intimacy (Hallam et al., 2008) and interpersonal difficulties (Hallam et al., 2008). These relationship difficulties may be linked to greater communication hardship within the marriage (Kelly & Atcherson, 2011; Scarinci et al., 2008; Scarinci et al., 2009a; Stephens et al., 1995).

Self Impacts

Family members have reported increased feelings of frustration (Hallam et al., 2008; Stephens et al., 1995), stress (Knutson et al., 2006), loneliness (Knutson et al., 2006), emotional reactions (Preminger & Meeks, 2012; Scarinci et al., 2008; Scarinci et al., 2009a), worry (Scarinci et al., 2009a), and anxiety (Knutson et al., 2006). These emotional reactions to the hearing impairment may not only be linked to increased communication difficulties, but also increased dependence and caregiving issues (Knussen et al., 2008; Stephens et al., 1995). For example, family members have reported experiencing burden over increased roles and responsibilities in the family including having to answer the telephone on behalf of the person with hearing impairment (Scarinci et al., 2008; Vas et al., 2017) and having to act as an interpreter during group conversation (Morgan-Jones, 1998; Scarinci et al., 2008).

As a result of the adult's hearing impairment, family members have also reported poor physical, mental, and social health (Wallhagen, Strawbridge, Shema, & Kaplan, 2004), reduced quality of life (Kelly & Atcherson, 2011), low mood (Preminger & Meeks, 2010a), distress (Tolson, Swan, & Knussen, 2002), and negative impacts on their social life (Kelly & Atcherson, 2011; Knutson et al., 2006; Lormore & Stephens, 1994; Preminger & Meeks, 2012; Scarinci et al., 2008; Scarinci et al., 2009a, 2009b). These impacts on the family member may result in family withdrawing from communication interactions with the individual with hearing impairment, leading to increased hearing disability and increased levels of frustration and isolation within the family (Scarinci et al., 2012).

Family Member Support for the Patient with Hearing Impairment

Involving family members in audiologic rehabilitation through a family-centered model of health care also has the advantage of empowering participants to provide support for the individual with hearing

impairment. This support can be provided from initial contact through to assessment and rehabilitation and counseling. Support from family is in no way meant to devalue the patient themselves nor take away their autonomy. Rather, FCC values both the patient and their family member equally; although family member support is strongly advocated, audiologists must ensure that nothing is taken away from the patient.

Indeed, some patients may indicate a clear preference for making health care decisions independently, while others may rely on the support and advice from others. A recent population-based survey conducted in the United States reinforced the heterogeneity of older adults' health care preferences, with participants indicating a wide range of preferences regarding how they manage their health care (Wolff & Boyd, 2015). Interestingly, the survey showed that while approximately two-thirds of older adults (69.4%) prefer to self-manage their health care, one-third indicated a preference for comanaging (19.6%) or delegating health care decisions (11%) to doctors and family or close friends (Wolff & Boyd, 2015). This study concluded that the implementation of FCC requires careful consideration and respect for individual decision-making preferences and the broader social and family context in which older adults manage their health (Wolff & Boyd, 2015).

Research has indicated that families play a key role in the uptake and successful use of hearing aids (Knudsen et al., 2010; Meyer & Hickson, 2012; Meyer, Hickson, Lovelock, Lampert, & Khan, 2014; Hickson, Meyer, Lovelock, Lampert, & Khan, 2014). Specifically, the decision to seek help for hearing impairment is influenced not only by the attitude of the individual, but also by the beliefs and support of their family and friends (Meyer & Hickson, 2012; Meyer et al., 2014). Further, family member input influences the rehabilitation decisions of patients (Laplante-Lévesque, Hickson, & Worrall, 2010a), and the positive support of family members has been shown to be strongly associated with successful hearing aid outcomes, with successful hearing aid users being more likely to think their family had a positive attitude towards their hearing aids (Hickson et al., 2014).

Implementation of Family-Centered Audiologic Rehabilitation

How Family Centered Is Current Practice?

Despite a clear rationale for FCC in adult audiologic rehabilitation, its implementation to date remains

inconsistent. For example, in a series of video observation studies exploring the implementation of FCC for adults with hearing impairment in Australia, family member attendance and the active engagement of family members was observed to be low (Ekberg, Meyer, Scarinci, Grenness, & Hickson, 2015; Grenness, Hickson, Laplante-Lévesque, Meyer, & Davidson, 2015a, 2015b). Specifically, family members were observed to be present in only 27% of appointments (Grenness et al., 2015a, 2015b). A qualitative study of audiologists verified this low attendance rate, with audiologists reporting that family member attendance typically ranged from 10% to 50%, with an overall estimate of 30% of appointments having a family member present (Meyer, Scarinci, Ryan, & Hickson, 2015).

Further analysis of the video observations showed that when present in appointments, family members only contributed to 13% of the total utterances in the appointments (Ekberg et al., 2015), with audiologists tending to address their questions only to the individual with the hearing impairment rather than also encouraging input from family members. This is despite family members demonstrating a desire to contribute to the discussion by answering questions on behalf of the individual with hearing impairment or directly asking the audiologist questions (Ekberg et al., 2015).

Despite this low rate of attendance and active contribution, audiologists have reported a favorable attitude towards FCC in adult audiology rehabilitation (Meyer et al., 2015). Specifically, audiologists have acknowledged the value of a true partnership with families such that a shared understanding and responsibility for managing the hearing impairment could be established and families can subsequently play an active role in the rehabilitation process (Meyer et al., 2015).

How Can Family-Centered Care Be Implemented in Adult Audiology Rehabilitation?

The well-acknowledged impacts of hearing impairment on family members highlight the importance of including them in audiology rehabilitation and emphasize the appropriateness of FCC in this population. In practice, this involves acknowledging and identifying the communication difficulties experienced by both the patient and their family and using these findings to inform goal setting and outcome measurement for both parties. Audiologists who work collaboratively with patients and families can help increase awareness of the impact

of the hearing impairment on their everyday lives and encourage both parties to view the disability as a family problem. This may encourage all involved to see themselves as a team, increasing the likelihood that they will engage in joint problem solving and employ active coping strategies and rehabilitation options (Scarinci et al., 2013). Once engaged in audiology care, audiologists have identified a range of possible family member roles in treatment, including: (a) having input into the assessment, decision making, and goal setting process; (b) receiving information provided during the consultation; (c) supporting the patient's rehabilitation goals; and (d) providing emotional support (Meyer et al., 2015).

Family Member Input into the Assessment, Decision Making, and Goal Setting Process

As active members of the family-centered rehabilitation team, family members can provide valuable information about the patient's hearing impairment, and the impact of the hearing impairment on the family as a unit. This perspective provides a valuable addition to the patient's self-report of communication difficulties and allows a more thorough assessment of the everyday impacts of the hearing impairment on family communication and dynamics (Steinberg, 1993).

Indeed, acknowledging the impact of hearing impairment on both the patient and family may be key to encouraging patients to seek help in the first place, with awareness and understanding of the perceived burden of hearing impairment on family members being shown to be a significant motivator to pursue hearing evaluation (Schulz et al., 2016). Therefore, seeking the perspective of family members on the impact hearing impairment has on their life may raise awareness and, therefore, potentially encourage them to seek rehabilitative options (Meyer et al., 2015; Schulz et al., 2016). Tools such as the Goal-Sharing for Partners Strategy (GPS) (Preminger & Lind, 2012) can be used to facilitate this process, with this and other approaches discussed further in Chapter 19.

Family members can also assist the individual in making decisions throughout the rehabilitation journey. This is particularly important in adult audiology rehabilitation, where complex decisions often need to be made about hearing aids and hearing technologies. Interestingly, there is mounting evidence within the field of adult audiology rehabilitation that involving families in audiology appointments increases the likelihood that the patient will act. For example, in a recent study, Singh and Launer (2016) found that patients with hearing impairment were significantly more likely to obtain hearing aids in appointments

when family was present, suggesting that family may be influential about the need for hearing rehabilitation. Further, as family support is a key predictor of help-seeking and hearing aid success, the audiologist can work with both the patient and their family when discussing concerns and expectations about hearing aids, supporting a shared decision-making process (Meyer et al., 2014).

Finally, the contribution of family members to the assessment process also means that they can be involved in setting mutually beneficial goals for not only the person with the hearing impairment, but also for themselves. Family participation in goal setting is a notion supported by the principles of FCC. Given the family is the unit of attention in FCC, shared goal setting is a means of addressing the needs of individuals within the family and the family unit as a whole.

Receiving Information

In addition to contributing to the assessment and goal-setting process, family members also bring a second set of ears to appointments and can assist patients in absorbing and recalling information after the appointment (Ellingson, 2002; Jansen et al., 2010; Wolff & Roter, 2008). Audiologists have described a range of specific information that is important for families to process, including feedback about hearing test results, recommendations for intervention and/or hearing aid options, instructional information about hearing aid management and use, and the expected outcomes of rehabilitation (Dillon, 2012; Meyer et al., 2015).

Supporting the Patient's Rehabilitation Goals

Following the assessment and decision-making process, family members play an ongoing role in supporting the patient with hearing impairment in achieving his or her goals. In addition to hearing aids, communication strategies are an important component of audiologic rehabilitation. The involvement of family members has key advantages to the training and implementation of these skills (Meyer et al., 2015).

In the field of adult audiologic rehabilitation, communication strategies are typically described as verbal, nonverbal, and maladaptive. Verbal and nonverbal strategies, such as clarification requests, requests for repetition, lipreading, and positioning strategies aim to improve communication and reduce the negative effects of hearing loss. However, maladaptive strategies, such as pretending to hear, avoiding communication situations and guessing what others say may also be used and could detract from

or inhibit the process of communication (Demorest & Erdman, 1986). Interestingly, research has shown that the use of maladaptive communication strategies contributes to lower quality of life in adults with hearing impairment (Hallberg, Hallberg, & Kramer, 2009). In many situations, family members play a key role in enhancing the communicative success of adults with hearing impairment due to the frequency of their communication and their ability to increase confidence and satisfaction with communication exchanges (Heine, Erber, Osborn, & Browning, 2002). Once engaged in audiologic rehabilitation, family members can support the patient not only in managing their hearing devices, but also in using communication strategies to improve everyday communication (Hickson, Worrall, & Scarinci, 2007a, 2007b).

Given the known relationship between the use of maladaptive communication strategies and general well-being, FCC provides a unique opportunity for the audiologist to work with the individual with hearing impairment and their family to discuss the use of current strategies, determine which are working effectively, and jointly develop new approaches to implement in everyday communication. Patients and family members can reflect upon the relative success of communication strategies being used within the family, and both parties can be trained in techniques to prevent or resolve communication breakdowns (Heine et al., 2002). For example, families could be encouraged to use a slower rate speech, modify their speech in response to background noise, reduce distances to ensure face-to-face communication, and gain the person's attention before communicating with them (Dillon, 2012; Hickson et al., 2007a, 2007b; Preminger & Meeks, 2010b). Active involvement of family in the adjustment and training process is an important contributor to successful audiologic rehabilitation (Hallberg et al., 2009).

In particular, research has indicated that providing specific training and opportunities to practice strategies with communication partners within the clinical setting are more effective than simply asking family to alter their speech (Caissie et al., 2005). Further, it is important to note that the broader family unit, including adult children, grandchildren, friends, and neighbors, could benefit from this training to maximize communicative success across different environments (Montgomery, 1993).

Emotional Support

Finally, family members can provide emotional support for patients as they adjust to a diagnosis of hearing impairment and/or the fitting of hearing aids (Meyer

et al., 2015). The family's support and communicative competence can be assessed by evaluating not only the behaviors of the adult with hearing impairment, but also the level of support provided by their family, including assessment of beliefs and expectations, and knowledge of hearing impairment, hearing aids, and the audiologic rehabilitation process (Meyer et al., 2014; Hickson et al., 2014; Shadden, 1988). Further, as a mutual understanding of the impact of the hearing impairment is important to family member support, the audiologist can work with both the patient and the family to increase understanding (Manchaiah, Stephens, Zhao, & Kramer, 2012). This is of particular importance given the known presence of third-party disability (Scarinci et al., 2012).

Benefits of Family-Centered Care for Stakeholders

For many decades, the general health care literature has demonstrated positive outcomes of FCC for patients, families, and health care professionals alike. Most recently, a review of systematic reviews in the field of patient-centered care and FCC demonstrated that patients experience improved health care knowledge, enhanced skills to manage self-care, and increased satisfaction and quality of life (Park et al., 2018). The review also showed that families experience reduced stress, anxiety, and depression, and improved satisfaction and relationships with health care providers as a result. Health care providers also experienced positive effects of FCC, including improved job satisfaction and confidence, higher quality of care, and reduced stress and burnout (Park et al., 2018). Benefits of FCC further extend to clinicians and businesses. Use of FCC has been linked with greater job satisfaction (Lein & Wills, 2007) and lower rates of medical malpractice claims (Levinson, Roter, Mullooly, Dull, & Frankel, 1997).

Outcomes of Family-Centered Audiologic Rehabilitation for Individuals With Hearing Impairment, Family Members, and Clinicians

Patients whose families are involved in their hearing rehabilitation are more likely to be successful hearing aid users (Hickson et al., 2014), have increased confidence in managing hearing (Meyer et al., 2014), and have an emotional-support person who can assist

with correct use and care of assistive hearing devices (Manchaiah & Stephens, 2013). Patients with family members involved in their hearing care also experience greater satisfaction with medical care and adherence to treatment (Stewart et al., 2000).

Similarly, family members of people with hearing impairment experience benefits from FCC; primarily, decreased third-party disability (Habanec & Kelly-Campbell, 2015), and increased quality of relationships and life (Seniors Research Group, 1999). Kamil and Lin's (2015) systematic review summarized the positive outcomes of FCC on family members, involving the provision of hearing aids, cochlear implants, and communication training programs. These benefits included an improved ability to enjoy life (Hickson, Worrall & Scarinci, 2006; Kramer, Allessie, Dondorp, Zekveld, & Kapteyn, 2005; Mo, Lindbæk, & Harris, 2005), improved quality of life (Brooks, Hallam, & Mellor, 2001; Kennedy, Stephens, & Fitzmaurice, 2008; Kramer et al., 2005; Stark & Hickson, 2004), reduced stress (Preminger & Meeks, 2010b), reduced perception of hearing hassles (Tolson et al., 2002), decreased negative affect (Preminger & Meeks, 2010b), and improved social life (Kennedy et al., 2008).

Positive outcomes have also been reported across the areas of improved empathy and emotions towards the individual with hearing impairment (Mo et al., 2005), more favorable attitudes towards hearing aids and communication education (Kramer et al., 2005), and greater understanding of the hearing disability (Preminger & Meeks, 2010b; Stephens, Kerr, & Jones, 2004). Family members have also reported improved patience (Stephens et al., 2004), reduced third-party disability (Preminger & Meeks, 2010b), and reduced hearing hassles (Tolson et al., 2002).

Finally, audiologists have also reported clear benefits to the involvement of families in audiologic rehabilitation, with family participation in rehabilitation seen as a facilitator of good hearing aid management and improved patient satisfaction (Meyer et al., 2015). Studies have also shown that as a result of family member involvement, patients experience reduced communication difficulties and communicative burden (Kennedy et al., 2008), including one-on-one communication and group conversation (Brooks et al., 2001), improved ability to communicate in background noise (Stark & Hickson, 2004), reduced need for repetition (Stark & Hickson, 2004), reduced difficulties watching television and listening to the radio (Brooks et al., 2001; Stark & Hickson, 2004), and enhanced communication skills (Stephens et al., 2004).

Businesses can also benefit from FCC: A recent study indicated that FCC is positively correlated with hearing aid uptake, with more patients purchasing a

hearing aid if their family member also attended the appointment compared to patients who attended alone (Singh & Launer, 2016).

Clinical Implications of Family-Centered Care

Adopting an FCC approach in the clinic has a range of implications for patients, family members, and clinicians alike. Chief among these is the consideration of psychosocial factors contributing to the success of hearing rehabilitation. FCC enables clinicians to consider their clients holistically, rather than from the traditional site-of-lesion perspective that focuses on type and severity of the hearing impairment (Singh et al., 2016). In doing so, clinicians can increase the quality of information they receive from patients and families, creating rehabilitation opportunities with greater suitability and likelihood of compliance, and therefore greater likelihood of success.

Given the known advantages of involving families in audiologic rehabilitation, there has been increasing focus on clinical tools and recommendations for maximizing the implementation of FCC in clinical practice. Namely, the Ida Institute, an independent nonprofit organization focused on applying person-centered care in hearing rehabilitation, has long acknowledged the importance of working with family members to maximize rehabilitation outcomes. The Ida Institute has developed and disseminated a number of clinical tools to facilitate the involvement of family members in the rehabilitation process. These tools can be readily applied to audiologic rehabilitation including before, during, and after audiology appointments. Table 16–1 provides a summary of the relevant “Communication Partner” tools developed by the Ida Institute and freely available on its website (<http://www.idainstitute.com>).

In addition, the Phonak Expert Circle on Family-Centered Care recently published a position statement on applying FCC in adult audiologic rehabilitation (Singh et al., 2016). This position statement highlighted 10 recommendations for implementing FCC in the clinic, which are summarized in Table 16–2.

Key Challenges and Solutions for Implementing Family-Centered Care

Although FCC offers considerable opportunities for clinicians to engage with their clients, there can be some barriers to implementation. Therefore, while

the recommendations provided by the Phonak Expert Circle on Family-Centered Care provide a plethora of ideas for how FCC could be implemented in clinical practice, before making changes to their practice, audiologists should consider how these suggestions could be implemented in their particular practice. For example, one of the most salient barriers to the implementation of FCC is low family member appointment attendance, as previously discussed. Families will often need to be explicitly invited to attend appointments (Singh et al., 2016), provided this is consistent with the client’s wishes. When booking appointments, clients can be advised of the benefits of family member attendance, with reassurance that the purpose is not to change their independence. Furthermore, clinicians should be mindful that for many family members, additional involvement may initially be perceived as a burden, and efforts must be made to convey that FCC can assist the family as a whole to manage the rehabilitation process, potentially reducing third-party disability.

A second barrier to implementation can include the physical space in which appointments are held. Singh et al. (2016) recommend setting up the appointment environment as inclusively as possible, to provide a forum in which all participants feel they may contribute equally. This will often involve ensuring that patients and family members are seated side by side or, ideally, in a triangle with the clinician.

A third implementation barrier can include the structure of talk during the appointment. Ekberg et al. (2015) noted that families typically initiated their own contributions to appointments by asking their own questions or responding on behalf of the patient. Clinicians can ensure talk is more inclusive by advising clients early in the appointment that both patients and family members will be asked to contribute at different times, and consciously asking family members for their perspectives throughout. Additionally, clinicians should ensure goals are set for all families. The GPS (Preminger & Lind, 2012) mentioned earlier and the Client-Oriented Scale of Improvement (COSI) (Dillon, James, & Ginis, 1997) can be used for these aims. In using the tools, valuable information can be received by asking participants for their opinion or reflection on others’ goals. Producing joint goals provides an opportunity for decision making that is shared between patients, family members, and the clinician (Laplane-Lévesque et al., 2010a) and a framework to which clinicians can apply rehabilitation options: Solutions should be proposed in the context of client-generated issues and goals (Singh et al., 2016). These can further be revisited in later appointments to ensure all participants’ needs are being met.

Table 16–1. Communication Partner Tools Developed by the Ida Institute to Encourage Family Member Participation (<http://www.idainstitute.com>)

Tool	Purpose	Description
Communication Rings	Can be used to help the patient with hearing impairment to identify the most important communication partners in their life, and how their hearing impairment might be affecting their relationship with these people.	The Communication Rings tool consists of a series of circles. The patient uses the rings by listing the people who make up their social networks. The inner circle is used to represent the people who are most important to the patient, typically the people they communicate with the most. The patient is asked to fill in the rings from the inner circle out. The outer circles represent the people with whom the patient might only interact with on occasion and others who are not as important in their life.
Goal Sharing for Partners (GPS)	Can be used to engage family members in appointments so that the patient, family member, and audiologist can set attainable goals together and develop strategies to work towards these goals.	The GPS tool describes a step-by-step process for facilitating discussions with the patient with hearing impairment and their family member. The patient and family member are asked to identify: (a) where they have successful communication; (b) what problems they both experience as a result of the hearing impairment; and (c) in which situations they would like to improve their communication.
Partner Journey	Provides family members with a visual way of expressing their own experiences with hearing impairment	The Partner Journey is used as a visual stimulus to trace the emotional reactions and practical activities that family members travel through from the onset of their significant other's hearing impairment to their successful management. It can be used to promote discussion with family members about their own experiences with hearing impairment.

Table 16–2. Phonak Expert Circle Recommendations for Implementing Family-Centered Care

Recommendation	Clinical Strategy
1. Invite a family member along to audiologic appointments.	When making appointments you could say, "Our experience is that it is very helpful if you can bring a friend or a loved one along to the appointment. Who would that be?" If the patient asks why you are recommending this, you could say, "There is a lot to discuss and it helps to include family and friends in the process." Ensure this information is reinforced in any written information provided to patients about the appointment (e.g., confirmation letter, email contact).
2. Set up the physical environment for FCC.	Consider how you set up the clinical room, including furniture placement. This means ensuring family members are included in the consultation rather than being seated behind the patient or at the back of the room. An inclusive physical environment encourages all participants to provide equal contributions to the rehabilitation process.

continues

Table 16–2. *continued*

Recommendation	Clinical Strategy
3. Start the appointment by letting the patient and family member know that you will be seeking input from both of them.	At the beginning of the appointment, you could say to the patient, “We are going to do a lot today. For the next 10 minutes, I want to find out about your hearing and communication” and then say to the family member, “Then I want to find out about this from your perspective.”
4. Set joint hearing and communication goals with both the patient and family member.	During the consultation, discuss what both the patient and the family member would like to achieve and prioritize the goals together. You could use the Client Oriented Scale of Improvement (COSI) (Dillon et al., 1997), setting both patient and family member goals, or the Goal Sharing Partnership Strategy (GPS) (Preminger & Lind, 2012).
5. Present options for rehabilitation that address the needs of both the patient and the family member.	When discussing rehabilitation options, acknowledge the potential role of technological solutions, as well as communication strategies. For example, although a hearing aid alone can improve communication with a family member at the dinner table, a hearing aid and remote microphone, along with the communication strategy of gaining the other person’s attention and face-to-face communication, can be used when communicating across greater distances.
6. Aim for shared decision making with the patient, family, and clinicians as equal partners in the process.	Decision aids can be used to guide discussions about options for hearing rehabilitation (Laplante-Lévesque et al., 2010b). Decision aids provide patients and families with a simple summary of each rehabilitation option and the advantages and disadvantages of each option.
7. Remember that the patient and the family are the experts.	In FCC, the patient and their family are considered to be the experts, not the clinician. Their lived experience of the hearing loss should be acknowledged; for example, they could be asked, “What do you want to do about your hearing loss?”
8. Actively encourage involvement of the family at all stages.	In FCC, the family should be actively involved in all stages of the audiologic rehabilitation process, including initial case history taking, rehabilitation planning, hearing aid fitting, follow-ups, and annual reviews. As such, they should be actively invited to participate in each appointment.
9. Measure outcomes of interventions for both the patient and family.	In addition to measuring outcomes for the patient, in FCC, the family is also acknowledged as the client, and their outcomes should also be considered. You could revisit the goals identified at the beginning of the process by both the patient and family and find out how much the treatment has affected these goals. Tools such as the Significant Other Scale for Hearing Impairment (SOS-HEAR) (Scarinci et al., 2009b) and the Hearing Impairment Impact–Significant Other Profile (HII-SOP) (Preminger & Meeks, 2012) can be used to measure family member outcomes.
10. Make the entire clinic family centered.	In order to ensure your entire practice is family centered, you will need to facilitate buy-in from all stakeholders, including executives, managers, clinicians, and front office staff. Put FCC on the agenda of regular staff meetings.

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Conclusion

Including family members in adult audiology rehabilitation provides a range of benefits to patients, family members, clinicians, and businesses. The best way to include family in the audiology rehabilitation process is through the implementation of FCC. FCC is an approach to health care that acknowledges the family as the unit of attention in service delivery and promotes the importance of family choice, family strengths, the family-professional relationship, and individualized family services (Epley et al., 2010). FCC is not a new concept in health care, but it has only recently been identified as a suitable approach to use with adults with hearing impairment and their family members. Implementation of FCC does not come without its challenges; however, there are a range of tools and strategies which can be used by audiologists to promote this model of service delivery. This and other chapters in this book provide a basis from which to consider how families can be embraced as key members of the rehabilitation team.

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17

NARRATIVES AND TEXT MEDIA IN ADULT AUDIOLOGIC REHABILITATION

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Introduction

A narrative is a story; an account of one's experiences. Incorporating narratives, also known as the patient's story, in medical practice is becoming more common. Narratives fill a gap in health care delivery by providing a more complete, holistic perspective of a patient's health condition. While focusing solely on the impaired structures and functions of the body is limiting, narratives communicate the individual's unique needs, beliefs, and perspectives of their health. Listening to patients' stories has been found to build empathy in practitioners' care and increase understanding of the patient experience, trust, and health outcomes (Charon, 2001a; Greenhalgh & Hurwitz, 1998). For example, Hojat and colleagues (2011) examined physician empathy involving 891 individuals with diabetes. They found that patients whose physicians scored high on an empathy scale had better control of their blood sugar levels than patients whose physicians scored low. Attar and Chandramani (2012) found that individuals with migraines were more likely to adhere to treatment plans resulting in better outcomes when they felt their physicians were empathetic. The inclusion of narratives in health care practice is consistent with the biopsychosocial perspective endorsed by the World Health Organi-

zation; specifically, to acknowledge and address the structural and functional, personal, environmental, and social factors of patients to increase intervention success (World Health Organization-International Classification for Functioning, Disability, and Health [WHO-ICF], 2001).

There is a growing interest in using narratives in audiology. To elicit the narrative, the audiologist must show a genuine interest in the patient as a person, the impact of hearing loss, and how they came to seek audiologic services. The audiologist's body language, facial expressions, and tone of voice need to convey curiosity and concern. Attention to the patient's word choice, their use of metaphors, body language, and tone of voice assist in understanding the patient's experience and perception of their hearing impairment on communicative functioning.

Thus, incorporating narratives has a place in adult audiologic practice. The quality of the interaction between the audiologist and individual is a main component of the rehabilitation process and dictates how the individual will respond to treatment (Gregory, 2013). Frequently, individuals with hearing loss and audiologists view this impairment differently. Specifically, individuals tend to see their hearing loss as an illness; their perception of their ability to live with this loss, as well as their response to it, is central to their perspective (Gregory, 2013). In contrast,

audiologists view this hearing loss as a disorder and/or disease with a focus on the anatomy and physiology of the auditory system. Thus, audiologists tend to rely on the audiogram peppered with jargon to explain the hearing impairment, frequently saturating the dialogue with information about hearing aids and assistive devices without taking into account the individual's concerns, feelings of vulnerability, identity, and self-perception (Gregory, 2013). By incorporating the individual's story into the audiologic rehabilitation plan, the audiologist can provide the kind of care most appropriate, helpful, and satisfying.

The narrative begins when we ask the individual with hearing loss, "What brought you here?" Asking the client this question allows the individual to ponder and reflect on the possible impact of their hearing loss in various environments (e.g., theatre, restaurants, at meetings, lectures, work, in larger groups with multiple conversations, watching television, on the telephone) and on their self-image, and quality of life. This question also begins the formation of a partnership between provider and client; the provider is attentively listening to absorb and interpret the client's story, leading to improved service delivery and satisfaction. The act of attentive listening demonstrates care and the valuing of the individual's well-being, enabling the provider to address and provide the necessary rehabilitation practices tailored to the individual's needs.

This client-centered approach is a preferred practice model in adult audiologic rehabilitation (DiLollo, 2014; DiLollo & DiLollo, 2014; Gagne & Jennings, 2011; Grenness, Hickson, Laplante-Lévesque, & Davidson, 2014a, 2014b). The narrative can be expanded to include family members and communication partners by asking, "What is the effect of your spouse's/parent's/grandparent's/friend's hearing loss on you?" Hearing loss often has an impact on the entire family and communication partners. This expansion of the narrative viewpoint is consistent with the current family-centered approach, which is to engage significant others in the audiologic rehabilitation process (Hickson et al., 2016; Montano, 2018; Singh et al., 2016).

In addition to incorporating patient narratives in the therapeutic process (Crowell, Hanenburg, & Gilbertson, 2009; DiLollo, 2014; DiLollo & DiLollo, 2014), exploring alternative forms of text media, such as memoirs, nonfiction, fiction, graphic novels, poems, quotes, films, documentaries, plays, YouTube videos, and TED Talks may help individuals, family members, significant others, and audiologists to empathize, recognize, reframe, and reflect on the influence of the

hearing impairment on communication and quality of life. This chapter will address the role of narratives and text media in the adult audiologic rehabilitation process and educational practices as well as provide potentially useful resources, found in the References and Appendix 17–A.

Narrative Competence and Components

Hearing loss is chronic and isolating. To treat effectively, the audiologist must learn as much as possible about the impact of the hearing loss on the individual, family, and significant others (Meyer, Scarinci, Ryan, & Hickson, 2015; Scarinci, Meyer, Ekberg, & Hickson, 2013). Incorporating narratives in audiologic rehabilitation practice challenges the narrow "medical model" often used. The medical model (Barnes, 1997) positions individuals with disability as having a problem or impairment whose solution or cure is dependent upon the medical community. This model is often characterized by a kind of "colonizing listening" that has traditionally organized the health practitioner-patient/client relationship in which individuals with disability are positioned as listeners and clinicians are positioned as experts (Goggin, 2009).

In contrast, narrative medicine prioritizes the patient's experience through the clinician's attention to the details of the story and the clinician's subsequent affiliative practice (Charon, 2004). The "telling" of the time course of the hearing loss, tone of voice, word choice, body language, facial expressions, and gestures used by the individual and family members all help to increase the audiologist's understanding of the influence of the deficit on their communication and quality of life. Narratives of older adults, for example, have demonstrated that because hearing loss is considered a natural part of the aging process, they are less likely to seek or accept hearing aids when they begin struggling to hear. Instead, older adults wait until they feel their hearing loss has a negative impact on their relationships and their sense of who they are as a person to readily use assistive listening devices (Espmark & Scherman, 2003). Middle-age women with moderate hearing loss tend to have a different experience and therefore different needs; where their male peers tend to feel sad about hearing loss, women, who are often tasked with maintaining relationships and caring for children and elder adults, generally express worry as their primary concern. In such cases group rehabilitation and the sharing of

experience can be an effective modality (Jonsson & Hedelin, 2018). Successfully employing narratives in practice endows clinicians with “the ability to acknowledge, absorb, interpret, and act on the stories and plights of others” (Charon, 2001b, p. 1897). Without hearing the individual’s, family members’, and/or significant others’ story, treatment will be limited, incomplete, and ultimately diminished in effectiveness.

In addition to telling their own story, reading others’ narratives can also be beneficial to the individual, family, and audiologist. For example, Kooser (2014), a clinical social worker, writes of her experience with progressive hearing loss that ultimately resulted in her receiving a cochlear implant as an adult. She not only shares her journey and personal encounters, but also offers therapeutic and counseling strategies for the professional to consider. As narratives are personal reflections, someone with a progressive hearing loss will have a different narrative than someone with a sudden, gradual, or age-related hearing loss. The offering of these stories would most likely be specific or similar to the client’s type and onset of hearing loss.

As narratives give meaning to lives, shape behaviors, and organize one’s sense of self (DiLollo & DiLollo, 2014), engagement with narratives is beneficial for clinicians, individuals with hearing loss, family members, and significant others of those with hearing loss. To develop narrative competence, the audiologist must first attend, suspend judgement, tolerate ambiguity of multiple perspectives, and “recognize, absorb, interpret, and be moved by the stories one hears or reads” (Charon, 2004, p. 863). The ability to attend demands the audiologist be free of distractions, personal thoughts, feeling, and beliefs (Charon, 2005). As soon as the professional begins to form an opinion, listening has stopped. With careful attention, audiologists can gather a more accurate representation of the client’s, the family’s, and/or significant others’ perspectives and beliefs. What follows from attention and representation is affiliation, where empathy and partnerships between audiologists and the client, their family, and significant others begin (Charon, 2005).

Framework and Strategies for Incorporating Narratives in Adult Audiologic Practices

The purpose of the narrative is to assist the individual to expand, reconstruct, and/or reframe their “domi-

nant” story; that is, in this case, the story of their hearing loss. While the audiologist is the technical expert on hearing loss, the individual is the expert regarding their experience. Thus, the audiologist must communicate openness and acceptance, being mindful that the individual’s narrative may be limited and narrow. For some adults, while the major component of the narrative may be hearing loss, the goal is to encourage those individuals to include additional personal attributes that may affect their adjustment. A significant number of adults with hearing loss, however, have not fully identified issues and often see an audiologist to appease their spouse or other family members. For these individuals, the goal is also to explore and expand their dominant story, to reframe and reconstruct their narrative, and to accept that their hearing loss is one attribute of their personhood. In other words, adults with hearing loss often provide “thin” descriptions due to limited examination of their beliefs, stereotypes, and expectations (e.g., aging). Broadening or “thickening” these descriptions invites the individual to reexamine their beliefs, stereotypes, and expectations to construct an alternative and richer narrative that more accurately represents the complexity of their life experiences (DiLollo & DiLollo, 2014; Wolter, DiLollo, & Apel, 2006).

The audiologist accomplishes this by listening for contradictions (e.g., “I can’t hear so I don’t bother anymore” followed by “They mumble”) and the “voice” of others in the prevailing narrative (McKenzie & Monk, 1997; Wolter et al., 2006). To help the individual reframe their narrative and make meaning of their experiences, the following three steps have been shown to be effective (White & Epston, 1990; Wolter et al., 2006).

- First, the individual is asked to externalize and detach from the hearing impairment by giving it a name that provides specific meaning to them. For example, the name could be the problem (e.g., hearing loss) or feeling (e.g., frustration, irritation, annoyance). By giving the problem a name, the individual can focus and gain a sense of control. Once an appropriate name is used to identify the problem (e.g., hearing loss), the audiologist then refers to the hearing loss in the third person. The audiologist might state, “It sounds like the hearing loss prevented you from enjoying the play.” This enables the individual to separate the problem from themselves and facilitate a

new perspective (Wolter et al., 2006). In other words, the hearing loss is “personified and becomes the entity in itself” (Wolter et al., 2006, p. 171).

- Second, the individual is asked to provide a comprehensive description of the impact of the hearing loss, the problem, to map its influence. For example, ask the patient, “Describe how the hearing loss prevented you from . . .” The objective is to assist the client to externalize, separate from, and gain control of the hearing impairment rather than the other way around.
- Third, recognition of the influence of the hearing loss on their participation in activities and social encounters, and its negative effects on their quality of life, allows the individual to reflect on, take better control of, and imagine possibilities for different outcomes. This stage is a reconstruction of their story and represents the person’s response to and influence on the effect of the hearing loss on them (White & Epston, 1990; Wolter et al., 2006). For example, ask the client to reflect on a time when they were able to lessen the impact, overcome it, or prevent their hearing loss from dominating the communication experience. The individual is asked to reflect on this “alternative” narrative to empower and reconstruct their story and capitalize on their strengths (Wolter et al., 2006).

Eliciting a Narrative

Several additional strategies have been shown to increase practitioner competence; that is, the ability to elicit, attend, interpret, and coconstruct the client’s narrative (Charon, 2006; Greenhalgh & Hurwitz, 1998; Peterkin, 2012). Investing in eliciting the client’s story at the beginning of the process can save time later.

- Ask and/or present open-ended questions and statements to allow the individual to present his/her concerns and perceptions (Charon, 2006; Peterkin, 2012). Some examples of open-ended questions include: “What brought you here?”; “Can you describe the situations and environments when the hearing loss is most limiting?”; “Can you tell me how you currently cope with your hearing loss and the strategies you use?”; “Can you share with me some of your feelings concerning your hearing loss?”; “Can you share with me the impact of your hearing loss on members of your family, significant others, and coworkers?”; “Can you describe how this hearing deficit has changed your participation in activities or your self-image?”; “How would others describe you?”; and “Please tell me what you think I should know about you and your hearing deficit.”
- Attend carefully not only to the individual’s words, but also their body language, facial expressions, and tone of voice. Attention by the audiologist includes a relaxed body position, not sitting behind a desk, a head nod, a verbal acknowledgment, rephrasing what the client said to be sure you understood correctly (e.g., “I heard this . . . Did I hear correctly? Is that what you meant?”).
- Ask the client and family members/significant others to write a brief one-page document on the impact of the hearing deficit on them. This activity opens an opportunity for dialogue and perspective taking. The individual with the hearing deficit may have limited or no knowledge of its effect on important people in their life.
- Remember the hearing loss is but one attribute of the client. Find out your client’s interests (e.g., hobbies, travel, film, theatre, books, etc.) and their family (e.g., spouse, siblings, children, grandchildren).
- Be alert to recurring themes, key words, and metaphors that emerge in your communications. They can be useful in reconstructing their narrative.
- Reflect on your exchange with the client. Use a parallel chart (Charon, 2006), separate from the client’s official medical chart, to record your thoughts and feelings that emerge, increase empathy, promote deeper understanding of the client’s issues and beliefs, and obtain patient information, leading to a more competent practitioner.
- Be mindful of your body language, posture (relaxed or tense), and facial expressions. This also includes chair placement (e.g.,

sitting behind a desk), interruptions, and distractions.

- Be mindful of your own beliefs, values, assumptions, and stereotypes that are often subconscious and unexamined.

Narratives Written by Health Practitioners

Although some health practitioners, such as nurses, medical students, and physicians, have written narratives about their practice to describe the emotional and personal aspects of patient care, this form of prose is extremely limited or nonexistent in audiology. The exploration of one's beliefs, values, biases, and self-awareness is a reflective activity (Verghese, 2001) necessary to listen and hear what the client is saying. The act of writing and reading one's personal narrative can help the graduate student or audiologist better empathize with their client as well as other health care practitioners. Profession-based narratives may also lead to an increase in career satisfaction and less burnout, and should be considered as an element of graduate education in audiology. Finally, audiology is a profession with expected knowledge, skills, and ethical standards. Participating in interprofession-based narratives with one's colleagues can increase awareness of shared common goals for the patient/client and reduce the isolation and disconnect that often occurs in service delivery (Charon, 2001a).

Types of narratives written by health practitioners (Charon, 2001a) include fiction (Verghese, 2001; Williams, 1984), books and essays specifically written for the lay public (Gawande, 2002; Groopman, 2007), autobiography and memoirs (Kalanithi, 2016; Nuland, 2008), narrative medical practice stories featured in medical journals (*Health Affairs*, *Annals of Internal Medicine*, and *Journal of the American Medical Association*), and medical training experiences (Branch, Pels, Lawrence, & Arky, 1993; Klass, 1992; Reifler, 1996). However, none of these venues explores the clinical experience of audiology rehabilitation.

The ethics of medical narratives bear consideration and are genre specific. Clearly, permission and consent from the patient/client is required if the narrative is to be considered for publication. If, however, the narrative written about the patient/client, is composed of private ruminations for the intent to clarify thoughts and to improve care, disclosure is unnecessary (Charon, 2001a). Thus, having the client write their initial, thin narrative followed by their thicker

narrative as a result of reflection and reexamination for their own use and to provide better care would not require consent.

Text Media: Form and Function

There are various forms of text media that may resonate with and provide new ways of thinking about hearing loss for the individual, family members, significant others, and the clinician providing adult audiology rehabilitation services. Thus, carefully selected and appropriate text media used in conjunction with personal narratives can assist in the representation and expansion of the client's voice.

Representation is a mechanism by which meaning is constructed in a culture (Hall, 1997). It is not just the process of reflecting on what one knows, what one has heard, nor is it simply the process of presenting something anew. Through our oral, written, and visual languages, we make meaning of our world, forge cultural understandings of concepts, and render certain ideologies common sense. It is through our production of and engagement with representations of hearing loss and deafness that a shared cultural understanding of these disabilities is created (Quinlan & Bates, 2009; Schuchman, 1988). The distinction between the medical and social models of disability elucidates the constructive nature of representation.

When represented through the medical model, adult hearing loss and deafness appears as an individual problem needing an individual solution (Goggin & Newell, 2003) and may increase patients' feelings of isolation (Norden, 1994). In contrast, when represented through the social model, the focus of hearing impairment turns to the social and cultural barriers that constrict full participation in public life (Altman, 2001). In this model, the experience of living with hearing loss and deafness shares the spotlight with clinical expertise and experience and is thus compatible and congruent with incorporating narratives and text media in adult audiology rehabilitation practices.

Films, television shows, songs, graphic novels, memoirs, blogs, and other forms of media do not simply reflect experiences of hearing loss, but also structure cultural understanding and social practices. It has been demonstrated that media portrayals shape our perceptions of disability as well as affect the self-identities of individuals with disabilities both positively and negatively (Zhang & Haller, 2013). Therefore, media narratives can provide individuals,

family members, significant others, and audiologists new and alternative ways of thinking about hearing loss as well as motivate them to challenge inaccurate or negative representations of this deficit and audiology rehabilitation. Moreover, representations as tools that shape cultural perceptions can have an impact on health policy and access to resources (Garland-Thomson, 2005). For instance, narratives that portray hearing loss as a humorous and laughable part of the natural aging process may create a society in which less social pressure is applied to government and insurance companies to cover the costs of hearing aids and other assistive devices.

Films such as *Music Within* (Sawalich, 2007), which narrates the experience of the disability activist Richard Pimentel, a veteran whose deafness was a result of his combat duty, may lead to awareness, advocacy and, ultimately, material resources for individuals with disabilities. Media narratives of hearing loss may also lead individuals to consult with audiologists. In the television series *Switched at Birth* (Weiss, 2011–2017), Cameron, a Deaf man who has a Deaf teenage son, Emmett, decides to obtain a cochlear implant while Emmett declines the technology for himself. In the scenes that follow, the audiologist is depicted not only as a clinical expert but also as a partner in the family dynamic surrounding this decision; likewise, the technology is portrayed as life changing for a Deaf adult but not without its challenges. This positive representation of the audiologist-patient-family relationship may reduce concerns of adults who are reluctant to seek audiology care.

Television has often been effective in delivering health information, and viewers frequently approach their health care professionals about information and treatments they have seen on fictional television (Brodie et al., 2001). In fact, since audiences tend to trust medical information on fictional television more than documentary sources (Davin, 2004), audiologists may consider using media narratives to share information or start conversations with clients.

Audiologists must be selective in identifying which media narratives to use. Disability is often represented in popular culture as a personality flaw in which individuals with disability are self-loathing and socially maladjusted (Longmore, 2003). As with depictions of other disabilities (Wolfson & Norden, 2000), Deaf characters are often used as plot devices where their deafness may be a “psychosomatic response to trauma” or a metonymic device where their deafness is shorthand for isolation (Lerner, 2010). Though there has been some progress in representation, individuals with hearing loss and deafness are still frequently

depicted as “comical, embarrassing, lonely, and threatening to one’s work” (Foss, 2014, p. 888). In contrast, persons with hearing loss may also be represented as heroic (Longmore, 2003) or having superpowers, with American Sign Language (ASL) portrayed as a hero’s tool (Lerner, 2010). The protagonist family in *A Quiet Place* (Krasinski, 2018), for example, is able to protect itself from murderous monsters because one family member is Deaf and the family has learned to communicate in ASL and without using sound. Representations of deafness, ASL, and cochlear implants are far more frequent in media than those of other adult audiological conditions including hearing impairment and hearing loss, and the challenges hard-of-hearing individuals encounter (Foss, 2014). When the experience of hearing loss is depicted, it is often temporary and, on television, resolved by the end of the episode “through time, magic, or treatment” (Foss, 2014, p. 895). When permanent, generally only hearing aids and no other accommodations, support communities, and/or adaptive technologies are present in the narrative (Foss, 2014).

Audiologists may also wish to discuss media narratives with their clients as the information provided in films may be misleading. For example, some viewers were excited to see a depiction of tinnitus in the recent film *Baby Driver* (Wright, 2017) in which Baby, who suffers from the disorder, is a calm, cool getaway driver (McPhillips, 2017). The representation of a young man with tinnitus who is successful and ultimately respected may be beneficial for patients’ self-esteem. However, the only “treatment” Baby receives is a self-prescribed nonstop musical soundtrack to “drown out . . . the hum in the drum.” In addition, women, people of color, and young people are less likely to be represented on broadcast television as experiencing hearing loss than older white males, which may have an impact on their likelihood to seek or readily accept rehabilitative treatment (Foss, 2014).

Media narratives can also give individuals with hearing impairment an opportunity to explore the intersectionality of disability and other oppressed identities. Boswell (2001) explores being silenced by her father both as a girl and as a child with a hearing impairment. The act of reclaiming her voice through writing her story has been central to her experience as an adult woman with hearing loss. In another example, the documentary film *Deaf Jam* (Lieff, 2011), tells the story of Aneta, a Deaf, Israeli-born, and a budding ASL poet who finds an artistic affiliation with Tahani, a Palestinian hearing poet who uses spoken word.

Varied forms of narrative offer different opportunities to explore hearing loss. While no one media

form has an exclusive ability to tell stories in a particular way, each form has its strengths and advantages. Good memoir, for example, while reflecting on the past, can help us imagine the future. As Sutherland (2016) suggests, through memoir “we try to understand who we are becoming by reference to where we have been” (p. 771). The visual flexibility of the graphic novel form offers a unique perspective on human embodiment (Brophy & Hladki, 2014; Squier, 2008) and the passage of time (Magnet & Watson, 2017). In *El Deafo* (Bell, 2014), Cece Bell’s account of her hearing loss, the first-person narrator is alternately represented by an anthropomorphic bunny and a superhero bunny whose hearing aids and frequency-modulated (FM) listening device give her special powers. Echo, a Native American-Hispanic Deaf superhero from the *Daredevil* series (Goddard, 2015–2018), is able to imitate any movement she sees. The graphic novel form affords new ways of imagining the embodied form of the represented individual as well as a challenge to the linearity of their experience. Books for young children rarely feature hard-of-hearing or Deaf adults; Deborah Abbott’s (1994) *One Blasting TV and a Pig Outdoors*, a didactic book written from the perspective of a hearing boy with a Deaf father, is a rare exception, as is Myron Uhlberg’s (2009) historical fiction picture book, *Hands of My Father*, set in the 1930s based on Uhlberg’s experiences as the hearing son of Deaf parents. However, books for youth with Deaf and hard-of-hearing child characters are becoming more common (Brittain, 2004; Golos, Moses, & Wolbers, 2012) and may be useful for introducing children of adults with hearing loss and deafness to their parents’ and grandparents’ experiences with physician and audiologist visits, hearing aids, and cochlear implants. Gallaudet University professor Sharon Pajka maintains a blog (<https://pajka.blogspot.com/>; the reader is referred to References and Appendix 17–A) that summarizes and reviews Deaf characters in young adult and easy to read adult literature. This collection has nearly 250 contemporary titles, some of which focus on the experience of hearing children of Deaf adults such as the novel *Singing Hands* (Ray, 2006) and the aforementioned memoir, *Hands of My Father* (Uhlberg, 2009). Still, there remains a scarcity of youth literature exploring adult audiologic rehabilitation for diagnoses other than deafness.

The unique qualities of film, including its range of camera angles, create an opportunity for multiple perspectives and rich studies of the interior life of character. Different camera angles can be used to evoke different emotions including empathy and affil-

iation from audiences (Plantinga, 1999) as can sound and music. *It’s All Gone, Pete Tong* (Dowse, 2004), based on the real experience of DJ Frankie Wilde, who loses his hearing due to excessive exposure to intense music, uses visuals of sound waves, extreme close-ups of pulsing objects, and intense club music to explore both Wilde’s hearing loss and his subsequent reconnection to music through vibration. Film can create and capture both the ordinary, which can humanize disability and position it as typical human experience (Brylla, 2017), and the fantastic, which can challenge stigmas and lead viewers to imagine worlds with “accessible and valued spaces for the disabled” (Fletcher & Primack, 2017, p. 344).

On television, the episodic nature of the medium affords longer and varied story arcs that include a range of experiences. *Switched at Birth* (Weiss, 2011–2017), which aired for 6 years, explored family, romantic relationships, friendships, hearing aids, cochlear implants, speech therapy, Ménière’s disease, and the right to an education at a Deaf school. Finally, the easy access to digital technologies and the ability to access free or inexpensive streams of distribution makes producing narratives of all forms possible for a greater number of individuals. TED Talks by Heather Artinian, Rachel Kolb, and Christine Sun Kim are representative of this wide distribution and narrative exploration of a range of experiences within deafness (the reader is referred to the References and Appendix 17–A). In 2015, Josh Feldman and Shoshannah Stern (2018) crowd-sourced funding to create a web series about two Deaf young adults. The project, now called *This Close*, ultimately received backing from Sundance and was renewed for a second season.

While there are an increasing number of representations of deafness in media, the emphasis is on individuals with severe and profound deficits. The majority of adults with hearing loss have lesser degrees of this impairment and the challenges they encounter are rarely addressed. Audiologists and graduate students can use media texts to explore the stories that are being told as well as those that are missing from mainstream media.

Implications for Adult Audiologic Rehabilitation and Educational Practices

In addition to the suggestions provided in the *Framework and Strategies for Incorporating Narratives in Adult Audiologic Practices* section, listed below are further

strategies to incorporate narratives and text media in adult audiologic and educational practices. The reader is encouraged to explore the References and Appendix 17–A for resources.

- **Personal narratives:** Encourage the individual with the hearing loss to provide a narrative as well as reflection about the impact of the deficit on their quality of life and self-image. Family members and significant others can also be encouraged to provide their perspectives of the hearing impairment. Have the individual and family members/significant others share their perspectives with each other. Sharing narratives in individual and/or group audiologic rehabilitation venues can give voice to common communication difficulties, lessen feelings of isolation and frustration, and promote understanding and empathy. Further, having the individual(s) with hearing loss write and share about the most frustrating aspects, as well as humorous facets, can provide a more balanced perspective and reduce feelings of isolation. In graduate education, having students write and share their personal narrative about a time when they felt “other” or “different” can promote empathy as well as reflection. For practicing audiologists, writing and sharing narratives can reinforce why they chose this profession, give meaning to their work, and possibly increase satisfaction and ameliorate professional burnout.
- **Nonfiction, fiction, essays, memoirs, graphic novels, poems, films, documentaries, TED Talks:** Including targeted forms of text media can promote discussion, give words to unarticulated thoughts and feelings and can be used in individual and group adult audiologic rehabilitation and in graduation education. Specifically, in an adult group rehabilitation session have participants contemplate on and share a specific passage or scene that did or did not resonate with their experience regarding hearing loss. This exercise could easily be adapted in graduate curriculum by asking students to reflect on one aspect of the experience of hearing loss that they had not previously considered. In addition, engaging health care practitioners from various disciplines (e.g., occupational therapists, physical therapists, physician assistants, nurses, physicians, speech-language pathologists) in addition to practicing audiologists can promote perspective taking, leading to improved service delivery and integrated health care.
- **Writing a graphic novel or poem, producing a video, and/or creating art:** Several platforms and apps are available to create comics, graphic novels, and/or videos. For example, Pixton and Zimmer Twins are two platforms for creating comics and graphic novels and can be used by students, educators, and the general public. Animoto is an example of a website used to produce videos. The Guardian provides information on 20 of the best apps for producing videos. This information appears fully in Appendix 17–A. These platforms can be used in individual and group adult audiologic rehabilitation, by all family members and significant others, and in graduate education. For example, each member of the family could construct a frame illustrating their perception of or recall of an event or aspect regarding the hearing loss. In an audiologic rehabilitation session, participants could be paired or grouped to construct frames illustrating some aspect (frustrating and/or humorous) of living with hearing loss.
- **Role-playing:** Have graduate students take turns role-playing client and audiologist. This type of forum can provide graduate students with opportunities to practice their skills in a safe environment to elicit the client’s story, reflect, and reframe. Thoughtful and constructive critique of graduate student technique can follow the role-playing.
- **Mapping the influence of a challenge:** To help graduate students include reframing and alternatives narratives into their practicum experience, have them externalize, detach from, and give name to a personal challenge to provide a comprehensive description of the impact of this challenge (mapping its influence). This first step will help the student recognize the influence of this challenge on their well-being to reflect on, take control of, and imagine possibilities for a different outcome (reframing to give rise to an alternative narrative). As an example, a student named her challenge

“paralyzed.” She described how “paralyzed” caused her to become immobilized and unable to study or complete her work (mapping the influence), resulting in stress and anxiety. The student was able to describe a time when she was able to harness her “paralyzed” and start her work. She noted that her fellow students also often expressed feeling overwhelmed and feared failing. She was then able to reflect on the strategies she used to overcome “paralyzed” (reframing), recognizing moving forward she could continue to use these strategies and develop others as well as talk more with her colleagues (alternative narrative).

In summary, narratives and selected forms of text media have an important place in adult audiologic rehabilitation and graduate education practices. Whether written independently or in community, narratives and targeted text media serve as a therapeutic practice for clients, family members, significant others, and as an affiliative practice for audiologists. The act of writing their narratives and reading, watching, and discussing text media may imbue clients with a sense of agency that combats a feeling of powerlessness against hearing loss as they create an outlet through which to voice their experiences. These forums provoke discussion and offer the individual with hearing loss, family members, significant others, graduate students, and practicing audiologists opportunities to increase empathy and personal satisfaction in services received by their clients and in themselves. These genres provide multiple perspectives leading to reflection and reframing of the impact of hearing loss on the individual, family members, significant others, graduate students, and audiologists providing adult audiologic rehabilitation services. Narratives and text media, when used in conjunction with amplification and assistive technologies, anticipatory and conversational strategies, a family-centered approach, and individual and/or group therapy, can lead to better care and increased satisfaction by the individual, their family, and the practicing audiologist.

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Appendix 17-A

NARRATIVE AND TEXT MEDIA RESOURCES

While certainly not exhaustive, the websites, memoirs, nonfiction, fiction, books, TED Talks, films, documentaries, and other resources for children and adolescents with adult family members with hearing impairment listed in this Appendix are interesting in their own right. They also provide a vehicle to initiate conversation and discussion of the various perspectives of hearing loss that leads towards greater acceptance, understanding, reflection, and reframing.

Websites and Blogs

Goodreads has a substantial number of fiction and nonfiction books on hearing loss and deafness. Retrieved from https://www.goodreads.com/list/show/61680.Books_about_Hearing_Loss

People Hearing Better. Retrieved from <http://phb.secondsensehearing.com/category/tags/personal-hearing-loss-story>

Deaf Lives Series. Retrieved from <http://gupress.gallaudet.edu/deaflives.html>

Quotes on Deafness. Retrieved from <https://www.brainyquote.com/topics/deaf>

The Poetry of Hearing Loss. Retrieved from <http://hearinghealthmatters.org/betterhearingconsumer/2014/poetry-hearing-loss/>

List of Films Featuring the Deaf and Hard of Hearing. Retrieved from https://en.wikipedia.org/wiki/List_of_films_featuring_the_deaf_and_hard_of_hearing

Deaf Characters in Fiction. Retrieved from <http://myshelf.com/deaf/miscfiction.htm>

Sudden Hearing Loss Narratives. Retrieved from <https://www.clickondetroit.com/health/hearing-loss-readers-share-their-stories>

Bettina's Story: Sudden Sensorineural Hearing Loss. Retrieved from <http://weareallears.com.au/bettinas-story-sudden-sensorineural-hearing-loss/#.WwWg1i-ZNo4>

Pajka, S. A blog that summarizes and reviews Deaf characters in young adult and easy to read adult literature. Retrieved from <https://pajka.blogspot.com/>

Sophie's Story: My Sudden Hearing Loss. Retrieved from <http://www.sophiesoundcheck.org/site/sophies-story-my-sudden-hearing-loss.html>

Sygrove, C. *My experience with sudden sensorineural hearing loss*. Retrieved from <https://www.hearingaidknow.com/my-experience-sudden-sensorineural-hearing-loss-sshl>

She also has a blog about her experiences at <https://myhearinglossstory.wordpress.com>

Deaf Characters in Fiction. This website has a substantial number of novels on deafness and hearing loss. Retrieved from <http://myshelf.com/deaf/miscfiction.htm>

20 Best Apps for Making Videos. Retrieved from <https://www.theguardian.com/technology/2016/jul/17/20-best-apps-making-videos>

Animoto. A website for making videos. Retrieved from <https://animoto.com>

Two Sites for Constructing Comics/Graphic Novels:

Zimmer Twins. Retrieved from <http://www.zimmertwins.com/splash>

Pixton. Retrieved from <https://www.pixton.com>

Patient Stories on Tinnitus

American Tinnitus Association. Retrieved from <https://www.ata.org/managing-your-tinnitus/patient-stories>

Gur, T. *Tinnitus success*. Retrieved from <http://tinnitussuccess.com>

Tinnitus Talk Support Forum. Retrieved from <https://www.tinnitustalk.com/forums/success-stories.47/>

Patient Tinnitus Stories. Retrieved from <http://neuromonics.com/category/patient-stories/>

The Mindbody Discussion Forum. Retrieved from <http://www.tmswiki.org/forum/threads/my-tms-success-story-anxiety-tinnitus-skin-crawling-back-pain.11587/>

Dr. Hubbard's Tinnitus Story. Retrieved from <https://www.cbtfortinnitus.com/dr-hubbards-story/>

Memoirs and Nonfiction

Chorost, M. (2006). *Rebuilt: How becoming part computer made me more human*. New York, NY: Mariner Books.

Hammond, M. (2016). *What did you say?: An unexpected journey into the world of hearing loss* (2nd ed.). Minneapolis, MN: Two Harbors Press.

Hoffman, B. (2013). *Listen to me*. Sarasota, FL: First Edition Design Publishing.

Kisor, H. (2010). *What's that pig outdoors? A memoir of deafness*. Champaign, IL: University of Illinois Press.

Myers, D. G. (2000). *A quiet world: Living with hearing loss*. New Haven, CT: Yale University Press.

Rodriguez, M. K. (2012). *Hear your life: Inspiring stories and honest advice for overcoming hearing loss*. Austin, TX: Greenleaf Book Group Press.

Sacks, O. (1989). *Seeing voices*. Visalia, CA: Vintage Press.

Silver, R. (2000). *Invisible: My journey through vision and hearing loss*. Bloomington, IN: iUniverse LLC.

Swiller, J. (1994). *The unheard: A memoir of deafness and Africa*. New York, NY: Holt Publishers.

Thompson, E. (2008). *Day by day: The chronicles of a hard of hearing reporter*. Washington, DC: Gaulladet University Press.

Uhlberg, M. (2009). *Hands of my father: A hearing boy, his deaf parents, and the language of love*. London, UK: Bantam.

Fiction

Flemming, J. H. (2014). *Songs for the deaf: Stories*. Orlando, FL: Burrow Press.

Lodge, D. (2009). *Deaf sentence*. London, UK: Penguin Press.

For Children and Adolescents

Abbott, D. (1994). *One TV blasting and a pig outdoors*. Park Ridge, IL: Albert Whitman & Company.

Bell, C. (2014). *El deafo*. New York, NY: Scholastic Books.

Children and Young Adult Novels on Deafness: Retrieved from https://www.goodreads.com/list/show/2558.Children_YA_Novels_About_Deafness

Davies Okimoto, J. (1996). *A place for grace*. Seattle, WA: Sasquatch Books.

Deaf Characters in Adolescent Literature. Retrieved from http://pajka.blogspot.com/2007/05/deaf-characters-in-comics-graphic_21.html

Deaf Characters in Fiction for Adolescents and Young Adults. Retrieved from http://myshef.com/deaf/teen_ya.htm

Jordan, S. (1999). *The raging quiet*. New York, NY: Simon Pulse, Division of Simon and Schuster.

Landau, E. (2009). *The sense of hearing*. New York, NY: Scholastic Books.

Pajka, S. A blog that summarizes and reviews Deaf characters in young adult and easy-to-read adult literature. Retrieved from <https://pajka.blogspot.com/>

Ray, D. (2006). *Singing hands*. New York, NY: Clarion Books.

Seeger, P., Dubois Jacobs, P., & Christie, R. G. (2006). *The deaf musicians*. New York, NY: G. P. Putnam's Sons Books.

Uhlberg, M. (2009). *Hands of my father: A hearing boy, his deaf parents, and the language of love*. London, UK: Bantam.

For the Audiologist

Harvey, M. A. (2018). A psychological tool for managing tinnitus: Creating useful narratives. *The Hearing Review*, 25(3), 22–24.

Iverson, S. M. (2016). Biographies of hearing loss: Understanding the social world of late deafened adults through life experience narratives. *UCLA*. ProQuest ID: Iverson_ucla_0031D_15150. Merritt ID: ark:/13030/m5644bqp. Retrieved from <https://escholarship.org/uc/item/5tm7n441>

Kooser, C. (2014). This journey of mine. *Perspectives in Aural Rehabilitation and its Instrumentation*, 21(2), 46–55.

Lockey, K., Jennings, M. B., & Shaw, L. E. (2010). Exploring hearing aid use in older women through narratives. *International Journal of Audiology*, 49(8), 542–549. doi:10.3109/14992021003685817

TED Talks and YouTube Videos

Artinian, H. (2013). *Not the hearing or deaf world* [Video file]. Retrieved from <https://www.youtube.com/watch?v=jhm5OaXJVMQ>

Kolb, R. (2018). *Navigating deafness in a hearing world* [Video file]. Retrieved from <https://www.youtube.com/watch?v=uKKpjuPd6Xo>

Sindell, D. (2014). *Being deaf in the 21st century* [Video file]. Retrieved from <https://www.youtube.com/watch?v=HdIYB9voeMQ>

TED Talks on Deafness: Retrieved from <https://www.ted.com/search?q=deafness>

Films and Plays (Limited to English-Language Films)

Aronson, J. (Director). (2000). *Sound and fury* [Documentary]. United States: Aronson Film Associates.

Aronson, J. (Director). (2006). *Sound and fury 6 years later* [Documentary]. United States: Aronson Film Associates.

Dowse, M. (Director). (2004). *It's all gone, Pete Tong* [Motion picture]. Canada: Vertigo Films.

Hines, R. (Director). (1986). *Children of a lesser God* [Motion picture]. United States: Paramount Pictures.

Kaplan, O. (Director). (2010). *The hammer* [Biography]. United States: Tapout Films Sawhorse Productions.

Lieff, J. (Director). (2011). *Deaf jam* [Documentary]. United States: PBS Independent Lens.

Riedelsheimer, T. (Director). (2004). *Touch the sound: A sound journey with Evelyn Glennie* [Documentary]. Germany: Skyline Productions.

Sawalich, S. (Director). (2007). *Music within* [Motion picture]. United States: Quorum Entertainment.

Sclar, H. (Director). (2010). *See what I'm saying: The Deaf entertainer's documentary* [Documentary]. United States: Worldplay.

Taylor Brosky, I. (Director). (2007). *Hear and now* [Motion picture]. London, UK: Vermillion Films.

Vass, R. (Director). (2013). *Lake windfall* [Motion picture]. United States: Rustic Lantern Films.

Wright, E. (Director). (2017). *Baby driver* [Motion picture]. United States: Working Title Films.

Television Shows

Goddard, D. (2015–2018). *Daredevil* [Television series]. United States: Netflix. Retrieved from <https://www.imdb.com/title/tt3322312/>

Feldman, J., & Stern, S. (2018). *This close* [Television series]. United States: Sundance Now. Retrieved from <https://www.imdb.com/title/tt8036816/>

Sorkin, A., Schlamme, T., Graves, A., O'Donnell, L., Misiano, C., & Noah, P. (1999–2006). *West wing* [Television series]. United States: NBC.

Weiss, L. (2011–2017). *Switched at birth* [Television series]. United States: ABC Family.

18

GROUP AUDIOLOGIC REHABILITATION FOR ADULTS

Current Practice and Future Directions

Jill E. Preminger and Laura N. Galloway

Introduction

Irene is a 71-year-old retired bookkeeper. She is divorced and lives alone; however, she spends a lot of time with her adult daughter, June, and her grandchildren, who live just a few miles away. Irene admits that she has difficulty hearing, especially in restaurants and when playing with her grandchildren, but she has refused to see an audiologist. She knows of two people (a friend and her deceased aunt) who had hearing aids and never wore them. She thinks that audiologists are primarily interested in selling hearing aids despite the fact that they don't work. In addition, Irene believes she is too young to wear hearing aids and does not want members of her social circle to see her wearing them.

Antonio is a 78-year-old man with a moderate to severe sensorineural hearing loss with speech-in-noise scores consistent with his degree of hearing loss. He was fit with bilateral receiver in the canal hearing aids 2 years ago. At his 1-year evaluation he reported being fairly satisfied with his hearing aids but at his 2-year evaluation he was increasingly frustrated. He reported difficulty hearing at his neighborhood asso-

ciation meetings and at the movies with his wife. He noted that he has stopped attending these events. James is a 56-year-old emergency medical technician. He has had tinnitus for a number of years and reports it is so loud that he has been having trouble hearing his patients and colleagues.

Whereas these three individuals seem to have different backgrounds and complaints, there is one thing they all have in common: They could each benefit from a comprehensive audiologic rehabilitation (AR) program with an audiologist. We will be referring back to Irene, Antonio, and James throughout this chapter.

Most audiologic appointments are spent discussing audiometric results or device-related concerns such as smartphone connectivity or battery life. While this knowledge is understandably important, its delivery reduces the amount of time clinicians can spend with patients on complementary methods necessary to enhance the rehabilitation process. These additional AR techniques and strategies include coping skills focused on solving communication difficulties or dealing with the emotional aspects of living with hearing loss. Due to their efficacy, as well as time and economic constraints, these methods are often taught

in a group setting over at least a few sessions and are known collectively as Group Audiologic Rehabilitation (GAR). Boothroyd (2007) described AR as a holistic approach promoting the “reduction of hearing loss-induced deficits of function, activity, participation and quality of life (QoL) through a combination of sensory management, instruction, perceptual training, and counseling” (p. 63). AR performed in a group setting allows individuals with hearing loss to share solutions to these deficits of function and their residual effects on QoL. GAR has been shown to reduce the hearing aid return rate and improve an individual’s psychological, social, and emotional adjustments to hearing loss (Hawkins, 2005; Northern & Beyer, 1999). The group setting, while being financially feasible for the clinician and the patient, allows for a direct way to address the psychological, social, and emotional adjustments to hearing loss that go beyond traditional hearing aid fitting techniques. A group setting for an AR program allows an open atmosphere of mutual support and sharing of ideas and strategies for alleviation of the psychosocial effects associated with hearing loss (Preminger & Yoo, 2010).

Recent focus groups at the University of Louisville, including adults with addressed and unaddressed hearing loss, demonstrated a general lack of trust in audiologists that was tied to the high cost of hearing aids (Preminger, Rothpletz, Galloway, Smith, & Glasheen, 2018). This is consistent with qualitative research that has demonstrated that in audiology care, low trust is associated with a commercialized approach whereas high trust is associated with a patient centered care approach (Preminger, Oxenbøll, Barnett, Jensen, & Laplante-Lévesque, 2015). An ideal way to increase trust is to offer a comprehensive AR approach via GAR. This chapter will discuss how and why GAR works by reviewing the current literature and recommend how to offer GAR to patients.

Rationale for Group Audiologic Rehabilitation

When considering group rehabilitation, many audiologists immediately think of psychotherapy groups and, as a result, may feel unqualified to lead a group. But in fact, rehabilitation groups exist for a myriad of chronic health conditions. In a supportive group, patients can learn to cope with a health condition and to manage its psychosocial impact (Stewart, Davidson, Meade, Hirth, & Weld-Viscount, 2001; Thaxton, Emshoff, & Guessous, 2005). Groups provide a forum for participants to learn about their medical condi-

tion, to meet others who face similar circumstances, and to practice coping strategies (Moskowitz, Hult, Bussolari, & Acree, 2009). Successful rehabilitation in a group setting has been shown to improve QoL, improve mood, and to increase knowledge about disease in individuals with heart disease (Wolfgang, Phillips, & Leclerc, 2007), stroke (Larson et al., 2005), breast cancer (Goodwin et al., 2001), and chronic pulmonary obstructive disease (Woo et al., 2006). Group visits for chronically ill patients have been shown to reduce repeat hospital admissions and emergency care use, lower cost of care, deliver certain preventive services more effectively, and increase patient and physician satisfaction (Beck et al., 1997). A myriad of evidence exists that demonstrates the effectiveness of group therapy in adults with chronic health conditions. Hearing loss should be no different.

Hearing Loss as a Stigmatizing Trait

GAR serves a multitude of purposes in the AR process. One of the most important is it directly addresses the stigma associated with hearing loss to debunk stereotypes and promote positive coping strategies. According to Crocker, Major, and Steele (1998), stigmatization can occur when a person possesses (or is believed to possess) “some attribute or characteristic that conveys a social identity that is devalued in a particular social context” (p. 505). Stigma has been well defined within the context of other chronic health conditions, including mental health, Alzheimer’s, and obesity as a means to advance research for use with clinical interventions (David & Werner, 2016). David, Zoiazer, and Werner (2017) describe how public stigma becomes self-stigma; initially, there is awareness of the public stereotype, then individual agreement with the stereotype, and finally self-concurrence with the stereotype. In other words, self-stigma results from a multistep process involving cognitive attributions (e.g., perception of being old or stupid), emotional responses (e.g., feeling shame or pity), and behavioral reactions (e.g., concealment of stigmatizing trait) (David, Zoiazer, & Werner, 2017). For example, an older adult with hearing loss, when questioned, may acknowledge that if someone admits they have a hearing loss or use hearing devices they may be thought of as old or less able. Despite openly expressing that they do not agree with these stereotypes, their emotional and behavioral responses often suggest otherwise. This may include avoidance of social situations or simply nodding along in agreement despite not having heard what was said. A final step toward self-stigma may result, and can be categorized as

harm. This would be when these responses result in reduced self-esteem or depression. It is important to note that not all end up reaching this step. Nevertheless, it is well established that stigma is a significant deterrent to managing one's hearing loss (Wallhagen, 2010). Stigma is one of the primary reasons Irene did not seek help for her hearing problems. (For further discussion of stigma, the reader is referred to Chapter 4 in this text.)

Case Study 18-1. Irene: Part A

Irene's daughter, June, saw an advertisement for GAR in a community newspaper and encouraged her mother to attend. Irene and her daughter attended three GAR sessions. In the sessions, they met others who were struggling to accept their hearing loss and others who had taken action. They learned more about hearing aids, communication strategies, and hearing assistance technologies from an audiologist. They also learned about benefits and limitations of these management strategies from both the audiologist and the other members of the group.

After attending the GAR sessions, Irene scheduled an appointment with the GAR audiologist for an audiologic evaluation and to discuss hearing aids. Irene reported that spending time with others with hearing loss gave her confidence to do something about her own problems. She also told the audiologist that after attending GAR she realized that the audiologist was truly a hearing expert and she now trusted her recommendations. Irene was so pleased with the benefit she received from hearing aids that she wanted to help others with hearing loss. She began attending meetings at her local Hearing Loss Association of America chapter. Plus, she volunteered to serve as a peer mentor in GAR groups led by her audiologist.

action helps individuals realize they are not alone in their experiences and feelings regarding their hearing loss. In addition, they can learn new ways to cope with communication breakdown or other difficulties they may encounter as a result of their hearing impairment. The second step of Héту's normalization process is utilizing these new skills and emotional awareness to interact with individuals with normal hearing to restore former social identity. This process sounds a lot like GAR, doesn't it? As everyone in the group shares the stigma of hearing loss, the stigma begins to fade. As a result, the members feel more comfortable practicing effective communication strategies and are better able to manage the consequences of hearing loss. For Irene, while attending GAR, hearing loss was no longer stigmatizing, and she finally felt comfortable enough to pursue hearing aids. After completing GAR, Irene continued to combat stigma by participating in volunteer activities with others who have hearing loss.

Case Study 18-2. Antonio: Part A

At Antonio's follow-up appointment, audiologic evaluation indicated a slight decrease in his hearing since the previous year and his Hearing Aids (HAs) were adjusted and verified appropriately. Antonio's audiologist believes he is doing as well as can be expected with his HAs. Rather than educating him again about realistic expectations, she invites Antonio and his wife to attend a GAR program that she offers on a monthly basis. There, Antonio could learn about hearing assistance technologies that he can use in addition to his HAs. In GAR, Antonio and his wife can practice more effective communication strategies, and they can learn strategies from others in the Group. Finally, in GAR, he can receive encouragement to keep participating in the activities that he enjoys, so he is not sitting at home alone.

To address hearing loss directly as a stigmatizing trait, Héту (1996) proposed a two-step normalization process. The first step involves interacting with individuals who share the stigmatizing trait of hearing loss. As the trait of hearing loss is no longer deviant and is instead shared, the negative stereotypes associated with it can be explored in a supportive environment. GAR participants can share difficulties they have experienced as a result of their hearing loss and methods that have been used with varying degrees of success to overcome the issues raised. This inter-

Reducing the Psychosocial Effects of Hearing Loss

GAR seeks to reduce the psychosocial effects of hearing loss. Psychosocial simply means the combined impact on an individual's psychological and social function. The impact of hearing loss is multidimensional and may include the emotional, cognitive, interpersonal, behavioral, and physical responses to hearing loss (Trychin, 2002). Thus, a psychosocial

approach addresses how hearing loss affects an individual in the psychological domain (e.g., self-image, stigma, denial, stress) as well as in the social domain (e.g., avoidance, family, social life, and work). Hearing loss usually impacts both domains, though it is important to note that the effect is individual and can vary along a wide continuum regardless of the degree of hearing loss. To avoid the stress involved in concealing one's stigmatizing trait, an individual with hearing loss may avoid social interaction, leading to depressive symptoms, feelings of loneliness, and a reduced social network (Kramer, Kapteyn, Kuik, & Deeg, 2002; Pronk et al., 2011; Strawbridge, Wallhagen, Shema, & Kaplan, 2000). Recent studies show a correlation between hearing loss and social isolation, depression, and dementia (Amieva, Ouvrard, Meillon, Rullier, & Dartigues, 2018; Deal et al., 2017). For adults with hearing loss it is often easier to just stay home than to try and communicate in a public or group setting, especially when there are a lot of people and background noise present. Amieva et al. (2018) found that older adults, specifically men, showed a negative association between hearing loss and social isolation. That is to say, the greater the hearing loss, the less social activity reported by the participants. Conversely, other studies have found that hearing loss has been associated with greater degrees of social isolation in women (Mick, Kawachi, & Lin, 2014). Emerging evidence (Amieva et al., 2018) suggests that hearing aid use may protect against the detrimental concomitant effects associated with untreated hearing loss in older adults; successful completion of GAR may also reduce social isolation and thus could lead to similar positive outcomes. Participation in a GAR program helped Antonio to reengage in the activities he used to enjoy, thus reducing his social isolation and possibly reducing risk for developing depression and dementia.

Depression and social isolation contribute to psychosocial status, which is an important component to consider when assessing QoL. The World Health Organization (WHO) defines health-related QoL as individuals' "perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns" (WHO, 1995, p.1403). As successful health care management has increased quantity of life, health care practitioners and researchers have looked toward methods to improve quality of life. Health related QoL can be measured with a variety of assessment scales, which fall into two categories: (a) generic areas of everyday life, which include both the physical and mental effects of a health condition; and (b) disease-specific areas, which for our purposes are the participation restrictions and activity limitations

that arise from hearing impairment (National Institute of Health, 1993). (For additional information on quality of life, the reader is referred to Chapter 7.)

It is the expectation that GAR will address the psychosocial impact of hearing loss, thus improving health-related QoL. One way this can be achieved is by sharing, discussing, and practicing effective coping strategies. When faced with a stressful situation, such as communicating in a difficult listening environment, an individual with hearing loss inevitably develops some combination of positive and or negative coping strategies. Lazarus and Folkman (1984) described two types of strategies for coping with a chronic condition: problem-focused and emotion-focused coping strategies. Problem-focused coping emphasizes developing solutions to manage a health condition, whereas emotion-focused coping is achieved by managing one's own emotional response to the health condition. A GAR program is an ideal setting to discuss and practice effective coping strategies. For example, most individuals with hearing loss report problems when going to the theater. In GAR programs, there is discussion about hearing-assistive technologies (HATs) that can be used in the theater (problem-focused coping). When group members hear about others' successful experiences with these technologies, it helps to alleviate the stigma that may have prevented technology use in the past (emotion-focused coping). Rehabilitation programs that utilize a combination of problem-solving strategies have been shown to be most effective (Duangdao & Roesch, 2008; Preminger & Yoo, 2010). For example, success of the Active Communication Education (ACE) program (Hickson, Worrall & Scarinci, 2007a), a GAR program that focuses on communication awareness, education, and training for adults with hearing loss, has been well documented in the literature (Hickson et al., 2007a; Hickson et al., 2007b; Öberg, 2017). One of the primary activities of this program is the use of problem-solving interactive approaches that encourage participants to discuss their communication difficulties with group members and facilitators as well as to practice communication solutions. The group setting allows individuals with hearing loss to learn, share, and generate both types of coping strategies.

GAR programs facilitate reduction of the psychosocial effects of hearing loss in accordance with the Social Comparison Theory. According to this theory, people with similar health problems inevitably compare the physical and behavioral effects of the condition (Martin, Suls, & Wheeler, 2002). Downward social comparison occurs when a group member believes that he or she is coping better than others; for example, "That guy must have a really bad hear-

ing loss because he keeps asking me to repeat myself. I can usually get it after the first repeat. Maybe I am doing OK.” Upward social comparison, on the other hand, occurs when a group member believes he or she is coping as well as others who appear to manage their condition in a satisfactory manner (Martin et al., 2002). Upward social comparison can inspire individuals to try new strategies or devices to manage as well as other group members. Individuals in a group setting find solace in realizing there are others like them. This comparison generates an understanding among participants that their stressful situations and reactions are not caused by some personal deficit. After comparing themselves to others, group members may feel that their problem is not so bad, that they are managing better than others, or that they can improve their situation. This comparison can have a huge impact on an individual’s psychological and emotional adjustments to hearing loss and is one of the keys to explaining the rationale for GAR.

Self-Management

As discussed in this textbook, a comprehensive view of AR takes a biopsychosocial approach rather than a biomedical approach. (For additional information on the biopsychosocial approach, see Chapter 11 of this text.) A biopsychosocial approach is consistent with

the view that hearing loss is a chronic health condition rather than an acute health condition (Convery, Hickson, Keidser, & Meyer, 2019). A chronic condition cannot be cured; rather, it can be managed. For successful management to be achieved, the patient must actively participate in the process. James can benefit from GAR by learning about the connection between hearing loss and tinnitus and how to manage these chronic conditions.

Case Study 18–3. James: Part A

James learned about a GAR program at his primary care doctor’s office and decided to attend. He thought that he could learn about techniques to help his tinnitus so that he could hear better. When he explained this during the first GAR session, the audiologist explained that he likely had hearing loss in addition to the tinnitus. Audiologic evaluation revealed a bilateral mild to moderate hearing loss that was consistent with his history of noise exposure. James was very surprised to learn that he actually had a hearing loss and remarked that he would not have figured this out if he had not attended GAR.

Case Study 18–4. James: Part B

James decided to trial hearing aids. His audiologist encouraged him to continue to attend GAR during the trial and explained that the group sessions could help him to adjust to the new hearing aids and learn about strategies and additional technologies to complement the benefit he would receive from the hearing aids. James attended two more GAR sessions during the trial period. One member of the group had hearing aids that synced with his phone and spoke positively about the advantages. Another group member explained that she did not like how her hearing aids sounded at first, but worked with her audiologist to “tune them up” every few weeks after she started to use them; she reported that she continues to visit the audiologist every 6 months for adjustments.

James really appreciated getting advice from others who wore hearing aids. During his trial period he decided to exchange his hearing aids to ones that were compatible with his cell phone. He reported that he now understood it would take time to adjust to the hearing aids. James’ audiologist found this somewhat ironic, as she had explained all of this to him during his initial hearing evaluation. She had explained different hearing aid technologies she thought he would be interested in (including cell phone-compatible hearing aids), plus she had discussed the hearing aid acclimatization period with James, as she did with all of her patients. However, she encouraged him to continue to attend GAR because she learned from experience that patients needed to hear this information from multiple sources and on multiple occasions before they really understood their management options and the rehabilitation process.

Convery et al. (2019) describe the Chronic Care Model (CCM), which can be applied to hearing loss. In the CCM, the patient and the audiologist collaborate, and it is up to the audiologist to guide the patient in how to obtain support from the health care system (e.g., how to participate in shared decision making, where to get self-management support) and from the community (e.g., participation in support groups, self-help groups) (Convery et al.,

2019). A GAR program is an ideal location to offer self-management support. Indeed, many self-management programs for individuals with chronic conditions such as diabetes or arthritis are delivered in small group workshops (Stanford School of Medicine, 2015). An effective self-management program includes the components in Table 18–1. Most of these components can be addressed effectively in GAR programs (Preminger & Rothpletz, 2016).

Table 18–1. The unshaded rows list the 10 self-management components described by Lorig and Holman (2003); column three describes the component and gives examples. The final two rows (shaded) are additional self-management components described by Pearce et al. (2015).

Tasks	Medical Management	Use of devices or techniques to manage HI: Learning to use a hearing aid, regular use of a hearing aid, auditory-visual speech perception training
	Role Management	Modifying life roles, as needed, to accommodate HI: Asking one's partner to handle social phone calls, hosting small family dinners rather than going to a restaurant
	Emotional Management	Managing the emotional consequences of living with HI: Developing coping strategies to manage frustration, loss, or stress
Skills	1. Problem Solving	Problem definition, generation of possible solutions (from self, peers, coaches), solution implementation, evaluation of results
	2. Decision Making	Select an appropriate solution for problems as they arise: Requires knowledge and information about HI, communication strategies, hearing aid technologies, and more
	3. Utilizing Resources	Knowledge regarding how to find and utilize resources related to HI: hearing health care information on the internet, an audiologist, and HI support groups
	4. Partnership With Hearing Health Care Provider	Learning how to communicate and practice shared decision making with one's hearing health care provider: making informed choices about treatment and strategies
	5. Taking Action	Solution implementation: developing a short-term action plan, selecting achievable/attainable strategies (e.g., turning down the TV when having dinner with spouse), executing the plan
Mechanisms	Self-Efficacy	Taking action requires self-efficacy: selecting strategies that one believes can be accomplished
	Self-Tailoring	Self-management requires: knowledge about one's condition, knowledge about available strategies and resources, the ability to problem solve and make decisions unique to one's own situation
	Education	Education about the condition and management of the condition
	Social Support	From peers with HI and/or communication partners

Note. HI = Hearing impairment.

Source: Reprinted from Preminger, J. E., and Rothpletz, A. M. (2016). Design considerations for internet-delivered self-management programs for adults with hearing impairment. *American Journal of Audiology*, 25(Suppl. 3), 272–277. doi:10.1044/2016_AJA-16-0010

Effectiveness of GAR

Since the theoretical basis for GAR is now clear, it is important to review its effectiveness in the literature. GAR seeks to remediate areas of activity limitations, participation restrictions, and adjustment to hearing loss. It also aims to increase use of communication strategies for all those with hearing loss including new hearing aid users, established hearing aid users, non-hearing aid users and Cochlear Implant (CI) recipients. Chisolm et al. (2004) found that a group of new hearing aid users who received routine hearing aid orientation plus GAR (consisting of informational lectures, communication strategies training, and assistive device use) showed better outcomes post-treatment, and 6 months later, on the Communication Profile for the Hearing Impaired (CPHI; a comprehensive assessment of communication problems related to hearing loss) (Demorest & Erdman, 1987) than a group receiving routine hearing aid orientation only. Heydebrand, Mauze, Tye-Murray, Binzer, and Skinner (2005) followed 33 adult CI users who participated in a 2-day GAR program and a follow-up session 4 weeks later. These participants reported significant improvements on measures of assertiveness, emotional well-being, and coping behaviors at both 3 months and 12 months post-GAR. The most recent systematic review on GAR (Chisolm & Arnold, 2012) was consistent with a previous review (Hawkins, 2005). Both showed evidence that GAR was effective in reducing activity limitations and participation restrictions in the short term but noted that more research is needed to determine the long-term benefits.

Subsequent articles continue to demonstrate benefits of GAR for adults with hearing impairment. Golder, Walsh, Buchanan, and Lind (2012) conducted semistructured interviews on 10 adults with hearing loss who participated in a GAR program. Qualitative evaluation of their responses showed the primary benefit reported by participants to be empowerment through improved self-image. This was demonstrated with themes including improved understanding of communication strategies, improved social relationships, decreased emotional isolation, and improved self-confidence. Habanec and Kelly-Campbell (2015) showed a significant improvement following GAR as measured by the CPHI and a significant reduction in third-party disability for frequent communication partners of working adults with hearing loss who did not use hearing aids. A study of 77 adults with hearing loss who participated in the ACE program showed statistically significant short- and long-term effects on communication strategy use, activity and

participation on the Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982) and the Communication Strategies Scale (CSS) of the CPHI (Öberg, 2017). In addition to measuring benefit with questionnaires, outcomes were measured on self-generated goals using the Client-Oriented Scale of Improvement (COSI) (Dillon, James, & Ginis, 1997). Participants including HA users, CI users, and individuals without technology reported improvement on goals related to preventing isolation and increasing self-confidence as a result of being more open about acknowledging their hearing loss to others and being braver when talking to strangers (Öberg, 2017). In a narrative review of the AR literature, Hashir and Moore (2017) found that individual differences likely play a part in the decision to include GAR into a patient's treatment. They noted that research looking at patients' attitudes toward hearing loss and hearing aids, as well as traits such as motivation and readiness, may play into the decision to add individual AR, GAR, and/or hearing aids to a patient's treatment program.

In summary, the psychosocial benefits of GAR programs arise from a reduction in stigma associated with hearing loss, the practice of problem- and emotion-based coping strategies, and the opportunity to experience social comparison. GAR has been demonstrated to be effective in reducing stigma and social isolation associated with hearing loss as well as increasing use of communication strategies. However, additional research is needed to determine how individual differences interact with successful GAR outcomes. The next sections review activities that have been developed to achieve these benefits.

Implementation of Group Audiologic Rehabilitation

Group Audiologic Rehabilitation Process

When planning a GAR program, it is important to consider not only the class content, but also the process involved in creating and leading the group. (See Table 18–2, which lists seven factors to consider when designing and implementing a GAR program.) The majority of these process points are based on the work of Dr. Sam Trychin (2014). The first process point to consider is the role of the audiologist. The audiologist leading the group should not act as an instructor, whose role is the teaching of content, but rather should structure the class so that participants can begin to address the negative feelings and behaviors

Table 18-2. Processes to Be Considered When Developing a Group AR Program

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1. The role of the group leader is to provide group structure and manage group flow. The leader is not there to teach.
 2. Class topics should focus on issues important to class members.
 3. Class members should work collaboratively to solve the communication problems identified by the group members.
 4. The AR class should be designed so that everyone in the group will understand what is being said most of the time.
 5. Hearing loss effects the person with hearing loss and his/her communication partners; therefore, communication partners are always welcome in the AR group.
 6. Psychosocial benefits of group AR arise from overcoming stigma and acknowledging feelings and experiences.
 7. It is difficult to change thoughts, feelings, and behaviors associated with hearing loss; change requires interaction, practice, and feedback.
-

Source: Based on Gagné et al. (2019).

they may have developed as a coping mechanism. As an example, consider Irene from earlier. It is unlikely that she would have benefited from listening to a lecture on “understanding your audiogram” or learning about hair cell loss in the cochlea. After all, will learning how to read an audiogram help to destigmatize hearing loss? Will learning the difference between conductive and sensorineural hearing loss assist one in utilizing effective communication strategies? Likely not. Instead, the authors believe most individuals with hearing loss would benefit from actively generating and practicing effective communication. The processes and strategies described here are appropriate for both new and experienced users of hearing aids and cochlear implants as well as for those considering device usage (Abrams, Hnath-Chisolm, Guerreiro, & Ritterman, 1992; Heydebrand et al., 2005; Öberg, 2017).

The importance of this first process point must be emphasized, as audiologists and students of audiology are typically most comfortable leading GAR as instructors rather than as facilitators. If the audiologist is in front of the group explaining the audiogram, then there is no need to encourage each member to contribute, acknowledge a comment made by an angry class member, or discuss an emotionally charged question. The audiologist’s role should not be to teach the participants; rather, it is to facilitate changes in attitudes and in behaviors (Kricos, 2000a). As a facilitator, the audiologist encourages group members to work cooperatively to actively solve com-

munication problems (Kricos, 2000a). In fact, Kricos (2000b) recommended that the audiologist speak no more than 30% of the time. As Abrahamson (2000) pointed out, experience has shown that AR class participants can teach each other more effectively than can the audiologist. After all, wouldn’t you be more receptive to learn and take advice from someone who knows firsthand what you are going through?

One way to manage the group flow is to consider the class rules shown in Table 18-3, which are based on the recommendations of Trychin (Gagné et al., 2019) and Hogan (2001). These rules may be considered in the implementation of a typical group activity in which the facilitator describes a problem and the participants collectively offer solutions. Typically, the

Table 18-3. Group Audiologic Rehabilitation Class Rules

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1. Only one person speaks at a time.
 2. While we value everyone’s contribution, everyone has a right to pass. You do not have to answer a question if you do not want to.
 3. We treat each other with respect and kindness.
 4. Inform the person speaking as soon as you are aware that you are not understanding what is being said. (No smiling and nodding allowed!)
-

Source: Based on Gagné et al. (2019) and Hogan (2001).

audiologist goes around the room and asks each participant to offer solutions. By following the first class rule (only one person speaks at a time) class members will learn an effective communication strategy: It is easier to hear one talker at a time. If a participant feels uncomfortable with the conversation, he/she should not be pressed to contribute (rule 2). Sometimes there are difficult class members who continually go off topic or who may present their thoughts in an unacceptable manner (rule 3). The facilitator can flag ineffective communication with a kind word, humor, or an example of a more acceptable way to contribute one's opinion. Finally, if a class member is having difficulty understanding, he/she should speak up so that adjustments can be made (rule 4). As with all of the class rules, this one demonstrates an important communication strategy that may be used outside of GAR.

With the facilitator now in a comfortable role and class rules in place, the rest of the process points listed in Table 18-2 can be implemented. The second process point is that class time should focus on topics that are of importance to the participants. Selecting these topics can and should be an activity in and of itself and will be discussed further in the next section on GAR content. Focusing on topics important to the participants' desired outcomes achieves two goals. First, chances are better that participants will report satisfaction with the class, and second, it will be easier to achieve the third process point, which is that class members should work collaboratively to solve the problems identified by the group. If the class members are actively solving communication problems, then the first point will also be achieved: The audiologist will be facilitating the group rather than instructing the group. Many of the activities described in the next section are flexible enough that they can easily be adapted to focus on problems and issues that are important to the individual class members. Plus, the authors' personal experience has shown that the problems typically identified by GAR participants are quite similar from group to group.

The third process point, encouraging collaborative problem solving, can be achieved with activities discussed in the next section on GAR content. The fourth process point recommends arranging the environment so that everyone in the group understands what is being said. This may be accomplished through the use of hearing-assistive technologies (HATs) such as personal FM systems, soundfield FM systems, and/or real-time captioning. There are obstacles to each of these methods as time, effort, and

money are needed for implementation; however, the benefits are worthwhile. Other techniques that may be considered: (a) seat everyone in a circle and maintain adequate lighting so that speechreading is always available, (b) keep background noise to an absolute minimum, (c) if showing slides, do not dim the lights so that the talker's face remains clearly visible, and (d) give handouts for each activity as the handout will provide keywords that will aid in synthetic speech processing.

The fifth process point explains that frequent communication partners are always welcome. This recommendation has been offered by professionals who lead GAR programs (Abrahamson, 2000; Kricos, 2000b) and confirmed by research that has demonstrated that outcomes for people with hearing loss improve when their communication partners are active participants in AR groups, as compared to outcomes for individuals with hearing loss who attend such groups on their own (Preminger, 2003). It is likely that communication breakdowns occur most often with frequent communication partners. Yet, experience and research has shown that people rarely discuss their hearing loss with their spouses or children (Abrahamson, 2000; Getty & Héту, 1991). See Part B of Irene's case—it is likely that the support she received from her daughter was instrumental in moving her forward in the rehabilitation process. In a GAR program, communication partners can learn why their loved one with hearing loss is having difficulty hearing despite the fact that he or she wears hearing aids. Additionally, the communication partners can learn effective communication strategies and practice them. (For additional information on communication partners, see Chapter 19 of this text.)

Case Study 18-5. Irene: Part B

Irene's daughter, June, saw an advertisement for GAR in a community newspaper and encouraged her mother to attend. She even told her mother she would attend with her so that they could figure out together what options were best for Irene to manage her hearing problems. Irene later reported that she would not have attended if her daughter had not encouraged her and joined her. Her daughter's support made Irene realize her hearing problems were important.

Referring back to Table 18–2, we now finish the discussion of GAR processes with the final two process points. Process point six was described previously; we explained why an AR group should be effective in overcoming stigma, negative feelings, and negative behaviors. The process points described here and much of the GAR content explained in the next section are designed specifically to improve psychosocial outcomes. Process point seven explains that it is difficult to change the longstanding negative thoughts, feelings, and behaviors associated with hearing loss. Positive change requires interaction with others who share similar problems, opportunities to practice new behaviors, and positive feedback on these new behaviors. The content in the next section is designed to offer suggestions for encouraging interaction, practice, and feedback.

Group Audiologic Rehabilitation Content

Table 18–4 shows a list of suggested activities to implement in a GAR program. Some of these activities are quite structured and require some preparation, whereas others require little advance planning.

Problem Identification

A good way to begin a group is to identify the problems that will be the focus of subsequent sessions. Two examples of activities are described below. Hogan (2001) developed the exercise “What’s the worst thing about having a hearing loss?” in which the facilitator asks this question and then moderates a discussion. When someone speaks up with a problem (e.g., “I have difficulty understanding over my cell phone”) or an emotion (e.g., “I feel so lonely even when I am in a group because I cannot participate in the conversations”), the group leader writes it down on a whiteboard. Typically, the discussion is unforced and supportive. Participants are eager to share their difficulties and others in the group are quick to empathize and share similar tribulations. If the group members are reticent, the leader can ask a group member to expand on a concern (e.g., “What is it like for you when this happens?”) or ask other group members if they share similar concerns. Often group members share humorous examples (e.g., “I have difficulty understanding what people say on the party bus to the university football games”) and this humor adds to feelings of camaraderie engendered by the discussion.

Table 18–4. Suggested Activities for Group AR Classes

Types of Activities	Example Activities
Problem Identification	▪ What’s the worst thing about having a hearing loss? ▪ Communication Needs Analysis
Communication Strategies Training	▪ Anticipatory Strategies: “15 things” ▪ Repair Strategies: Clarification exercise
Self-Advocacy Training	▪ Assertiveness training ▪ Overcoming stigma
Speech Perception Training	▪ Lipreading: Learning to look ▪ Synthetic Training: Learning to use context
Stress Reduction Exercises	▪ Focused breathing ▪ Right way/wrong way
Informational Demonstrations	▪ Hearing Assistive Technology ▪ Emergency Preparedness
Structured Psychosocial Discussion	▪ What’s the worst thing about having a hearing loss? ▪ What’s a positive aspect of living with hearing loss?

In addition to identifying problems for the group to address, an additional purpose of this exercise is to make the participants aware of the difficulties associated with hearing loss and to demonstrate that they are not alone in facing these problems (Hogan, 2001). That is why this exercise is also listed as a “structured psychosocial discussion” in Table 18–4. While most individuals know that their hearing loss causes communication problems, they may not realize the broad range of accompanying psychosocial reactions (Trychin, 2002). Individuals may experience frustration, embarrassment, anger, and depressive symptoms as a result of their hearing loss, but may not realize this, or may choose not to acknowledge these unpleasant feelings. Individuals with adult-onset hearing loss can feel stigmatized by the shame or disgrace associated with hearing impairment that is often regarded as socially unacceptable (Hétu, 1996). As a result of this stigma, individuals are often unwilling to use hearing aids, assistive listening devices, or communication strategies. When AR group participants have the opportunity to share with a like-minded group, they may begin to recognize their own experience and feelings. Acknowledgement of hearing loss, and hearing loss-related feelings and problems, helps to alleviate the associated stigma and is necessary before individuals can take the required steps for successful AR.

Hickson, Worrall, and Scarinci (2007b) described another technique for problem identification: a two-part exercise consisting of a communication needs analysis followed by a nominal group technique, in which the members of the group work cooperatively to select communication problems to address during subsequent GAR sessions. When implementing this technique, participants initially work individually to complete a worksheet in which they identify their personal communication goals (Hickson et al., 2007b). This written record may be used as an outcome measure when the GAR program is finished, as participants can then consider if they have achieved their goals. Audiologists should be very familiar with this type of goal-setting technique as it is quite similar to the COSI. In the communication needs analysis, participants are asked to consider, “What communication difficulties do you experience in everyday life as a result of your hearing impairment?” (Hickson et al., 2007b, p. 12). Note that this question limits the discussion to communication-related problems, whereas in the exercise discussed previously (“What’s the worst thing about having a hearing loss?”), participants may respond with either communication

problems or problems related to poor communication (e.g., emotional concerns, work-related problems, impact on relationships). During the course of a GAR program it would be useful to implement each of these approaches. Whereas communication problems are vitally important, the discussion of related issues must also be addressed to improve psychosocial outcomes.

Communication Strategies Training

There are two basic types of communication exercises: anticipatory strategies and repair strategies. Anticipatory strategies are ones in which techniques are discussed to prevent a communication problem from occurring in the future. A useful method to accomplish this is the 15 Things technique developed by Trychin (Gagné et al., 2019). In this exercise, the leader describes one of the problems identified by a participant during the problem identification exercise and asks the GAR members to generate 15 possible solutions. It is typically a pleasure to lead this exercise as the group members offer common solutions as well as unusual or creative fixes. When group participants generate their own anticipatory solutions, it increases the chance that they will actually employ these techniques in everyday life. In this way, we acknowledge that the participants are experts about their own hearing loss and thus are the most qualified to select the best management techniques. Research has shown that when patients with chronic conditions work collaboratively with their clinicians to generate goal-oriented solutions, they become better at self-management of their condition and thereby improve general outcomes (Bodenheimer, Lorig, Holman, & Grumbach, 2002; Henry, Zaugg, Myers, & Kendall, 2010).

Repair strategies are used to resolve a communication breakdown once it has already occurred (Kaplan, Bally, & Garretson, 1985). An effective method is the one described in Table 18–5. This procedure is a three-step process in which an individual must admit to having a hearing loss, explain his/her communication problems, and then offer a solution. Although this may appear to be simple, its implementation is actually complex. Due to stigma, many individuals with hearing loss feel uncomfortable admitting their disability (Hétu, 1996). With support from the group, however, members may generate the courage to admit to their hearing problem. The next steps of this technique involve describing the reason for the communication breakdowns and generating possible solutions. GAR participants need time and

Table 18-5. Steps for Personal Self-Advocacy

Explain that you have a hearing loss	"I have a hearing loss"	"I wear hearing aids, but I still have difficulty hearing"
Explain the communication problem	"I have difficulty understanding when I cannot see someone's face"	"I especially have trouble understanding in noisy places"
Offer a solution	"Can you make sure that you are looking at me when you speak?"	"Can we go into the hallway and see if it is quieter there?"

Source: Based on Kaplan, Bally, and Garretson (1995).

Case Study 18-6. Antonio: Part B

Communication Strategy Training: 15 Things

During the problem identification exercise, Antonio contributed the following dilemma: "I can't hear well at restaurants. My wife, Marie likes to arrange dinner dates with two other couples, but I hate it when she does this because it is typically at a restaurant and I just can't hear what anyone says due to all the background noise. I end up sitting there like a buffoon staring into space because I just can't participate in the conversations." During the 15 Things exercise, the class members offered lots of creative solutions, including:

- Sit at a round table (rather than a square or rectangle) so you can easily lipread anyone who is speaking with you.
- Create a subtle signal to Marie when you can't hear what is being said so she can rephrase it for you with slow, clear speech.
- Ask Marie to call the restaurant beforehand and request a quieter table, perhaps one in a back corner of the restaurant or an area with soft carpeting or drapes to help absorb all the background noise.

After completing this exercise, Antonio agreed to try some of these techniques and report back during the next class.

Case Study 18-7. James: Part C

Structured Psychosocial Discussion: Worst Thing

When asked, "What's the worst thing about having a hearing problem?" James joked "Not knowing that I had a hearing loss, I just thought I was walking around with a bell choir in my head." He continued in a more serious manner and explained, "I often need to speak with doctors and nurses in confidence when transferring a patient, but I can't hear when people whisper to me." He was surprised when his contribution was met by a huge laugh, after which every single group participant said (in one form or another), "No kidding! I can't hear when people whisper either." After that GAR session, when he realized that his communication problems were typical and there was nothing wrong with him, James reported that he felt better about his hearing problems.

tantly, in the AR group, the participants must be given the opportunity to practice the implementation of strategies. This can be done during structured activities or informal discussions.

There are a number of resources that offer prepared worksheets with formal activities to practice both anticipatory and repair strategies. Some are specifically designed for groups, whereas others are designed for individuals and can easily be adapted for groups. Examples are the Communication Tactics Questionnaire, described by Hickson et al. (2007b) and the "Placement" and "Using Receptive Repair" exercises described by Kaplan, Bally, and Garretson

repetition so they may consider the components of effective communication and discover useful repair strategies. Finally, and perhaps even more impor-

(1985). With the placement exercises, the participants are given a handout showing a room—such as a restaurant, lecture hall, or living room—and asked to consider the best place to sit in that room for effective communication. Additionally, the group may discuss how to get a preferred seat in a particular room (e.g., arrive early at the restaurant or lecture, which is an anticipatory strategy) and how to arrange a room for better communication (e.g., turn down the television, which is a repair strategy).

Self-Advocacy Training

During these activities, participants learn how to advocate for their communication needs in a variety of situations. This can vary from personal communication with familiar or unfamiliar communication partners, to public venues such as the workplace and large auditoriums. Self-advocacy training often leads to a discussion of the Americans With Disabilities Act and the rights of people with hearing loss. The activity described in Table 18–5 was described as a repair strategy, but it is also a type of self-advocacy strategy, as delineating one's needs is a version of self-advocacy. Additional self-advocacy activities are ones in which participants determine the necessary steps to obtain HATs and practice requesting them. For example, during James's GAR participation group, members suggested that he use an amplified stethoscope at work. Self-advocacy training involves the discussion of the steps to figure out, in James' case, how to identify an appropriate stethoscope and how to request that one's employer purchase the stethoscope. During the problem identification exercise, other situations will likely arise in which participants can request special accommodations to improve communication. Topics for discussion may include how to request amplification devices for places of worship or how to negotiate closed captioning on the television with family members.

Speech Perception Training

Speech perception training is typically described as being analytic (syllable- or phoneme-based) or synthetic (sentence- or conversation-based). It can be conducted auditory-alone, visual-alone, or in auditory-visual format, performed in either quiet or noise. While research suggests that individual speech perception training can result in improved outcomes (Henshaw & Ferguson, 2013), Preminger and Ziegler (2008) showed that speechreading and auditory speech perception

training did not result in improved speech recognition when implemented in the group setting. Yet in the group setting, individuals who received speech perception training reported benefits. As a result, group activities can include time spent on synthetic training (e.g., using context, filling in the blanks, considering key words) and on the value of speechreading. The "Concentration" exercise (Preminger, 2011) is one in which participants are shown the benefit of speechreading and listening effort to improve understanding. Irene learned that by seeing her grandchildren's faces she could understand them better. She learned to ask them to say her name first before speaking and to make sure she could see their faces.

Stress Reduction Exercises

It is well established that hearing loss is associated with increased listening effort and fatigue (Alhanbali, Dawes, Lloyd, & Munro, 2018). A full day of effortful listening is stressful. Additionally, individuals with hearing loss may experience stress due to the impact of their hearing loss on their relationships and communication at home, at work, and in the community (Hétu, Jones, & Getty, 1993). It is important to address stress as it can manifest both physically and behaviorally (Trychin, 2002). Therefore, discussion about stress and exercises to relieve it should be integrated into a GAR program. Stress-reduction exercises also help participants manage their emotional response to their chronic condition (Lazarus & Folkman, 1984). One stress reduction activity that can be employed is the practice of controlled breathing. Henry et al. (2010) provide a CD that can be used to teach controlled breathing in a GAR program. While the exercises on the CD are designed for patients with tinnitus, the instructions can be modified to accommodate individuals without tinnitus. (As it is likely that many GAR participants do experience tinnitus, it is a good idea to discuss this fact and consider how tinnitus can contribute to stress.)

Another method for dealing with stress to introduce through GAR programs is the "Right Way/Wrong Way" exercise developed by Trychin (Gagné et al., 2019). This exercise is typically used for communication strategy practice, but this method may also be used for a guided group discussion about methods to deal with stress. Participants can be asked to discuss negative actions (e.g., alcohol or arguments) and positive actions (e.g., exercise or hobbies) and to consider how more time can be spent using positive techniques.

Informational Demonstrations

Informational lectures can be used in GAR, as this is an ideal opportunity to review material for which there is limited time in the typical audiology encounter. Several topics can be prepared for discussion and relevant areas addressed based on the problems identified by the particular AR group. A topic such as HATs can be discussed and various styles of technology can be demonstrated. Topics of interest typically include listening to the television, listening in large rooms (e.g., houses of worship, movies, theaters), and listening over the telephone (e.g., mobile phones, captioned telephones, relay services, state programs for low-cost or no-cost communication technologies). Other topics may include emergency preparedness alarms (e.g., tornado warnings), tinnitus, cochlear implants, and dizziness.

Structured Psychosocial Discussions

There is no need for structured psychosocial discussions to be elaborate. Tucker, Compton, Mankoff, and Rulison (2011) describe a simple weekly exercise that they use in their cochlear implant support programs called “Roses and Thorns.” GAR participants are asked to keep a daily journal documenting their communication successes (roses) and challenges (thorns). During class they share their thoughts and feelings about the high and low points that occurred during the past week. Tucker et al. (2011) report that this activity is useful as it may help generate practical solutions to common communication dilemmas, but more importantly, builds trust among the group members.

Case Study James: Part C demonstrated humor when James participated in the exercise, “What is the worst thing about having a hearing problem?” Based on the work of Stephens and Kerr (2003), it is worthwhile to investigate some positive aspects of living with hearing loss. Participants frequently discuss the humorous situations that may arise when they misinterpret what was said. This often results in a useful discussion on the use of humor to cope with communication breakdowns and family relationships affected by hearing loss. Group members typically are able to come up with other positive aspects of living with hearing loss; for example, not be woken up by thunderstorms or being able to relax at the end of a work day by turning off one’s hearing aids.

It takes practice and feedback to change behaviors. Homework assignments can be used to adapt these new behaviors to everyday life. Participants can be asked to use a specific communication strat-

egy at least three times during the coming week. The next class can begin with a discussion of the homework. Did the strategy result in successful communication? Did admitting that you have a hearing loss make you feel uncomfortable? How did the hostess at the restaurant react when you requested that the volume of the music be lowered? The ensuing discussion is an opportunity to celebrate successful communication and to commiserate over communication breakdowns. This could lead to the generation of new communication strategies, a discussion of the need for self-advocacy, or identification of the next class activity.

Multiple goals may be achieved with a small number of group activities. A single activity can address many of the content areas listed in Table 18–4. For example, the 15 Things activity can be a communication strategies exercise, self-advocacy training, and a psychosocial discussion. Table 18–4 may be used like a cookbook to ensure that each ingredient is included in a GAR program; however, the actual recipe is determined by the facilitator and the problems identified by the group members.

Some research has tried to determine whether choice of GAR content will influence outcomes. Preminger and Yoo (2010) evaluated the effectiveness of four types of activities listed in Table 18–4: communication strategies training, psychosocial activities, stress reduction activities, and informational lectures. Using hearing loss-related QoL as an outcome measure, they found that session content had only a small influence on outcomes; however, there was some evidence that sessions with psychosocial exercises (which included stress reduction activities) resulted in the best outcomes. Larger effect sizes were measured from participants who participated in a GAR program that included training or information plus psychosocial exercises, as compared to participants whose classes included only training or information. This suggests that GAR should include a mix of content, principally activities that focus on problem management (e.g., communication strategies training, informational lectures, and/or identifying and solving problems) and managing one’s emotional response to hearing difficulties (e.g., psychosocial exercises and/or stress reduction). Interestingly, research has suggested that the composition of the GAR members is even more important than the activities in determining outcomes. Preminger and Rothpletz (2016) showed large effective sizes when communication partners participated in GAR, and only small or medium effect sizes when groups consisted only of people with hearing loss.

Measuring Clinical Efficacy of Group Audiologic Rehabilitation

It is important to demonstrate the effectiveness of GAR to determine whether the treatment is having the desired effect on a patient's psychological, social, and emotional adjustments to hearing loss. This instills an underlying justification that it is worth everyone's time and money ("everyone" may include the patient, the audiologist, and potentially the employer). Demonstrating effectiveness of GAR is also important for determining where to spend health care dollars for things such as insurance reimbursement or research funding. In addition, an outcome measure such as a self-assessment tool can in itself be a positive experience. Kricos (2000c) postulated that "the administration of the self-assessment itself could have a treatment effect by making the subject more aware of problems, thereby helping them problem solve themselves" (p. 77).

To show effectiveness of GAR, it is important to understand how to assess outcomes and the obstacles associated with these measurements. Since emotion- and problem-focused coping strategies are critical in GAR, it makes sense to measure abilities in these areas before and after treatment. Assessment areas can include activity limitations and participation restrictions related to hearing loss, adjustment to hearing loss, and increased use of communication strategies. To demonstrate effectiveness clinically, outcome measures designed for hearing loss specifically, such as the HHIE or the Hearing Handicap Inventory for Adults (HHIA) (Newman, Weinstein, Jacobson, & Hug, 1990) are often used as they are sensitive to the psychosocial effects of hearing loss (Abrams, Chisolm, & McArdle, 2005). Other hearing loss-specific outcome measures include the International Outcome Inventory-Alternative Interventions (IOI-AI) (Noble, 2002). It measures seven items, including activity limitation, participation restriction, and QoL, and it

has been shown to have good internal consistency for individuals undergoing a GAR program (Hickson, Worrall, & Scarinci, 2006). As discussed in the section on problem identification exercises, the use of a modified COSI can be helpful in setting goals for GAR treatment. One scale similar to a modified COSI that has been used successfully is the Goal-Sharing for Partners Strategy (GPS) (Preminger & Lind, 2012). The GPS is a goal-setting tool in which an individual with hearing loss and a family member are asked together to consider various situations that are conducive to successful communication as well as problems that each person experiences due to hearing loss. The two participants are also asked to set specific goals they would like to achieve in regard to communication and the steps that they can take to achieve their goals. Case Study: Irene Part C demonstrates how Irene and her daughter used the GPS successfully as a part of their GAR program.

As noted previously, the psychosocial aspects of hearing loss are multidimensional and may include the emotional, cognitive, interpersonal, behavioral, and physical responses to hearing loss (Trychin, 2002). Thus, benefits of GAR can be easily missed with a measure of hearing loss related QoL (Preminger, 2007). Successful GAR may improve an individual's self-efficacy or self-advocacy, or it may convince a participant to purchase a hearing aid. Thus, a qualitative assessment of the GAR (open-ended questions) is also important. Case Study: Antonio Part B demonstrates a quick qualitative assessment of Antonio and Marie's experience with GAR. A simple form with one or two open-ended questions that can be filled out at the last session can help to show the Audiologist that GAR was effective; additionally, it can help the participants to reflect on the experience and understand its impact. Alternatively, one can put together a test battery (Barker et al., 2015); however, this may not be necessary for clinical purposes.

Case Study 18–8. Irene (and June): Part C

Effectiveness of GAR

During the initial GAR session, Irene and June completed the GPS (Table 18–6). This experience gave each of them important "aha" moments, as they had never considered the other's point of view. June did not realize that her mother was embarrassed by her hearing loss and Irene did not realize the extent of June's frustration with Irene when she had to repeat herself so many times. This discussion helped Irene to understand that her inaction could cause problems with her relationships. As a result, she started GAR with an open mind.

Table 18–6. Goal-Sharing for Partners Strategy Completed by Irene and Her daughter June at Their First Group Audiologic Rehabilitation Session

Person With Hearing Loss (Irene)	Communication Partner (June)
<i>Where do you find communication most easy?</i> When the grandkids are in the basement and the TV is turned off.	<i>Where do you find communication most easy?</i> When the two of us are at the kitchen table having a cup of coffee.
<i>What problem do you experience because of hearing loss?</i> I worry that if I wear hearing aids people will think I have dementia.	<i>What problem do you experience because of hearing loss?</i> I have to keep explaining to my kids how to talk to Nana so that she will understand them.
<i>What problem does your partner experience because of hearing loss?</i> She seems to get irritated when I ask her to repeat something more than once.	<i>What problem does your partner experience because of hearing loss?</i> I am frustrated that she won't do anything about her hearing problems. It feels like she doesn't value our relationship enough.
<i>What problems do you both experience?</i> We want to be able to have easy conversations when it is just the two of us and when we have a family dinner.	
Shared goals	
1. Be able to communicate successfully during spur of the moment conversations at home 2. Communicate successfully with grandchildren at the dinner table	
Steps to achieve the goal	
1. Go to GAR and learn what our options are 2. Get a hearing test at the audiologist's office	

Case Study 18–9. Antonio (and Marie): Part C

After the GAR program, Marie and Antonio were contacted by email and asked to each write down what they thought about the benefits and drawbacks of attending GAR. Marie wrote the following:

Last week Antonio and I discussed the Communication Classes. I told him I noticed his hearing seems to be less of a problem, and this is my theory. After the classes with you, he seemed to relax about his hearing problem, especially because he realized his problem was not as bad as the others in the class; that made us both feel better. He seems to hear more because he's not stressing about what he hears and doesn't hear. He has been more social when we're out with friends, although background noise is still a problem, but he interacts more than he did. He said he noticed himself more at ease and even at the neighborhood association meeting, things are better. He felt he had a block and now was more relaxed, not so bothered about the deafness.

Based on Marie's comments it seems that both Antonio and Marie's ability to employ social comparison during the GAR session influenced the positive outcomes. This demonstrates the importance of open-ended feedback with GAR participants.

Future Directions

A few investigators have been evaluating GAR on the internet. Thorén et al. (2014) evaluated a 5-week internet-based AR program for experienced hearing aid users who reported residual hearing handicap as measured by the HHIE (Ventry & Weinstein, 1982). Most of the program was individually based and included weekly readings about the anatomy of hearing, hearing aids, communication strategies, hearing assistance technologies, relaxation techniques, and information for frequent communication partners. The individual components also included listening training and communication strategy training plus weekly coaching emails from an audiologist. These activities are quite similar to those described here, but they were done individually and with asynchronous communication and coaching by an audiologist (Thorén et al., 2014). There was a pseudo group component to the program, however, as all (experimental) participants were invited to participate in an internet-based discussion forum that was open only to the participants in this study. There was no leader or audiologist participation in the forum; rather, participants were invited to remark on a new discussion topic each week. For example, in week 4 the discussion topic was "Describe a situation where you have used a strategy that has worked and describe a situation where you have used a strategy that has not worked" (Thorén et al., 2014, p. 456). The experimental group showed significant improvements on scales of hearing loss-related QoL that were not seen in a control group that received a reading assignment about the history of hearing aids. This is interesting, as most of the activities were similar to those done in a typical group; however, it was only the discussion forum that had a grouplike aspect to it. It is difficult to assess how important the discussion forum was to the participants as only approximately 50% posted each week and 40% replied to other posts. Malmberg and colleagues (2018) contacted all 36 participants from the Thorén et al. (2014) study and 20 agreed to participate in individual qualitative interviews in which they were asked to discuss their thoughts about the internet-based AR program. Qualitative analysis of the data revealed the overarching theme: "Increased self-esteem as a process of becoming more knowledgeable about [hearing loss] and having confidence in the ability to deploy knowledge practically" (Malmberg et al., 2018, p. 571). Several of the participants reported that they did not participate in the discussion forum; some found it technologically difficult to participate in the online forum. Whereas a minority of participants reported that it was ben-

eficial to share experiences in the discussion forum, a majority found value in reading others' postings and to learn "that you're not alone with a feeling" (Malmberg et al., 2018, p. 573).

Can unmoderated discussion forums offer the same type of support as that found in in-person AR groups? The authors are skeptical. Recently, the authors met with members of Hearing Loss Association of America and asked if they would be willing to offer peer support in an online forum for individuals who are deciding whether or not they wish to receive hearing health care. The authors posed questions to the group that they expect to see in such a forum, and were surprised to see that some experienced hearing aid users had many questions about hearing aid options and use themselves. Some authors have looked into the value of unmoderated discussion forums and the results are equivocal. Cummings, Sproull, and Kiesler (2002) sent questionnaires to subscribers of "Beyond Hearing," a subscription distribution list that served as an electronic support group for people with hearing loss. The results showed that participants who had little support for their hearing loss-related problems at home were most likely to participate in the forum and most likely to report emotional and informational benefits from participation in the online group. Interestingly, some participants reported having family members who also participated in the group, and those individuals were more likely to report having supportive family members (Cummings et al., 2002). This is another example of the value of frequent communication partners even in an online, unmoderated support group. Ainscough et al. (2018) took a more recent look at online support groups and performed a qualitative analysis of postings from four open discussion forums for individuals with tinnitus. These sites did have forum moderators who answered questions and monitored acceptability of posts. The authors noted both positive and negative consequences of participation in the forums. Positive consequences included a place to exchange knowledge, to share experiences, and to express emotions. Negative consequences included the exchange of messages that were sarcastic or damaging. Other negative aspects were when there was disagreement among participants about information or opinions and the offering of inaccurate information (Ainscough et al., 2018).

The presentation of GAR within a multidisciplinary program is another new direction. In their study protocol for the "Walk, Talk 'n' Listen" (WTL) project, Lambert and colleagues (2017) point out the well-documented association between hearing loss and functional fitness. For example, hearing loss or

reported hearing handicap in older adults has been associated with increased sedentary behavior, increased difficulties with activities of daily living (ADL), diminished functional status, decreased walking speed, and increased falls (Lin & Ferrucci, 2012; Loprinzi, 2013; Tomioka et al., 2015). Thus, the 10-week (20-session) WTL program was created. It includes socialization with other group members, health education, exercise, walking, and GAR (as based on the GROUP program, Montano, Preminger, Hickson, & Gregory, 2013). The WTL program is designed to improve functional fitness, increase social connectedness, reduce social isolation and improve hearing loss related QoL (Lambert et al., 2017). Preliminary qualitative evidence of the WTL program (without the GAR component) indicates that participation in the program decreased feelings of loneliness, improved psychosocial well-being, and increased physical well-being (Hwang, Wang, Siever, Medico, & Jones, 2019). We look forward to seeing the results of future investigations.

Conclusion

In addition to realizing the financial and psychosocial benefits of GAR, by providing these services, an audiologist is promoting acceptance of hearing loss. A group setting promotes reduction of stigma associated with hearing loss, not only in individuals with hearing loss, but also in the people these individuals interact with and encounter in their daily lives. These individuals become advocates for themselves as they spread some level of this new knowledge and empowerment in their community.

Communication plays an essential role in the human condition and it has been clear for some time that there is no simple fix for hearing loss-induced communication problems. GAR is a cost-effective way to increase patient acceptance and adjustment to the social, emotional, and psychological effects of hearing loss. GAR can be an ideal way to improve and increase audiologist/patient communication. Many individuals with hearing loss desire a forum for their hearing loss story to be heard. While it may be difficult to find time for this in the clinic setting, GAR can be the perfect opportunity for everyone to share his or her experience. GAR can be personally rewarding and can create a venue to foster improved relationships between audiologists, patients, and communication partners, often leading to new insights about hearing impairment and communication.

Additional information about practical issues to consider when implementing GAR, including cost effectiveness and patient recruitment, are available through the Ida Institute (<http://www.idainstitute.com/group>). The GROUP sample was developed by members of the Ida Institute and several of its Fellows (Jean-Pierre Gagné, Patricia Kricos, Jane Madell, Joseph Montano, Jill Preminger, and Sam Trychin) to assist audiologists in the implementation of GAR. The GROUP webpage contains sample activities and video clips of some of these activities in action.

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19

INCORPORATING COMMUNICATION PARTNERS INTO THE AUDIOLOGIC REHABILITATION PROCESS

Jill E. Preminger and Joseph J. Montano

Audiologic rehabilitation extends well beyond the fitting of technology to improve hearing ability. The journey from identification of problems associated with hearing loss to acknowledging the need for assessment, considering the treatment options and, finally, living with and adjusting to hearing loss is an individual experience. Gregory (2012) highlights the importance of the patient journey as a tool to better understand the process a person undertakes toward acceptance of hearing loss. This is seldom a solo journey, affecting not only the patient but also all those involved in day-to-day interactions, from a spouse and family to friends, work colleagues, or acquaintances (Manchaiah, Stephens, & Meredith, 2011). Trychin (2010) states “it is not hearing loss but rather a communication loss.”

As this journey progresses and the patient begins to seek treatment, he/she is often encouraged and supported by the people in his/her life. The impact of hearing loss is far reaching and affects participation at home, at work, and in social situations. In a study of older adults with hearing loss, Kramer et al. (2002) identified an association between hearing loss and significant psychosocial issues such as depression, self-efficacy, social networking, and loneliness. As a result of their study, the authors recommended that a significant other be included in the AR process to lessen the impact of hearing loss on social function-

ing and to promote maintaining relationships. Scarinci, Hickson, and Worrall (2011) report that hearing loss not only affects the individual, but spouses also encounter an increase in limitations and restrictions in activities and participations. As a result of their research, the authors advocate for family-centered intervention in audiologic rehabilitation (AR) with the inclusion of spouses and/or supportive individuals in the process. The term *family* has been broadly defined to include anyone with whom the person with hearing loss (PHL) closely aligns. This could be a spouse, a child, a friend, or a work colleague (Singh et al., 2016).

As hearing loss is one of the most prevalent chronic conditions for older adults and the presence of a significant other in the management of health-related issues has been shown to have a positive impact (Gallant, 2003; Seeman, 1996), it stands to reason that family participation can influence success with AR interventions. Management of hearing loss may, therefore, involve much more than working only with the actual individual with the loss, but rather expands the role of the audiologist to include counseling and education of the people involved in the patient's life. The significant other, particularly the spouse, can be an important catalyst for moving the patient forward on his/her journey toward living well with a hearing loss (van den Brink, Wit, Kempen,

& van Heuvelen, 1996). Involving others in the care and treatment necessitates rethinking the manner in which audiology services are provided.

Traditional medical or biomedical models of treatment imply that the audiologist assess hearing and provide a curative philosophy to the identified loss. Essentially, the treatment decision is provided by the audiologist and the patient is informed of the solution—in most cases, amplification. In some situations, this includes deciding upon the style and model of hearing aids (Erdman, 2014; Erdman, Wark, & Montano, 1994; Montano, 2012). Inclusion of others in this process is limited to informing them of a decision already made. In contrast, service delivery using a rehabilitation or biopsychosocial model implies a shared decision process whereby treatment is recommended after discussion of options and an exploration of the issues related to the life of the person with hearing loss, including the communication partners and the environments in which communication takes place (Erdman, 2014; Erdman et al., 1994; Montano, 2012). (*Editors' note:* For further discussion of the biopsychosocial model, the reader is referred to Chapter 11 of this text.)

Significant Other/ Communication Partner

The term *significant other* (SO) is seen throughout health care literature and used frequently by professionals to represent “an individual who has an important role and influence in the shaping of behavior of another” (Montano & Al Makadma, 2012, p. 47). A significant other can be anyone of importance in a person's life, but usually refers to a spouse, partner, child, sibling, or close friend. There is an abundance of literature on the impact of hearing loss on the significant other, with that person most often being a spouse. Recently, new research is beginning to surface on the impact of a parent's hearing loss on their adult children (Heacock, Montano, & Preminger, 2018; Preminger, Montano, & Tjornhoj-Thomsen, 2015). Moreover, hearing loss has an impact well beyond a person's home environment to other areas where communication takes place, such as work and social settings (Noble & Héту, 1994). For this reason, the term *communication partner* (CP) has gained popularity in the audiology literature. Montano and Al Makadma (2012) refer to communication partnerships as the “relationships between individuals and groups involved in activities within a communication

environment” (p. 48). With the use of this terminology, a person with hearing loss visiting an audiologist with his/her spouse would engage in communication within the audiology office environment and the communication partners would include the husband/wife and the audiologist. The audiologist is a vital partner to both the patient and the spouse. During this communication exchange, the audiologist has the opportunity to explore both the audiology and nonaudiology issues related to the hearing loss and develop a keener awareness of the impact on the communication partner. The term *family-centered care* has begun to be more commonly used in literature and practice. Evidence has shown that outcomes are improved when family is involved in the rehabilitation process (Rathert, Wyrwick, & Boren, 2013; Singh et al., 2016). (For additional information of family-centered care, the reader is referred to Chapter 16 of this text.)

Impact of Hearing Loss on Communication Partners

Prior to considering the impact of hearing loss on a CP, it is useful to consider terminology. Differing language has been used to describe the experience of having a partner with hearing loss. Stephens and Héту (1991) used the term *secondary handicap*, which they described as the adjustments employed by CPs to compensate for the hearing disability. As the World Health Organization (WHO) language regarding functioning and disability has changed, so has the terminology used to describe the experience of the CP (Scarinci, Worrall, & Hickson, 2009b). The impact of dealing with a partner with hearing impairment is now termed *third-party disability*, which can be described as the activity limitations and participation restrictions that result from a partner's hearing disability (Scarinci et al., 2009b).

In one of the first studies to consider the impact of hearing impairment on CPs, Héту, Lalonde, and Getty (1987) developed a questionnaire to measure the awareness of hearing disability and the use of coping strategies by men with noise-induced hearing loss (NIHL) and by their spouses. They noted that the entire family had the responsibility of coping with the hearing disability; however, it was observed that both the PHLs and their spouses did not necessarily develop good coping strategies. In a follow-up study (Héту, Riverin, Lalonde, Getty, & St-Cyr, 1988), the authors noted CPs had unique needs that could be addressed via AR. Later, Héту and colleagues

reported CPs may actually impede this AR process by minimizing the problems caused by the hearing loss (Héту, Riverin, Getty, Lalande, & St-Cyr, 1990).

Héту, Jones, and Getty (1993) described three types of communication breakdowns that may significantly impair a couple's relationship in which one member has a hearing loss. First, there are the misunderstandings on the part of the CP when his/her PHL does not answer a question or responds inappropriately. In cases such as this, the CP may attribute the PHL's misunderstanding to a lack of attention or a lack of interest rather than due to hearing loss. Second is the reduction in interactions between the CP and the PHL; this includes both a reduction in intimate communication and in everyday companionship. Third is the restriction in the content of the communication; as so much effort is required for communication, conversation may be restricted to the essentials.

More specifically, authors have commented on the impact of hearing loss on CPs in dyads in which the partners are just discovering the hearing impairment versus in dyads in which the hearing loss is well established. When the hearing loss is just emerging, there are frequent misunderstandings in communication and the partners may struggle to come up with reasons for these breakdowns. CPs may (wrongly) believe their PHL doesn't care enough to listen carefully (Héту et al., 1993). In these couples, the PHLs may minimize the presence of the hearing loss so as not to acknowledge the stigma (Héту et al., 1993). CPs may contribute to this denial of hearing loss by minimizing the problem and its consequences (Héту et al., 1993). Alternatively, adult children of parents with HL have reported uncertainty related to the emerging HL (Preminger, Montano, & Tjornhoj-Thomsen, 2015); they are uncertain whether their parents' lack of understanding is due to a true HL or another health condition such as cognitive decline or dementia.

Hallberg (1996) interviewed men with severe NIHL and their spouses. It was noted that these men were often unwilling to acknowledge their hearing difficulties, yet their loss did cause them to change behaviors, such as avoiding social activities. The avoidance of activities affected both the PHLs and their CPs; thus, it was concluded that family involvement in AR was necessary to assist PHLs in acknowledging and coping with their hearing disability (Hallberg, 1996). Once the hearing loss is acknowledged, however, its impact continues and the effects may be wide-ranging. The effect of hearing loss on the CP may be categorized by the emotional, social, and behavioral effects as well as a direct impact on the relationship between the CP and the PHL.

The emotional effects are listed in Table 19-1 and have been confirmed by a variety of researchers. These findings are based on qualitative research studies in which the CPs were interviewed and allowed to speak freely about living with a partner with hearing loss. In some studies, the partners are spouses (Héту et al., 1993; Scarinci, Worrall, & Hickson, 2008); in one they were adult children of parents with HL (Preminger et al., 2015), and in two they included family members and friends (Hallam, Ashton, Sherbourne, & Gailey, 2008; Manchaiah & Stephens, 2013). The emotional effects are numerous and include reports of feeling frustrated, irritated, angry, and resentful. As a result, many CPs also feel guilt for having negative emotions, for being impatient with their spouse, or for employing negative communication strategies such as yelling (Héту et al., 1993; Preminger et al., 2015). CPs may also feel sympathy for their spouses due to the need to cope with the hearing loss and subsequent communication difficulties (Scarinci et al., 2008). Finally, CPs often feel concern for their partner's safety due to the inability to hear warning signals (Hallam et al., 2008; Preminger et al., 2015; Scarinci et al., 2008). It is interesting to note that the emotion of uncertainty only arose in the study of adult children of parents with HL (Preminger et al., 2015); adult children wondered whether hearing loss or some other factor could explain their deteriorated communication with their parent. In these two examples, one adult daughter wondered if it is hearing loss or a lack of interest in her life that may explain her mother's communication: "It's hard to know if she's just bored and ready to move on to the next thing or, if she really can't hear what I'm saying." Another adult daughter wondered if cognitive decline or hearing loss can explain her mother's communication: "I think the hearing really had become a big problem for her; I was really worried that she was becoming senile and I'm not concerned about that now."

Some research has specifically focused on positive emotions of CPs. Manchaiah and Stephens (2013) asked CPs to list any positive feelings or events that they experienced due to their partner's hearing loss. CPs reported increased patience and tolerance, increased awareness and knowledge of hearing loss, reduced stigma related to hearing loss, and improved communication abilities. Heacock, Montano, and Preminger (2018) followed up on this study by asking 48 adult children if they experienced the positive feelings and events described by Manchaiah and Stephens (2013). The most commonly reported experiences were increased understanding and awareness of hearing loss, feeling more helpful or useful, improved

Table 19–1. The Impact of Hearing Loss on Communication Partners

▪ **Emotional responses reported by CPs**

- Stress associated with tolerating loud television volume¹
- Tension and worry because PHL cannot hear emergency signals or telephone¹
- Irritation and embarrassment at social gatherings¹
- Fatigue associated with:
 - having to continually repeat oneself and having to put forth extra effort to be understood^{1–3}
 - having to act as an interpreter in public¹
 - always being the one to answer the telephone¹
- Frustration, anger, resentment, and anxiety associated with:
 - restriction of social activities¹
 - reduction in intimate conversations¹
 - the need for frequent repetitions²
 - being the one to accommodate the hearing loss²
 - communication difficulties and the need to change one's communication behaviors^{4,5}
- Guilt over:
 - feeling impatient due to repeated misunderstandings¹
 - believing PHL is not making enough effort to understand¹
 - poor use of communication strategies⁵
 - lack of involvement in the hearing health care process⁵
- Uncertainty associated with:
 - whether or not partner with HI truly understood⁵
 - whether hearing loss or a separate health condition could explain poor communication⁵
- Concern for partner with hearing loss due to:
 - the PHLs difficulty in social situations²
 - inability to respond to warning signals^{2,3,5}

▪ **Behavioral responses reported by CPs**

- The need to tolerate a high television volume^{2,3}
- The extra effort of being the only one to answer the telephone²
- The need to speak for one's partner over the telephone³
- The need to speak loudly and to repeat oneself frequently^{1–3}
- The effort needed to make face visible²
- The extra effort required for communication³
- The need to take over public activities such as shopping³

▪ **Social impact reported by CPs**

- Tendency to stay home, reduction in social activities^{1–3}
- Less enjoyment of social activities when need to interpret³
- The need to act as an interpreter in social activities¹
- Giving up certain activities such as dining with a group³

¹Hétu, R., Jones, L., & Getty, L. (1993). The impact of acquired hearing impairment on intimate relationships: Implications for rehabilitation. *Audiology*, 32(6), 363–381.

²Scarinci, N., Worrall, L., & Hickson, L. (2008). The effect of hearing impairment in older people on the spouse. *International Journal of Audiology*, 47(3), 141–151.

³Hallam, R., Ashton, P., Sherbourne, K., & Gailey, L. (2008). Persons with acquired profound hearing loss (APHL): How do they and their families adapt to the challenge? *Health*, 12(3) 369–388.

⁴Manchaiah, V. K. C., & Stephens, D. (2013). Life consequences and positive experiences reported by communication partners of people with hearing impairment: A pilot study. *Speech, Language, and Hearing*, 16(1), 2–8.

⁵Preminger, J. E., Montano, J. J., & Tjornhoj-Thomsen, T. (2015). Adult-children's perspectives on a parent's hearing impairment and its impact on their relationship and communication. *International Journal of Audiology*, 54(10), 720–726.

communication skills, and increased patience and tolerance related to communication.

Here, and in Table 19–1, we describe the behavioral impact of having a partner with hearing loss. CPs must continually put forth extra effort to ensure successful communication. This includes the need to: get up and walk over to one's PHL to make speechreading available, repeat oneself frequently, and speak loudly (Brooks et al., 2001; Héту et al., 1993; Scarinci et al., 2008). This can become exhausting when these behaviors are required recurrently. CPs may also need to take over their partner's previous duties, such as speaking on the telephone or shopping in the community (Hallam et al., 2008; Scarinci et al., 2008).

Hearing loss has a great impact on the social life of the CP (see Table 19–1). Their social lives are reduced and restricted as their PHLs avoid social situations such as parties and restaurants (Brooks et al., 2001; Hallam et al., 2008; Héту et al., 1993; Scarinci et al., 2008). CPs experience the added stress and burden of having to act as an interpreter at social gathering and in other public situations, such as at the doctor's office (Héту et al., 1993).

Table 19–2 shows how CPs describe the impact of hearing loss on their relationship with their PHL. CPs spend less time with their PHL and less time communicating (Brooks et al., 2001; Héту et al., 1993; Scarinci et al., 2008). When the CP finds that is it too much effort to share a funny story that happened at work or it is just too difficult to relate a phone conversation with a family member, CPs experience a loss of intimacy and spontaneity in the relationship with their PHL (Héту et al., 1993; Scarinci et al., 2008). Finally, CPs express a feeling of loss due to changes in their relationship due to the hearing loss. Adult children discuss the loss of the relationship with their parent that existed prior to the hearing loss. One participant noted "We are just like two people going someplace, we are not engaging, and that's not the way it was" (Preminger et al., 2015, p. 725).

Measurement of Hearing Loss-Related Quality of Life (HLQoL) in Communication Partners

We can quantify the effects of hearing loss described in the previous section in a number of ways. Here, we review three ways in which hearing loss-related quality of life (HLQoL) can be measured within members of a couple in which one of the partners has hearing loss. The participation restrictions and activity limita-

Table 19–2. The Impact of a Hearing Loss on the Family Relationship as Reported by the CP

■ The impact on the relationship

- Reduced communication²
- Increased tension²
- Spend less time together^{2, 3} and less time conversing¹
- Increased vulnerability to failure of the relationship³
- Loss of spontaneity³
- Loss of intimacy^{1, 2, 4}
- Loss of the relationship that existed prior to the hearing loss including shared activities⁴

¹Héту, R., Jones, L., & Getty, L. (1993). The impact of acquired hearing impairment on intimate relationships: Implications for rehabilitation. *Audiology*, 32(6), 363–381.

²Scarinci, N., Worrall, L., & Hickson, L. (2008). The effect of hearing impairment in older people on the spouse. *International Journal of Audiology*, 47(3), 141–151.

³Hallam, R., Ashton, P., Sherbourne, K., & Gailey, L. (2008). Persons with acquired profound hearing loss (APHL): How do they and their families adapt to the challenge? *Health*, 12(3) 369–388.

⁴Preminger, J. E., Montano, J. J., & Tjornhoj-Thomsen, T. (2015). Adult-children's perspectives on a parent's hearing impairment and its impact on their relationship and communication. *International Journal of Audiology*, 54(10), 720–726.

tions associated with hearing loss can be measured in the PHL. This is often measured by the Hearing Handicap for the Elderly (Ventry & Weinstein, 1982) in individuals 65 and older and by the Hearing Handicap for Adults in those younger than 65 (Newman, Jacobson, Hug, Weinstein, & Malinoff, 1991). As these scales are very similar (three out of 25 questions are different between the two scales), they will be referred to here as the HHI. These 25-item scales measure the emotional effects and the social/situational effects of living with hearing loss. For example, an item from the emotional subscale is "Does your hearing problem cause you to feel embarrassed when meeting new people?" while an item from the social subscale is "Does your hearing problem cause you to avoid groups of people?" PHLs respond yes (4 points), sometimes (2 points), or no (0 points) to each question; a total score can range from 0 (no reported hearing handicap) to 100 (extreme hearing handicap).

In the health care literature, proxy ratings of health-related quality of life (QoL) are typically given by family members when the person with a specified condition cannot complete a report. For example, in

patients with dementia, a caregiver can report QoL (Hickey, Barker, McGee, & O'Boyle, 2005). However, in the hearing impairment literature, proxy reports are also used to determine how the CP perceives the activity limitations and participation restriction experienced by their PHLs. Proxy reports of HLQoL can be measured with the HHI-SO, where the SO indicates "significant other" (Newman & Weinstein, 1988). To reiterate, in the HHI the PHL may respond to a question such as "Does a hearing problem cause you to feel embarrassed when meeting new people?" while in the proxy version (HHI-SO) the CP responds to the question "Does a hearing problem cause your SO to feel embarrassed when meeting new people?" (Newman & Weinstein, 1988; Ventry & Weinstein, 1982).

It is often informative to consider the similarities or differences in HLQoL reports of PHLs and proxy reports of their CPs. Figure 19-1 shows the results when PHLs completed the HHI and CPs (mostly spouses) completed the HHI-SO across six different studies. Newman and Weinstein (1986) were the first to measure reports of HLQoL within members of a couple and they found that PHLs reported significantly more hearing handicap than that reported by their CPs. The authors noted that "Although an individual may be affected by a family member's hearing loss, the spouse may not be able to understand the total impact of hearing loss on a hearing-impaired person's psychosocial well-being" (p. 113). These authors raised two important points. The first

is that proxy reports of HLQoL do not measure the third-party disability that arises from being married to a PHL (this will be discussed in greater detail in a later section). The second point is that HHI scores in the PHL are often incongruent with the HHI-SO scores in the CP (Preminger & Meeks, 2010a). Here, we will assume that scores within 14 points are congruent, based on 90% critical differences for PHLs on the HHI as reported by Chisolm et al. (2005).

Across the six studies shown in Figure 19-1, four of the studies found significantly greater reports of hearing handicap in the PHLs and two of the studies reported significantly greater proxy reports of hearing handicap in the CP. Numerous reasons have been proposed for these discrepant findings: personality factors (Newman & Weinstein, 1986; Preminger & Meeks, 2010b), degree of hearing loss in the PHL (Newman & Weinstein, 1986), and degree of hearing handicap reported by the PHL (Preminger, 2002). Another important factor to consider is that in the two Chmiel and Jerger studies (1993, 1996), the majority of PHLs had not yet sought amplification for their hearing loss, while the majority of PHLs in the other four previously cited studies were hearing aid users; thus incongruities in HLQoL within some couples may be explained by the stage of the PHL in the patient journey (Manchaiah, Stephens, & Meredith, 2011).

Based on the data shown in Figure 19-1, it is evident that in some couples a CP may underreport a partner's HLQoL, while in other couples a spouse

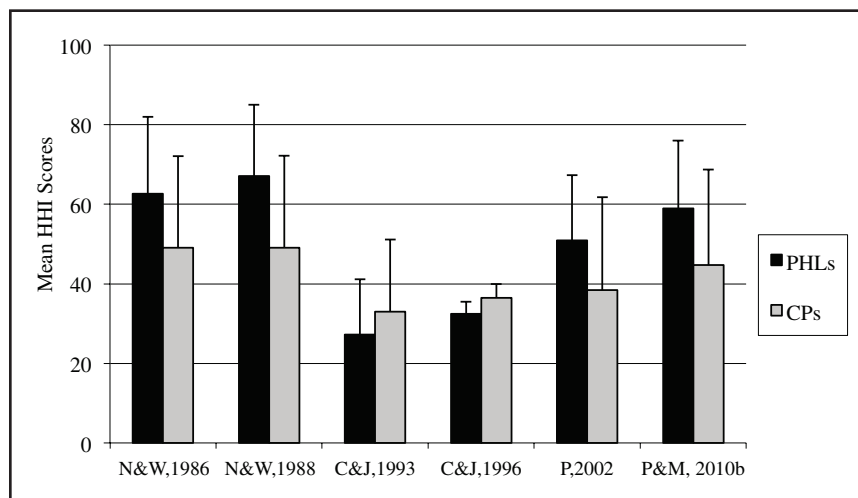


Figure 19-1. HHI scores in PHLs and proxy reports in CPs. *Note.* This figure shows the Hearing Handicap Index (HHI) scores in PHLs (black bars) and proxy reports in CPs (gray bars) across six different studies: Newman and Weinstein (1986), Newman and Weinstein (1988), Chmiel and Jerger (1993), Chmiel and Jerger (1996), Preminger (2002), and Preminger and Meeks (2010).

may overreport it. It is useful to consider congruence in perceptions of HLQoL as this may have implications for treatment (Preminger & Meeks, 2010b). For example, a CP whose proxy report underestimates his/her spouse's HLQoL may be less likely to practice effective communication strategies and may not encourage his/her partner to seek AR. In fact, difference scores between HHI and HHI-SO can be measured before and after AR to measure AR effectiveness (Newman & Weinstein, 1988; Preminger & Meeks, 2010a).

Earlier in the chapter, third-party disability was defined as the participation restrictions and activity limitations that arise from being partnered to a PHL. These effects are outlined in Tables 19-1 and 19-2. There are two known scales available to measure this third-party disability: the Significant Other Scale for Hearing Disability (SOS-HEAR) (Scarinci, Worrall, & Hickson, 2009a) and the Hearing Impairment Impact—Significant Other Profile (HII-SOP) (Preminger & Meeks, 2012). The HII-SOP is a 20-item scale with three subscales and is shown in Appendix 19-A. Relationships and Emotions is an 11-item subscale that describes the impact of the hearing loss on the marital relationship and the emotions associated with hearing loss reported by the spouse; for example, “Do you get irritated when you try to talk with your SO

but she/he cannot understand you?” and “Because of your SO's hearing loss, do you talk less often than you used to?” Communication Strategies is a five-item subscale that describes the strategies employed by the CP to improve communication; for example, “Do you have to make sure your SO is looking at you when you speak to him/her?” Social Impact is a four-item subscale that describes the effect of the hearing loss on the social life as experienced by the CP; for example, “Do you and your SO avoid going to restaurants because of your SO's hearing loss?” CPs can respond yes (5 points), sometimes (2.5 points), or no (0 points) to each question. As the HII-SOP was modeled on the HHI and HHI-SO, scores can range from 0 to 100. The HII-SOP has adequate internal consistency reliability, construct validity, and test-retest reliability (Preminger & Meeks, 2012). The 95% critical difference value is 19.7 points; accordingly, scores of 20 to 39 reflect mild third-party disability, scores of 40 to 59 reflect moderate third-party disability, and scores greater than 60 reflect severe third-party disability associated with hearing loss (Preminger & Meeks, 2012).

We can consider the HLQoL scores for two couples to understand the value of making these measurements in both the PHL and the CP. The black bars in Figure 19-2 show the HLQoL before the PHL

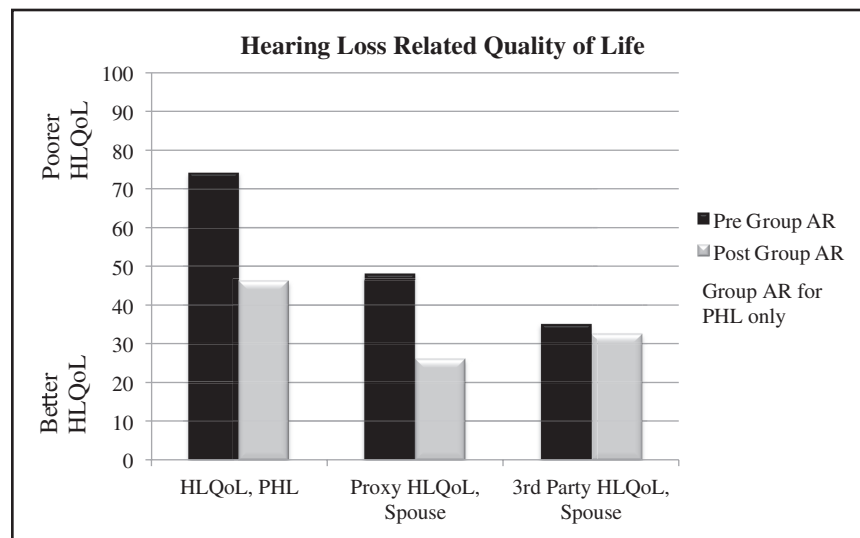


Figure 19-2. HLQoL scores measured before and after GAR program participation—Case No. 1. *Note.* This figure shows the HLQoL scores measured in a 63-year-old female with hearing loss and reported by her husband before the person with hearing loss completed a group audiologic rehabilitation (GAR) program (*black bars*) and 6 months after completing the program (*gray bars*). HLQoL in the PHL loss is measured with the HHI. Proxy reports in the spouse are measured with the HHI-SO. Third-party disability reported by the spouse is measured by the HII-SOP.

completed a four-session group AR (GAR) program. HLQoL in the PHL was measured with the HHI, the proxy report in the CP was measured with the HHI-SO, and the third-party disability report in the CP was measured with the HII-SOP. In this case the PHL is a 63-year-old female with symmetrical hearing loss (three-frequency pure tone average [PTA] = 78 dB HL). As measured by the Quick Speech in Noise test (QuickSIN) (Etymotic Research, 2001), this PHL needed sentence materials at a 19 dB signal to noise ratio (SNR) to understand 50% of the key words correctly. In other words, this PHL has a severe hearing loss with poor speech understanding in noise. At the time of the first evaluation, this PHL had worn binaural hearing aids for 25 years. Prior to participating in GAR, reports of HLQoL were incongruent. The wife with hearing loss reported an HHI score of 74, and her husband's proxy report was 26 points lower. Additionally, the husband reported only a mild third-party disability with a HII-SOP score of 35. After completing the group AR program, the PHL's HLQoL score improved by 28 points, a significant improvement (Newman, Weinstein, Jacobson, & Hug, 1991). Additionally, the proxy report of the CP also showed an improvement in HLQoL; consequently, the reports of the PHL and CP remained incongruent post GAR

with a difference score of 20. Note that the third-party disability reported by the CP did not change post-GAR. This is understandable, as the CP did not participate in the program. In summary, while the HLQoL reported by the PHL did improve, this individual is still reporting a mild to moderate hearing handicap. The fact that the post-treatment proxy report is still 20 points lower in the CP suggests that this CP still does not understand the participation restrictions and activity limitations experienced by his wife. Thus, he may not understand that his communication behaviors could influence his wife's HLQoL.

Figure 19-3 shows findings for another couple before and after each member of the couple completed a GAR program. The PHL, a 71-year-old female, attended a traditional AR program and her husband attended a four-session program designed specifically for CPs (Preminger & Meeks, 2010a). This PHL's binaural PTA is 48 dB HL and her QuickSIN score is 6 dB. She had worn binaural hearing aids for 10 years when the results shown here were measured. Prior to GAR participation, we see a very large incongruence between the HLQoL reported by the PHL and the proxy report of her husband, as the difference between these ratings was 60 points. This CP did not report that his wife had any participation restric-

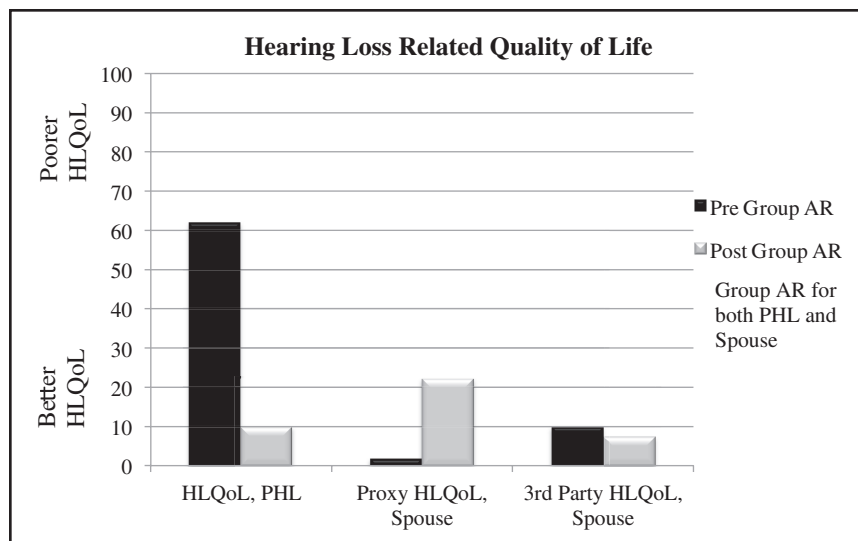


Figure 19-3. HL QoL scores measured before and after GAR program participation—Case No. 2. *Note.* This figure shows the HLQoL scores measured in a 71-year-old female with hearing loss and reported by her husband before each member of the couple completed a group audiologic rehabilitation (GAR) program (*black bars*) and 6 months after completing the programs (*gray bars*). HLQoL in the PHL loss is measured with the HHI, proxy reports in the spouse are measured with the HHI-SO, and third-party disability reported by the spouse is measured by the HII-SOP.

tions or activity limitations as a result of her hearing loss, nor did he report experiencing any third-party disability (HII-SOP = 10). Six months after participating in GAR, there were large changes in HLQoL reports. The PHL reported a significant improvement in HLQoL, while the spouse showed a small increase in his proxy report. The important finding here is that, after AR, the HLQoL reports are now congruent between the wife with hearing loss and her husband. It is possible that, as a result of participating in the group AR program for CPs, the husband may have started to communicate more effectively with his wife, thus explaining her large improvement in HLQoL. If this is the case, it occurred without any deleterious effects on the husband, as his HII-SOP score remained quite low after the AR program.

Recent research has used yet another tool to measure HLQoL in PHLs and CPs: proxy measures of third-party disability. Schulz and colleagues (2016) investigated both third-party disability in CPs and proxy reports of third-party disability in adults with hearing loss between 55 and 75 years of age as a means to predict which adults would seek out audiologic care. The tool to measure third-party disability was the SOS-HEAR (Scarinci, Worrall, & Hickson, 2009a). In the SOS-HEAR, CPs rate 27 items in six domains of third-party disability: communication changes, communicative burden, relationship changes, going out and socializing, emotional reactions to adaptations, and concern for partner (Scarinci, Worrall, & Hickson, 2009a). The CP rates each item on a 5-point scale with 0 equal to *no problem* and 5 equal to *a complete problem*. For example, as a measure of third-party disability, CPs rate the item “Because of my partner’s hearing difficulties we do not communicate as often as I would like” in the domain of “Changes to Communication” (Scarinci, Worrall, & Hickson, 2009a). For the proxy version of third-party disability, PHLs would rate the item as follows: “Because of my hearing difficulties, my communication partner and I do not communicate as often as she/he would like.” Schulz et al. (2016) reported that PHLs who believed their hearing loss was “burdensome” (high proxy third-party disability scores) were most likely to seek out audiology care.

Impact of Communication Partners on Audiologic Rehabilitation

There is an increasing body of literature on the influence of CPs on the uptake of, use of, and satisfaction with hearing aids in PHL; for a review, see Table 19–3.

PHL who perceive social support from their CPs are more likely to visit an audiologist (Duijvestijn et al., 2003; van den Brink et al., 1996; Wallhagen, 2010), similarly those with unsupportive CPs are less likely to visit an audiologist and take up hearing aids. There is also evidence of improved hearing aid outcomes, including increased hearing aid satisfaction, when PHLs receive positive social support from their CPs (Hickson et al., 2014; Meyer et al., 2014; Singh, Lau, & Pichora-Fuller, 2015) and increased hearing aid uptake (Singh & Launer, 2016). Audiologists appear to be aware of the benefits of including CPs in the AR process as they report there is value in family-centered care (Meyer et al., 2015); that is, they believe it is important to include CPs in audiologic appointments to expand understanding of the steps necessary for successful AR and for CPs to understand that they are an important part of the AR process.

Despite the increased understanding of the value of family-centered care, many audiologists do not include CPs in the AR process. Some studies have indicated that CPs are rarely invited to audiology appointments (Grenness et al., 2015; Heacock, Montano, & Preminger, 2018). Plus, a number of studies have noted that when CPs are in the audiologic appointment, audiologists rarely include them in the conversation, and in fact, may actively exclude them from the conversation (Ekberg et al., 2015). With the understanding that actively involving CPs improves the uptake of, use of, and satisfaction with AR, the authors believe audiologists will begin to seek active participation of CPs in audiology appointments.

Impact of Audiologic Rehabilitation on Communication Partners

It is well established that amplification improves quality of life in people with hearing impairment (Chisolm et al., 2007). It is also well known that amplification improves quality of life and reduces third-party disability in CPs. From interviews with couples both pre- and post-hearing aid fitting, Brooks and colleagues (2001) determined that post-amplification, spouses of PHLs noted improved communication with the PHL, a reduction in distress as a result of decreased repetition, decreased embarrassment in social situations, and less disagreement involving television viewing habits. (See Table 19–3 for a review of the impact of AR on CPs.) This benefit is not limited to hearing aids, as CPs of individuals who received cochlear implants also reported benefits (Kennedy, Stephens,

Table 19–3. Impact of Communication Partners on the Audiologic Rehabilitation Process

▪ Uptake of hearing aids

- Those with supportive CPs were more likely to visit a hearing health care professional and those with unsupportive CPs were less likely.^{1,2,4}
- Individuals who perceive third-party disability in their CPs are more likely to visit a hearing health care professional.⁸
- Individuals who attend a hearing health care appointment with a CP are more likely to adopt hearing aids.⁷

▪ Hearing aid use

- Social pressure from CPs was associated with increased hearing aid use.¹

▪ Hearing aid outcomes

- Individuals who perceived positive support from CPs had successful outcomes, whereas unsuccessful hearing aid users perceived negative support from CPs.^{4,6}
 - Social support from CPs is correlated with hearing aid satisfaction.⁵
-

¹van den Brink, R. H., Wit, H. P., Kempen, G. I., & van Heuvelen, M. J. (1996). Attitude and help-seeking for hearing impairment. *British Journal of Audiology*, 30(5), 313–324.

²Wallhagen, M. I. (2010). The stigma of hearing loss. *Gerontologist*, 50(1), 66–75.

³Duijvestijn, J. A., Anteunis, L. J., Hoek, C. J., van den Brink, R. H., Chenault, M. N., & Manni, J. J. (2003). Help-seeking behaviour of hearing-impaired persons aged > or = 55 years: Effect of complaints, significant others and hearing aid image. *Acta Otolaryngologica Supplementum*, 123, 846–850.

⁴Meyer, C., Hickson, L., Lovelock, K., Lampert, M., & Khan, A. (2014). An investigation of factors that influence help-seeking for hearing impairment in older adults. *International Journal of Audiology*, 53(Suppl. 1), S3–S17.

⁵Singh, G., & Launer, S. (2016). Social context and hearing aid adoption. *Trends in Hearing*, 20, 1–10.

⁶Hickson, L., Meyer, C., Lovelock, K., Lampert, M., & Khan, A. (2014). Factors associated with success with hearing aids in older adults. *International Journal of Audiology*, 53(Suppl. 1), S18–S27.

⁷Singh, G., Lau, S. T., & Pichora-Fuller, M. K. (2015). Social support predicts hearing aid satisfaction. *Ear and Hearing*, 36(6), 664–676.

⁸Schulz, K. A., Modeste, N., Lee, J. W., Roberts, R., Saunders, G. H., & Witsell, D. L. (2017). Burden of hearing loss on communication partners and its influence on pursuit of hearing evaluation. *Ear and Hearing*, 38(5), e285–e291.

& Fitzmaurice, 2008). Following cochlear implantation, CPs reported a decreased communication burden, increased participation in social activities, and an improved relationship with their PHL.

Quantitative research has also demonstrated the benefit of amplification in CPs. Newman and Weinstein (1988) measured HHI scores in male veterans and proxy reports (HHI-SO) in their wives pre- and 1 year post-hearing aid fitting. While both groups showed a reduction in perceived hearing handicap, the PHLs showed more improvement than the CPs. Another interesting finding emerged from the data; at the pre-hearing aid fitting appointment, the PHLs reported significantly lower HLQoL reports than the proxy reports of the CPs. Then at 1 year post-hearing aid fitting, the HLQoL scores were more congruent and no longer statistically different.

As PHLs rely on effective communication from their CPs to improve their quality of life, GAR has typically been recommended for both members of a couple. While a PHL may ask his/her partner to please move closer to make speechreading available, the CP often continues to hold a conversation from a different room. It has been suggested that CPs may finally understand the need for changes in their communication behaviors if they are fully integrated into the AR process and the GAR program (Getty & Héту, 1991). Getty and Héту (1991) offered GAR for men with hearing loss and their wives. After participating in the program, the wives reported that they did indeed try to stop speaking from another room and that they were less angry about having to deal with communication breakdowns. Interestingly, several of the wives also reported that an additional benefit

of GAR was the opportunity to discuss hearing loss and communication, a topic they had not addressed previously. Hallberg and Barrenas (1994) found that, when spouses were included in GAR, CPs became more aware of problems related to hearing disability in their PHLs. These findings are summarized in Table 19–3.

The benefit of including CPs in GAR has been confirmed in two additional studies (Preminger, 2003; Preminger & Meeks, 2010a). In 2003, Preminger conducted a quasiexperimental study to determine whether involvement of CPs in GAR could reduce hearing handicap further than that provided by the PHL attending alone. Thirteen “experimental” PHLs attended GAR with their CPs and 12 “control” PHLs attended GAR on their own. The GAR consisted of six 90-minute sessions that included informational lectures and training in communication strategies. Both experimental and control PHLs reported improvements in HLQoL as measured by the HHI following participation in GAR; however, the greatest improvement in HLQoL was measured for subjects who attended the classes with their CPs. This clearly shows that, when CPs participate in GAR, PHLs benefit. A summary of these findings is shown in Tables 19–4A and 19–4B. Preminger (2003) also found that the experimental couples showed more congruent perception of HLQoL post-GAR, while the control couples did not. It is likely that participation in the GAR provided the experimental CPs with an opportunity to understand their PHL’s communication difficulties better.

In a related study, Preminger and Meeks (2010a) measured the benefit of GAR programs designed specifically for CPs (in this study, all CPs were spouses). PHLs in the control group participated in a GAR while their CPs received no treatment. PHLs in the experimental group also participated in a traditional GAR program while their CPs participated in a separate GAR program designed for spouses. GAR consisted of 90-minute sessions meeting once a week for 4 weeks. The classes for the CPs included psychosocial discussions about what it is like to live with a PHL and communication strategy exercises aimed at improving communication with a PHL. Based on proxy reports of HLQoL, measured by the HHI-SO, results indicated improved awareness of participation restrictions and activity limitations experienced by partners with hearing loss in the experimental CPs. As a result, reports of HLQoL became congruent in the couples in which the CPs attended the Group AR program.

In summary, a number of authors have stressed the importance of the involvement of CPs in AR to

improve outcomes for each member of the couple. It is believed that PHLs need the support of their CPs to acknowledge the hearing loss and to recognize the resultant communication problems (Donaldson, Worrall, & Hickson, 2004; Wallhagen, 2010; Getty & Héту, 1991). Without this support from the CP, PHLs may minimize their hearing disability and will not present for AR. CPs may not understand the associated communication problems, yet they are the ones who most often must adjust to the hearing impairment (Getty & Héту, 1991). Negative views about hearing aid use in PHLs may be reinforced by their CPs. For example, many older adults are reluctant to try hearing aids due to ageist attitudes of others; if their CPs share these negative views, the PHLs may be even more reluctant to pursue AR (Wallhagen, 2010). On the other hand, negative views by PHLs about hearing aids may be counteracted by their CPs, who may disagree with their partner and actively support the uptake of hearing aids during an audiology appointment (Ekberg et al., 2014). It is generally believed that CPs should play an active role in AR as this will aid in the acceptance of amplification and improve the outcomes.

Models Representing Communication Partnerships

Hearing loss has been shown to have an impact on relationships, as the hearing loss of one affects the behavior of another. Difficulties arise in conversation and psychosocial behaviors (Donaldson, Worrall, & Hickson, 2004; Hallam et al., 2008; Hallberg, 1999), resulting in modification of activities and participations in both members of a communication partnership (Manchaiah, Stephens, Zhao, & Kramer, 2012). A seminar held at the Ida Institute in Denmark focused on the enablement of communication partnerships in the audiologic rehabilitation process. Certainly, the presence of a CP in the hearing aid delivery system has long been associated with positive outcomes (Brooks & Hallam, 2001; Duijvestijn et al., 2003), and an early tool developed to assess possible successful hearing aid use rated the presence of a CP as the most influential factor necessary to predict positive results (Rupp, Higgins, & Maurer, 1977). It is, therefore, valuable to understand the role CPs play in the daily lives of people with hearing loss. As noted in the previous sections, the CP discussed in research is almost exclusively the spouse, but in reality, PHLs may have a variety of other important CPs

Table 19–4A. Impact of Audiologic Rehabilitation on Communication Partners

▪ Hearing aids/cochlear implants ▫ Improvement in communication with PHL^{1, 2} ▫ Less of a need to repeat¹ ▫ Decreased communication burden² ▫ Television less a source of argument¹ ▫ More participation in social activities as a couple² ▪ Group audiologic rehabilitation ▫ First time had the opportunity to discuss hearing impairment and communication³ ▫ Stopped trying to speak to PHL from another room³ ▫ Felt less responsible for PHL in groups³ ▫ Less anger about communication breakdowns³ ▫ Felt less burdened with coping with partner's hearing loss³ ▫ Better understanding of participation restrictions and activity limitations due to hearing loss experienced by their partners^{4, 5} ▫ Reported satisfaction from attending group AR⁶ ▫ Reduced third-party disability⁷

¹Brooks, D. N., Hallam, R. S., & Mellor, P. A. (2001). The effects on significant others of providing a hearing aid to the hearing-impaired partner. *British Journal of Audiology*, 35(3), 165–171.

²Kennedy, V., Stephens, D., & Fitzmaurice, P. (2008). The impact of cochlear implants from the perspective of significant others of adult cochlear implant users. *Otology and Neurotology*, 9(5), 607–614.

³Getty, L., & Héту, R. (1991). Development of a rehabilitation program for people affected with occupational hearing loss. 2. Results from group intervention with 48 workers and their spouses. *Audiology*, 30(6), 317–329.

⁴Preminger, J. E., & Meeks, S. (2010). Evaluation of an audiological rehabilitation program for spouses of people with hearing loss. *Journal of the American Academy of Audiology*, 21(5), 315–328.

⁵Hallberg, L. R., & Barrenas, M. L. (1994). Group rehabilitation of middle-aged males with noise-induced hearing loss and their spouses: Evaluation of short- and long-term effects. *British Journal of Audiology*, 28(2), 71–79.

⁶Hickson, L., Worrall, L., & Scarinci, N. (2006). Measuring outcomes of a communication program for older people with hearing impairment using the International Outcome Inventory. *International Journal of Audiology*, 45(4), 238–246.

⁷Habanec, O. L., & Kelly-Campbell, R. J. (2015). Outcomes of group audiological rehabilitation for unaided adults with hearing impairment and their significant others. *American Journal of Audiology*, 24(1), 40–52.

Table 19–4B. The Influence of CP Involvement in Audiologic Rehabilitation on PHLs

▪ Group audiologic rehabilitation ▫ Improved HLQoL in the partners with hearing loss (as compared to PHLs who attend classes on their own)¹ ▫ Improved HLQoL in both the PHL and the CP³ ▪ Home-based educational program for PHLs and CPs ▫ Increased awareness of benefit of speechreading in PHLs² ▫ Improved interactions between PHLs and CPs²
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¹Preminger, J. E. (2003). Should significant others be encouraged to join adult group audiologic rehabilitation classes? *Journal of the American Academy of Audiology*, 14(10), 545–555.

²Kramer, S. E., Allesie, G. H., Dondorp, A. W., Zekveld, A. A., & Kapteyn, T. S. (2005). A home education program for older adults with hearing impairment and their significant others: A randomized trial evaluating short- and long-term effects. *International Journal of Audiology*, 44(5), 255–264.

³Habanec, O. L., & Kelly-Campbell, R. J. (2015). Outcomes of group audiological rehabilitation for unaided adults with hearing impairment and their significant others. *American Journal of Audiology*, 24(1), 40–52.

and engage in conversation in multiple environments. These environments, people, and circumstances make up the social networks that all individuals experience, and knowledge of these networks can contribute to the rehabilitation decisions PHLs make with their audiologists, including but not limited to the need for hearing-assistive technology systems or communication training.

Social networks can be defined as the “social relationships that surround a person, their characteristics, and the individual’s perceptions of them”; while the social support garnered can be described as “the interactive process in which emotional, instrumental, or financial aid is obtained . . .” (Bowling, 1994, p. 41). Social networks vary from person to person in dimensions such as size, number of personal interactions, and the amount of support available (Berkman & Glass, 2000). Given the dynamic nature of social networks, changes are likely to occur over time. Antonucci and Akiyama (1987) explored social networks of older adults and noted that persons within the closest proximity and with the highest number of personal contacts had more influence and support for this population. Lind et al. (2003) found similar results in a study involving older Australians with dual sensory impairments of hearing and vision, though their contacts were fewer outside immediate family and close friends.

An advantage of the social network is the availability of social support. Gallant (2003) found that this support had a positive relationship to self-management of behaviors.

Successful management of chronic conditions requires individuals to understand the issues related to the conditions and have an understanding of the decisions related to treatment. They must be able to manage the condition to maintain psychosocial functioning (Clark et al., 1991). When thinking about PHLs, the inclusion of partners and an understanding of the communication networks that exist within the person’s environments may have a direct impact on their own management of hearing loss and the ability to improve communication abilities. CPs’ knowledge may have implications for AR recommendations for PHLs.

The Communication Rings: A Tool to Explore Social Networks

A product developed out of the Ida Institute 2010 seminar “Enabling Communication Partnerships” was the Communication Rings (Figure 19–4) (Mon-

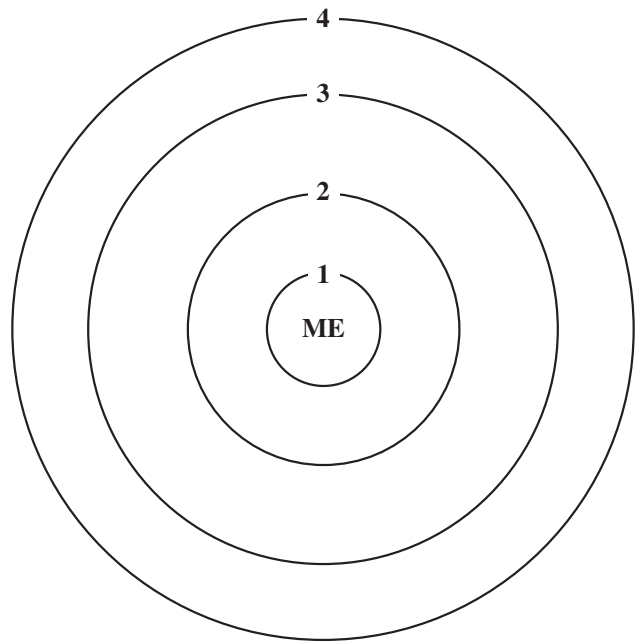


Figure 19–4. The communication rings. *Source:* Used with permission of the Ida Institute (2012).

tano & Al Makadma, 2012). The tool was based on the work of Antonucci and Akiyama (1987) and was designed to identify the people within communication environments allowing “individuals with hearing loss to reflect on the people who are part of their social network and help identify the challenges that hearing loss may present for successful communication” (Montano & Al Makadma, 2012, p. 49).

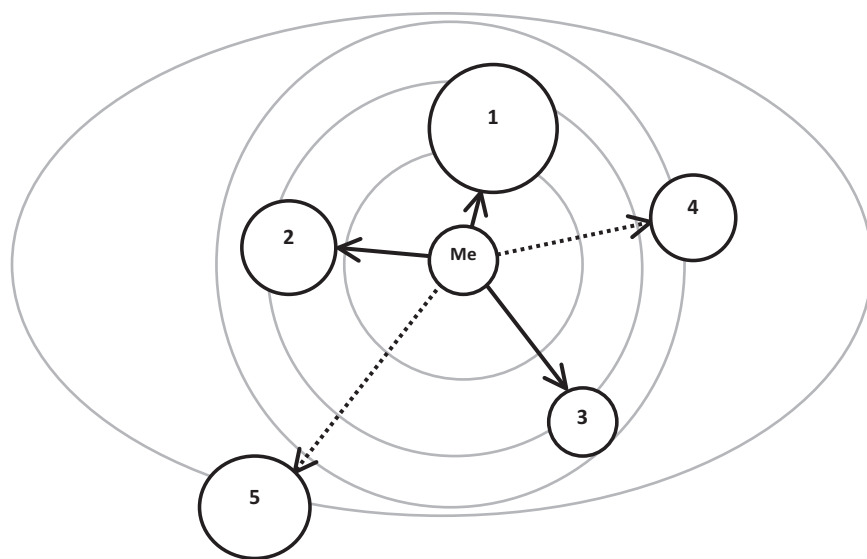
The Communication Rings is a simple figure consisting of a series of concentric circles with the word “me” presented at the core or Ring 1. This is meant to represent the individual completing the exercise; most often, in this case, the PHL. Rings 2 to 4 extend beyond the core (Ring 1) and denote the communication partners who influence the person in the core. Those people identified in Ring 2, closest to the core, represent those most important to the patient: critical communication partners and frequently close family or friends. As the rings extend out, names are inserted to denote other communication partners who may not be as influential or significant in the day-to-day life of the PHL. For example, an older adult with hearing loss might assign his/her spouse, son or daughter, or perhaps physician to Ring 2; the name of his/her brother who visits on occasion in Ring 3; and the waiter at the restaurant in Ring 4. Patients are encouraged to think about the people in their lives and how the hearing loss might affect their communication

efforts and even their relationships. The information obtained can be used for counseling, deciding upon amplification options, and communication strategies. An important benefit of the Communication Rings is that it opens a dialogue between the audiologist and patient about effective communication strategies and how they may vary among CPs. The exercise can be a good icebreaker in the clinical arena while providing valuable communication information. When family is encouraged to participate in the AR session, their input into the discussion elicited by the use of the Communication Rings tool can be enlightening not only to the PHL, but also to the audiologist when planning rehabilitation strategies.

Manchaiah and Stephens (2011) introduced a different model, the Communication World (Figure 19-5), to represent the more complex nature of communication partnerships to the PHL. While acknowledging the significance of the Communication Rings, the authors believed it did not adequately characterize the intricacy of interpersonal relationships. The authors identified three important factors they believed should be considered when trying to

understand relational communication. Factors such as the frequency or number of communication events, emotional closeness, and ease of communication are directly linked.

The Communication World model uses a solar system analogy rather than the concentric systems of the Communication Rings. The creators of this model felt it would allow the PHL to identify not only the communication partners but also the frequency of communication, the emotional closeness, and ease or difficulty of communication. Figure 19-5 illustrates the clinical interpretation of the Communication World. The circles represent people present within the social network of the PHL. The size of the circle denotes frequency of communication, distance from the person with hearing loss denotes importance or emotional closeness, and the connecting lines represent communication ease. In the example provided, the spouse has the most frequent, effective communication and is closest to the PHL. In addition, the work colleague also has significant contacts, but he/she has less emotional closeness and is the person with whom the PHL has the most problems communicating.



1 – Spouse; 2 – Mother; 3 – Cousin; 4 – Friend; and 5 – Colleague

Size of the circle: Frequency of communication

Distance from the center: Emotional closeness/importance

Type of line: Communication ease (Solid line easier, dotted line more challenging)

Figure 19-5. The communication world. *Source:* Used with permission of V. Manchaiah (2012).

The Communication Rings and Communication World tools not only have clinical application, but can also be useful in audiology student education, as an AR group exercise, and as part of a community awareness program. The use of the tool is particularly effective as a role-playing activity for students and consumer group meetings.

Goal Sharing for Partners

While research has shown that outcomes for both the PHL and the CP improve when both are actively involved in the AR process, audiologists are sometimes faced with uncooperative CPs who either don't see the need for expensive hearing aids or who expect those aids to solve all of their partner's hearing and communication needs. At the same time, the CPs may often be aware of their importance in the AR process. Consequently, members of the Ida Institute devel-

oped the Goal Sharing for Partners Strategy (GPS) as a tool for audiologists to use with PHLs and their CPs (Preminger & Lind, 2012). The GPS provides a framework to assist both the PHL and their CP to (a) acknowledge the hearing loss; (b) acknowledge the hearing loss-related activity limitations and participation restrictions that they each face as a result of the hearing loss; (c) acknowledge that they are partners in communication; and (d) realize their shared responsibility in managing the hearing loss (Preminger & Lind, 2012). (See Figure 19-6 for a step-by-step procedure to implement the GPS.)

The first step involves asking both the PHL and the CP to identify listening situations where they achieve successful communication. The authors chose to begin the GPS discussion by focusing on successful communication rather than to immediately begin by identifying communication problems because it is useful to emphasize the PHL's abilities rather than his/her disabilities (McKenna, 1987). Additionally, the successful communication strategy described here

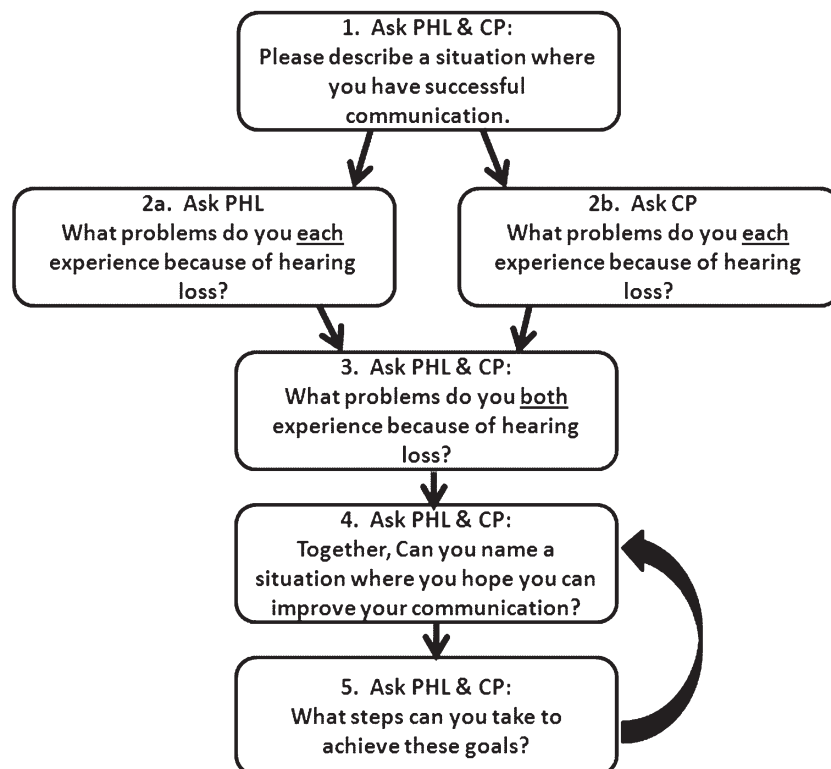


Figure 19-6. Visualization of the steps included in the Goal Sharing for Partners Strategy (GPS). *Note.* The figure depicts the steps included in GPS, where CP signifies communication partner and PHL signifies person with hearing loss.

may assist the partners in developing the steps that they need to take to achieve their goals in final step of the process (Step 5).

Step 2 is broken down into four separate questions:

- A. To the PHL: What problems do you experience because of your hearing loss?
- B. To the PHL: What problems does your communication partner experience because of your hearing loss?
- C. To the CP: What problems do you experience because of your partner's hearing loss?
- D. To the CP: What problems does your partner experience because of his/her hearing loss?

The purpose of these questions is to enable each of the CPs to reflect on their experience related to hearing loss and to also consider their partner's point of view. The hope is that the partners come to realize that the hearing loss is a shared condition, and thus a shared responsibility to manage. Please note that, as written in Figure 19–6, these questions are quite broad as they ask about what problems are experienced due to the hearing loss. Most participants will likely discuss communication problems, but they are also free to bring up emotional problems, marital problems, and the like. If the audiologist administering the GPS believes that the conversation is moving out of the scope of audiology practice, he/she may instruct the participants to focus specifically on communication problems.

After considering the hearing loss for multiple points of view, the participants are asked to consider the problems that experience together because of the hearing loss (Step 3). At this point, the audiologist should steer the conversation to problems that can be addressed within a comprehensive AR program. For example, audiologists can help to solve problems such as:

- "We never go out to restaurants anymore because of my husband's hearing loss."
- "The TV is very loud and annoys me, and yet my wife still keeps asking me to repeat what was said during the television program."

However, participants may bring up a problem that seems to be out of the scope of AR practice. Perhaps a wife of a PHL says, "He never calls me and this is a problem because he travels quite often and sometimes I have issues with the kids that I really need to

discuss." Clearly, there are a number of reasons why the PHL may not be calling home, so the audiologist can ask, "Is this a problem related to communication and hearing loss? If yes, tell me how?" At this point, the CP and the audiologist can turn this into a solvable problem such as, "Whenever my husband is traveling for work, he calls me over his cell phone, but he has trouble understanding me so the conversations become very frustrating for us."

In Step 4, the audiologist works with the participants to turn the problems into achievable goals, and then in Step 5, all participants brainstorm the steps that can be taken to achieve these them. These steps are adapted from the Goal Attainment Strategy (GAS) procedure, which was first described by Kiresuk (1973) and was first applied to audiology by McKenna (1987). The active involvement of the CPs in goal setting is extremely important as research has shown that patients involved in goal planning have a greater sense of motivation and self-efficacy (Duncan & Pozehl, 1971; Scott, Setter-Kline, & Britton, 2004; Webb & Glueckauf, 1994) and show improved adherence to treatment recommendations (Levack et al., 2006). See Table 19–5 for goals and steps to achieve these goals for the problems described in the last paragraph. Future research is necessary to determine the efficiency and effectiveness of the GPS in the clinical environment.

Conclusion

Throughout this book, it is evident that the hearing test and resultant audiogram are only a small component of the journey people with hearing loss take on their way to improve communication. Along the journey, they interact with many people in a multitude of environments in varying communication conditions. Self-management of hearing loss is a worthy goal for any PHL. This can be best achieved with the support of those within their social networks who are important to them. Understanding the impact of the hearing loss on daily function is critical not only for that PHL, but also for the audiologist who works with him/her. For this relationship to thrive, the audiologist becomes a new, important communication partner. This chapter focused on the current research on communication partnerships including spouses, family, and friends and introduced the reader to three tools that can be used to enable the involvement of partners in the rehabilitation partners. These tools can help

Table 19–5. Possible Goals and Steps to Achieve the Goals for the Three Problems Identified During Administration of the GPS

-
- Problem: We never go out to restaurants anymore because of my husband's hearing loss.
 - Goal: Go out to restaurant once per month; possible steps to achieve the goal:
 - Only go with one other couple with whom we feel comfortable discussing hearing loss and communication.
 - Do research ahead of time to select a quiet restaurant and dine early in the evening during quieter times.
 - Select a restaurant with good lighting and sit where speechreading is available.
 - Use an FM microphone with the hearing aid.
 - Explain effective communication techniques to dining partners.
 - Problem: The TV is very loud and annoys me, and yet my wife still keeps asking me to repeat what was said during the television program.
 - Goal: Figure out a way to reduce the volume of the TV and improve understanding; possible steps to achieve the goal:
 - Try an infrared system or induction loop system for the TV.
 - Use closed captioning.
 - Refrain from conversations until times when the TV is turned off.
 - Problem: Whenever my husband is traveling for work we try to communicate over the cell phone, but he has trouble understanding me so the conversations become very frustrating for us.
 - Goal: Communicate at least once per day when one partner is out of town; possible steps to achieve the goal:
 - Text or email prior to the phone call so PHL is aware of the conversation topics.
 - Try videochat (e.g., Skype) rather than phone calls so that speechreading is available.
 - Work with audiologist to select a new cell phone with better hearing aid compatibility.
 - Consider Bluetooth device to transmit cell phone signal directly to hearing aids.
-

the audiologist create a dialogue and develop an ongoing therapeutic relationship conducive to counseling. Engaging patients and family in the rehabilitation process can help guide the audiologist to developing appropriate communication goals with improved outcomes.

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Appendix 19–A

THE HEARING IMPAIRMENT IMPACT—SIGNIFICANT OTHER PROFILE

1	CS	Do you feel like you are shouting all the time because of your SO's hearing loss?	Yes	Sometimes	No
2	CS	Do you have to make sure your SO is looking at you when you speak to him/her	Yes	Sometimes	No
3	R&E	Do you get irritated when you try to talk with your SO but she/he cannot understand you?	Yes	Sometimes	No
4	SI	Do you feel that your SO's hearing loss hampers your social life?	Yes	Sometimes	No
5	R&E	Does having to repeat what you say to your SO all the time make you feel tired?	Yes	Sometimes	No
6	CS	Do you have to make sure your SO can see your face when you talk with him or her?	Yes	Sometimes	No
7	R&E	Does your SO's hearing loss make you feel frustrated?	Yes	Sometimes	No
8	SI	Do you and your SO avoid social gatherings because of his/her hearing loss?	Yes	Sometimes	No
9	R&E	Do you think your SO's hearing loss has made your relationship less satisfying?	Yes	Sometimes	No
10	R&E	Do you think that communicating with your SO requires a lot of effort because of his/her hearing loss?	Yes	Sometimes	No
11	R&E	Do you find yourself annoyed because your SO turns the TV up too loud?	Yes	Sometimes	No
12	SI	Do you and your SO avoid going to restaurants because of your SO's hearing loss?	Yes	Sometimes	No
13	R&E	Do you feel that your SO's hearing loss has a negative effect on the intimate communication between the two of you?	Yes	Sometimes	No
14	R&E	Does your SO's hearing problem make you feel angry?	Yes	Sometimes	No
15	CS	Do you have to repeat what you say to your SO because of his/her hearing loss?	Yes	Sometimes	No
16	R&E	Does your SO's hearing loss cause the two of you to argue?	Yes	Sometimes	No
17	R&E	Because of your SO's hearing loss, do you talk less often than you used to?	Yes	Sometimes	No
18	SI	Do you find that it is difficult to enjoy social gatherings because of your SO's hearing loss?	Yes	Sometimes	No
19	CS	Do you have to get up and go over to your SO when you need to talk with him/her?	Yes	Sometimes	No
20	R&E	Do you think your SO's hearing loss has created tension in your relationship?	Yes	Sometimes	No

CS = Communication Strategies, R&E = Relationship & Emotions, SI = Social Impact.
Yes = 5 points, Sometimes = 2.5 points, No = 0 points.

20

COMMUNICATION PARTNERSHIP THERAPY IN AUDIOLOGIC REHABILITATION

Christopher Lind

Prologue

Everyday communication is at once a sensory-perceptual, linguistic, and social activity, and access to a speaker's message at each of these levels is required for successful participation in conversation. It is commonly held that the effect of adult acquired hearing impairment (HI) on the individual and his or her communication partner(s), henceforth, CPs, arises primarily as loss of sufficient sensory-perceptual access to another's message and that this ultimately manifests as reduced access to social participation in social situations. In response to this multidimensional nature of spoken communication, practices in audiology rehabilitation (AR) vary and in each case address some component of the sensory-perceptual, linguistic, social, interpersonal, and/or psychological domains.

It is a premise of this chapter that across the range of AR therapies, the ultimate goal for the adult with HI is to better manage everyday interaction than he or she did prior to seeking audiology assistance. To this end, the hearing health clinician (henceforth, "clinician") establishes the needs of the client and assesses and provides intervention in response to these stated needs; that is, they become the goals for intervention and, by extension, the yardsticks by which the success

of intervention may be measured. It is also a premise of this chapter that adult-acquired HI manifests as a communication disorder that affects patterns of talk with family members or others close to the adult with HI and, as a consequence, family members have some vested interest in (re-)establishing successful communication. Partners and family members may often require support and guidance from AR intervention to assist in dealing with the consequences of the HI on their communication.

Evidence about the relative merits of therapy programs in addressing communication difficulties remains inconsistent and incomplete, and the reasons for undertaking one or more of these areas are not well defined. While this reduces the confidence with which different models of therapy are taken up, it should not limit their use.

Adults with HI who are able to maintain conversational success in their day-to-day talk are less likely to present for AR services. Those who find their HI results in these difficulties are more likely to wish to address the perceived loss of ease, fluency, or success in their everyday talk. This chapter aims to present some guidelines for the decision-making process in selecting and conducting AR therapies, with particular focus on conversationally oriented therapy involving both the HI adult and his or her frequent CP. This chapter will focus on the content, role, and purpose

of communication partnership therapy in the context of current models of audiologic service delivery, particularly family-centered care (FCC).

Introduction

Audiologic rehabilitation practices are in the midst of a certain crisis of identity. While evidence grows for the nature of the impact of adult-acquired HI on everyday talk (Lind, Hickson, & Erber, 2004, 2006; Skelt, 2007, 2012; Pajo, 2012, 2013), the importance of communication-based AR (Laplante-Lévesque, Hickson, & Worrall, 2010; Scarinci, Lind, Hickson, Meyer, Tulloch, & Hall, 2018) and for person-centered and family-centered care as models of service delivery (Scarinci et al., 2018), there seems to be little confidence among practitioners about some of the clinical decisions to be made in offering these services. Solid research evidence for AR practices grows but remains patchy and incomplete, and the place of nontechnological AR remains unclear to many practitioners.

Pressures on patterns of practice arise from various sources; most commonly, government funding models, commercialism, and professional standards of practice (Tuohy, 1999). These complex and very real pressures on clinical practice drive clinicians to shape their practices toward certain activities. An analysis of international practices in audiology suggests the trends away from offering AR as an integral part of the adult rehabilitative process are commonplace (Egbert et al., 2012). In the absence of either appropriate funding models or stronger evidence for the practice of adult AR (as opposed to technological intervention, primarily hearing aid fitting or cochlear implantation), the clinical role of adult AR remains uncertain and its offering in clinical settings is infrequent and inconsistent.

The focus on adults' perceptions of their HI as an issue in social participation is reflected in qualitative research findings that adults with HI report results from intervention (technological or otherwise) as "coming back to life" (Hallberg & Ringdahl, 2004, p. 117) or "becoming a new person" (Hogan, 1997, p. 239). Similar studies have reported that the key elements of intervention are increased social participation and, especially, increased interaction with family and friends, leading to improved quality of life and decreased isolation (Faber & Grontved, 2000; Hawthorne et al., 2004; Mo, Lindbaek, & Harris, 2005; Vermeire, Brokx, Wuyts, Cochet, Hofkens, & van de Heyning, 2005). So, the question arises: How do clini-

cians align clients' perspectives (i.e., what they say they want) with the available intervention techniques (i.e., what they have to offer) and the appropriate assessment tools and outcome measures?

Over the last 25 or so years, rehabilitation practices have changed in both content and process. The clinical and research hegemony of lipreading/speechreading as the dominant rehabilitative paradigm has been replaced by a variety of therapy approaches, including hearing and environmental tactics (Kaplan, Bally, & Garretson, 1985), conversation tactics (Erber, 1996, 2002; Tye-Murray, 2009), and assertiveness skills (Trychin, 1995).

Most recently, patient- and family-centered care models (Gregory, 2012; Grenness, Meyer, Scarinci, Ekberg, & Hickson, 2016) have reflected the move away from hearing rehabilitation as a purely sensory-focused pursuit toward practices that contextualize the goals of rehabilitation and focus on the personal, interpersonal, and social consequences of HI. For example, the Active Communication Education (ACE) program (Hickson, Worrall, & Scarinci, 2007; Scarinci, Worrall, & Hickson, 2008) provided evidence for the effectiveness of group-based AR programs for the influence of adult HI on the spouse and for the effectiveness of the client's informed selection of therapies to meet his/her perceived needs (Laplante-Lévesque et al., 2010). Saunders and Forsline (2012) have advocated for a model of short-term, individually based information counseling along similar lines to those espoused in this chapter. These developments may be seen to mirror the concepts underlying the World Health Organization's (WHO) (2001) International Classification of Functioning, Disability and Health (ICF) model, in which problems of structure and function may be distinguished from limitations to activity and restrictions in participation. These models reflect the increased attention paid to overcoming problems of activity and participation by designing therapy that addresses the situation-specific interactional consequences of adult-acquired HI, as they might impinge on both the adult who has HI and his or her frequent or familiar CPs.

Process in Audiologic Rehabilitation

As mentioned above, the relationship between technical and communication-based intervention is an uneasy one, reflecting the various clinical, professional, regulatory, and commercial attitudes and exigencies in the delivery of these services. In the

previous edition of this chapter (Lind, 2014), Lind distinguished between four models of delivery of AR services that reflect this relationship. The first, separated intervention, arises when the planning and execution of technological and communication-based activities occur entirely independently from each other. The clinical activities may be separated in time and even in place (i.e., delivered by different service providers). As a consequence, only a small subset of clients attending clinics in which this model is practiced may be offered communication-based AR services. In the second, overlapping model, the technical and communication-based activities addressing these goals may be offered in one setting and some assessments may inform both services, although they are still seen as distinctly different activities and are thus practiced quite separately. Often in these first two service delivery models, communication therapy is only considered after the completion of device fitting should the client report specific residual difficulties. Again, only a few clients may be offered communication-based intervention, as it remains largely dislocated from the technological intervention that is typically offered first.

The third model, shared intervention, may see a common assessment arising from a single point of planning for therapy offered by one clinician in some cases, but the communication-based therapy still sits at the edge of this planning. Despite a greater degree of integration between the technology and communication-based interventions to address situation specific needs, in which the success of the hearing aid fitting may be evaluated against the client's communication goals, communication-based therapy is typically offered only after the completion of technological intervention and is seen as a secondary activity.

Finally, the integrated intervention model combines both technical and communication-based therapy approaches and coordinates their execution according to the client's stated needs and in an order that reflects those needs. By contrast with the separated and the overlapping models, the ordering of therapy is defined by the individual with the HI and by the ability of the chosen therapy to address that difficulty. By contrast with all three prior approaches, it takes both forms of therapy to be of equal importance. It is sometimes the case that communication therapy alone addresses the client's needs and technological intervention is not required.

Both the clinic's preferred service delivery model and the clients' needs dictate which of these models might be most appropriate. From a service delivery perspective, clinicians who do not feel well equipped

to undertake communication-based therapy may refer clients to another agency. Alternatively, from the viewpoint of the client's needs, the more widespread and the less situation specific the client's reported communication difficulties arising from the acquired HI, the more likely that technological rehabilitation will be the appropriate first (and often only) action. Across these four intervention approaches, the key elements are the focus on situation-specific needs, the commonality of assessment across intervention types, and the selection of intervention activities to meet clients' stated needs. Finally, the need for intervention-specific outcome measures is implied across all four service delivery models.

Content in Audiologic Rehabilitation

This section gives a brief overview of two models of AR therapy, specifically the distinction between audiovisual intervention, and communication-based therapies.

Audiovisual Speech Reception Training

Audiovisual speech reception training has been the predominant nontechnological intervention strategy for most of the last 60 years. It continues to be the preferred functional outcome measure in assessing hearing aid fitting and cochlear implantation. Speech reception tests abound and speech perception training has been developed as a tool to directly address the perceptual consequences of acquired HI on everyday talk. The intervention implies massed practice as a strategy and a long period of therapy via both direct (i.e., training in the enhanced use of residual auditory speech reception skills) and compensatory (i.e., training in the enhanced use of visual speech reception skills) treatments. Audiovisual speech reception training traditionally distinguishes analytic (typically sound-by-sound processing of a speech signal) from synthetic (typically speech reception combining all elements of a spoken message to arrive at its meaning) processing paradigms, as originally suggested by Jeffers and Barley (1971) (see also Erber, 1981). While measureable improvement in speech reception test scores in comparison of pre- and post-intervention measures in both analytic and synthetic training programs has been demonstrated over many years, attempts to apply speechreading principles to larger units of talk have not borne fruit as clinical activity.

In part, this reflects the complexity of the linguistic models that underpin verbal communication outside of the confines of the well-constructed sentence.

The promise of research findings into speechreading across turns at talk (Erber, 1992; Flynn & Dowell, 1999; Gagné, Tugby, & Michaud, 1991; Lind, Erber & Doyle, 1999) have not translated well into clinical assessment or intervention. The singular exception remains continuous discourse tracking (deFilippo & Scott, 1978), although Okell and Lind (2012) identified several shortcomings in this approach from a conversational perspective.

Ross (1987, 1997) and Gagné (1994) noted the qualified success of audiovisual speech reception therapy models in AR over its history. These limitations arise from several sources, but predominantly from the lack of functional post-intervention assessment by which training might be judged to have benefited the participant in everyday talk. It is important to note that this shortcoming is not specific to speech reception training. The greater the decontextualized nature of the intervention, however, the more adaptive work the client must undertake to apply skills learned in therapy to everyday interaction. This has been addressed in the revival of auditory and audiovisual speech reception paradigms by Sweetow and Sabes' (2006) Listening and Communication Enhancement (LACE) program; by Levitt and colleagues (Levitt, Oden, Simon, & Lotze, 2011); and by Mauze, Schroy, and Tye-Murray's (2011) computer-based auditory speech training exercises. All address HI adults' perceived changes in everyday communication as outcome measures.

In the absence of more direct evidence of the link between speechreading training and advantages to the client in everyday conversation, speech reception training may be best offered as structured listening practice to adults with sudden and significant changes in speech reception; that is, either a substantial drop in speech reception as a result of sudden-onset loss or substantial improvement in hearing thresholds as a result of perceptual advantages following hearing aid fitting or cochlear implantation. Synthetic speech reception tasks, such as the Speech Perception in Noise (SPiN) test (Kalikow, Stephens, & Elliott, 1977) and Erber's (2002) sentence predictability exercise, demonstrate the complex relationship between adults' ability to use linguistic context and perceptual information in decoding individual words presented in the presence or absence of within-sentence grammatical and semantic cues. Patterns of results in speech reception tests such as these suggest that emphasis in therapy is best placed on the role

of linguistic context in speech reception and may be expanded to include the influence of conversational context in speech reception (Erber & Lind, 1994; Flynn & Dowell, 1999). (*Editors' note:* For more information on audiovisual speech perception training, the reader is referred to Chapter 14 of this text.)

Communication Partnership Therapy

Prior to outlining the communication partnership therapy model, the question arises as to what behaviors HI adults exhibit in conversation and, as a result, what might their CPs expect from them. Research has indicated that it is possible to observe directly the influences of adult-acquired HI on everyday interaction. For example, HI adults have been demonstrated to take fewer and shorter turns at talk with less semantic content than their CPs (Johnson & Pichora-Fuller, 1994; Stephens, Jaworski, Lewis, & Aslan, 1999). On the other hand, Wilson, Hickson, and Worrall (1998) noted the presence of longer turns and monologues by the HI adults in talk with strangers. HI adults' topic management was marked by more topic changes and less topic elaboration or discussion (Pichora-Fuller, Johnson, & Roodenburg, 1998). As listeners, HI adults were noted to use more general fillers and to undertake more backchanneling (Pichora-Fuller, Johnson, & Roodenburg, 1998). Finally, HI adults have been noted to initiate repairs much more often than their CPs (Lind et al., 2004, 2006).

Con conversationally oriented research indicates that conversation breakdowns typically occur when one person notes that a portion of another's talk is unclear in the context of the conversational topic, and chooses to request that some or all of a previous turn be clarified to resolve the miscommunication. The evidence in research literature suggests that the strategies that the HI adult uses to mark their need for repair and those the communication partner uses to resolve the miscommunication are in essence the same as those adopted for the same purposes in interactions between people where HI is not present (Lind, Okell, & Golab, 2009). That is, the actions people take to mark the need for repair and to clarify the communication breakdown are not specific to adult HI. As such, the content or process of repair is seldom the goal of AR intervention. By contrast, the frequency of occurrence with which adult HI disrupts conversation marks the perceived communication difficulty for HI adults and their CPs. Work by Lind and colleagues demonstrates the substantial differences in

occasions of repair initiation undertaken by adults with severe to profound hearing loss (a) prior to and following cochlear implantation and (b) by contrast with the number of initiations undertaken by the HI adult's CP (Lind, Okell, & Golab, 2009). That is, it may be the asymmetry in instances of requests for repair rather than the patterns of their resolution that strikes to the heart of the communication difficulty arising as a consequence of adult acquired HI. To this end, conversation-focused adult AR may set a broad goal to ameliorate the effects of the HI by offering strategies to reduce the instances, rather than manipulate the content of breakdown and repair in everyday conversation. In the context of family-centered care, the therapy implies direct intervention involving both the individual who has a HI and their frequent CP/significant other toward this end.

Prior to describing elements of the therapy, it might be worthwhile to consider who the clinician might expect the CP to be. In response to questions, almost all clients will readily identify people who are important to them, with whom they talk regularly, or particular occurrences of talk that are important to them that are threatened by their HI and thus warrant attention in therapy. These interactions arise across entire social networks of individuals in the HI adult's life. Analysis of social networks (Antonucci & Akiyama, 1987) highlights the importance to HI adults of communication patterns with the various people who populate them as a function of the immediacy or importance of the interaction with them (Lind, Hickson, Worral, Lovie-Kitchin, Yiu, & Barnett, 2003; Montano & Al Makadma, 2011).

Manchaiah and Stephens (2011) tailored Antonucci and Akiyama's original concept for use especially in communication disorders by depicting frequency of communication and emotional proximity, as well as the presence or absence of communication problems as links between the HI individual and his/her contacts. If conversation is an activity whose content and purpose are shared by participants, then it is reasonable to assume both partners have equal or at least similar commitment to its successful outcome. As such, it is reasonable to assume both partners have equally vested interests in the success of the interaction and, if that is the case, then they may both wish to address the issues at risk as a result of one partner's acquired HI.

As a starting point in therapy, a combination of clinical work in hearing, environmental, and communication tactics (Kaplan et al., 1985) may be designed to assist the HI client and CP in the conduct of their interaction. In some instances, clients might not be able to articulate or respond to the communica-

tive needs imposed by the HI when they interact. It is useful in these cases to assist the client and CP in addressing the various elements of the communicative environment that impinge on his/her interaction. These clients might be described as having poor metacommunication skills (i.e., people who are poor at talking about talking). This type of therapy is also valuable for adults who report specific situations to be problematic without strategies to resolve them or who require situation-specific needs that focus on the setting rather than the person.

The emphasis on the CP in therapy is particularly pertinent when they report doing most of the compensating for the other's HI. This therapy is designed for adults who see that both partners need to work on conversation skills. This therapy model implies the adult client knows of the tactics they might use but is not sure how to implement them or does not know how to shape the management of their needs leading to successful conversation. This work may include assertiveness training (Trychin, 1995), giving clients the skills to address issues concerning their HI with others, or Tye-Murray's (1992) anticipatory strategies by which, for instance, clients might work to prepare for contingencies in upcoming interactions. Pedley, Giles, and Hogan (2005) provided a range of useful exercises to assist clients in addressing the interpersonal aspects of their HI.

These AR practices ideally reflect the integrated model outlined above, combining device- and communication-based intervention in some balance, selected to address client-, partner-, and situation-specific daily needs. This chapter will not address device fitting in any detail, and will assume that AR is able to be offered as a combination of services with a clear and defined relation to the individual client's stated and observed needs. The remainder of this chapter addresses aspects of participation, specifically everyday talk, as influenced by an individual's acquired HI. In so doing, the chapter will reinterpret the application of the interactionally motivated therapy models (specifically, hearing and environmental tactics, conversation strategies, and psychosocial intervention) from a conversation partnership perspective.

Communication partnership therapy incorporates the current AR therapies with a focus on an understanding of the WHO's two ICF categories—activity and participation—and their manifestation in a clinical setting in family-centered care. The approach has the adult with HI and his or her chosen CP attend the AR session together and discuss situations in which HI impinges on their interaction with each other or with others. Thus, within any particular

situation, physical environmental factors, personal interaction styles, and/or differences in understanding the impact or consequences of the HI may all conspire to reduce the effectiveness of interaction. The aim of the therapy is to address these in any combination as they are indicated by the adult with HI and/or the CP according to several premises:

- If everyday talk is the most common site of difficulty arising from an acquired HI, then therapy that attempts to ameliorate the conversational consequences of an individual's HI is an appropriate and valid clinical activity. Furthermore, while this might be the primary goal of all (or almost all) AR, the more directly the therapy assesses and addresses the problem, the more immediate will be its application and potential benefit.
- Conversation difficulties arising from an adult's acquired HI have consequences for his/her CP(s) also, and those most often communicating with the HI adult are likely to be most affected by these difficulties. As a result, CPs have as much of a vested interest in the amelioration of the consequences of the acquired HI as do the adults who have HI.
- The interactional goal of everyday conversation is the maintenance of ongoing shared or mutual understanding. From a conversational perspective, there are only two goals for those participating in a conversation: to establish and maintain this shared understanding, and to restore it via repair when it is compromised. Thus, for HI adults and their partners, reducing the number of breakdowns (and their subsequent repair) arising as a result of mishearing and in some cases reducing

the time (specifically the number of turns) taken to resolve repairs when they do arise are reasonable and valid therapy goals. To this end, conversational fluency and the establishment and maintenance of shared understanding are tasks shared equally by CPs and HI adults. Each participant has roles to play to promote conversational success and fluency.

- Conversational fluency, while a reasonable goal for therapy, remains a poorly defined and understood term. It is not to be seen as a consequence of improved auditory or audiovisual reception alone. It is embedded in the constellation of social, interpersonal, linguistic, perceptual, and environmental influences. Ultimately, it is embedded in the co-construction of shared meaning. That is, it is embedded in the way participants in a conversation take their turns at talk and the way they construct the meaning of their turns with their CP in mind.
- Direct assessment of conversational fluency is the primary and most ecologically valid means by which to judge the success of any rehabilitative activity aimed at influencing conversational behavior. There is little precedent for the assessment of communicative behaviors other than by self-reported measures with their attendant complexities and indirectness, and/or by reference to clinical tests that typically sample component skills rather than the activity itself.

The scenario set out below takes the reader through a typical clinical interview between the clinician, a HI adult, and his CP. The section that follows then addresses, from the clinician's perspective, each of the issues that have been raised by the two clients.

Assessing Rehabilitation Needs Using Communication Partnership Therapy

Referral

Mr. A. (67 years) is referred to the AR clinic following a recent upgrade of his bilateral hearing aid fitting subsequent to a noticeable change in his hearing thresholds and his report of increased difficulties, particularly in conversations where there is background noise. Client-Oriented Scale of Improvement (COSI) (Dillon, James, & Ginis, 1997) goals were identified as part of the hearing aid refitting, and background noise consistently arose as his major concern. Although the hearing aids had been adjusted to his and the clinician's mutual satisfac-

tion with respect to sound quality and his loudness tolerance issues, the problems identified by the COSI persisted for Mr. A. and, as a result, the audiologist fitting Mr. A.'s new aids suggested referral to the AR clinic conducted by her colleague. The audiologist requested that Mr. A. invite his wife to accompany him to the AR appointment that was scheduled for 6 weeks following the final appointment for the adjustment of the new hearing aids.

Hearing and Communication History—Mr. A.

Mr. A. and Mrs. A. attend the AR appointment together, not really sure what to expect. It had been explained only that they would talk about communication strategies that might assist with his everyday hearing difficulties. They say that they were led to expect that this might last one or two sessions only. The appointment commences with the clinician asking Mr. A. to give a brief history of his hearing loss and hearing aid use.

Mr. A. tells the clinician that his hearing problems started gradually about 10 years earlier, but that he had not done anything about it until about 3 years ago, when he and his family noticed he was missing things others said to him or in general conversation. He had first attended another clinic for assessment and hearing aid fitting but had found the aids unhelpful and had shelved the idea of wearing aids until about 6 months prior, when he came to this clinic on a friend's recommendation. He reported that he had been happy with the hearing aid fitting service he had received; however, he gave the hearing aids "about 6 out of 10" for the help they provided.

The clinician asks Mr. A. to identify both the situations in which conversation is successful and those in which he finds his hearing lets him down.

Mr. A. lists several situations in which he feels hearing is an issue for him:

- Mr. A. notes that he manages well in one-on-one conversation, especially when he is face to face with the person to whom he was talking and when there is little or no competing noise. He notes that he manages well in conversation with his wife and with a couple of his close friends.
- Mr. A. reports that, although it is quiet at home most of the time, he and his wife experience some difficulty in everyday talk. He says he tends to miss the first words of the things she says. Further, Mr. A. reports that these difficulties are worse when they try to converse from different rooms, when Mrs. A. does not get his attention before starting to talk, or when background noise such as the television or kitchen noise compete with her voice. Mr. A. says that these miscommunications are resolved often once he is attending and that they deal with these miscommunications with relatively good humor, although sometimes there can be some tension between them.
- Mr. A. indicates that he struggles to keep up with conversation at restaurants, family dinners, or other sit-down occasions. He says that, typically, Mrs. A. sits on his right side as he feels he hears slightly better on the left and thus prefers to have that ear toward the less familiar person. He feels that Mrs. A. "acts as his ears," especially for things said to the right of him and at a distance. Most commonly, difficulties arise when Mr. A. does not hear a person speaking to him so either does not respond or speaks over the top of them. He notes that at family events, where typically 12 to 15 people attend, he manages well, unless "everyone talks at once." His family members know about his hearing loss, but he finds some adult voices and most of his young grandchildren's voices very hard to follow. Mr. A. is less keen to go out to a café or restaurant for fear of not being able to join in. He feels he can manage reasonably well if there are only four people at the table (he, his wife, and another couple, for instance) and as long as there is only one conversation at a

time. If there is any background noise he struggles but is aware that others might also struggle in the noise. He notes there are particular people whom he struggles to understand more in conversation than others. He notes that quiet voices, rapid speech, high-pitched talkers, and people with accents cause most difficulties for him. Mr. A. reports feeling disconnected and isolated in some of these situations but is pragmatic about the loss of hearing and the consequent loss of enjoyment at his time of life. He is sure many others are in a similar position.

- When Mr. A. attends monthly meetings of his local council, he reports struggling to hear all the voices in the room. Even though some of the others at the meeting have hearing problems as well, they do not seem to compensate for his needs, which he finds distressing. He reports positioning himself at the front of the meeting if the speaker is at the front of the room or either beside or across from the speaker if the meeting is around a table. He tries to arrive early to get the best seat, but if he arrives later, he often finds himself sitting in a poor position and missing out on a lot of the meeting.

Hearing and Communication History—Mrs. A.

Following the interview guidelines of the Goal Sharing Partnership Strategy (GPS) (Preminger & Lind, 2012), the clinician invites Mrs. A. to comment on Mr. A.'s concerns and to add her own perspective as the person who shares (or at least is present for) much of his daily communication. Mrs. A. raises several issues in response to what she has heard so far:

- Mrs. A. agrees with Mr. A.'s perspectives on most matters related to his hearing. She agrees that they find social events with large groups, lots of talk, and thus lots of background noise, quite difficult and while she is willing "to be his ears," she finds this restricts her enjoyment of the event. She notes that Mr. A. is less keen to attend social events such as these now and, as a result, her social life is also being curtailed.
- Mrs. A. says her primary concern when they socialize together relates to the times when she knows Mr. A. has difficulty in noisy places. She is constantly on alert to ensure that Mr. A. does not talk over others and she says that she has to always be ready to "jump in" if he responds to someone by saying something that "doesn't make sense or has missed the point." Mrs. A. feels as if she is being controlling in these situations and it distresses her. In any case, both Mr. A. and Mrs. A. are embarrassed by these events and are at pains to explain to people who do not know about Mr. A.'s HI that he missed what they said but was not ignoring them.
- Mrs. A. says that the television is a problem also, although Mr. A. does not always realize that he has the volume up so loud. Mrs. A. says the TV is up too loud for her if Mr. A. sets the volume. Mr. A. says that he can hear the news but not the movies or dramas because so much is spoken off screen, in the dark, or too rapidly for him. They agree that it limits their enjoyment of the TV when Mr. A. is constantly saying, "What did they say?" Mrs. A. says, "We don't talk about the shows we watch like we used to do."

The clinician reflects on the care and detail with which they have responded to the questions and asks if there are any other situations that the two of them feel are influenced by Mr. A.'s hearing loss. Mr. A. and Mrs. A. suggest that they have covered all the major things they can think of. Mr. A. and Mrs. A. have been talking for about 35 minutes and the clinician has taken careful note of the issues they have raised. He feeds back to them the major problems they have reported to ensure that he has understood them properly and clearly.

Assessing Client and CP Needs

The following section identifies the major hearing-related issues raised by Mr. A. and Mrs. A. in the hypothetical assessment scenario above, interprets them in light of the communication partnership therapy model, and suggests a range of clinicians' responses that might follow. The therapy is largely focused on informational counseling addressed at both people attending the session. Readers may note that much of the clinician's advice combines elements of more than one of the therapy models outlined in the first section of this chapter, reflecting the possible artificiality of some of the implied borders between these therapy models.

In addition to Mr. A.'s comments, readers might also note the richness of functional assessment information available to the clinician as a result of having Mrs. A. give her views on the impact of her husband's HI. Mrs. A.'s perspective, as the most frequent other half of Mr. A.'s daily communication, provides critical data by which therapy goals might be set. A review of the text in the scenario indicates that, if Mrs. A. had not attended the clinic and her remarks had thus been not included, intervention would have lost an important dimension and artificially reduced the array of strategies that might solve or at least ameliorate several of the communication difficulties the couple identified.

The key elements of this assessment protocol in the communication partnership therapy model are the identification of (a) situations in which the client is having success in communication, (b) those situations in which their HI causes problems, (c) the perspectives of the CP about both the successes and the difficulties, and (d) which of the issues might be goals for AR therapy. This assessment model reflects the GPS (Preminger & Lind, 2012), which, in conjunction with client-driven outcome assessments, provides multi-layered information about the environmental, communicative, interpersonal, and psychosocial aspects of the difficulties arising from a client's acquired HI. (*Editors' note:* For further information about incorporating communication partners into the AR process, the reader is referred to Chapter 19 of this text.)

A substantial period of time was spent assessing Mr. A.'s communication needs. Many clinicians balk at the time spent in gathering hearing and communication history from both CPs in a setting such as this. Spending 30 to 40 minutes focusing on the everyday consequences of a client's HI seems out of reach to them, particularly when they report the time

pressures placed upon them to get formal assessment data (e.g., audiograms, speech reception results). Much of this perceived pressure arises from the regulatory, commercial, and professional pressures on clinical staff, which often bring stress to the clinical setting when these forces conflict with ideals of client management.

Successful Communication as a Point of Comparison

Each of the seven excerpts that follow revisit the key points reported above, followed by commentary about the assessment and/or intervention that might best suit their investigation.

Excerpt 1

Mr. A. notes that he **manages well in one-on-one conversation**, especially when he is **face to face** with the person to whom he was talking and when there is **little or no competing noise**. He notes that he manages well in conversation with **his wife** and with a couple of **his close friends**.

Mr. A.'s comments about successful communication are not remarkable but they are useful. Essentially, he reports that when he is in an environment conducive to good communication, in face-to-face interaction or talking with certain people he knows well, he feels he manages interaction successfully. It is not surprising that he manages well in quiet, with access to visual speech cues and in conversation with familiar people. Later in the assessment interview, when talking about more difficult interactions, it might be worthwhile to contrast these conditions with those in less successful conversational venues.

Preminger and Lind's (2012) GPS provides a structured approach to include the CP in assessment and goal setting. It implies that assessing communication difficulties arising from HI along with client reports on the settings in which they perceive they are having communicative success has two important consequences. First, it provides the clinician with critical information about the quality of settings in which successful communication arises, as a point of contrast with the settings the client reports to be difficult. Second, it provides some potential relief to the client from the unintentionally harsh focus on

the difficulties he/she may be experiencing. This works to counteract the inference of communicative (in)competence (see Epilogue) that such focal interviewing may evoke. Whether used in conjunction with outcome measures such as the COSI (Dillon et al., 1994), IOI-OA (Cox et al., 2000; Stephens, 2002), or the GPHAB (Gatehouse, 1999), or simply as an interview protocol depends in part on the clinician's confidence and client responsiveness.

Communication at Home

Excerpt 2

Mr. A. reports that, although it is quiet at home most of the time, he and his wife **experience some difficulty in everyday talk**. He says he tends to **miss the beginning of the things** she says. Furthermore, Mr. A. reports that these difficulties are **worse when they try to converse from different rooms**, when Mrs. A. **does not get his attention** before starting to talk, or **when background noise** such as the television or kitchen noise compete with her voice. Mr. A. says these **miscommunications are resolved readily once he is attending** and that they deal with these miscommunications with **relatively good humor**, although sometimes there can be **some tension** between them.

Mr. A. and Mrs. A. are having some communication difficulties at home although they report that their interaction is largely successful. Mr. A.'s comments on the circumstances in which difficulties arise reflect poor use and awareness of environmental tactics by both partners when talking to each other from different rooms or when the background noise is sufficient to reduce the intelligibility of Mrs. A.'s speech.

Andersson, Melin, Scott, and Lindberg (1995); Erdman, Crowley, and Gillespie (1984); and Tye-Murray (1992) all address attitudes toward and selection of appropriate tactics in situations such as these. Kaplan, Bally, and Garretson (1985) also provide some excellent exercises for the clinic. The communication partnership therapy model takes these exercises a step further by suggesting that the discussion center not only on the choice of relevant hearing and environmental tactics, but also on the social milieu in which they arise. Clients often see the importance of

sitting in the best place, but struggle with the social act and consequences of asking to sit in a particular place or even asking someone to move to accommodate their hearing needs. It can be helpful to decide in clinic how such a situation might be tackled and to discuss the way the HI adult and/or the CP might address this.

Finally, Mr. A. reports that breakdowns do arise on occasion, but when they do, they are resolved easily. Lind et al. (2006) indicated there were two common characteristics in repair behavior across dyadic conversations involving adults with acquired HI and their familiar CPs. First, breakdowns were more commonly initiated by HI adults than by their CPs. Second, and more pertinent to Mr. A.'s comments, when conversing in quiet, these sequences were overwhelmingly resolved in the minimal number of turns and thus had as little impact on the conversation as possible. It is of note that Skelt (2007, 2012), in her research on repair behavior and adult HI, found that many breakdowns in these conversations went unrepaired. As such, it may be of use to ask clients how many times they need to ask for repetition in these settings, as it might be expected that if more than two requests are required, practice of repair sequences might be a target for therapy, for example using Erber's (2002) Questions for Aural Rehabilitation (QUEST?AR).

Social Talk in Multitalker Background Noise

Excerpt 3

Mr. A. indicates that he struggles to "keep up" with conversation at restaurants, family dinners, or other **"sit-down occasions."** He says that, typically, Mrs. A. sits on his right side as he feels he hears slightly better on the left and thus **prefers to have that ear toward the less familiar person**. He feels that Mrs. A. **"acts as his ears,"** especially for things said to the right of him and at a distance. Most commonly, difficulties arise when Mr. A. does not hear a person speaking to him so **either ignores them or speaks over the top of them**. He notes that at family events, where typically 12 to 15 people attend, he manages well unless "everyone talks at once." His family members know about his hearing loss, but **he finds some adult voices and most of**

his young grandchildren's voices very hard to follow. Mr. A. is less keen to go out to a café or restaurant for fear of not being able to join in. He feels he can manage reasonably well if there are only four people at the table (he and his wife and another couple, for instance) and as long as there is **only one conversation at a time**. If there is **any background noise**, he struggles, but is aware that others might also struggle in the noise. He notes there are **particular people whom he struggles to understand** more in conversation than others. He notes that quiet voices, rapid speech, high-pitched talkers, and people with accents cause most difficulties for him. Mr. A. reports **feeling disconnected and isolated** in some of these situations but is "pragmatic" about the loss of hearing and the loss of enjoyment as a consequence at his time of life. He is sure many others are in a similar position.

The problems Mr. A. mentions he has in social situations most likely arise from issues in managing communication in background noise, especially when that background noise comprises multiparty talk. Mr. A. reflects on the physical elements of the settings he finds himself in (possibly addressed via environmental tactics when seated in social settings), asking others for repetitions of misheard segments of talk (addressed via hearing tactics and repair when asking others for repetition of misheard talk), and psychosocial consequences reflected in his reduced desire to socialize (addressed by counseling and/or assertiveness skills). He also makes it clear that Mrs. A. plays a large role in assisting him in overcoming the social impact of his HI in these settings. To this extent, she acts an advocate for him.

From a conversational viewpoint, research tells us CPs can vary in the clarity and intelligibility of their speech (Kricos & Lesner, 1982, 1985; Picheny, Durlach, & Braida, 1985, 1986) and that CPs are able to improve the intelligibility of their speech either on demand (Schum, 1996) or in some cases without instruction (Michael, Seigel, & Pick, 1995). Furthermore, speakers will alter the clarity of their speech in response to perceived changes in the complexity of the setting in which they are talking (Pedlow & Wales, 1987). As such, talker intelligibility cannot be separated out from message and environment (Gagné, Masterton, Munhall, Bilida, & Querengesser, 1994).

Transactional Talk in Group Settings

Excerpt 4

When Mr. A. attends **monthly meetings of his local council**, he reports struggling to hear all the voices in the room. Even though some of the others at the meeting have hearing problems as well, they do not seem to compensate for his needs, which he finds distressing. He reports **positioning himself at the front of the meeting** if the speaker is at the front of the room or beside or across from the speaker if the meeting is around a table. He tries to **arrive early to get the best seat**, but if he arrives later he often finds himself sitting in a poor position and missing out on a lot of the meeting.

Mr. A. endeavors to apply good environmental tactics in this setting by seating himself in advantageous positions in the room. This situation may also be aided by the use of an FM system or similar to provide improved access to the key speakers at the meeting. Furthermore, by working with Mr. A. on assertiveness skills by which he might be able to ask people to allow him to sit in a particular place to have good access to the proceeding of the meeting, communication may be improved. Perhaps more pertinent to this setting is the distinction between interactional and transactional talk (Cheepen, 1988; Pichora-Fuller et al., 1998). *Interactional talk* is talk for the purposes of establishing and maintaining social contact. It implies equal speaker rights, the negotiation of topic, and the freedom to take the conversation where either participant wishes. *Transactional talk*, by contrast, focuses on more formal talk—the exchange of information with one party holding more power than another. Clients report greater pressure on their hearing when they are required to listen to information-laden talk rather than when they are engaged in social chat. All things being equal, Mr. A. may feel a greater pressure in this setting simply because of the transactional nature of the talk that comprises the agenda of meetings.

The CP as Client

The advantage in a communication partnership therapy-focused assessment is being able to incorporate the CP's views of the difficulties his/her HI partner is having as well as addressing the problems that arise

for the CP him/herself as a consequence of the partner's HI. Mrs. A. has provided instances of both types of difficulties in her history.

Shared Perspectives Between Client and CP

Excerpt 5

Mrs. A. agrees with Mr. A.'s perspectives on most matters related to his hearing. She agrees that they find social events with large groups, lots of talk and thus lots of background noise quite difficult and while **she is willing "to be his ears,"** she finds this **restricts her enjoyment** of the event. She notes that Mr. A. is less keen to attend social events such as these now and, **as a result, her social life is also being curtailed.**

Mrs. A. is clearly sensitive to many of Mr. A.'s needs, and they work together to solve them. In some cases, the solutions are best addressed by the two of them working together (e.g., by agreeing not to call each other from different rooms). Other times, the better solution is one handled by Mr. A. alone (e.g., ensuring he lets Mrs. A. know when he is not hearing her). Finally, it may take their work in concert to resolve some situations (e.g., at sit-down dinners where they work together when Mr. A. misses something said by another and is alerted to this by Mrs. A., but deals with it himself using an assertive approach via apology and/or brief explanation).

Television Viewing as a Social Event

Excerpt 6

Mrs. A. says her primary concern when they socialize together relates to the times when the **TV is up too loud** for her if Mr. A. sets the volume. Mr. A. says he can hear the news but not the movies or dramas because so much is spoken off screen, in the dark, or too rapidly for him. They agree that it **limits their enjoyment of the TV** when Mr. A. is constantly saying, "What did they say?" Mrs. A. says that the television is a problem also, although Mr. A. does not always realize that he has the volume up so loud. Mrs. A. says, "We don't talk about the shows we watch like we used to do."

Clients are often well informed about television devices such as infrared and RF systems and personalized or room loops. It is widely held that their use often reduces the difficulties associated with two (or more) people with very different hearing capacity watching television together. Readers might agree that the tendency to talk during television programs is forbidden by some and incessant in others. Either way, the right to enjoy television as a social event—to discuss the news, to evaluate the acting in the movie, to discuss the reality or the plot in a drama, or perhaps simply to talk about events of the day—is an important aspect of television viewing. In all cases, the ability to converse should not be lost and often the use of these devices, given how people watch television, does limit their social enjoyment of the event. All advice about these device-focused solutions should take into account the willingness or desire of partners to communicate with each other while enjoying their favorite programs. Devices that isolate the HI adult (and thus the partner also) from each other during this activity are perhaps less desirable than those that leave them free to talk, while providing independent volume control to each party.

Advocacy and Interpersonal Meaning of Managing One's HI

Excerpt 7

Mrs. A. says **she** knows Mr. A. has difficulty in noisy places. She is **constantly on alert** to ensure that Mr. A. does not talk over others and she says she has to be ready always to "jump in" if he responds to someone by saying something that "doesn't make sense or has missed the point." Mrs. A. feels as if she is being **controlling in these situations and it distresses her.** In any case, both Mr. A. and Mrs. A. are embarrassed by these events and are at pains to explain to people who do not know about Mr. A.'s HI that he missed what they said but was not ignoring them.

Mrs. A. makes a surprising remark when giving her perspective on her role in assisting Mr. A. in the social setting mentioned above. She says that she feels like she is being "controlling" when she jumps in to explain to others that her husband has not heard what they have said to him. It is clear that she feels a great

deal of social pressure arising from her husband's HI in what is always a difficult situation for communication. She feels she imposes herself on the setting and on the talk and this is creating some stress for her. Some find this burdensome and may need encouragement and reframing of their self-image (e.g., as controlling or perhaps as overbearing) to ensure the CP understands that, in most cases, this is a most appropriate and supportive action.

An assertiveness or psychosocial viewpoint marks the importance of advocacy and the psychology of interpersonal issues in adult HI that are pertinent to dealing with Mr. A.'s concerns (Trychin, 2005). The psychosocial consequences of acquired HI are powerful and well studied. As an area of practice in which many feel ill equipped, addressing these issues depends on the clinicians' comfort in dealing with counseling at this level of personal and emotional support. Not only is it clear that Mr. A.'s HI has a substantial impact on Mrs. A., but this also highlights that her personal, emotional, and psychosocial needs might also be the focus of therapy.

In summary, this simple scenario comprising commonly identified communication difficulties has raised many of the typical issues rehabilitation-oriented clinicians have heard repeatedly. Clinicians will know that a well-structured interview can raise the key aspects of an individual's HI that are important to one or both partners. In turn, this information becomes the basis for both information counseling and/or strategy-based intervention as time and service delivery styles permit.

Some Issues in Clinical Activity in AR

This section addresses three additional aspects of intervention that may be of benefit in considering the design and selection of therapy activities.

A Note on Family-Centered Care

There is a commonality in approach to therapy practices between communication partnership therapy and family-centered care. To this point, the therapy model exemplified above has focused on the communicative needs of the HI client and his partner. Both partners have been consulted on the goals for therapy and have been engaged in the strategies by which to address these goals. Epley, Summers, and Turnbull (2010) suggest that family-centered care

engages in services individualized to the family unit and are responsive to the family members' chosen goals of intervention. As a consequence, intervention plays to the (in this case) communicative strengths of the family members. The combination of family-centered care and conversation-focused therapy provide clear principles and guidelines for the conduct of AR intervention (Scarinci et al., 2018). (*Editors' note:* For more information on family-centered counseling, the reader is referred to Chapter 16 of this text.)

Assessment and Intervention

Some of the key elements of intervention in communication partnership therapy include: (a) assessing using tools and techniques that reflect the purpose of the intervention; (b) assessing and responding to clients' metacommunication abilities (i.e., their ability to talk about talking); (c) only undertaking massed practice following the overt discussion of the strategies the client might apply to that practice; (d) involving the client's CPs, as they have as much to learn and to offer as do their HI partners; (e) viewing oneself as a facilitator with the task to assist clients in developing skills with everyday CPs; and (f) ensuring generalization of therapy tasks to everyday life such that any skill developed in clinic can be readily applied to the client's life.

Informational Counseling and Massed Practice

Pressures conspire to give clinicians precious little time with clients in AR, and the provision of short-term intervention is seen as a desirable and fundable therapy option. Given the time required to undertake massed practice of communication strategies, most often short-term therapy is in the form of information counseling. The scenario set out above and the clinician's responses are mostly in the form of informational counseling of communication difficulties. By contrast, massed practice of conversational behaviors might include (a) varying the type of conversational task, (b) varying the (in)equality of the interaction, (c) recognizing that repair is a common conversational activity regardless of hearing status and that it does not always imply mishearing, and (d) implying that the sequence of turns in a conversation is as important as the content of the conversation.

Generalization

Generalization is the transfer of clinically learned skills to everyday life situations. As a general rule, the more real life the therapy task, the easier it is to apply the learned skills/generalize to the real world. Clinicians may draw clients into therapy by clarifying the link between what we do in therapy and its real-world applications and implications. By extension, clinicians need also to assess the real-world consequences of intervention. The onus falls to the clinician to (a) identify everyday consequence of changes implied by therapy model, (b) address client- and family member-centered issues/concerns that are most commonly pitched as activity limitations or participation restrictions, and (c) assess the outcome of therapy with respect to the client's concerns and goals and assume these to be embedded in everyday activities, especially conversation/talk in its various forms. The generalization potential of the intervention influences the ease with which the client is able to interpret and impose the tasks undertaken in therapy into their everyday lives.

Epilogue—Communicative (In)Competence

How might clients, or any adults, judge their communicative competence? It might be suggested that there are two elements together that underpin this judgment: that one achieves one's communication goals and that one interacts in a manner that maintains one's relationships on terms acceptable to others involved. As adult communicators, we take these matters for granted until they are violated, in this case by the superimposition of an acquired HI, but equally this might be any communication disorders and/or many psychological disorders that influence everyday talk. In life, our identities are situated within the tasks we undertake and the people with whom we spend time. In practice, clinicians deal with people who may perceive themselves as incompetent and, unwittingly, may further infer incompetence by evaluative remarks, such as, "You did very well, considering your limitations" (Kovarsky, Kimbarow, & Kastner, 1999). The suggestion is that clinicians avoid classifying individuals in clinical context and make tasks intrinsically motivational by ensuring that they have immediate or at least observable consequences in everyday life and interaction. This might offer

motivational reinforcement and Kovarsky et al. (1999) suggest this may go some way to restore client attitude and self-image.

This chapter has endeavored to set out some principles of communication partnership therapy offered to individuals with HI and their chosen CPs. To conduct this and any therapy, clinicians require a range of intervention techniques that complement the clinical and professional environments in which they work. In turn, they require a range of assessments that complement the range of intervention techniques. Assessments will benefit from the inclusion of direct clinical evaluation of communication abilities to complement self-reports. In turn, this implies (a) further research and evidence-based outcomes for the various rehabilitation intervention techniques, (b) development of a common vocabulary by which we might discuss conversation, and (c) being mindful of the way we communicate about communication with our clients. Ultimately, the decision about how to conduct AR remains intimately tied to the professional orientation of clinicians and the daily functioning of each clinic.

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21

ADVANCED PRACTICES

Assistive Technology in the Age of Smartphones and Tablets

Linda M. Thibodeau

Introduction

With advances in wireless hearing technology, the term *Hearing Assistive Technology Systems* (HATS) must be considered in a broader context. In the past, HATS referred to analog devices used by persons with hearing loss to facilitate reception and identification of speech and nonspeech signals. Although HATS technically encompass all technology used by individuals with hearing impairment, the personal hearing aid has generally been considered the primary technology, while HATS have been considered only when communication difficulties still remain. Perhaps it is time to designate these analog devices as A-HATS. With the increased digital connectivity options for personal hearing aids, many benefits of A-HATS are already available in the personal device when connected wirelessly. The time has come to consider Digital Hearing Assistive Technology Systems, D-HATS, as a continuum of benefits from features within the personal hearing aid to wireless connections to devices or the internet. Of course, there will always be some need for the traditional analog devices such as the flashing lamp to signal a door knock because of the inevitable range of experience with digital technology. However, as baby boomers continue to age with increased vitality and life expect-

tancy, the diversity in technology experience among those seeking solutions for hearing challenges may diminish. The active senior citizen is likely to have a cell phone that connects to his/her car for hands-free phone conversations. In addition, they have probably experienced some type of real-time video exchange to stay in touch with grandchildren whose parents are adept at wireless technology, smart homes, and useful applications (apps) for almost everything from "Amplifiers" to "Zumba Fitness."

The audiology profession continues to grow with millennials who have never fit an analog hearing aid. Because they have considerable experience with wireless technology that monitors activity, safety, diet, and sleep; streams music, audio, and video signals; and facilitates instant communication, the likelihood that they will incorporate HATS, specifically D-HATS, is rising. The gradual merger of the digital experiences of the audiologic service provider and the audiology consumer will undoubtedly stimulate the need for new auditory rehabilitation assessment tools, increased communication partner interaction, and appropriate individualized informational counseling for the consumer to reach maximum benefit of the available options.

The beneficial features of D-HATS can be incorporated in the initial stages of the rehabilitation process. With the use of a comprehensive assessment

scale, the possible situations where hearing assistance is needed may be determined. The results of the latest MarkeTrak IX survey showed an increased satisfaction with hearing aids (from 74% to 81%) and a decreased number of in-the-drawer hearing aids (from 12% down to 3%) relative to MarkeTrak VII (Abrams & Kihm, 2015). It is possible that this is related in part to the hearing aid being used for more than just amplification as connectivity options have expanded over the past 10 years.

It is interesting to note some historical influences on the terminology. Vaughn, Lightfoot, & Arnold (1981) argued that the advent of smaller hearing aids resulted in certain communication limitations and that it was imperative that greater attention be focused on communication centered environments and effective listener and talker devices. As the industry moved toward smaller hearing aids, the distinction between hearing aids and other devices such as hearing-assistive technology developed. There evolved a focus on fitting hearing aids by the audiologist and assistive technology received less attention in clinical practice. The technology became so separate, in fact, that in 1982, the notion of an Assistive Device Demonstration Center was recommended (Fellendorf, 1982). Nearly 15 years later, Sandridge and Lesner (1995) encouraged audiologists to no longer ask whether to provide assistive listening devices (ALDs), but rather to ask how to incorporate ALDs within one's service delivery. Wayner (2004) was probably the first to recommend that assistive technology be considered part of the fitting process. It was an underlying belief that most persons with hearing loss could receive some benefit from amplification and that hearing aids were only one of a number of devices for successful rehabilitation and improved communication. Now, the connectivity options with hearing aids are viewed as a major "door opener" for the person with hearing loss (Leavitt et al., 2016). Hearing aids are capable of receiving signals from remote tablets, cell phones, and microphones. Therefore, the notion that assistive devices are a separate category that require a special focus in an audiology practice may disappear. To explore these benefits with patients, an audiology practice does not need a separate room with examples of HATS to demonstrate benefits available for phone conversations, television, or alerting signals like those recommended in the past (Wayner, 2004). In fact, the most basic wireless connectivity in hearing aids that existed even in analog hearing aids, i.e., the telecoil also known as t-coil, is now joined by an array of wireless protocols including near-field mag-

netic induction, 2.4 GHz Industrial Scientific Medical (ISM) band, and 900 MHz ISM band. Regardless of the sophistication of the technology or programming options, the basic process for obtaining and successfully using D-HATS remains the same. Initially, there must be an assessment of the communication difficulties that will lead to recommendations. There must be verification of the technology to document appropriate settings and, finally, validation must occur to ensure that the user receives the intended benefits. To summarize the benefits of meeting an individual's HATS needs when devices are moving from analog to more digital options, lighthearted illustrations are provided in Appendices 21-A and 21-B. Of course, the overall challenge is to integrate these steps into the routine clinical practice, much like checking one's blood pressure during a physical exam, so that D-HATS are considered a continuum of options from a single personal device to a network of devices and apps to address communication of both verbal and nonverbal signals. Before a review of the assessment, verification, and validation of D-HATS, a discussion of the major wireless transmission protocols will provide a foundation upon which to compare the various wireless technology options. These range from longstanding methods such as magnetic induction (telecoils) to more recent very high-frequency protocols (Bluetooth Low Energy, LE).

Wireless Transmission Protocols

A variety of wireless transmission protocols have been used to benefit those with hearing loss. These include magnetic induction, low-frequency radio signals such as near-field magnetic induction (NFMI); midfrequency radio signals such as FM 72–76 MHz, 216–217 MHz, and 900 MHz; and high-frequency transmission 2.4 GHz Bluetooth, Bluetooth LE, and proprietary protocols. All wireless transmissions have a transmitter and a receiver. How the audio signal from the transmitter gets to the receiver defines the type of wireless system.

Electromagnetic Transmission

The oldest wireless system used by those with hearing challenges involves electromagnetic transmission, which has been in use since the late 1930s. The first wearable hearing aid that contained an induction

receiver, i.e., a telecoil, was reported to be the Multitone VPM in 1938 (Bauman, 2015). In an induction loop system, an audio signal from a source such as a microphone creates an electromagnetic signal that induces current to flow through an induction receiver such as the telecoil in a hearing aid. The transmission is direct rather than depending on a carrier signal. Therefore, there is no tuning required but simply positioning the induction receiver, i.e., hearing aid with a telecoil, within the induction loop. The distance from the telecoil to the loop needed for adequate transmission depends on the strength of the electromagnetic signal. Some are low-level such as a loop worn around one's neck that transmits a signal to the telecoil in the hearing aid from a receiver attached to the neck loop. A room loop, however, is a much stronger electromagnetic field that may be installed in the ceiling or floor; thus, one may be sitting several feet from the actual loop. The signal is universal in that any manufacturer's hearing aid with a telecoil can receive the induction signal from a loop.

The input to an induction loop is typically from a microphone and amplifier or another audio source, and the current fluctuations that travel through the loop of wire create a magnetic field that mirrors the frequency and intensity of the original signal. The telecoil in the hearing aid is sensitive to the changes in the magnetic field, which causes a current to flow through the amplifier of the aid. The listener with hearing loss can switch his/her hearing aid to telecoil to receive the audio signal from the source and improve speech perception in background noise, over distance, and in reverberant environments.

Despite the simplicity and generally low cost, there can be several negative effects of telecoils such as spillover of the signal from adjacent room loops and the pickup of stray electromagnetic energy from power lines, computer monitors, and even some smart watches. Furthermore, signals transmitted via telecoils tend to be noisy, have reduced low-frequency energy relative to the original signal, and vary in intensity with head movements (Thibodeau, McCaffrey, & Abrahamson, 1988). Because of the universality of telecoils, there is a significant movement to have loop systems installed in many public areas throughout the U.S., hence efforts have been called "Loop America" or "Time to Loop America". These efforts include web-based resources that allow the visitor to learn about loops and find public venues, such as churches or theaters, that are looped for the hearing impaired. The symbol used to alert the public of loop systems is shown in Figure 21-1.



Hearing Loop

Switch hearing aid to T-coil

Figure 21-1. Signage used in public places to alert visitors that a loop system may be accessed by using the telecoil in their hearing aids. *Source:* <http://www.hearingloop.org/logo.htm>

FM Transmission

Perhaps the next significant advancement in wireless transmission was the use of frequency-modulated (FM) transmission. In FM systems, the signal from the audio source is transmitted via a carrier frequency, such as 72–76 MHz or 216–217 MHz. This carrier frequency can be selected by the user on either a wideband or narrowband range of frequencies. For example, a wideband carrier frequency may be 72.1000 MHz to 72.1250 MHz and designated by the manufacturer as Channel "A" and a narrowband carrier frequency may operate on 216.0125 MHz to 216.0375 MHz and be designated as Channel "1." The signal of interest picked up by a microphone is used to modulate the frequencies of the carrier wave in a pattern to correspond to the original signal. The receiver must be tuned to the same channel so that the transmitted FM signal can be demodulated to recover the original signal.

The first FM receivers operated on wideband carrier frequencies and were body worn. In 1996, FM receivers were introduced into behind-the-ear hearing aids, which were limited to receiving only one or two channels and had somewhat cumbersome antennae.

With advances in technology, the transmitter microphones became more sophisticated with noise reduction and directional pickup techniques, which resulted in significant improvements in speech recognition in noise (Thibodeau, 2010). Furthermore, the receivers became small enough to be attached to most behind-the-ear hearing aids with a connector called an audio shoe. The audio shoe is specific to the manufacturer of the hearing aid, but all have the universal 3-pin Euro plug to accept the miniaturized FM receiver. One manufacturer has designed receivers that integrate with their specific behind-the-ear models so that it always remains on the hearing aid.

When applied as a solution for communication challenges for those with hearing loss, the signal is higher quality, less noisy, and has less interference than the electromagnetic transmission systems. Although FM systems have been successfully used in school classrooms to improve reception of the teacher's voice, some major drawbacks are the limited number of channels available for transmission, depending on how close the transmitters are located; and the fact that the receivers draw power from the hearing aid, thus shortening the battery life. Reddy and Thibodeau (2018) reported battery life was reduced about 50% when used in a wireless arrangement for 12 hours per day. FM systems frequently used in public venues are typically body-worn receivers offered to visitors with headphones, earbuds, or a neck loop, which would require the visitor to have a personal device with a telecoil.

Near-Field Magnetic Induction

The transmission protocols discussed thus far involve analog signals. The remaining transmission protocols to be discussed all involve a digital signal being encoded and transmitted wirelessly to a receiver that decodes the signal. The first application of wireless digital signal used in hearing aids was introduced in 2009 and is known as Near-Field Magnetic Induction (NFMI). It operates on carrier frequencies even lower than those used for FM and typically falls between 3 and 15 MHz (Galster, 2010). This frequency band easily travels through and around the human head and body, making it ideal for communication between bilateral hearing aids for synchronizing program or volume changes and spatial processing of acoustic signals to enhance the signal-to-noise ratio. This type of transmission involves a proprietary code developed by each manufacturer.

Because the transmission range is generally about 1 meter, NFMI may also be used to send signals from an intermediary device that can be small enough to fit in one's pocket or can be connected to a loop worn around the neck, as shown in Figure 21–2. These intermediary devices are called streamers, so as not to be confused with induction loops. Although both can be worn around the neck and communicate with the personal hearing aid, the induction neck loop requires a universal telecoil in the aid so it can be used with an aid by any manufacturer. However, the streamer requires a hearing aid with NFMI capability by the same manufacturer. Superior to the induction loop, the streamer allows for greater consistency of the signal across the frequency range and with head movement.

The streamer is ideal for housing other wireless receivers that require greater power and space than can occur within an ear-level hearing aid, such as a 2.4 GHz Bluetooth connection to a cell phone. They can even be interfaced with FM or digital-modulated (DM) receivers for streaming signals from FM/DM wireless microphones to one's hearing aids. Because the NFMI streamer can act as a receiver and a transmitter, it is important to use clear terminology with clients about wireless transmission. Most streamers are capable of connecting with cell phones through Bluetooth 2.4 GHz universal protocol, but they transmit the audio information via a NFMI proprietary protocol to the hearing aid.

900 MHz Transmission

To avoid some issues with NFMI such as the limited transmission range, wireless transmission protocols on the 900 MHz frequency band were developed for long-distance audio streaming that could operate on typical hearing aid power levels. Starkey developed this protocol in 2012, called SurfLink, to allow hearing aids to connect directly to a television or other media source up to 20 feet away without any intermediary device on the user. The hearing aid can be programmed to detect the 900 MHz signal so that no pairing is necessary. Users can enter a living room and automatically hear the television if it is connected to a SurfLink 900 MHz transmitter. The 900 MHz transmission protocol has also been used to transmit signals from remote microphones that can be worn by a communication partner in noisy environments. In addition, the SurfLink family of accessories includes a device that can establish a Bluetooth connection with a cell phone and then transmit the call to the hearing



Figure 21-2. Streamers that communicate with hearing aids via near-field magnetic induction. *Sources:* A. Photo courtesy of Oticon A/S. B. Image © Sonova AG. Reproduced here with permission. All rights reserved.

aid via the 900 MHz protocol. It is important to recognize that the line of hearing aids compatible with SurfLink are different than the “made for iPhone” aids described in the next section although they both allow for wireless phone communication. Recall that the hearing aids compatible with SurfLink require communication with the 900 MHz transmitting device, which provides the wireless connection to cell phones through the Bluetooth protocol described next.

2.4 GHz Transmission

The 2.4 GHz frequency band is designated by the Federal Communications Commission for public use. Since the launch of the 2.4 GHz frequency band in 2001, many technologies utilize this technology, such as Bluetooth, Wi-Fi, phones, and video game controllers. Each of these technologies uses a communication protocol for sending and receiving data that may be standardized or proprietary.

The 2.4 GHz transmission method was first used in 2005, when Starkey introduced the first wireless connection to cell phones with Bluetooth capability (*Audiology Online*, 2005). The Ear Level Instrument (ELI) was an attachment to the base of a behind-the-ear hearing aid that essentially doubled the hearing aid’s size and weight while significantly reducing the battery life. For these reasons there was limited interest in the ear-level 2.4 GHz option until a new protocol was developed, referred to as Bluetooth LE.

The first aids that were capable of communicating with cell phones through Bluetooth LE were announced by G. N. Resound in 2013, referring to them as “made for iPhone” or MFi aids. At this time, there are seven manufacturers that have hearing devices that communicate directly with cell phones without an intermediary device as shown in Table 21-1. This means that persons can send and receive phone calls and stream audio signals from Bluetooth-enabled devices that are paired with their hearing aids.

Table 21–1. Manufacturers of Hearing Aids and Cochlear Implants That Provide Direct Connectivity With Smartphones

Manufacturer	Hearing Instrument Style				Cochlear Implant
	BTE	ITE	ITC	RIC	
Phonak MFA	Bolero M T/R			Audeo Marvel T/R Audeo B direct	
Oticon MFi	Oticon Opn BTE PP T Oticon Siya BTE PP T Xceed Play	Oticon Opn T Oticon Siya T		Oticon Opn T/R Oticon Siya Series T	
Widex MFi	Evoke			Beyond Fusion T/R Evoke Fusion 2 T/R	
ReSound MFi	LinX 3D Enzo 3D	LinX 3D	LinX 3D	LinX Quattro R LinX 3D R	
Starkey MFi	Livio Series Livio AI Halo i Series Halo			Livio Series Livio AI Halo iQ Series Halo 2 Series	
Signia MFi	Motion 13 Nx Motion P 13 Nx Motion Charge & Go R	Insio Nx	Insio Nx	Pure 13 Nx T Pure 312 Nx Styletto R Styletto Connect R	
Cochlear MFi					N7

Note. MFA = Made for all phones; MFi = Made for iPhones; BTE = behind-the-ear; ITE = in-the-ear; ITC = in-the-canal; RIC = receiver-in-the-canal; T = telecoil; R = rechargeable iPhones.

As the name implies, the MFi aids only work with Apple products. Given that 88% of cell phones sold as of 2018 were Android phones (Statista, 2019), there was certainly a need for direct wireless connectivity to these products. In 2017, Phonak announced a new protocol based on a proprietary chip that allowed pairing with Apple or Android phones (Hearing Review, 2019). The SWORD chip allowed running protocols in parallel so that the chip allowed connectivity to the hearing aids and compatibility with both Bluetooth LE and Bluetooth Classic protocols. The hearing aids with the SWORD chip were referred to as “made for all” or MFA aids. However, unlike the MFi aids, the connection was only between the phone and a single designated hearing aid. In 2019, bilateral phone connection to both iPhone- and Android-based phones was possible in the MFA aids by sending the signal from the connected hearing aid to the second hearing aid via NFMI (Figure 21–3).

Although a great solution for wireless connectivity to cell phones that are typically close by but not in the same room as the listener, the Bluetooth protocol is not ideal for transmitting in face-to-face conversation because the transmission delay can be more than 100 ms. Therefore, other digital protocols utilizing the 2.4 GHz band for wireless transmission of signals to aid those with hearing challenges were developed. In 2013, Phonak introduced a proprietary adaptive digital wireless transmission technology called Roger, where audio signals are digitized and coded in small bursts that are transmitted repeatedly in the 2.4 GHz band. This is known as *frequency hopping*, which avoids interference issues and has audio delays less than 25 ms. Unlike Bluetooth receivers, Roger systems have an unlimited number of connections and wider transmission bandwidths up to 7300 Hz. The Roger transmission protocol allows hearing aids to be connected to coordinated receivers; thus,



Figure 21-3. Current Made-for-iPhone hearing aids that communicate directly with phones through Bluetooth Low Energy transmission protocols. *Source:* Photo courtesy of Starkey Hearing Technologies.

the user benefits from improved wireless reception from remote microphones and other audio sources hardwired to Roger transmitters such as tablets, televisions, and MP3 players. Roger systems may also be referred to as DM systems in contrast to the older FM systems.

Summary of Transmission Protocols

Hearing aids have been able to communicate wirelessly with other audio signals generated outside the device since the 1930's when telecoils were introduced. However, within the past 20 years, there has been exponential growth in the development of wireless connectivity options for persons with hearing loss. It is critical for the hearing care professional to have knowledge of the array of options and describe them accurately to their patients. The most appropriate type of transmission system for an individual depends on many factors. Therefore, a comprehensive assessment is needed to determine if the user will need to connect wirelessly to devices to solve multiple communication challenges or if they only have a single issue, such as hearing a smoke alarm when unaided. By exploring the patient's current use of both hearing aids and wireless systems, their acceptance of advanced connectivity options with amplification may be estimated as the various solutions are explored.

Needs Assessment

There is a variety of assessments that evaluate the communication difficulties facing a person with hearing loss. One common, efficient tool is the Client Ori-

ented Scale of Improvement (COSI) (Dillon, James, & Ginis, 1997). This assessment is particularly useful because the individual provides the five most difficult communication situations, which are rated before and after amplification. Although this and other scales, such as the Abbreviated Profile of Hearing Aid Benefit (APHAB) (Cox & Alexander, 1995), are very helpful in documenting the benefits of amplification, they only indirectly evaluate the need for D-HATS. There are no questions to prompt exploration of hearing alarms like the smoke detector or alarm clock. In addition, traditional scales do not assess one's knowledge of laws that provide these accommodations in hotels while traveling. A tool that was developed in a convenient format for the audiologist to use with every client to ensure a comprehensive assessment of communication needs is known as the TELEGRAM. Tools are most useful if the name in some way implies their function. In this case, the novel acronym was chosen to convey the desire to improve communication across distances. The TELEGRAM was designed to be completed following the routine audiologic evaluation (Thibodeau, 2004). As shown in Figure 21-4, the TELEGRAM is intended to be a prompt for the areas that must be considered: Telephone, Employment, Legislation, Entertainment, Group communication, Recreation, Alarms, and Members of the family. Obtaining information regarding one's functioning in each of these areas will lead to recommendations for HATS or other rehabilitative strategies. The questions provided in Table 21-2 were based on critical areas recommended by Ross (2004) to be explored with every patient. Associated rating scales are suggested to quantify the difficulties. By providing a graphic form that is analogous to the audiogram, the audiologist should find it easy to document a patient's current

Table 21–2. TELEGRAM Rating Scale Key

Topic	Question	Rating
T	Are you having difficulty with communication over the telephone ?	Difficulty 1 = None, 2 = Occasional, 3 = Often, 4 = Always, 5 = Can't use the phone <i>Use "L" to designate Landline and "C" to designate Cell Phone.</i>
E	Are you having any difficulty with communication in your employment or educational environment?	Difficulty 1 = None, 2 = Occasional, 3 = Often, 4 = Always, 5 = Stopped working
L	Do you know about legislation that provides assistance for you to hear in public places or in hotels when you travel?	Knowledge 1 = Vast, 2 = Considerable, 3 = Some, 4 = Limited, 5 = None
E	Are you having difficulty with hearing during entertainment activities that you enjoy such as television, movies, or concerts?	Difficulty 1 = None, 2 = Occasional, 3 = Often, 4 = Always, 5 = Stopped Going
G	Are you having difficulty with communication in group settings?	Difficulty 1 = None, 2 = Occasional, 3 = Often, 4 = Always, 5 = Can't hear at all in groups
R	Are you having difficulty with hearing during recreational activities such as sports, hunting, or sailing?	Difficulty 1 = None, 2 = Occasional, 3 = Often, 4 = Always, 5 = Stopped the activity
A	Are you having difficulty hearing alarms or alerting signals such as the smoke alarm, alarm clock, or the doorbell?	Difficulty 1 = None, 2 = Occasional, 3 = Often, 4 = Always, 5 = Can't hear alarm <i>Use "S" for Smoke Alarm, "D" for Doorbell, and "A" for Alarm Clock</i>
M	Are you communicating with members of your family?	1 = Live with Normal Hrg Adult, 2 = Live with Young Children, 3 = Live with Teenagers, 4 = Live with Adult with Hrg Loss, 5 = Live Alone <i>Check all that apply</i>

dix 21–C so that current information regarding federal and state programs may be obtained when considering solutions for a particular individual's needs.

Telephone

When considering options for telephone communication, one must account for the fact that 95% of Americans now own a cell phone of some kind. The

ownership of smartphones is now up to 77% from 35% in 2011 (Pew Research Center, 2019a). This increase even applies to the use of technology by baby boomers, among whom usage has increased from 25% in 2011 to 67% in 2018 (Pew Research Center, 2019b). However, it is still important to distinguish problems with landline phones versus cell phones because the solutions may vary depending on the features of the personal hearing aid. For example, difficulty with landline phones may be addressed by a

phone amplifier or by increasing the gain of the telecoil, whereas difficulties with cell phone communication may be addressed by setting up direct audio input or a Bluetooth connection. Several resources for solutions for phone communication are provided in Appendix 21–D.

For someone with mild communication difficulties on the landline phone who does not wear hearing aids, a simple phone amplifier that fits over the handset may be sufficient. Others may need a phone that provides an amplified handset or a visual indicator. The design features of the phone, such as cordless or large numbers, will need to be considered relative to the patient's needs.

Persons with MFi and MFA hearing aids will most likely use the connectivity features to enhance phone communication. The seven manufacturers shown in Table 21–1 provide devices that are capable of direct connectivity to phones through the 2.4 GHz transmission protocol (either Bluetooth LE for the iPhone connectivity or proprietary protocol for the Android connectivity). This feature is available in styles that fit behind or in the ear. Some of these are rechargeable and/or have a telecoil option. Unlike using a landline phone with older hearing aids, these MFi and MFA hearing aids that communicate wirelessly with smartphones allow bilateral reception of the signal from the phone, resulting in significantly improved speech recognition (Picou & Ricketts, 2011).

For those users without direct cell phone connections, the next consideration might be the acceptance of a body-worn connection to the cell phone, such as an induction loop that can interface with a telecoil in the aid or a proprietary streamer. These options will involve remembering not only to wear the accessory but also to charge it regularly. Each of these can connect to a smartphone via the classic Bluetooth protocol and have the option for increasing volume of the signal from the phone. The phone must be paired with the neck loop device to establish communication, which is a short process of turning both devices on and following “seeking” instructions on the cell phone. When paired, the audio signals from the cell phone will be sent to both hearing aids.

If the TELEGRAM assessment indicates difficulty hearing in noise, then a system with a remote microphone might be considered; such a device can be paired via the classic Bluetooth protocol to convey phone communication directly to the hearing aid, as shown in Figure 21–5, or to a neck loop or receiver connected to a hearing aid or cochlear implant. Although these microphones also require a dedicated charging cord, some have charging docks that can be



Figure 21–5. A remote microphone that connects via classic Bluetooth to a cell phone. Photo courtesy of ReSound.

interfaced with the television to send the audio signal to receivers on the hearing aids while charging.

For those with more severe hearing loss who rely on visual communication, the older devices known as the Telecommunication Device for the Deaf (TDD) have often been replaced by use of instant messaging between cell phones or caption phones. With text messaging, the information is typically typed in abbreviated format because of the limitations of the small keyboard and displays on cell phones. In addition, the conversation is not in real time and the sender or receiver may not be able to respond immediately. Therefore, many persons prefer to have a landline phone that provides captions where conversations can be stored and voice mail can be captioned. Because these phones involve both computer speech recognition and human oversight, the accuracy may be greater than in text messaging between phones. These devices, referred to as captioned telephones, are provided for those with hearing loss at no cost other than the fee required for internet access. Further information regarding these cell phone options may be obtained through the links provided in Appendix 21–D.

Another telephone accommodation made possible via smartphone technology allows the person with hearing loss to see the other person in the conversation. There are several ways to do this through cell phone applications or internet access to websites. The links for programs such as Skype, FaceTime, and Tango are provided in Appendix 21–D. Skype is a video and instant messaging application that is available with any smartphone as long as both users are connected to the Internet. Another application, FaceTime, allows video messaging between iPhones (model 4 or later) over a Wi-Fi connection. Tango

and SnapChat are also video messaging applications that are accessible from a variety of smartphones and that utilize mobile internet as well as Wi-Fi. Despite the increasing popularity of these video messaging applications, there may be visual delays or degraded images, the effects of which are unknown (Cromartie et al., 2012).

The ability of the listener to hear the phone ring should also be considered. A common solution for the inaudibility of the typical high-pitched ring of a landline phone is to select a phone with an amplified ringer and/or visual alert. An accommodation that does not involve HATS is to recommend the user purchase a cordless phone with a high-volume ringer and with multiple receivers that can be placed in rooms of frequent use. There are also applications that can run on cell phones that will result in flashing or vibrating signals when the phone rings. Most users of cell phones will use the vibrate feature to be alerted to incoming calls.

Employment

Communication difficulties at the workplace can be minimized in a variety of ways. This section of the TELEGRAM is included to prompt discussion of chal-

lenges at work that can be addressed through specific programs on the personal hearing aids or perhaps through group amplification systems. In addition, there may be challenges with phones at work that can be reduced through HATS. A referral to a state government program that provides services to assist with employment may be necessary. Every state in the U.S. has government programs that provide services such as assistive technology, college tuition, and interview and résumé coaching to facilitate gaining employment. HATS required in the workplace depend on the type of job. Through exploring the challenges in the workplace, many possible solutions can be offered. In addition to phone solutions mentioned above, communication issues may be addressed through the use of a remote microphone system. Such a system includes a wireless microphone that communicates directly with the hearing aid or implant or that communicates with a receiver that connects to the hearing aid or implant as shown in Figure 21–6. The advantage is significant improvement in speech recognition in noise as a direct result of microphone location (Lewis et al., 2003; Schafer & Thibodeau, 2006; Thibodeau, 2010, 2014, 2019). Some microphones can be worn by the speaker, whereas others can be held by the listener and pointed directly at the speaker for optimum reception. A particularly useful feature in



Figure 21–6. A DM transmitter (A) and integrated receiver attached to base of a hearing aid (B). *Source:* Images © Sonova AG. Reproduced here with permission. All rights reserved.

small meetings is the use of a conference microphone that can be placed in a central location to pick up conversation ranging around the table. Thibodeau (2019) found that the Roger Select microphone resulted in 61% improvement in sentence recognition in noise compared to the use of hearing aids alone for persons with moderate to severe hearing loss when the source originated randomly from one of five speakers around a conference table.

Legislation

As part of a comprehensive service to persons with hearing impairment, audiologists should determine the person's awareness of laws and state programs that provide assistive devices. Persons with hearing impairment can use this information to become advocates for their needs in employment, travel, or educational situations. Probably the most useful legislative action is the American with Disabilities Act (ADA). The ADA requires employers and/or businesses that provide services to the public to make accommodations for persons with disabilities such as hearing loss. These services could include amplified telephones, infrared devices in theatres, vibrating alarm clocks in hotels, or an FM system with a conference microphone in staff meetings. Knowledge of legislation regarding phone communication is also important. The Telecommunications Relay Service (TRS) mentioned earlier allows persons with hearing or speech disabilities to place and receive telephone calls and is available in all states for local and/or long-distance calls. The Federal Communications Commission (FCC) has adopted a convenient code, 711, for access to TRS regardless of the state. However, ADA requires equal access in the case of an emergency. Therefore, a person who relies on a TDD should dial 911 directly and not make a TRS call via 711. More information regarding 711 services may be found in the websites provided in Appendix 21–C.

Many states have legislation that provides telephone assistance for persons with hearing loss. In Texas, for example, the state provides the Specialized Telecommunications Assistance Program (STAP) for people who are deaf or hard of hearing. Eligibility requirements for the program include proof of Texas residency and an application signed by a physician or audiologist documenting that the person has impaired hearing. A STAP voucher is mailed to the recipient's home and then brought to a certified vendor. A variety of devices is offered at these sites, such

as amplified telephones, signaling devices, and telephone listening systems.

Entertainment

Entertainment is often provided in group formats, such as theatrical productions or concerts, thus requiring the assistance of a large-area system. Three main systems may be used in large auditoriums, including infrared, FM, and induction loop. The first two transmission methods involve receivers offered by the establishment at no cost to the person with hearing impairment. Whether the client watches television at home or attends concerts, the audiologist should determine the need and recommend possible solutions. Many hearing aid manufacturers provide an accessory to connect to the television that will send the audio signal via digital streaming to the hearing aid through the 900 MHz or 2.4 GHz transmission protocols. These are typically designed specifically to transmit to the matching receivers within the hearing aids by the same manufacturer. Another solution for improved reception of the sound from the television is to purchase a TV band radio that can be placed next to the chair of the person with hearing loss. By having the radio speaker close to the listener, the signal-to-noise ratio is improved. In addition, personal earphones may be used with the radio to reduce unwanted ambient sound.

Many persons with significant hearing loss can only enjoy movies at the theatre when there is open captioning. Information can be accessed on a smartphone about which movies are showing in local theatres that have captioning. An application can be downloaded onto one's phone, iPad, or iPod Touch that will provide information about the types of captioning available and show times at the theatres. The information is also available through a website link, shown in Appendix 21–D.

Groups

Communication in groups is often difficult because of distance from the speaker and noise and/or reverberation in the background. Remote microphone systems have been shown to reduce the deleterious effects of these factors and may be interfaced with the personal hearing aid in two primary ways. First, a small FM or DM receiver may be attached to the hearing aid or implantable hearing device processor through a

direct audio input connection. The receiver may be secured to the aid with an audio shoe or directly attached to the aid as shown in Figure 21–6. Second, the FM or DM signal may also be received via an induction neck loop or a streamer with an appropriate receiver attached and worn on the body. In group situations, such as a lecture where there is one primary speaker, the remote microphone can be worn by the presenter. For small group conversations, the remote microphone may simply be pointed toward the speaker of interest. New developments in remote technology allow the use of multiple transmitters to one receiver. This will allow members of a group to use their own microphone and communicate directly with the person with hearing loss. Conveniently, one manufacturer provides a wireless option compatibility search called the “Roger Configurator” on their webpage. Connectivity options for Roger technology are provided for several hearing aid and implant manufacturers (Phonak, 2019).

The use of a neck loop accommodation to improve hearing in group public settings has received much attention through a national campaign known as “Get in the Hearing Loop” supported by the Hearing Loss Association of America (HLAA) and the American Academy of Audiology (AAA). A large percentage of hearing aids and cochlear implants dispensed today include a telecoil. Many public places, such as theaters and airports, provide induction loops, which send an electromagnetic signal from a source to the telecoil of the hearing aid or implant. The plan for the campaign was launched in June 2010 with the hope that audiologists would dispense a greater number of hearing aids that include telecoil functions. In addition to increasing consumer awareness, the HLAA and AAA have also educated owners of public facilities about the benefits and relative ease of providing additional amplification to patrons with hearing impairment through the use of induction loops.

Another option for improving speech recognition in groups is to use the cell phone as a microphone to transmit to the MFi hearing aids through the feature called “Live Listen.” Because of the noticeable transmission delay associated with the Bluetooth transmission protocol, the user will want to mute the hearing aid microphone if possible. Other options with the cell phone include apps listed in Appendix 21–D that will amplify sound picked up by the microphone on the cell phone and deliver it to the user via wired earphones. Some of these may be downloaded at no cost to the user and provide considerable benefit as a backup or precursor to hearing aid use.

Recreation

Hobbies, of course, vary with personal preference and the dependence on hearing for full enjoyment should be considered. For example, if a retired couple enjoys traveling in a recreational vehicle across the country, communication may be limited because of road noise. An 8-hour trip could be significantly more enjoyable with a hardwired amplifier with an extended microphone cord to be worn by the speaker. Some persons prefer to use the FM/DM system described earlier for more freedom of movement. A less expensive option may be a mini microphone that is designed for short range transmission and communicates directly with the hearing aid through digital streaming without the need for additional specialized receivers as shown in Figure 21–5. These mini microphones are paired with manufacturers’ hearing aids and are not universal like FM technology. Providing a tour guide with an FM transmitter can significantly improve the quality of an organized vacation for a traveler with hearing loss. Assistive technology should always be explored for those with hobbies that involve noise such as woodworking or boating, depending on their communication needs. For example, an amplified phone and intercom system may reduce frustrations for a wife trying to communicate with her husband, who spends considerable time in a woodworking shop. Individuals may be wearing ear protection in these noisy environments, further necessitating assistive technology, such as the vibrating signals described in the next section.

Alarms

Hearing-assistive technology is often a necessity to detect alarms, phone ringers, and warning signals in the environment. Most common alarm-type signals include doorbells, alarm clocks, and smoke alarms. A single device can be used for all these alarms that converts the auditory signal into some type of notification. These alerts can be conveyed through a band worn on the wrist, a dedicated display, or a flashing lamp pattern. Recently, apps have been developed for the smartphone to alert the user to alarms in the environment. Some apps are specific to a single sound such as a doorbell or a smoke alarm and others are designed to alert for multiple signals based on the ability of the smartphone’s microphone to detect and identify the sound. The accuracy may vary with sensitivity of the microphone on the phone and noise in the

environment so it is best to read reviews and arrange specific trials before relying solely on a smartphone app for an alerting system.

With cameras that can be placed at the front door or in a baby's room, alerts can be conveyed via the associated app on the cell phone, signaling that movement or sound has occurred. The apps can be set to record the activity over a period of time or send an alert when certain activity has occurred, such as a doorbell being rung.

Although not considered technology, the benefit of hearing dogs or hearing-ear dogs should not be overlooked when considering options for hearing alarms. Hearing dogs are specially trained to help an individual hear the specific alarms in their environments. The dogs are selected from animal shelters and are trained over several months before being matched with an individual. They then receive more specific training for that person's environment.

Members of the Household

Members of the household certainly provide hearing assistance and should be considered when addressing the needs of the person with hearing impairment. If there is a spouse or life partner, he or she should be an integral part of the technology and communication strategy training. Small children or teenagers present a special set of challenges with high-pitched voice and rapid speech. HATS may be an integral solution for these challenges, but simple communication rules, such as not talking until in the same room or facing the listener, will often be of great assistance. Solutions can also be creative, such as wearing a button or t-shirt with a printed reminder like "Thank you for facing me when you speak!"

Fitting and Verification

After determining the need for assistive technology, the particular device should be verified in some way. Just as considerable time and expertise may be spent on achieving the optimal gain and output of an ear-level device for face-to-face communication, the optimal output when connected wirelessly for communication should also be verified. The techniques to do this vary with the device. Ideally, verification can include both electroacoustic and behavioral measures. Those that deliver the signal via the personal hearing

aid, such as an FM or DM system with a direct audio input FM or DM receiver, can be evaluated using existing electroacoustic test equipment and couplers. When the coupling does not allow use of standard couplers, behavioral evaluation should be performed.

There are two documents that relate to the verification of HATS. The ANSI standard for "Specification of Hearing Assistance Devices/Systems" (ANSI, 2014) addresses the electroacoustic evaluation of technology that is packaged as a personal system (rather than large-area group systems). Recommended electroacoustic measurements are similar to those for hearing aids in the ANSI S3.22 standard, but have specific requirements for placement of the transmitting microphone and the receiver. These procedures allow comparison to manufacturer specifications, as well as comparison across equipment models, because prescribed input levels and equipment arrangements are used (Schafer, Thibodeau, Whalen, & Overson, 2007). Many devices can be evaluated with existing couplers used to test hearing aids whereas other devices, such as those with headphones, may not. The manufacturer will then report the coupling method used to obtain the electroacoustic results.

The second document is the "AAA Clinical Practice Guidelines: Remote Microphone Hearing Assistance Technologies" (AAA, 2011). This comprehensive guide focuses on hearing-assistive technology for individuals from birth to 21 years. It is an expansion of the original guidelines developed by an American Speech-Language-Hearing Association (2002) Task Force, which focused on real-ear, electroacoustic, and behavioral evaluation procedures. As technology has expanded, the need for a more comprehensive document emerged. The AAA guidelines include information to consider in determining candidacy, device selection, fitting and evaluation, and staff in-service. A supplement is included with specific protocols for the evaluation of ear-level FM systems when used with children who wear hearing aids or cochlear implants or who have normal hearing. Additional supplements will be forthcoming to address evaluation of soundfield and induction loop systems.

Although written for fitting children, these protocols are applicable to adult candidates. When fitting wireless systems, it is important that electroacoustic verification be performed to measure that the wireless signal is received at a level above those of the environmental signals processed through the hearing aid, thus resulting in a favorable signal-to-noise ratio (S/N) (Bondurant & Thibodeau, 2011). This is determined by comparing the output of the hearing

aid alone to that of the combined hearing aid and wireless microphone system when tested with a 65 dB SPL complex input. The two curves should be closely aligned. Then, when the FM microphone receives the typical input of 80 dB SPL from the speaker, the optimal S/N will be accomplished. If the two curves are not similar, adjustments may be made in the receiver to compensate for the offset. The AAA task force has proposed a taxonomy that includes the type of measure (electroacoustic), the system receiving the input (hearing aid or FM), and the input level (65 dB SPL). For example, the first measurement is EHA65 (electroacoustic hearing aid 65 dB SPL input), which is an electroacoustic measure performed with the hearing aid in the test box and a 65 dB SPL input. The next measure is EFMHA65 or EDMHA65, which is performed with the FM or DM microphone in the test box to receive the input and the hearing aid outside the test box. In summary, the difference between the output curves for EHA65 and EFMHA65 or EDMHA65 should be close to zero.

With various options for using smartphones with hearing aids, there will undoubtedly be new protocols for verifying performance. For example, for those who choose to use the smartphone in a traditional way by holding it close to the microphone of their hearing aid, the impact of the smartphone case on the intensity of the voice of the caller should be evaluated. Fray and Thibodeau (2018) devised a procedure to evaluate two different cases for smartphones and found that, in some cases, the signal was reduced by 37 dB relative to the signal picked up by the hearing aid microphone. The output measured from the hearing aid when connected to the phone without the case that is placed in the test box is compared to the output of the hearing aid when connected to the phone with the case that is placed in the test box as shown in Figure 21–7.

Although the benefit of assistive technology is best measured in one's real-life environment, the AAA Guidelines include procedures for behavioral verification of remote microphone technology. The

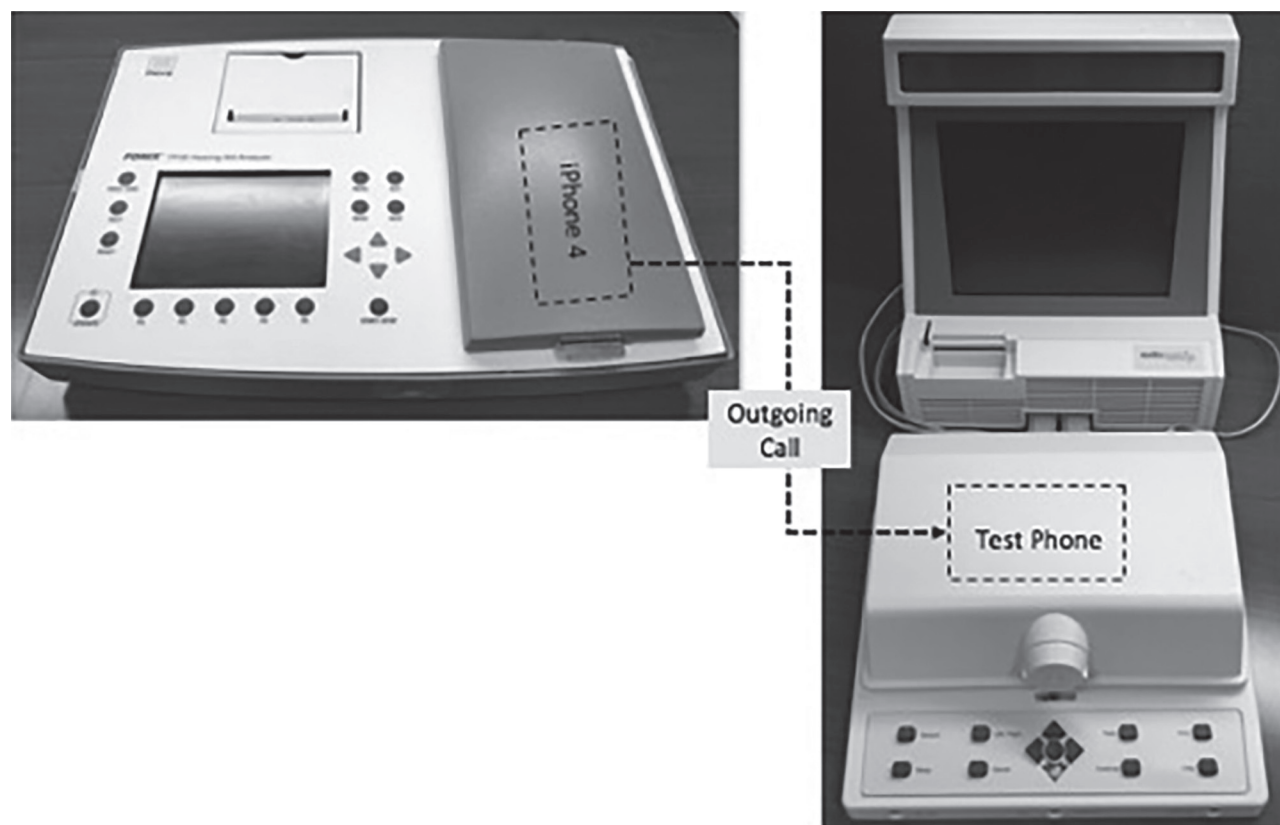


Figure 21–7. Electroacoustic analysis of the effect of phone cases where one phone is placed inside a test box to receive the stimulus (*left*) and transmit to a second phone that is inside another test box (*right*). The second test box is also used to measure the output of the hearing aid when paired with the second phone.

listener with the technology is seated in the sound booth at 0 degrees azimuth, while the examiner with the microphone is seated at the audiometer outside the booth. Using appropriate speech recognition materials, the first score is obtained via live-voice presentation in a 0 dB S/N. Following the proposed terminology, this condition is BHA50/50, which designates a behavioral evaluation with the hearing aid alone (BHA) with speech and noise presented at 50 dB HL (50/50). The next condition is BFM/HA50/50 or BDM/HA50/50, which is similar to the first measure except now the examiner has turned on the remote microphone. When BHA50/50 does not result in a score below 80%, the noise may be increased to create a poorer S/N so that the benefit from the assistive technology can reach significance. The 20% change in scores required for significant difference is based on a 25-word list. The average benefit for 10 adults when tested using this protocol with FM technology was 34% (Thibodeau, 2007).

Following verification, the individual and possibly a family member will need instruction on the care and use of the chosen technology. Although some HATS can be very simple to operate, such as a vibrating alarm clock, others may involve multiple components, such as an FM or DM system that interfaces with one's personal hearing aids or implantable device. Not only is there technology added to the hearing aid, but there is also a remote microphone, which may also have several features. In addition to the verbal instruction and practice at the auditory rehabilitation appointment, written materials should be provided for review at home.

Validation

Once HATS have been appropriately fit, the final step is validation. This is necessary to determine if the individual receives the intended benefits. Although a hearing aid with wireless microphone connectivity could be precisely fit and significant benefit shown in the FM fitting evaluation, if the individual forgets how to operate the device when he/she gets home and does not use the system, then the intended benefits are not realized. Therefore, the TELEGRAM or COSI should be administered 3 to 6 months following the fitting to determine if there is reduction in the degree of difficulty. In a study using the TELEGRAM, it was determined that all individuals showed improvement in six of the seven areas (Thibodeau,

2007). By reviewing the TELEGRAM at the annual audiologic evaluation, the benefit received through the technology that was recommended the previous year can be documented and new recommendations made if necessary.

Integrating Into Clinical Practice

Prendergast and Kelley (2002) reported that more than 80% of audiologists were providing information regarding HATS to their patients. Given all the currently available connectivity options, audiologists would be more likely to mention D-HATS than before, when these options were considered part of a separate program. There is no longer a need for a separate room for incorporating HATS into one's practice like that suggested by Lesner and Klinger (1995) who offered considerations such as room size, furnishings, environment, and organization of special equipment. Rather, once the speech and nonspeech communication needs and the level of technology are determined, the audiologist can determine possible network options. All D-HATS can be demonstrated or illustrated through the use of a small tablet that could be available in every fitting room. Some rooms are already equipped with wall-mounted screens for patient education during the fitting process. For those who still need access to A-HATS, many resources can be offered with minimal effort by providing a simple resource brochure of commonly used A-HATS for the telephone, television, and alarm clocks (Oaktree Products, 2019). These brochures can be customized with contact information and provided to each patient following the recommendations provided on the TELEGRAM.

In addition to incorporating these HATS resources into clinical practice, the benefits of group sessions focused on HATS should be considered. Offering information sessions regularly to review hearing aid connectivity features and D-HATS could not only lead to greater satisfaction with a purchase, but also to new referrals. Consumers may increase their acceptance of HATS when sharing communication frustrations and HATS solutions in group discussions (Thibodeau & Cokely, 2003). Furthermore, overall clinical efficiency is increased when device explanation occurs in group formats (Chisolm, McArdle, Abrams, & Noe, 2004). One format involves four weekly, 1-hour meetings focused on trials with wireless technology. The program is referred to as Appli-

cation of Advanced Listening Technology in Adults (AALTA) and is offered with support from manufacturers of wireless technology so that individuals may have personal experiences, which can then be shared in the weekly AALTA meetings (Thibodeau, 2007).

Summary

The needs of the adult with hearing loss extend into many aspects of life. The first solution that is considered is to provide amplification that can restore audibility of acoustic information. There are many instances, however, in which noise, reverberation, and distance render the acoustic signal inaudible, regardless of the amplification technology. In addition, hearing aids are not worn all hours of the day, and yet there are alarms that can convey life-saving information, such as a smoke detector, that may need to be recognized at any time. Therefore, it is part of basic audiologic care to consider the assistive technology that a person may need for everyday life. Despite the basic needs of persons with hearing loss, audiologists can no longer view assistive technology as just accessories. A routine tool is proposed as part of the basic audiologic evaluation to prompt the audiologist to address these needs. The format of the TELEGRAM allows for a quick review of problems that can lead to recommendations for technology or acquiring knowledge (such as learning about legislative issues). Once the needs are identified, the next challenge is for the audiologist to provide the technology or guidance to use existing technology, such as their smartphone or a tablet, by which patients can find solutions. After the appropriate technology is received, the audiologist's responsibility includes follow up to ensure that it is being used properly and is effectively meeting patient needs. The progress can be noted on the TELEGRAM so that areas may be reassessed at subsequent evaluations.

Hearing aids alone are sufficient for some persons with hearing loss, but many patients can also benefit from the many features that now connect those aids to solve hearing and communication challenges. Audiologists are best suited to see that the process of acquiring and fitting assistive technology occurs efficiently without weeks or months of frustration. With this comprehensive audiologic care, those with impaired hearing may reduce their communication challenges and continue to live enjoyable and productive lives.

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Appendix 21-A
**ILLUSTRATION OF BENEFITS OF ANALOG HEARING-ASSISTIVE
TECHNOLOGY (A-HAT)**

THREE DEAF PIGS

The first pig built a house of straw.

The wolf came.

He yelled for the pig to come out.

The pig could not hear him.

The wolf blew down the house and ate the pig.

The second pig built a house of sticks.

The wolf came.

He yelled for the pig to come out.

The pig could not hear him.

He thought it was a tornado.

The wolf blew down the house and ate the pig.

The third pig built a brick house with flashing lights and all the necessary deaf devices.

The wolf rang the bell and the lights flashed.

He called the zoo, using the TTY relay service.

A zoo keeper came and put the wolf in a sign language class.

The wolf learned to communicate and became friends with the pig.

Author Unknown

Appendix 21-B
**ILLUSTRATION OF BENEFITS OF DIGITAL HEARING-ASSISTIVE
TECHNOLOGY (D-HAT)**

THREE DEAF PIGS

The first pig built a house of straw.

The wolf came.

He yelled for the pig to come out.

The pig could not hear him.

The wolf blew down the house and ate the pig.

The second pig built a house of sticks.

The wolf came.

He yelled for the pig to come out.

The pig could not hear him.

He thought it was a tornado.

The wolf blew down the house and ate the pig.

The third pig built a brick house with a front-door webcam connected to his cell phone and all the necessary apps.

The wolf rang the bell and the pig received an alert on his smartphone while he was watching a captioned movie on Netflix.

He immediately sent a text to the zoo.

A zookeeper came and put the wolf in a computer class.

The wolf learned to type and became Facebook friends with the pig.

Author Unknown

Appendix 21–C

RESOURCES FOR SERVICES FOR PERSONS WITH HEARING LOSS

Legislative and State Programs Regarding Accommodations for Those With Impaired Hearing

<https://www.fcc.gov/accessibility>

<https://www.hearingloss.org/hearing-help/financial-assistance/state-agenices/>

Professional Organizations With Information Regarding HATS

Hearing Loss Association of America

<https://www.hearingloss.org/>

American Speech-Language-Hearing Association

<https://www.asha.org/public/hearing/hearing-assistive-technology/>

American Academy of Audiology

<https://www.audiology.org/>

Alexander Graham Bell Association

<https://www.agbell.org/>

Academy of Rehabilitative Audiology

<https://www.audrehab.org/>

National Deaf Center

<https://www.nationaldeafcenter.org/topics/assistive-technology>

National Institutes of Health

<https://www.nidcd.nih.gov/health/assistive-devices-people-hearing-voice-speech-or-language-disorders>

Appendix 21-D

RESOURCES FOR HEARING-ASSISTIVE TECHNOLOGY SYSTEMS

General Hearing-Assistance Technology

<http://www.oaktreeproducts.com>

<http://www.dogsforbetterlives.org>

<http://www.harc.com>

<http://www.hearingloss.org/programs-events/get-hearing-loop>

Hearing-Assistive Technology via Smartphone Applications or Internet Access

Captionfish—Guide to movie theatre showings with captioning.

<http://www.captionfish.com/>

AVA—An app to transcribes conversations in real time.

https://www.ava.me/download/?_branch_match_id=603694227730547044

Live Transcribe—An app for Android platforms to transcribe conversations in real time.

<https://www.android.com/accessibility/live-transcribe/>

Otter—An app for IOS platforms to transcribe conversations in real time.

<https://apps.apple.com/us/app/otter-voice-meeting-notes/id1276437113>

Ear Machine—Amplifier of conversation with tone and pitch adjustments.

<http://www.earmachine.com>

Hear Boost—Amplifier of conversation with record feature.

Android link: <https://play.google.com/store/apps/details?id=com.audiofix.hearboost>

iPhone link: <https://itunes.apple.com/us/app/hear-boost-enhanced-recorder/id1437159134?mt=8>

Frequency/Digital-Modulation Systems

Audio Enhancement—<https://audioenhancement.com/audio/>

Comfort Audio—<http://www.comfortaudio.com/us/>

Comtek—<http://www.comtek.com/tour-guide>

Etymotic—<http://www.etymotic.com>

Oticon—<http://www.oticon.com/solutions/opn-wireless-connectivity>

Phonak—<http://www.phonak.com/com/en/hearing-aids/accessories/accessories-overview.html>

Williams Sound—<http://www.williamssound.com/personal-listening-products>

Captioned Phones Available at No Cost for Persons With Hearing Loss

CaptionCall—<https://captioncall.com>

Clear Captions—<https://clearcaptions.com>

CapTel—<https://www.captel.com>

Apps for Phone Communication

Video Messaging via Internet Access

Skype—for all phones via Internet access:

<https://www.skype.com/en/get-skype/>

FaceTime—for iPhones (model 4 or later):

<https://support.apple.com/en-us/HT204380>

Tango—for all phones:

iPhone link: <https://itunes.apple.com/us/app/tango-live-video-broadcast/id372513032>

Android link: <https://play.google.com/store/apps/details?id=com.sgiggle.production>

Internet Protocol Services for Phone Conversations

IP-Relay for Cell Phones

iPhone link: <https://itunes.apple.com/us/app/sprint-ip-relay/id542802329>

Android link: https://play.google.com/store/apps/details?id=com.sprint.trs&hl=en_US

Hamilton Mobile CapTel for Cell Phones

iPhone link: <https://itunes.apple.com/us/app/hamilton-mobilecaptel/id370615084?mt=8>

Android link: https://play.google.com/store/apps/details?id=com.hamilton.ui&feature=nav_result#?t=W251bGwsMSwxLDMsImNvbS5oYW1pbHRvbi51aSJd

Sprint CapTel for Cell or Landline Phones

<https://sprintcaptel.com>

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TINNITUS MANAGEMENT

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Introduction

Tinnitus has become an increasing global health concern affecting approximately 278 million people worldwide (Araujo & Lório, 2016; Swain, Nayak, Ravan, & Sahu, 2016; Tunkel et al., 2014). In the United States alone, as many as 50 million Americans report some degree of tinnitus, with 42 million experiencing chronic tinnitus (Henry, Dennis, & Schechter, 2005; Hoge et al., 2008; Shargorodsky, Curhan, & Farwell, 2010). Approximately 10 million Americans have tinnitus severe enough to seek professional assistance from a hearing health care provider (Saunders, 2007).

The disabling effects of tinnitus have been associated with a wide range of distressing psychosocial symptoms that impact a patient's health-related quality of life (HRQoL), much like chronic pain (Magnusson, 2010; Møller, 2007). In this connection, the negative consequences of tinnitus include sleep disturbance, anxiety, depression, irritation, frustration, loss of control, concentration difficulty, and reduced cognitive functioning (Handscomb, Hall, Shorter, & Hoare, 2017; Langguth, Kreuzer, Kleinjung, & de Ridder, 2013; Tegg-Quinn, Bennett, Eikelboom, & Baguley, 2016; Trevis, McLachlan, & Wilson, 2017). Moreover, tinnitus has been recognized as a substantial societal economic burden on the health care system considering the service-connected disability compensation costs for veterans, the loss of workplace productivity, and high resource utilization (Fagelson, 2016; Maes,

Cima, Vlaeyen, Anteunis, & Joore, 2013; Steinmetz, Zeigelboim, Lacerda, Morata, & Marques, 2008; Tunkel et al., 2014; Vio & Holme, 2005).

Perusal of the literature suggests there is an increase in the momentum of clinical and research interest in the topic of tinnitus. Yet, relatively few audiologists provide services to patients who suffer from tinnitus, perhaps because many audiologists feel ill-equipped to evaluate patients with tinnitus and to develop and execute a treatment plan. There may be several reasons for this:

- There is no single unifying model describing the mechanisms underlying the symptoms of tinnitus, making it difficult to develop specific treatment protocols.
- Clinical practice guidelines for the assessment and treatment of tinnitus are just beginning to emerge worldwide with diverse recommendations that contribute to the variations that currently exist in tinnitus intervention options (Fuller et al., 2017; Tunkel et al., 2014).
- There remains a lack of comprehensive training about tinnitus and its management even in today's Doctor of Audiology (Au.D.) graduate programs (there are a few exceptions to this). Typically, students may only receive a few lectures on the evaluation and audiologic management of tinnitus, coupled with limited clinical exposure (Henry, Zaugg, & Schechter, 2005a).

- There is considerable variability across patients in regard to their tinnitus and their ability to cope with its psychosocial consequences. Accordingly, the rehabilitative management of patients with tinnitus varies depending upon the impact it has on the individual's HRQoL—creating a less than straightforward treatment approach.

Taking all of these issues into consideration, it is understandable that so many audiologists are reluctant to provide clinical management for this patient population in need of clinical services.

Tinnitus management, however, does fall within the scope of practice for audiologists (American Academy of Audiology, 2004; American Speech-Language-Hearing Association [ASHA], 2018). Furthermore, it is the audiologist who is best positioned to function as the primary health care provider on an interprofessional collaborative practice tinnitus management team. As audiologists, we have the necessary preparation: (a) knowledge of auditory disorders and their impact on auditory system function; (b) understanding of acoustic and psychoacoustic principles underlying tinnitus assessment (e.g., pitch and loudness matching) and intervention (e.g., sound therapy including tinnitus masking and habituation); (c) competency in using clinical instrumentation necessary to evaluate hearing and conduct the tinnitus psychoacoustic evaluation; (d) familiarization with self-assessment methodology used to quantify HRQoL; (e) expertise in selecting and fitting hearing aids (HAs), which remains a primary treatment method for providing tinnitus relief; (f) professional working relationship with physicians, psychologists, and other health care practitioners; and (g) patient-centered counseling skills that serve as the cornerstone for all contemporary audiologic rehabilitation programs (ASHA, 2018; Henry, 2004; Henry et al., 2005a).

The purpose of this chapter is to provide an overview of the management of patients with tinnitus using an interprofessional collaborative practice care path model that can be implemented in a busy clinical practice. The chapter begins by providing the basics such as definitions, epidemiologic factors, perceptual attributes, perceived severity, classification schemes, and the mechanisms of tinnitus generation and perception. The latter fundamental concepts are followed by a description of the evaluation process and treatment options, concluding with a discussion of the roles of other health care practitioners and alternative nonaudiologic treatment options for patients with tinnitus.

Underlying Principles of Tinnitus

Definitions

Tinnitus has been defined as the perception of sound in the absence of an external acoustic stimulus (Folmer, Griest, & Martin, 2001). Some have described it as a “phantom” auditory perception resulting from tinnitus-related neuronal activity within the auditory pathways (Jastreboff, 1990, 1995). It has been argued, however, that the use of the term *phantom* may not be helpful in a clinical context because it suggests that the perception is nonexistent (Noble & Tyler, 2007). Further, tinnitus should be recognized as a symptom of various causes and mechanisms but not as a disease in and of itself (Tunkel et al., 2014).

Some authors have defined tinnitus in terms of its temporal course (Coles, 1984; Dauman & Tyler, 1992; Davis, 1995; Hazell, 1995; Meikle, Creedon, & Griest, 2004). Most recently, the American Academy of Otolaryngology—Head and Neck Surgery (AAO-HNS) published clinical practice guidelines for tinnitus (Tunkel et al., 2014) and defined *recent onset tinnitus* as occurring less than 6 months in duration as reported by the patient, whereas *persistent tinnitus* has a duration of 6 months or greater.

Epidemiology

Obtaining a clear picture of the prevalence of tinnitus (i.e., number of individuals who experience tinnitus at a given time) is difficult because it has many forms and levels of severity, causing different studies to yield widely different outcomes (Møller, 2007). Furthermore, methodological techniques to gather prevalence data have varied using diverse sampling methods (e.g., questionnaires sent by mail or asked during an interview) and study populations (e.g., clinical samples having otologic pathologies or lack of differentiation between “normal” ear noises from “pathologic” tinnitus). In addition, such factors as general health status, degree of hearing loss, age, and noise exposure have not been well controlled, if at all, in most epidemiologic studies (Hoffman & Reed, 2004).

With an understanding of the aforementioned caveats, data estimate that at least some form of tinnitus occurs in approximately 25% of American adults (50 million people), and is a frequent occurrence in 8% (16 million people) of the adult population (Sharгородsky et al., 2010). Although most epidemiological studies are based on data from the U.S. (Hoffman

& Reed, 2004; Sindhusake et al., 2003) and European countries (Krog, Engdahl, & Tambs, 2010; Quaranta, Assennato, & Sallustio, 1996), recent studies investigating the extent of tinnitus in other regions of the world are emerging, providing evidence that tinnitus is a significant global health burden (Khedr et al., 2010; Kim et al., 2015; Lasisi, Abiona, & Gureje, 2010; Michikawa et al., 2010; Xu et al., 2011).

Generally speaking, the prevalence of tinnitus increases with age (peaking in the 60–69 age range), recreational firearm usage, military or occupational noise exposure, lower socioeconomic class, ototoxic drugs, and hearing loss (Bhatt, Lin, & Bhattacharyya, 2016; Fagelson, 2016; Radziwon, Hayes, Sheppard, Ding, & Salvi, 2016; Shargorodsky et al., 2010; Tunkel et al., 2014;). The effects of gender on the prevalence of tinnitus are inconsistent across several studies (Davis & El Rafaie, 2000). The latter factors, however, are not mutually exclusive and a complex relationship among the variables exists (Baguley, 2002; Davis & El Rafaie, 2000; Hoffman & Reed, 2004).

Perceptual Attributes of Tinnitus

The characteristics of tinnitus can be described in a multitude of ways: intermittency, perceived location, quality descriptors, loudness magnitude ratings, loudness and/or pitch fluctuations, and number of sounds comprising the tinnitus, to name a few. Meikle and colleagues created an extensive database containing information about the perceptual attributes of tinnitus reported by 1,630 patients with clinically significant tinnitus (Meikle et al., 2004). The reader is encouraged to visit the Tinnitus Archive™ (2nd edition) website <http://www.tinnitusarchive.org>, created initially by Meikle and colleagues (2004) and currently maintained by the Oregon Hearing Research Center.

Classification Schemes

Tinnitus is often categorized as being either *subjective* (i.e., heard only by the patient) or *objective* (i.e., produced by an internal source that can be heard by another person). It has been argued (Dobie, 2004b; Hazell, 1995), however, that the latter term is inappropriate in that some internally generated acoustic sounds may not be heard by anyone except the patient. Accordingly, the term *somatosound* (“body sound”) has been proposed (Dobie, 2004b) as more suitable than the term *objective* for all sounds with internal acoustic sources whether or not they are audible to

an external listener. The etiology of somatosounds may include such conditions as vascular pulsations, patulous Eustachian tubes, palatal or intratympanic myoclonus, and jugular outflow syndrome (Perry & Gantz, 2000).

The site of generation has also been proposed as a classification scheme, similar to hearing loss categorization. In this manner, the terms *middle ear/conductive*, *sensorineural*, or *central tinnitus* (Tyler & Babin, 1986; Zenner & Pfister, 1999) are used to describe tinnitus. Further, sensorineural tinnitus has been subclassified into four types based on involvement of the outer hair cells, inner hair cells, auditory nerve, and microvascular or osmotic endolymphatic mechanisms (Zenner, 1998). Although the latter classification scheme further delineates the potential tinnitus generator site, it most likely lacks clinical utility.

Using a broad classification scheme, the AAO-HNS (Tunkel et al., 2014) categorized tinnitus as being either *primary* or *secondary*. Primary tinnitus may be idiopathic in nature and may or may not be associated with sensorineural hearing loss. In contrast, secondary tinnitus is associated with a specific underlying cause (other than sensorineural hearing loss) or identifiable from organic auditory (e.g., otosclerosis, Meniere’s disease, vestibular schwannoma) or nonauditory (e.g., myoclonus, intracranial hypertension) conditions.

On the other hand, Tyler and Baker (1983) proposed that tinnitus be described as a function of associated HRQoL problems. Using the data obtained from 72 members of a tinnitus self-help group completing the Tinnitus Problem Questionnaire (TPQ), problems were assigned to one of four general categories: effects on hearing, effects on lifestyle, effects on general health, and emotional problems. Table 22–1 displays the 15 highest ranked tinnitus-related difficulties associated within each of the four general categories.

Using a cluster analysis statistical technique, Tyler and colleagues (2008) derived subgroups of tinnitus patients based on a number of variables involving different data collection mechanisms: outcome on questionnaires including the Tinnitus Handicap Questionnaire (Kuk, Tyler, Russell, & Jordan, 1990), Tinnitus Activities Questionnaire (Tyler et al., 2006), and portions of the Tinnitus Intake Questionnaire (Stouffer & Tyler, 1990); biographical factors; and psychoacoustic tinnitus measures including pitch and loudness matching. The rationale for developing tinnitus subgroup classifications is that they might provide insight into the underlying mechanisms of tinnitus and be helpful in determining a specific

Table 22-1. Highest Ranked Differences (Percentage of Total Respondents) Associated With General Functional Consequence Categories for Individuals Attending a Tinnitus Self-Help Group ($n = 72$)

Effects on Lifestyle	Effects on Hearing	Effects on General Health	Emotional Problems
Sleep (56.9)	Understanding speech (37.5)	Drug dependence (23.6)	Despair/frustration/depression (36.1)
Persistence (48.6)	Understanding TV (11.1)	Pain/headache (18.0)	Annoyance/irritation/inability to relax (34.7)
Worse on awakening (16.6)		Giddiness/balance/fuzzy head (13.8)	Concentration/confusion (33.3)
Avoid noise (15.3)			Insecurity/fear/worry (16.6)
Withdrawal/avoid friends (13.8)			
Avoid quiet (11.1)			

Source: Adapted from Tyler and Baker (1983).

intervention approach or combination of approaches. Based on their preliminary attempt to classify patients with tinnitus using cluster analyses they proposed four major subgroups:

- Constant distressing tinnitus;
- Varying tinnitus exacerbated by noise;
- Patients who are “copers” and whose tinnitus is not somatically modulated (tinnitus that cannot be changed by touch or position); and
- Patients who are copers but whose tinnitus is exacerbated in quiet environments.

Our preference for classifying tinnitus, especially within the context of audiologic rehabilitation, is based on a *biopsychosocial* model. The World Health Organization’s International Classification of Functioning, Disability, and Health (WHO-ICF) (2004) provides a conceptual framework for developing a tinnitus classification taxonomy (Henry et al., 2005; Manchaiah et al., 2018; Newman & Sandridge, 2004; Ramkumar & Rangasayee, 2010; Tyler, 1993, 2000). In short, the ICF addresses functioning and disability related to a health condition affecting an individual’s activities and participation in everyday life within the context of environmental and personal factors. Table 22-2 summarizes the individual domains of the WHO-ICF schema describing how these categories pertain to tinnitus and their application in the assessment process.

Mechanisms and Models

The precise origins of tinnitus and associated mechanisms involved in the generation of tinnitus are not well-understood. Based on current research findings, it seems most likely that multiple mechanisms (including the auditory periphery, central auditory nervous system, and nonauditory sensory and motor systems) are involved in the generation of tinnitus, either simultaneously or interactively (Baguley, 2002; Bartels, Staal, & Albers, 2007; Georgiewa et al., 2006; Henry, Roberts, Caspary, Theodoroff, & Salvi, 2014; Kaltenbach, 2011). In general, there is agreement that the mechanisms of tinnitus involve deprivation of input or abnormal input from the peripheral auditory system to the central auditory nervous system. Reduced neural activity from the cochlea (e.g., due to noise exposure or ototoxicity) may result in increased spontaneous neural activity (hyperactivity) in the central auditory nervous system (Salvi, Lockwood, & Burkard, 2000) possibly due to a reduction in inhibitory influences (Kaltenbach et al., 2002; Kaltenbach, Zhang, & Zacharek, 2004). Accordingly, tinnitus is currently viewed as a systemic problem arising from imbalances in the excitatory and inhibitory inputs to auditory neurons occurring at multiple levels of the auditory system (Kaltenbach, 2011). Because of neural plasticity, changes in neural input may result in structural and/or functional changes in the central pathways (e.g., cortical and tonotopic reorganization), thus causing the perception of tinnitus (Eggermont

Table 22-2. Tinnitus Classification Taxonomy, Associated Functional Domains, and Representative Assessment Tools Based on the World Health Organization's International Classification of Functioning, Disability, and Health (WHO-ICF, 2004)

Category	Definition	Functional Domains	Assessment Tools
Impairment	Dysfunction of the auditory system resulting in the perception of tinnitus	Perception of tinnitus: loudness; pitch; quality; location; maskability; timbre	Psychoacoustic measures (e.g., pitch matching; loudness matching; minimum masking level; residual inhibition); direct estimation/scaling techniques (e.g., numeric rating scales; visual analog scales)
Activity Limitation	Effect of impairment on reducing an individual's ability to function in a normal manner	Emotional effects (e.g., depression, anxiety, annoyance); hearing effects (e.g., receiving spoken messages; understanding conversations); sleep disturbance; intrusiveness; cognitive effects (e.g., reduced concentration; impact on working memory); loss of control (e.g., handling stress and other psychological demands); intrusiveness	Open-ended questionnaires (e.g., TPQ); tinnitus-specific self-report measures (e.g., TFI, THI); dimension-specific self-reported measures (e.g., HADS; PHQ-9; GAD-7; PSQI); diary/journal entries
Participation Restriction	Psychosocial manifestations of impairment and activity limitation resulting in the need for extra effort and reduced independence	Interference with: rest/relaxation; work; socialization; recreation and leisure activities; family, friend, and coworker interpersonal relationships	Open-ended questionnaires; tinnitus-specific self-reported measures; dimension-specific self-reported measures; diary/journal entries; generic health-related quality of life self-reported measures (e.g., SF-36; WHO-DAS II)

HADS = Hospital Anxiety and Depression Scale (Zigmond & Snaith, 1983); GAD-7 = Generalized Anxiety Disorder-7 (Spitzer, Kroenke, Williams, & Löwe, 2006); PHQ-9 = Patient Health Questionnaire 9 (Kroenke, Spitzer, & Williams, 2001); PSQI = Pittsburgh Sleep Quality Index (Buysse, Reynolds, Monk, Berman, & Kupfer, 1989); SF-36 = Medical Outcomes Study Short Form 36 (Ware & Sherbourne, 1992); TFI = Tinnitus Functional Index (Meikle et al., 2012); THI = Tinnitus Handicap Inventory (Newman, Jacobson, & Spitzer, 1996; Newman, Sandridge, & Jacobson, 1998); TPQ = Tinnitus Problems Questionnaire (Tyler & Baker, 1983); WHO-DAS II = World Health Organization Disability Scale (McArdle, Chisolm, Abrams, Wilson, & Doyle, 2005)

Note. Each category may be affected by personal (e.g., lifestyle; coping style; habits) and/or environmental (e.g., exposure to sound; technology use; climate) factors (WHO-ICF, 2004; Manchaiah et al., 2018).

Source: Adapted from Newman and Sandridge (2014).

& Roberts, 2004; Kaltenbach, 2000; Lockwood et al., 1998; Noreña, 2011; Roberts et al., 2010).

Moreover, abnormal somatosensory afferent input from the neck, jaw, and face can increase activity in the central auditory pathways, contributing to the generation of tinnitus (Roberts et al., 2010; Sanchez, 2016). The complex interactions among auditory, sensory, and motor systems are seen clinically when some patients with tinnitus can modulate its pitch, loudness, or both by forcefully contracting muscles in the head, neck, or limbs; moving eyes in

the horizontal or vertical plane; applying pressure on myofacial trigger points; moving the face or mouth; and/or applying pressure to the temporomandibular joint (Roberts et al., 2010; Sanchez, 2016; Sanchez & Kii, 2008; Simmons, Dambra, Lobarinas, Stocking, & Salvi, 2008).

As is evident, there is no single convergence of view regarding tinnitus mechanisms. In fact, several models, theories, and hypotheses of tinnitus have been proposed, based on anatomic sites, physiologic mechanisms, and psychologic influences. Figure 22-1

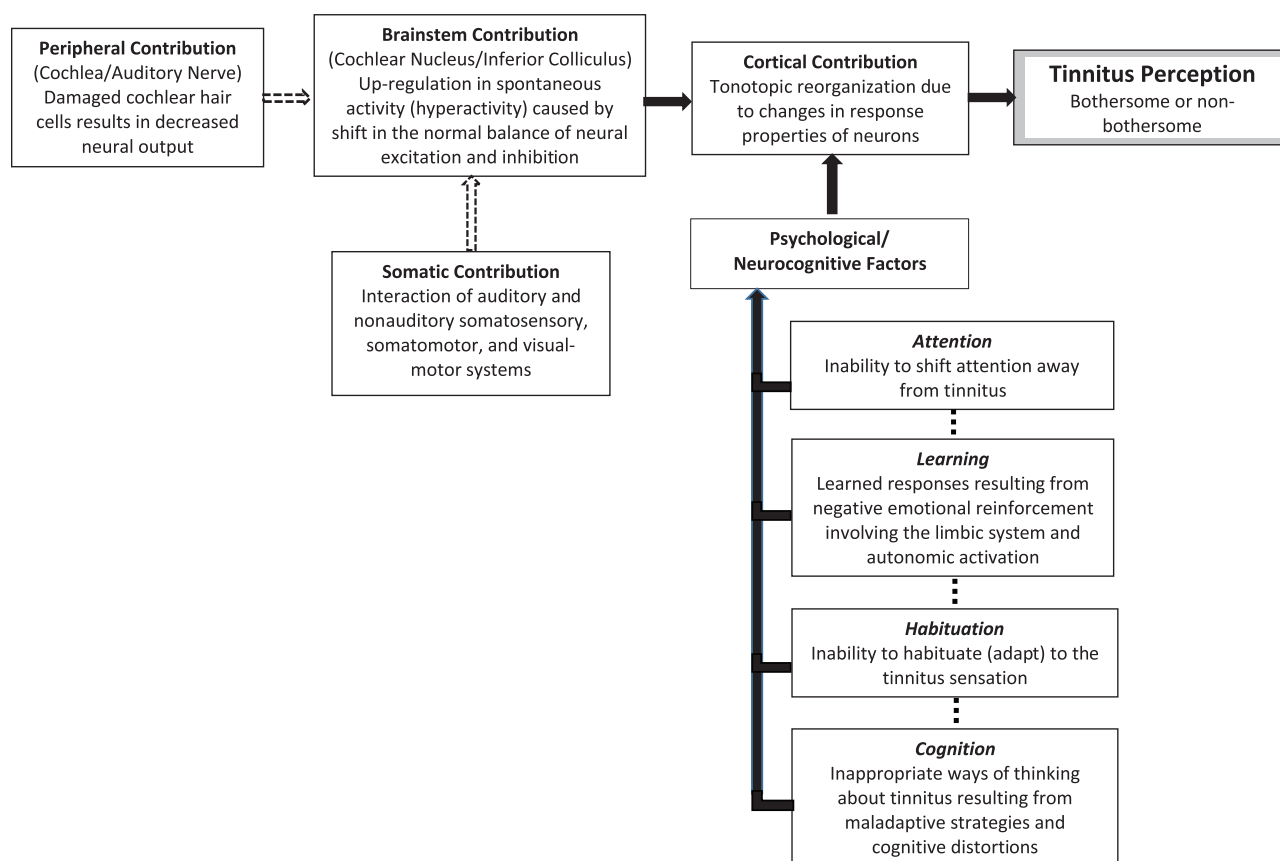


Figure 22-1. Integrative model illustrating mechanisms underlying the generation and perception of tinnitus.

provides a summary of current models and mechanisms of tinnitus involving the cochlea, auditory nerve, and central auditory nervous system. Note that the figure includes both potential anatomic and physiologic sites of tinnitus generation as well as general psychological models of tinnitus development.

Components of the Care Path

The next section of this chapter describes an approach designed to assist in clinical decision making targeted at promoting tinnitus relief and empowering the patient. Empowerment is a critical factor in the rehabilitation process in that it not only improves patient satisfaction with care, but also increases compliance with and adherence to the treatment plan (Greenhalgh & Meadows, 1999). Figure 22-2 illustrates a model that can guide the clinician in the management of patients with tinnitus. It is a multistep interprofessional col-

laborative care path process allowing patients to be referred to several health care service providers and be discharged at various points within the clinical pathway. In general, care paths foster value-based, patient-centered treatment plans that provide a schema of sequenced actions necessary to achieve best practice outcomes (Schrijvers, van Hoorn, & Huisjes, 2012). The presentation of the Tinnitus Management Clinic (TMC) model used at the Cleveland Clinic will provide the framework for discussion of an interprofessional collaborative management approach.

TMC Clinical Path Entry

Patients with tinnitus can be referred to the TMC from any health care specialist (e.g., primary care physician, gerontologist). The initial step before a patient is scheduled into the TMC is an evaluation by both an audiologist and otolaryngologist. The importance of each service is described below.

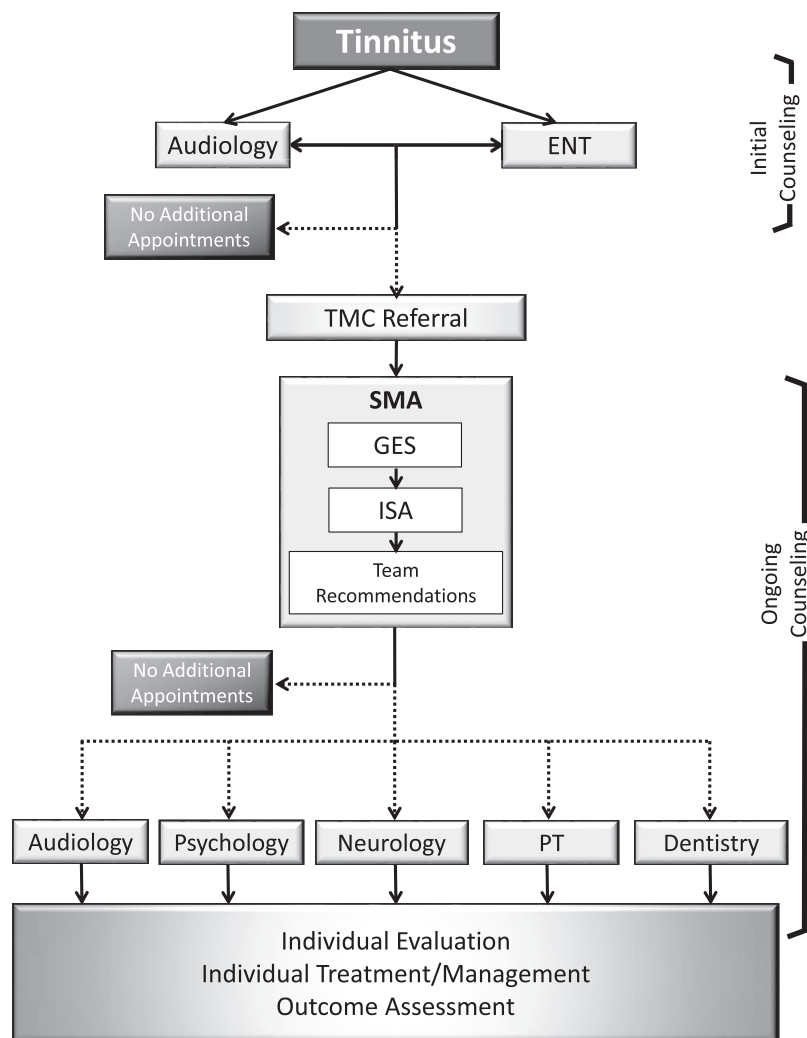


Figure 22-2. Interprofessional Tinnitus Management Clinic (TMC) Care Path. ENT = Ear, Nose, and Throat (Otolaryngology); SMA = Shared Medical Appointment; GES = Group Education Session; ISA = Interprofessional Screening Assessment; PT = Physical Therapy. *Note.* The dotted lines represent alternate options in the TMC Care Path. *Source:* Adapted from Newman and Sandridge (2016).

Audiology

Results from the comprehensive audiologic evaluation (i.e., pure-tone and speech audiometry, immittance measurements, and otoacoustic emissions) provide information about the severity and type of hearing loss, audiometric configuration, and functional integrity of the auditory system. The latter clinical data are critical to the medical diagnosis of any ear disease that may underlie the tinnitus symptoms. It is important to note, however, that many patients

who have tinnitus also experience some degree of hyperacusis (i.e., hypersensitivity to sound). Thus, the clinician must be aware that the administration of specific tests in the audiologic battery, such as acoustic reflex threshold determination and reflex decay testing, may need to be abandoned to prevent patient discomfort due to loudness intolerance. (*Authors' note:* The reader is referred to Fagelson and Baguley [2018] and a two-part review article appearing in the *American Journal of Audiology* [Pienkowski et al., 2014; Tyler et al., 2014] for a complete discussion of hyperacusis

and sound intolerance mechanisms, diagnosis, and treatment.)

The results of audiometric testing also serve an important role in the rehabilitation process by providing a basis for selection of the particular sound therapy in the treatment phase of the care path. For example, a patient with hearing loss and tinnitus may improve communication function and gain relief from tinnitus by using HAs or combination devices (HA and sound generator [SG] housed in the same unit). In contrast, an individual with normal or near normal hearing may benefit from SGs alone without the need for amplification.

It would be prudent to incorporate a tool to screen for clinically significant tinnitus into every audiologic evaluation, not just when evaluating patients who are referred with tinnitus as their chief complaint. A quick and easy method to screen for clinically significant tinnitus is shown in Figure 22–3. Using this flowchart, the patient responds to each question guiding the clinician toward follow-up treatment recommendations. For example, if a patient indicates that his or her tinnitus is “not a problem” or a “small problem,” the provision of some basic counseling within the context of the audiologic evaluation may be sufficient. This minimal level of counseling might include information

about the causes of tinnitus and hearing loss, hearing loss prevention, some strategies to manage his or her reactions to tinnitus, and the need to maintain a sound-enriched environment. For many patients, this brief counseling is sufficient and negates the need to return for further intensive tinnitus treatment. Also, consider directing the patient to the Tinnitus First Aid Kit website (<http://www.tinnituskits.com>), which was developed by the Ida Institute to provide some essential information about tinnitus and how to manage it effectively. In contrast, if a patient indicates that his or her tinnitus is chronic and causing him or her at least moderate problems, referral to further tinnitus management may be warranted.

A more formalized assessment of tinnitus severity can be accomplished using standardized screening tools. The 10-item Tinnitus Handicap Inventory–Screening Version (THI-S) (Newman, Sandridge, & Bolek, 2008), shown in Table 22–3, is a shortened version of the 25-item THI (Newman, Jacobson, & Spitzer, 1996) using the same questions and same scoring process. Patients indicate “yes” (4 points), “sometimes” (2 points), or “no” (0 points) for each question asked, yielding a possible total score ranging from 0–40 points, with higher scores representing greater perceived psychosocial problems as a conse-

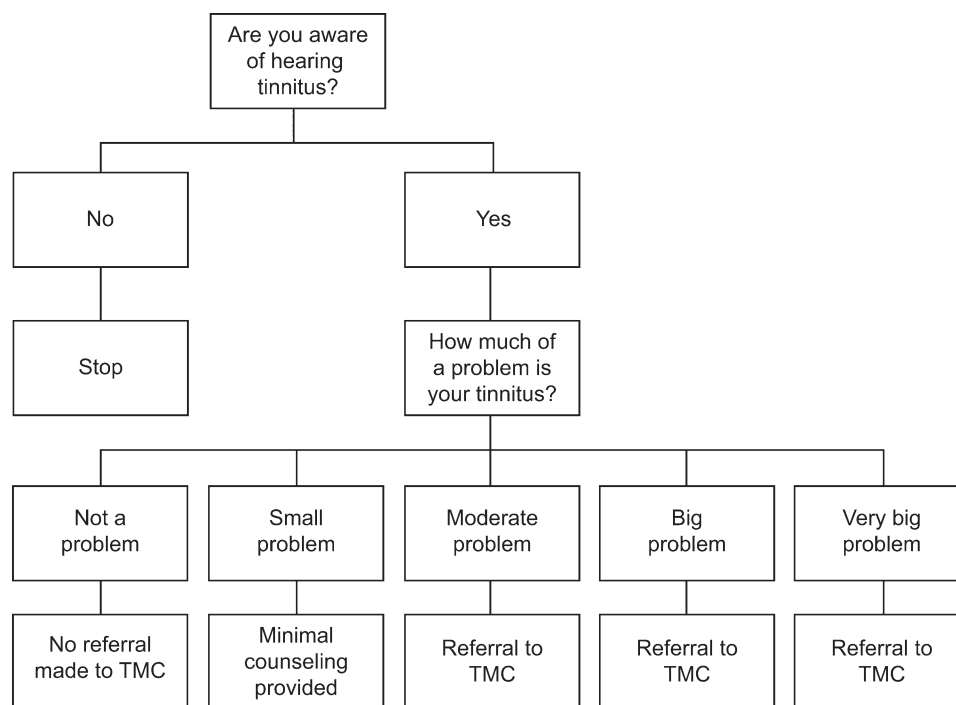


Figure 22–3. Flowchart used as a screening model to determine need for enrollment in the Tinnitus Management Clinic (TMC).

Table 22–3. Ten Questions Comprising the Screening Version of the Tinnitus Handicap Inventory (THI-S)

Because of your tinnitus, is it difficult for you to concentrate?
Do you complain a great deal about your tinnitus?
Do you feel as though you cannot escape your tinnitus?
Does your tinnitus make you feel confused?
Because of your tinnitus, do you feel frustrated?
Do you feel that you can no longer cope with your tinnitus?
Does your tinnitus make it difficult for you to enjoy life?
Does your tinnitus make you upset?
Because of your tinnitus, do you have trouble falling asleep at night?
Because of your tinnitus, do you feel depressed?

Source: Adapted from Newman, Sandridge, and Bolek (2008).

quence of tinnitus. The scores on the THI-S have been shown to be comparable ($r = .90$) to the full 25-item version of the THI and yielded adequate test-retest reliability ($r = .81$). Furthermore, a cutoff value of more than 6 points (out of a possible 40 points) was established as a fence for recommending follow up. Another option is the Tinnitus and Hearing Survey (THS) screening tool (Henry et al., 2015). This 10-item questionnaire is especially helpful to quickly differentiate between hearing difficulties (including sound tolerance issues) and bothersome tinnitus. The THI-S and THS are attractive screening tools to use in a busy clinical practice because they are brief (i.e., they take less than 2 minutes to complete) and easy to administer, score, and interpret.

Otolaryngology

The primary goals of the examination conducted by the otolaryngologist are to: (a) rule out any health condition that requires further medical or surgical intervention; (b) alleviate the patient's fear that they might have a serious medical problem; and (c) provide medical clearance for audiologic and other nonmedical (e.g., cognitive behavioral therapy) management.

The evaluation of tinnitus by a physician should include a thorough history, physical examination, and referral for appropriate diagnostic testing (e.g., imaging studies, bloodwork panels) geared toward identifying any underlying condition that may cause or

contribute to the tinnitus (Shi, Robb, & Michaelides, 2014). Table 22–4 summarizes the main components of the medical evaluation of tinnitus.

The medical or surgical management of an underlying pathology may help to alleviate the symptom of tinnitus for some; however, for the majority of patients, the cause for the tinnitus cannot be identified, or if it is, it cannot be reversed (e.g., noise-induced hearing loss) or successfully treated. When this is the case, physicians need to encourage their patients to seek nonmedical treatments (e.g., sound therapy, cognitive behavioral therapy) and reassure them that help is available from other health care providers. Unfortunately, all too often patients are told to “learn to live with it.” The latter advice can lead to further distress and discourages the patient from seeking appropriate nonmedical services that offer tinnitus relief. On the other hand, the physician's positive comments about the benefits of nonmedical treatments set the stage for successful management by the audiologist and other health care providers. Accordingly, our role as audiologists is to educate physicians regarding the benefits of nonmedical intervention so that they encourage patients to seek assistance from other knowledgeable providers.

Shared Medical Appointment

The next major component for patients in the care path is attendance at the Shared Medical Appointment (SMA). The goal of the SMA is to promote patient access and clinician productivity. As shown in Figure 22–2, the TMC is a 3-hour SMA divided into a 1.5-hour Group Education Session (GES), followed by a 1.5-hour Interprofessional Screening Assessments (ISA). TMC providers include an audiologist, dentist, neurologist, physical therapist, and psychologist.

Prior to attending the SMA, each patient is mailed information about the TMC and several questionnaires to be completed before arriving to the appointment. These include an in-depth tinnitus case history and standardized disease- and condition-specific questionnaires (i.e., THI [Newman et al., 1996; Newman, Sandridge, & Jacobson, 1998]; Tinnitus Functional Index [TFI] [Meikle et al., 2012]) and dimension-specific screening measures for anxiety (Generalized Anxiety Disorders-7 [GAD-7] [Spitzer, Kroenke, Williams, & Löwe, 2006], depression (Patient Health Questionnaire-9 [PHQ-9] [Kroenke, Spitzer, & Williams, 2001]), neck issues (Neck Disability Index [Vernon & Mior, 1991]), and dizziness (Dizziness Handicap Inventory [Jacobson & Newman, 1990]).

Table 22–4. Components of the Medical Evaluation of Tinnitus

History Characteristics of tinnitus (e.g., duration, character, location, constant or intermittent) Otologic symptoms (e.g., hearing loss, vertigo, otalgia, otorrhea) Head trauma Use of medication/ototoxic agents Central nervous system infections Family history of otologic disease Otologic surgery Social history (e.g., tobacco use, alcohol use, noise exposure—occupational or recreational) Review of systems Physical Examination Otomicroscopy/otoscopy Tuning fork examination Tests of cranial nerve function Oral cavity examination Palpation of temporomandibular joint and inspection of dentition Audiologic Studies Pure-tone and speech audiometry Immittance measurements (i.e., tympanometry, acoustic reflex testing, reflex decay) Otoacoustic emissions Auditory brain stem response	Additional Testing As Needed: Radiologic Evaluation Carotid ultrasound Gadolinium-enhanced magnetic resonance imaging Contrast-enhanced computerized tomography Magnetic resonance angiography Magnetic resonance venography Additional Testing As Needed: Laboratory Evaluations Blood work (e.g., blood count, chemistry panel, blood glucose, lipid panel) Thyroid function testing Screens for ototoxic drugs and pollutants (e.g., solvents, heavy metals, carbon monoxide) Syphilis serology Autoimmune workup (e.g., rheumatoid factor, C reactive protein, serum antibody to inner ear antigens) For Pulsatile Tinnitus Need to also Assess Auscultation of ear canal, periauricular region, orbits, and neck to assess for bruits Rate of tinnitus compared to patient's pulse rate Effect of light digital pressure over internal jugular vein Effects of turning head (causes tinnitus of venous origin to subside)
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Source: Adapted from Perry and Gantz (2000); Schleuning, Shi, and Martin (2006); Shiley, Folmer, and McMenomey (2005); Wackym and Friedland (2004).

Moreover, a Shared Medical Appointment Acknowledgment form is included. This form is signed by each patient prior to the GES acknowledging that he or she is choosing to participate in a group session where confidentiality may be lost. Following are descriptions of the activities comprising each component of the TMC.

Group Education Session (GES)

Consistent with the SMA philosophy (Bronson & Maxwell, 2004), the GES offers a number of advantages. Briefly, from the clinician's viewpoint, conducting group sessions is very cost- and time-efficient, allowing the same information to be delivered to more patients in less time and maximizing available

resources. From the patient's viewpoint, the group setting allows him or her to realize they are not alone in that others are experiencing the same negative consequences of tinnitus and have similar HRQoL difficulties. The group situation promotes a safe and supportive environment to share experiences with tinnitus and how to cope—or not cope—with the handicapping nature of the tinnitus.

The ideal size for an educational group is eight to 12 members, with the session lasting approximately 1.5 to 2 hours in length (Childers & Couch, 1989). It is important to control the group size because a group that is too large or too small may inhibit individual participation and reduce opportunities for interaction. Spouses/significant others should be encouraged to attend the session so that they have a better

understanding of the patient's problems and are better prepared to provide emotional support. The meeting room should be quiet and inviting, with chairs arranged in a circle so that everyone feels a sense of equality with one another and the exchange of information among participants and providers is enhanced.

The overarching goals of the GES are to:

- Demystify tinnitus by providing fundamental information about hearing, hearing loss, and tinnitus.
- Remove fears and concerns about tinnitus and clarify misconceptions (e.g., "Will I go deaf because of my tinnitus?").
- Provide reassurance and hope for tinnitus relief balanced with realistic expectations for treatment outcome.
- Offer self-help coping strategies and practical suggestions for immediate relief (e.g., sound therapy using environmental enhancement devices such as tabletop SGs, fans, music, etc.).
- Empower the patient to control his or her tinnitus rather than being controlled by the tinnitus.
- Establish trust and rapport between the patient and provider to promote adherence to and compliance with further treatment recommendations.

At the beginning of the GES, participants are asked to provide a brief description of their tinnitus including quality descriptors, onset, duration, and etiology, if known. These introductions allow each participant to realize that he/she is not the only person to experience specific tinnitus difficulties and that their situation is not unique. This introductory portion of the session allows each member to feel more comfortable with the other participants. In addition, it is important that the audiologist leading the GES controls the session so that individual members do not dominate the group by using it as a sounding board for personal problems. It is critical to the success of the group that all members understand this important ground rule.

The AAO-HNS evidence-based clinical practice guidelines for tinnitus endorse the value of education and counseling in the overall management process (Tunkel et al., 2014), supporting the importance of the GES component of the TMC care path. The specific topics addressed during the education session should be selected carefully to include appropriate informa-

tion for patients representing a wide range of tinnitus severity. The following topics serve as an outline for areas to be addressed during GES:

- definitions of tinnitus and related sound tolerance problems (e.g., hyperacusis, misophonia);
- prevalence of tinnitus in the general population (e.g., age and gender distribution) along with examples of well-known individuals who suffer from tinnitus such as entertainers, musicians, politicians, and historical figures;
- anatomy and physiology of hearing and hearing loss;
- mechanisms underlying the generation and perception of tinnitus;
- common psychosocial and emotional reactions to tinnitus;
- lifestyle factors affecting tinnitus (e.g., occupational and recreational noise exposure, diet, medications);
- benefits of sound therapy in promoting tinnitus relief through masking and habituation;
- application of different types of sound therapy devices (e.g., tabletop sound generators, computer applications [apps], ear-level HAs and SGs, combination instruments) and delivery systems (e.g., smartphones, tablets, personal sound players, pillows, headbands);
- hearing loss protection for occupational and/or recreational sound exposure including the appropriate use of earmuffs and earplugs;
- overview of nonaudiologic treatment options (e.g., diet; cognitive behavioral therapy, mindfulness, acceptance therapy; relaxation therapy) and the role of other health care professionals in the treatment process;
- realistic expectations for tinnitus management strategies and outcomes; and
- clarification of the remaining steps in the care path for patients interested in pursuing follow-up treatment.

One way to convey the aforementioned information is through the use of a computerized, picture-based slideshow presentation. The presentation, incorporating a series of animated picture sequences and sound clips (e.g., simulated tinnitus sounds),

serves to guide the flow of the counseling session, organizes complex information in a logical sequence, keeps group discussion focused, and provides visual and auditory interest to the patients. Figure 22–4 displays some examples of the pictures contained in the slide show presentation.

In addition to the formal presentation and the informal dialogue that may occur among the group

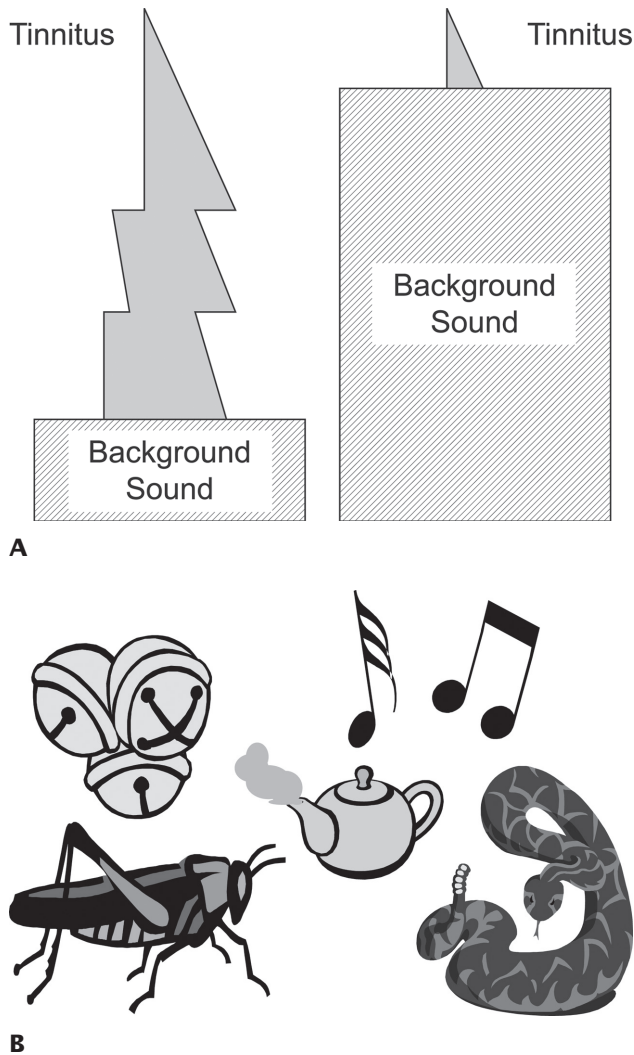


Figure 22–4. Examples of pictures from slide show presentation used during the Group Education Session (GES). **A.** Illustration depicting the benefits of sound-enriched environment in providing tinnitus relief (e.g., partial masking of tinnitus). **B.** Picture examples of perceptual tinnitus attributes played with associated audio sound files (e.g., snake and teapot represent hissing percept; musical notes represent tonal tinnitus quality).

members, take-home information should be provided. Resource materials reinforce and expand upon topics areas presented in the GES. The take-home packet may include, but may not be limited to, the following items:

- *The Consumer Handbook on Tinnitus* (Tyler, 2016) is an excellent patient-oriented resource covering a variety of tinnitus-related topics (e.g., underlying mechanisms and causes, reactions to tinnitus, sound therapy options, improving sleep and hyperacusis) written by leading clinicians and researchers.
- Fact sheet describing different forms of sound therapy, including HAs and SGs. List of budget-friendly options that can be used to maintain a sound-enriched environment (e.g., fans, radios, television, talk radio).
- Website addresses for environmental enhancement devices such as tabletop generators and other delivery systems (e.g., pillows and headbands with embedded speakers).
- List of tinnitus apps for smartphones or tablets designed to generate a variety of tinnitus relief sounds.
- Contact information for the American Tinnitus Association (<http://www.ata.org>; 800-634-8978).

Interprofessional Screening Assessments (ISA)

Following the GES, each participant of the SMA meets individually with the providers for approximately 15 minutes. The purpose of this brief encounter is to determine if the patient would benefit from a scheduled individual appointment with the provider, at which time a more thorough evaluation would be conducted, resulting in appropriate follow-up recommendations and treatment plans. Following is a brief summary of the activities performed by each clinician during the ISA.

- **Audiologist:** Reviews purposes of and demonstrates various sound therapy options to determine which form of sound therapy provides relief and which may be most acceptable to the patient over the long term. Observed reactions to the use of different broadband sounds and/or amplification

help to gauge the patient's motivation for pursuing device use.

- **Dentist:** Screens for specific dental problems such as temporomandibular disorders and teeth clenching/grinding that may be associated with the generation of tinnitus.
- **Neurologist:** Screens for potential jaw, neck, or cervical disorders and conducts a physical examination to determine if tinnitus is somatically modifiable (see Figure 22–5 for physical screening maneuvers used with each patient) and potentially amenable to physical therapy intervention.
- **Physical therapist:** Screens for biomechanical problems associated with neck, spine, or jaw injuries, strains resulting from occupational requirements (e.g., time spent at computer), and poor posture.
- **Psychologist:** Reviews results from the PHQ-9 and GAD-7 to address depression

and/or anxiety and introduces the concept of the mind-body connection. For patients not requiring formal follow-up psychological counseling, *The Mindfulness and Acceptance Workbook for Anxiety* (Forsyth & Eifert, 2016) is an excellent resource and recommended for self-help to overcome generalized anxiety, fear, and worry.

TMC Disposition and Case Staffing

Once the ISAs are completed and the patients are discharged, the team gathers to discuss each participant's case history and the screening observations to develop an individualized treatment plan. A summary report is generated and mailed to each patient along with specific recommendations from each provider on the team (e.g., hearing devices, bite modification appliances, physical therapy, cognitive behavioral therapy). It is critical that the report is



Figure 22–5. Examples of head, neck, and jaw physical maneuvers used by the neurologist to provoke somatic tinnitus during the Interprofessional Screening Assessment (ISA) Portion of the Tinnitus Management Clinic (TMC). **A.** Cervical flexion with resistance. **B.** Cervical extension with resistance. **C.** Side bend with resistance. **D.** Sternocleidomastoid right. **E.** Teeth/jaw clench. **F.** Jaw open with resistance. **G.** Jaw protrusion with resistance. **H.** Jaw left. **I.** Jaw left with resistance.

written without extensive medical/clinical jargon so that it can be understood by a wide range of patients.

The authors have found that many patients are able to manage their tinnitus successfully after participation in the TMC where they receive reassurance and some simple suggestions about how to handle their tinnitus at home, in social situations, and at work. For this subset of patients, the information and strategies offered during the TMC are sufficient to satisfy their curiosity about tinnitus and further management is unnecessary. Continued participation in the care path requires the patient to be motivated, have realistic expectations for treatment outcomes, understand the nature of long-term management programs, and be willing to comply with treatment requirements from one or more of the other health care providers on the team.

TMC Patient Satisfaction and Benefit

Following the TMC, patients are mailed a survey to assess satisfaction with and benefit from the experience. Figure 22–6 summarizes selected responses from the survey. As illustrated, the bulk of participants ($n = 100/103$) reported that the SMA was beneficial and would recommend the TMC to friends experiencing tinnitus ($n = 97/103$). Further, the survey examined the value of attending the TMC using the willingness-to-pay (WTP) econometric. Knowing that the cost of the TMC was \$300 (self-pay), the mean WTP was \$287 (standard deviation [SD] = \$164; range \$0–\$1,500), showing nominal equivalence between the cost of the TMC and average WTP. These findings suggest that the patients perceived value in attending the TMC.

Individual Tinnitus Evaluation

The next two sections will highlight the audiologic tinnitus evaluation and management components of the care path, respectively. Subsequently, an overview of the assessment and treatment options offered by the other providers on the team (i.e., dentistry, neurology, physical therapy, psychology) will be presented.

The tinnitus evaluation protocol should include the following areas of investigation: (a) an in-depth tinnitus case history; (b) assessment of tinnitus impairment; (c) evaluation of tinnitus activity limitations and participation restrictions; (d) selection of the most appropriate form of sound therapy for a given patient; and (e) determination of treatment outcome.

Case History

The intake case history provides information about the nature of the patient's complaints that cannot be measured objectively and supports quantitative self-reported measures. In addition, spending time to obtain a detailed case history facilitates rapport and the development of the patient-audiologist relationship. For medicolegal cases, completion of the detailed history is especially critical because it supports the consistency and plausibility of the patient's report, or the absence of true symptoms, as well as motivation for seeking treatment.

Several published tinnitus case history questionnaires are available (Henry et al., 2005a; Henry, Jastreboff, Jastreboff, Schechter, & Fausti, 2002; Schechter & Henry, 2002). For example, the Iowa Tinnitus Questionnaire (Stouffer & Tyler, 1990) is a two-part history form. Part A, completed by the patient, focuses on the patient's description of tinnitus, problems associated with tinnitus, and other aspects of tinnitus from the individual's perspective. Part B, completed by the audiologist, consists of three questions regarding the primary diagnosis, patient's primary complaint, and air-conduction thresholds for each ear at 1000 and 4000 Hz.

Many audiologists develop their own intake questionnaires. Table 22–5 summarizes the salient points that should be addressed during the history. In general, the history should include inquiry about: (a) descriptive attributes of the tinnitus; (b) specific behavioral, social, emotional, and interpersonal consequences of tinnitus; (c) factors that may increase or reduce tinnitus disturbance; (d) hearing, hearing loss, and sound tolerances issues; (e) previous tinnitus treatments; (f) prioritization of concerns (e.g., hearing loss, tinnitus sound tolerance problems); and (g) expectations about treatment.

Tinnitus Impairment

Tinnitus impairment (i.e., perceptual characteristics of the tinnitus) may be quantified using two general approaches: scaling techniques or psychoacoustic measurement. Scaling techniques for tinnitus typically involve the application of a direct estimation method. The latter approach is designed to elicit from the patient a direct quantitative estimate of the magnitude of a tinnitus attribute such as pitch and/or loudness. Figure 22–7A illustrates examples of direct estimation numeric rating scales. Using a numeric scale technique, Stouffer and Tyler (1990) indicated

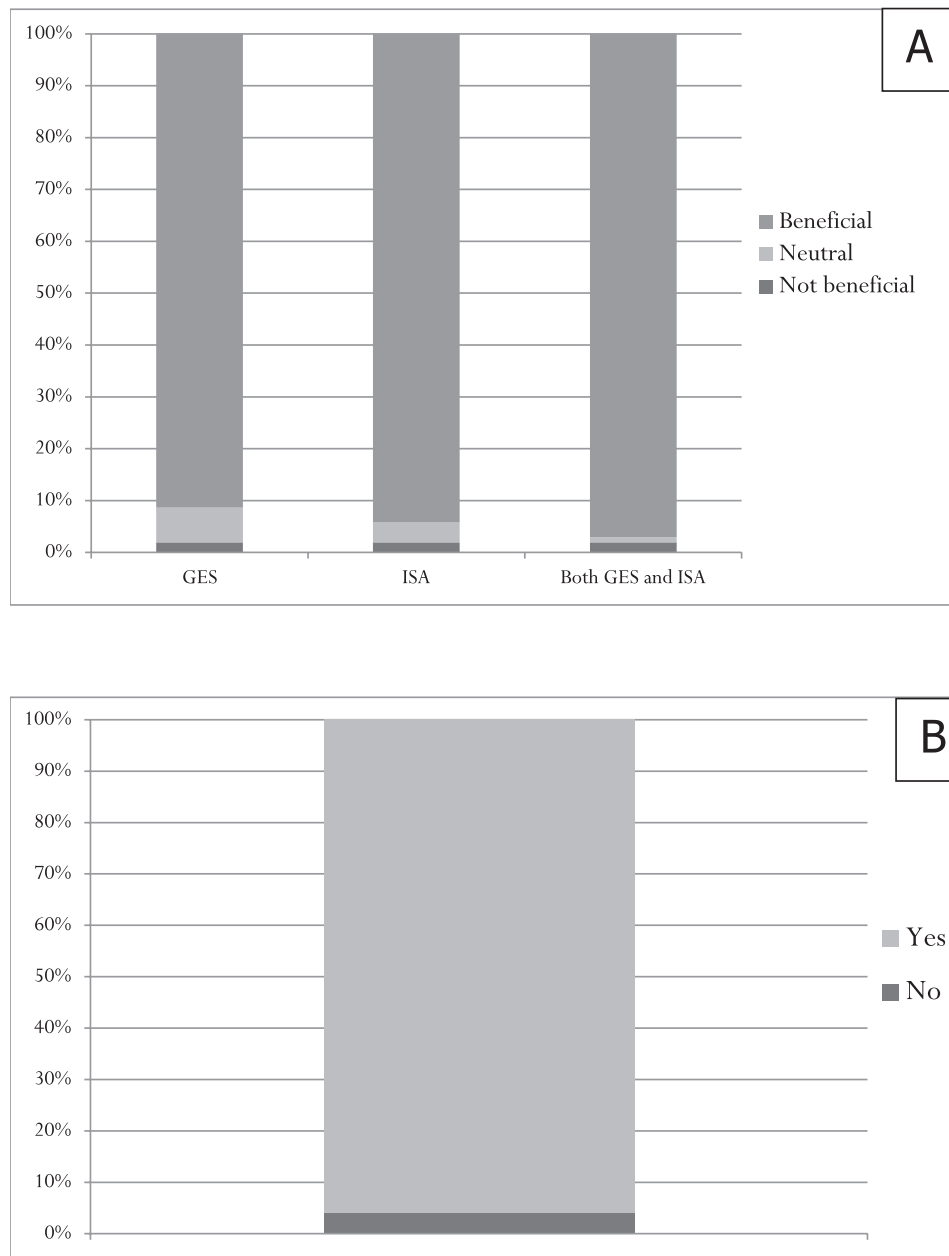


Figure 22-6. Representative outcome data illustrating benefits received from attending the Tinnitus Management Clinic (TMC). **A.** Patient survey responses ($n = 103$) showing perceived benefit from attending the Group Education Session (GES), Interprofessional Screening Assessments (ISA), and overall experience (combined GES and ISA). **B.** Patient survey response ($n = 103$) to an item querying likelihood of recommending the TMC to persons with bothersome tinnitus.

the average rating for subjective pitch estimates for their sample on a 10-point scale was 7.12 (SD = 2.3), with 65% of the subjects rating their pitch as 7 points or higher. Stouffer and Tyler further noted that the

average loudness rating was 6.3 on the 10-point scale (SD = 2.3). An alternative approach is the visual analog scale (VAS) that often employs a line of fixed length, usually 100 mm, with anchors appropriate

Table 22–5. Elements of a Tinnitus Case History Questionnaire

Nature of Tinnitus Perceptual features (e.g., location, pitch, loudness) Duration (e.g., onset and presence, percentage of time disturbed, percentage of time aware) Tinnitus quality descriptors (e.g., ringing, hissing) and timbre (e.g., sharp, piercing, tense) Recent experiences/changes in pitch, loudness, and/or quality Tinnitus History Noise exposure (e.g., vocational, recreational) Drug use (e.g., ototoxic medications) Dental problems (e.g., jaw pain, grinding) Ear surgery/disease Exacerbating factors (e.g., sound environment; diet; stress; medication; activity level; alcohol; smoking; posture; changes in neck and shoulder biomechanics such as computer use, number of pillows used, neck restriction, injury) Reducing factors (e.g., same as above) Hearing Difficulties Evaluation of impairment (e.g., audibility), activity limitation (e.g., difficulty hearing in background noise), and participation restriction (e.g., relationships with family, friends, coworkers) Hearing aid use Hearing protection device uses (e.g., earplugs, earmuffs) Sound tolerance disturbance (e.g., hyperacusis, phonophobia, misophonia)	Psychosocial/Functional Consequences Level of annoyance Sleep disturbance Depression Concentration difficulty Suicidal ideation Stress Interference/avoidance of social situations Positive/negative thoughts about tinnitus Treatment History Medical (e.g., prescribed medications, diet) Surgical Sound therapy Psychological counseling Dietary supplements and/or homeopathic remedies Expectations and benefits from past/current treatments Health/Other Overall general health status Prioritization of concerns and primary tinnitus-related problems Seeking compensation, disability, or other legal actions related to tinnitus
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Source: Adapted from Newman and Sandridge (2004); Newman and Sandridge (2016).

A Describe the LOUDNESS of your tinnitus by circling ONE of the numbers below.

Zero indicates your tinnitus is **Very Faint**

Ten indicates that it is **Very Loud**

Very Faint 0 1 2 3 4 5 6 7 8 9 10 *Very Loud*

B Indicate to what degree your tinnitus interferes with your sleeping by placing an X on the line below.

Not At All ----- *Cannot Sleep At All*

Figure 22–7. Magnitude estimation scaling techniques used to quantify perceptual attributes of tinnitus and impact on health-related quality of life (HRQoL). **A.** Example of a numeric rating scale used to quantify loudness. **B.** Example of visual analog scale used to estimate sleep disturbance.

for the attribute being measured (Figure 22–7B). In a sample of patients with chronic tinnitus, Adamchic and colleagues (2012) demonstrated that the VAS has good test-retest reliability and convergent validity for measuring self-perceived tinnitus annoyance and loudness.

Psychoacoustic measurement techniques provide the researcher and clinician a method for quantifying the sensory and perceptual characteristics of the tinnitus experience. These procedures have a long history (Henry & Meikle, 2000), focusing on the development of specific techniques, test-retest reliability of the obtained data, and the extent to which different protocols might reflect differences among patient populations. In the clinical setting, psychoacoustic tinnitus measurements (Table 22–6) include pitch and loudness matching, minimum masking levels, residual inhibition, and loudness discomfort levels (used to quantify sound tolerance problems that often accompany tinnitus). The reader is referred to Henry and Meikle (2000) for a discussion on how to perform these measurements.

Tyler (2000) proposed several reasons for conducting the aforementioned measures, including: reassuring the patient that the tinnitus is real; demonstrating the acoustic characteristics of the tinnitus to the patient's significant others; distinguishing different subcategories of tinnitus; determining potential treatment benefit and selecting specific treatment options; quantifying treatment outcome/effectiveness; and assisting in medicolegal issues. The clinical utility of pitch matching was suggested by Schaette, König, Horning, Gross, and Kempter (2010), who

demonstrated that sound therapy was more effective when tinnitus was matched to frequencies less than 6000 Hz. Moreover, the Tinnitus Research Initiative consensus statement (Langguth et al., 2007) proposed that loudness matches and tinnitus maskability should be considered primary psychoacoustic outcome measures in tinnitus treatment clinical trials.

Although measurement protocols for assessing the psychoacoustic attributes of tinnitus are not currently standardized and may require specialized instrumentation (Henry et al., 2009; Henry, Rheinsburg, Owens, & Ellingson, 2006), methods using a diagnostic audiometer have been recommended (Henry, 2004; Henry et al., 2005a; Lentz, Walker, Short, & Skinner, 2017). Currently, a clinically available computer-based tinnitus assessment tool has been developed to measure psychoacoustic attributes of tinnitus (e.g., frequency, bandwidth, tempo, and slope; intensity threshold; maskability) more precisely than can be accomplished using a standard audiometer (Figure 22–8; Tinnometer Main Screen, courtesy of MedRx). The unit of measurement for each component for the psychoacoustic battery varies depending upon the stimulus parameter. For example, pitch (psychoacoustic correlate of frequency) is measured in hertz (Hz), and loudness (psychoacoustic correlate of intensity) has typically been measured in dB sensation level (SL; amount, expressed in dB, above the patient's threshold).

It has been argued, however, that SL may not accurately reflect the patient's perception of tinnitus loudness (Newman, Wharton, Shivapuja, & Jacobson, 1994; Tyler, Aran, & Dauman, 1992; Tyler & Conrad-Arnes,

Table 22–6. Psychoacoustic Measurements Conducted as Part of the Tinnitus Assessment

Measurement	Definition
Pitch match	Ability of the patient to equate the pitch of an externally generated pure-tone (or narrowband noise) to the most prominent pitch of his or her perceived tinnitus
Loudness match	Ability of the patient to equate the loudness of an externally generated pure-tone (or narrowband noise) to the overall loudness of his or her perceived tinnitus
Minimum masking level	Minimum level of a broadband noise (BBN) required to completely mask the patient's tinnitus. If the tinnitus cannot be masked, the minimum level of a BBN that changes the perception of the tinnitus (e.g., louder or softer)
Residual inhibition	Period of time when the patient's tinnitus has been partially or completely suppressed after the externally produced masking stimuli has been turned off
Loudness discomfort level	The threshold level of discomfort for pure-tones, BBN, or cold running speech

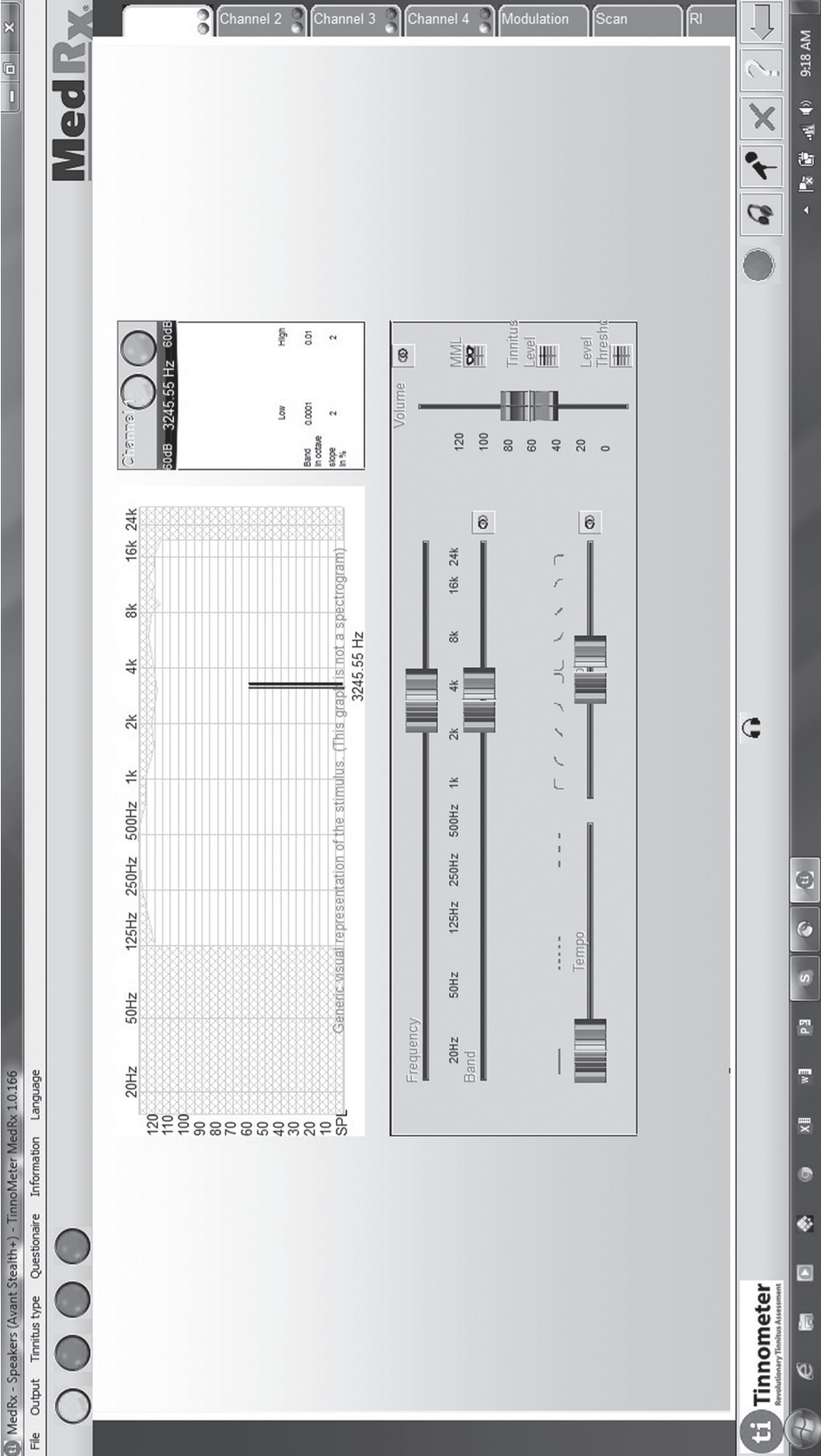


Figure 22-8. Computer-based psychoacoustic tinnitus matching assessment software (Tinnometer). *Note.* This figure displays the Tinnometer screen illustrating application of clinically available instrumentation designed to assess perceived tinnitus pitch, loudness, shape (bandwidth and slope), and tempo (steady and pulsed). The software also has the capability to conduct minimum masking levels (MML) and timed signal delivery of MML used to perform residual inhibition measurement. *Source:* Courtesy of MedRx, Incorporated.

1983). Tyler (2000) advocates that the use of sones, the conventional psychoacoustic unit for measuring loudness, would be a more appropriate measure than dB SL. The following example illustrates this point. Figure 22–9 displays the loudness growth function for a patient with normal hearing at 1000 Hz and a moderate sensorineural hearing loss at the pitch-match frequency. The predicted growth in loudness in sones is based on the equation described by Tyler and Conrad-Arnes (1983):

$$L = k(P - P_0)^{0.6}$$

where: L = loudness in sones

k = constant that depends on the unit of measurement (with P measured in micropascals, $k = 0.01$)

P = equal-loudness match in dB sound pressure level (SPL)

Note that, in Figure 22–9, SPL is converted into dB hearing level (HL) on the abscissa with loudness in sones on the ordinate. As can be seen, this patient's threshold at 1000 Hz was 20 dB HL. Tinni-

tus was determined to have an equal loudness match of 34 dB HL, which was 14 dB SL and calculated to have a loudness judgment value of 0.88 sones. At the pitch-match frequency, however, this same patient had a threshold of 48 dB HL with an equal loudness match of 61 dB HL. The sensation levels for both frequencies were essentially equal, at 14 and 13 dB SL, respectively. Yet, when the sone value was calculated for the pitch-matched frequency, it was determined to be 5.68 sones—more than 5 times greater than the sone value at 1000 Hz. This case illustrates the appropriateness of estimating loudness with sones, rather than simply reporting SL (Newman et al., 1994).

Self-Reported Measures

The use of self-reported measures to assess the impact of tinnitus on the patient's HRQoL is a critical component in the evaluation of activity limitation and participation restriction. In fact, the AAO-HNS tinnitus guidelines strongly recommend the use of self-reported measures to help distinguish patients with bothersome tinnitus from those with nonbothersome tinnitus (Tunkel et al., 2014). Newman and Sandridge

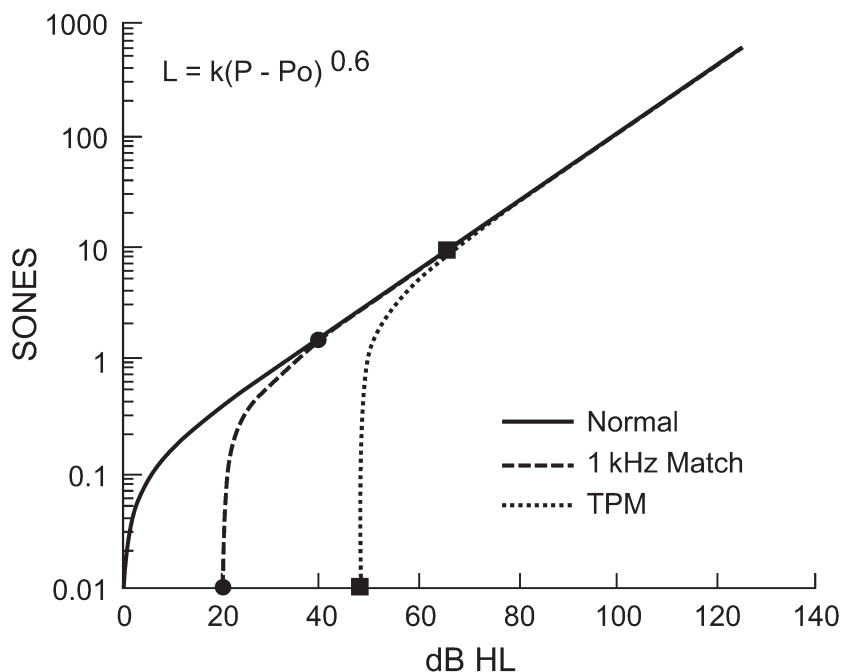


Figure 22–9. Loudness growth functions (sones) for an individual patient. *Note.* Loudness growth functions (sones) for an individual patient with normal hearing (20 dB HL) at 1000 Hz (●) and moderate hearing loss (48 dB HL) at the tinnitus pitch-matched (TPM) frequency (■). *Source:* Adapted from Newman, Wharton, Shivapuja, and Jacobson, 1994, p. 56.

(2004) outlined a number of reasons for inclusion of these measures in the evaluation process:

- There is a low correlation among psychoacoustic measures of tinnitus (e.g., loudness and pitch rating) with perceived severity and annoyance (Jakes, Hallam, Chambers, & Hinchcliffe, 1985; Meikle, Vernon, & Johnson, 1984), indicating that the sole utilization of psychoacoustic assessment is inadequate in assessing the patient's reactions to the perceived tinnitus sensation(s). For example, a loudness rating of 9 on a 10-point scale (indicating very loud) may be associated with a minimal annoyance rating or vice versa—a low loudness rating (3 on the 10-point scale) may be causing significant stress in the patient's life.
- Tinnitus is fundamentally a self-reported phenomenon and is not readily apparent to others except through the complaints of the sufferer, similar to chronic pain. That is, others cannot see it, feel it, hear it, or touch it, and therefore cannot appreciate the impact that tinnitus has on one's life.
- Self-report tinnitus questionnaires allow the patient to self-realize the extent that the tinnitus is impacting his or her day-to-day life. The questionnaires also allow the audiologist to identify those patients who are especially bothered by their tinnitus and who may benefit from treatment.
- Specific problem areas may be identified by conducting an item analysis of the self-reported measures.
- Treatment efficacy can be quantified when questionnaires are administered in a pre- and post-treatment paradigm, promoting evidence-based practice for quantifying outcomes.

Prior to the selection of a particular instrument for inclusion in the tinnitus assessment, the clinician needs to have a good understanding of the psychometric characteristics of that instrument. When evaluating a candidate tool, the clinician should be aware of the following psychometric standards (Hyde, 2000; Langguth, Searchfield, Biesinger, & Greimel, 2011): overall methodology of development, validation, and norming; appropriateness of content domains; reliability; validity; feasibility of practical application such as scoring and ease of administration; and responsiveness, which is especially important when

selecting a tinnitus instrument that will be used to detect treatment-related effects.

Over the years, several questionnaires have been developed. Table 22-7 summarizes some of the readily available questionnaires that evaluate a broad spectrum of activity limitation, participation restriction, cognitive reaction to tinnitus, and coping. As shown, for each questionnaire there is a short description of its content domains, scoring and interpretation, psychometric properties, and strengths and weaknesses as viewed by the current authors. The self-reported measures summarized in Table 22-7 are all quantitative in nature or score driven (i.e., responses can be summed and/or averaged, resulting in a total score or subscale scores).

In contrast, qualitative tinnitus questionnaires are not score driven, and provide the clinician with information that is more descriptive in nature. For example, the TPQ developed by Tyler and Baker (1983) employs an open-ended format. The patient is simply asked, "Please make a list of the difficulties that you have as a result of your tinnitus. List them in order of importance, starting with the biggest difficulties. Write down as many of them as you can." The primary advantage of the open-ended approach (e.g., TPQ), in contrast to structured measures (e.g., THI; Tinnitus Reaction Questionnaire [TRQ] [Wilson, Henry, Bowen, & Haralambous, 1991]), is that they allow the patient to list their own specific tinnitus problems instead of limiting the patient to a fixed set of responses that may or may not be of relevance. This is similar to the benefits offered by the HA tool; namely, the Client-Oriented Scale of Improvement (COSI) (Dillon, James, & Ginis, 1997), where the patient reports his or her own specific hearing difficulties rather than select from among a closed-response list.

Maintaining diaries has been used as a technique for assessing the patterns of tinnitus complaints over time (e.g., changes in pitch, loudness, annoyance, sleep disturbance) and factors that may be associated with those changes (e.g., noise exposure, medication use, diet). Daily diaries have been helpful in designing cognitive behavioral therapy (CBT) sessions by identifying unhelpful and negative thought patterns (Hallam & McKenna, 2006; Henry & Wilson, 2001); however, diary/journal entries may also be counterproductive as they increase the patient's attention to the tinnitus (i.e., self-checking behavior). Continued shifts in attention toward the tinnitus may not be advantageous for sound therapy treatments where the goal is to direct attention away from the tinnitus. Further, outcome assessment is limited by the lack of psychometric data regarding diary/journal validity, reliability, and responsiveness.

Table 22-7. Examples of Self-Report Tinnitus Questionnaires

Questionnaire/Authors	Content	Scoring/Interpretation	Psychometrics	Strengths/Weaknesses
Tinnitus Questionnaire/ Tinnitus Effects Questionnaire (TQ/TEQ; Hallam, Jakes, and Hinchcliffe, 1988)	52 items: sleep disturbance; emotional distress; auditory perceptual difficulties; inappropriate or lack of coping skills ("absolutist" beliefs)	Level of agreement to each statement: true = 2 points partly true = 1 point; not true = 0 points Score range: 0–104 (points with higher scores reflect greater tinnitus complaints)	High internal consistency reliability: Cronbach's $\alpha = .91-.95$ High test-retest reliability: $r = .91-.94$ Convergent validity with the THQ and TRQ	Stable measure over time making it a useful outcome measure No data available regarding what is considered a clinically or statistically change in score following intervention for a given patient
Tinnitus Handicap Questionnaire (THQ; Kuk, Tyler, Russell, and Jordan, 1990)	27 items: Factor 1: physical, emotional, social consequences of tinnitus Factor 2: effects on hearing Factor 3: patient's view of tinnitus	Level of agreement to each statement 0–100 0 = strongly disagrees 100 = strongly agrees Mean scores calculated with higher scores reflecting greater handicap Individual scores can be compared against normative percentile data	High internal consistency reliability for total scale: Cronbach's $\alpha = .95$ (with factors ranging from .47–.95) High test-retest reliability: Total ($r = .89$) Factor 1 ($r = .89$) Factor 2 ($r = .90$) Low test-retest reliability: Factor 3 ($r = .50$) Adequate construct validity: $r > .50$ (with tinnitus loudness judgments, life) satisfaction, hearing, depression, and general health	Percentile ranking for given patient is useful in determining individual severity Application of Factor 3 as an independent measure is questionable Scoring on a 100-point scale may be difficult for some individuals

continues

Table 22-7. *continued*

Questionnaire/Authors	Content	Scoring/Interpretation	Psychometrics	Strengths/Weaknesses
Tinnitus Severity Scale (TSS; Sweetow and Levy, 1990)	15 items: intrusiveness; distress; hearing loss; sleep disturbance; medication	Multiple choice format for each item range in score: 1 = no impact 4 = most impact Each item weighted 1 to 3 points: Total = 39 points Total score calculated by multiplying item score X weight and summing products	Adequate test-retest reliability: $r = .86$	Item analysis may be useful in determining treatment goals Limited psychometric data available
Subjective Tinnitus Severity Scale (STSS; Halford and Anderson, 1991)	16 items: severity defined as intrusiveness, and prominence, and distress	YES/NO response format 10/16 items earn 1 point for YES response 6/16 items earn 1 point for NO response Scores range: 0–16 (high scores reflect greater overall severity)	High internal consistency reliability: Cronbach's $\alpha = .84$ Criterion validity determined by significant correlations with clinician rating of severity	Brief and simple to administer and score No severity classification schemes developed No test-retest reliability available required for measuring outcome
Tinnitus Reaction Questionnaire (TRQ; Wilson, Henry, and Bowen, Haralambous, 1991)	26 items: distress consequences including anger, confusion, annoyance, helplessness, activity avoidance, and panic	5-point scale: 0 = not at all 4 = almost all of the time Score range: 0–104 (higher scores reflect greater distress)	High internal consistency reliability: Cronbach's $\alpha = .96$ High test-retest reliability: $r = .88$ Construct validity supported by moderate to high correlations with clinician ratings and self-report measures of anxiety and depression Four factors emerged using principal components analysis: general distress, interference, severity, avoidance	Global measure of distress Brief and simple to administer and score Test-retest data helpful in determining treatment outcome; however only short- term (3 days to 3 weeks) data available No data available regarding what is considered a clinically or statistically significant change in score following intervention for a given patient

Questionnaire/Authors	Content	Scoring/Interpretation	Psychometrics	Strengths/Weaknesses
Tinnitus Handicap/Support Scale (TH/SS; Erlandsson, Hallberg, and Axelsson, 1992)	28 items: Factor 1: perceived attitudes or reactions of others; Factor 2: social support; Factor 3: personal and social handicaps	5-point scale 1 = strongly disagree 5 = strongly agree	Construct validity assessed with the TSQ showing significant relationships between tinnitus severity and perceived attitudes (Factor 1) and between severity and personal and social handicaps (Factor 3)	Assesses influence of significant other in management process Lacks test-retest reliability data Factor 2 is not sufficiently sensitive to quantify complexity of social support
Tinnitus Handicap Inventory (THI; Newman, Jacobson, and Spitzer, 1996; Newman, Sandridge, and Jacobson, 1998)	25 items: Functional subscale: role limitations in mental, social/occupational, physical functioning; Emotional subscale: anger, frustration; irritability; depression; Catastrophic subscale: desperation; loss of control; inability to cope and escape; fear of grave disease	Response format: YES = 4 points SOMETIMES = 2 points NO = 0 points Total score range: 0–100 points with higher scores reflecting greater handicap Handicap severity categories: no handicap (0–16 points) mild (18–36 points) moderate (38–56 points) severe (58–100 points)	Excellent internal consistency reliability: Cronbach's $\alpha = .96$ High test-retest reliability: total ($r = .92$) subscales ($r = .84-.94$) 95% confidence interval = 20 points Convergent validity with THQ and TQ Construct validity with Beck Depression Inventory, Modified Somatic Perception Questionnaire, symptom rating scales (e.g., sleep, annoyance), tinnitus pitch and loudness	Brief, easy to administer and score Useful as a companion to the Hearing Handicap Inventory for the Elderly/Adults and Dizziness Handicap Inventory Available in at least 15 languages other than English (e.g., Chinese, Danish, Spanish) Psychometrical robust; however, only short-term (3 week) retest data available Should be considered unifactorial (distinctness of subscales questioned)

continues

Table 22-7. *continued*

Questionnaire/Authors	Content	Scoring/Interpretation	Psychometrics	Strengths/Weaknesses
Tinnitus Coping Style Questionnaire (TCSQ ; Budd and Pugh, 1996a, 1996b)	33 items: Maladaptive coping subscale; Effective coping subscale	7-point scale: 1 = never 7 = always Higher scores on maladaptive subscale reflect poor coping skills Higher scores on effective coping subscale reflect better acceptance of tinnitus	Excellent internal consistency Reliability for maladaptive subscale: Cronbach's $\alpha = .90$ coping subscale: Cronbach's $\alpha = .89$ Coping subscale significantly associated with measures of tinnitus severity, depression and anxiety Effective coping subscale significantly associated with tinnitus adjustment measures	Helpful in identifying behaviors and through patterns affect by tinnitus and coping strategies used Test-retest data are not available limiting application for evaluating treatment outcome
Tinnitus Cognitions Questionnaire (TCQ; Wilson and Henry, 1998)	26 items: positive and negative thoughts	5-point scale: 0 = never 4 = very frequently Negative items are scored 0 to 4 Positive items reverse scored 4 to 0 Total score range: 0-104 (higher scores reflect greater tendency to engage in negative thoughts and low engagement in positive thoughts)	Excellent internal consistency reliability: Cronbach's $\alpha = .90$ Excellent test-retest reliability: $r = .88$ Factor analysis supported positive and negative subscales Construct and convergent validity assessed: TCQ showed moderate correlations with TRQ, THQ and TQ TCQ negative subscale showed high correlations with tinnitus (tinnitus related distress, handicap, complaint behavior) and non-tinnitus (general depression, automatic thoughts, locus of control) measures	Useful index of cognitive responses Psychometrically robust Useful as an outcome measure Self-assessment instrument and does actually measure the extent to which patients actually engage in the reported cognitions/thoughts

Questionnaire/Authors	Content	Scoring/Interpretation	Psychometrics	Strengths/Weaknesses
Tinnitus Functional Index (TFI; Meikle et al., 2012)	25 items and 8 domains/subscales: intrusive, feeling, thinking, hearing, relaxing, sleeping, managing, quality of life	11-point scale: 0 to 10 All responses summed, divided by number of questions answered, multiplied by 10 Total score range: 0–100	High test-retest reliability; short-term, $r = .82$ long-term, $r = .76$ Excellent internal consistency reliability: Cronbach's $\alpha = .98$ Good construct validity with THI: $r = .87$ Visual analogue scale for severity: $r = .75$ THQ: $r = .83$ High responsiveness to treatment-related changes in the severity of tinnitus (.79 effect size for overall TFI score)	Developed in a large sample of patients: $n = 327$ for Prototype 1 $n = 347$ for Prototype 2 Excellent measurement and psychometric properties Valid for measuring treatment-related changes (i.e., responsiveness)
Tinnitus Primary Function Questionnaire (TPFQ; Tyler, Ji, Perreau, Witt, Noble, & Coelho, 2014)	12 items: based on items with the highest loadings from a factor analysis of the 20-item version of the TPFQ; emotions, hearing, sleep, concentration	Level of agreement to each statement 0–100 0 = completely disagree 100 = completely agree Mean scores calculated for total and subscale scores with higher scores reflecting greater perceived primary tinnitus handicaps	High internal consistency reliability for total scale: Cronbach's $\alpha = .89$ (with subscales ranging from .86–.93) Adequate construct validity with the THQ ($r = .77$), Pittsburgh Sleep Index ($r = .68$), Beck Depression Inventory ($r = .66$), and the Trait Anxiety Questionnaire ($r = .67$) Sensitivity demonstrated to treatment changed based on statistically significant pre- and post-treatment paired t-scores	Items constructed to represent the primary consequences Brief and useful as a screening measure Scoring on a 100-point scale may be difficult for some patients Overall, good psychometric adequacy, but lacks calculations of effect size, which would strengthen application for determining responsiveness/sensitivity to treatment change

Assessment of self-efficacy (confidence individuals have in their abilities to perform specific tasks, goals, and behaviors) may play an important role in the overall evaluation of tinnitus patients' ability to engage in and carry out a successful treatment plan. To this end, Smith and Fagelson (2011) created the 40-item Self-Efficacy for Tinnitus Management Questionnaire (SETMQ) that has been shown to have adequate internal consistency reliability, test-retest reliability, and construct validity. Specifically, the SETMQ was developed to determine the level of confidence patients with tinnitus believe they have in selected abilities (i.e., routine tinnitus management, emotional response to tinnitus, internal thoughts and interactions with others, tinnitus concepts, and use of devices) that will allow them to manage and control the negative consequences of tinnitus. This questionnaire provides a unique perspective by evaluating the patient's belief system, potentially resulting in improved counseling strategies and better management outcomes.

The administration of a self-reported measure during the initial assessment appointment serves as the baseline, which can then be compared against subsequent measures to quantify intervention outcome. That is, a reduction in self-perceived HRQoL, which may or may not be accompanied by a change in tinnitus impairment, may be reflective of the subjective benefit derived from a particular audiologic treatment approach (e.g., HAs, SGs). In contrast, no change or an increase in perceived problems may suggest that the treatment is not providing sufficient benefit, and that additional or alternative options should be considered. For a comprehensive review of measuring tinnitus treatment outcome, the interested reader is directed to Newman, Sandridge, and Jacobson (2014).

The THI is one of the most widely used self-reporting measures. It has been popular among clinicians and researchers alike because it is brief; easily scored, administered, and interpreted; and psychometrically adequate. The authors have found utility in the THI in that it assesses the domains of function that can be remediated by the TMC care path model. Newman et al. (1998) determined that a change in the total score of at least 20 points suggested that treatment is clinically effective. Furthermore, the four severity categories (0–16 points indicates no handicap; 18–36 points indicates mild handicap; 38–56 points indicates moderate handicap; 58–100 points indicates severe handicap) provide a method for clinicians to quantify changes in perceived participation restriction. McCombe et al. (2001) have proposed a fifth, catastrophic category for scores over 76 points.

It is possible to monitor movement from one category to another over time with or without intervention. In addition, the THI, can be used with its companion scales, the Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982), Hearing Handicap Inventory for Adults (HHIA) (Newman, Weinstein, Jacobson, & Hug, 1990), and the Dizziness Handicap Inventory (DHI) (Jacobson & Newman, 1990) to quantify HRQoL for a variety of patient populations, such as those with Menière's disease (Kinney, Sandridge, & Newman, 1997). Furthermore, a spouse version of the THI (THI-Spouse) has been developed to quantify a spouse's perception of his/her partner's tinnitus handicap (Newman & Sandridge, 2012a). The THI-Spouse may be helpful in identifying differences in perception between spouses and thus potentially useful for counseling purposes.

Usage of the Tinnitus Functional Index (TFI) (Meikle et al., 2012) has increased over the past few years, undergoing evaluation in several different patient samples (Fackrell, Hall, Barry, & Hoare, 2016; Henry et al., 2016; Rabau, Wouters, & van de Heyning, 2014). This psychometrically robust questionnaire comprised of 25 items was developed originally by a team of 17 tinnitus researchers and clinicians led by Mary Meikle. The TFI was designed to assess the severity and negative impact of tinnitus, to be used as an integral component of the tinnitus intake assessment, and to measure treatment-related changes (responsiveness as demonstrated by large effect sizes) to tinnitus intervention. Based on statistical analysis, the authors indicated that a 13-point reduction in the overall TFI score is considered a clinically significant reduction in tinnitus HRQoL severity. Most recently, Chandra and colleagues (2018) confirmed that the TFI is a reliable and valid measure of tinnitus severity and is responsive to treatment-related changes, based on a sample of 871 patients attending the University of Auckland Hearing and Tinnitus Clinic.

Sound Therapy Needs Assessment

A number of sound therapy devices (to be discussed later in the chapter) are available to provide relief from the annoying and bothersome consequences of tinnitus. These include assistive sound enrichment devices, HAs, ear-level SGs, combination units, and other personal listening devices. The overall objectives of sound therapy are to alter tinnitus perception and reduce the negative emotional reactions associated with bothersome tinnitus, thereby promoting clinical benefit in the form of tinnitus relief (Tunkel et al., 2014). Moreover, the rationale for using any

device is to decrease the perception of the tinnitus, thus reducing the signal (tinnitus)-to-background noise ratio. That is, the goal is to reduce the contrast between the environment and the patient's tinnitus percept. In this way, the brain interprets the tinnitus as less noticeable, and therefore, less annoying and troublesome. Henry, Zaugg, Myers, and Schechter (2008) propose that sounds can provide not only soothing relief from the tinnitus, but can also act as a distracter either through passive background sound or active interesting sound.

A challenge faced by clinicians during the sound therapy needs assessment is the selection of the specific sound therapy device for a given patient. As an aid in this process, Henry, Zaugg, and Schechter (2005b) developed a sound treatment worksheet. The worksheet requires the patient to list three specific situations when the tinnitus is most bothersome and is followed by a checklist that recommends specific types of sound treatment such as tabletop sound machines, wearable CD player, and ear-level SGs. On the worksheet, the patient reports how helpful (based on a 5-point checklist ranging from *not at all* to *extremely*) he or she found the device, providing feedback about treatment effectiveness to the clinician. Furthermore, the audiologist can note if the sound therapy was used primarily for enriching the sound environment or was used as a distracter to the tinnitus.

The Sound Therapy Option Profile (STOP) (Newman & Sandridge, 2006) is an 11-item tool (see Appendix 22–A) that guides the audiologist in the selection of the optimal sound therapy device for the patient's individual needs. More specifically, completion of the STOP creates a profile of the patient including his or her motivation, willingness to seek treatment and pay for that treatment, preference for specific types of SGs, expectation level (i.e., realistic or not), and the need for referral to psychology. Within a few short minutes, the profile directs the clinician to the selection of broadband noise versus music, ear-level devices versus body-worn or handheld devices, or custom versus noncustom products, for example. The STOP is a useful tool during the tinnitus evaluation; however, it is important to note that the profile is best administered after the patient has been counseled about tinnitus treatments and different forms of sound therapy. The patient must be an informed consumer to answer the queries regarding specific devices or treatments.

As part of the device selection needs assessment, it is a valuable experience for patients to listen to the various available sound therapy options. Clinic-stock demonstration devices can be used to determine if tinnitus relief can be achieved from environmental

enrichment devices (e.g., tabletop sound machines, fan), personal listening devices (e.g., recordings or apps with a variety of broadband sounds, environmental sounds, and music downloaded on smartphones or tablets), HAs, ear-level SGs, or combination devices.

The wide range of available open-fit behind-the-ear (BTE) combination devices now allows for efficient demonstrations of sound options. Some patients may experience immediate relief during the demonstration because of complete or partial masking; however, this is not the sole purpose of the demonstration. The in-clinic trial also allows the patient an opportunity to listen to different types of sounds and indicate which form of sound therapy might be more acceptable over the long-term.

If the patient has hearing loss in addition to the tinnitus, it is important to determine which is more problematic to the patient: the hearing loss or tinnitus. As the patient's response will influence the device selection (or activation of the microphone in combination devices), it is important to add a tool that will assess the psychosocial impact of the hearing loss. The HHIE/A can determine to what degree the hearing loss impacts the patient's communication and/or psychosocial function. For example, if the patient's hearing loss has little or no impact (e.g., less than 18 out of 100 points on the HHIE/A), the use of SGs alone may be an appropriate first choice. In contrast, if the HHIE/A exceeds 18 points, reflecting significant activity limitation and participation restriction, the use of HAs or combination units would be more appropriate options.

Patient Disposition

At the conclusion of the tinnitus evaluation, recommendations are made for specific sound therapy devices. If ear-level devices are selected, the patient is scheduled for a return fitting and counseling appointment. If no further treatment is indicated or chosen, the patient may return on an as-needed basis for further counseling.

Individual Audiologic Management

As shown on the bottom of Figure 22–2, the next step in the care path following evaluation is a discussion about appropriate intervention options with the patient. The focus of audiologic management is the provision of the selected sound therapy device which must be accompanied by ongoing and intensive

counseling. It is critical to be cognizant that devices are merely a tool and only one component in the overall management of tinnitus. That is, audiologic rehabilitation should not simply revolve around the technology provided, but a treatment plan should be put into place ensuring that patients have a complete understanding regarding the underlying rationale for device application as well as other techniques to help reduce the consequences of tinnitus (e.g., mindfulness, importance of continued participation in activities; hearing loss prevention strategies; techniques for improving sleep). In this vein, the audiologic management of tinnitus cannot and should not be one size fits all. Although the following sections describe specific treatment approaches and techniques, it is critical to understand that intervention should be personalized and target patient-centered functional goals.

Sound Therapy: Advantages and Benefits

In general, sound therapy has been defined as the “use of sound where the intention is to alter the tinnitus perception and/or reaction to tinnitus in a clinically meaningful way” (Hoare, Searchfield, El Rafaie, & Henry, 2014, p. 63). Specifically, inclusion of sound therapy devices within the management plan is important for a variety of reasons. They can:

- provide immediate total or partial masking relief for many patients, thereby reducing the emotional consequences of tinnitus such as frustration, anxiety, and even depression;
- increase the level of background sound to create less contrast between the tinnitus and acoustic environment, resulting in reduced perception of the tinnitus;
- facilitate patient control and empowerment over the tinnitus rather than allowing the patient to feel controlled by the tinnitus;
- promote habituation to the tinnitus perception and subsequent negative reactions by neutralizing the threatening quality of the tinnitus;
- act as an attention-getting sound that distracts/diverts the listener away from the tinnitus;
- offer a noninvasive approach with no adverse side effects;
- allow some patients to experience residual inhibition (i.e., tinnitus suppression or temporary disappearance after exposure to external sound); and

- contribute to the reorganization of the central auditory nervous system pathways and centers responsible for tinnitus generation and perception (Folmer, Martin, Shi, & Edlefsen, 2006b; Henry et al., 2005b, Noreña, 2011; Noreña & Chery-Croze, 2007; Shiley, Folmer, & McMenomey, 2005; Shore, Roberts, & Langguth, 2016).

Sound Therapy: Device Options

Regardless of the type of sound treatment, each device provides a sound-enhanced environment by maintaining a low level of background sound, whether it is delivered via soundfield or coupled directly to the ears. Accordingly, the use of sound therapy is to minimize the patient’s perception of the tinnitus by essentially reducing the signal-to-noise ratio between the tinnitus (signal) and ambient or environmental sounds (noise). Following is a summary of device option categories. Moreover, there are several patient-centered factors that need to be considered when selecting any sound therapy options in the treatment plan (Newman & Sandridge, 2016). These include:

- severity of hearing loss and communication needs;
- lifestyle (e.g., ability to incorporate the use of sound therapy);
- manual dexterity;
- acceptability of the selected sound stimuli (e.g., broadband sounds, nature/environmental sounds);
- cosmetics (e.g., personal ear-level devices, headphones, body-worn streaming devices);
- daily dosage of auditory signals (e.g., hours per day of sound therapy use); and
- budget for purchasing devices.

Environmental Enrichment Devices

A variety of simple-to-use, inexpensive techniques and devices can be implemented to increase the level of background sound to decrease the tinnitus signal-to-noise ratio. The exposure to pleasant external sounds promotes relief, provides distraction, and decreases the patient’s awareness to tinnitus. Many patients find it beneficial, for example, to play the television or radio at home in the background. A number of other environmental enrichment devices are available and include: (a) tabletop sound machines that can generate different types of sounds (e.g., rain, wind, waterfall); (b) personal sound players that con-

tain recordings of music and nature/environmental sounds and/or apps that can stream similar tinnitus relief-producing sounds; (c) tabletop water fountains; (d) fans; and (e) water purifiers (Folmer et al., 2006b). For patients with sleep disturbance, the sound generated by tabletop sound machines, personal sound players, and apps can be delivered (e.g., hardwired or streamed) effectively to commercially available pillows (e.g., Sound Pillow® Sleep System) or headbands (e.g., SleepPhones®) with integrated small, flat speakers designed for comfort.

Hearing Aids

The use of HAs in the management of tinnitus is a common and acceptable practice (Kochkin & Tyler, 2008, 2011; Sheldrake & Jastreboff, 2004; Trotter & Donaldson, 2008; Vernon & Meikle, 2000). In fact, the AAO-HNS tinnitus guidelines suggest that clinicians should recommend a hearing aid evaluation for patients with aidable hearing loss and persistent, bothersome tinnitus (Tunkel et al., 2014). It is estimated that up to 90% of patients experience a reduction in the loudness of the tinnitus from the use of HAs (Johnson, 1998; Schechter, Henry, Zaugg, & Fausti, 2002). Although the exact mechanisms involved in the success of HAs for tinnitus treatment are unknown, there are a few proposed theories. The most common theory suggests that HAs provide sufficient amplification of ambient sound to mask the tinnitus. In this connection, it has been suggested that HAs increase neural activity in the auditory pathway, thereby reducing the difference between the hyperactivity caused by the tinnitus and the ongoing background neuronal activity (Searchfield, 2006). Furthermore, the amplification provided by HAs decreases listening effort (Hornsby, 2013), potentially reducing overall stress and anxiety associated with the tinnitus (Sheldrake & Jastreboff, 2004).

Regardless of the exact mechanisms responsible for providing benefit from the use of HAs, it is important for patients to understand the relationship between hearing loss and tinnitus and to appreciate the differences between the two auditory symptoms. For patients with both hearing loss and tinnitus, Shiley et al. (2005) suggested that the following points should be addressed:

- Tinnitus does not cause hearing loss; however, hearing loss may increase the chances for individuals to have tinnitus.
- Even when relief is provided for tinnitus, residual communication breakdown may remain due to hearing loss.

- HAs do not amplify the tinnitus. Improvement in communication function due to HA use will help the individual feel less frustrated, isolated, and withdrawn.

It is interesting to note that use of most advanced HA technology incorporating sophisticated noise-reduction algorithms may be potentially counterproductive for patients with tinnitus. Some amplification of environmental noise is, indeed, helpful for these patients. On the other hand, today's HA technology provides flexibility in manipulating the signal. For example, soft sounds can be enhanced by lowering the compression kneepoint in the HAs, creating a masking effect which, in turn, may reduce the perception of tinnitus. Open-fit HAs will be especially beneficial as they permit normal entry of environmental sounds into the ear canal.

Sound Generators

In support of audiologic management, the AAO-HNS clinical practice guides recommend that clinicians consider sound therapy for patients with persistent bothersome tinnitus (Tunkel et al., 2014). In this connection, ear-level SGs are a choice for patients who do not benefit from or require amplification because they have normal or near-normal audiometric thresholds. BTE or in-the-ear (ITE) SGs are the most portable and inconspicuous method for patients to receive a stable broadband signal to the ear. This contrasts with the use of tabletop sound machines, for example, where the effectiveness of the sound is lost when the patient leaves the room. During the fitting of SGs, it is important for the clinician to describe the signal in positive terms (e.g., "gentle shower sound" or "therapeutic sound") and to avoid negative terms such as *noise* or *static*. In the authors' experience, the use of positive terminology when describing sound therapy signals helps to promote greater initial acceptance of the devices.

Combination Devices

Combination devices contain a HA circuit and a sound-producing circuit in the same instrument, allowing patients who have both hearing loss and tinnitus to use one device. Today's combination devices have the flexibility of turning on or off the hearing aid microphone and have multiple program capabilities so that the devices can be used as HAs only, SGs only, or HA-SG combinations. A critical feature in the combination device is the ability to control the volume of the SG portion independent of the

volume control of the HA, as the perception of the tinnitus fluctuates. Further, most commercially available devices have several built-in tinnitus relief sound options (e.g., broadband sounds, modulated sounds, fractals) contained within the same unit. Figure 22–10 displays examples of magnitude transfer functions of

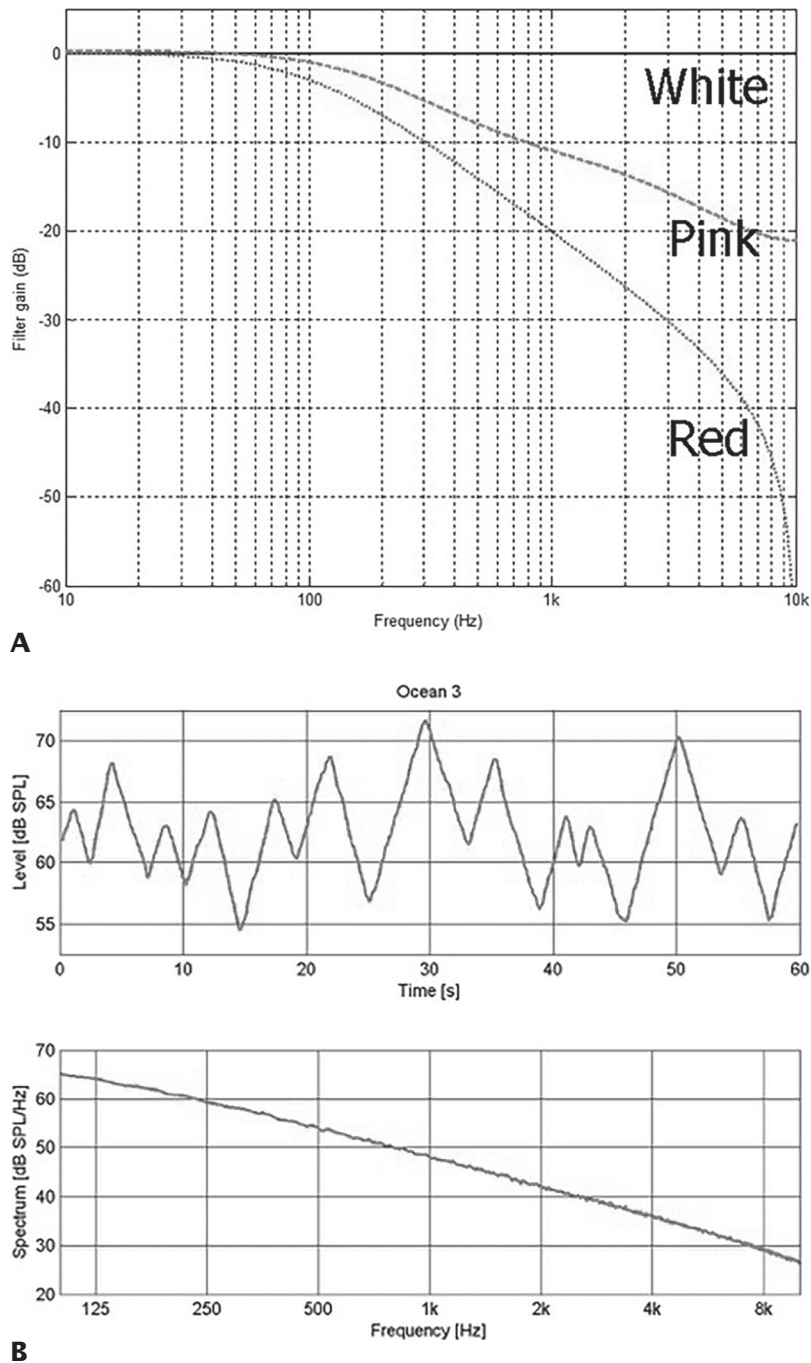


Figure 22–10. Examples of sound therapy stimuli generated by an ear-level device. **A.** Magnitude transfer functions of filters used for white, pink, and red broadband sound options in a combination device. **B.** Example of an “ocean sound” based on proprietary variations of modulation patterns that mimic rhythms of the ocean. *Source:* Courtesy of Oticon A/S.

filters used for three broadband sound options (A) and modulation patterns that mimic rhythms of the ocean (B) in a commercially available combination device. Moreover, tinnitus relief apps downloaded to smartphones/tablets permit an almost endless range of sounds (e.g., broadband, environmental/nature sounds, music, layered and complex sounds) that can be directly streamed to the combination devices.

Sound Therapy: Device Applications

Sound therapy devices can be used in a variety of approaches. No matter which application is selected, it is important that the patient be actively involved in sound therapy selection decision making and understand the rationale for its use and role as a tool in the overall management process (Searchfield, Dural, & Linford, 2017). Recent evidence suggests that the probability of achieving a reduction in perceived tinnitus loudness may be greater when the patient's tinnitus pitch is located within the frequency range of the treatment device used, whether HAs, SGs, or music (Schaette et al., 2010). As mentioned earlier, Schaette et al. (2010) reported that patients who pitch matched their tinnitus to frequencies of 6000 Hz or below received benefit from SGs; however, if the pitch match was greater than 6000 Hz, little benefit

was received from the SGs. The latter research should be viewed with caution based on a small sample size without replication from other studies.

Masking. Masking therapy has a long history, and readers are referred to the following works for historic reference: Hazell et al., 1985; Vernon, Griest, and Press, 1990; Vernon and Meikle, 2000; and Vernon and Schleuning, 1978. The key features of a tinnitus masking protocol are summarized in Table 22–8.

For many patients, SGs can provide total or partial masking, resulting in immediate relief. Total masking involves the use of an external sound to make the masking inaudible. In contrast, partial masking interferes with the perception of tinnitus, yet the external sound and tinnitus are still both audible to the patient (Hoare, Searchfield, El Rafaie, & Henry, 2014). For many patients, the broadband sound is found to be more acceptable than their own tinnitus. Some patients use maskers on an as needed basis when they are most bothered by the tinnitus. Moreover, Hazell (1987) suggested that long-term use of masking may produce beneficial effects even when the masker is removed.

There are a few caveats regarding SGs as maskers that must be discussed with the patient. It is important that patients are cautioned that SGs should never

Table 22–8. Comparison of Salient Features Underlying Masking Therapy and Habituation/Retraining Therapy

Feature	Masking Therapy	Habituation/Retraining Therapy
Rationale	To mask or partially mask perception of tinnitus	To promote habituation of tinnitus
Counseling	Not to attend to tinnitus; provide control	Do not fear tinnitus
Types of ear-level devices used	Maskers or tinnitus instruments (masker and hearing aid housed in same device). Used monaurally or binaurally. Trial-and-error procedure in clinic to determine most effective device	Sound generator or combination unit (sound generator and hearing aid housed in the same device). Used binaurally. Device selection based on prescribed category (0, 1, 2, 3, or 4)
Spectral characteristics of sound	Sound varied depending on patient's perception of most effective	Stable broadband noise
Sound therapy regimen	Use of ear-level device with masking sound adjusted to patient's preference, typically at the lowest level to mask (or partially mask) tinnitus. Wear as desired	Use of ear-level device with sound adjusted just below the "mixing point" at the beginning of the day. Use during waking hours
Follow-up visits	Typically at 6 months and 1 year to ensure proper use of maskers	Typically at 3 weeks, 6 weeks, and 3, 6, 12, and 18 months
Long-term use	Use as long as necessary to provide masking relief	Use until habituation is achieved, typically 1 to 2 years

Source: Adapted from Henry, Schechter, Nagler, and Fausti (2002).

cause discomfort and, if set too loudly, might interfere with communication. They might even cause damage to hearing if used for extended periods of time at too intense a setting. Current HA fitting software often shows a warning on the screen when the masking stimuli are generated at high SPLs. In addition, SGs can increase the perceived tinnitus loudness so it is suggested that patients give themselves rest periods from using the devices (Tyler & Bentler, 1987).

Habituation and Retraining. In contrast to immediate relief produced by total or partial masking, SGs can be used to promote habituation of the tinnitus, resulting in a decline in the reaction to and the perception of tinnitus over time (Hallam, Rachman, & Hinchcliffe, 1984; McKenna, 2004). SGs provide a continuous and monotonous sound stimulus, one identified as “nonthreatening.” The brain habituates to the nonthreatening sound, as well as the tinnitus. Jastreboff (1990) and Jastreboff and Hazell (1993) defined habituation as an adaptation process of the auditory system to reduce perceived signal intensity. By reducing the perceived signal intensity, the negative consequences of tinnitus, including emotional and automatic reactions, will decrease. Accordingly, the application of SGs in this type of habituation treatment focuses on the disappearance of tinnitus as it becomes an essentially irrelevant signal. Hallam and McKenna (2006), however, proposed several roadblocks to the habituation process, including an elevated arousal state, avoidance of exposure to external sound and/or tinnitus, and negative beliefs about tinnitus.

Tinnitus Retraining Therapy (TRT), a protocol-driven strategy developed by Jastreboff and colleagues (Jastreboff, 2000; Jastreboff & Hazell, 2004; Jastreboff & Jastreboff, 2004), is based on the concept of promoting habituation along with neuroplasticity and may involve up to 24 months of treatment (see Table 22–8). The treatment is comprised of two major components: (a) the use of broadband SGs or HAs and (b) directive counseling (Jastreboff & Hazell, 2004; Jastreboff & Jastreboff, 2004). Based on the protocols recommended by Jastreboff, SGs are used for approximately 6 to 8 hours per day and set specifically below the *mixing point*. This is the level where the external sound is just below the patient’s perception of his or her tinnitus and is considered important for facilitating habituation (Jastreboff & Hazell, 1993), although more recent evidence (Tyler, Noble, Coelho, & Ji, 2012) suggests total masking may be just as effective in promoting habituation as the “just below the mixing point” approach. Directive counseling includes ongoing explanations and education about the causes of

tinnitus and identification of specific anxieties relating to tinnitus to challenge the patient’s perception of harm or threat.

Criticism of the original description of TRT included failure to acknowledge that there may be a need for additional psychological intervention, such as cognitive behavioral therapy or cognitive restructuring techniques (Kroener-Herwig et al., 2000; Wilson, Henry, Andersson, Hallam, & Lindberg, 1998), as well as separating the beneficial effects of sound stimulation from counseling (McKenna & Irwin, 2008). In addition, there remains a paucity of data from well-controlled studies supporting the efficacy of TRT and its superiority over other standard-of-care tinnitus approaches (Bauer, Berry, & Brozoski, 2017; Phillips & McFerran, 2010).

Combined Approach. Another treatment option the authors have found to be beneficial involves a combination of masking and habituation approaches. That is, if a patient requires immediate relief from the tinnitus, complete masking (if possible) is attempted during the initial phases of treatment. The relief provided by the masking promotes a reduction in stress and thereby helps to reduce the perception of the tinnitus, which in turn promotes a sense of hope. Yet, masking may not be effective for long-term treatment; therefore, after approximately 1 month, the patient is transitioned into a more long-term management approach using the habituation model (Hallam et al., 1984). In this approach, patients are counseled not to cover up the tinnitus (total masking) but to set the device to deliver the signal at a level slightly below the tinnitus (partial masking or below the mixing point) to facilitate habituation. The salient differences between conventional tinnitus masking and TRT are summarized in Table 22–8.

Music Therapy

Music has been used therapeutically for a number of different health care-related applications and has been shown to be successful in addressing the physical, emotional, cognitive, and social needs of individuals of all ages (Standley, 1995). In this connection, music therapy interventions have been designed to promote wellness, manage stress, alleviate pain, and promote physical rehabilitation. As with other forms of sound enhancement, music may be very helpful to patients with tinnitus and can be delivered using a personal listening device (e.g., MP3/MP4 player) or streamed from smartphones or tablets directly to wireless earbuds or HAs. Keeping in mind the patient’s pref-

erence, any music that distracts from the tinnitus, evokes positive feelings, and/or provides relaxation may be beneficial in the management process (Hann, Searchfield, Sanders, & Wise, 2008). The reader is referred to <http://www.musictherapy.org> for further information from the American Music Therapy Association regarding music therapy benefits.

Davis (2006) extended the known benefits of music into the arena for treatment of tinnitus and developed an acoustic desensitization protocol, known as the Neuromonics Tinnitus Treatment (NTT). The Neuromonics sound processor (Oasis™ and Oasis Pro™) delivers the modified music (compensating for hearing loss and providing less low-frequency bias than unmodified music) to the patient through headphones or via a cloud-based application using the patient's iOS device (i.e., iPhone or iPad). The treatment is composed of two stages: Stage 1, high tinnitus interaction using the spectrally modified music and an embedded broadband sound; and Stage 2, lower-level tinnitus interaction using only the peaks and valleys of the music to promote a masking effect (Hoare et al., 2014). The sound processor is used approximately 2 hours per day for both Stages 1 (longer, if preferred or needed) and 2, followed by use on an as-needed basis during the maintenance phase of treatment.

An alternative form of music therapy is the use of fractal-generated musical chimes, originally developed for audiologic application to promote relaxation and stress reduction for patients with hearing loss using HAs (Kuk & Peeters, 2008). A combination unit (HA and fractal SGs housed in same device) offers predefined music programs (commercially available as the Zen feature) with unique combinations of tonal pitch, tempo, and intensity that can play along with amplification or alone. The fractal notes (based on a mathematical algorithm that generates a series of sounds) have an overall chime or bell choir quality and do not repeat themselves. In addition to appropriate counseling, the application of fractal music delivered through HAs has been found to provide relief for some tinnitus sufferers (Sweetow & Sabes, 2010).

Music application for tinnitus reduction has been attempted using manipulated music samples to promote lateral inhibition between tinnitus and nontinnitus areas of the cortex (Hoare et al., 2014). For example, a technique designed to reduce tinnitus loudness was developed using patient-chosen music that eliminated ("notched") the frequencies in the range surrounding the patient's pitch-matched frequency (Okamoto, Stracke, Stoll, & Pantev, 2010). Using a small sample size ($n = 8$), Okamoto et al. (2010) demonstrated that tinnitus loudness was significantly

diminished following 12 months of regular listening to the custom-tailored notched music treatment. Further, using functional neuroimaging techniques (i.e., magnetoencephalography), the auditory steady state and N1m responses were significantly reduced in the notched-music sample in comparison to the controls. Stracke, Okamoto, and Pantev (2010) hypothesized that the latter findings may reflect reduced neural populations and/or less neural synchrony following the notched music intervention, potentially reversing maladaptive cortical reorganization caused by the tinnitus.

Sound Therapy: Efficacy of Options

Unfortunately, evidence to support most treatment strategies used in current practice is limited or of poor quality (Fernández, Shin, Scherer, & Murdin, 2015; NHS Evidence, 2011), with emerging support for sound therapy (Henry et al., 2017; Hoare, Kowalkowski, Kang, & Hall, 2011; Hobson, Chisholm, & El Refaie, 2012; Tunkel et al., 2014). With this in mind, it is imperative that future research be conducted using rigorous methodological designs that should include the following characteristics: utilization of randomized controlled trials or systematic reviews (including assessment of the external validity of the latter analyses); appropriate statistical testing that includes "dropouts" is used in an "intention to treat" analysis, rather than being eliminated from the data set; long-term follow-up of treatment effects; use of validated outcome measures; and appropriate inclusion criteria and sample sizes based on a power analysis (Dobie, 2004b; Plein et al., 2016). From a practical perspective, however, research evaluating the efficacy of sound therapy devices is problematic. It is difficult to have a placebo group or conduct a blinded study. For example, subjects clearly know that they are using a SG when they hear the white noise or that listening performance improved with the use of amplification.

Despite the paucity of evidence-based research showing the clinical effectiveness of sound devices and therapy approaches (Hobson et al., 2012; Tunkel et al., 2014), the body of literature is extensive and describes the underlying rationale, clinical methodologies, and success rates for different sound therapy approaches. In this connection, Hobson and colleagues (2012) suggested that the "absence of conclusive evidence should not be interpreted as evidence of lack of effectiveness" and that "optimal management may involve multiple strategies" (p. 2). Although it would be beneficial to have a body of research to support evidence-based practices, different forms

of sound therapy should not be abandoned on that basis alone. In fact, evidence-based practice's central message is one of flexibility and blending past clinical experiences with new research and knowledge as it becomes available to the practitioner (Law, 2002).

In general, HAs have been shown to be an effective treatment option (Ibarra, Tavira-Sanchez, Recuero-Lopez, & Anthony, 2018; Kochkin & Tyler, 2008), perhaps because they improve communication function, which in turn reduces stress and consequently reduces the annoyance factor from the tinnitus (del Bo & Ambrosetti, 2007; Surr, Kolb, Cord, & Garrus, 1999; Surr, Montgomery, & Mueller, 1985). From a neurophysiologic perspective, they may downgrade tinnitus generation by moderating the drivers of central gain and hyperactivity (Moffat et al., 2009). Clinically, HAs have been shown to be effective in reducing perceived tinnitus intensity and handicap (Araujo & Lório, 2016), especially when used in conjunction with intensive counseling (Searchfield, Kaur, & Martin, 2010). Furthermore, it has been observed that patients with greater degrees of hearing loss preferred the use of HAs over maskers (Mehlum, Grasel, & Fankhauser, 1984). Shekhawat and coworkers (2015) conducted a scoping review (study inclusion/exclusion criteria and quality are not strictly controlled) where the majority of studies evaluated (18 research studies; 11 reviews) support the practice of fitting hearing aids for tinnitus management, providing reassurance that there is mounting evidence supporting the use of HAs. The latter authors caution, however, that there is still a need for stronger methodology and application of randomized controlled trials in future studies.

Masking devices have been shown to be superior to placebo devices (Erlandsson, Ringdahl, Hutchins, & Carlsson, 1987) and provide greater improvement in sleep over HAs (Stephens & Corcoran, 1985). Results have been inconclusive, however, when maskers were compared to nonacoustic treatments such as self-hypnosis (Attias et al., 1993), acupuncture (Jansson, Warfvinge, Edensvärd, & Wiberg, 1996), and cognitive behavioral therapy (Jakes, Hallam, McKenna, & Hinchcliffe, 1992).

Henry and colleagues (Henry et al., 2006a, 2006b) conducted a controlled clinical study using a repeated measures design to evaluate the clinical efficacy of tinnitus masking and TRT in samples of veterans with clinically significant tinnitus. Several standardized outcome measures were used, including the THI and Tinnitus Handicap Questionnaire (THQ) (Kuk et al., 1990). Qualifying patients were quasirandomly

(alternating placement) placed into the two groups: tinnitus masking group (TM) or TRT treatment. Outcome measures were administered at 0 (baseline), 3, 6, 12, and 18 months. Of the 123 patients enrolled in the study, 118 were included in the final analysis. Based on effect sizes, both the TM and TRT groups showed improvement, with the TRT group showing greater improvement over the long term. Greater improvement for TRT when compared to TM occurred more strongly when the patients began treatment with a very big tinnitus problem. In contrast, when patients began treatment with a moderate tinnitus problem, benefits were not as great. Figure 22-11 displays representative outcomes from the study using data from the THI.

In further study of the question, Parazzini, del Bo, Jastreboff, Tognola, & Ravazzani (2011) compared the effectiveness of TRT between SGs and open-fit HAs. Treatment outcome was measured following baseline administration of the THI at 3, 6, and 12 months. Statistically significant improvements were observed for both sound therapy approaches starting from the first 3 months and continuing through the first year. It is noteworthy that TRT was equally effective, whether subjects were treated with SGs or open-fit HAs.

Recent efforts from the National Center for Rehabilitative Auditory Research (Veterans Affairs Rehabilitation Research & Development Service, Portland, Oregon) have focused on comparing different forms of technology applications for tinnitus management. Henry and colleagues (2015) evaluated the effectiveness of combination devices ($n = 15$) when compared to HAs alone ($n = 15$). All participants had bothersome tinnitus and were HA candidates, yet had no previous HA use within the previous 12 months. Using the TFI as the primary outcome measure, the authors found that both groups received significant benefit in reducing the negative HRQoL consequences associated with tinnitus; however, there were no statistically significant differences between groups. In a related study, Henry et al. (2017) examined the relative efficacy among HAs, combination devices (same HA with the SG activated), and extended-wear, deep-fit hearing aids. Again, the TFI was used as the primary outcome measure along with other secondary hearing-specific questionnaires (e.g., HHIE/HHIA; Speech, Spatial, and Qualities of Hearing) and the QuickSIN. A clinically significant improvement in reaction to tinnitus (≥ 13 -point reduction on the TFI) was observed across all device conditions with no significant differences on TFI scores between conditions.

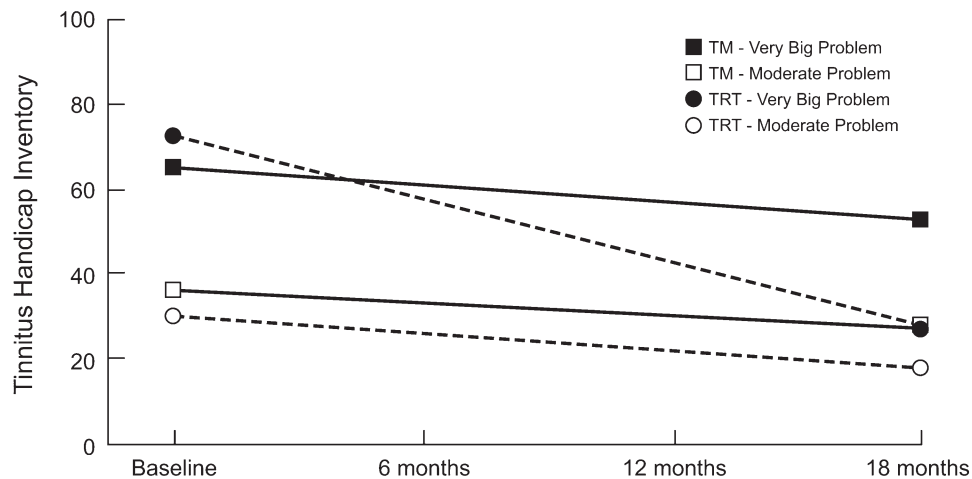


Figure 22-11. Representative outcome data for tinnitus masking (TM) and Tinnitus Retraining Therapy (TRT). *Note.* Outcome data trajectories for Tinnitus Handicap Inventory (THI) responses across 18 months. The scores illustrate differences between tinnitus masking TM (for reported Very Big Problem [■] or Moderate Problem [□]) and TRT (for reported Very Big Problem [●] or Moderate Problem [○]). *Source:* Adapted from Henry et al., 2006b, p. 120.

In the Audiology Research Laboratory at the Cleveland Clinic, the authors completed a study comparing benefits received between a hearing aid (Oticon, Alta2 Pro) and a combination device (Oticon, Alta2 Pro TI) also using the TFI as the primary outcome measure. Participants completed the TFI over a 6-month period. As shown in Figure 22-12 (preliminary data analysis) including 37 participants (HA only, $n = 20$; combination device, $n = 17$), both groups demonstrated a statistically and clinically significant (≥ 13 points) reduction in TFI scores at 1, 3, and 6 months postfitting in comparison to baseline. Moreover, no significant differences were found between groups at any postfitting interval. These findings are consistent with those of Henry and colleagues (2017), in that studies from both laboratories support the clinical practice of fitting patients with HAs with and without the addition of the sound generator; however, from the authors' clinical experience they believe that it is critical to have the flexibility in technology to offer patients what is most acceptable and provides the greatest amount of tinnitus relief for any given individual.

NTT has also been the subject of several peer- and nonpeer-reviewed clinical studies (Davis, 1998; Davis & Wilde, 1996; Davis, Wilde, & Steed, 1999, 2001; Távora-Vieira, Eikelboom, & Miller, 2011). Davis, Paki, & Hanley (2007) evaluated 35 subjects randomly allocated to one of two treatment groups, corresponding

to the two stages of the NTT. The TRQ was used as the primary outcome measure. Patients were followed at 2, 4, 6, and 12 months after the initiation of treatment. After 2 months of treatment, both groups displayed clinically and statistically significant improvement on the TRQ. After 6 months of treatment, 91% of the subjects exceeded the criteria considered as a significant improvement (i.e., reduction of at least 40% on the TRQ). Furthermore, the percentage of time that subjects reported being aware of their tinnitus progressively decreased as the mean usage of the device also decreased, suggesting a permanent treatment effect.

In 2010, Henry and Istvan questioned the merits of NTT based on the lack of methodological transparency of previously published papers, limited independent investigation (i.e., studies conducted by investigators without ties to the manufacturer), and comparisons to other treatment approaches. To address the issue of independent study, Wazen and colleagues (2011) reported the findings from nine tinnitus treatment centers across the United States, revealing statistically significant improvements in perceived improvement on the TRQ and THI at 12 and 24 months postfitting with the Neuromonics device. With respect to comparisons of NTT to other treatments, Newman and Sandridge (2012b) observed equivalent benefit between NTT and binaurally fit ear-level SGs at 6 months postfitting based on THI scores. Although NTT clearly provides tinnitus relief

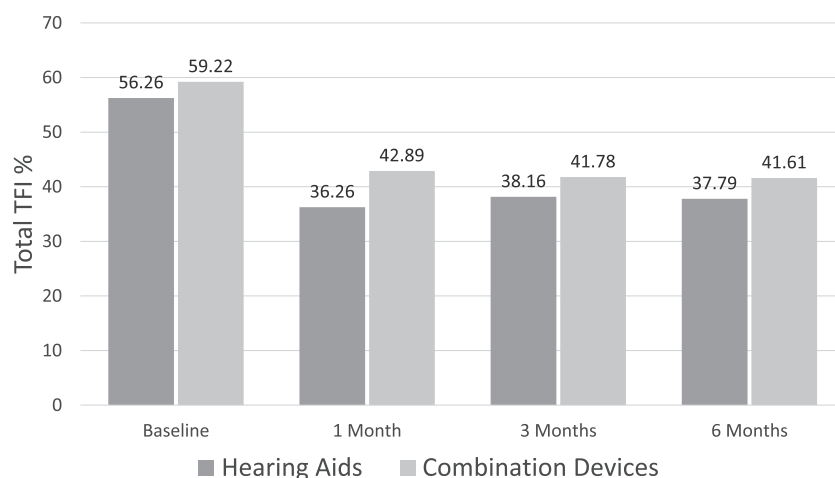


Figure 22-12. Pre- and post-fitting Tinnitus Functional Index (TFI) scores over a 6-month interval for a sample of patients ($n = 37$) with bothersome tinnitus fitted with hearing aids only and combination devices (hearing aids + sound generator in same unit).

for many patients, further study regarding its clinical efficacy is warranted.

Emerging Audiologic Options

Cochlear Implants. The prevalence of tinnitus among cochlear implant candidates is high, consistent with the hypothesis of increased central auditory system gain and hyperactivity resulting from decreased peripheral afferent input (Klooster, Arnold, & van Dijk, 2016). Cochlear implants have been shown to improve or eliminate tinnitus in 37%–95% of cases in several different studies (Amoodi et al., 2011; Baguley & Atlas, 2007; Pan et al., 2009). Mirz, Mortensen, Gjedde, and Pederson (2002) demonstrated the positive effect of the cochlear implant on the suppression of tinnitus through the use of positron emission tomography (PET). The PET scan revealed a reduction of tinnitus-related activity in the primary auditory and associative cortices, as well as in areas related to emotion (limbic system) and attention (dorsolateral prefrontal cortices). Quaranta, Wagstaff, and Baguley (2004) conducted a literature search evaluating the effects of cochlear implants on tinnitus. They identified 32 papers meeting their search criteria. Although results were varied, the use of a cochlear implant was associated with a reduction in tinnitus intensity and awareness for approximately 86% of patients; an exacerbation of the tinnitus occurred in only 9% of the patient population. Based on a sample of 174 cochlear implant recipients, Kompis and his colleagues (2012)

suggested that if both ears are equally suited for implantation, the ear with tinnitus or with the louder tinnitus should be chosen.

Unfortunately, tinnitus can also become worse or begin after implantation in a small proportion of patients (Baguley & Atlas, 2007; Klooster, Arnold, Hofman, & van Dijk, 2015; Klooster et al., 2016; Kompis et al., 2012). Accordingly, patients should be counseled preoperatively about potential tinnitus outcomes, both positive and negative. Most recently, an additional application for cochlear implants is for patients with single-sided deafness and disabling tinnitus. That is, studies (Macías et al., 2015; Mertens, de Bodt, & van de Heyning, 2016) have demonstrated improvements in tinnitus for patients with single-sided deafness when the primary objective of implantation was tinnitus suppression.

Acoustic Coordinated Reset (CR®) Neuromodulation Therapy. This treatment involves delivery of patterned sounds designed to target primarily tonal tinnitus, which is theorized to be the result from pathological neural synchrony. Tass and Popovich (2012) developed a mathematical model to generate a sound sequence that could be used to interrupt and unlearn pathological synchronous neural network activity, leading to a permanent reduction in tinnitus due to neuroplastic changes (Hoare & Sereda, 2016). Clinically, the patterned tonal stimulation is based on the patient's dominant tinnitus pitch match. A reduction in the perception of the tonal tinnitus is achieved by

repetitive delivery of the tonal sequence comprised of different frequencies centered around the dominant pitch with the purpose of counteracting the abnormal neuronal synchronization. Recently, a systematic review (evaluating eight eligible publications for review, comprising a total of 329 patients) was conducted to evaluate the current evidence on acoustic CR neuromodulation as an efficacious tinnitus treatment (Wegger, Ovesen, & Larsen, 2017). Overall, the authors reported that the level of evidence was low with the exception of one randomized control trial. Although the studies indicated the technique was safe and well tolerated, with patients reporting reduction in tinnitus symptoms, the authors concluded that the available evidence at the time of publication was inconclusive for clinical implementation.

Levo System. Typically, sound therapy is recommended for use during waking hours (Hoare et al., 2014); however, Pedemonte, Testa, Diaz, and Suarez-Bagnasco (2014) evaluated the unique application of sound stimulation during sleeping hours wherein subjects matched their tinnitus perception as closely as possible to an acoustic signal. The goal of this approach is to alter neural activity and networks underlying information processing that might be responsible for generating tinnitus during waking hours. Recently, a randomized controlled study (Theodoroff et al., 2017) was conducted to evaluate the efficacy of a commercially available in-ear device (Levo System, version 2.2.6, Otoharmonics Corp.) that uses a customized tinnitus-matched stimulus during sleep. TFI scores and measures of tinnitus loudness were evaluated among three randomized groups of participants (i.e., Levo System with pitch-matched stimuli; Levo System with a broadband noise [sham]; and a bedside sound enhancement device [Marsona, model 1288] regarded as the standard of care in this study). The authors found greater average improvements with reactions to tinnitus, with both the custom and broadband noise devices in comparison to the bedside unit and the custom pitch-matched stimuli showing the greatest reduction for tinnitus loudness using a numerical rating scale. Although encouraging, Theodoroff and colleagues are cautious about the magnitude and generalizability to a variety of clinical populations, as well as the need for future controlled studies of the system.

Auditory Perceptual Training. A number of auditory perceptual training tasks and stimuli have been developed to “renormalize tinnitus-related neuroplastic changes” (Langguth, 2015). Such train-

ing has focused on frequency discrimination tasks, adaptive loudness matching, and identification and localization of auditory objects (Roberts & Bosnyak, 2011). A systematic review of the efficacy of auditory perceptual training for tinnitus (Hoare, Stacey, & Hall, 2010) found that evidence at the time of publication was of low to moderate quality, indicating a need for randomized controlled studies to determine the effectiveness of auditory perceptual training in clinical practice.

Alternative Treatment Options

A variety of alternative tinnitus treatment options exist, with a wide range of reported success rates. It is important for audiologists to be familiar with the various alternative and complementary treatments because patients often ask for advice about these available options.

Dietary Supplements

There are a number of dietary supplements on the market that have been touted to provide relief to the tinnitus sufferer. These include minerals such as magnesium or zinc, herbal preparations including ginkgo biloba, homeopathic remedies, melatonin, bioflavonoids, and B vitamins (DiSogra, 2017; Keate, 2010; Latino, 2010; Seidman & Babu, 2003). Although there are many anecdotal reports associated with the aforementioned options, there is no conclusive evidence to support the use of supplements based on the dearth of well-controlled trials (Rojas-Roncancio et al., 2016; Smith et al., 2010). Accordingly, dietary supplements are not currently recommended to treat tinnitus (Coelho et al., 2016; Tunkel et al., 2014). In fact, the AAO-HNS clinical practice guidelines for tinnitus recommends that clinicians should carefully counsel patients about dietary supplements and include the following educational points (Tunkel et al., 2014):

- No dietary supplements or herbs have been approved for the treatment of tinnitus.
- Although they are readily available to consumers, none have been approved by the Food and Drug Administration (FDA) for the intended use of tinnitus treatment.
- Dietary supplements can actually have negative side effects, especially when taken with other medications or supplements (e.g., ginkgo biloba can interact with blood thinner medications and cause serious

bleeding or worsen bleeding risk in patients with blood clotting disorders).

- Patients should discuss the use of any dietary supplement with their physician.

Pharmacology Treatments

No drugs have been specifically developed to treat tinnitus, *per se*; however, medications are often prescribed by physicians to treat insomnia, anxiety, depression, phobias, obsessive-compulsive behaviors, or other psychological problems associated with the negative consequences of tinnitus (Shiley et al., 2005). The major classes of drugs that have been associated with the treatment of tinnitus specifically include antiarrhythmics (lidocaine), anticonvulsants (carbamazepine), benzodiazepines (diazepam, alprazolam), antidepressants (nortriptyline, amitriptyline, fluoxetine, paroxetine); and N-methyl-D-aspartate (NMDA) receptor antagonists (memantine, flupirtine). Results of randomized clinical trials with the aforementioned drugs have limited, if any, benefit and in fact may have serious side effects (Dobie, 2004a; Smith, Zheng, & Darlington, 2010). The most promising drugs affect the emotional state of patients through the reduction of anxiety, depression, and sleep disturbance (Brummett, 1989). It is noteworthy that the FDA has not approved any specific drug for the purpose of tinnitus treatment (Schleuning, Shi, & Martin, 2006). In fact, the AAO-HNS clinical practice guidelines (Tunkel et al., 2014) recommended against the use of medications as a primary medical treatment for patients with persistent, bothersome tinnitus.

Electrical and Magnetic Stimulation Approaches

A variety of electrical tinnitus treatment options currently exist, with a wide range of reported success rates. These include: transcranial magnetic stimulation (TMS); transcranial current stimulation (transcranial direct current stimulation [tDCS] and transcranial alternating current stimulation [tACS]); and vagal nerve stimulation. In general, the goal of electrical stimulation is to modulate pathological neural activity (i.e., neuromodulation) in an attempt to reduce and/or eliminate the tinnitus percept (Hoare & Sereda, 2016). Following are brief descriptions of the major types of electrical stimulation currently being investigated.

Transcranial Direct Current Stimulation (tDCS). This form of electrical stimulation involves the de-

livery of a noninvasive weak direct (constant polarity) current to the patient using electrodes applied to the scalp with the aim of altering (i.e., potentiation or suppression) cortical neuronal activity. Recently, Wang and colleagues (2018) conducted a meta-analysis and systematic review evaluating the effects of tDCS in patients with tinnitus. The primary outcomes evaluated were tinnitus loudness (i.e., magnitude estimates), tinnitus-rated distress (i.e., visual analog scale), and tinnitus handicap (i.e., THI). A total of eight studies met the inclusion criteria for the meta-analysis. The outcome of the systemic review revealed that tDCS did not result in decreased loudness or significant changes on the THI; however, the pooled data demonstrated a greater reduction in distress for the tDCS treated groups in comparison to subjects administered a sham treatment. Although tDCS holds promise as a neuromodulation treatment technique, it has been recommended that future studies should entail randomized control trials using local multiple stimulations sites, optimized parameters, and objective outcome measures before it is accepted as a mainstream clinical tool (Shekhawat et al., 2015).

Transcranial Alternating Current Stimulation (tACS). Similar to tDCS, tACS involves the use of alternating current (constant polarity changes) to affect neuromodulation. Studies have been limited involving small sample sizes (Claes, Stamberger, van de Heyning, de Ridder, & Vanneste, 2014; Helfrich et al., 2014) without replication. Accordingly, there is scant evidence supporting its current clinical application (Hoare & Sereda, 2016).

Vagal Nerve Stimulation (VNS). This more invasive electrical stimulation approach includes the pairing of the electrical stimulus via implantation of stimulating electrodes into the vagus nerve with an acoustic signal (excluding the tinnitus frequency) to “drive auditory cortex plasticity in a controlled and therapeutic direction” (de Ridder, Vanneste, Engineer, & Kilgard, 2014, p. 171). In a proof-of-concept study, the authors concluded that VNS paired with pure tones (excluding tinnitus frequency-matched tones) is a safe and feasible procedure with no side effects or complications. In a subsequent case report using VNS paired with tones (de Ridder, Kilgard, Engineer, & Vanneste, 2015), it was found that the patient experienced a reduction in tinnitus perception lasting 2 months after treatment accompanied by clinically meaningful reductions on the THI and TRQ. More recently (Tyler et al., 2017), a double-blind controlled pilot study revealed significant improvement on the

THI for the paired VNS group ($n = 16$) in comparison to the controls ($n = 14$). Moreover, participants with tonal and non-noise-induced tinnitus received greater benefit from the treatment in comparison to controls. The authors concluded that VNS paired with tones holds promise as an effective treatment for a subgroup of patients with tinnitus.

Transcranial Magnetic Stimulation (TMS). TMS is conducted using coils that come into contact with the patient's head that deliver magnetic stimulation through the skull to specific underlying areas of neural tissue and the cortex. Because low frequency repetitive TMS (rTMS) (e.g., ≤ 1 Hz) is known to suppress neural activity, it has been applied in the treatment of tinnitus, which is associated with hyperactivity in the auditory system. Individual studies comprising a Cochrane review (Meng, Liu, Zheng, & Phillips, 2011) and systematic review with meta-analysis (Soleimani, Jalali, & Hasandokht, 2016) are highly variable in term of results, owing to variety of methodological issues associated with the studies including small sample size, inadequate placebo controls, variable patient inclusion and exclusion criteria, stimulation parameters, short follow-up periods, and outcome measures. At the time of its writing, based on available evidence, the AAO-HNS clinical tinnitus guidelines did not recommend TMS for the treatment of patients with persistent, bothersome tinnitus (Tunkel et al., 2014).

Counseling at Each Care Path Stage

Although the focus of audiologic intervention is often directed toward the provision of the selected sound therapy devices, it is important to remain cognizant that use of any device technology is only one factor in the management process. In fact, McKenna and Irwin (2008) suggest that it may be the counseling component of sound therapy treatment that serves as the mediating variable in patient improvement and not necessarily the acoustic stimulation itself. Accordingly, ongoing education, guidance, and support to individuals and their families are all central to the primary goal of improving the patient's overall HRQoL.

It is noteworthy that interactions with patients create expectations that ultimately influence the treatment. In this connection, Tyler and colleagues (Tyler, Haskell, Preece, & Bergan, 2001; Tyler, Noble, Preece, Dunn, & Witt, 2004) developed a model that can be used to enhance patient expectations, with the goal of improving treatment outcome. Certainly, expectations

about the course of treatment for sound therapy must be realistic. The patient must understand that there is no singular cure for tinnitus and that the long-term goal is to provide relief from the distressing nature of the tinnitus. Table 22–9 summarizes the attributes that influence patient expectations based on the nurturing patient expectations model.

In an interesting related study by Husain, Gander, Jansen, and Shen (2018), the authors found that patients and audiologists differed in regard to their expectations for successful treatment. That is, audiologists tended to focus on the patients' reactions to tinnitus (e.g., stress/anxiety relief). In contrast, patients were more concerned about changes in the perceptual characteristics of tinnitus (e.g., loudness; elimination of tinnitus). The authors suggested that clinicians should provide realistic expectations and information about tinnitus options so that the focus of counseling is on the patient's reactions to tinnitus rather than the tinnitus percept itself.

In addition to developing realistic expectations regarding treatment, it is critical that patients are committed, motivated, and engaged to participate in a long-term course of management. Accordingly, motivational interviewing (MI) is a counseling technique using a collaborative, person-centered, goal-oriented style of patient interaction with the intention

Table 22–9. Attributes Comprising the Nurturing Patient Expectations Model

Being perceived as a knowledgeable professional using jargon and examples that are understandable by the patient
Being sympathetic to the patient by listening and acknowledging problems
Demonstrating an understanding of the problem by providing relevant information about hearing, hearing loss, and tinnitus
Providing a clear treatment plan with realistic goals and expectations
Showing that you sincerely care about the patient
Providing feelings of mastery to the patient so that they gain a sense of control over the tinnitus
Providing hope which is not typically offered by other health care professionals to patients with tinnitus (e.g., too often a patient is advised, "learn to live with it")
Instilling confidence in your intervention efforts by believing in the treatment plan

Source: Adapted from Tyler, Haskell, Preece, and Bergan (2001); Tyler, Noble, Preece, Dunn, and Witt (2004).

to motivate changes in behavior and instill a sense of commitment in the patient (Miller & Rollnick, 2009; 2012). Recently, Zarenoe and collaborators (Zarenoe, Söderlund, Andersson, & Ledin, 2016) applied MI techniques when fitting hearing aids in a sample of 50 patients (MI group, $n = 25$; standard practice [SP] group, $n = 25$) experiencing tinnitus and sensorineural hearing loss. MI interviewing techniques were employed in patient counseling and included the use of open questions, reflective listening, summaries, and affirmations. Although both groups were found to have significantly decreased levels of tinnitus annoyance following HA fitting, the MI group demonstrated a statistically significant greater reduction in THI scores in comparison to the SP group. The latter findings support the value of using MI coupled with amplification when managing individuals with tinnitus and hearing loss.

It is also important to consider the social support system when counseling patients, taking into account the need to provide family-centered management. Social support by family members and significant others often requires education as to the nature of both hearing loss and tinnitus and their negative biopsychosocial consequences. Because spouses are central to the support network, they can significantly influence how patients perceive their HRQoL and treatment outcome (Leonard, Cano, & Johansen, 2006). Accordingly, insight about marital congruence

(i.e., level of agreement between partners' perception, assessments, or beliefs about a health state) or incongruence (i.e., level of discrepancy between partners) is important to providing appropriate family-centered counseling.

In this connection, Newman and Sandridge (2012a) evaluated spousal congruence in a sample of 20 married couples attending the TMC using the THI-Spousal Version (THI-SP). The scatter plot shown in Figure 22-13 illustrates overall strong agreement ($r = 0.84$, $p < 0.0001$) between THI and THI-SP scores. The diagonal line on the scatter plot represents equivalent scores between the THI and THI-SP. Data points below the diagonal line indicate that the patient perceived his or her tinnitus as more handicapping than did the spouse. In contrast, data points falling above the diagonal line indicate that the spouse perceived the patient's tinnitus as more handicapping than did the patient. Although there is overall good agreement, special attention needs to be paid to the outliers (e.g., those partners falling the greatest distance from the diagonal line). These data points highlight the need to incorporate assessment of both partners to determine whether a spouse may overestimate or underestimate the patient's perceived HRQoL. The observation that a spouse overestimates a patient's perceived disability may suggest that the patient has good coping skills or possibly may not be vocal about his or her problem (i.e., "suffers in silence"). Alter-

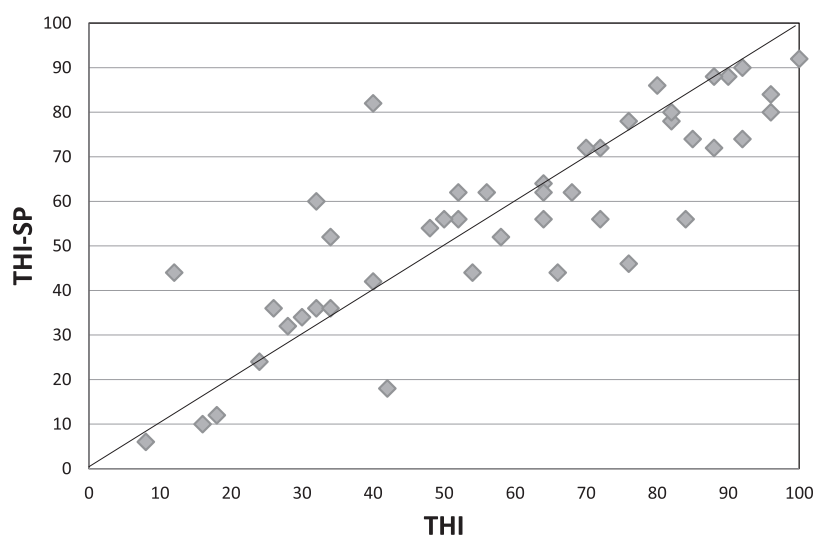


Figure 22-13. Scatter plot depicting the relationship between responses to the Tinnitus Handicap Inventory (THI) and Spousal Version of the THI (THI-SP). *Note.* The diagonal line represents equivalent scores between the THI and THI-SP. Outlying data points represent greater incongruence between spouses' judgments.

nately, overestimation of the patient's disability by a spouse may reflect excessive complaint behavior by the patient. Either scenario, identified by using the THI and THI-SP, would signal the clinician that further counseling is required to help establish appropriate and constructive coping skills for both partners. For example, the invisibility of tinnitus makes it difficult for social support networks to understand the degree of inference and distress that can be caused by the tinnitus. To assist in partner counseling, computer sound files (e.g., <http://www.hearing.nihr.ac.uk/public/auditory-examples-sounds-of-tinnitus>; <https://www.hear-it.org/Impressions-of-hearing-loss-and-Tinnitus>) and YouTube videos (e.g., <https://www.youtube.com/watch?v=2mV-Z54fiBo>; <https://www.youtube.com/watch?v=82Eo9ZfjXbI>) that simulate tinnitus are helpful in conveying to significant others the annoyance caused by tinnitus.

Following device fitting, patients should be scheduled for follow-up counseling appointments at regular intervals. These contacts may be face to face, virtual (e.g., telehealth), or through email, depending upon the specific needs of the patient and treatment plan. As summarized in Table 22–10, several authors (Andersson & Kaldo, 2006a, 2006b; Folmer, Martin, Shi, & Edlefsen, 2006a, 2006b; Henry et al., 2005; Sizer & Coles, 2006; Husain, Tai, & Finnegan, 2016; Tyler et al., 2006) have offered areas of counseling that need to be addressed on a continuing basis during the audiologic management process (i.e., before, during, and after the fitting of sound therapy devices). In addition, there are a number of online resources that can be used with patients. For example, the Ida Institute has a suite of tools that offer patients a sense of hope and encouragement, information about the impact of tinnitus on their everyday life, and specific counseling and coping strategies (<http://idainstitute.com/toolbox/tinnitus/>). Audiologists, however, must be mindful that the internet, including social media (i.e., Facebook, Twitter, and YouTube) contains much misinformation and that patients should be counseled accordingly (Deshpande, Deshpande, & O'Brien, 2018).

As indicated earlier, it is well documented that patients with bothersome tinnitus often experience significant emotional distress (Langguth et al., 2013). If audiologists do not have easy access to services offered by a psychologist, it is critical that available screening tools are used to help identify those patients in need of further psychological/psychiatric assessment and treatment (Pattyn et al., 2016). For example, the Hospital Anxiety and Depression Scale (HADS) (Zigmond & Snaith, 1983) has been evalu-

ated as a screening tool for anxiety and depressive disorders specifically in patients with tinnitus (Zöger, Svedlund, & Holgers, 2004). The HADS is composed of 14 items (seven items on the anxiety subscale and seven items on the depression subscale) with ratings on a scale ranging from 0 to 3. Zöger et al. (2004) recommended using a cutoff score for each subscale of ≤ 5 points for patients with tinnitus. Other simple screening tools that can be useful to assess quickly a patient's emotional response to tinnitus and to help enter into a conversation with the patient about this topic include the PHQ-9 (Kroenke et al., 2001) and the GAD-7 (Spitzer et al., 2006). Both the PHQ-9 and the GAD-7 questionnaires and guidelines for interpretation and referral are available on the internet.

In extreme cases, audiologists must be prepared for the patient who communicates suicidal and self-harm ideations (Aazh & Moore, 2018), although a direct causation between the presence of tinnitus and suicide was not observed in a veteran sample (Martz, Jelleberg, Dougherty, Wolters, & Schneiderman, 2018). In this vein, Jacobson and McCaslin (2001) indicated that it is not necessarily the tinnitus *per se* that results in suicide, but concomitant psychiatric disorders that amplify the effects of tinnitus. Regardless, the authors have had patients voice suicidal thoughts, and believe audiologists should be prepared to address suicidal ideas, plans, means, intentions, and safety measures with the patient (Pridmore, Walter, & Friedland, 2012). This requires the clinician to be comfortable conducting these conversations in a direct and forthright manner and having access to an immediate and reliable referral to a qualified mental health practitioner. Asking a patient to commit to calling 911 or going to the nearest emergency room if he or she has an impulse for self-harm is a simple strategy that very distressed patients may have never considered.

Interprofessional Care Following the SMA

The TMC care path (see Figure 22–2) reflects the importance of interprofessional collaborative practice among audiology and other health care practitioners, including otolaryngology at the outset, followed by dentistry, neurology, physical therapy, and psychology. Based on recommendations following the SMA, a patient may be seen by one or more clinicians on the team. The following is an overview of services offered by other TMC team members.

Table 22–10. Example Topic Areas for Ongoing Counseling

Importance of Lifestyle Changes Considerations of diet (e.g., potentially limiting sodium, caffeine, alcohol, tobacco) Stress management techniques Continued participation in enjoyable activities and hobbies Benefits of regular physical exercise Improving sleep patterns (e.g., keeping regular sleep hours, keeping bedroom cool, adopting a “worry time” outside of bedtime, taking a warm bath or shower before going to bed)
Informational Counseling Hearing, hearing loss, and hearing loss prevention including use of hearing protection devices (e.g., earmuffs, earplugs) for occupational and recreational use
Reduction of “Checking Behaviors” Monitoring tinnitus by keeping a diary of changes in perceived tinnitus pitch, loudness, and/or quality can be counterproductive as it continues to bring attention to the tinnitus Reducing or stopping searches on the internet, as this also brings attention to the tinnitus as well as exposure to information and/or management options that have no evidence to support or information is incorrect
Refocusing Attention on the Task Diverting attention away from tinnitus onto other tasks
Reassurance About the Lack of a Serious Underlying Cause Based on medical clearance for treatment program provided by otolaryngology
Realistic Expectations and Perspectives About Treatment Outcome Including discussion that tinnitus management is a process and there is not an “overnight” cure
Promotion of Relaxation Use of relaxation exercises using a number of available YouTube videos and apps
Benefits of Sound Therapy Exploration of different sounds that may be soothing, provide background stimuli that reduce the contrast between the tinnitus and acoustic environment, and/or interesting things that distract from the tinnitus
Candid Discussion About Benefits/Limitations of Alternative Treatment Options Including pharmacologic intervention, dietary supplements, electrical stimulation (e.g., cochlear implants), and alternative approaches (e.g., acupuncture, transcranial magnetic stimulation)
Actively Listen to Patient
Provide Patients With Encouragement and Hope

Source: Adapted from Henry, Dennis, and Schechter (2005); Sizer and Coles (2006); Andersson and Kaldo (2006a, 2006b); Folmer, Martin, Shi, and Edlefsen (2006a, 2006b); Tyler, Gehring, Noble, Dunn, Witt, and Bardia (2006); Newman and Sandridge, 2016.

Psychology

The major focus of intervention by the psychologist is to educate the patient about the mind-body

connection. In this vein, Figure 22–14 illustrates the relationship among the negative thoughts experienced by patients with tinnitus, the emotional consequences of those thoughts, and the impact on HRQoL

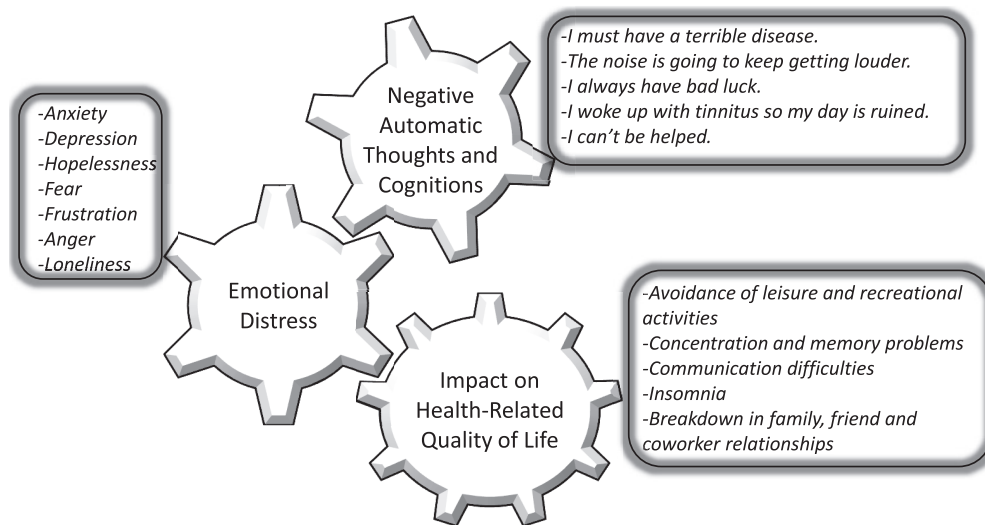


Figure 22-14. Relationship among negative thoughts, emotional responses, and impact on health-related quality of life (HRQoL).

in terms of participation restriction and activity limitation. Patients seeking tinnitus treatments also typically report a fear of the long-term impact that tinnitus may have on their HRQoL. Specifically, the primary complaints that patients attribute to their tinnitus include sleep disturbance, realization that there is no quick cure for tinnitus, difficulty adjusting to having a chronic condition, and the inability to relax secondary to attentional focus on internal tinnitus perception.

Personality traits and/or sociocultural roles and expectations may influence a patient's construction of meaning and formulation of a model of their own chronic condition (Pachter, 1994). Using a web-based personality survey, Durai, O'Keeffe, and Searchfield (2017) compared personality traits between individuals with distressing tinnitus and nontinnitus controls. The personality profiles revealed that those suffering from tinnitus displayed: higher levels of stress reaction (tendency to experience frequent and intense negative emotions, including overreaction to minor events); lower social closeness (less outgoing and less likely to turn to others for support); lower self-control (reflective with greater cautiousness); and higher alienation (feeling of being victimized or having bad luck) in comparison to the control group.

The AAO-HNI clinical practice guidelines for tinnitus (Tunkel et al., 2014) support the use of CBT as an effective intervention for tinnitus-related distress in that it addresses problematic cognitions, emotions, and behaviors experienced by the patient. This approach provides patients with tools that help

them (a) modify errors in logic; (b) recognize dysfunctional thoughts (i.e., beliefs, attitudes, and/or attributions); and (c) identify hypotheses that can be tested in relation to these thoughts. The psychologist helps the patient challenge or test the validity of his or her thoughts as a description of themselves, the world, or the future (Wilson et al., 1998). This approach focuses on helping patients identify cognitive distortions such as personalization ("Why did this happen to me?"), jumping to conclusions ("I woke up with tinnitus this morning so the rest of my day will be ruined"), or catastrophizing ("Because I have tinnitus, I must have a terrible disease"), and to alter those reactions and behaviors. The goal of treatment is not only to identify the cognitive distortions, but also to change the way in which the patient interprets, feels, and thinks to modify the patient's actions and remove inappropriate belief, anxieties, and fears (Sweetow, 2000). Fortunately, cognitive distortions respond well to behavior interventions such as scrutiny for evidence that supports or does not support worrisome beliefs. CBT protocols for tinnitus treatment have also been developed using self-help books (Henry & Wilson, 2001, 2002) and audiologist-supported internet-delivered (iCBT) formats (Aazh & Moore, 2018; Beukes, Allen, Baguley, Manchaiah, & Andersson, 2018; Beukes, Allen, Manchaiah, Baguley, & Andersson, 2017; Beukes, Baguley, Allen, Manchaiah, & Andersson, 2017). Using high levels of evidence, both systematic and scoping reviews of the literature (Andersson & Lyttkens, 1999; Hesser, Weise, Westin, & Andersson, 2011; Martinez-Devesa, Waddell, Perera, & Theodou-

lou, 2007; Thompson, Hall, Walker, & Hoare, 2017) support the application of CBT in the treatment of tinnitus patients and confirms that this approach has a positive effect on the patient’s HRQoL.

In addition to CBT, other forms of behavioral therapy, including imagery training (focusing thoughts on something pleasant, thereby directing thoughts away from the tinnitus), attention control (learning to switch attention away from the tinnitus when it is bothersome), relaxation training (progressive muscular relaxation using a guided protocol), and acceptance and commitment therapy (strategies for promoting acceptance, living in the here and now, and separating thoughts from emotions) are helpful in the treatment process (Hesser, Westin, Hayes, & Andersson, 2009; Tyler et al., 2004). Table 22–11 summarizes approaches used in CBT, mindfulness-based therapy, and acceptance and commitment therapy for tinnitus.

Psychological treatment, although initially viewed with apprehension and reluctance by some, is very helpful for many patients. Noble and Tyler (2007) highlight the importance of, and support the need for, an ongoing relationship between audiology and psychology. The current authors believe this interprofessional collaboration is critical to creating an optimal model for tinnitus management.

Neurology

Evaluation by a neurologist is especially appropriate when patients report somatic modulation of their tinnitus. Recall that somatic modulation is screened for during the initial assessment by the neurologist during the TMC (refer to Figure 22–5 for physical maneuvers used to elicit somatic modulation during the screening assessment) and is the phenomenon where some aspect of tinnitus, most typically perceived loudness, can be transiently modulated (increased or decreased) by different inputs from somatosensory, somatomotor, and visual-motor systems. That is, somatic modulation of tinnitus can be generated by: muscle contractions of the head, neck, and limbs; eye movements in the vertical/horizontal planes that put pressure on myofacial trigger points (i.e., hypersensitive spots within bands of skeletal muscle fibers that can cause pain or change tinnitus); cutaneous stimulation on the hand or fingertips; electrical stimulation of the median nerve; orofacial movements; and pressure applied to the temporomandibular joint (Sanchez, 2016). Accordingly, somatic modulations are a good example of central nervous system integration wherein an auditory percept is generated through nonauditory stimulation/manipulation (Levine,

Table 22–11. Strategies Used in Cognitive Behavioral Therapy (CBT) and Acceptance and Commitment Therapy (ACT) to Help Manage Bothersome Tinnitus

Strategy	Consequence
Cognitive restructuring	Alters automatic thoughts, beliefs, and cognitions about tinnitus so that it can be perceived as nonthreatening, resulting in acceptance rather than resistance
Reducing avoidance behaviors	Restores participation and quality of life, facilitating the process of tinnitus habituation
Reducing checking behaviors	Limits misinformation about tinnitus and self-attending to the tinnitus percept, facilitating the process of tinnitus habituation
Mindfulness	Assists patients to notice and label their thoughts, beliefs, and experiences nonjudgmentally using a meditative approach, limiting future-oriented tension and disturbance by focusing on the present moment
Cognitive defusion	Assists patients to respond to their own thoughts about tinnitus from an external observational perspective, resulting in decreased emotional distress
Relapse prevention	Addresses potential stressors or changes in tinnitus that may challenge the patient’s coping strategies and resources in the future, providing effective management strategies to be able to overcome and mobilize in these instances
Guided muscle relaxation	Progressive muscle relaxation reduces a state of hyperarousal and reactivity in response to tinnitus, facilitating tinnitus acceptance and habituation

Source: Adapted from Newman et al. (2011); Newman and Sandridge (2014).

2004; Levine, Abel, & Cheng, 2003; Sanchez, 2016; Sanchez, Guerra, Lorenzi, Brandão, & Bento, 2002). From a management perspective, treatment at reducing muscular tension in the jaw (Wright & Bifano, 1997) or resolving vertebral subluxations can be effective in reducing tinnitus (Alcantara & Plaughner, 2002; Whedon, 2006). Therefore, referral to physical therapy for exercises designed at reducing somatic modulation of tinnitus may be of clinical benefit (Newman et al., 2011).

In addition to the evaluation of somatic modulation, the neurologic examination serves to identify accompanying symptoms—vestibular (e.g., dizziness, imbalance), headache (e.g., tension or migraine), pain—or disorders (e.g., cranial nerve disorders such as facial nerve injury or trigeminal neuralgia) that may help to localize and ultimately diagnose an underlying neurologic condition causing the tinnitus. If and when a treatable disorder is identified, symptom-based intervention should be deferred until treatment has been completed or discontinued. Other treatments/regimes to address the tinnitus symptoms may be recommended on a case-by-case basis, including various oral medications (e.g., antiarrhythmics, anticonvulsants, benzodiazepines, and antidepressants) or anesthetic blocks (e.g., intravenous anesthetic-plus steroid injections) (Newman et al., 2011).

Physical Therapy

The goals of the physical therapy evaluation are to identify biomechanical problems of the head, neck, and jaw that may contribute to the generation of tinnitus. Subsequent therapy is designed to focus on motion (e.g., repeated cervical and thoracic movements), manual manipulation (e.g., mobilizations to upper, mid, and lower cervical spine and thoracic spine including extensions and rotations), soft tissue massage (e.g., self-messaging of myofascial trigger points), exercise (e.g., strengthening of neck flexors/extensors), multimodal treatment (e.g., combination of manual mobilizations, exercise therapy and home-based exercises), and patient education (e.g., body mechanics, ergonomics such as sitting at a computer, posture, and spine care). Studies supporting the aforementioned therapy approaches are mounting, providing evidence that minimization of biomechanical and somatosensory problems results in a reduction of tinnitus severity in select groups of patients (Cherian, Cherian, Cook, & Kaltenbach, 2013; Latifpour, Grenner, & Sjö Dahl, 2009; Michiels et al., 2016; Mich-

iels, van de Heyning, Truijen, Hallemans, & de Hertogh, 2016, 2017; Tullberg & Ernberg, 2006).

Dentistry

The dentist participating in the authors' TMC is an integral member of the interprofessional team because tinnitus is often associated with temporomandibular disorders (TMD) involving the temporomandibular joints, the muscles of mastication, and the teeth (Lam, Lawrence, & Tenenbaum, 2001; Steigerwald, Verne, & Young, 1996; Vielsmeier et al., 2011). A variety of dental interventions are offered by the dentist. For example, a dental orthotic (i.e., a bite split) may be recommended to reduce jaw contractions, thereby decreasing hypercontraction. Bite realignment may be beneficial for a select group of patients who experience TMD pain (Morgan, 1996). The goal of dental treatment is to improve abnormal jaw mechanics and tracking, as well as normalize the internal and external soft tissue. Treatment is aimed at reducing tension in jaw musculature. For example, Wright and Bifano (1997) report that TMD therapy reduced tinnitus in individuals whose tinnitus worsened with teeth clenching. A number of self-help dental therapies may also be recommended (Newman et al., 2011; Newman & Sandridge, 2015), each focusing on decreased teeth clenching and overloading of the masticatory system to promote myofascial trigger point deactivation. Patients are instructed to avoid gum chewing, cheek biting, nail biting, and eating "hard chewy" foods. Further, they may be counseled about proper tongue and teeth posture (i.e., lips together, teeth apart, and tongue resting lightly on palate). Correcting cervical posture and changing sleep positions may also decrease many of the tinnitus-associated symptoms of TMD (Newman et al., 2011).

Quantifying Treatment Outcome

The final stage in the TMC care path (see Figure 22-2) is to monitor treatment outcomes. To quantify treatment effectiveness, measures of both tinnitus impairment and tinnitus activity limitation or participation should be administered on a regular basis to assess the changes that have occurred secondary to treatment. As mentioned earlier, information about the psychoacoustic attributes of tinnitus (obtained from impairment measurements) and HRQoL consequences of tinnitus (obtained from self-reported

measures) provide fundamentally different information and have somewhat different utility in assessing user benefit. Assessing tinnitus outcome has several implications in the rehabilitation process including: tracking clinically meaningful changes in patient function over time; improving audiologist-patient communication; setting realistic expectations regarding various intervention options; and incorporating the patient’s perspective of clinical treatment, which helps to empower the patient in the management process (Newman & Sandridge, 2004).

As indicated earlier, when selecting outcome measurement tools to gauge treatment effects, the psychometric properties of test-retest reliability (ensures that changes in scores before, during, and after treatment, reflect true changes in performance rather than instrument measurement error) and responsiveness (ability to detect treatment effects) are especially critical when determining changes in performance over time. The application of the statistical technique of effect size (Cohen’s *d*) should be evaluated for your outcome measures as this approach is used to assess the magnitude of treatment effects for group data. More specifically, the following interpretation guidelines should be considered when selecting a tool to be used for quantifying treatment outcome: *d* < .20, inconsequential or insignificant; *d* = .20 to .50, small; *d* = .50 to .80, moderate; and *d* > .80, large. Accordingly, larger *d* values reflect outcome measures yielding greater responsiveness. For a complete review of outcomes measurement techniques, including calculation of 95% confidence intervals and percentage change scores, the interested reader is directed to Newman et al. (2014).

Patients are discharged from the TMC once the tinnitus has become manageable; however, periodic contact with the patient is recommended in that it offers support and evaluation of longitudinal benefit of the treatment program. Moreover, the importance of quantifying treatment outcome is imperative to demonstrate treatment effectiveness not only to our patients, but to accreditation bodies, payers, and policy makers alike.

Economic Considerations

Coding and Reimbursement

When establishing a tinnitus clinic, a number of practice management issues need to be considered, including the use of appropriate diagnostic and bill-

ing codes and insurance reimbursement for service provision. The current 2019 International Classification of Diseases and Related Health Problems, 10th revision, Clinical Modification (ICD-10-CM) diagnostic codes for tinnitus, most typically used by audiologists, are listed in Table 22–12. For a full list of the most up-to-date ICD-10-CM codes, descriptors, and instructions, refer to the official ICD-10-CM publication at: <http://www.cdc.gov/nchs/icd/icd10cm.htm>

The Current Procedural Terminology coding system describes how to report procedures or services and is maintained and copyrighted by the American Medical Association (CPT®) (American Medical Association, 2019a). The CPT® code 92625 can be used for reimbursement of the psychoacoustic assessment of tinnitus including pitch matching, loudness matching, and masking (minimum masking level and testing for residual inhibition). The modifier “-52” (reduced services) should be used when only one ear is tested or if only part of the psychoacoustic tinnitus assessment battery is performed. Further, the modifier “-59” can

Table 22–12. Current ICD-10-CM Codes Commonly Used by Audiologists for Tinnitus and the Related Symptom of Hyperacusis

H93.1 Tinnitus	
H91.11	Tinnitus, right ear
H93.12	Tinnitus, left ear
H93.13	Tinnitus, bilateral
H93.19	Tinnitus, unspecified ear
H93.A Pulsatile Tinnitus	
H93.A1	Pulsatile tinnitus, right ear
H93.A2	Pulsatile tinnitus, left ear
H93.A3	Pulsatile tinnitus, bilateral
H93.A9	Pulsatile tinnitus, unspecified ear
H 93.23 Hyperacusis	
H93.231	Hyperacusis, right ear
H93.232	Hyperacusis, left ear
H93.233	Hyperacusis, bilateral
H93.239	Hyperacusis, unspecified ear

Note. This is not a comprehensive listing of codes. Other categories of codes (e.g., specific ear diseases; types of hearing loss) may be more extensive, contain additional instructional notes, and require higher levels of specificity.

be used to establish that the tinnitus assessment is distinct from another procedure billed on the same day.

There are no current CPT® codes that can be used specifically for the provision of tinnitus treatment. Medicare and most other third-party payers still consider sound therapy (tinnitus masking) an experimental therapy and, therefore, will not cover this management option. Tinnitus devices such as sound generators and Neuromonics could be coded using the Level II Healthcare Common Procedure Coding System (HCPCS) (American Medical Association, 2019b), which is used for reporting most medical services and supplies provided to Medicare and Medicaid patients and by most private payers. Although there is no specific code for a tinnitus device, the V5299 code for “hearing services, miscellaneous” may be considered. HCPCS instructions indicate this latter code should be used only if a more specific code is unavailable. Furthermore, clinicians should consider the use of other CPT codes when additional services, such as HA evaluations, are conducted or when amplification is used to improve communication function and provide tinnitus relief. Establishing direct patient fee-for-service reimbursement options (e.g., unbundling charges for follow-up tinnitus counseling visits) is appropriate when developing fee structures for individual work settings.

Valuation of Treatment

Another perspective on the economics of tinnitus management is valuation of treatment benefit, especially in this era of cost containment and shrinking health care dollars where clinicians need to be both clinically and fiscally accountable. Accordingly, tinnitus management protocols must not only be evidence-based, but should also demonstrate economic value to patients, other providers, third-party payers, and policy makers. Yet, limited data are available regarding the economic valuation of tinnitus treatment outcome.

In this connection, Tyler (2012) administered a WTP questionnaire to a sample of individuals attending a tinnitus self-help group. The purpose of the questionnaire was to explore the amount of money patients would be willing to spend to reduce their perceived tinnitus loudness and annoyance by either “half” or “completely.” On the questionnaire, subjects selected from among the following cost categories that represent the largest amount of money they would be willing to spend to alleviate tinnitus annoyance: \$0, \$100, \$500, \$1,000, \$5,000, \$10,000, or \$25,000. To reduce their tinnitus by half, participants ($n = 189$)

most commonly selected to pay at least \$1,000; to reduce their tinnitus completely, participants ($n = 187$) most commonly selected to pay \$5,000, with 20% ($n = 40$) of the individuals willing to pay up to \$25,000. A limitation of the study, as suggested by the author, was that responses to a WTP questionnaire may be very different from what a patient might actually pay for treatment.

To address the latter clinically, the STOP (Newman & Sandridge, 2006), previously described in this chapter and available in Appendix 22–A, explores an individual patient’s willingness to pay for sound therapy treatment by providing several different cost category options from which the patient can select. This component of the STOP provides a starting point for the clinician to discuss related charges for sound therapy devices and services.

Newman and Sandridge (2012b) incorporated two types of standard economic analyses to help determine the value associated with two forms of sound therapy; namely, ear-level broadband noise SGs and NTT. A cost-effectiveness analysis (CEA) was used to examine the incremental cost per unit of benefit on a standardized outcome measure achieved when one treatment option is chosen over the other. A cost-utility analysis (CUA) was used to capture the cost per quality of life years (QALY) gained as a result of the chosen treatment.

More specifically, the CEA calculations shown in Table 22–13 consist of three separate incremental treatment comparisons (Gold, Siegel, Russell, & Weinstein, 1996): treatment cost, treatment effectiveness, and the cost-effectiveness ratio (C/E) between the two treatment alternatives. The C/E ratio reflects which treatment option (SG versus NTT) produces the most benefit for a given expenditure. CUA incorporates quality of outcome and survival information resulting in QALY gained. The following formula was used to calculate QALY gained (Abrams, Hnath Chisolm, & McArdle, 2002; Hnath Chisolm & Abrams, 2008):

$$\text{Cost per QALY gained} = \text{Cost of treatment} \div (\text{Obtained benefit [derived from THI]} \times \text{life expectancy [from actuarial table]})$$

Newman and Sandridge (2012b) found that both sound therapy methods yielded significant and equivalent benefit to the patient (between baseline evaluation and at 6 months post-treatment), based on serial administrations of the THI; however, CEA and CUA suggested that the SGs were a more cost-effective alternative, especially for those patients with greater perceived initial tinnitus handicap. For example, the

Table 22-13. Formulas Used to Calculate the Cost Effectiveness of Sound Generators (SG) Versus Neuromonics Tinnitus Treatment (NTT)

Cost Calculation: Incremental Cost of Treatment (IC)		
Variables	Total SG cost (i.e., the cost of the devices and follow-up services)	TC ₁
	Total NTT cost (i.e., the cost of the devices and follow-up services)	TC ₂
Formula	IC = TC ₁ – TC ₂	

Effectiveness Calculation: Incremental Effectiveness of Treatment (IE)

Incremental effectiveness of treatment (i.e., the benefit of therapy as determined by the difference in Tinnitus Handicap Inventory (THI) scores pre- and postfitting)

Variables	Total effectiveness of SG (i.e., benefit of SG based on improvements in THI scores)	E ₁
	Total effectiveness of NTT (i.e., benefit for NTT based on improvements in THI scores)	E ₂
Formula	IE = E ₁ – E ₂	

Cost Effectiveness Calculation: Cost Effectiveness Ratio (C/E)

Variables	Incremental cost of treatment (see above)	IC
	Incremental effectiveness of treatment (see above)	IE
Formula	C/E = IC ÷ IE	

Source: Adapted from Gold, Siegel, Russell, and Weinstein (1996); Newman and Sandridge (2012b).

cost per unit of improvement (treatment utility) on the THI was \$74 per point for the SGs in comparison to \$189 per point for NTT. Moreover, the CUA revealed that the overall SG treatment cost \$604 per QALY gained, while the NTT cost \$1,771 per QALY gained. In short, the SGs appear to be just as effective as NTT, while being much less expensive. (*Editors' note:* For further information about health-related quality of life assessment, the reader is referred to Chapter 7 of this text.)

Other studies have employed econometrics to evaluate the costs associated with various combinations of tinnitus treatment options, clinical pathways (Cima et al., 2012; Maes et al., 2013; Stockdale et al., 2017) and patient satisfaction (Goldstein et al., 2015). The aforementioned studies provide insight into overall health care costs of tinnitus intervention and outcome; however, economic analyses cannot be used in isolation when making clinical decisions including the recommendation of specific treatment options including the selection of specific sound therapy devices and modes of sound delivery (e.g., tinnitus relief apps, ear-level combination units). That is, a

number of patient-focused factors must be considered when developing best clinical practices including such variables as: the patient's lifestyle; manual dexterity; acceptability of the auditory stimulus cosmetics; daily dosage (hours per day of sound therapy usage recommendations); and personal finances. Audiologists must begin to consider the application of economic valuation of intervention strategies an essential part of clinical practice and decision making. In view of the current economic environment, clinicians must be able to justify their recommendations about the value of one treatment option over another in relation to observed benefit.

Conclusion

Tinnitus is a distressing symptom that affects the HRQoL for many individuals. As audiologists, we have an obligation to help these patients overcome the psychosocial consequences of tinnitus impairment. Of course, not all patients require intensive

audiologic management. Many patients will benefit considerably from minimal counseling about hearing, hearing loss, hearing loss prevention, and tinnitus. For those patients requiring further treatment, the care path described in this chapter provides a best-practices model to guide the audiologist through the management process. The sequencing of strategies provides differing levels of treatment to promote tinnitus relief for a wide variety of patients with tinnitus. In the authors' interprofessional collaborative TMC model, they are fortunate to work hand in hand with a variety of providers, including otolaryngologists, psychologists, neurologists, physical therapists, and dentists. It may be necessary for some audiologists to develop working relationships with other health care practitioners to make appropriate referrals as needed.

Regardless of work setting and available resources, the main contributions an audiologist can make when providing audiologic rehabilitation services to patients with tinnitus are to listen actively to their concerns, offer evidence-based interventions targeted at improving HRQoL, maintain patient-centered and ongoing counseling, monitor outcomes, and provide realistic expectations. At the same time, it is most imperative to offer patients hope, reassurance, and encouragement.

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Appendix 25-A

SOUND THERAPY OPTION PROFILE (STOP)

Name: _____ Date: _____

CCF #: _____ Audiologist: _____

Our goal is to help you find a method or combination of methods that will provide you with relief from your tinnitus. Treatments provided by our audiologists may include sound therapy using assistive devices, hearing aids, ear-level sound generators, and/or an acoustic desensitization protocol using a form of music therapy. Other options may include behavioral modification therapy provided by our Tinnitus Management Clinic team members in the Department of Psychology. To reach our goal of providing you with tinnitus relief, it is important that we understand your perceived tinnitus and hearing problems, your personal preferences, and your expectations. By having a better understanding of your needs, we can use our expertise to recommend the form of sound therapy most appropriate for you. By working together, we will find the best tinnitus treatment option for you.

Please complete the following questions. Be as honest as possible. Be as precise as possible. Thank you.

1. How much does your tinnitus affect your overall quality-of-life? Mark an X on the line.

Not very much _____ *Very much*

2. How important is it for you to hear better? Mark an X on the line.

Not very important _____ *Very important*

3. How motivated are you to use some form of sound therapy to help provide tinnitus relief?
Mark an X on the line.

Not very motivated _____ *Very motivated*

4. Are you willing to use sound therapy:

only at those times when your tinnitus is bothering you? _____ Yes _____ No

at least 2–3 hours per day for at least six months? _____ Yes _____ No

at least 6–8 hours per day possibly up to 12 to 18 months? _____ Yes _____ No

5. Do you expect the recommended sound therapy to be effective in providing you relief from tinnitus?
Mark an X on the line.

I expect sound therapy to:

Not be helpful at all _____ *Provide a great deal of relief*

6. What is your most important consideration regarding sound therapy treatment? Rank the following factors with 1 as the most important and 4 as the least important. Place an X on the line if the item has no importance to you at all.

___ Improved hearing
 ___ Tinnitus relief
 ___ Improved hearing along with tinnitus relief
 ___ Cost of treatment

7. Answer this question ONLY after you have listened to the demonstration sounds. How acceptable did you find the each one of the following sounds?

Nature sounds (e.g., waterfall, surf, wind, etc.)

Not very acceptable _____ *Very acceptable*

Gentle white noise

Not very acceptable _____ *Very acceptable*

Music

Not very acceptable _____ *Very acceptable*

8. Would you be willing to pursue any of the following forms of treatment?

Wearable ear-level device for one or both ears _____ Yes _____ No

Wearable device that looks like an iPod or MP3 player _____ Yes _____ No

Assistive sound generating device (e.g., bedside masker, masking tapes/CDs, etc.) _____ Yes _____ No

Psychological forms of treatment (e.g., additional counseling, cognitive behavioral modification training, biofeedback) _____ Yes _____ No

9. How confident do you feel that you will be successful in using some form of sound therapy in the treatment of your tinnitus?

Not very confident _____ *Very confident*

10. In the past, have you tried any of the following forms of tinnitus treatment:

Wearable (ear level) sound generators _____ Yes _____ No

Assistive (tabletop) sound-generating device _____ Yes _____ No

Medical (medications, sleep therapy, surgery, other medical intervention) _____ Yes _____ No

Psychological _____ Yes _____ No

Other form/s of treatment _____

11. There is a wide range of available sound therapy options that may be used to help provide relief from your tinnitus. Devices are typically not covered by insurance. The selection of a particular form of sound therapy depends on a variety of factors including the type of device preferred (for example, assistive device versus customized wearable device), level and sophistication of device technology (for example, hearing aid versus personal music players), length of time required for treatment (for example, six months versus 1 year), and personal finances. This information is helpful for us to select the most appropriate hearing device technology based on your budget. The costs for Categories B through D include the devices and related audiology appointments over the first year. Please check the cost category that represents the maximum amount you are willing to spend for your tinnitus treatment.

- | | |
|--|-----------------------------------|
| ___ Category A
Assistive sound generating devices | Cost is between \$XXX and \$XXXX |
| ___ Category B
Noncustom wearable sound generators | Cost is between \$XXXX and \$XXXX |
| ___ Category C
Custom wearable sound generators | Cost is between \$XXXX and \$XXXX |
| ___ Category D
Custom hearing aids
Custom combination devices (sound generator and hearing aid) | Cost is between \$XXXX and \$XXXX |

Source: Adapted from Newman, C. W., & Sandridge, S. A. (2006). Sound Therapy Option Profile (STOP): A tool for selecting devices used in tinnitus treatment. *Audiology Online*, Article 1687. Available via the Articles Archive on <http://www.audiologyonline.com>. Direct access URL located at http://www.audiologyonline.com/articles/article_detail.asp?article_id=1687

Part IV

EXPANDING THE SCOPE OF AUDIOLOGIC REHABILITATION: SPECIAL ISSUES

23

E-HEALTH IN ADULT AUDIOLOGIC REHABILITATION

Alessia Paglialonga

Words such as *e-health* (electronic health) and *m-health* (mobile health) have gradually become part of our daily vocabularies. An increasing number of novel digital solutions and services is being integrated into clinical practice and the daily life of patients and citizens—and the area of audiology rehabilitation is no exception. As a general introduction to these themes, in the first part of this chapter the general definitions of e-health and m-health and the core characteristics that make them unique compared to earlier solutions are presented. Also, the potential benefits and challenges of these novel solutions are discussed, as well as the multifaceted concept of quality, that is imperative for use with these solutions in the clinic. In the second “part of this chapter, a detailed insight into the latest trends in e-health in the area of audiology rehabilitation is presented and the available evidence discussed, as related to the various platforms for service delivery (offline, internet-based, and mobile-based platforms) and the several solutions and services available throughout the client/patient journey (education and counseling; screening and assessment; and hearing rehabilitation).

E-health and M-health: Definitions and Core Characteristics

The term *e-health* was introduced in the 1990s in relation to other e-words used on the internet in health

care (reviewed by Eysenbach, 2001). Della Mea (2001) described e-health as an umbrella term that incorporates telehealth, telemedicine, teleaudiology, and other fields of medical informatics. The World Health Organization (WHO) introduced two definitions that describe e-health as broad in scope and are able to capture the various aspects of this new area, including the newest ones (World Health Organization, 2005, 2016). The 2005 E-health Resolution (World Health Organization, 2005) defined e-health as the use of information and communication technologies (ICT) locally and at a distance to support health and health-related fields, including health care services, surveillance, and health promotion. Therefore, it is clear that e-health is a broad term and includes not only the concept of remote services (telehealth, telemedicine), but also local use of ICT such as personal computer (PC)-based applications or digital video disks (DVDs). The 2005 document by the WHO emphasized the enormous potential of e-health for the development of public health. It is reported that e-health can reinforce fundamental human rights by improving equity, solidarity, quality of life, and quality of care. In 2016, the Status of e-health in the WHO European Region Fact Sheet introduced new perspectives on e-health scope and services (World Health Organization, 2016). This document described e-health as the use of electronic means to deliver information, resources and services related to health. According to this document, e-health covers many domains, including electronic health records, m-health, and health analytics, among

others and, as such, it can enable provision of more services, to a wider population, and in a personalized manner.

The area of m-health, in particular, is gaining momentum. A general definition of m-health was introduced by the World Bank in 2012, i.e., the use of mobile technology for health care (Qiang et al., 2012). As such, m-health encompasses the use of mobile and wearable devices to deliver health-related information, to access or analyze data, to provide clinical services, or to support health care delivery in clinical as well as in nonclinical settings. Ozdalga, Ozdalga, and Ahuja (2012) described m-health as a novel facilitator to address key health care challenges such as access to care, quality of services, affordability of technology, or matching of resources. The U.S. Food and Drug Administration (FDA) (Food and Drug Administration, 2015) in the Digital Health Innovation Action Plan highlighted the impact of m-health and emphasized that mobile applications (apps) and devices can help people manage their own health, promote healthy living, and provide access to useful information when and where needed. The National Broadband Plan by the U.S. Federal Communications Commission (Federal Communications Commission, 2010) highlighted that m-health uses mobile networks and devices in supporting e-care, leveraging health-focused applications on general-purpose tools such as smartphones or tablets to drive active health participation by consumers and clinicians. Compared to earlier e-health solutions, m-health devices have unique features. Specifically, the core characteristics of m-health have been described in terms of: (a) rapid propagation into populations; (b) widespread availability of apps; (c) wireless broadband access to the internet, communication speed, and connectivity; and (d) direct connection with individuals due to increasing capability of locating, measuring data, monitoring function, and communicating with others (Davis, DiClemente, & Prietula, 2016). Increasing smartphone capabilities, functionality, and sensor integration, along with a widespread use of consumer devices, opened new possibilities for health and well-being. Data recorded by multiple sensors can be combined, for example, with behavioral and clinical data and can be used to address and monitor the health status and well-being of an individual (Cornet & Holden, 2018). Due to ubiquitous use of personal mobile devices (e.g., smartphones and tablets) and the related increase in popularity of m-health apps, the area of m-health has grown rapidly (IMS Institute for Healthcare Informatics, 2013; Statista Portal, 2016). Recently, it has been estimated that about 325,000 health apps are

available on the leading app stores overall (Research2 guidance, 2017).

The range of uses of e-health is large in several medical specialties as well as for health management and disease prevention. Evidence in the area of audiology is growing. Recently, the term *eAudiology* has been proposed as a concept that encompasses technologies and services for remote provision of audiology care at each stage along the patient journey, including screening, assessment, coaching, adjustments, monitoring, assistance, rehabilitation, and aftercare (Montano et al., 2018). Throughout this chapter, the term e-health is used as broader in scope than remote delivery of health services, according to the above WHO definitions (World Health Organization, 2005, 2016). Several e-health-enabled services have been introduced for audiology rehabilitation throughout the years, and there is growing evidence that these services can deliver significant value for patients as they can help them overcome challenges of distance and access to care (e.g., Bright & Pallawela, 2016; Gladden, Beck, & Chandler, 2015; Paglialonga et al., 2018a; Montano et al., 2018; Paglialonga, Pinciroli, & Tognola, 2015; Saunders & Chisolm, 2015; Swanepoel & Hall, 2010; Tao et al., 2018). As this is a growing field, however, it is characterized by a number of potential benefits for the users and the health care systems along with several possible challenges and risks, as outlined in the following section.

Benefits and Challenges of E-health: The Complex Nature of Quality

The area of e-health in audiology is dynamic and shows promise. Yet, it is an emerging area and, therefore, the actual benefits and potential challenges of introducing these novel solutions into the clinic are still a matter of debate. In fact, there is ample interest about the benefits and challenges of e-health in general.

As said above, e-health apps have been seen since the beginning as a possible way to support health systems and improve access to care, quality, and cost efficiency to enable provision of high-quality, personalized services to the widest possible population, including those living in underserved communities (Ozdalga et al., 2012; World Health Organization, 2005). In fact, it is known that socioeconomic context, educational level, age, gender, and ethnicity can strongly influence access to high-quality care delivered through conventional clinic-based models. Moreover, low income and education levels are typically associated with reduced capability to acquire

and understand health information and to prevent or manage pathologies (Nutbeam, 2008). In this context, digital technology can support health care delivery and help health workers, caregivers, and patients living in resource-limited settings to access higher-level medical knowledge. Services delivered remotely, either on conventional or on mobile platforms, are potentially a valuable approach to limit health inequalities as they can enable access to care and reduce intervention time and cost for vulnerable patients who live in rural regions (Akter & Ray, 2010; Asamoah-Odei et al., 2012).

The potential benefits of e-health for the individual user (including patients, significant others, citizens, and professionals) have been widely studied. For example, digital technologies can open novel ways to address health and well-being that can translate into preventive behaviors and behavior change (Zhao, Freeman, & Li, 2016). Mobile applications, for example, can support management of chronic conditions and risk factors through instant tailored feedback on the smartphone, allowing detailed patterns to emerge in the outcomes of interest (Scherer et al., 2017). Tailored use of m-health can promote patient engagement with treatment, which includes behavioral, affective, and cognitive components that can contribute to maximize treatment outcome. More generally, significant benefits of m-health in patient empowerment, person-centered care, and effective patient-clinician relationships have been identified (Kim & Lee, 2017; Paglialonga et al., 2019a; Paglialonga et al., 2019b).

In the area of audiology, these novel technologies promise to pave the way to more effective adult hearing care and might lead to improved patient outcomes and satisfaction by bringing technology closer to patients. As outlined later in this chapter, an increasing number of functions and services can now be implemented by using modern platforms, e.g., data collection, teleconsultation, remote monitoring, remote fitting, delivery of educational content, and personalized programs for auditory and cognitive training (Paglialonga et al., 2018a; Tao et al., 2018).

Along with the several advantages of introducing digital technology into the clinic, the ongoing digital transformation in health care can also bring along specific challenges and risks that need to be understood and, whenever possible, addressed. A prominent aspect associated with the most recent developments in e-health and m-health is related to

clinical evidence. Clinical evidence, in turn, encompasses diverse elements that complement each other. First of all, for digital solutions to be effective in the clinic, adherence to available guidelines and recommendations is essential. This needs to be taken into account throughout the design and development process, and needs to be verified by the clinician interested in adopting digital technologies into the clinic. Similarly, it is necessary to address the degree of scientific verification and clinical validation as available in the professional literature and related databases (e.g., EMBASE, MEDLINE, Global Health, Web of Science, CINAHL, and mHealth Evidence). However, in the area of m-health apps, the evolution of the market is so rapid that apps are typically put on the market well before any published article is available on scientific journals. As a result, the newest health apps, even if scientifically sound and reliable, might be penalized due to insufficient evidence. Therefore, specific criteria are needed for mobile apps evaluation and adoption (Aungst et al., 2014; Paglialonga, Lugo, & Santoro, 2018b). Potential app users cannot rely solely on the information reported on vendor markets due to the huge number of applications and to the inherently fragmented nature of the information provided. Moreover, app information as reported on the app stores' websites is at the discretion of developers so it is not necessarily a reliable indication of quality and usefulness. In this context, potential app users (patients, clinicians, medical students, and other consumers) may have difficulty in keeping an overview of all the health apps that are available. Some recently developed methods for automated app classification and characterization promise to compensate, at least in part, for the effects of app overload on the market (Albrecht, Hillebrand, & von Jan, 2018; Paglialonga, Schiavo, & Caiani, 2018c; Paglialonga et al., 2017a; Paglialonga et al., 2017b). These novel automated methods are able to extract information from the web and, as such, they may become the basis for user-oriented support tools able to highlight meaningful knowledge about health apps, especially if they are used to extract information from reliable sources.

Considering the difficulties associated with app discovery and quality verification on the vendor markets, specific databases and web resources have been created to index, comment, and review health-related apps, such as the U. K. National Health Service (NHS) Apps Library,¹ the Organisation for the Review of Care and Health Applications (ORCHA),²

¹<https://www.nhs.uk/apps-library/>

²<https://www.orchaco.uk>

the RANKED Health Project,³ or MyHealthApps⁴ by PatientView. Trustworthy resources about m-health apps can be helpful to support potential users in identifying the most promising means to answer specific needs. However, it is of note that these resources, although more reliable and accurate than the vendor markets, are not exempt from limitations. As the review process is inherently subjective, there is potential for bias. Moreover, there is an inherent delay from app release to assessment as reviews can take a significant amount of time, effort, and resources to be effective, so the app reviewed may receive updates and substantial revisions before a review is released.

In addition to information from the scientific literature and web resources, potential users can also assess aspects related to quality. These factors include the credentials of who developed the e-health or m-health tool, the sponsors and their reliability, the scientific methodology, or the documented involvement of the relevant stakeholders in the development process (i.e., scientists and health care professionals as well as the target user groups according to the principles of participatory design) (Clemensen et al., 2017).

Other important challenges of digital health are related to privacy, data protection, and security. These issues are especially important whenever services are delivered remotely over the internet or through smartphone apps. However, despite its importance, data protection is frequently overlooked in the area of digital health. Whereas existing electronic systems and servers in health care operate behind secure firewalls, in novel digital solutions and mobile health apps, data protection regulations are rarely fulfilled. Gaps in compliance with data protection principles in accredited health apps have been reported. For example, a study on 79 health apps certified as clinically safe and trustworthy by the U.K. NHS Health Apps Library showed that personal information stored locally was not encrypted and that identifying information sent over the internet was not encrypted in two-thirds of apps. Some apps (20%) did not have a privacy policy and, among the apps having some form of privacy policy, the majority (78%) transmitted information without describing the nature of personal information included in transmissions, and two apps put the users at risk of data theft (Huckvale et al., 2015). The issue of data protection, so crucial in health, is still largely unresolved in m-health and is challenging, as it is difficult to identify the appropri-

ate basis for data processing and the requirements of explicit, informed consent. In addition, complexity of information systems and system networks make the exercise of data subjects' rights more difficult (Mantovani & Quinn, 2014). Access to patient data through mobile devices would require careful consideration of the ethical, security, and governance aspects of data handling (Charani et al., 2014). Addressing data protection is part of the broader concept of trustworthiness, which is just one of the main reported concerns when dealing with digital health technology. In general, aspects related to trust can be difficult to address so some frameworks have been introduced to try to assess trustworthiness in a comprehensive way by evaluating, for example, elements of transparency, data management and reuse, data protection and privacy, purpose, reliability of content, and imprint (Albrecht, Noll, & von Jan, 2014; Lewis, 2013).

All of the abovementioned aspects (i.e., clinical evidence, quality of developers, scientific soundness, involvement of professionals, participatory design methodology, data protection, transparency, and reliability of content) contribute to the broad, complex concept of quality in digital health. However, for a comprehensive evaluation of quality, one more component needs to be addressed: the users' perspective. Examples of user-oriented elements of quality are usability, quality of experience, functional design and aesthetics, ease of use, perceived quality of content and function, and perceived effectiveness (Brown et al., 2013; Lemon et al., 2018; Martínez-Pérez et al., 2013; Moumane, Idri, & Abran, 2016; Paglialonga et al., 2018b).

Overall, it is clear that the assessment of quality in e-health is an open debate, as the concept of quality is complex and subjective in nature. It is difficult to define standards for digital health assessment as it is difficult to identify the core components of quality as well as appropriate measures to assess them. Generally, there is a need for comprehensive recommendations on how to identify, characterize, and evaluate e-health and m-health solutions (Larson, 2018; Paglialonga et al., 2018b). Related to this, in the heterogeneity of hearing data and devices in the area of e-health, there is a need to move toward interoperability and standardization of data types, collection, exchange with communication protocols, handling, storage, and interpretation. Interoperability refers to health information systems that work together within and across organizations. Standardization is

³<http://www.rankedhealth.com>

⁴<http://myhealthapps.net>

the process of developing and implementing technical standards and, as such, it is the necessary basis for securing interoperability, compatibility, quality, safety, and repeatability (Laplante-Lévesque et al., 2016). Research in this direction is needed and important as it can help raise the quality of solutions available to clinicians.

E-health in Audiologic Rehabilitation

In the field of audiology, use of e-health was introduced as early as in 1994 by Cherry and Rubinstein, who suggested remote follow-up by telephone following face-to-face hearing aid (HA) fitting. Nowadays, the range of digital technologies for hearing is wide and several services are available for different target users and age groups, including children with hearing loss (HL) and their parents, adults fitted with HAs or implantable devices, older adults and their significant others, or tinnitus sufferers. This section outlines the latest trends in the area of adult audiologic rehabilitation, the evolution of platforms for service delivery, the various solutions available throughout the patient journey, and the research needs and open questions.

Evolution of E-health Platforms in Audiologic Rehabilitation

Since the earliest attempts to deliver remote audiologic services through the telephone, the area of e-health for hearing has evolved in a way that services can now be delivered over a variety of platforms. These platforms can be categorized into three broad categories: offline, internet-based, and mobile-based.

Offline Platforms

Offline platforms include telephone, DVDs, or PC-based programs that do not use the internet. The telephone was introduced early as a straightforward, accessible means to complement face-to-face clinical encounters. Examples range from clinician-initiated follow-up calls to address questions related to HA fitting (Cherry & Rubinstein, 1994) to telephone-based consultations and weekly discussions between clinicians and HA users as complements to a home-based educational program on hearing and HAs (Lundberg, Andersson, & Lunner, 2011). DVD- and PC-based applications have been used to deliver educational content or auditory and cognitive training programs

(e.g., Ferguson et al., 2016; Henshaw, McCormack, & Ferguson, 2015; Olson, Preminger, & Shinn, 2013).

DVD- and PC-based programs have represented an important advancement in audiology as they can be used to implement user-tailored exercises for auditory and cognitive training or to deliver personalized educational content. As such, these programs can enable clients to train or learn skills such as communication strategies and to cope with hearing-related problems in their home environment (Kramer et al., 2005; Sweetow, 2006). Even though DVD- and PC-based programs were introduced well before the advent of the internet, they are still widely used in audiology. The success of offline platforms is due to a combination of factors, such as full maturity of technology, availability of clinical evidence, and the fact that these offline platforms, differently than those relying on the internet, are easier to use and can function under less than ideal conditions (e.g., poor connectivity, underserved settings, or untrained personnel with low technical skills) (Paglialonga et al., 2018a).

Internet-Based Platforms

The broad category of internet-based platforms includes any application and service delivered over the internet (except the native mobile solutions that are described in the next section). Internet-based platforms have increased in popularity due to the growing rate of internet usage. Interestingly, high prevalence of internet use has also been reported in community-living older adults with HL (Gonsalves & Pichora-Fuller, 2008), with elderly people with HL using the internet more than those with normal hearing (Henshaw et al., 2012; Thorén et al., 2013).

Various audiological services have been introduced on internet-based platforms. For example, web portals for hearing-related information or virtual communities (Rothpletz, Moore, & Preminger, 2016; Shoham & Heber 2012), rehabilitation programs and auditory training (Abrams, Bock, & Irey, 2015; Malmberg et al., 2015; Thorén et al., 2014), remote control of clinical equipment for HA fitting (Ferrari & Bernardes-Braga, 2009), and audio- and videoconferencing for teleconsultation (Pross, Bourne, & Cheung, 2016). Overall, compared to earlier offline platforms, the use of internet-based services in audiology has opened novel opportunities for effective dissemination of information on hearing and postfitting care, easier exchange among peers, and delivery of remote services. Complementing audiologic rehabilitation in the clinic with internet-based services has been shown to contribute to increased participation, easier data

collection, reduced time and cost, and improved patient outcomes (Malmberg et al., 2015; Thorén et al., 2014; Thorén et al., 2011).

Mobile-Based Platforms

This category includes m-health technologies; that is, solutions that make use of mobile and wearable devices for audiology rehabilitation. The m-health branch in audiology is relatively new but the interest among researchers and clinicians is growing rapidly, especially in the area of mobile apps (Bright & Pallawela, 2016; Clark & Swanepoel, 2014; Galster, 2012; Paglialonga, Pinciroli, & Tognola, 2019c; Paglialonga et al., 2015).

Thanks to their portable and instant nature, m-health technologies can enable on-demand information and data sharing. A variety of personalized solutions can be implemented to connect different hearing care actors such as patients, audiologists, and manufacturers. An increasing number of apps for hearing are made available on the market, providing a wide range of functions and services for the different target user groups throughout the whole journey: from professional and patient education to hearing screening, from remote counseling to auditory training, and from personal sound amplification to support tools for patients and their significant others (Paglialonga et al., 2019c).

Overview of E-health Services Along the Patient Journey

The area of e-health in audiology has grown not only in digital platforms as described above, but also in type of solutions that can support the prefitting, fitting, and postfitting phases of the HA adult patient journey (Paglialonga et al., 2018a). A wide range of services related to adult audiology rehabilitation have been developed and tested on digital platforms throughout the years albeit with varying degrees of clinical implementation and uptake. Overall, these services can be categorized into three broad areas: education and counseling; screening and assessment; and hearing rehabilitation (Paglialonga et al., 2019c; Paglialonga, Pinciroli, & Tognola, 2017c).

Education and Counseling

The area of education and counseling includes any function that delivers information to patients, significant others, audiologists, or health care profession-

als. As such, this area includes services that aim to improve general knowledge and awareness about hearing, HL, and HAs or cochlear implants (CIs); support tools for patients and significant others (e.g., educational material, counseling, motivational interviewing, or psychological interventions); peer-to-peer support groups and online discussions; and professional education resources.

Using e-health for patient education and counseling in the prefitting phase can support patients in their needs; increase awareness, motivation, and attitude to change behavior; and engage patients by having them share information about their choice of interventions and their needs and expectations. Recent literature shows that adult patients show growing interest in the internet as a means to improve knowledge and awareness about HL and hearing rehabilitation options. For example, Peddie and Kelly-Campbell (2017) interviewed a sample of adults and older adults with HL in New Zealand and showed that all but one had used the internet as a source of health-related information, and most of them had searched for hearing health information. Interestingly, all of the participants reported they had not discussed which websites to read with a hearing professional—typically, they read websites in the top 10 listed results in the search engine. It is therefore important that audiologists are aware that not only do patients feel the need for accessible knowledge about hearing, but they also tend to search for information on their own. Audiologists need to take this into account during clinical encounters and, if possible, recommend which resources to look at or discuss with their clients about their current knowledge and beliefs. To address the needs of people with HL, several online resources have been created and launched recently (e.g., information websites, virtual communities, and online forums for patients) (Ng et al., 2015; Shoham & Heber, 2012).

Counseling programs over the internet can be useful to bridge the gap between hearing screening and rehabilitation. For example, motivational interviewing over the internet has been proposed to promote movement in the stages of change and help seeking in people who failed an online hearing screening (Weineland et al., 2015). The study adapted the intervention to the internet format by delivering videos, written exercises, and texts in a fully automated way, and by including automatic feedback to the user. Fully automated counseling programs over the internet are currently under investigation and show promise as a possible means to influence the behavior of people with hearing impairment who

may be reluctant to seek help initially, but may accept person-centered online psychological approaches as a way to find motivation. Acceptance of such interventions may be higher than expected, including in the older age groups. For example, a study demonstrated acceptance of internet resources for hearing health care in a sample of older adults who failed a hearing screening (Rothpletz et al., 2016). The study showed that participants in general did not perceive themselves to be knowledgeable about how to manage their hearing problems, and seemed to believe that it was important to become informed about management strategies. Even though participants in the study sample were fairly diverse in age, education attainment, computer skills and experience, and personal access to the internet, overall they were generally accepting of computers and hearing health care websites. The study also pointed out the importance of website design (e.g., simple user interfaces, reminders) and short-term training to increase acceptance and ultimately use of internet delivery of hearing information among older adults. This stage (seeking information after failing hearing screening) can be particularly important for people with hearing problems to become aware of their needs and begin to self-manage their HL until they are ready to seek conventional clinical services.

The approaches to patient education through e-health in the prefitting phase are diverse. For example, Ferguson et al. (2015) introduced an educational program to improve knowledge of HAs and communication in first-time HA users. The program was based on the concept of reusable learning objects; i.e., short, highly visual chunks of interactive multimedia learning to support specific learning goals. A randomized controlled trial published in 2016 demonstrated the benefits of this program, delivered on PC, DVD, or the internet, to first-time HA users. Use of reusable learning objects led to significantly greater HA use for suboptimal users and, in general, to significantly better knowledge of practical and psychosocial issues and significantly better practical HA skills (Ferguson et al., 2016). Another example of online counseling in the prefitting phase comes from the study by Manchaiah et al. (2014), who introduced an online prefitting counseling program for individuals with hearing disability and their communication partners. The program was person centered and focused on the lived experiences of hearing disability with significant emphasis on patients' self-reflection regarding their journey and their interaction with communication partners. Such programs, tailored to the patient as enabled by digital technology, can be

used to support patients in their social and emotional needs, help them modify attitudes, and engage them by discussing their experiences and choices regarding intervention. Online resources for group audiologic rehabilitation have also been developed. For example, the Group Rehabilitation Online Utility Pack (GROUP), an online tool developed by Montano et al. (2013) to support hearing health care professionals to develop and offer group audiologic rehabilitation in their clinical practice. The tool includes the evidence for implementing group audiologic rehabilitation, content for preparation and planning, instructions for facilitating groups, and group activities.

Interestingly, psychological interventions such as cognitive behavior therapy and acceptance and commitment therapy have also been proposed for online delivery. For example, internet-based delivery of acceptance and commitment therapy to people with hearing impairment was also introduced recently as a potentially effective approach to psychological symptoms associated with hearing problems. Acceptance and commitment therapy is a form of cognitive behavior therapy that aims to help individuals accept their symptoms, feelings, and thoughts to live a fulfilling life. A pilot randomized controlled trial showed improvements in perceived psychosocial handicap and quality of life, and reduced depressive symptoms with internet-based acceptance and commitment therapy that combined text readings, mindfulness exercises, and interaction with therapists (Molander et al., 2018). Similarly, the positive effects of internet-based cognitive behavior therapy for tinnitus have been demonstrated. Recent randomized controlled trials showed that online cognitive behavior therapy was more effective on tinnitus-related distress and associated symptoms compared to weekly monitoring or moderated online discussion forums (Beukes et al., 2018; Kalle et al., 2018; Weise, Kleinstäuber, & Andersson, 2016).

As shown by the examples above, several studies have introduced services on offline and internet-based platforms in the area of education and counseling. On the other hand, only a few studies about educational services on mobile-based platforms can be found to date. Wong and Fung (2015) reviewed 75 educational apps relating to otolaryngology-head and neck surgery and classified them into four main categories: (a) education; i.e., educational resources for medical students, residents, physicians, and health care professionals; (b) journal/conference; i.e., apps containing abstracts and articles from journals and apps designed for conferences and meetings; (c) clinical; that is, apps to support clinical activities

and patient-doctor interactions; and (d) patients; that is, patient-oriented information. Broader reviews of the market of m-health apps for hearing showed that in the area of education and counseling, the available apps offered services such as lay audience-modified scientific information; service finder tools; educational support tools for clinical encounters; professional resources (interactive anatomy atlases, videos of clinical procedures, reference guides, continuing education programs and question banks); awareness tools (audio simulations of HL and hearing amplification solutions); and tools for HL prevention and noise monitoring (Paglialonga et al., 2015; Paglialonga et al., 2017c).

In addition to the examples in the prefitting phase outlined above, there is also increasing evidence about use of e-health for education and counseling in the fitting and postfitting phases of the patient journey. In fact, the educational component is one of the core elements of hearing rehabilitation whereby HA and CI fitting can be complemented by counseling and tailored educational programs for overall better benefit and improved patient outcomes. Improved knowledge of HL and communication strategies can help increase social interaction and reduce hearing disability (Ferguson & Henshaw, 2015a; Kramer et al., 2005; Lundberg et al., 2011; Sweetow, 2006; Thorén et al., 2011). Audiologic counseling on e-health platforms can be a powerful enabler of successful intervention as even simple strategies such as daily emails between patients and audiologists can improve understanding of day-to-day experiences. This can facilitate more timely responses compared to face-to-face encounters (Laplante-Lévesque, Pichora-Fuller, & Gagné, 2006). Online discussion groups and interaction with peers have also been investigated as possible means to complement HA rehabilitation. For example, randomized controlled trials demonstrated that a comprehensive online intervention program including self-studies, professional coaching, HA fitting, communication strategies, and peer-to-peer discussions can significantly reduce perceived hearing handicap in experienced HA users (Thorén et al., 2011; Thorén et al., 2014). Other investigators suggested similar programs, such as a combination of reading material, home training assignments, online and telephone interaction with an audiologist, and participation in a discussion forum (Malmberg et al., 2015); a support system comprising messaging tools, individual client tasks, agendas for next face-to-face visits, memo possibility, and a library with educational videos (Brännström et al., 2015); and a personalized online rehabilitation system for HA users able

to provide content based on each user's own individual needs and guide the user throughout the entire rehabilitation process (Denev et al., 2008).

In general, internet-based delivery of educational and counseling programs can be advantageous in data collection, participant access to content, personalization of programs, feedback, and clinician participation in terms of personnel, training, time, and cost. It can also translate into improved patient knowledge and awareness, increased motivation, and stronger self-management capabilities compared to conventional delivery in the clinic. However, the clinician willing to use internet-based services for education and counseling needs to be aware that these tools can be effective to the extent that individuals are in a position to use them appropriately.

Factors such as digital skills, internet competency, literacy, and quality of online material play an important role in determining the uptake, adherence, and benefits of e-health interventions. For example, a randomized controlled trial on about 200 first-time HA users showed that internet competency predicted practical knowledge about HAs and handling skills, controlling for factors such as age, hearing sensitivity, educational status, and gender. Better internet competency led the study participants to use reusable learning objects fewer times because of improved self-efficacy (Maidment et al., 2016). Previous knowledge of the e-health tools is also an important factor related to acceptance of remote delivery of education and counseling services (Eikelboom & Atlas, 2005) or, in the absence of previous knowledge, brief training to use online resources can help (Rothpletz et al., 2016). Digital skills and proficiency with the internet can be key to rehabilitation as effective use of online educational tools can, in turn, enable more effective management of HL. Literacy and health literacy also determine the way individuals use and understand health information from the internet (Norman & Skinner, 2006). For example, Laplante-Levesque et al. (2012) and Laplante-Lévesque and Thorén (2015) demonstrated that between 9 and 14 years of education may be required to be able to understand the online information available. Low literacy can be a barrier to adoption of internet-based educational tools, especially for older adults and for people in disadvantaged socioeconomic groups (Carroll et al., 2017; Joseph et al., 2016; Kontos et al., 2014; Moore, Rothpletz, & Preminger, 2015). The structure of the websites and navigation features are also important. For example, Thorén, Pedersen, and Jørnæs (2016) found that users of an online rehabilitation system reported difficulty navigating the system and com-

pleting the designed hands-on tasks. Possible solutions to improve usability of educational websites may be simplification of website structure and navigation options; limitation of the hierarchical structure; and establishment of an effective learning flow that will allow users to become aware easily of their progress. In addition to complex language and structure of online resources, other barriers to effective online delivery of information may include outdated or poorly maintained websites or disabilities such as visual or cognitive impairment that may make it difficult for patients to read and understand the information provided.

Hence, for effective user involvement and empowerment, it is important that e-health services developed for the patients meet the expected skills and literacy of the target groups and that the information provided can be understood easily. Collecting users' feedback is important for improving existing systems, whereas involving users in the initial design process should be considered. Participatory design of e-health-enabled educational resources can be key to successful rehabilitation as patients can contribute to original and effective innovations of health technology (Kanstrup et al., 2017a; Kanstrup et al., 2017b). Recognizing the role of the end users and audiologists through codesign and involving them across all stages of development can help create educational content that is user friendly, easy to understand, and accurate, ensuring the end product is fully aligned to the users' needs (Ferguson et al., 2018).

Screening and Assessment

This area includes e-health services for pure-tone and speech audiometry, speech in noise testing, tinnitus measurement, and self-reported measures. As such, services in this area may support the prefitting, fitting, and postfitting phases of the patient journey. Technologies in this area range from the earlier examples of remotely connected clinical equipment to more recently introduced internet-based systems and mobile apps for remote testing.

Remote hearing testing is one of the pillars of teleaudiology. Several studies have introduced remote testing as part of the provision of rehabilitation services, such as in underserved areas or in community settings (Tao et al., 2018). Implementing a remote testing session requires careful evaluation of the equipment needed, the personnel involved, and the methodology used (Pearce, Ching, & Dillon, 2009). Moreover, the potential barriers to distance service provisions in the various settings need to be

addressed. For example, it is important to evaluate the capacity and acceptance of the staff involved; the availability of specialized equipment enabling remote data transfer and control; the quality and reliability of the telecommunication network and infrastructures; issues related to reimbursement; or barriers related to acceptance, age, and costs (Coco, Champlin, & Eikelboom, 2016; Saunders & Chisolm, 2015; Tao et al., 2018). Studies have shown that, if properly implemented, remote testing (and, more generally, teleaudiology) is feasible and effective (Blamey, Blamey, & Saunders, 2015). For example, the U.S. Veterans Health Administration developed a wide teleaudiology program to provide HA services to veterans residing in rural settings. The program is based on synchronous, real-time video conferencing between an audiologist on one end and the patient and audiology technician on the other end. Remote diagnostics is based on a comprehensive audiology assessment that may include video-otoscopy, air conduction and/or bone conduction thresholds, otoacoustic emissions, immittance, and speech testing. The program was shown to be as effective as in-person encounters, as measured by the International Outcome Inventory for Hearing Aids (IOI-HA) (Jacobs & Saunders, 2014; Pross et al., 2016). Diagnostics is one of the components of teleaudiology; the others being education and counseling, HA fitting and assessment, and perceptual training, whose peculiar features, benefits, and barriers will be discussed in more detail later in this chapter, in the section "Hearing Rehabilitation."

In the past few years, the interest around remote hearing testing has grown significantly. This is related not only to the opportunity to test in remote areas, but also to the opportunity to test people outside the clinic to improve access to care. Hearing testing methods for hearing screening and assessment have been introduced on offline, internet-based, and mobile-based platforms. Interestingly, among the several mobile apps introduced in the area of audiology, many apps provide services in the area of screening and assessment. Examples include, but are not limited to: pure-tone tests; speech audiometry; speech in noise tests; video otoscopy; self-reported measures of hearing handicap, disability, and outcomes; and measures of tinnitus pitch, loudness, and its effects (Bright & Pallawela, 2016; Kalle et al., 2018; Paglialonga et al., 2015; Paglialonga et al., 2017c).

The introduction of user-operated methods to measure pure-tone thresholds has represented the basis for the implementation of automated pure-tone hearing tests. A systematic review of automated pure-tone threshold audiometry showed that it can be an

accurate measure of hearing threshold, with average differences between manual and automated air conduction audiometry being comparable with test-retest differences for manual and for automated audiometry (Mahomed et al., 2013). Several smartphone- and tablet-based pure-tone hearing tests for screening or assessment have been developed recently (Abu-Ghanem et al., 2016; Larrosa et al., 2015; Livshitz et al., 2017; Thompson et al., 2015). User-operated methods for tinnitus matching have also been introduced on offline platforms (e.g., the computer-automated tinnitus assessment tool by Henry et al., 2006), internet-based platforms (e.g., the web-based pitch matching protocols introduced by Mahboubi et al., 2012), and mobile-based platforms (e.g., the iPod®-based tinnitus pitch and loudness matching by Wunderlich et al., 2015).

This is an emerging area and further research is still needed to assess the accuracy of mobile apps for ear and hearing assessment (Bright & Pallawela, 2016). Specifically, efforts are still needed in the area of clinical validation, calibration, and use in nonclinical settings such as noisy environments or underserved contexts. For example, to limit the effects of noisy environment on hearing screening, Na et al. (2014) developed a system that combines a smartphone, a custom app, consumer earphones, and an algorithm that corrects hearing thresholds to compensate for the environmental noise, as measured by the smartphone microphone. Other studies validated a smartphone audiometry tool with calibrated low-cost headphones for use in primary health care clinic environment and underserved primary health care contexts, and demonstrated accuracy and time-efficiency (Louw et al., 2017; Sandström et al., 2016). The accuracy of hearing testing apps using consumer earphones in different settings (soundproof and waiting room settings) was also assessed as a means to explore the possible application in developing countries and underserved contexts (Peer & Fagan, 2015). Possible improvements in pure-tone testing by using consumer technology have also been suggested. For example, Corry, Sanders, and Searchfield (2017) conducted a preliminary evaluation of an iPad® app with consumer earphones. Good test-retest reliability was observed but the study showed that the app with Apple® earbud earphones measured different hearing thresholds than a clinical audiometer. To try to address the issue of calibration, procedures for biological calibration of reference sound levels for mobile devices with bundled headphones have been introduced, specifically the reference sound levels are determined in relation to the subjects' normal-hearing threshold. Comparison with

audiologist-assisted pure-tone audiometry in a sound booth showed that hearing screening using biological calibration reached 98% sensitivity and 79% specificity, suggesting potential application of biological calibration in pure-tone testing for hearing screening and monitoring (Masalski, Grysiński, & Kręcicki, 2018).

An important milestone in the area of e-health in audiology has been the introduction of automated speech tests for adults and older adults. For example, a telephone-based speech in noise screening test based on digits was originally introduced in the Netherlands and then translated to several languages. The test uses digit triplets in speech-shaped noise and an automated adaptive procedure controlled by a remote computer and delivered by telephone (Smits, Kapteyn, & Houtgast, 2004). The test was used in nationwide screening campaigns in various countries, including the Netherlands (Smits & Houtgast, 2005), France (Jansen et al., 2010), and the United States (Watson et al., 2012), and was recently adapted for delivery over the internet and through the smartphone (Folmer et al., 2017; Potgieter, Swanepoel, & Smits, 2018). Similarly, other investigators introduced self-administered speech in noise tests for hearing screening and assessment in adults and older adults, such as the Speech Understanding in Noise (SUN), an automated vowel-consonant-vowel screening test on a portable, battery-operated device (Paglialonga, Tognola, & Grandori, 2014); an automated, internet-based speech-in-noise screening test based on multiple-choice word recognition (Molander et al., 2013); and online consonant-vowel-consonant speech perception tests based on either open-set or multiple-choice automated procedures (Blamey et al., 2015; Sheikh, Rashid, & Dreschler, 2018).

The advent of e-health has enabled novel ways to deliver self-reported questionnaires as an alternative to the conventional paper-and-pencil format to save resources and try to record more complete answers online compared to the paper ones. In principle, any self-reporting tool can be implemented on internet-based or mobile-based platforms. Thorén, Andersson, and Lunner (2012) compared online versus paper-and-pencil administration of the Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982), the IOI-HA (Cox & Alexander, 2009), the Satisfaction with Amplification in Daily Life (SADL) (Cox & Alexander, 2001), and the Hospital Anxiety and Depression Scale (HADS) (Snaith, 2003). The study showed that the scores remained consistent across administrations and formats for the IOI-HA, SADL, and HADS, whereas the HHIE scores were higher with the online format compared to paper-and-pencil

administration, indicating that participants reported more hearing handicap with the online format. Online administration can be a valuable opportunity to save resources. Also, it may lead to more complete answers when compared with conventional formats. However, audiologists and researchers willing to use online formats to deliver questionnaires need to be aware that there may be slight differences in patient self-reported hearing handicap compared to conventional formats, so consistent administration format across surveys and in follow-up visits is recommended (Thorén et al., 2012).

In the area of self-reported measures, Ecological Momentary Assessment (EMA) is regarded as a powerful tool to overcome the limitations of static retrospective self-reported methods, which can be vulnerable to reporting errors and recall biases. EMA, sometimes referred to as *ambulatory assessment* or *experience sampling*, is an umbrella term encompassing a group of techniques that allow the subjects to self-report on detailed information about their status, activity, and experience in real time and in their own natural environment (Timmer, Hickson, & Launer, 2018a). Therefore, EMA requires regular collection of information several times throughout daily life, typically in the form of a diary (Shiffman, Stone, & Hufford, 2008). Nowadays, mobile and wearable sensors technology pave the way to new approaches to EMA, enabling frequent self-assessment sessions and real-time monitoring of symptoms, feelings, activities, and behavior, thus making it easier to collect individual information that can inform the rehabilitation process in a way that can be person centered.

Galvez et al. (2012) examined the communication difficulties of HA users and demonstrated the viability of EMA delivered through personal digital assistants to address the experienced hearing problems, especially in challenging listening situations. The study participants also reported increased awareness regarding HL and HAs after wearing the personal digital assistants 12 hours a day for 2 weeks. Interestingly, this study shows an example of how mobile-based technology may enable collection of real-life outcome measures so that the within- and between-day variability in outcomes and associated factors can be addressed effectively. Similarly, Henry et al. (2012) used EMA on personal digital assistants in a group of participants with chronic tinnitus and demonstrated the feasibility of EMA in the study population. Moreover, the study participants reported positive feedback about the methodology and reached high compliance, at 90%. Use of digital technology for EMA in subjects with tinnitus is particularly powerful com-

pared to conventional retrospective questionnaires as the latter are not able to reflect daily fluctuations in tinnitus (Henry et al., 2012). In the attempt to increase further the ability of EMA to capture changes in real time, mobile apps have been developed to track tinnitus. For example, recent studies describe an app developed to measure tinnitus along several dimensions, including subjective perception and loudness, distress, emotional valence and arousal, stress and concentration levels, and general degree of symptoms compared to the user's worst experience (Probst et al., 2016; Schlee et al., 2016). The study by Probst et al. (2016) assessed emotion dynamics in about 300 tinnitus sufferers and showed that emotion dynamics are associated with tinnitus-related distress and with the time course of tinnitus loudness. The study by Schlee et al. (2016) involved approximately 850 patients and demonstrated the viability of the app as a safe instrument for longitudinal assessment of tinnitus perception in daily life.

Timmer, Hickson, and Launer (2017) demonstrated the feasibility, compliance, and construct validity of EMA delivered through a mobile app on adults with bilateral mild hearing impairment who did not own or wear HAs. Their approach included multiple surveys with questions about the listening situation and environmental acoustic characteristics, speech understanding, evaluation of listening effort, and the impact of hearing difficulties on everyday listening situations. The surveys were either time based or triggered by sound environment or by the subjects themselves. The study showed high participant compliance (i.e., 93%) and demonstrated that participants' rating of environmental loudness was in agreement with objective measures of the overall noisiness of the listening event. Moreover, the use of separate surveys for listening effort and self-reported speech understanding showed that the effect of listening effort on overall hearing performance was particularly evident for the older adults with mild hearing impairment. Successful use of the EMA for the measurement of HA outcomes in older adults has also been documented. For example, a pilot study involved older adults with mild hearing impairment with no previous amplification experience using a single-subject experimental design (1 week without HAs, 2 weeks with HAs, and 1 week without HAs) (Timmer et al., 2018a). Participants were given a mobile app, instructions, and brief training on how to use the smartphone and perform surveys. The study showed that participants reported significantly better speech understanding, less listening effort, and greater enjoyment of listening events when using HAs compared to baseline conditions. In

general, EMA can be useful to quantify self-reported aided benefit and can provide deeper insight into the individual response to amplification, providing audiologists with meaningful information about speech understanding as well as listening effort, hearing difficulties hampering communication, and enjoyment.

Mobile technologies can be a powerful means to support EMA in research and clinical practice. It is an emerging area in audiology but it has been tested widely in various fields of social and health sciences. Currently, EMA can be performed via smartphone apps and integrated monitoring sensors and can be used to combine qualitative and quantitative data. However, it is important to implement it correctly and maximize its quality and effectiveness by following best practice guidelines (reviewed by Timmer, Hickson, & Launer, 2018b). It is also important to find strategies that maximize patients' feedback, engagement, and compliance. In fact, a risk associated with EMA and, more generally, with the use of mobile tools for tracking patients, is a possible increase in drop-out rate compared to face-to-face or PC-based procedures (Paglialonga et al., 2019a). In summary, the use of digital technologies for self-testing shows potential to support increased awareness, earlier diagnosis, help-seeking behavior, and person-centered care. However, for these novel solutions to bring benefit for the patients, it is important to consider possible issues as outlined above and to minimize them whenever possible.

Hearing Rehabilitation

This area is related to the dimensions of sensory and cognitive treatment and includes services in a wide range, e.g., tools for HA or CI fitting and control, smartphone-based sound enhancement solutions, e-health enabled training programs, and tinnitus relief apps.

Remote delivery of HA fitting and follow-up procedures is a core component of teleaudiology. It includes physical management of devices and related components as well as sensory management consisting of fitting, programming, fine adjustments, and verification with real ear measurements (Bush et al., 2016; Ferrari & Bernardez-Braga, 2009; Pearce et al., 2009; Penteado et al., 2012; Tao et al., 2018). Various studies demonstrated that remote HA fitting consultations and procedures require about the same amount of time compared to face-to-face consultations (Campos & Ferrari, 2012; Reginato & Ferrari, 2014). For remote rear ear measurements, negligible differences were observed compared to face-to-face

measurements, within the variability for probe microphone measurements (Ferrari & Bernardez-Braga, 2009). Client satisfaction, as measured by the SADL questionnaire, was shown to be comparable for remote and face-to-face HA fittings (Penteado et al., 2014).

Overall, studies in the area of remote HA fitting and follow up procedures demonstrated the feasibility of delivering these services at a distance, enabling for better services in underserved context or remote areas. However, a recent systematic review of teleaudiology services for HA rehabilitation showed there is still limited recommendation for translation into clinical practice due to current lack of strong or even moderate quality of evidence (Tao et al., 2018). Moreover, the study pointed out that there are still some key procedures that remain unexplored and some potential issues in the implementation of remote HA fitting and follow-up. Potential challenges include ear impressions, troubleshooting the physical and acoustic aspects of earmolds (e.g., grinding, drilling, retubing), and physical management of HA fitting (e.g., technical problems, changing the microphone cover, inspecting the ear canal, choosing the size of the dome, verifying appropriate fitting and position of domes in the ears). In general, the experience of clients is also an important aspect that may influence the feasibility of remote procedures.

Evidence in the area of remote CI rehabilitation is also growing, especially related to intraoperative implantation telemetry and CI activation and programming (reviewed by Bush et al., 2016). A study reported that intraoperative CI telemetry through remote connection can be significantly timesaving for audiologists in the setting of a densely populated urban area where the surgery center is geographically separated from CI audiologist—provided that operating room staff are given technical training to perform on-site setup and address possible technical difficulties (Shapiro et al., 2008). Remote CI activation and programming have been addressed in various studies using similar clinical setup with an assistant, patient, and testing equipment/computer at the remote site and the audiologist at the primary site who uses CI programming software and video conferencing tool on an internet-based connection. Overall, remote CI programming was shown to be feasible, to require the same or slightly more time than face-to-face mapping, and to be able to provide high satisfaction of providers and patients (Bush et al., 2016; Eikelboom et al., 2014; Kuzovkov, Yanov, & Levin, 2014). Studies showed that measures of electrode impedances, maximum comfort levels, neural response imaging, patient map preference, and auditory performance

were not significantly different between remote and face-to-face methods (Eikelboom et al., 2014; Ramos et al., 2009). To date, some CIs on the market offer FDA approved remote features for remote follow-up programming (Muio, 2017). However, the audiologist interested in providing remote CI programming services needs to be aware of potential issues. For example, compromise of audio or video quality or lack of synchronization that may affect the quality of the programming session, or the fact that the programming venue and the connection method have the potential to affect CI programming and hearing testing outcomes (such as depending on whether or not a sound treated booth is used and the background noise levels) (Hughes et al., 2012).

Recent progress in ICT and devices paved the way to novel opportunities for rehabilitation. Nowadays, increased HA and CI capabilities enable new features that promise to improve client satisfaction and empowerment and, as such, may support person-centered care; e.g., direct wireless audio streaming, integration of multiple sensors, and user-operated fitting and device control.

Wireless audio streaming has been an important feature in HAs for decades since the introduction of the telecoil. Nowadays, the Bluetooth™ standard for digital wireless communication in the 2.4 GHz band is used, either through intermediary connected devices or as a direct connection with the HA or CI by low-energy Bluetooth™ protocol. Wireless streaming can provide support in everyday situations such as talking on the phone, watching TV, or listening to an audio stream from a mobile device. Direct wireless streaming capabilities incorporated into the HA or CI can be advantageous as no intermediary connected device is required in the signal chain, leading to better usability and improved sound quality and, as such, can contribute to the overall client satisfaction. However, there are some steps in the signal path that may degrade the sound quality of some wireless streams (e.g., artifacts and distortions originating from encoding-decoding, digital-to-analog conversion, and signal amplification). It is important, therefore, that audiologists assess the perceived sound quality of the streamed signal in addition to traditional HA/CI processing quality, and that the usability of wireless connectivity is addressed, especially among older users (Debevč, Kožuh, & Meier 2013; Ramsgaard et al., 2016; Ramsgaard, 2016).

Integration of multiple sensors into the HA has been proposed recently as a possible means to turn the HA into a multipurpose hearing device. For example, inertial sensors such as accelerometer and gyroscope

devices can be integrated into the HA for physical activity tracking or for fall detection. Moreover, possible integration of sensors for vital signs monitoring has been proposed recently, following recent studies that demonstrated the feasibility of ear-worn sensors to monitor heart and brain function (von Rosenberg et al., 2017; Winokur, He, & Sodini, 2012). Sensor integration is currently an area of lively research and promises to drive changes in HA technology that may improve the patients' life experiences with HAs.

User-operated fitting (also referred to as self-fitting) is a relatively new concept introduced to improve access to HA rehabilitation for people in remote areas, in low- and middle-income countries, or people with physical disability. The basic principle of self-fitting HAs is to conduct the whole fitting process, including physical fitting, without the direct supervision of a professional. The initial fitting is done to approximate user preferences as the starting point for the user and is based on automated evaluation of hearing thresholds to yield an initial hearing aid response and derive a prescription. Then, the users receive HA training by learning how to change the parameters in various listening situations so that the HAs are able to automatically adapt to the listening environments according to client preferences (Convery et al., 2017; Wong, 2011). Typically, the process is managed from a software application ("assisted" devices), but recently the concept of self-contained devices has been introduced, in which onboard buttons are used without requiring access to other hardware and connection to the internet (Keidser & Convery, 2016). In general, for self-fitting to be correctly performed and effective, it is important to provide knowledgeable support to clients, possibly tailored according to previous HA experience and preferably delivered by trained personnel (Convery et al., 2017). Features such as user operated HA controls and training are becoming more and more common and easy to use, thanks to the recently introduced integration with smartphone apps (Aldaz et al., 2014; Paglialonga et al., 2015). The employment of mobile-based platforms can be a valued means to empower patients as the user becomes part of the HA fine-tuning process. This brings along important implications for app developers as the cognitive, functional, and psychological aspects of the user experience need to be taken into account in app design and development (Aldaz et al., 2014).

In the area of mobile-based platforms, delivery of sound enhancement solutions through the smartphone has raised interest recently. Smartphone-based sound enhancement can be useful to improve speech

intelligibility and can help individuals with hearing problems develop awareness about the need for rehabilitation. In this vein, Galster (2012) described apps designed for personal sound amplification through the smartphone or tablet. The apps used either the audiogram measured through the smartphone or a list of preset configurations to amplify and filter the acoustic input. Some apps also included optional volume adjustment and output limiting. Assessment of subject performance with sound amplification apps and HAs, as measured via behavioral (speech recognition in noise) and self-reported (benefit, quality of life, and satisfaction) measures, showed that speech recognition was similar, but that the subjects favored the apps in terms of benefits but reported higher satisfaction with conventional HAs (Amlani et al., 2013). Possible inclusion of adjustable controls in sound amplification apps has been suggested to improve engagement and satisfaction. For example, Grebel et al. (2015) developed a sound enhancement app on Android™ that combined gain adjustment and frequency compression features as well as user-operated control to set an individualized target compression bandwidth. The study showed that sound processing features may pose problems when implemented in a smartphone app. In this regard, the investigators observed distortions and high output latency, beyond the desired limit. In general, this is an interesting area of investigation, but caution should be taken in considering smartphones as potential substitutes of HAs. In fact, sound amplification apps show potential as a tool to provide temporary assistance to individuals with HL but should be used as a starter solution rather than as a substitute for HAs (Amlani et al., 2013).

One of the most interesting developments in the area of e-health for hearing rehabilitation is related to auditory and cognitive training. Ample literature can be found in the area of training programs delivered via offline, internet-based, and mobile-based platforms (Chisolm et al., 2013; Olson et al., 2013; Olson, 2015; Sweetow, 2006). A variety of programs for auditory and cognitive training have been introduced throughout the years, such as training with speech, speech in noise, audiovisual stimuli, time-compressed stimuli, visuospatial tasks, and cognitive-linguistic exercises, often in the form of interactive games using audio, video, graphics, and written materials (Abrams et al., 2015; Chisolm et al., 2013; Ferguson & Henshaw, 2015a, 2015b; Henshaw et al., 2015; Olson et al., 2013; Olson, 2015; Rishiq et al., 2016; Tognola et al., 2015). In general, e-health-enabled training programs have facilitated adaptive auditory training and

have the advantage of being interactive and able to provide immediate feedback to the user (Saunders & Chisolm, 2015). In addition, more complex programs can be delivered via e-health platforms compared to conventional face-to-face encounters, such as programs that combine auditory and cognitive training for improved benefit, where cognitive enhancement can be embedded within auditory tasks (Ferguson & Henshaw, 2015b). However, these programs may also present some potential issues that should be minimized whenever possible by design or by appropriate use of the programs in the clinic, and by taking into account the targeted user group. For example, physical memory for storage can be a limitation if a large number of stimuli need to be used, especially for mobile-based platforms compared to offline or internet-based programs. Also, unwanted noise within the smartphone or loss of connectivity for internet-based programs might distort stimuli or create artifacts. Moreover, the small visual layout on the smartphone may not be adequate for older adults, even if a large font is used (Olson, 2015).

Last but not least, recent advances in the development of smartphones have made tinnitus relief therapies easily accessible. Various forms of sound therapy (e.g., masking, tailor-made notched music, nature sounds) can be delivered by means of smartphone apps via earphones or via HAs or CIs controlled via Bluetooth™ (Barozzi et al., 2016; Kalle et al., 2018; Linan et al., 2015; Tyler et al., 2018; Yoo et al., 2015).

Overall, it is clear that the introduction of e-health-enabled services in the area of hearing rehabilitation can be helpful to make patients more engaged in their own experience with treatment and more conscious about the rehabilitation options available. At the same time, it is important that audiologists become aware of the possible obstacles to effective adoption of e-health services and consider patient candidacy, digital skills, technological infrastructure, and any other potential issue that may limit the benefits of e-health enabled hearing rehabilitation services.

Conclusions

The ongoing digital transformation in health care has raised great expectations for patients and clinicians. Clinical implementation of e-health-enabled services in audiologic rehabilitation has the potential to improve adoption, outcomes, and satisfaction, and to increase the range of services available and the degree of personalization. However, as out-

lined in this chapter, this area brings along not only potential benefits but also possible challenges and risks. Although the feasibility of delivering e-health services has been demonstrated in several contexts, there may be barriers to adoption, lack of confidence in how to access services remotely, or difficulties implementing services in daily practice. Also, it may be difficult to assess the effectiveness and the quality of digital services in its several components. Research is still needed in this direction to ensure that clinicians provide effective evidence-based audiological rehabilitation to their patients. State-of-the-art technology is only one of the components of effective e-health services as it is also important to put in place the proper context for implementation of digital services. From a provider's point of view, it is important to consider elements such as patient candidacy, patient literacy and digital skills, clinician education and training, digital infrastructure, regulation, data protection, and data security, just to name a few (Montano et al., 2018; Paglialonga et al., 2018a; Paglialonga et al., 2018b). It is likely, however, that the greatest benefit from e-health services will not be obtained by replacing face-to-face interactions but rather by complementing conventional approaches, especially for those patients that are likely to benefit most. This way, e-health may keep the promise to become a facilitator of increased access to care and enhanced patient benefit.

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ONE-TO-ONE SPEECH COMMUNICATION TRAINING FOR ADULTS WITH COCHLEAR IMPLANTS

Geoff Plant

Introduction

The services that children and adults receive following cochlear implantation are markedly different. Children and their families are provided with extensive support and training in an attempt to ensure that optimal use of the device is achieved. These services are provided by speech-language pathologists, teachers, audiologists, and other clinicians, and support the development of effective spoken language skills and listening skills. The provision of these services is supported by funding agencies, and is seen as an integral part of the cochlear implant (CI) process.

The services for adults, on the other hand, are often restricted to programming of the CI(s), instructing her/him how to manage the external components of the CI, and testing her/his ability to understand words and sentences. Although some facilities do provide additional rehabilitative services, these are usually restricted to clients who are experiencing extreme difficulties in speech understanding. Most clients receive very little training aimed at optimizing her/his ability to access the acoustic speech signal. For example, Moberly et al. (2016) note that, “the

role of intensive postoperative aural rehabilitation has been emphasized in the pediatric population but has primarily been limited in adult CI users to patient-driven computerized auditory training” (p. 1526).

It is difficult to obtain information on the services provided by CI clinics, and little has been published on the type and scope of training provided. I recently sought to obtain some insights into this by asking CI users who are members of the “Cochlear Implant Experiences” group on Facebook whether they had received any specific listening training following implantation, and, if so, what form it had taken. I stressed that I was also very interested in hearing from those who had received no training.

I received more than 70 responses, and found that the scope and type of services provided varied greatly. Only 14 reported receiving one-on-one listening training. Twenty-six respondents reported that the training they had undertaken had been home based, and many had used online courses such as i-Angel Sounds (angelsound.tigerspeech.com), AB Clix (Advanced Bionics, 2013), and Listening and Communication Enhancement (LACE) (Sweetow & Henderson Sabes, 2009). Others in this group of respondents listened to recorded books, TED talks,

and other videos, which, they reported, were of considerable help. Finally, almost half (33) reported they had received no training at all. Several of the respondents expressed concern that they had not received any specific listening training even though they were experiencing difficulties, or were not satisfied with their performance level.

In part, the lack of training is due to the great success that many CI users have with this bare-bones approach. Dorman and Wilson (2004), for example, reported average scores for adult CI users of 80% to 100% for sentences and 45% to 55% for words in isolation. There are, however, some adult CI users who are performing poorly, and the needs of this group cannot be dismissed. For example, Lenarz et al. (2012), in a study of more than 1,000 adult CI users, found that 13% of the group were unable to identify more than 10% of words in sentences. Unfortunately, as Moberly et al. (2016) note, the response provided by many clinicians is “restricted to reassurance and recommending that the patient ‘keep working at it.’ In some settings, a struggling CI patient may be referred to a speech-language pathologist who focuses on aural rehabilitation, although this strategy is limited by a lack of evidence-based methodologies or support from insurance providers” (p. 1523).

It can be seen that despite overall high performance levels, many experienced CI users express dissatisfaction with the current situation. In discussions with CI users in the U.S., the U.K., and Australia, and in the recent Facebook “survey,” I have found that many believe their performance could have been considerably enhanced if they had received systematic speech perception training in the period immediately following the activation of their CI’s. Perhaps surprisingly, those expressing dissatisfaction with the current approach include many CI users with very high performance levels. They express the belief that their progress would have been greatly accelerated if they had been provided with ongoing support and training. They also believe there are many other CI users who continue to struggle because they have not been provided with appropriate intervention.

This chapter is concerned with approaches to speech communication training that I have used with a large number of adults after they have received a CI or CI’s. The training was provided one on one, and the clients were attending the Hearing Rehabilitation Foundation (HRF) in Somerville (1996–2017) or Woburn (2018), MA. This experience provided an excellent opportunity to provide and evaluate the effectiveness of long-term training for adults with

CIs. As part of my work with the HRF, I have also been able to trial these approaches with clients with severe to profound hearing loss who have been fit with hearing aids (HAs). The chapter will provide an overview of the materials and approaches that I use in speech communication training, and present case studies of two adult clients.

Historical Background

The first systematic attempts to alleviate the effects of an acquired hearing loss on speech perception date from pioneering lipreading (speechreading) teachers in the U.S. and Europe around the turn of the 20th century (see Jeffers & Barley, 1971 for an excellent overview). Effective hearing aids were still several decades away, and the approaches developed focused on the use of vision as a substitute for hearing. The visual signal is, however, notoriously ambiguous, and offers limited access to the speech signal. To extract meaning from this limited sensory signal, lipreaders need to take “advantage of linguistic redundancy—the constraints imposed by the rules and function of language— . . . to fill in the gaps left by the visual stimulus” (Boothroyd, 1988, p. 78). Nitchie (1912) recognized the importance of training his students to take advantage of these synthetic processes, but he did not achieve his goal of developing effective materials and approaches before his premature death in 1916.

Two significant events in the 1940s contributed greatly to the development of speech communication training. The first was the introduction of the one-piece hearing aid in 1944 (Dillon, 2001), resulting in the first truly wearable hearing aids. The second was the “birth” of audiology in the military rehabilitation centers set up in the U.S. to provide assistance to service personnel who had developed hearing losses due to war service. These centers provided up to 3 months of direct intervention, designed to help overcome the effects of an acquired hearing loss (Schow & Nerbonne, 1996). Although these rehabilitation centers were closed soon after the end of World War II, this intensive approach to rehabilitation continued at other facilities such as the Walter Reed Hospital in Washington, D.C.

Ross (2001) has provided an account of the 2-month course at Walter Reed Hospital that he attended in 1951, following diagnosis of a hearing loss. The program included hearing aid fitting and testing, lipreading instruction, and auditory training,

as well as information sessions on various aspects of hearing loss. Informal exchanges between patients also contributed greatly to an awakening understanding of how best to cope with their hearing losses. Ross obviously believes that this was an extremely worthwhile experience, and 3 years later, prior to discharge from the Air Force, asked for and was granted a second 2-month course. He likens this second course to a “booster shot,” and believes that it helped him to “truly come to terms with the fact that [he] had a permanent hearing loss” (p. 63).

Ross recognizes that this “Camelot” period in aural rehabilitation can never return. It would be far too costly to replicate this service, much of which, he admits, “was overkill” (p. 66) but he feels that we can learn a great deal from it. In summary, he calls for an approach that focuses on the individual person with hearing loss, and her/his needs, and recognition that technology, the CI and/or HA, is “only one of the tools that can be used to reduce the communicative and handicap impact of a hearing loss” (p. 67).

Over the following half-century or so, there has been a decline in the amount of audiologic rehabilitation, particularly auditory training, provided to adults with an acquired hearing loss. In part, this has been due to advances in technology, leading to clinicians having “the unsubstantiated view that modern technology is sufficient to negate the need for additional auditory training” (Sweetow & Henderson Sabes, 2009, p. 273). Sweetow and Henderson Sabes (2009) also point out that clinicians may feel that there is little evidence to support the provision of such services, and they recognize that valuable professional time spent providing such training is unlikely to be reimbursed. Sweetow and Henderson Sabes believe auditory training does provide benefit for people with hearing loss, but there is a need to gather more objective and subjective data that supports this view.

Auditory Training

Sweetow and Henderson Sabes (2009) define auditory training as “a process designed to enhance the ability to interpret auditory experiences by maximally utilizing residual hearing” (p. 267). This definition, although accurate for persons fit with HAs, needs to be modified slightly to include the reality of CI hearing. Although there is now recognition of the need to preserve the residual hearing of CI users, the electrical signal provided by CIs directly stimulates the

auditory nerve. A slight change in wording to include “residual (*acoustic*) and *electrical* hearing” would more accurately reflect the current situation.

The terms *analytic training* and *synthetic training* are used to describe the two basic approaches to auditory training.

Analytic training involves bottom-up processes and focuses on the context-free acoustic-phonetic signal. Examples include contrasting words with differing syllabic structures (*cat* versus *apple* versus *football*), different vowels (*hard* versus *heed* versus *who’d*), initial consonant differences (*pot* versus *tot* versus *cot*), etc. Such tasks are usually presented using a closed-set format. The listener’s task is to listen for the salient acoustic cues and indicate which of the contrasted words was presented as the stimulus.

Synthetic training involves top-down processes, where the listener uses her/his linguistic knowledge to fill in the gaps in the sensory information provided by her/his CI(s) and/or HA(s). Boothroyd (1988), in a discussion of linguistic factors in lipreading, lists six sources of linguistic redundancy—pragmatic, topical, semantic, syntactic, lexical, and phonological—that can be used to assist in deriving meaning and understanding.

Analytic or Synthetic?

Some researchers, such as Rubinstein and Boothroyd (1987), debate the value of analytic training and suggest that the primary focus of training should be on synthetic training exercises, which concentrate on sentence level materials, and attempt to exploit the “uses of linguistic and situational redundancy” (Rubenstein & Boothroyd, 1987, p. 152). Native speakers of a language bring a vast store of knowledge to any communication situation. They know, at least implicitly, the rules of their language—how words are formed (phonological rules), how sentences are put together (syntactic rules), which words tend to occur in close proximity to each other (collocational rules), and so forth—and it’s this knowledge that enables us to fill in the gaps and understand speech even when we can only partially hear what is being said.

Others, such as Sweetow and Henderson Sabes, argue for an approach that combines both synthetic and analytic training. Support for the inclusion of analytic training can be found in the work of Walden, Erdman, Montgomery, Schwartz, and Prosek (1981), who provided a group of adult subjects with 7 hours of training in the auditory recognition of nonsense

syllables. They found that the subjects' post-training scores for an auditory-visual sentence test presented in noise were significantly better ($p < 0.01$) than those obtained before training.

I believe training should reflect, as much as possible, the realities of everyday communication, and advocate for an approach that has synthetic training as its primary focus, but does include some analytic training. This has been motivated by the recognition that there are many occasions where listeners are forced to rely exclusively on acoustic/phonetic (analytic) cues to derive understanding. Consider, for example, the possible range of responses to the question, "What is your last name?" The U.S. Census Bureau includes 889,799 last names in its database, ranging from the common, and, perhaps predictable, "Smith," "Jones," "Williams," and "Garcia" to the more obscure and far less predictable "Argenti," "Barnak," and "Muckenfuss." In all of these cases, the listener must use acoustic/phonetic information to determine a person's name.

Speech understanding using analytic (bottom-up) processes presents problems for almost all people with hearing loss. Adult EARS (Plant, Ng, & Amann, 2011) included a sentence test, which used the Speech Perception in Noise (SPIN) Test format developed by Kalikow, Stevens, and Elliott (1977). Fifty sentences were presented, with the listener's task being to repeat the last word in the sentence. In half of the sentences the identity of the last word can be predicted to a large extent, based on the rest of the sentence. Examples of these high-predictability sentences include, "I asked my boss for a raise" and "I heard someone knocking at the door." In the remaining half of the sentences, the identity of the final word cannot be predicted using contextual cues. Examples of these low-predictability sentences are "We need to talk about the vase" and "Do you think she said, 'Stop'?" Two lists of these sentences were presented to 20 experienced adult CI users over a 3-month period. The mean score for the high-predictability (HP) sentences was 90.6% correct, while that obtained for the low-predictability (LP) items was 72.3%. The results of the nonparametric Wilcoxon signed-rank test showed that the difference between the HP and LP scores was significant ($p < 0.001$).

Warm-Up Exercises

One of the most important parts of any training session occurs at the beginning of the session. Many clients, at least at first, approach training with some

degree of trepidation, and need some short exercises to set the scene for the remainder of the session. In some ways, the beginning of each training session is the warm-up time, where the client gets ready for the rest of the session by listening attentively and responding as quickly as possible.

At this time, I like to use simple closed-set materials, which start at a relatively easy level and then, if appropriate, become progressively more difficult. One exercise that I have found particularly useful is presenting numbers for identification, at first in isolation and then in progressively longer strings. For example, I will tell the client that s/he will hear the numbers 0 to 9 presented in a random order. Her/his task is to repeat what s/he hears. I usually present the isolated number in the carrier phrase, "The number is . . .," but occasionally vary the procedure and embeds the target in a short sentence such as, "I saw . . . cars," "I have . . . brothers," etc. If the client is able to identify the numbers with few errors, I then instruct her/him that I am now going to present two numbers at a time. If the client is able to perform this task with little difficulty, I extend the number of items to three, four, five, or six number strings.

Miller (1956), in a study of digit recall, found that the limit of short-term memory was seven plus or minus two. Later studies (for example, Baddeley, Thomson, & Buchanan, 1975) have found that while this result appears to hold for college-age students, it varies for other populations. A study by Souza, Boike, Witherall, and Tremblay (2007) of three groups of listeners with moderate to severe hearing losses aged 50 to 65 years, 67 to 75 years, and 77 to 82 years, however, found no significant differences in the auditory digit recall of the three groups, although there was a slight decline for the older listeners. The mean scores for the three groups were 7.5, 6.8, and 6.7, respectively. The mean score for a group of younger listeners with normal hearing (mean age = 23.5 years) was 7.0. As a result of these studies, I rarely present number strings of more than six items.

If a client is able to perform this task easily, I ask her/him to repeat the string of numbers backwards. For example, if the numbers presented are 5 7 3 4, the client's response should be 4 3 7 5. I can also ask clients to put the numbers in a string in their numerical order. For example, the correct response for the string 5 7 3 4 would be 3 4 5 7. Such exercises also provide insights into the working memory capacities of individual clients.

I can also vary the items presented for identification. For example, I will sometimes present strings of color names, animals, or the names of U.S.

presidents. I provide the client with a sheet showing the set of alternatives, and repeat the procedure described above: items presented in isolation, and then in increasing number, with the client asked to repeat what s/he heard forwards, and, if appropriate backwards.

Another approach that I sometimes use as a warm-up activity is based on the test procedure developed by Hagerman (1982). The client is shown a set of words arranged in a matrix. An example is presented in Table 24–1. I tell the client that I will select one word from each column to form a short sentence such as “Joan took five blue taps,” “Jan had two new caps,” etc. At first, I present the sentences using Clear Speech with a short pause between each word. If the client is able to perform this task easily, I start to use a more normal speaking style, and if the client is still able to perform at a high level, I will use a rapid speaking style. This can result in the client’s performance declining rapidly as the word boundaries become blurred, and many clients have commented that this is what happens to them in real-life conversations. Another modification I sometimes use is to present sequences from the matrix in the presence of background noise. This is particularly useful with clients who are not confident in their ability to cope with competing noise. The use of closed-set material provides them with a degree of security that encourages them to attempt the task.

Analytic Training

In my work with adult CI and/or HA users, I try to include some analytic training exercises at the beginning of each session. I recognize, however, that analytic training can be quite stressful for clients, and try to ensure the training targets are at an appropriate level. Clients rarely complain that any analytic mate-

rials are too easy, and I feel it is best to err on the side of caution in determining a starting point for training. This type of training requires considerable effort on the part of the client and, if not presented appropriately, can be boring. As a result, I try to ensure that I don’t use analytic training exercises for more than 10 to 15 minutes per session.

The following example presents the sequence I would use to develop and consolidate a client’s ability to reliably identify the consonant [s]. I start with a word list that contrasts word pairs that differ only in that one word contains an initial [s] and the other does not. Examples of this type of contrast include *sit/it*, *say/A*, and *sigh/eye*. The client is presented with each contrast in turn and asked to identify which word was presented as the contrast. I usually present the items using a short carrier phrase such as, “Number 1 is . . .,” “Number 2 is . . .,” and so on.

If the client is able to perform this task in both the initial and final (*peace/pea*, *cat/cats*) position, I then move up to contrast the target with consonants that differ considerably in their acoustic structure. Examples include word pairs contrasting [s] with nasals (*my/sigh*, *mass/man*), semivowels (*way/say*, *right/sight*), or voiced stops (*bend/send*, *Rob/Ross*). If the client is able to complete those items successfully, potentially more difficult contrasts could then be attempted such as word lists that contrast [s] with voiceless stops (*pet/set*, *miss/mitt*), the affricates (*jet/set*, *match/mass*), or nonsibilant consonants (*fine/sign*, *myth/miss*). Only after passing through a sequence such as that outlined above would I attempt contrasting [s] with the voiceless sibilant [sh] in word pairs such as *sigh/shy*, *see/she*, *gas/gash*, and *mass/mash* or its voiced cognate [z] in word pairs such as *zoo/Sue*, *zap/sap*, and *pays/pace*.

My primary source for analytic training materials is *Analytika* (Plant, 2004), which is a collection of almost 600 word lists presenting consonant or vowel contrasts in meaningful words. I have also developed some materials that present the contrasts in closed-set sentences.

Examples of this approach include:

“Are you sure that she SIPPED it?”

“Are you sure that she SHIPPED it?”

“Where did you get these new SEATS?”

“Where did you get these new SHEETS?”

These materials are presented using a closed-set format, with the client asked to identify which of the contrast words was presented.

Table 24–1. An Example of a Simple Sentence Matrix Used in Training

JOHN	TOOK	TWO	RED	CAPS
JAN	SHOOK	THREE	GREEN	CUPS
JANE	HAS	FOUR	BLUE	TAPS
JOAN	HAD	FIVE	NEW	PUPS

Note: One word is chosen from each column to form a short sentence.

The materials are presented auditory alone using an “acoustic shield” consisting of an embroidery frame with speaker cloth stretched over it, which I hold in front of my face to prevent access to lipreading cues. I strongly caution against covering the face with a sheet of cardboard or the hand, as these can lead to a significant reduction of high-frequency speech energy.

I show the client each contrast printed on a sheet of paper, or as a PowerPoint slide presented via a computer monitor. I ask her/him to tell me which of the words was presented, and provide immediate feedback as to the correctness of her/his response. If an error is made, I provide feedback such as, “No, it was ___, not ___.” Wherever possible, I present lists of 20 contrast pairs. A score of 15/20 is significantly above chance at the 5% level, and I use this as the “pass/fail” criterion for each list.

If a client scores less than 15/20, I attempt some limited training that focuses on the contrasts in a variety of word and/or nonsense syllable pairs. If the client still experiences difficulty, I will often represent the list with both auditory and visual cues, to see if access to lipreading results in improved performance. My aim is to provide the client with an awareness of her/his “problem” areas, and then work on solutions to the problems. These can, of course, include training in the perception of the contrast, but can also involve developing an awareness of which contrasts create difficulty, so that the client can be aware of potential problems in real life and act accordingly.

I’ll give a personal example of such a compensatory strategy. As a result of a moderate hearing loss, I have a degree of difficulty in reliably hearing the difference between the numbers “two” and “three” in situations where I do not have access to lipreading cues. I’m aware of this problem, and try to ensure that I reduce the chance of making an error by asking the speaker if the number was “two,” thus eliciting a “yes/no” response, or, in the case of phone numbers, for example, repeating the number for verification.

Another approach that is often used in auditory training programs is the presentation of consonants in a vowel-consonant-vowel syllable such as [aCa] or [iCi]. I used this procedure for many years, but stopped when I realized I was having difficulty reliably differentiating between the clients’ responses for consonants such as [p] and [t], and [m] and [n]. I felt, however, that the exercise provided valuable insights into the sensory information available to individual CI users, and eventually developed an alternative response that enabled me to easily differentiate between the items. The system uses a modification of

the NATO phonetic alphabet used by emergency services personnel when spelling out a word. The alphabet uses distinct words for each letter of the alphabet: *bravo* for “b,” *hotel* for “h,” etc. I provide clients with a chart listing the word to be used for each of the 20 consonants (p, t, k, b, d, g, f, v, s, z, sh, h, ch, j, m, n, w, r, l, y) presented in testing. The modifications made to the existing NATO alphabet are *Charlie* for the affricate [ch], *kangaroo* for the stop [k], and *shadow* for the fricative [sh].

The task is presented using the Kungliga Tekniska Högskolan (KTH) Tracking Procedure (Gnosspeilius & Spens, 1992), which is described in detail later in this chapter. The syllables are presented for two minutes, and immediate feedback is provided if the client’s response is incorrect: “No, it was [repeat syllable].” If the client does not recognize the syllable correctly after three repeats, s/he is shown the correct response on a computer monitor. After 2 minutes, the program calculates the number of syllables correctly repeated per minute. The number of syllables correctly perceived is also recorded, but the syllable rate provides insights into how quickly the client is able to recognize the test items.

Useful insights into a client’s performance can also be obtained by asking them to provide subjective ratings related to a particular task or contrast. I often ask clients to rate the amount of effort that a task requires, using a 7-point scale ranging from 1 (*extreme amount of effort*) to 7 (*no effort at all*). I also ask clients to give an estimate of how confident they are that their response is correct. Again, I use a 7-point scale with the rating items ranging from *no confidence at all* (1) to *completely confident* (7). Such rating scales can be used to track the amount of work required to complete a task, and this can be measured over time.

Synthetic Training

Synthetic training provides clients with practice using materials that more closely replicate everyday communication. Sweetow and Henderson Sabes (2009) believe synthetic training provides opportunities for the listener “to use various communication strategies, including attention, use of contextual cues, repair strategies, knowledge about linguistics and communication, and the redundancies therein to communicate effectively” (p. 269).

In presenting synthetic exercises, I try to tailor the materials to meet the individual client’s specific needs and skills, and I have developed a number of materials to help achieve this goal.

Questions and Answers exercises (Plant, 1996a, 2001a) are an example of materials that can be tailored to fit the needs of individual clients. The client is provided with a list of questions related to a particular theme or topic, while the clinician has the answers provided. The topic is usually relatively obscure, to prevent the client from using her/his preexisting knowledge to understand the answer. The client is asked to ask each question in turn, and the clinician provides the answer. I always advise clinicians to become familiar with the answer to each question before attempting to use these exercises with clients, so that they are able to modify their response to meet the needs of the individual client.

One Question and Answer exercise in *Auditrain* (Plant, 2001a) has "Denmark" as its theme, and I'll use it as an example of the approach. Question 1 in this exercise is, "Where is Denmark?" Depending upon the skills of the client, the clinician might respond with a short answer such as, "Denmark is in the north of Europe" or with a more complex response such as, "Denmark is located in the area of northern Europe known as Scandinavia. Its neighbors include Germany to the south, and Sweden and Norway to the north." Similarly, the clinician's response to the question "Is Denmark a large country?" might range from, "No, it's a very small country" to "No, Denmark is not a large country. It's one of the smallest countries in Europe. It has an area of only 27,280 square miles." The client is not expected to repeat every word of the answer, but rather to provide a brief summary of its salient points.

Because s/he asked the question, the client has some idea of the form of the response, and this helps reduce the complexity of the listening task. Clinicians can present the information in relatively long sentences, or in several short chunks, depending upon the listening skills of the client. This "expansion" technique is particularly useful, and will be covered in detail later in this chapter.

The Question and Answer topics often provide opportunities for conversational practice. Clients often show particular interest in a specific topic or answer, and this can be used as the basis for a more extended conversation. Over the years, I have built up a large stock of materials such as maps, photographs, and other materials to use in such circumstances. These can be used to provide more information and to heighten the interest level of the training session. Pictures and maps also provide contextual information that can help reduce the level of difficulty of a particular task.

Topic-Centered Sentences consist of lists of sentences based around themes such as "Things people

say about the weather" and "Things people say about food" (Plant, 2001a). Each list contains a series of sentences that are related to a particular theme. For example, sentences for "Things people say about clothes" include:

- "That's a nice dress."
- "He always wears a suit to work."
- "What size do you think would be best for their new baby?"
- "Their new catalogue has a lot of nice clothes, but they are far too expensive for me."

Note that the sentence length in this selection ranges from four to 17 words. Many people with hearing loss report they have difficulty with longer segments, so it may be necessary to adjust the style of presentation to meet the needs of individual client. For example, it may be necessary to present a longer sentence in two shorter chunks such as, "Their new catalogue has a lot of nice clothes" and "but they are far too expensive for me." Another way to help clients become more confident when confronted by longer sentences, is to take a short sequence that has been correctly repeated, and gradually increase its length through the addition of words and phrases. For example, the sentence, "That's a nice dress," could be extended in the following way.

- "That's a nice dress."
- "That's a nice dress. Is it new?"
- "That's a nice dress. Is it new? I haven't seen it before."
- "That's a nice dress. Is it new? I haven't seen you wear it before."

This expansion technique is also used in Plant (2000) with short segments being made progressively longer, and clients are asked to listen for additions to recognized sentences. I have found that clients respond very positively to this technique, and are often surprised by their ability to perform the task.

Sentence lists consisting of the most frequently occurring words in spoken American English. Several years ago (Plant, 2001b), I developed a number of resources based on the 500 most frequently occurring words in spoken American English (Dahl, 1979). These words form more than 80% of the words used in everyday conversation, with the 10 most-frequently-occurring words—I, and, the, to, that, you, it, of, a, know—representing around 25% of the total number of words used. I believe these 500 words represent the "core items" of spoken English, and I have developed

a large number of word and sentence lists based on these items. Although it may sound constraining, it is relatively easy to generate sentences using only these 500 words. Examples include:

- "It's too late now!"
- "Do you know anything about them?"
- "I'm afraid I don't know the answer to your question."

Each list (Plant 2001b) consists of 25 sentences, with a total of 200 words in each list. Average sentence length is eight words, but there are several short (four to five words) and long (12–14 words) sentences in each list. Unlike the topic centered sentences discussed above, these sentence lists are unrelated, and have no common theme. As a result, they can be a little more difficult for some listeners. I usually present one list per session, and encourage the client to repeat as many words as possible.

I have recently started to present these sentences using the KTH Tracking Procedure (Gnosselius & Spens, 1992) to determine how quickly clients are able to recognize and repeat these items. I have also introduced a new set of sentences, all of which are 14 or more words in length. Clients often report difficulty maintaining focus with longer sentences, and this approach provides practice with these more difficult materials. The duration of the task is 3 minutes, and the client has to correctly repeat all words in the sentence before moving on to the next. If the client fails to identify a word after three repeats, s/he is shown the word on a monitor. At the end of the time period, a tracking rate (TR) is automatically calculated in words per minute. This approach provides additional information on a client's speech recognition skills. For example, some clients are able to correctly repeat the words in a sentence, but their response is drawn out and requires a great degree of cognitive effort. Thus, although the client is able to score quite highly for the number of words repeated, the time taken to achieve this indicates that s/he would have difficulty following continuous speech, and would soon start to falter. This is especially obvious when the sentences are longer. Some clients seem to "turn off" at some point in a longer utterance, and this training encourages them to maintain focus on what is being said.

Promoting conversational skills should be the primary aim of speech communication training. I try to provide opportunities for two-way conversations in all training sessions. This can be difficult if the client is uncertain about her/his skills, or if s/he has developed a pattern of communication where s/he

dominates the conversation and greatly reduces the contributions of others. I try to set aside at least a little time during a training session to have a chat with the client. The topic can be about what's happened since our last meeting, either personally or in the wider community, or might be stimulated by the topic of one of the exercises. I travel a great deal as part of my work, and I always try to keep a photographic record of places that I visit. I will often show a selection of these to the client, and encourage him/her to ask questions, comment, etc. I find these activities to be good starting points for conversation. I usually use the acoustic shield during these conversations to eliminate the possibility of the client using lipreading cues, and do have some reservations about how natural this approach is, but find that clients rarely comment on it. Most accept it as a part of the training approach, and understand that the focus is on auditory speech perception.

Speech tracking (de Filippo & Scott, 1978) is an important part of almost all training sessions. In speech tracking, there is one talker (usually the clinician) and one receiver (usually the person with hearing loss). In the original form of this approach, the talker read from a text, segment by segment, and the receiver had to repeat each segment without error. If an error occurred, the talker had to re-present the segment until it was repeated verbatim. To reduce the level of difficulty, the talker could give clues, reword, rephrase, etc., to help the receiver correctly identify the original segment. The procedure continued for a specified time period—often 10 minutes—but I strongly recommend going for only 5 minutes. At the end of the time period, the talker counted the total number of words presented and correctly repeated. This figure was then divided by the time elapsed, to calculate the receiver's TR in words per minute (wpm). For example, if a receiver was able to repeat 250 words in 5 minutes, her/his TR would be 50 wpm.

I use a modified form of the KTH Tracking Procedure in my work with adults. This computer-controlled approach was developed by Gnosselius and Spens (1992) at the *Kungliga Tekniska Högskolan* (Royal Institute of Technology) in Stockholm, Sweden. The original program ran in MS-DOS, and a subsequent Windows-compatible version was developed by the Rehabilitation Engineering Research Center on Hearing Enhancement at Gallaudet University in Washington, D.C. (Plant, Bakke, Bernstein, Levitt, & Oden, 2007).

The KTH Tracking Procedure uses live-voice presentation, but with predetermined segment length, and only one repair strategy: repetition. The test is

entered line by line onto the computer, and is displayed on a computer monitor for the talker. The talker uses the right-hand button of the computer mouse to start the program and move from line to line, and the left-hand mouse button to mark any word that is not repeated correctly. If a word is not correctly identified after two repeats, the receiver is shown the written word on a separate monitor, and this is recorded automatically on the computer. At the end of the specified time period, the receiver's TR is automatically calculated and displayed. Separate measures, including the receiver's ceiling rate (based on lines repeated without error) and the proportion of blocked words, which is based on the number of times words have to be repeated as a proportion of the total number presented, are also automatically calculated.

I have written two long stories, which I use with almost all of my adult clients. The first story, "Kumanjayi," contains more than 160,000 words and is set in central Australia. It is told in the first person, and is written in a conversational style in an attempt to provide training that replicates, at least in part, everyday communication. The second story, "The Old Ones," consists of three parts of around 100,000 words each. This story is written in the third person and is designed to be more complex than "Kumanjayi." It is set in Sweden and Norway and has a fantasy/science fiction theme. Clients report that they enjoy the stories, and this helps them to maintain focus as the want to find out what happens next!

In any training session, I present up to 10 (five Auditory-Visual [AV] and five Auditory-only [A]) 5-minute speech tracking sessions, and I find that clients respond extremely positively to this approach. A client once told me that he regarded his TR in much the same way as his golf handicap, with the only difference being that he wanted his TR to rise, and his golf handicap to fall! Many clients report initially that they derive no benefit from lipreading, but an examination and discussion of their scores in the two conditions and their ratings of effort and confidence quickly reveal to them that visual cues make the task much easier. The materials are usually presented in quiet, but if a client's scores indicate it will not be too stressful, the materials can also be presented with a competing speaker. I use several different recorded books as the source for the competing speech. Over time, it has been determined that there is a hierarchy of difficulty for the various recordings. The easiest is a woman with a Scottish accent, the next most difficult is a male speaker with a general American accent, and the most difficult is a male speaker of southern British English. The competing speech is set at a level

that creates difficulty for the listener, but not so great that s/he has to struggle too greatly to understand what is said.

Bernstein et al. (2012) looked at the use of speech tracking using both de Filippo and Scott's original procedure (TRAD) and the KTH Speech Tracking Procedure. They found that training with both approaches lead to significant gains in TR and sentence recognition scores. The study also looked at between trial variability for the TRAD and KTH approaches and found that the KTH approach was significantly less variable than the TRAD approach.

Modified Speech Tracking

Speech Tracking, especially via audition alone, is a difficult task for many people with severe or profound hearing losses. My experience with the procedure over the past 30 years indicates that if a client has a TR of 25 wpm or less, the approach should not be used. At this level, the receiver is struggling to perceive single words, so context and story line are no longer of much assistance, and the task is extremely stressful. Clients with poor receptive communication skills need to be encouraged to use their language skills to take the fragmentary information available to them, and to synthesize meaning. Attempts at forcing such clients to relentlessly strive to identify every word may be self-defeating, and lead the client to have an inappropriately low opinion of her/his receptive communication competence.

Commtrac (Plant, 1989, 1996b) was developed for use with those clients for whom speech tracking was unsuitable. In this procedure, a story is presented line-by-line auditory alone, and the client is asked to repeat as many words as possible. After the client has responded, s/he is shown the written line, and the next line is presented auditory alone. There are no time constraints imposed; clients can take as long as they require before they respond, and they are also encouraged to ask for the line to be repeated if necessary. Each story in *Commtrac* was divided into 200-word parts, which were, in turn, segmented into separate lines for presentation. At the end of each part, the client's percent correct score is calculated.

I have subsequently prepared a number of different stories that use this approach. One of these, *TeenTrax* (Plant, 2007), is a story that is divided into more than 40 parts, ranging in length from 200 to 250 words. Feedback is provided to the client using a PowerPoint presentation, with each line being presented on a separate slide. Although originally designed for use with teenagers and young adults,

I have used *TeenTrax* with a number of adult clients, and all have responded favorably to it.

I find this modification of speech tracking particularly useful with clients who are reluctant to attempt auditory only training. Conventional speech tracking, with its insistence on every word being identified, can be intimidating for many clients, and it is better, at least initially, to use this modified approach, which allows for incomplete responses and provides immediate feedback to ensure that the client is able to keep up with the story and use contextual information to assist in understanding.

Case Studies

There are two case studies in this section. The two subjects are a middle-aged woman with a congenital hearing loss, and an older man with a long-term acquired hearing loss. Both attended the Hearing Rehabilitation Foundation (HRF) for training and made a donation to the HRF for each session attended. It should be noted that this fee was voluntary and the service would have been provided regardless of the client's ability to make a donation.

Case Study 24-1

The client was a female in her late 40s with a congenital hearing loss. She has two younger siblings who also have congenital hearing losses and who are both successful CI users. The client had worn hearing aids since early childhood, but reported that, unlike her siblings, she was unable to understand speech through listening alone. Her hearing aids did, however, serve as a very useful adjunct to lipreading, and she has very well-developed speech and language skills. Her decision to receive a CI was influenced by the excellent performance of her siblings, and her desire to be able to communicate better with her two children.

I first saw the client in late October 2011, approximately 1 month after her CI was activated. At that time, she said she was quite happy with the outcome, but reported that she was unable to understand speech via listening alone. She had opted to continue wearing a hearing aid in her other ear, and all subsequent testing and training was conducted in this combined/bimodal (CI + HA) mode. The client attended training sessions at approximately 2-week intervals for just over 3 years.

The first training sessions focused mainly on closed-set tasks that contrasted word syllable patterns, and used word lists that presented word pairs that differed only in that one contained a target consonant such as [s] and the other didn't. She was able to perform these tasks very well, and after the first five sessions, I decided to introduce some open-set materials. These included expansion sentences and simple sentences that required the client to fill in the missing words. At first, the client found these tasks very challenging, but over time, she became far more confident in her ability to understand speech through listening alone. She particularly enjoyed the warm-up exercises using closed sets of numbers, colors, vegetables, and fruits presented in isolation and in strings of two, three, or four items. As her ability to perform these tasks improved, I asked her to repeat the items backwards, and she was able to perform this task very well.

In late January 2012, I introduced the *TeenTrax* tracking program. The first segment of the program was presented auditory-visually to familiarize the client with the procedure. Her score was almost 100% correct, which was a realistic measure of her face-to-face communication skills. All subsequent segments were presented via audition alone. The subject's performance for this task is shown in Figure 24-1. This shows a steady improvement over a period of approximately 6 months.

As the client's auditory alone performance for the *TeenTrax* materials improved, I introduced the sentence lists drawn up using only the 500 most-frequently occurring words in spoken American English (Plant, 2001b). These were presented auditory alone, and each sentence was repeated before the subject was asked to repeat what she had heard. One sentence list was presented at each training session, and the results of this testing is shown in Figure 24-2.

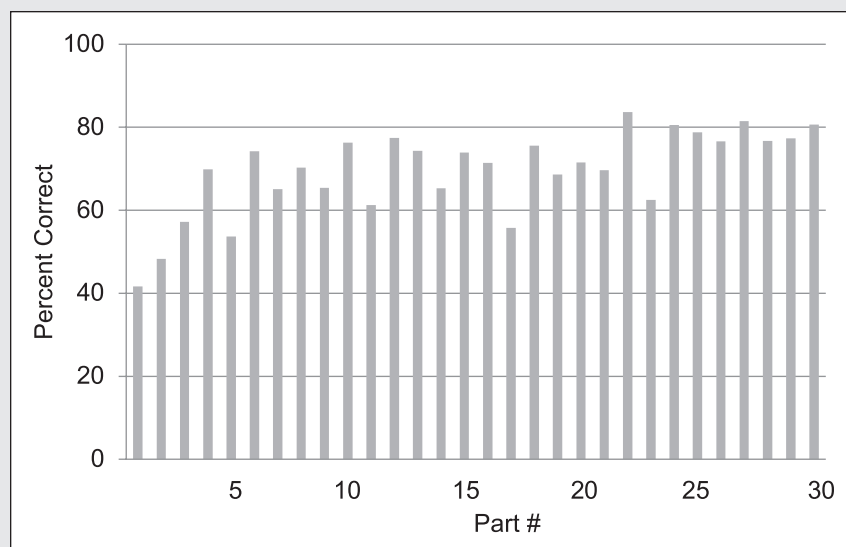


Figure 24-1. Percent correct scores for TeenTrax materials obtained by the subject in Case Study 24-1.

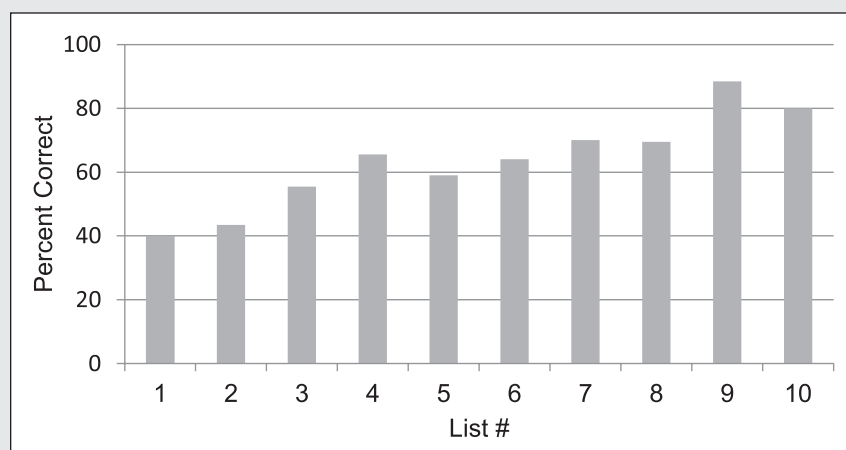


Figure 24-2. Percent correct scores obtained by the subject in Case Study 24-1 for lists of sentences presented via audition only.

The results are similar to those obtained with the *TeenTrax* materials, a steady rise in performance across the course of training, which was surprising given that the lines in the *TeenTrax* are related and provide the opportunity to use contextual cues to assist in understanding.

I then started to use the KTH tracking procedure, in a modified form, with the client. I presented each line once with my face covered, but if any errors were made, all subsequent presentations of that line were made auditory-visually. Over time, it became apparent that her skills had developed to the point where she would be able to perform the task via audition alone.

Speech tracking was then used in each training session for the next 23 months. Five-minute sessions were conducted in both the auditory-visual and auditory alone conditions. The client's performance over this period is summarized in Figure 24-3, which presents her

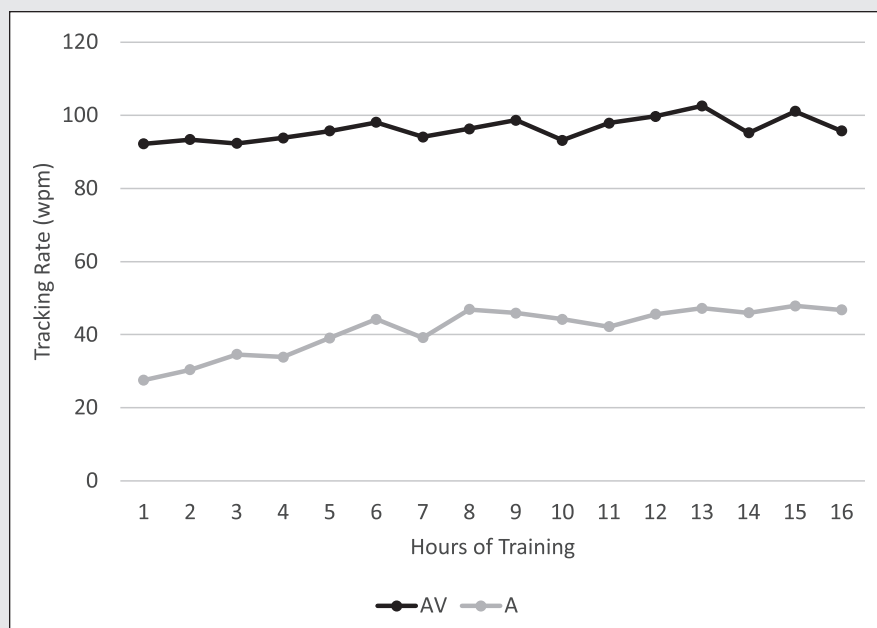


Figure 24-3. Tracking rates (TRs) (in words per minute) obtained by the subject in Case Study 24-1 for the auditory-visual and auditory-alone conditions. Each datum point represents the subject's mean score for 12 × 5-minute (1-hour) tracking sessions.

mean TRs for the materials for each hour of training conducted in the AV and A-only conditions. A total of 16 hours training was presented in each condition over this period. The client's AV TRs are in the range 90 to 105 wpm over the training period. It should be noted that this almost certainly represents the client's maximum (ceiling) performance level. In common with many people with congenital hearing loss, the client has a relatively slow articulation rate and this imposes a limit on her overall TR.

In the A-only condition, the client's mean TR rose from 27.5 wpm in the first hour to 47.9 wpm in the 15th hour. There was a steady improvement over the first 10 hours of KTH training, at which point the client's performance reached an apparent asymptote of 46 wpm. Although there was no apparent improvement in her speech tracking performance in the final 6 hours of training, the client continued to attend. In part, this was motivated by her performance in individual 5-minute sessions where her TR was 50 wpm or better. For example, in three of the sessions in Hour 13, her TR exceeded 50 wpm. Similarly, in Hour 15 she had four sessions of better than 50 wpm. It was also quite apparent that she was far more relaxed in these sessions and reported that the amount of effort required in the A-only condition was considerably lower.

The training concluded only when the client was diagnosed with a health condition that made travel to and from training too difficult. Plant, Bernstein, and Levitt (2015) presents a detailed analysis of this client's performance across the course of the training period.

Case Study 24-2

The client was a 70-year-old male with a long-standing profound hearing loss. He had been wearing HAs for around 30 years, and had recently been referred as a candidate for cochlear implantation. He reported that he experienced great difficulties speaking on the phone, but was able to cope relatively well face to face unless the speaker “had a foreign accent.” Live-voice word (lists of 50 words) testing yielded scores of 16% for the right ear only, and 38% for binaural presentation. Despite the obvious difference between these two scores, the client declined testing using the left ear only, as he felt this would be too difficult. Two 5-minute speech tracking sessions were presented in the auditory-visual and auditory-only conditions. The client was aided binaurally for these sessions, which yielded mean scores of 85.1 wpm (AV) and 70.8 wpm (A).

The decision for the client to receive a CI in his left ear was made soon after this visit. He then attended several training sessions in which the primary focus was on training him to use his right HA. This training continued until the activation of the CI. Figure 24-4 shows the TRs obtained in nine 5-minute tracking sessions in the AV and AV conditions using the right HA only. The mean scores for the AV and A only conditions were 94.3 wpm and 53.6 wpm, respectively.

Following activation of the CI, the focus shifted to training in the CI-only and CI+HA conditions. At this point, the subject started to attend for two or three 2-hour training sessions per week. The materials used for CI only training included “*homeheARing*” (Plant, 2017), a training program presenting both analytic and synthetic materials, and an adaptation of *The Emperor’s New Clothes* for modified speech tracking. The client’s scores for the modified speech tracking task over the first 19 parts of the story are shown in Figure 24-5. This shows a rapid improvement, resulting in scores approaching 100%. At this point, the decision was made to discontinue the use of the material as it was no longer challenging for the client.

Speech tracking was the major training provided for the CI+HA condition. The subject’s TRs in the AV and A-only presentation conditions for twenty 5-minute speech tracking sessions are shown in Figure 24-6. These show a rapid performance growth, especially in the A-only condition.

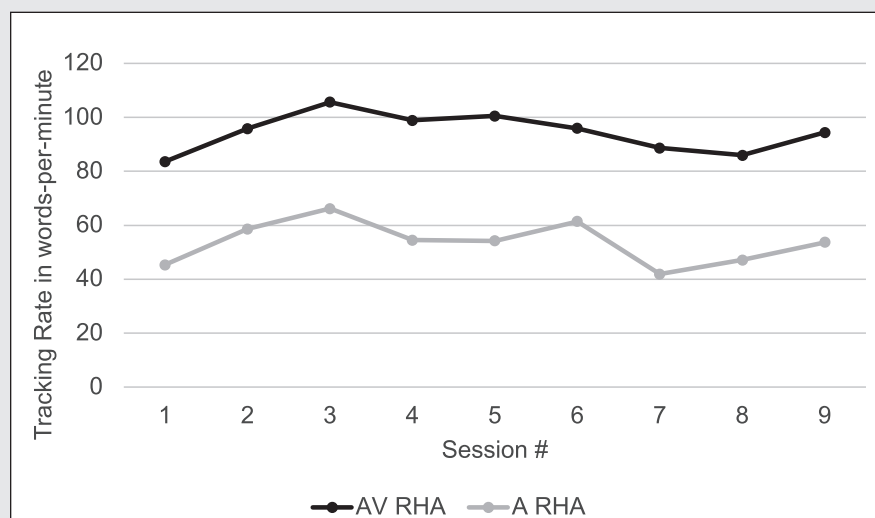


Figure 24-4. Auditory-visual and auditory-only TRs for the subject in Case Study 24-2 using his right HA only.

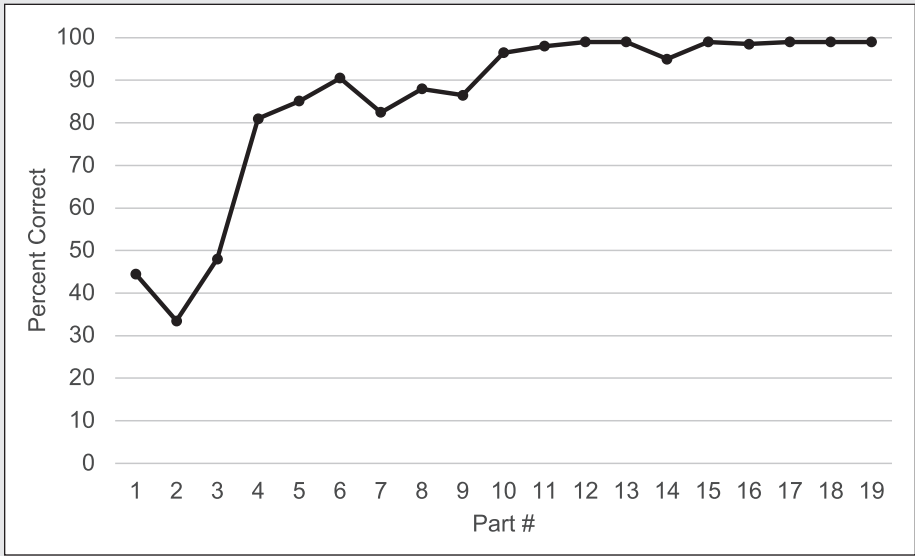


Figure 24-5. Auditory-only results obtained by the subject in Case Study 24-2 for a modified speech tracking task using only the CI in his left ear.

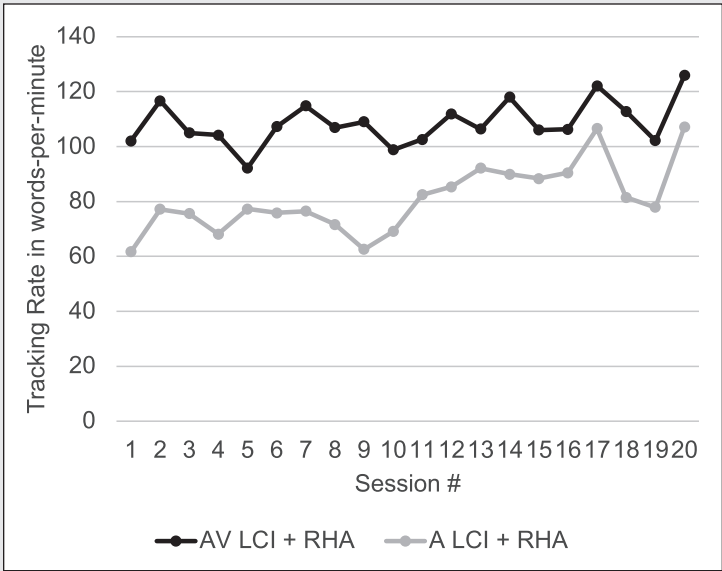


Figure 24-6. Auditory-visual and auditory-only TRs obtained by the subject in Case Study 24-2 for 20 × 5-minute tracking sessions. The materials were presented with the subject in the CI and HA condition.

Figure 24-7 shows the client's AV and A-only performance using the CI only. The client's performance improves rapidly over the first 3 hours of training, and then remains at around 80 wpm for the next 4 hours. There was a slight decline in performance in the sixth and seventh hours, and this period coincided with the client receiving a CI in his right ear. The client noted that speech sounded "a little lower than normal" following the operation on his right ear. He had previously reported that he could hear a talker on the phone in

his unaided right ear, although the signal was not sufficient to allow him to understand what was said. It is postulated that he was receiving some low-frequency speech cues in his right ear when the hearing aid was removed, and this information was lost following the insertion of the CI.

Once the CI in the client's right ear was activated, the training program focused on the bilateral condition. Figure 24–8 shows the client's performance over twenty 5-minute speech tracking sessions. The information provided by the two CIs resulted in a higher performance level than that seen for the left CI only.

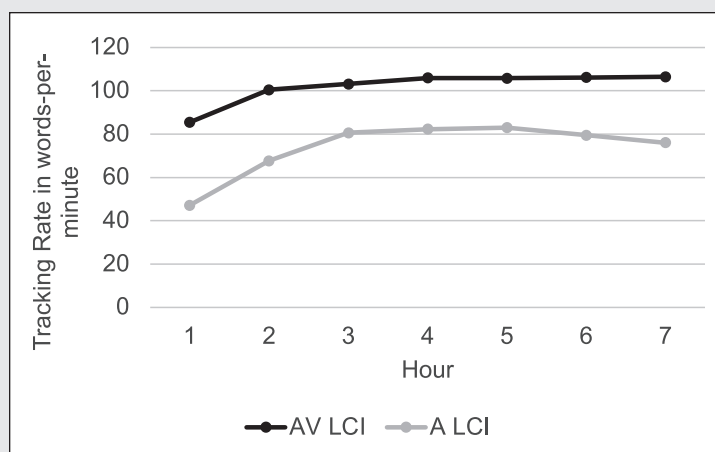


Figure 24–7. Auditory-visual and auditory-only speech tracking scores obtained by the subject in Case Study 24–2 with the CI only. Each datum point represents the subject's mean score for 12 × 5-minute (1-hour) tracking sessions.

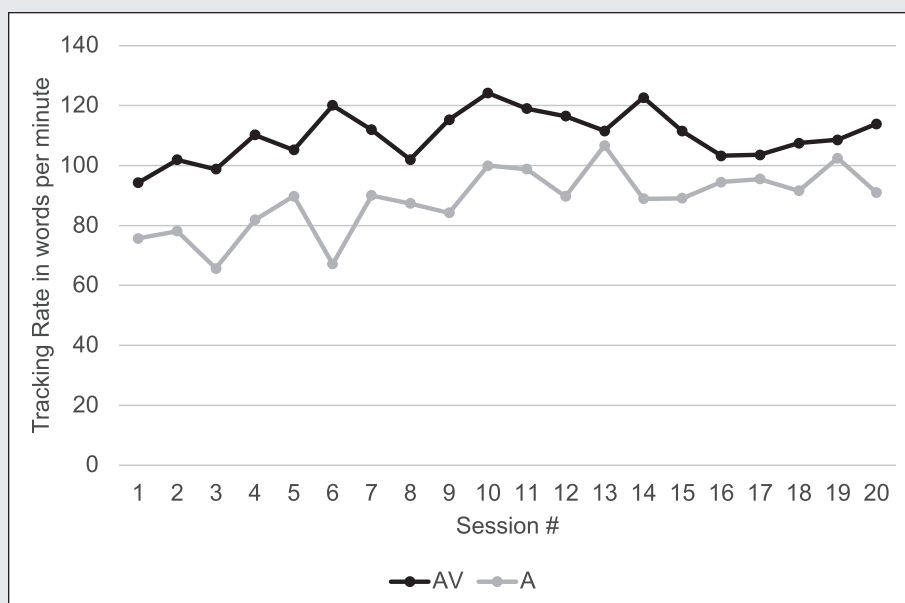


Figure 24–8. Auditory-visual and auditory-only scores obtained by the subject in Case Study 2 using bilateral CIs.

Conclusion

Speech communication training provides clients with hearing loss the opportunity to develop their ability to understand speech via CIs and/or HAs. The two case studies presented in this chapter each showed improvement in their speech perception skills, and their results support the value of training. The major problem, however, is that these services are time intensive and long term. It is recognized that it may not be possible to institute such programs in most publicly or privately funded institutions, but these results do show the value of the approach.

In a previous version of this chapter (Plant, 2014), I mentioned my interest in examining the effects of short-term intensive training. I took part in several such programs at the National Acoustic Laboratories in Sydney, Australia, in the late 1980s and early 1990s. Clients attended for 5 hours a day for 5 days, and, during that time, took part in group activities as well as receiving one-on-one training from a number of different clinicians. Almost all of the people with hearing loss who attended these sessions reported that they gained a great deal from the experience, but no formal attempt was made to evaluate the outcome. Since that time, I have taken part in three 4-day intensive training programs held at the University of Connecticut and at the HRF between 2016 and 2018. The responses of the clients who attended these courses were extremely positive, and indicated that they had derived a great deal of benefit from the experience (Cienkowski et al., 2017). One of the participants (Bouton, 2016) has written about her experience at the first intensive training course.

Computer-based training programs, which clients can use either in the clinic or in their own homes, should also be evaluated to determine their overall utility and effectiveness. There are a number of such programs currently available, but Listening and Communication Enhancement (LACE) (Sweetow & Henderson Sabes, 2006), which provides both synthetic and analytic training exercises, has been extensively used and evaluated.

Sweetow and Henderson Sabes (2006) reported results obtained with a group of adult subjects who worked with LACE for 30 minutes per day, 5 days a week, for 4 weeks, a total of 10 hours training time. Pre- and post-training testing showed improvements in a number of objective and subjective test measures.

A further study reported by Taylor and Shrive (2008) looked at the effect of providing LACE training on hearing aid returns. Clients were encouraged to complete nine LACE sessions at the hearing aid

dispensing office, and were also offered the opportunity to continue this training via the internet. The results were very positive, with almost all (93/96) of the group opting to keep the HA, but it was found that around one-third of the group did not complete the nine sessions of LACE training. A later study by Sweetow and Sabes (2010) found that only around one-third of subjects completed the home training program for LACE.

Compliance issues aside, it would appear that there are major obstacles to the use of programs such as LACE in a client's home. These include her/his access to a computer, suitable speakers, and so on, and, as McCarthy and Schau (2008) point out, LACE "requires the client to have a certain degree of computer literacy" (p. 171).

Mention should also be made of the cLEAR, an online listening training resource developed by Nancy Tye-Murray and colleagues at Washington University in St. Louis, MO (Tye-Murray et al., 2012, 2016). This program provides listening training using a variety of materials and talkers, and can be tailored to include the listener's family and friends as presenters of the exercises.

At this time, I believe that programs such as LACE and cLEAR are an extremely valuable part of an overall approach to speech communication training, but should not be seen as a replacement for one-on-one training conducted by an experienced clinician. "Live" training provides opportunities for the person with hearing loss to engage in communication situations that closely mirror real-life conversations, and offer the opportunity to practice and improve skills in a supportive environment.

Another approach that merits further examination is that of providing opportunities for family members and friends of the person with hearing loss to act as trainers. Programs such as "Hear at Home" (Plant, 2000) offer materials and suggestions for significant others to provide training for people with hearing loss. Schum's (1996) work on teaching family members and friends how to use Clear Speech (Picheny, Durlach, & Braida, 1985, 1986, 1989) has shown the value of providing training for significant others, and such intervention can result in improved communication opportunities for people with hearing loss. *homeheARing* (Plant, 2017) was developed to help family members who wish to provide training opportunities for individuals fit with CIs and/or HAs. The materials include both analytic and synthetic training exercises with PowerPoint slides used to provide feedback, orient the listener to the task, and provide feedback as to correctness of response.

All of these approaches recognize that the mere provision of technology, be it HAs, CIs, or other form of assistive technology, does not solve all of the receptive communication problems of people with hearing loss. Many people with HAs and CIs are dissatisfied with an approach that implies technology is all that is needed to overcome the communication difficulties. Speech communication training is not a new idea, and it does have several drawbacks, not the least of which are related to funding. But it provides a valuable service to many people with hearing loss, and needs to be continued and, if possible, expanded to help people with hearing loss maximize the benefits they receive from CIs and/or HAs.

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25

VOCATIONAL ISSUES FOR PERSONS WITH HEARING LOSS

Sophia E. Kramer and S. Theo Goverts

Introduction

Work has a central role in most people's lives. People need to work to provide an income, but they may also choose to work because they value the social and psychological benefits of it, such as satisfaction, sense of achievement, avoidance of boredom, social functioning and support, social identity, and social status (Shepherd, 1989). The World Health Organization (WHO, 2001) identified participation in work as one of the major areas in life (see also Stephens & Kramer, 2010). Although audiologists may be well aware of the importance of work for persons with hearing loss, they may be less certain about the particular challenges encountered by persons with hearing loss, the demands of a realistic work setting, and what aspects to cover in the problem-solving process. This chapter presents relevant demographics related to hearing and work and illustrates how the workplace can be a challenging context for adults with hearing impairment. Also, examples of vocational rehabilitation (VR) programs to facilitate optimal participation of workers with hearing impairment in the workplace that were developed over the last 30 years are presented and discussed.

Prevalence

In 2016, age-related and other hearing loss was identified as one of the 10 most prevalent health conditions

worldwide (Vos et al., 2017). According to the WHO (2018), there are 466 million persons in the world with moderate to severe disability levels of hearing loss. The vast majority of these (93%) are adults. The prevalence of disabling hearing loss in adults aged 15 years and over is 7.6%, and this proportion is rapidly increasing to almost one in three adults (33%) in adults older than 65 years. These figures indicate that hearing loss affects many people who are still active in the workforce. This is confirmed by population survey data provided for several countries. For example, in Canada the mean age of adults reporting hearing loss is 51 years (Fok, Shaw, Jennings, & Cheesman, 2009). In the United States, communication problems (hearing, language, and speech) are estimated to have a prevalence of 5% to 10% in the labor force (Ruben, 2000). The National Center for Health Statistics (NCHS; <https://www.cdc.gov/nchs/index.htm>) in the United States reports a still larger percentage rate. According to the NCHS, 17% of the population aged 21 to 64 years has some degree of hearing loss (Mascia & Mascia, 2008). In the Netherlands, 3% of the labor force report hearing difficulties (Cuijpers, Lautenbach, & Kösters, 2007). A recent study in Sweden revealed that approximately 32% of the Swedish working population suffer from either hearing problems or tinnitus (Hasson, Theorell, Westerlund, & Canlon, 2010).

It is expected that hearing impairment in the workplace will become much more common in the near future due to changes in the workforces in modern Western economies. First, there is the rapid

growth in the elderly populations worldwide. An important consequence of this development is a drastic decrease in the number of workers per retiree in the society, putting high strains on national pension funds, government support, and health care systems. One of the measures taken by governments to manage this problem is increasing the retirement age for pensioners. This will unavoidably lead to larger number of adults who continue to work at a later age, which, in turn, will lead to a higher prevalence of hearing impairment in the workforce in the near future. Another change is related to a revolutionary shift in the labor tasks, which was observed during the last half of the 20th century. According to Ruben (2000), at the beginning of the 20th century, at least 80% of the American workforce was employed in tasks that depended on manual skills, whereas this proportion drastically decreased at the end of that century. By that time, the proportion of workers holding white collar jobs composed 62% of the U.S. workforce. Ruben argued that in the communication-age societies of today, occupations rely more and more on communication skills such as hearing, voice, and language. Other, more recent changes in the labor market include trends toward greater mobility and to more casual, freelance, and contract work (Punch, 2016). The increasing dependence on cloud-based technology has influenced the shift toward remote work and a more independent and self-motivated workforce. Whereas remote work may offer great opportunities for people with hearing loss, it also brings challenges. The skills and flexibility to manage and advance one's own career, seek new work, and motivate and promote oneself are more important than ever before, and these issues may imply a greater burden on groups of people suffering from communication disorders (Punch, 2016). Several studies demonstrated that people with hearing impairment in the labor force comprise a vulnerable group for whom the conditions at work are more challenging than for normally hearing colleagues (for reviews, see Danermark, 2005, and Punch, 2016). Issues such as unemployment, underemployment, sick leave due to stress, low income, and early retirement seem to be more pervasive among workers with hearing difficulties than workers without this disability (Stam et al., 2013). The following paragraphs will discuss these work-related issues of hearing impairment.

Unemployment

Studies have found that hearing impairment is correlated to a greater likelihood of unemployment.

Specifically, a study in Denmark revealed that about 30% of young adults (20–35 years of age) with hearing impairment were unemployed, whereas the level of unemployment was much lower (12%) in the age-matched reference group without hearing impairment (Parving & Christensen, 1993). Rydberg, Gellerstedt, and Danermark (2010) explored the position of deaf and hard-of-hearing people in the Swedish labor market and found a 63% employment rate for those who had attended a special school for the deaf (including hard-of-hearing adults), a rate significantly smaller than the employment rate for the normal hearing reference population (78%). Similar patterns were reported by Schroedel and Geyer (2000), with the deaf and hard-of-hearing alumni experiencing more unemployment (15%) than their similarly educated peers without hearing problems (10%) 15 years after college. Data from the U.S. Bureau of Labor Statistics (BLS) revealed an unemployment rate of 41.9% among adults with communication disorders as opposed to 29.5% for the working-age population without this disability (Ruben, 2000). Also Hogan, O'Loughlin, Davis, and Kendig (2009) in Australia found that people with all levels of measured hearing loss (mild, moderate, or severe) were less likely to be employed when compared to the general population. It must be noted, however, that the proportion of people without work in the Australian study was much larger for those reporting a main disability other than hearing loss (back problems, arthritis and related problems, and psychiatric disorders). When hearing loss itself was associated with employment difficulties, the effects appeared to be more prominent when people were also rendered more vulnerable in the labor market by other factors such as communication difficulties, older age, gender (being female), or low educational attainment (Hogan et al., 2009). In a longitudinal study among 1,888 adults, Stam et al. (2013) observed that participants with poorer hearing were less likely to have paid work and more likely to look for work or to be unfit for work.

Underemployment or Employment Restrictions

Whereas unemployment is shown to be prevalent among workers with hearing impairment, it is likely that issues such as underemployment or employment restrictions are even more common. Population survey data on the prevalence of underemployment are limited in availability. Clogg (1979) defined an underemployed person as "one whose abilities or educational credentials are higher than those usu-

ally required for the job in which he or she presently works" (pp. 222–223). Many of the professionals in the study performed by Tye-Murray, Spry, and Mauze (2009) indeed recognized this condition. The professionals with hearing impairment expressed a sense of having lost their competitive edge. They believed they had lost out on promotions or lost job opportunities due to their hearing loss. Maintaining job competency and/or a competitive edge emerged as a central component in their focus group data.

Restrictions in the type of work one can undertake and difficulty with changing jobs or getting the preferred job are also reported in the literature. Among the group of workers with reported hearing loss and communication difficulties, 33.8% reported being able to work only a restricted number of hours per week (Hogan et al., 2009). Studies of people who are deaf-blind also support the notion that having a physical disability is negatively associated with competitive employment (Ehn et al., 2016). Nachtegaal, Festen, and Kramer (2012) reported a significant association between hearing ability and experienced limitations in the kind or amount of work one can do. With every dB worsening of hearing ability, the odds of experiencing limitations at work increased significantly.

Sick Leave

A study by Kramer, Kapteyn, and Houtgast (2006a) revealed that normal hearing workers and employees with hearing impairment differed significantly in the amount of sick leave they take. In the group with hearing impairment, the proportion of employees reporting sick leave during a period of 12 months was significantly higher (77%) than in the normal hearing group (55%). This appeared to be exclusively due to a higher proportion of people reporting sick leave due to stress-related complaints (fatigue, strain, and burn-out) in the group with hearing impairment (26%) than in the normally hearing group (7%). When examining the proportion of sick leave due to usual reasons (flu, a cold, or a hospital appointment) the proportions in both groups were equal, indicating that people with hearing impairment are not different from their normally hearing colleagues in that respect.

Low Income

In Australia, more than one-quarter (26%) of people with self-reported communication difficulties due to hearing loss reported being in the bottom three income groups (up to AUD\$224 per week) compared

with 16% of people without hearing loss (Hogan et al., 2009). Schroedel and Geyer (2000) noted that in all educational levels, the deaf and hard-of-hearing alumni in their survey earned less than the college graduates without hearing loss. According to Kochkin (2007), individuals with severe hearing loss earn USD\$12,000 less per year than individuals with mild hearing loss. Stam et al. (2013) found that participants between 18 and 64 years old with poorer hearing ability were less likely to be found in the upper categories of educational level and income than people with better hearing.

Early Retirement

Finally, hearing impairment seems to be related to a greater likelihood of early retirement. In Sweden, the burden on groups of people with communication disorders led to a disproportionate percentage of employees taking early retirement. Danermark and Gellerstedt (2004) reported that, whereas only 5.4% of the general population reported having a hearing impairment, approximately 12% of those taking early retirement were hard of hearing. This indicates an overrepresentation of hearing impairment in that group. The option of early retirement from work as a way to overcome weariness and a marker for becoming resigned to the limitations imposed by hearing loss in the workplace was also seen as a real possibility by the participants in the studies by Fok et al. (2009) and Tye-Murray et al. (2009). In Stam et al.'s study (2013), however, a positive association between hearing impairment and taking early retirement was not found.

Costs

Aside from prevalence figures of hearing impairment in the workforce and the accompanying factors characterizing the vulnerable position of individuals with hearing loss in the workplace, there is the issue of societal and organizational costs of hearing impairment. The adverse work-related consequences of hearing impairment described in the previous paragraphs may be related to high costs for companies and for the society. A few studies provided data related to this issue. For example, Ruben (2000) estimated the combined costs of communication disorders (i.e., costs for special education for children with communication problems from birth to age 18, unemployment, and underemployment) as 2.5% to 3% of the Gross National Product in the United States, which means

a significant loss to the U.S. economy. Mohr et al. (2000) demonstrated that the largest proportion of the societal costs related to hearing impairment is due to reduced work productivity.

The facts and numbers described above clearly illustrate the burden of hearing loss in the workforce. These data, together with the changes being observed in workforces, indicate that the number of clients in clinics requiring help for work-related difficulties may increase in the near future. The challenge for professionals in audiology is to foster better understanding of the specific barriers for workers with hearing loss and to provide rehabilitation services that address their specific needs. The next sections describe how hearing impairment may affect participation in work and present examples of rehabilitation programs to support workers with hearing loss, thereby improving their working situations.

The Workplace as a Challenging Context

The relationship between hearing and work is mostly associated with the risks of exposure to occupational noise and the effects on workers' health. Loud noise can cause temporary or permanent hearing impairment. The underlying mechanisms of noise-induced hearing loss have been studied extensively, both in animal models and in human populations (Job et al., 2009; Ohlemiller, 2008). Ample knowledge and information exists on how to diagnose and monitor noise-induced hearing loss (e.g., Lutman, Coles, & Buffin, 2016), reduce excessive noise at work, institute hearing conservation programs, and provide hearing protection devices (Nelson et al., 2005). There is no doubt that attention to these issues is relevant because hazardous occupational noise levels remain a problem in all regions of the world (Concha-Barrientos, Campbell-Lendrum, & Steenland, 2004; Flamme et al., 2012; Tikka et al., 2017). However, the number of people with noise-induced hearing loss is relatively small compared to the number of employees with other types of hearing impairment.

Consequences of Hearing Impairment at the Workplace

Existing or acquired hearing impairment (even mild degrees), regardless of its origin, may adversely affect

occupational performance and well-being. People with hearing impairment are more sensitive to background noise than people with normal hearing, predominantly due to a distorted peripheral processing of signals (Houtgast & Festen 2008; Plomp & Mimpen, 1979). This may lead to unfavorable outcomes in speech-in-noise testing with stationary maskers but also with more realistic fluctuating maskers (Soli et al., 2018). With any background noise, people with hearing impairment may experience greater difficulties understanding their colleagues than those with normal hearing do. It should be noted that this may occur in a lot of realistic workplace scenarios and not only in work environments where the ambient noise level exceeds the critical level of 80 dB(A) (Sorgdrager et al., 2006). More generally, the peripheral hearing impairment may lead to reduced monaural and binaural processing of sounds, which in turn may pose more strain on the top-down processing (Goverts & Houtgast, 2010). To date, the effect of hearing aids is evident in restoring audibility but is known to be limited in restoring processing of sounds in noise. Thus, individuals with hearing loss need enhanced signal-to-noise ratios at work. Unfavorable signal-to-noise ratios may compromise their work performance or performance may be achieved at the cost of more mental effort (Pichora-Fuller et al., 2016). This is further discussed in the paragraph below entitled "Mental Distress, Fatigue, and Need for Recovery After Work." In addition to these audiologic factors causing increased sensitivity to background noise and reverberation, there is a range of other factors that may cause difficulties for people with hearing loss in the workplace.

What Specific Difficulties Do Individuals With Hearing Loss Encounter in the Workplace?

The precise implications of hearing loss in the workplace are not fully understood yet (Jennings & Shaw, 2008), and remains a relatively unexplored area of research. With a growing number of studies in recent years, however, some evidence is beginning to emerge. Both qualitative and quantitative studies have been performed and evidence accumulated through repeated observations of phenomena. The most pertinent problems of workers with hearing loss relate to job control, mental distress, fatigue, need for recovery after work, and lack of knowledge. Results

of studies on each of these issues are summarized in the following paragraphs.

Job Control

Kramer et al. (2006a) compared the occupational performance of a group of 151 employees with hearing impairment with that of a group of 60 workers with normal hearing. The groups were matched for age, gender, educational level, and type of job. A questionnaire was used to assess the participants' self-reported environmental conditions at work: type of job, type of contract (permanent versus temporary; full-time versus part-time), general working conditions (job demand, control, support), and activities required for the performance of the job (including communication activities). An analysis of group differences revealed that the reported effort in hearing needed during listening was significantly higher in the group with hearing impairment. No differences between the two groups were found for the general working conditions (demand, support, career satisfaction), except for job control (the possibility to interrupt work or take breaks when wanted). Participants with hearing loss perceived themselves to have significantly less control at work compared to their normally hearing colleagues. It is worth noting that these findings very much support the results of comparable investigations, such as a study by Danermark and Gellerstedt (2004) in a group of 445 employees who completed a questionnaire on the psychosocial environment at work. The scores were compared with reference data obtained from more than 8,000 people employed by local municipalities. The results showed that the psychosocial environment at work was much more demanding for the hard-of-hearing group than for the normally hearing reference group. The most important difference appeared to be the lower self-assessed degree of control at work for those with hearing impairment.

Perceived sense of control also appeared an important factor in the disclosure of hearing loss in the workplace (Southall, Jennings, & Gagné, 2011). This sense of control was mostly influenced by the predictability of work-related tasks and the balance of power between the worker with hearing impairment and his/her coworker. People with hearing impairment apparently feel a greater need to have control over their work (i.e., organizing their own schedules so as to be able to take breaks or interrupting work after having had auditory demanding activities) (Backenroth, Ohasko, Wennberg, & Klinte-

berg, 2003). Providing control over their work for employees with hearing impairment enhances their well-being.

Mental Distress, Fatigue, and Need for Recovery After Work

Higher sick leave rates among workers with hearing impairment in the study by Kramer et al. (2006a) appeared to be exclusively due to a higher prevalence of stress-related complaints (fatigue, strain, and burnout) in the group with hearing impairment (26%) compared to that in the normally hearing group (7%). Along the same line, Nachtegaal et al. (2009) observed higher levels of need for recovery after work among those with poor hearing ability compared to individuals with good hearing ability. Need for recovery is an indicator of mental distress. It refers to the degree to which employees are able to recover from fatigue and distress after a day of work. It is an important factor influencing an individual's physical and mental health status. In occupational health care, need for recovery after work is seen as an acute short-term reaction to work. Exposure to stressful psychosocial working conditions, combined with repeated inadequate recovery after work, may lead to psychosomatic health problems (Sluiter, de Croon, Meijman, & Frings-Dresen, 2003). The extended Job Demand Control (JDC) model by Karasek and Theorell (1990) offers a helpful framework for understanding the mechanisms behind the development of stress, fatigue, and strain in the workplace. According to the JDC model, high psychological job demands combined with a lack of control are associated with psychological strain and health problems (Karasek et al., 1998). Nachtegaal et al. (2009) found that for every dB signal-to-noise ratio (dB SNR) point scored more poorly on a hearing test, the need for recovery increased by 1.4 percentage points in a group of 925 workers, of whom 430 had limited hearing ability. In their later study, Nachtegaal et al. (2012) found the need for recovery to be a mediator in the relationship between hearing ability and sick leave. Examples of items describing the need for recovery after work are "At the end of the day I am really feeling worn out" and "I find it hard to relax at the end of a working day."

The findings of Nachtegaal et al. (2009, 2012) confirm what has been documented in a range of other studies. For example, the workers in the study by Backenroth et al. (2003) reported that they were striving to do a good job and felt that they did so to

an even higher extent than did the normally hearing colleagues, but that this was at the cost of great fatigue at the end of the working day. Similarly, a few participants in the study by Tye-Murry et al. (2009) explicitly indicated they had to work twice as hard to compensate for having hearing loss and to “go above and beyond.” They described how tiring such a “double duty” can be. The participants in Grimby and Ringdahl’s (2000) study expressed their concern about stress and strain resulting from conversations in the coffee room or their offices that contributed to exhaustion both at and after work.

Hasson et al. (2011) used physiological measures to investigate the relation between stress and hearing loss. They found that symphony orchestra musicians with hearing problems exhibited lower heart rate variability than those without hearing problems. A lower heart rate variability indicates an impaired ability to unwind from stress due to the inability to activate the parasympathetic nervous system.

It is encouraging to see that the mounting number of studies all point in the same direction. Apparently, mental distress and fatigue among workers with hearing loss cannot be neglected in clinical practice and vocational rehabilitation services. All studies unanimously emphasized the relevance of addressing these matters. Stakeholders (e.g., occupational physicians, otolaryngologists, general practitioners, audiologists, employers, and the employees themselves) should also be aware of the fact that hearing loss may be the underlying cause of fatigue.

In most cases hearing loss develops gradually over time; therefore, monitoring an individual’s need for recovery after work in clinical practice or in the workplace may be considered a useful measure. Thus, if employees present themselves to their general practitioner (GP) or occupational physician with complaints of fatigue and mental distress, hearing loss should be considered as a possible cause. Early detection of enhanced levels of need for recovery will identify patients at risk and enable stakeholders to take timely actions. Such measures can contribute to the prevention of long-term sick leave.

Lack of Knowledge or Understanding

Whereas mental distress and auditory fatigue are critical correlates of hearing loss in the workplace, employees with auditory difficulties expressed additional factors that limit their participation in work. One is the lack of knowledge among colleagues, supervisors, employers, and others about the manifestations and

impact of hearing loss in the workplace or appropriate workplace accommodations (Detaille, Haafkens, & van Dijk, 2003; Héту & Getty, 1991; Héту, Getty, & Waridel, 1994; Punch, 2016; Schoffstall, Cawthon, Tarantolo-Leppo, & Wendel, 2015). This phenomenon is aptly expressed in an article by Southall et al. (2011). One of the interviewees in that study reported a profound social exclusion in the workplace due to the fact that the general population does not understand the concept of partial hearing loss. While both normal hearing and clinical severe to profound deafness are apparent, partial hearing loss is not. Additionally, interviewees reported that coworkers simply do not know intuitively how to communicate with people who have a hearing loss. They easily forget about the hearing loss, partly because it is invisible.

Lack of knowledge or awareness of suitable workplace accommodations also exist among workers with hearing loss themselves. This was one of the outcomes of an integrative review of the literature on experiences of barriers, accommodations, and stress in the workplace (Punch, 2016), as also found by Tye-Murray et al. (2009). In some of the focus groups in the latter study, asking the question “What services are available to you through work?” led participants to talk about the Americans With Disability Act (ADA). Twenty of the 48 interviewees indicated they were familiar with the ADA. However, when they were asked the follow-up question “What do you know about the ADA?” many of them knew little about it or were misinformed about its provisions. Furthermore, a survey among workers with hearing impairment in the United States revealed that most consumers lack information regarding available technology (Mascia & Mascia, 2008). It must be noted that reluctance to request accommodations on the part of workers with hearing loss has also been reported, particularly among those who are older, those with lesser degree of hearing loss, those who perceive their work environment to be unsupportive, and those in part-time work (Punch, 2016). Reluctance to request appropriate accommodation may also occur when persons prefer not to disclose their hearing loss at work because they fear being stigmatized and perceived as incompetent.

Stigmatization

Data on stigmatization in the workplace are less straightforward or unanimous than those on mental effort. According to Southall et al. (2010), stigma is defined as “the possession of, or belief that one possesses an attribute or characteristic that conveys a

social identity that is devalued in a particular social context” (p. 804). Héту (1996) and Héту et al. (1994) have extensively studied stigma in the workplace. Stigma may be related to embarrassment, lack of confidence, a sense of inferiority, and shame. Focus group interviews of Héту et al. (1994) revealed that at least some coworkers of people with noise-induced hearing loss perceived hearing impairment as a weakness and associated it with a lack of capacity. Whereas most of the coworkers in that study showed compassionate understanding about the difficulties posed by the hearing impairment, they appeared to be unfamiliar with the manifestations and consequences of it (see previous paragraph). In contrast to the results of Héту et al. (1994), Tye-Murray and colleagues (2009) did not find much evidence that the professionals in their study were experiencing stigmatization in the workplace. (For further information on stigma, the reader is referred to Chapter 4 of this text.)

A recent study of Koerber, Jennings, Shaw, and Cheesman (2017) is also relevant to mention in this context. The goal of this study was to analyze how the Canadian newspapers frame workers with hearing loss. The results of their thematic analysis revealed a rather positive framing. Newspapers writing about workers with hearing loss most frequently presented an image of workers cheerfully striving toward a good work life.

Lack of Support or Adequate Services

Another barrier that people with hearing impairment frequently experience in the workplace is an overall lack of timely coordination of services. Using narratives describing the lived experiences of persons in their 50s with hearing loss, Jennings and Shaw (2008) identified the lack of support to help workers with hearing loss manage barriers and the lack of recognition of the extent of the hearing problem due to its invisibility. Just as in Kramer’s study (2008), lack of recognition of the multidimensional needs and lack of timely coordination of services was reported. In identifying the gaps in existing services, Jennings and Shaw (2008) emphasized the need to begin questioning traditional approaches in aural rehabilitation for workers with hearing loss. A similar conclusion was drawn by de Graaf and van Bijl (2002), who conducted a large survey among Dutch deaf and severely hearing-impaired people on the use and need of health care. Counseling on how to handle one’s own hearing loss in the workplace was one of the most frequently reported care needs.

Concern and Anxiety

Throughout the range of studies focusing on the consequences of hearing impairment in the workplace and the possible barriers that professionals with hearing loss experience, there was one overriding factor: concern. It seems as if a troubled or anxious state of mind is predominant among workers with hearing loss. Backenroth et al. (2003) observed significantly higher levels of anxiety among workers with hearing loss compared to their control group. Concern about a range of issues was expressed. To illustrate, in Tye-Murray et al. (2009), the participants expressed concern about maintaining their job competency. The employees’ primary concerns in Morata et al. (2005) were about job safety as the result of a reduced ability to hear ambient sounds and warning signals in a noisy environment. Also, their inability to monitor essential equipment sounds was a concern. Other concerns encountered in the literature and through personal observation are about future employability, perceptions of coworkers and supervisors, stress and fatigue, reduced work productivity, reduced communication abilities, and social isolation in the workplace (Grimby & Ringdahl, 2000; Kramer, 2008; Morata et al., 2005; Tye-Murray et al., 2009).

Implications for Vocational Adult Aural Rehabilitation

As outlined in the previous paragraphs, the research findings so far indicate that, for many workers with hearing impairment, even after hearing rehabilitation, the conditions at work are more challenging than for normally hearing colleagues. This brings us to the question of what is needed to accommodate people with hearing loss in the workplace. Several issues that ought to be addressed in vocational rehabilitation for workers with hearing impairment have already been mentioned in this chapter and include the following examples:

1. Equipment and environmental management:
 - Appropriate hearing aid fitting
 - Appropriate hearing-assistive technology
 - Environment management strategies, including reducing background noise and reverberation
 - Timely and integrated approach
2. Awareness for hearing colleagues:
 - Lack of knowledge among employers, employees, and coworkers

- Support from employers, employees, and coworkers
 - Strategies on disclosing hearing loss in the workplace
3. Psychosocial issues:
- Need for some time off from work
 - Ongoing supervision or assistance
 - Need for control over work
 - Mental distress and fatigue
 - Maintaining and building self-esteem, self-confidence, and self-advocacy skills
 - Empowerment and assertiveness training
 - Communication strategies
4. Information on current legislation

The list above clearly illustrates that the suggested issues requiring attention in vocational rehabilitation refer not only to the content of care, but also to its logistics and management. It is an illustration of the complexity of vocational rehabilitation. To handle vocational problems successfully for persons with hearing loss, a multidisciplinary, integrated approach is needed.

What Is Needed Is a Multilevel, Interdisciplinary, Integrated Approach

The workplace is a very complex environment with many different actors and factors playing a role. Hence, for an adequate assessment of a hard-of-hearing employee's limitations and the consequences for work performance, a multifactorial understanding is required, just as outlined in the International Classification of Functioning Disability and Health (ICF) framework (WHO, 2001) and applied to the domain of vocational auditory adult rehabilitation in Figure 25–1 (Kramer et al., 2009). A person's disability and functioning should always be evaluated within context. In other words, functioning and disability are viewed as outcomes of interactions between an individual's health condition (e.g., hearing impairment) and contextual factors. Among contextual factors are environmental factors (e.g., room acoustics, colleagues, tasks, schedules) and personal factors (e.g., age, cognitive capacities, visual ability, coping styles, educa-

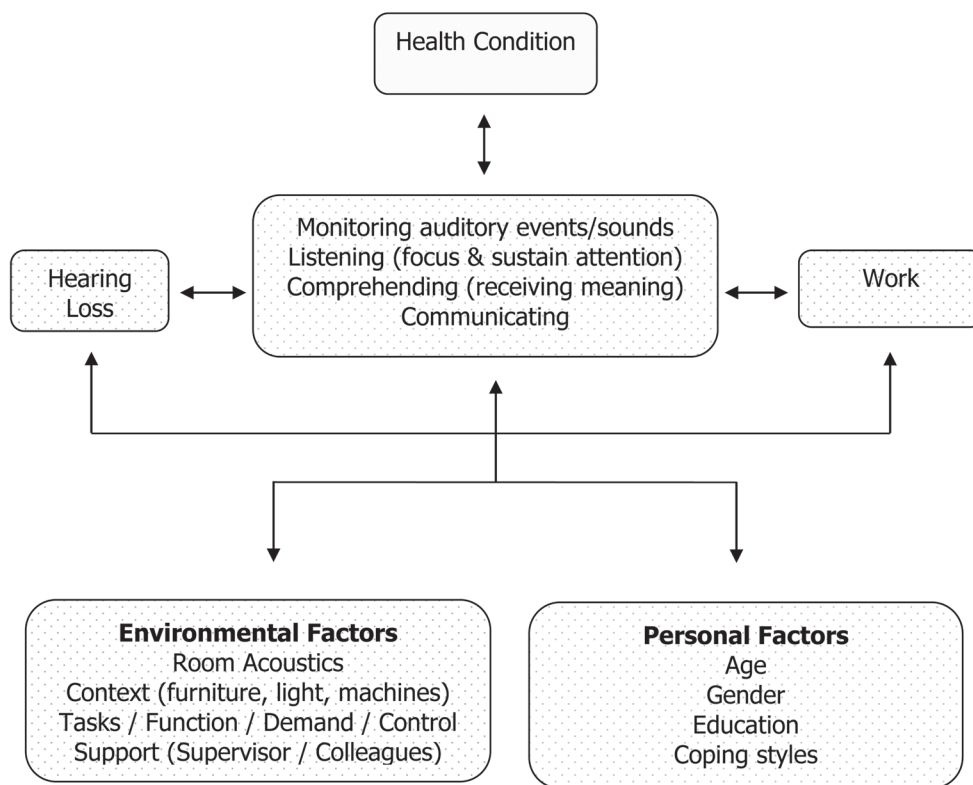


Figure 25–1. ICF framework illustrating factors acting on work functioning. *Note.* The figure illustrates how the functioning at work of a person with hearing loss is viewed as an outcome of interactions between the person's health condition and contextual factors according to the ICF framework.

tion). A compelling example showing the importance of taking personal factors into account is the case of Dual Sensory Loss (DSL). Lacking visual information reduces the amount of nonauditory information that may be needed to reconstruct the auditory scene; specifically, speechreading is hampered.

In considering such an integral ecological context, not only an individual's health condition, but also other personal factors play a prominent role and should be addressed in the design of diagnostic test protocols and vocational programs aimed at improving and facilitating the employment situation. (For further information on the ICF, the reader is referred to Chapter 3 of this text.)

This is also acknowledged in the audiologic-ergonomic (AE) framework of Jennings et al. (2010) that can be considered a further expansion of the ICF with additional levels of influence: the macro, meso, and micro level, each with its specific implications. The *macro level* refers to the hearing and communication systems in which the worker with hearing impairment participates. It is made up of all the situations where a person with hearing loss participates. The macro level also covers the availability of financial resources to support the accommodations of (re) design of the work contexts (Fok et al., 2009), as well as the political, legal, and economic organization of a country. The *meso level* refers to the immediate workplace context or situation of the occupational tasks being performed that require hearing demands. There are four hearing components identified in the meso level of the AE framework: monitoring, listening, comprehending, and communicating. In addition, one can identify the culture in the workplace as a meso level factor. Workplace culture refers to the formal and informal rules set there, as well as the company's or institution's infrastructure, safety and security issues, deadline pressures, and exposures to unanticipated emotional, confrontational, or situational distractions (Fok et al., 2009). Finally, the AE framework identifies the *micro level*. This level refers to the personal resources, skills, knowledge, perceptions, and attitudes relevant to auditory perception, processing, sensing, and behavior in the workplace (Jennings et al., 2010).

Both the ICF and the AE framework draw attention to the need to evaluate the hearing and sensory demands in the workplace and resources of a person within context. It is important to be aware of the existence of the different levels as they may influence the barriers or difficulties workers with hearing loss experience. They may provide opportunities for solutions as well, as acknowledged by Héту and Getty (1991),

who argued that the different causes of a health problem should always be analyzed first, so as to be able to identify the different levels of intervention that can reduce the influence of the causative factors. The problem-solving process—including a functional status assessment, goal setting, and treatment planning—should not only focus on the individual's abilities, but also reveal to what extent the environment can be changed to facilitate the individual's performance. Actions on a single level (e.g., restricting the enablement process to the prescription of hearing aids) would have little effect.

Interdisciplinary Team Approach

Similarly, restricting the enablement process to the involvement of one single profession potentially ignorant of the demands of a realistic work setting may also be ineffective. Hence, in addition to the multilevel approach as proposed by the ICF (WHO, 2001) and AE frameworks (Jennings et al., 2010), a multidisciplinary approach is indispensable. Such a model of rehabilitation brings together medical, audiologic, cognitive, psychological, social, linguistic, and occupational expertise, with each team member appreciating and knowing each other's expertise and contribution, rather than seeing one another's expertise as distinctly peripheral (Shepherd, 1989). The main feature of an integrated approach is that the clinicians make contact with the stakeholders in the workplace to ensure that the proposed accommodations are actually implemented. Close collaboration is a prerequisite for the success of an integrated program.

Examples of Vocational Rehabilitation Services for Workers With Hearing Impairment

Over the last several years, several attempts have been made to develop rehabilitation programs or services for workers with hearing impairment, dating back to Lalande's work in 1988 (Lalande, Riverin, & Lambert, 1988). Gussenhoven et al. (2013) provided an exhaustive summary of the literature on services described in the literature. In all, nine programs were found and are listed in Table 25-1. A summary of each of the programs follows in the next paragraphs and information available on the effectiveness of the programs is described.

Table 25–1. Vocational Rehabilitation Programs Developed Over the Last 30 Years

Name	Country	Reference
Aural rehabilitation program for workers and their spouses (ARP-Ca)	Canada	Lalande et al. (1988)
Deaf Employment Assistance Network (DEAN)	USA	Myers & Danek (1990)
Rehabilitation program for people affected with occupational hearing loss (RP-OHL)	Canada	Getty & Héту (1991), Héту & Getty (1991)
Audiologic Rehabilitation Program (ARP-Sw)	Sweden	Ringdahl et al. (2001)
Return-to-Work Program (RTW)	Hong Kong	Li et al. (2006)
Vocational Enablement Protocol (VEP)	Netherlands	Kramer (2008)
Vocational Rehabilitation (VR-US)	USA	Mascia & Mascia (2008)
Universal Design for Hearing (UDH)	Canada	Fok et al. (2009)
Canadian Hearing Demands Tool (C-HearD)	Canada	Jennings et al. (2010)

Aural Rehabilitation Program for Workers and Their Spouses (ARP-CA, Canada)

Lalande et al. (1988) developed a 7-week course for workers with noise-induced hearing loss (NIHL) and their spouses, consisting of weekly 2-hour group meetings. The standard format of each meeting included a discussion on the disadvantages caused by NIHL, a formal presentation by the group leader followed by an open discussion, testing amplification devices, role playing, assertiveness training, and presentation of homework assignments. Each meeting had a central theme with the following topics in order of presentation: (a) nonauditory effects of noise, (b) types of auditory impairment and disabilities associated with NIHL, (c) communication strategies, (d) attitudes toward hearing disability and coping strategies, (e) hearing aids, (f) assistive listening devices, (g) tinnitus, and (h) prevention of further NIHL. During each meeting, training was provided regarding relaxation techniques, as well as speechreading. Homework assignments focused on recognizing stress, practicing relaxation exercises, identifying difficult listening situations in the workplace, understanding the spouse's perspective, and training exercises to practice assertiveness and speechreading. The impact of the program was evaluated via a questionnaire. In addition, semistructured

interviews with the participants and their spouses were conducted in the participants' homes. Three out of five participants completed this evaluation procedure. The reported benefits were an increased understanding of hearing impairment and the disabilities due to noise exposure, a better spousal relationship, a greater use of communication strategies in everyday life, purchase of amplification devices, less concern about auditory fatigue, and increased willingness to wear hearing protection at work.

Rehabilitation Program for People With Occupational Hearing Loss (RP-OHL, Canada)

The rehabilitation approach developed by Getty and Héту (1991) was designed as a group program to initiate active search for solutions to listening and communication problems experienced by workers with hearing impairment. The program was developed for workers with hearing loss and their spouses. It entailed three elements:

- psychosocial support to the workers and their spouses,
- provision of knowledge, and
- the development of new skills in coping with hearing loss.

The program was based on Blum's model (1981) about social changes that included two principles. The first is that hearing impairment affects the person experiencing it, but also anyone with whom he or she interacts. Therefore, rehabilitation must consider several levels of coordinated intervention. The second principle is that a change in the attitudes and behaviors of the individuals with hearing impairment, as well others around them, is required (Hétu & Getty 1991). Two modes for the group sessions (with 48 workers and their spouses) were held: an intensive session during the weekend, lasting 1.5 days; or four 2-hour weekly sessions, with the first three sessions occurring on successive weeks, and the last one 2 weeks later. At the final meeting, participants were asked to invite communication partners (relative or friends) to widen the circle of people informed about the effects of occupational hearing loss.

The first session provided information about the functioning of the ear, the audiogram, and stress effects related to hearing impairment. During the second meeting, participants could try out hearing aids and assistive listening devices. Relaxation therapy was also part of that session. Practice in speechreading and hearing tactics was the main element of the third session. During the fourth meeting, review of what had been learned during the previous sessions occurred, as did role play by the participants on assertive behavior, use of a video to help communication, relaxation practice, and extensive feedback from the group leaders. During all meetings, there was the opportunity for discussion.¹ Three months after the participation, the intervention was evaluated. The group intervention generally provided a better understanding of the hearing difficulties and consequently an increased awareness of the problems, together with more confidence in dealing with them. Although Hétu and Getty (1991) acknowledged the potentially unfavorable physical barriers in the workplace (e.g., excessive reverberation, loud ambient noise levels) as obstacles to listening and communication for those with hearing impairment, they felt that handling these issues fell outside the influence of the person with hearing loss. Changing the physical environment as an option to improve the working situation, therefore, was not actively addressed in their program. Also, the program focused entirely on workers with hearing loss and their spouses. The program did not follow an integrated approach by involving stakeholders in the workplace to help find solutions.

Multidisciplinary Audiologic Rehabilitation Program (ARP-Sw, Sweden)

The program described by Ringdahl and his colleagues (2001) encompassed a 4-week, full-time program. The ultimate goal of the program was to enhance the individual's self-image and boost his/her self-confidence. The content of the program was highly similar to that of Getty and Hétu (1991). It started with education about the anatomy of the ear, hearing and hearing loss, hearing aids, assistive listening devices, aural rehabilitation, and legal issues. Selection and fitting of hearing aids were also part of the service. Excursions to restaurants, museums, and theatres were organized to try out assistive listening devices and hearing aids. Participants also received training in relaxation techniques and speechreading. Training in performance occurred with video support, and the role playing was videotaped. As in the program by Getty and Hétu (1991), spouses were invited to participate on two occasions. In contrast with the program by Getty and Hétu (1991), the program by Ringdahl et al. (2001) also included a workplace visit to optimize the situation at work according to the participant's needs. Issues that were evaluated were the ambient noise level, reverberation, lighting, content of the work, and relationships with coworkers and supervisors. If applicable, workplace accommodations were suggested. Disciplines involved in the program were: an audiologist, a sociologist, and an audiological physician (i.e., otologist). Evaluation of this program was done using questionnaires, including the Communication Strategies Scale of the Communication Profile for the Hearing Impaired (CSS-CPHI) (Demorest & Erdman, 1986) and the System Check List 90(R) (SCL90-R; Derogatis & Fitzpatrick, 2004). Thirty-nine participants completed the questionnaires. A significant improvement on the maladaptive behaviors subscale of the CPHI-CSS was observed. No significant effects were found for the remaining CPHI scales or the SCL90-R.

Return-to-Work Program (RTW, Hong Kong)

Li and colleagues (2006) described a tailored return-to-work program specifically designed for workers with occupational hearing loss. The program was structured according to Prochaska and DiClemente's model of readiness for change that assumes a linear

¹For a detailed list and the ordering of all activities, see Getty and Hétu (1991).

process of change that follows five stages: precontemplation, contemplation, preparation, action, and maintenance. The focus of the program was on returning to work after long-term leave and on maintenance of job competency. The program consisted of one to three individual 1.5-hour counseling sessions conducted by a rehabilitation counselor, followed by four structured group sessions. The spouse attended two of the three individual sessions to improve issues within the marital relationship. The group sessions were 3 hours each, and a guest speaker with hearing impairment, who shared his or her successful experiences, was invited for one of the sessions. Li et al. (2006) did not provide a detailed list of activities for the individual and group sessions, but the aim for participants was to review their work values, identify barriers encountered, make decisions on the course of the return-to-work action, and identify their own strengths and weaknesses in preparing for new job demands. After the group sessions, each participant received ongoing support in job matching and job placement for the following 3 months from two occupational therapists.

Six participants completed the return-to-work program. After following the program, the participants increased their work readiness by either showing confidence in returning to work (contemplation stage), getting prepared to look for jobs (preparation stage), or taking action to search for new jobs (action stage). Furthermore, an improvement was found in eight out of nine domains of the SF-36, in particular, on the mean score of the mental health subscale. Three months after the program, three participants had a job and two had received job offers.

Vocational Rehabilitation Program, VR-US (United States)

In the vocational rehabilitation program (VR) as designed by Mascia and Mascia (2008), the Functional Occupational Communication Assessment (FOCA) plays a key role. FOCA provides an analysis of the job tasks and what job tasks are difficult as a result of hearing loss. Five questions are central in the FOCA: (a) Is background noise an issue in the work environment? (b) Is communication difficult? (c) What are the potential workplace hazards? (d) What, if any, accommodations have already been implemented? (e) Is education of supervisory personnel or coworkers

necessary? Based on the answers, the different types of assistive technology available should be demonstrated and considered. Other elements that Mascia and Mascia regarded as essential in VR for adults with hearing loss were educational services, vocational assessment, evaluation and counseling, job training, orientation and mobility training, and job placement. Adequate technology, including hearing aids, should first be provided, followed by information counseling. Depending on the individual needs, the content of such an individualized plan should be determined. Typical components of such a plan are coping with hearing loss, social support, evaluation of economic status, educational background, work demand, and the presence of other disabilities.²

Deaf Employment Assistance Network (DEAN, United States)

The Deaf Employment Assistance Network (DEAN) model, as presented by Myers and Danek (1990), uses an ecologically based framework for rehabilitation interventions for deaf persons. Four dimensions were identified in the model: (a) employment acquisition services, including assessment, pre-employment adjustment training, and placement; (b) employment maintenance, enabling the deaf person to maintain employment, including employment counseling, employee advocacy, and interpreting; (c) employment accommodations services (e.g., sign language, accessibility assistance) that enable the deaf person to achieve maximum effectiveness and efficiency in relation to his/her job environment; and (d) employment enhancement services that assist the deaf person with job mobility and advancement. It includes job (re)training and professional development. The DEAN model³ is a conceptual framework that implicitly acknowledges the importance of addressing both environmental and individual factors. It encompasses a continuum of services to both hearing disabled/deaf people, as well as employers.

Vocational Enablement Protocol VEP (The Netherlands)

The Vocational Enablement Protocol (VEP) developed and locally implemented in the Netherlands

²Information on the effectiveness of this program is not available.

³For a detailed description of the dimensions of the DEAN model, we refer the reader to Myers and Danek (1990). Information on the effectiveness of the DEAN model is unknown.

addresses the specific needs of those who are occupationally involved and who report problems at work due to hearing loss (Kramer, 2008; Kramer et al., 2009). The main aim of VEP is to facilitate the process of maintaining and improving the employment situation of the worker with hearing loss and facilitate this process. It covers five key elements:

1. A comprehensive evaluation of the difficulties and workplace from various perspectives (e.g., employee, employer, occupational physician), including a questionnaire to be completed by the employee before his/her visit to the audiology clinic. Issues addressed in the questionnaire are auditory demands, general working conditions (e.g., job control, job demand, support at work, and career satisfaction), workplace characteristics, need for recovery, and coping. If indicated, the workplace is visited.
2. Diagnostic examination of the patient's auditory performance and coping abilities. This includes a half-day assessment in the clinic, including a semistructured interview by a psychologist and an occupational physician. Audiologic assessment goes beyond standard pure-tone and speech audiometry and also includes speech-in-noise tests, as well as localization tests, if indicated.
3. A multidisciplinary team meeting during which the possibility of technical, acoustical, speech therapeutic, psychosocial, and/or vocational interventions are discussed. Disciplines involved in the multidisciplinary team depend on the nature of the problem reported by the worker with hearing loss, but can include audiology, psychology, social work, speech pathology, otolaryngology, and occupational health. The VEP takes a patient-centered approach in that it actively involves the patient in the problem-solving process.
4. Following the multidisciplinary team meeting, a written report is compiled, including specific recommendations for the patient and the management plan. The report is then sent to the referrer. To clarify the diagnostic findings and to elucidate the reasonableness of recommendations proposed, each referrer is telephoned by the occupational physician of the team. The duty of the occupational physician on site is to communicate with the employer and to monitor and supervise the implementation of the recommendations.
5. Provision of an extensive information package presented on a CD-ROM (Kramer, 2006b). The package provides information on a large number of matters related to hearing and work, includ-

ing communication strategies. It focuses on the understanding of difficulties of working when the individual has a hearing impairment and the various options available to reduce the occupational problems of such people.

Experiences with the protocol in a clinical sample of 86 workers are described in Kramer (2008) and Stephens and Kramer (2010). Briefly, recommendations provided by the VEP were hearing aid (re)fitting, counseling, lipreading training, assistive listening devices, and work-related accommodations (e.g., modifications of environment; redelegation of assignments; restructuring of time schedules; occupational retraining, redeployment, or ending employment; and hearing protection). In most cases, combined advices were offered; for example, hearing aid fitting plus workplace modifications plus additional listening devices.

The effectiveness of the VEP was evaluated in a group of 136 employees with hearing impairment who were randomly assigned to either the VEP or the control group. "Need for recovery after work" was chosen as the primary outcome measure. Secondary outcome measures were communication strategy subscales (e.g., self-acceptance and maladaptive behavior), distress, and self-efficacy (Gussenhoven et al., 2012). The results did not support an effect of the VEP on need for recovery after 12 months, but a significant marginal effect on self-acceptance was found (Gussenhoven et al., 2017). Self-acceptance is seen as the basis for an individual's willingness to recognize and accept limitations at work associated with their hearing loss. Lower levels of self-acceptance are associated with greater likelihood of concealing or "masking" an experienced stigmatized identity in the workplace (Spiegel, de Bel, & Steverink, 2016). Southall et al. (2011) and Hannan (2013) emphasized the need for audiological rehabilitation programs that address identity management strategies specific to concealing and disclosing hearing loss in the workplace. Effects of auditory rehabilitation programs on self-acceptance have rarely been shown.

Universal Design for Hearing (UDH, Canada)

Rather than choosing a traditional approach that focuses on the provision of hearing-assistive technology only, Fok et al. (2009) introduced a more comprehensive approach to address the needs of older workers across the spectrum of work tasks: barrier-

free access or universal design (UD). They said, “barrier-free access is defined as a modification of the built environment such that barriers are removed for individuals with specific disabilities” (Fok et al., p. 367). UD involves designing all products, buildings, and exterior spaces to be usable by all people to the greatest extent possible without special adaptations (Fok et al., 2009). Taking the UD ideology as a starting point, Fok et al. (2009) propose a more comprehensive approach to address the gaps in supporting workplace changes, allowing workers to create a more hearing impairment-friendly workplace. The first key element in that approach is the occupational perspective, as opposed to the hearing protection approach, where reduction of noise levels, environmental controls, and hearing protection are used to enable safe work performance. The occupational perspective entails a detailed description of the occupation itself, the requirements and demands, the available resources to support individuals or groups, and the work contexts (i.e., workplace, tasks). The occupational approach also includes an examination of the barriers and facilitators within the immediate, community, and societal contexts to fully assess the quality and degree of occupational participation, as well

as opportunities for change. In other words, all levels, including the micro, meso, and macro work contexts, need to be explored to identify the feasibility of implementing changes. This was described in more detail in the previous paragraph of this chapter entitled, “What Is Needed Is a Multilevel, Interdisciplinary Integrated Approach.” In their approach, Fok et al. (2009) also addressed the employers by defining five UDH principles (shown in Figure 25–2). The principles are aimed at helping employers achieve a better appreciation of safety and productivity concerns as they relate to occupations or tasks that require hearing.

Canadian Hearing Demands Tool (C-HearD, Canada)

As did Mascia and Mascia (2008), Jennings et al. (2010) argued that a specific tool to evaluate and identify critical auditory perception and communication demands to carry out work tasks is highly needed. Such a tool could assist clinicians, workers, and workplace parties in identifying and describing workplace demands. In response to this need, Jennings and his colleagues developed the Canadian

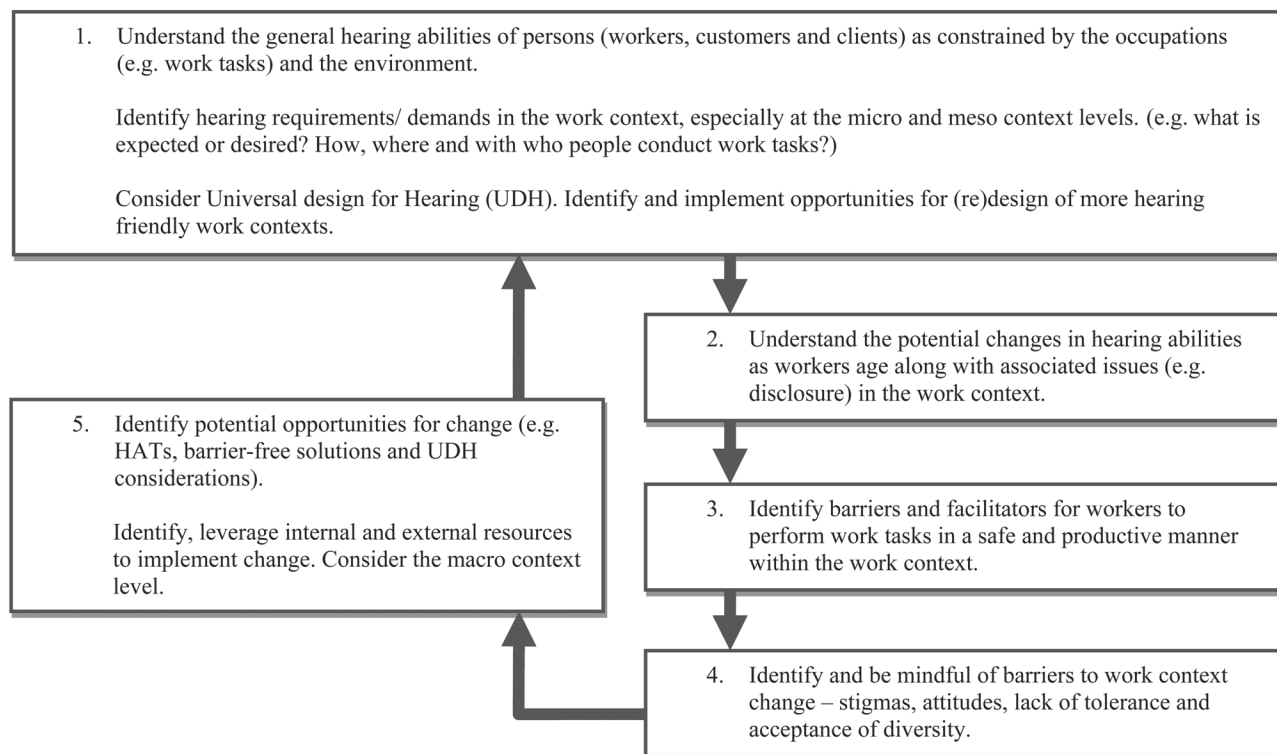


Figure 25–2. The UDH principles for management and workers in supporting safe and productive occupations that involve hearing. *Source:* Adapted from Fok et al. (2009).

Hearing Demands Tool (C-HearD). There are four areas identified in the C-HearD. In order of complexity, they are: monitoring (e.g., monitoring auditory events, detection, sensation, localization); listening (e.g., attending to auditory events, focus, sustain and switch attention, concentration, behavioral perception, selective attention); comprehending (e.g., receiving information, meaning, or intent from auditory events; knowledge of the lexical and semantic constraints of the language being spoken); communicating (perceive and interpret the need for individual or team-based action and timing of action; ability to respond to information). With the development of the C-HearD, Jennings et al. (2010) began to promote a common language and discourse for identifying and describing work demands among stakeholders and interdisciplinary professionals. Work to refine further and test the tool is currently in progress.

From Design to Process Evaluation and Implementation

The design of vocational enablement programs using a multilevel, interdisciplinary, integrated approach is key for enabling individuals with impaired hearing to participate optimally in the workplace. However, these programs will only become effective if their implementation is guided with care. This aspect has been given only limited attention in vocational auditory adult rehabilitation practice and in audiology in general. Gussenhoven et al. (2015) evaluated the process of implementing the VEP (see above) in audiologic care among employees with hearing impairment. Seven process parameters were analysed: recruitment, reach, fidelity, dose delivered, dose received, dose implemented, satisfaction and perceived benefit. Of the recommendations delivered by the professionals, 31% were actually implemented according to the patients' reports. Compliance rate was highest for hearing aid uptake (51%). Those who had specifically indicated that they needed help for work-related problems were most satisfied with the VEP. Hence, a person's awareness of his/her specific problem and need for care is a crucial determinant for the satisfaction with the rehabilitation they receive. The key finding of this study was that it is one thing to design appropriate vocational rehabilitation programs, and quite another to ensure they are fully implemented in clinical practice. Only when implemented successfully can all stakeholders including the workers with hearing loss optimally benefit from it.

Vocational Rehabilitation for People With Hearing Loss: Final Considerations

It is encouraging to see that the issue of vocational rehabilitation for persons with hearing loss has been of interest for over 30 years. Programs and tools have been developed in a number of countries. As outlined above, four programs are broadly similar, both in the elements incorporated and the structure. These are the programs by Lalande et al. (1988), Getty and Héту (1991), Ringdahl et al. (2001), and Li et al. (2006), all presented in the form of a course with education, group meetings, individual sessions, and training as the core elements, similar to the VR elements advocated by Mascia and Mascia (2008). Another common factor is the focus of these programs on the person with hearing loss without explicitly addressing the employer or any other stakeholder from within the workplace (integrated approach); the return-to-work program proposed by Li et al. (2006) is an exception. Also, Myers and Danek's DEAN model (1990) included services addressing the employer as did the VEP proposed by Kramer (2008), which requested clinicians to make contact with the stakeholders in the workplace to stimulate and ensure implementation of the requested workplace accommodations. Table 25-2 compares the programs according to some components of vocational rehabilitation.

Jennings et al.'s C-HearD tool (2010) and Fok et al.'s UDH principles (2009) are a bit different in nature in that these instruments are primarily designed to provide a method to assess carefully and describe workplace demands across the different levels of influence. Such tools are crucial in VR programs. Detailed evaluations and descriptions of the hearing demands and challenges in the workplace allow the design of tailored VR programs to support the worker with hearing impairment in meeting his or her personal goals for change and promoting greater confidence in being able to perform their job.

To our knowledge, none of the programs have been implemented on a nationwide level or been incorporated into standard audiologic care. Some are available only locally. Others were developed for the purpose of a research project or are still under development. Reasons for VR programs being poorly implemented may vary. First, clinicians may be largely uninformed about the complex nature of the workplace and underestimate the necessity of VR. Given that possibility, publication of existing programs and the experiences of stakeholders (e.g.,

Table 25-2. Comparison of the Vocational Rehabilitation (VR) Programs*

Program	Individual Sessions	Group Sessions	Spouse Participation	Education and Training	Employer Addressed	Inter-disciplinary Approach	Integrated Approach
ARP-Ca	✓	✓	✓	✓			
DEAN	✓			✓	✓		✓
RP-OHL		✓	✓	✓		✓	
ARP-Sw	✓	✓	✓	✓		✓	✓
RTW	✓	✓		✓	✓		
VEP	✓			✓	✓	✓	✓
VR				✓	✓		
UDH					✓	✓	✓
C-HearD					✓		✓

*Some components may relate to the lack of financial resources, legislation, or lack of statistical evidence for the effectiveness of VR programs.

clinicians, employers, workers with hearing impairment) with using these programs, is highly relevant and adds to our knowledge about what elements of VR really do matter, which subgroups benefit most, and what barriers still hinder broad implementation. Other reasons may relate to the lack of financial resources, legislation, or lack of statistical evidence for the effectiveness of the VR programs.

The last point is an important one. Empirical evidence (preferably at the level of a randomized controlled trial) for the effectiveness and efficiency of these programs, especially in terms of cost, is largely lacking. Such evidence is required to provide knowledge about the benefit of these programs and persuade policy makers, health insurance funds or companies, and other stakeholders to allow implementation of this type of rehabilitation in standard care. Empirical research should investigate the short and long-term effectiveness of the VR programs and explain to what extent programs significantly reduce barriers at work, enhance well-being, and improve work pleasure, performance, and safety. This issue deserves more attention in future research. Nevertheless, in spite of the dearth of empirical evidence, anecdotal evidence tells us that people with hearing impairment are highly motivated to work, provided that appropriate interventions and accommodations in the workplace are offered.

Conclusion

Due to changing demographic trends worldwide, revolutionary shifts in our labor tasks, and trends toward greater mobility and a more independent and self-motivated workforce, it is expected that the number of adults with hearing impairment in the workforce seeking help for their problems at work will increase in the near future. Ample evidence shows that people with hearing impairment in the labor force comprise a vulnerable group for whom the conditions at work are more challenging than for normally hearing colleagues. Unemployment, underemployment, early retirement, and low income are more pervasive among workers with hearing impairment than among workers without this disability. Common problems in the workplace are reduced performance in auditory communication, increased mental effort, lack of control, mental distress, fatigue, concern, and lack of knowledge. It is argued that a timely, interdisciplinary and integrated approach is required to successfully manage the difficulties of professionals with hearing loss. According to the ICF and AE frameworks, VR should not only focus on the individual's abilities, but also identify the different levels of influence (i.e., personal factors and environmental factors). Over the last 30 years, VR programs have been developed in several

countries in the world. Knowledge about and experiences with implementing these programs in clinical practice help us in further development, refinement, and optimization of (effective) use of VR programs for persons with hearing impairment.

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26

AUDITORY AND COGNITIVE PROCESSING IN AUDIOLOGIC REHABILITATION

M. Kathleen Pichora-Fuller

A New Era for Auditory Processing and Cognition in Adult Rehabilitation

Over at least the last half-century, the main rehabilitative approaches used to train listeners to improve their speech understanding abilities have focused on building analytic and/or synthetic skills (Jeffers & Barley, 1971). Analytic skills emphasize the correct identification of phonemes using signal-based acoustic-phonetic cues or what today might be called *bottom-up information processing*. Synthetic skills emphasize the use of knowledge and context, or what today might be called *top-down information processing*. Another way of thinking about the concepts embodied in these approaches is that the analytic approach is concerned more with the details of the surface information perceived by the listener, whereas the synthetic approach is concerned more with the overall meaning of the message that is understood. Yet another way to think about the two approaches is that the analytic approach builds on the contributions of the sensory systems (hearing and vision), while the synthetic approach builds on the contributions of the brain (cognition). In general, the synthetic approach has been more successful than the analytic approach, but combining these two approaches seems to be best.

Despite the enduring appeal of the analytic and synthetic approaches, much still remains to be learned about the best ways for audiologic rehabilitation (AR) to combine them to help adults maintain or improve their communication functioning in everyday life (for discussions, see Boothroyd, 2010; Humes, Pichora-Fuller, & Hickson, 2019; Kiessling et al., 2003; Laplante-Lévesque, Hickson, & Worrall, 2010). Importantly, emerging epidemiological evidence that hearing loss in older adults increases risks for cognitive decline as well as other comorbid health issues presents new imperatives for rehabilitation (Besser, Stropahl, Urry, & Launer, 2018; Deal et al., 2018; Pichora-Fuller, Mick, & Reed, 2015; Wei et al., 2017; Whitson et al., 2018). Beyond the traditional focus on communication functioning, the scope of AR is broadening in light of new findings concerning the importance of sensory-cognitive links for participation in everyday life and for aging well (Davis et al., 2016). This chapter will consider how the historical concepts underlying the analytic and synthetic approaches are being reborn in an era of AR in which there is renewed interest in how to harness the contributions of central auditory processing (CAP) and cognition, especially for those with age-related hearing loss (ARHL). New knowledge regarding sensory-cognitive links is reshaping communication interventions, motivating new developments in the interprofessional delivery

of health care for older adults with comorbidities, and advancing the shift in AR from treating hearing impairment to promoting health and improving quality of life for listeners and their partners as they participate in everyday activities.

Age-Related Hearing Loss Is Special

Most audiologic rehabilitation for adults focuses on the use of hearing aids, with recommendations regarding amplification being driven primarily by audiometric thresholds. Even though various diseases or genetic or environmental factors may cause hearing loss in adults, the etiology of sensorineural hearing loss is not often considered when rehabilitation is provided. It is reasonable to rely heavily on the audiogram, given that most sensorineural hearing loss involves damage to cochlear outer hair cells (OHC), with noise-induced hearing loss being a common example. Current knowledge of OHC pathology, its effects on perception, and many related rehabilitative issues can be generalized across etiologies that cause the same type of damage and across the adult age range because OHC damage is not age-specific. Of course, many older adults have sustained OHC damage over the course of their life, and a typical feature of ARHL is the elevation of high-frequency audiometric thresholds. Nevertheless, in ARHL, multiple structures in the cochlea and central auditory system can be damaged in ways that do not involve OHC damage and that are not typical in younger adults (Yamasoba et al., 2013). There is heterogeneity in the hearing abilities of older adults because different individuals may have sustained one or more types of damage to the auditory system. Importantly, some of these types of damage affect suprathreshold auditory processing, but such age-related auditory declines may not be obvious from the audiogram. It is important to understand the nature of age-related changes and their functional consequences if progress is to be made in tailoring rehabilitation to help individuals to prevent and overcome the unique difficulties that affect their communication function in everyday life (Gates & Mills, 2005).

Heterogeneity of Age-Related Hearing Loss and Variability in Everyday Communication

The elevation of audiometric high-frequency thresholds often seen in ARHL may be due to OHC damage;

however, less precipitous high-frequency audiometric loss may result from a decrease in the endocochlear potential as a result of damage to the stria vascularis that supplies blood to the cochlea (Humes & Dubno, 2010; Mills, Schmeidt, Schulte, & Dubno, 2006). In addition, neural degeneration may occur. Notably, research in animals suggests permanent neural damage may be the consequence of cochlear synaptopathy (Liberman & Kujawa, 2017) and such damage may occur in the absence of changes in the audiogram (Pienkowski, 2017). There is growing interest in finding better clinical tests to differentiate subtypes of presbycusis because determining the specific type(s) of damage to the auditory system in individual cases may clarify, at least to some extent, why there is such variability in the ability of older adults to perceive speech in noise and to communicate in complex listening situations (Anderson & Kraus, 2010; Anderson, Pabery-Clark, White-Schwoch, & Kraus, 2012; Anderson, Parbery-Clark, Yi, & Kraus, 2011). It may even be possible in the future to tailor interventions better according to the subtype or combination of subtypes of presbycusis affecting older individuals.

It is not surprising that older adults with clinically significant hearing loss have problems in everyday communication situations, but even older adults who have nearly normal audiograms can experience significant communication problems. As early as the fourth decade of life, difficulties are first noticed in challenging listening environments where there is noise or when listeners converse in groups or engage in multiple simultaneous activities (Bergman, 1980; CHABA, 1988). In general, adults in their 60s or 70s, who have audiometric thresholds within normal clinical limits below 4 kHz and are considered to have normal hearing for their age (ISO, 2017), need about a 3- to 4-dB better signal-to-noise ratio (SNR) to match the performance of younger adults on speech-in-noise tests (Smith, Pichora-Fuller, Wilson, & MacDonald, 2012).

More insight into age-specific difficulties can be gained by comparing the responses of younger, middle-aged, and older adults with normal hearing for their age (Banh, Singh, & Pichora-Fuller, 2012; Demeester et al., 2012) on self-report measures such as the Speech, Spatial, and Qualities of Hearing Scale (SSQ) (Gatehouse & Noble, 2004). As shown in Figures 26-1 and 26-2 (adapted from Banh et al., 2012), the responses of older adults on the SSQ are not well predicted from either the pure-tone average threshold or the SNR threshold at which listeners achieve 50% correct word recognition on the Words in Noise test (WIN) (Wilson, McArdle, & Smith, 2007). It seems

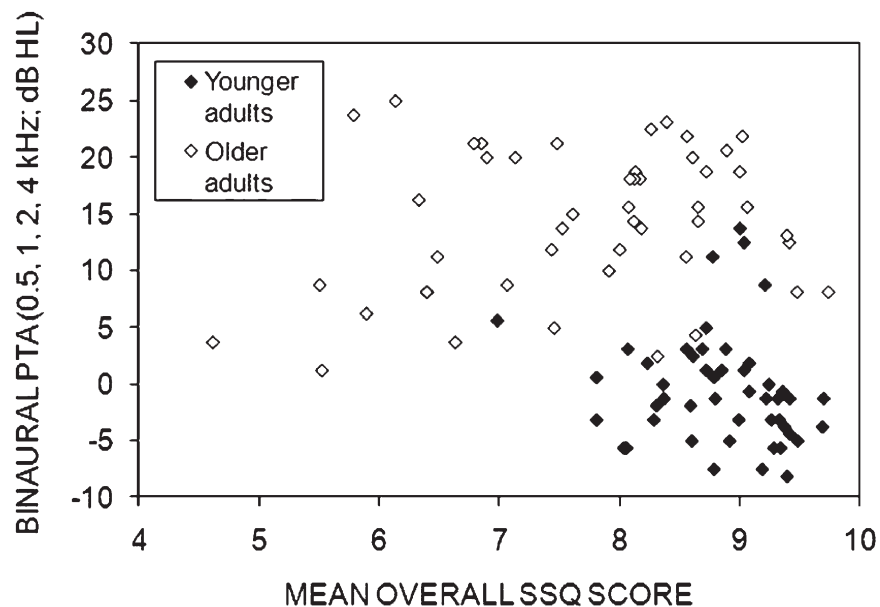


Figure 26-1. Pure-tone average score as predictor for SSQ score. *Note.* Mean binaural pure-tone average (0.5, 1, 2, and 4 kHz; dB HL) plotted as a function of mean overall SSQ score for younger and older adults. *Source:* Banh, J., Singh, G., and Pichora-Fuller, M. K. (2012). Age Affects Responses on the Speech, Spatial, and Qualities of Hearing Scale (SSQ) for Adults with Minimal Audiometric Loss. *Journal of the American Academy of Audiology*, 23, 87. Copyright 2012 by the American Academy of Audiology. Reproduced with permission of the American Academy of Audiology.

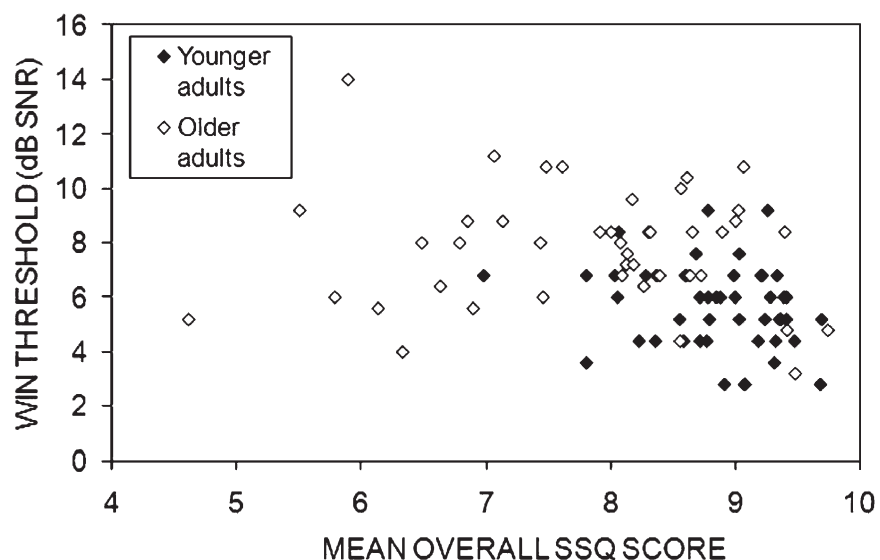


Figure 26-2. SNR score as predictor for SSQ score. *Note.* Words-in-noise threshold (dB SNR) as a function of mean overall SSQ score for younger and older adults. *Source:* Banh, J., Singh, G., and Pichora-Fuller, M. K. (2012). Age Affects Responses on the Speech, Spatial, and Qualities of Hearing Scale (SSQ) for Adults with Minimal Audiometric Loss. *Journal of the American Academy of Audiology*, 23, 87. Copyright 2012 by the American Academy of Audiology. Reproduced with permission of the American Academy of Audiology.

that typical audiometric measures such as pure-tone thresholds or SNR thresholds are not sufficient to gauge all of the suprathreshold auditory abilities that may be important for successful listening in the everyday situations depicted in some of the questions on the SSQ.

Even in those with relatively good audiograms, age-related neural degeneration may manifest in declines in coding the monaural and binaural temporal properties of dynamic and complex sounds that are important for listening in everyday life. Experiments conducted in controlled laboratory conditions have shown that there are age-related differences on a range of psychoacoustic measures of suprathreshold auditory processing, especially measures of temporal processing (Fitzgibbons & Gordon-Salant, 2010; Füllgrabe, Moore, & Stone, 2014) and binaural processing (Eddins, Ozmeral, & Eddins, 2018). Laboratory experiments indicate that many older adults are not as able as younger adults to detect differences in the fundamental frequency of two talkers' voices when they are presented monaurally, and declines in this auditory ability seem to be important in helping listeners to segregate a target voice from a competing voice. Furthermore, age-related declines in binaural processing may explain why it is more difficult for older adults to take advantage of the spatial separation between different sound sources in the complex auditory scenes encountered in everyday life (Pichora-Fuller, Alain, & Schneider, 2017). Indeed, using the Listening in Spatialized Noise-Sentence Test (LiSN-S) (Cameron, Glyde, & Dillon, 2011), older adults did not gain as much advantage as younger adults in using voice and spatial separation cues to understand speech in noise, and these age-related differences were attributable to both auditory and cognitive abilities (Besser, Festen, Goverts, Kramer, & Pichora-Fuller, 2015).

For most adults with cochlear pathology, the analytic approach to rehabilitation has been transformed over the last decade because advanced signal processing algorithms in hearing aids have increasingly improved the audibility of speech in quiet, and even in some noisy situations. Nevertheless, once audibility has been restored by sufficient amplification, older individuals vary in their ability to understand speech because of individual differences in auditory temporal and cognitive processing (Füllgrabe et al., 2014; Pichora-Fuller, 2003; Humes, 2007). These factors may also explain some of the variation in how well individual listeners learn to understand speech in noise when it is processed by hearing aids (Gatehouse, Naylor, & Elberling, 2006; Humes, 2003; Lunner & Sundewall-Thorén, 2007; Rudner, Foo, Rönnberg, & Lunner, 2009).

Research in cognitive science has provided new insights into how speech understanding can be altered when listening is challenged by a mismatch between the reduced abilities internal to an individual (e.g., ARHL) and the demands imposed by external conditions, such as environmental noise or multiple voices in a crowd (Mattys, Davis, Bradlow, & Scott, 2012). Perhaps even more importantly, progress has been made in realizing that the motivation, goals and strategies of individuals influence when and how they harness their residual auditory abilities and complement them with other sensory, cognitive, and social abilities to achieve participation in everyday life. When listening becomes effortful in challenging situations, an individual who is motivated to persist in participating (rather than giving up) may draw on a combination of internal auditory and cognitive resources to meet the demands imposed by demanding external listening conditions. An international expert group, following classical ideas from cognitive psychology (Kahneman, 1973), defined *effort* as the deliberate allocation of mental resources to overcome obstacles in goal pursuit when carrying out a task, with *listening effort* applying more specifically when tasks involve listening, as illustrated in the Framework for Understanding Effortful Listening (FUEL) (Pichora-Fuller et al., 2016). A framework like the FUEL provides a new way for rehabilitative audiologists to understand how and when auditory processing and cognitive processing combine to affect listening in everyday situations that vary in the demands they impose on a listener with particular auditory and cognitive abilities. A better understanding of the aspects of central auditory and cognitive processing involved in speech understanding in adverse conditions should enable new rehabilitation interventions to be designed that are more tailored to meet the needs and goals of individual older adults and their listening lifestyles. Using this framework, audiologists will be poised to enter a new era of rehabilitation that complements amplification and signal-based solutions with a more synthetic cognitive approach that pivots on individuals' personal and social goals. In the long term, this combined analytic-synthetic approach is expected to support compensatory brain reorganization, optimize social participation, and promote aging well.

Cognitive Aging in Healthy Older Adults

There is growing evidence that audition and cognition interact, even in healthy older communicators who have clinically normal or near-normal audiograms

and no clinically significant cognitive impairment (Peelle, 2018; Pichora-Fuller et al., 2017; Schneider, Pichora-Fuller, & Daneman, 2010). Some aspects of cognition decline with age, but others may actually even continue to improve. In general, there are declines in the dynamic or fluid processing of information, whereas static or crystallized linguistic and world knowledge are well preserved in healthy aging (Cunningham, Clayton, & Overton, 1975; Hartshorne & Germine, 2015). Accordingly, older adults can often use knowledge and contextual support as strengths to compensate for weaknesses in rapid information processing (Pichora-Fuller, 2008, 2010). Age-related declines in cognitive processing that could affect communication include slower speed of information processing, reduced working memory, and difficulty dividing attention or attending to relevant information and inhibiting distraction (Pichora-Fuller & Singh, 2006).

In everyday activities, these three aspects of cognitive processing—working memory, attention, and speed of processing—may be highly interrelated. Working memory is a limited capacity system for temporarily storing and processing the information required to carry out complex cognitive tasks such as learning, reasoning, and comprehension. An individual's working memory capacity, or span, is measured in terms of ability to simultaneously store and process information. For example, in the well-known working memory span task (Daneman & Carpenter, 1980; Pichora-Fuller, Schneider, & Daneman, 1995), after listening to or reading a set of sentences and performing some kind of judgment task after each sentence to ensure it has been comprehended (i.e., processed), the individual is asked to recall the sentence-final words for all of the sentences in the set (i.e., what has been stored). The number of sentences in the sets is increased and the working memory span is determined to be the largest set size for which the person can correctly recall a minimum specified proportion of the sentence-final words. The assumption is that, as the processing demands increase, there will be a corresponding decrease in how much can be stored in this limited capacity system; the working memory span is used to gauge the trade-off in the allocation of the limited capacity of working memory to processing versus storage. Working memory span can be used to compare different individuals when the task is held constant or to evaluate how the efficiency of processing shifts with variations in the task for a given individual (Pichora-Fuller, 2007).

It has been argued that there is an interface between working memory and attention insofar as complex working memory tasks reflect differences

in how well individuals can (1) control attention to activate and keep the information relevant to the current goal or task available and (2) dampen or inhibit irrelevant information under conditions where there is risk of interference or distraction (Barrett, Tugade, & Engle, 2004; Engle, 2010). In general, speed of processing or the amount of time required to process information is assumed to reflect the amount of processing demanded by a task. If a task is easy and can be done very automatically without needing much control or effort, then it will impose a very low processing demand and take little time. In contrast, if a task is difficult and requires attentional control or effort, then the processing demand will be high and it will take more time to complete the task. Slowing of information processing is a hallmark of cognitive aging, but more specifically, it seems that perceptual processing speed accounts for much of the age-related variance in working memory capacity (Redick, Unsworth, Kelly, & Engle, 2012). In challenging conditions, the processing demand imposed by a task will require more attentional effort and performance will be slowed. These conditions include situations involving listening to speech in noise. It often takes more effort for an older person to listen in conditions with levels of noise that do not require similar effort on the part of younger listeners (Anderson, Gosselin, & Gagné, 2011). Of course, two of the most common complaints of older listeners or of people who are hard-of-hearing are that even if they can understand what has been said, it takes more effort and time for them when they have to follow ongoing speech.

Interactions Between Auditory and Cognitive Aging in Healthy Older Adults

On the one hand, age-related hearing problems can increase cognitive processing demand, thereby exacerbating what may seem to be age-related cognitive decline (Pichora-Fuller, 2007). For example, the extra effort required to concentrate when following speech in noise or during multitasking may consume working memory processing capacity with the result being that, even if words are correctly recognized, they may not be stored in memory or understood as well as if they had been heard in quiet (Pichora-Fuller et al., 1995). On the other hand, cognitive processing demands may contribute to the listening problems of older adults when there are multiple talkers or when attention may need to be divided between simultaneous tasks, such as conversing while driving a car or crossing a street using a walker. Thus, auditory and

cognitive challenges can combine to make it much more difficult for listeners to function in everyday communication situations.

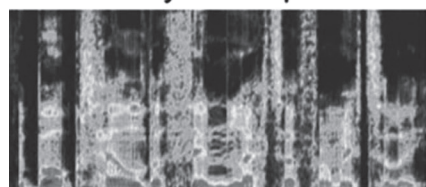
In contrast to the compounding of auditory and cognitive declines, there is also the possibility that there could be compensatory benefits from interactions between auditory and cognitive processing. Listeners with better hearing may be less vulnerable to cognitive overload and listeners with better cognition may be more successful than peers with lower cognition when it comes to attending in complex listening situations (Heyl & Wahl, 2012; Sörqvist, Stenfelt, & Rönnerberg, 2012) or even when learning how to use fast-acting signal processing in hearing aids (Gatehouse et al., 2006; Humes, 2003; Lunner & Sundewall-Thorén, 2007; Rudner et al., 2009).

It is now widely accepted that the brain continues to be plastic over the course of adult development and aging (Grady, 2012). Much has been learned about how parts of the brain not traditionally considered to be involved in hearing contribute to the per-

formance of complex listening tasks (Zatorre, 2007). Brain imaging and evoked-response potential (ERP) studies have shown that there are auditory-cognitive interactions during speech understanding, both in bottom-up and top-down processing, from the brain stem to the prefrontal cortex (Peelle, 2018; Peelle & Wingfield, 2016; Pichora-Fuller et al., 2017). As shown in Figure 26-3 (Peelle, 2018), when speech is acoustically degraded, various nonauditory cognitive areas are activated in addition to the core auditory speech areas of the brain.

Apart from shedding light on the mechanisms of auditory-cognitive interactions, recent advances in cognitive neuroscience have significant implications for rehabilitation. Brain imaging studies have shown that when younger and older adults perform perceptual or cognitive tasks at the same level of accuracy, there can be more widespread brain activation in older compared to younger brains. The increase in symmetric hemispheric activation (Cabeza, 2002) and the shift from posterior to anterior brain activation

Acoustically clear speech



Acoustically degraded speech

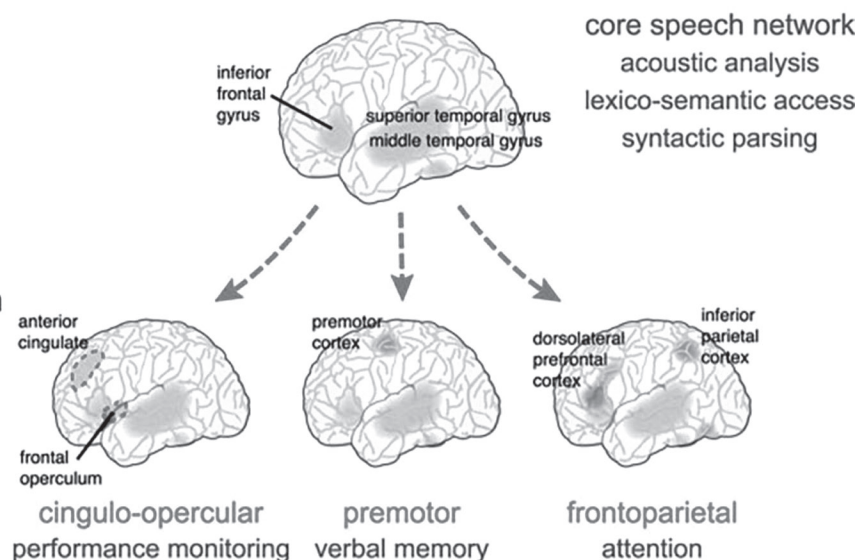


Figure 26-3. SNR brain networks activated during listening to clear and degraded speech. *Note.* Illustration of brain networks involved in processing clear and degraded speech. When the acoustic signal is clear (represented by the upper spectrogram), a core speech network is engaged consisting of bilateral temporal cortex and often left inferior frontal gyrus. This core speech network supports acoustic, phonological, lexicosemantic, and basic syntactic processing. When speech is degraded (represented by the lower spectrogram), the core speech network is still engaged but is complemented by additional activity required to deal with the degraded speech signal. The additional regions engaged depend on the specific type of cognitive support required and will likely differ as a function of the specific acoustic challenge, task demands, and cognitive and auditory ability of an individual listener. *Source:* Peelle, J. E. (2018). Listening effort: How the cognitive consequences of acoustic challenge are reflected in brain and behavior. *Ear and Hearing*, 39(2), 210. Copyright 2017 by Wolters Kluwer Health, Inc. Reproduced with permission of Wolters Kluwer Health, Inc.

with aging (Davis, Dennis, Daselaar, Fleck, & Cabeza, 2008) has been interpreted as evidence of compensatory reorganization of brain networks in response to perceptual and cognitive declines, including impaired speech perception (Du, Buchsbaum, Grady, & Alain, 2016). According to the Cognitive Compensation Hypothesis, declining sensory and sensorimotor functions are compensated by higher-level cognitive and attentional processes (Li, Krampe, & Bondar, 2005). Furthermore, the Compensation-Related Utilization of Neural Circuits Hypothesis (CRUNCH) (Reuter-Lorenz & Cappell, 2008) proposes that additional brain regions are recruited by older adults when capacity limits are reached in a given task or combined tasks. Importantly, the success of compensation varies with the demands of the task. By recruiting more cognitive resources, older adults can succeed in conditions when tasks are relatively easy and younger adults do not incur a cognitive load or need to expend effort, but recruiting more resources may not enable them to overcome demands when tasks are more difficult (Rypma, Eldreth, & Rebbelch, 2007). By extension, the Scaffolding Theory of Aging and Cognition (STAC) (Park & Reuter-Lorenz, 2009) suggests that there is the potential to enhance such compensation by training.

The implication of this research in cognitive neuroscience is that new approaches to AR could facilitate compensatory brain reorganization as older listeners with hearing loss are engaged in relearning how to listen and comprehend speech using signals delivered by hearing technologies. Evidence of brain plasticity is very encouraging for rehabilitation professionals because it suggests that older adults can compensate by finding new ways to perform complex tasks successfully, such as listening to speech in noise (Pelle, Troiani, Wingfield, & Grossman, 2010). A recent systematic review and meta-analysis examined the results of nine studies in which cognitive outcomes were measured following auditory and cognitive training in adults with hearing loss; the overall certainty of the evidence of benefit based on these few preliminary studies was low for auditory training and very low for cognitive training, but a combined auditory-cognitive approach seemed to be the best (Lawrence et al., 2018). Insofar as a goal of new auditory-cognitive interventions would be to enhance cognitive compensation so that it can be deployed to overcome demands in a wider range of challenging listening situations, more research on combined auditory-cognitive interventions should be undertaken and evaluated in terms of a wider range of outcome measures to assess auditory, cog-

nitive, communication and participation functioning in everyday life. Furthermore, combined auditory-cognitive interventions will likely be of even greater importance when older adults must contend with hearing loss in combination with clinically significant cognitive impairment.

Combined Auditory and Cognitive Impairments

Impairments in hearing and cognition both increase markedly with age, such that the majority of those over 75 years of age have hearing loss (Bainbridge & Wallhagen, 2014) and about 20% have mild cognitive impairment (MCI), with the rate of progression from MCI to Alzheimer's disease (AD) being roughly 10% per year (Petersen et al., 2009). Since hearing impairment and cognitive impairment are both highly prevalent in older adults, dual impairments should be common.

Hearing loss is found in up to 90% of those who have dementia (Gold, Lightfoot, & Hnath-Chisolm, 1996) and it is more prevalent in those with dementia than in matched controls (Uhlmann, Larson, Rees, Koepsell, & Duckert, 1989). In addition to the relationships found between audiometric hearing loss and dementia, relationships have also been found between central auditory and cognitive impairments. For example, clinical cases diagnosed with MCI have worse performance on the dichotic digits tests compared to controls, and cases diagnosed with AD have worse performance than those with MCI (Idrizbegovic et al., 2011). Reduced cognitive executive functioning has been associated with performance on tests of CAP (Gates et al., 2010). Even more striking are epidemiological findings that audiometric thresholds (Lin et al., 2011; Lin et al., 2011, Lin et al., 2013) and scores on dichotic speech tests (Gates, Anderson, McCurry, Feehey, & Larson, 2011; Gates, Beiser, Rees, Agostino, & Wolf, 2002) are predictive of the future manifestation of dementia. A recent meta-analysis showed that the overall combined relative risk of people with hearing impairment to develop Alzheimer's disease was almost five times greater compared to controls with normal hearing (Zheng et al., 2017). Based on one longitudinal study, the cognitive reduction associated with a 25-dB loss in hearing was calculated to be equivalent to the reduction associated with an increase of 7 years in age (Lin, 2011). In another study, hearing loss was associated with 10 out of 11 cognitive tests measuring speed of processing, executive function, memory, and global cognitive status (Harrison

Bush, Lister, Lin, Betz, & Edwards, 2015). Importantly, hearing loss has been identified as the most potentially modifiable midlife risk factor for dementia based on a systematic review of factors contributing to dementia prevention, intervention, and care (Livingston et al., 2017).

Compensatory benefit from vision may modulate auditory-cognitive links in aging (Wettstein, Wahl, & Heyl, 2018). However, audiovisual processing can be compromised in people with hearing loss (Musacchia, Arum, Nicol, Garstecki, & Kraus, 2009). Notably, a 2015 report on the Global Burden of Disease estimated that hearing loss and vision loss, respectively, were the second and third most common impairments worldwide (Vos et al., 2016), and dual sensory (hearing and vision) loss affects about 20% of people over 80 years of age (Smith, Bennett, & Wilson, 2008). Compared to hearing loss alone, dual sensory loss is associated with an even greater likelihood for cognitive decline, functional decline, and even mortality (Brenowitz, Kaup, Lin, & Yaffe, 2019; Laforge, Spector, & Sternberg, 1992; Lin et al., 2004). At an exploratory workshop on "Sensory and Motor Dysfunctions in Aging and Alzheimer's Disease" convened by the National Institute on Aging in the USA, experts concluded that there was a meaningful interface among sensory, motor, and cognitive dysfunctions that warranted more research (Albers et al., 2015). The possibility that AR could modify the risk of dementia is a topic of active research (Deal et al., 2017). AR designed to address sensory-cognitive links in aging adults will need to consider vision as a potentially compensatory sensory modality or a potentially comorbid sensory loss that could increase risk of cognitive decline.

The explanation for the correlations between hearing and cognitive impairments remains to be determined (for a discussion, see Lin et al., 2013; Uchida et al., 2019). It is possible that central presbycusis and executive dysfunction may result from similar neurodegenerative processes (Gates et al., 2010) or even that age-related sensory declines drive cognitive declines (Humes, Busey, Craig, & Kewley-Port, 2013). Alternatively, the association between hearing impairment and cognitive impairment may be so strong because individuals with better hearing simply exhibit less apparent cognitive decline, even though their brains are changing in a way similar to those who have hearing loss. Or maybe those with better hearing experience slower cognitive decline because they are better able to maintain healthy active lifestyles that are cognitively stimulating, thereby slowing cognitive decline in line with the "use it or

lose it" perspective. In any case, it seems clear that it is time for audiologists to become more involved in the diagnosis and management of cases in which hearing and cognitive health may both be in question. Some individuals may begin AR with clinically normal cognition and later develop cognitive impairments while others may already have cognitive impairments when AR begins.

New diagnostic guidelines for MCI and dementia were released in 2011 (Albert et al., 2011; McKhann et al., 2011), with an emphasis on the diagnostic reality that there is a continuum from normal age-related changes in cognition to advanced dementia. As new approaches to diagnosing and managing dementia emerge, there is an opportunity to develop best practices for how hearing should be factored in when cognitive tests are interpreted and recommendations for care are made by other health professions. For example, the Montreal Cognitive Assessment (Nassreddine et al., 2005) is a cognitive screening test that is often used in primary care, but its accuracy may be reduced by hearing or vision impairments (Al-Yawer, Pichora-Fuller, & Phillips, 2019; Dupuis et al., 2015) or noisy test environments (Dupuis, Marchuk, & Pichora-Fuller, 2016). Within audiology practice, in addition to the audiogram, it seems that there is value to incorporating central auditory testing (e.g., with dichotic digits or sentences) into assessment (Gates et al., 2011; Rosenhall, Hederstierna, & Idrizbegovic, 2011). New practice guidelines are also needed to inform audiologists as to the possible need to screen cognition to determine if referral to other health professionals is indicated. New partnerships with neuropsychologists and psychogeriatricians will be critical as audiologists develop more effective approaches to the rehabilitation of older adults with comorbid sensory and cognitive impairments (Dupuis, Reed, Bachmann, Lemke, & Pichora-Fuller, 2019). In particular, older adults with combined sensory and cognitive impairments who were receiving home care or residing in long-term care had greater functional difficulties in activities of daily living, less social engagement, increased loneliness, and reduced independence compared to those with only cognitive impairment (Guthrie et al., 2018). In general, AR should be a key component in health planning to promote aging well and to prevent older adults from becoming less active and socially isolated. With the anticipated rapid increase in cases of AD, there is great interest in promoting active lifestyles to stave off or slow down dementia (Fratiglioni, Paillard-Borg, & Winblad, 2004), but the role of ARHL in this endeavor remains to be determined (Pichora-Fuller, 2010).

Rehabilitation of Peripheral Versus Central Auditory and Cognitive Problems

The traditional basic audiometric evaluation is used as the first step in developing a rehabilitative plan and is most successful in defining the degree of hearing loss, rather than differentiating peripheral from central pathology. Individuals with audiometric hearing loss are approached as having primarily peripheral pathology; their rehabilitation usually focuses on hearing aid fitting. The prevalence of clinically significant hearing loss increases markedly with age (ISO, 2017). Using the screening criterion proposed recently in the United Kingdom (Davis et al., 2007), people whose pure-tone threshold at 3 kHz is in excess of 35 dB HL would be considered potential candidates for hearing aids. Accordingly, the majority of those 75 years of age or older would be referred for a hearing aid evaluation. Using the World Health Organization (WHO) grades of hearing impairment based on pure-tone averages (Stevens et al., 2013), about one-third of adults who are 65 years of age or older would be categorized as having disabling hearing loss defined as a better-ear four-frequency (0.5, 1, 2, and 4 kHz) pure-tone average exceeding 35 dB HL (WHO, 2013). With the aging of the population and the baby boom generation reaching retirement age, there will be a steadily increasing number of older adults who will seek help for hearing problems. Indeed, initial help seeking for age-related hearing problems occurs earlier now than it did a decade ago, with a shift in the average age of a first-time hearing aid user from about 70 to 63 years (Abrams & Kihm, 2015). The inaudibility of speech due to pure-tone hearing loss fully explains the speech-communication problems of about half of older adults, suggesting that amplification by hearing aids would be a good solution for their problems; however, other solutions would be needed to address the problems of the other half of the older adult population for whom audibility does not fully explain their speech-communication difficulties (Humes, 2019; Humes et al., 2019). Apart from audibility, the speech-communication difficulties of the other half of the older adult population may be attributable to age-related changes in auditory processing and/or cognition (Humes, Kidd, & Lenz, 2013). It is easy to overlook the more central auditory or cognitive factors that may contribute to the complaints of older listeners. Whether a person has peripheral auditory pathology or not, it is possible that there could be neural degeneration of a sort

that would be considered to involve primarily CAP. To date, it is uncommon for older adults with good audiometric thresholds who have complaints about auditory functioning to be assessed for CAP disorders and, if any rehabilitation is provided, it usually involves technologies other than conventional hearing aids (e.g., FM), environmental modifications, listening training, and/or behavior change to manage challenging listening situations.

Recently, the American Academy of Audiology Task Force on Central Presbycusis published a report and in defining *central presbycusis*, the task force used “an historical narrow structural definition . . . focused on modality-specific changes to auditory portions of the central nervous system from above the cochlear nucleus to the auditory cortex” (Humes et al., 2012, p. 637). Using that definition, it was concluded that there was

insufficient evidence to confirm the existence of central presbycusis as an isolated entity. On the other hand, recent evidence has been accumulating in support of the existence of central presbycusis as a multifactorial condition that involves age- and/or disease-related changes in the auditory system and in the brain. (Humes et al., 2012, p. 636)

In contrast to the current view of cognitive neuroscience, the traditional audiologic view of CAP has relied on an anatomical framework and a perspective that has viewed an array of auditory functions as modular processes which could be delineated and assessed independently from each other (Pichora-Fuller & Singh, 2006). Crucially, the task force did point out that the definition they used was “in contrast to a broad functional definition of central presbycusis, one that might encompass any age-related changes in the central nervous system beyond the auditory periphery that might impact communication, including cognitive changes” (Humes et al., 2012, p. 637).

Rather than following the traditional approach based on defining peripheral auditory, central auditory, and cognitive processing problems as if they were mutually exclusive, rehabilitative audiologists may find a functional and integrative approach more useful. Such a framework is provided by the World Health Organization International Classification of Functioning, Disability, and Health (ICF) (WHO, 2001, 2002) that has already been so influential in rehabilitative audiology (Humes et al., 2019; Stephens & Kramer, 2010; Worrall & Hickson, 2003). The ICF has functional health as its primary focus,

thus emphasizing the importance of the interaction between an individual's health conditions or status and the contextual factors around him/her. This more functional approach was taken in the recent Canadian Guidelines for Assessment and Management of Auditory Processing Disorder in Children and Adults (Millett et al., 2012). (*Editors' note:* For further discussion of the ICF, the reader is referred to Chapter 3 of this text.)

Everyday Function and the WHO ICF

In order to understand the communication needs of older adults, rehabilitative audiologists must take a broad view that goes beyond the problems arising from hearing loss due to peripheral disorders and extends to central auditory and cognitive processing issues. Age-related auditory changes can occur at one or many levels from cochlea to cortex, and changes in the auditory system often co-occur with nonauditory changes that will affect communication functioning (Kiessling et al., 2003). Table 26-1 summarizes the peripheral auditory, central auditory, vision, cognitive, and general health factors and mechanisms that may underpin the functional problems of older adults. It also illustrates how these factors can affect the functions of hearing, listening, comprehending, and communicating that are relevant for AR (Kiessling et al., 2003). The functions are based on ICF concepts and were defined by the consensus group who developed the table.

Hearing was defined to be

essentially a passive function that provides access to the auditory world via the perception of sound. It concerns sensing the presence of sounds and discriminating the location, pitch, loudness and quality of sounds. In the ICF classification system, hearing is located in the impairment domain. (Kiessling et al., 2003, p. S93)

Cognition comes into play for *listening*, which was defined as

the process of hearing with intention and attention. Listening is an activity. People engage in hearing for a purpose. There is intent of the part of a listener to experience auditory stimuli, and listening requires the expenditure of effort. (Kiessling et al., 2003, p. S93)

Comprehending involves the connection between signal and meaning and was defined as

an activity undertaken beyond the processes of hearing and listening. Comprehending is the reception of information, meaning or intent. Often this takes place as part of discourse, although the concept is not necessarily limited to discourse. It is a uni-directional concept. (Kiessling et al., 2003, p. S93)

Communicating involves social interaction and was defined as requiring

the bi-directional transfer of information, meaning or intent between two or more people. Although non-verbal and non-auditory communication is possible, in the current context, communicating requires that both participants are hearing, listening and comprehending. Communication involves both activity and participation. (Kiessling et al., 2003, p. S93)

Rehabilitation should provide potential solutions for each of the four functions: hearing, listening, comprehending, and communicating. This chapter, however, will focus on the functions of listening, comprehension, and communicating, which must rely on a combination of peripheral auditory, central auditory, cognitive abilities. Selected research concerning auditory-cognitive interactions suggest new directions for rehabilitative audiology.

Listening

Ease of listening depends on an individual's abilities, but also on the demands of the task(s) the person is trying to perform and how adverse the situation is in terms of noise and distraction. Research on working memory and attention could be applied in the rehabilitation process to address the need to improve ease of listening when task demands and adverse environments make it difficult.

Working Memory

As explained earlier, there are individual differences in working memory capacity and there are indications that these differences are related to speech-in-noise performance, listening effort, and success with hearing aids (Arehart, Souza, Baca, & Kates, 2013; Rudner & Lunner, 2013). Measures of working memory could be used when considering the selection of hearing aid processing options and/or in deciding who may need more training and support while learning to listen with a new hearing aid.

Table 26–1. Age-Related Changes in Hearing, Listening, Comprehending, and Communicating

Function	Cochlea/Periphery	Postcochlear Auditory and Associated Centers	Vision	Cognition	Other Health Conditions
Hearing: Availability of cues at input for higher-level processing	▪ Thresholds are elevated resulting in reduced audibility, especially at high frequencies. ▪ Dynamic range is reduced (e.g., recruitment and changes in intensity difference limens). ▪ Spectral processing is impaired (e.g., reductions in frequency selectivity, increases in frequency difference limens, and upward spread of masking). ▪ Temporal processing is impaired (e.g., increases in gap detection thresholds and difference limens for gap size and duration). ▪ Tinnitus may be peripherally generated.	▪ Binaural processing is impaired due to ageing of the central auditory system. ▪ Binaural processing is impaired when peripheral threshold, spectral and/or temporal impairments diminish interaural difference cues. ▪ Detection of signals in noisy and/or reverberant situations is reduced. ▪ Hyperacusis and/or tinnitus may be centrally generated.	▪ Visual segmental and suprasegmental articulatory cues are altered by visual impairment(s). ▪ Visual cues about objects and events in environment are altered by visual impairment(s).	▪ Increased demands due to auditory, visual and/or motor impairments divert mental resources from cognitive functions such as memory and comprehension. ▪ Sensory impairments diminish cues used in attention. ▪ Perception, attention, memory, and comprehension are reduced by transitory cognitive impairments in conditions (e.g., depression and/or the use of mind-altering medications). ▪ Impairments in perception, memory and planning occur in diseases such as Alzheimer or Parkinson diseases.	▪ Comorbid cardiovascular, haematological (e.g., blood lipids, cholesterol), and/or endochronological conditions increase fatigue, discomfort, and/or pain. ▪ Medications reduce perceptual and cognitive abilities. ▪ Demands of other interventions alter rehabilitative and help-seeking choices. ▪ Dexterity limitations influence choice of hearing technologies. ▪ Mobility issues necessitate modification in technological and/or therapeutic audiologic rehabilitation (e.g., visual cues may not be available to a person who is being pushed in a wheelchair).
Listening: Using cues for a purpose	▪ Recognition and/or identification of phonemes and other sound objects and events are reduced or slowed due to impoverished cues.	▪ Pattern perception based on envelope cues is preserved. ▪ Localization is reduced or slowed.	▪ Good vision compensates for impoverished cues for recognition and/or identification.	▪ World knowledge stored in long-term memory is preserved. ▪ Information processing slows. ▪ Distraction or informational masking increases.	▪ Fatigue, discomfort, stress, and/or pain alter motivation or attitude.

continues

Table 26-1. *continued*

Function	Cochlea/Periphery	Postcochlear Auditory and Associated Centers	Vision	Cognition	Other Health Conditions
Listening <i>continued</i>		- Perception in noise and/or reverberation is reduced. - Segregation of sound sources (e.g., voices) is reduced.	- Good vision compensates for localization and segregation difficulties. - Performance in low light conditions (night vision) is reduced. - Motion, color, depth perception are reduced. - Orienting to objects in periphery may be reduced or slowed.	- Selective attention (inhibition of irrelevant stimuli) decreases. - Ability to focus, sustain, and switch attention (e.g., from talker to talker in a group) declines. - Working memory capacity becomes overloaded when input is ambiguous. - Coordination of multiple tasks becomes more challenging (e.g., walking and talking).	- Other health conditions alter listening goals (e.g. degree and type of information or social support needed). - Environments in which care is provided may be acoustically challenging. - Settings may be under-stimulating or over-stimulating.
Comprehending: Creating meaning by mapping cues to stored knowledge	- Misperceptions or ambiguities due to failures in detection, discrimination, recognition, or identification of single auditory objects because cues are impoverished during peripheral processing. - Fast rates of input are more difficult.	- Preservation of prosody enables interpretation of affect and syntax. - Misinterpretations or confusions due to failures in localization or segregation when an auditory scene is composed of multiple auditory objects.	- Visual scene interpretation compounds or offsets auditory errors. - Articulatory, facial and/or gestural cues compensate for impoverished auditory cues.	- Use of contextual supports is preserved. - Slowing rate of input alleviates processing demands. - Reducing competing tasks alleviates processing demands. - Comprehension of facts is reduced when ambiguous or misperceived input is incorrectly mapped to world knowledge. - Compensation by use of world knowledge increases working memory load.	- Familiar settings may support comprehension. - Other health conditions may increase exposure to unfamiliar settings. - Service providers may not have same language as client. - Linguistic function may be compromised by neurological damage (e.g., aphasia).

Function	Cochlea/Periphery	Postcochlear Auditory and Associated Centers	Vision	Cognition	Other Health Conditions
Comprehending <i>continued</i>				- Integration of meaning over time (e.g., in discourse) may be reduced because of working memory overload or slowed information processing. - World knowledge may be compromised by neurological damage (e.g., dementia).	
Communicating: Meaningful interaction mediated by cues	Misperceptions and misinterpretations increase need for conversational repairs.	- Preservation of prosody enables speech accommodation and provides cues to topic and turn changing. - Ability to extract the target signal from the background will influence the success of communication.	- Facial and/or gestural cues guide interaction and support attention to and comprehension of both content and affect. - Eye contact compensates for impoverished auditory cues to turn-taking in conversation.	- Speech accommodation reduces demands of coordinating talker and listener roles. - Longer turns reduce demands of coordinating talker and listener roles. - Use of context reduces need for conversational repairs. - Longer turns reduce need for conversational repairs.	- Goals shift to less emphasis on getting information and more to giving information and to social interaction. - Setting in which care is provided may induce social isolation or interaction depending on compatibility and communication abilities of partners, as well as on the supports or challenges of the physical environment. - Disclosure of hearing loss or use of amplification may trigger stereotyped responses.

Source: Kiessling, J., Pichora-Fuller, M. K., Gatehouse, S., Stephens, D., Arlinger, S., Chisolm, T., . . . von Wedel, H. A. B. (2003). Candidature for and delivery of audiological services: Special needs of older people. *International Journal of Audiology*, 42(Suppl. 2), S94-S95. Copyright 2004 of International Journal of Audiology by British Society of Audiology, International Society of Audiology, & Nordisk Audiologisk Selskab. Reproduced with permission of Taylor & Francis Inc.

Working memory may also become a new measure for evaluating the outcomes of hearing aid fittings and rehabilitation. Some researchers have chosen to measure reading rather than listening working memory span because they reasoned that the effects of hearing loss and increased auditory processing demands would be avoided if stimuli were presented visually; however, others have argued that it is important to measure auditory working memory span, especially if the purpose of the test is to determine if an intervention has been effective in reducing listening effort (Besser, Koelewijn, Zekveld, Kramer, & Festen, 2013; Smith & Pichora-Fuller, 2015). In the auditory modality, measures of working memory span can be affected by the SNR or even the level of presentation of speech in quiet (Baldwin & Ash, 2011). Recall that even older listeners without clinically significant audiometric loss need about a 3 dB SNR advantage to match the word recognition accuracy of younger listeners. To achieve that the same level of performance in speech understanding as a younger listener, an older listener may be engaging brain networks in a compensatory fashion that requires more effort (Peelle, 2018). Even if a listener correctly recognizes words, as it becomes more effortful to listen, there should be a corresponding decrease in working memory span or the number of words that are recalled. If technology improves the quality of the signal or the SNR then listening effort should be reduced so that more of the words that were recognized should be remembered later and working memory span should increase. Similarly, if compensatory listening skills are honed by training, then listening effort should be reduced and there should also be an increase in how many words are remembered.

New measures of auditory working memory have been developed to test working memory in people who are hard of hearing, including the Word Auditory Recognition and Recall Measure (WARRM) test in English (Smith, Pichora-Fuller, & Alexander, 2016), the Sentence-Final Word Identification and Recall (SWIR) test in Swedish (Ng, Rudner, Lunner, Pedersen, & Rönnberg, 2013; Ng, Rudner, Lunner, & Rönnberg, 2015), and a Danish version, the Sentence-Final Word Identification and Recall Test in a New Language (SWIRL) (Lunner, Rudner, Rosenbom, Ågren, & Ng, 2016). Tests of executive function or attention have also been developed (Rudner, 2016). However, further research is needed to standardize these tests for clinical use in AR.

Attention: Segregation and Informational Masking

In many situations, including situations in which there is background noise, and depending on the

task, the listener may need to either divide attention or selectively attend to one sound source and ignore others. Research concerning sound segregation or streaming and research on informational masking have both produced new knowledge about listening in adverse environments that could inform new approaches to rehabilitation. The segregation of auditory streams depends on the bottom-up use of specific auditory cues as well as top-down attention-dependent processing of auditory and nonauditory cues (Carlyon, 2004; Synder & Alain, 2007). Energetic masking occurs whenever the spectra of the two sounds overlap; informational masking occurs when the two maskers are similar in terms of nonspectral characteristics (e.g., both are meaningful English sentences with similar linguistic characteristics). Auditory and/or cognitive interference or distraction from informational masking can increase the difficulty of following one sound while inhibiting or ignoring the other. Therefore, listening effort will depend on the type of masker and not simply on the SNR such that those who have better cognitive abilities will find it easier to cope with the higher demands of informational compared to energetic masking (Desjardins & Doherty, 2013). Studies of informational masking suggest that bottom-up temporal cues, especially pitch cues, as well as top-down cues based on knowledge, such as semantic or other linguistic knowledge, contribute to successful listening when speech is masked by other speech (Ezzatian, Li, Pichora-Fuller, & Schneider, 2015). In addition, when two competing sounds originate from different spatial locations, release from informational masking is greater and it is easier to segregate the sounds.

New Directions for Listening Rehabilitation

Based on recent research concerning sound segregation and informational masking, new measures could be developed to assess a listener's ability to segregate and attend to sounds based on their ability to use specific auditory and nonauditory cues. Temporal cues, especially pitch cues, play a key role in auditory stream segregation and release from informational masking, yet there are no standardized clinical measures available to measure this auditory ability. Many speech-in-noise tests are available for use in audiology, but only recently have tests such as the LiSN-S (Cameron et al., 2011) been designed to evaluate how well listeners can segregate voices using pitch or spatial cues. Future tests could systematically explore performance in different types of informational masking. The resulting profiles from these tests might guide the selection of signal processing options in hearing aids

depending on the listener's abilities to use particular signal cues. They might also guide the design of individualized training programs to improve attentional control and the allocation of listening effort.

Comprehension

Recognizing words is only part of comprehension. A listener needs to integrate multiple words to understand the meaning of an entire sentence and, more often, he/she will need to understand meaning across many sentences to comprehend discourse (e.g., a podcast or a conversation with a friend). A common complaint of older people and people who are hard of hearing is that they cannot keep up, especially when the talkers use a fast rate of speech. Hearing speech and successfully allocating attention to listen to a target talker while ignoring a distracting talker is not enough. The listener must engage in linguistic and cognitive processing to extract the meaning of what has been heard and listened to. To address the common complaint that it is hard to keep up in a conversation, new rehabilitative approaches might be informed by research on the speed of speech and language processing, how it is modulated by priming, and the compensatory use of contextual support.

Speed of Speech and Language Processing

Considerable research using time-compressed speech has shown that, at fast speech rates, it is more difficult for listeners to comprehend, especially if the syntax and linguistic structure of the material is complex (Vaughan, Storzach, & Furukawa, 2006; Wingfield, Tun, Koh, & Rosen, 1999). Similarly, using reaction time measures, speech understanding in noise has been found to be slower when syntactic complexity is greater (Carroll & Ruigendijk, 2013). Older listeners are also slower in segregating different competing sound streams (Ben-David, Tse, & Schneider, 2012). When processing demands are increased by greater syntactic complexity, it becomes even harder for a listener to keep up, even though the audibility of the speech may not be affected.

Priming and Compensatory Use of Contextual Support

Priming, or the effect of prior information on the processing of later information, can either facilitate (speed up) or inhibit (slow down) comprehension. When prior context is congruent with the upcoming information, the listener has the advantage of being

able to anticipate what will be said next. Fortunately, older adults retain their ability to use congruent context and this confers an advantage when they must listen in adverse listening conditions (Pichora-Fuller, 2008). When the prior context is incongruent, however, then the listener may need more time to recover because what they are hearing does not match what they were expecting. Unfortunately, older adults may be more disadvantaged than younger adults when the context is incongruent (Rodgers, 2017; Rodgers, Jacoby, & Sommers, 2012).

In an experiment, the authors measured the time required for a listener to decide if a target word was a real word or not; although the target was always heard in quiet without distortion, the preceding sentence context was sometimes acoustically distorted or presented in background babble (Goy, Pelletier, Coletta, & Pichora-Fuller, 2013). When the sentence context was congruent, lexical decision was facilitated by about 250 ms (about the duration of a syllable) when there was no distortion. That facilitation, however, was reduced as more acoustic distortion was applied to the sentence context, regardless of the type of degradation. Older adults were always slower to respond than younger adults, but their lexical decisions were facilitated by more than 300 ms, an advantage even greater than that of younger adults when the context was congruent compared to when it was neutral or incongruent. It is interesting that, like those of the younger adults, the responses of older adults were slowed when the context was acoustically degraded, but distortion by time-compressing speech had a more deleterious effect on the performance of older adults than on that of younger adults.

New Directions for Comprehension Rehabilitation

In everyday life, different strategies, both technological and behavioral, could address the listener's need for more time to comprehend auditory information. Rather than relying on traditional offline measures such as word recognition accuracy, it seems that it would be valuable to develop online measures or reaction time measures to evaluate how particular individuals perform depending on the rate of speech and the acoustical and linguistic characteristics of the speech to be understood. Insofar as the time required for processing information reflects effort, such new measures may also provide a way of measuring listening effort. Listening could be made easier and faster by ensuring signal quality is clear, thereby improving bottom-up processing of the signal by the auditory system. Listeners could also potentially benefit from

learning to make better use of congruent semantic context and expectations to guide listening, thereby improving top-down processing to harness the knowledge and expertise of older adults. Signal processing strategies in hearing aids might be developed that are more dependent on the rate of speech. Training exercises to practice listening to different rates of speech might be considered. In addition to the typical advice to ask talkers to slow down when speaking, other behavioral strategies to control the flow of information during discourse could be emphasized in training.

Communicating

Interacting with one or more other talkers in a conversational situation is even more cognitively demanding than is comprehension in a noninteractive situation such as watching television. In addition to understanding the linguistic meaning of what is said, the listener must at the same time allocate cognitive processing capacity to plan what to say when it is time to talk. As long as there is only one talker to attend to and the topic of conversation is known, the listener can capitalize on context and expectations, thereby easing processing demands. Shifting attention from one talker to another and from one topic to another, however, can increase processing demands, especially when an upcoming shift is unexpected (Singh, Pichora-Fuller, & Schneider, 2013). Furthermore, as well as comprehending the linguistic meaning of the conversation, the listener must also monitor other events in the situation, especially if there is a need to multitask (e.g., conversing while preparing dinner or crossing a busy street). The listener must also allocate processing resources to interpret the emotional and social information that is conveyed by how the talker is speaking and behaving. In addition to remembering the preceding material in a conversation, communication requires the listener to remember prior shared experiences and conversations with conversational partners. Our knowledge of the effects of hearing loss on conversation and the use of conversational therapy in AR for those with normal cognition could be extended to consider how additional cognitive processing demands affect conversational communication, especially for those who have both hearing and cognitive impairments.

Memory and Conversation

One of the most commonly reported changes in the conversational patterns of older adults, especially

those who have cognitive impairment, is that they stop telling stories because they fear retelling the same story, or they may retell stories without being aware that they are repeating themselves. In a recent experimental study that tried to approximate how destination memory might affect conversation, Gopie and his colleagues (2010) found that healthy older adults were less able than younger adults to remember the person (e.g., a picture of Oprah Winfrey) to whom they told facts (e.g., a dime has 118 ridges around it), even though the two age groups had similar memory for the individual components (people and facts). Nevertheless, the older people were more confident that they had *not* told a specific fact to a specific person even though they had indeed done so. In a companion experiment on source memory, in which participants were told facts by famous people, both age groups remembered who had told them the facts equally well, but the older people were more confident than the younger people that they had been told facts by specific people when other people had actually been the source of the particular facts. Thus, miscommunications during conversation could easily arise from age-related declines in memory and these problems could compound hearing-related communication problems.

New Directions for Communication Rehabilitation

Conversational therapy could be used to develop strategies to ease communication by alleviating cognitive overload. Such strategies could include conversational techniques to influence or control the speed of information flow, ensure that supportive context is maximized and distractions are minimized, manage transitions when topics and turns change, manage multitasking, and improve associative memory about who said what to whom.

Conclusion

The long-standing analytic and synthetic approaches to rehabilitation provide a foundation for a new era of rehabilitative audiology. Research on auditory-cognitive interactions and age-related changes in those interactions has provided knowledge to stimulate the development of new technological, behavioral, and environmental solutions to improve the listening, comprehension, and communication functioning of older adults.

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27

EVIDENCE-BASED PRACTICE AND RESEARCH OPPORTUNITIES

Louise Hickson and Barbra Timmer

Introduction

The rise of evidence-based medicine (EBM) in the 1990s had a profound effect on all areas of health care, including audiology. Sackett, Rosenberg, Gray, Haynes, and Richardson (1996) defined it as “the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients” (p. 71). Now referred to as *evidence-based practice* (EBP), thus incorporating disciplines beyond medicine, it involves the integration of the best available research evidence with clinical expertise and the patient’s preferences and goals (Hoffmann, Bennett, & Del Mar, 2017). The specific application of EBP to a range of audiologic interventions for children and adults with hearing impairment (e.g., cochlear implants, hearing aids, tinnitus treatments) has also been explored in a textbook by Wong and Hickson (2012a). In this chapter, EBP is illustrated using a case example of a new patient with mild hearing impairment considering hearing aid fitting, the evidence about the benefits of hearing aid fitting for such a case is summarized, and some areas of need for future research in adult audiologic rehabilitation are highlighted.

The EBP Process

The process of applying EBP is typically described as having five steps, each beginning with A:

- Ask a question
- Access the information
- Appraise the articles found
- Apply the information
- Audit the process

EBP Step 1: Ask a Question

The first step involves forming a clinical question. To assist with this, Needleman (2003) recommended breaking down the question into four components indicated by the acronym *PICO*: patient/problem, intervention, comparison, and outcomes. The best way to illustrate this is by example: A 56-year-old man presents to your audiology clinic complaining of difficulties hearing in meetings at work, problems hearing the television when the volume is set at a level that suits those he lives with, and anxiety when attending social events for fear of not understanding what is said. Your assessment indicates he has a mild bilateral sensorineural hearing impairment. There is a range of possible interventions available through the clinic that would improve his communication: provision of hearing aids, provision of an assistive listening device for television, auditory training, and/or group communication education. Options are discussed with him and he expresses most interest in hearing aids, but he wishes to know more about them and their likely benefits. The clinical PICO question in this case is: “For patients with acquired mild sensorineural

hearing impairment (patient), is hearing aid fitting (intervention) beneficial compared to no intervention (comparison) for improving self-reported hearing and communication difficulties in everyday life (outcome)?” Other PICO questions would be generated if he were particularly interested in one of the other intervention options or if he were completely unsure about which option to choose and wanted to know about the evidence for all options.

EBP Step 2: Access the Information

The second step in the EBP process is to assemble and evaluate the best evidence. This is accomplished by searching relevant databases. A good starting point is the Cochrane Database of Systematic Reviews (<http://www.cochranelibrary.com>), a resource for systematic reviews in health care and health policy. Cochrane Reviews uses a defined protocol to identify and select studies that meet specific quality criteria, ensuring the resulting review provides the best available evidence. Research evidence about audiology interventions can also be found in citation search engines. Wong and Hickson (2012b) suggest the use of Scopus (<http://www.scopus.com>), PsychINFO (<http://www.apa.org/psychinfo>), the Cumulative Index to Nursing and Allied Health Literature (CINAHL) (<http://www.cinahl.com>), and PubMed (www.pubmed.gov). It is possible to search a number of databases at once by using systems such as Web of Science or Medline. Most of these databases require a subscription, but if you do not have access, then PubMed and Google Scholar are popular freely available search engines.

A good starting point for search terms to use in the databases is to take the keywords from the PICO question. For example, search terms to start with based on the PICO question above would be: *mild hearing impairment*, *hearing aids*, *benefit*, and *outcomes*. Alternative terms, such as *mild hearing loss*, should be included in the search, and limits can be placed on the search to refine the results obtained. It is also important to specify criteria such as the language of publication (e.g., English only), age of participants (e.g., adults only), the year of publication (e.g., from 2010 to present), and type of publication (e.g., systematic review and peer-reviewed journal article). A PubMed search using these suggested search terms and limits was conducted and identified 99 references that could provide evidence for the example PICO question. Some of the relevant systematic reviews and peer-reviewed journal articles identified in this search are discussed later in the chapter. There is a useful tuto-

rial at the PubMed website to learn how to search in the database (<http://www.nlm.nih.gov/bsd/disted/pubmedtutorial/>). Databases such as Embase (<http://www.embase.com>) allow searches using PICO question formulation. It is also recommended to discuss search approaches initially with a librarian, as this can save valuable time in the long run.

EBP Step 3: Appraise the Articles Found

The next step involves determining the relevance of the articles you find to your clinical question, and then if they are relevant, critically reviewing them. Rating the quality of a particular piece of research is central to EBP. Table 27-1 shows a hierarchy of research evidence specific to the provision of hearing aids developed by Cox (2005).

Levels of Evidence

Level 1 evidence is available from systematic reviews, which are comprehensive evaluations of the literature that try to limit bias by systematically identifying, appraising, and summarizing research on a particular topic. Primary data sources are evaluated, and criteria for inclusion in such a review are predetermined and explicitly stated in the review. Such synthesis of research is important because single studies can and do provide conflicting evidence, and a systematic review can help establish why this occurs. Systematic reviews are also extremely useful for clinicians, as they provide a summary of the evidence on a par-

Table 27-1. Levels of Evidence Relevant to Hearing Aid Research

Level	Type of Evidence
1	Systematic reviews and meta-analyses of randomized controlled trials or high-quality studies
2	Randomized controlled trials
3	Nonrandomized intervention studies
4	Nonintervention studies: cohort studies, case-control studies, cross-sectional surveys
5	Case reports
6	Expert opinion

Source: Adapted from Cox, 2005.

ticular clinical question. For example, the question proposed earlier was, “For patients with acquired mild sensorineural hearing impairment (patient), is hearing aid fitting (intervention) beneficial compared to no intervention (comparison) for improving self-reported hearing and communication difficulties in everyday life (outcome)?” In relation to this PICO question, a PubMed search identified the two highly relevant reviews: a Cochrane Review by Ferguson et al. (2017) and another recent systematic review, by Johnson, Danhauer, Ellis, and Jilla (2016). Findings of both reviews are discussed later in this chapter.

Systematic reviews aside, there are various types of experimental studies mentioned in Table 27–1 which may involve an intervention (randomized controlled trials and nonrandomized intervention studies) or be noninterventive or observational, such as cohort studies, case-control studies, and cross-sectional surveys. It is important to understand the differences between them. Many textbooks discuss these types of studies, and Cox (2005) provides an excellent review relevant to audiology. The first and most rigorous, from a scientific perspective, is the randomized controlled trial (RCT). In an RCT, participants who meet stated inclusion criteria are randomly allocated to groups receiving different forms of treatment or to a no-treatment or placebo treatment group. Humes et al. (2017) provide an example relevant to this chapter’s PICO question in an RCT where participants were randomly allocated to receive hearing aids with amplification programmed by an audiologist; hearing aids with amplification programmed by the participant themselves; or hearing aids programmed to provide no amplification. In an RCT, participants and the researchers who assess them should be blind to group membership, that is, not aware of which treatment a participant is receiving. This step is necessary to reduce any potential bias. If both participants and researchers are unaware of the treatment, the study will be described as a *double-blind RCT*. If the researcher knows the nature of the treatment but the participant does not, this is referred to as a *single-blind RCT*. Occasionally the term *triple-blind* is used. This refers to the fact that the participant, the researcher, and the person doing the final analysis of the data are all blind to the different types of treatment. Randomization of individuals to groups should mean that the different groups will be equal on all variables except the type of treatment, although this equality needs to be checked statistically. RCTs vary in quality depending on how careful the researchers have been to follow the guidelines for such studies (e.g., <http://www.consort-statement.org>).

A crossover study is another type of RCT and has some advantages because participants can act as their own controls and all participants receive the treatment in the end. In this type of study, participants are first randomly allocated to different groups as they are for an RCT and then one group begins a particular type of treatment and the other groups begin other treatments. At a certain point in time, those who have completed one form of treatment begin the other, so that all participants complete all treatments. A crossover study design can also be used to trial different hearing aids or different hearing aid features, one example being Cox, Johnson, and Xu (2016).

The main aim of the Cox, Johnson, and Xu’s (2016) study was to test the effectiveness of different hearing aids for adults with mild to moderate hearing impairment. Forty-five participants with an average age of 70 years, of which 26 had never worn amplification, trialed four pairs of hearing aids in a single-blind, crossover design. The hearing aids differed in features (basic versus advanced) and were from two different manufacturers. The group results showed that hearing-related quality of life was at least moderately better with the hearing aid compared with not using any amplification. The participants also reported the hearing aids were effective in speech clarity, listening fatigue, and sound comfort. However, data analysis did not show significant outcome differences between the four different hearing aids, and the participants’ subjective preferences were in line with this, as there was no clear trend preferring advanced features over basic features, or one manufacturer over another.

A lower level of evidence is available from cohort studies. These are common in hearing aid studies and involve the observation of the effects of a particular intervention in a single cohort or group of participants. Results for the experimental cohort are compared to a group who did not receive the intervention. The major difference between a cohort study and an RCT is that random allocation to groups does not occur, so there is the potential for bias and for results to be influenced by variables other than the type of intervention. One example of a cohort study is that of Collins, Souza, O’Neill, and Yueh (2007). The researchers retrospectively investigated hearing aid outcomes from 74 participants who attended either individual or group hearing aid fitting and follow-up visits. Collins et al. (2007) found that there was no statistical difference between the participants who attended group appointments in terms of a reduction in self-reported hearing difficulties (as measured by the Hearing Handicap Inventory for the Elderly [HHIE]; Ventry &

Weinstein, 1982) when compared to the participants who were provided individual appointments. Additionally, the participants who attended group visits reported greater hearing aid satisfaction and hours of hearing aid use.

In case-control studies, participants with particular outcomes are grouped and studied retrospectively to investigate factors that influence outcomes. One group (cases) could be people who achieve high levels of benefit from hearing aids and another group (controls) could be people who do not achieve benefit. Thus, the outcome is measured after the intervention and then the groups are compared on different variables to identify those that are predictive of group membership. There are numerous examples of studies of this kind in the hearing aid literature (Helvik, Wennberg, Jacobsen, & Hallberg, 2008; Hickson et al., 2014; Kelly-Campbell, Thomas, & McMillan, 2015). Roup and Noe (2009) described such a study relevant to hearing aid outcomes for mild hearing impairment, in which they compared hearing aid outcomes for participants with two different types of pure-tone audiogram configurations. The 53 participants in this study had an average mild hearing impairment and were grouped according to their hearing threshold at 2000 Hz (at or below 30 dB hearing level [HL] versus at or above 40 dB HL). All were fitted bilaterally with hearing aids in the 4 years prior to the study. Perceived hearing aid benefit was measured using the Abbreviated Profile of Hearing Aid Benefit (APHAB) (Cox & Alexander, 1995), and hearing aid satisfaction was measured with the Satisfaction With Amplification in Daily Life (SADL) (Cox & Alexander, 1999) questionnaires. The study found that both groups reported benefit from, and satisfaction with, hearing aids and that there was no significant difference in these outcome measures between the two audiogram configuration groups. Relevant to the PICO question, this study suggests hearing aids can be beneficial for adults with mild hearing impairment. As Roup and Noe (2009) point out, the two groups in this study design were selected based on degree and configuration of hearing loss, and other factors were not controlled. There was, for example, a significant difference in age between the two groups and, had hearing aid benefit or satisfaction differed between the two groups, it would have been difficult to ascertain if those differences were due to hearing loss configuration or age. This is an inherent limitation in nonrandomized studies such as case-control studies.

Cross-sectional survey studies involve analyzing data, often collected through the use of question-

naires or demographic surveys at one point in time, to determine the relationship or association between various factors in a population or group. Cross-sectional study designs are common in prevalence studies but cannot, by design, infer any causation between factors. One example of a cross-sectional study pertinent to this chapter's PICO question is that of Mick and Pichora-Fuller (2016), who investigated the association between unaddressed hearing loss and poor physical and mental health in older adults. The study found that adults between 60 and 69 years who had not addressed their hearing loss by, for example, wearing hearing aids, had a higher incidence of lower cognitive ability and social isolation compared to those who did not have a hearing loss.

Case series or case reports are considered a very low level of evidence in that they do not provide information about the effectiveness of one kind of intervention over another. Such studies are quite common in audiology rehabilitation for the description of the outcomes of hearing device fitting (Bishop et al., 2017; Távora-Vieira et al., 2013; Uriarte et al., 2005; Vestergaard, 2006) and their value is that they can provide a baseline for comparison of outcomes across studies. In recent times, the availability of "big data" has allowed for the description of outcomes for some very large data participant samples, which increases the value of such studies as baselines for expectations about outcomes. For example, in a recent case series study, Timmer, Hickson, and Launer (2017b) used data extracted from 8,489 bilateral hearing aid fittings. Participants were retrospectively allocated to groups based on hearing impairment averaged across 500, 1000, 2000, and 4000 Hz to investigate if objective data-logging measures of hearing aid use are different for adults with mild hearing impairment compared to adults with moderate hearing impairment. Participant age was significantly different between the hearing impairment groups, with participants with a mild hearing impairment (average thresholds between 25 and 40 dB HL) being 7.8 years younger than those with a moderate hearing impairment (average thresholds between 40 and 55 dB HL) and 11 years younger than those with a moderately severe hearing impairment (average thresholds between 56 and 70 dB HL). The study found average daily hearing aid use was 8.5 hours, with no statistically relevant differences in use between adults with mild or moderate hearing impairment. Hearing aid use was significantly lower for participants with mild hearing impairment compared to those with moderately severe hearing impairment, but only for the use of the left hearing

aid. Age, gender, or hearing aid style were not related to objective hearing aid use and the hearing aids for this large group of participants were typically used in quiet for 83.3% of the wearing time and in noisy listening situations for 12.4% of the time.

The lowest level of evidence is expert opinion and sometimes a number of experts collaborate to produce a consensus statement or a guideline. The latter may or may not be evidence based, and it is important to scrutinize such reports carefully to determine if they are evidence based. An example of a guideline relevant to the PICO question that is the focus of this chapter is the National Institute for Health and Care Excellence (NICE) guideline on “Hearing loss in adults: Assessment and management” (<https://www.nice.org.uk/guidance/ng98>). The full guideline is some 280 pages, but the main messages are summarized as a series of recommendations based on the detailed evaluation of the best available evidence as analyzed by an expert panel. For example, in relation to offering hearing aids to adults, the recommendation is “Offer hearing aids to adults whose hearing loss affects their ability to communicate and hear, including awareness of warning sounds and the environment, and appreciation of music” (NICE, 2018, p. 16).

More detailed descriptions of levels of evidence, along with a number of other very useful tools, such as tips for finding the best evidence and critical appraisal worksheets, are available from the Center for Evidence-Based Medicine (<http://www.cebm.net>) and in Wong and Hickson (2012a). It is also important to point out that, although the level of evidence hierarchy is commonly used, it is not always the only way to rate the value of a particular piece of evidence. Sackett et al. (1996) make the point that “Evidence-based medicine is not restricted to randomized trials and meta-analyses. It involves tracking down the best external evidence with which to answer our clinical question” (p. 72). In the case of the question proposed initially in this chapter, “For patients with acquired mild sensorineural hearing impairment, is hearing aid fitting effective compared to no intervention for improving self-reported hearing and communication difficulties in everyday life?” an RCT will provide the most appropriate evidence to answer the question. If, however, the question were “For adults with mild hearing impairment, what are the barriers and facilitators to hearing aid uptake and success?” a qualitative research methodology would be most appropriate (Knudsen et al., 2012). Hannes and Bennett (2017) point out that qualitative research is a valid means of creating evidence about “the feasi-

bility, appropriateness or meaningfulness of a certain intervention or activity” (p. 226). Generally in qualitative research, the PICO question is adapted not to have a comparison and the “I” in PICO refers more to “issue” or “interest” rather than “intervention.” An example of a qualitative study investigating hearing aid use and adoption can be found in Gallagher and Woodside (2018), who conducted face-to-face semi-structured interviews with hearing aid owners and nonowners with varying degrees of hearing impairment including mild, and utilized thematic analysis to answer their research question. For example, they found that perceived complexity of hearing aid use, attitudes toward both hearing impairment and hearing aids, and inadequacy of audiology services influenced hearing aid adoption. Qualitative studies, like quantitative studies, vary in quality and should be critically appraised to determine the credibility of the evidence presented (Hannes & Bennett, 2017).

EBP Step 4: Apply the Information

After the clinical question has been developed, and the evidence sought and appraised, the next step in the EBP process is to discuss the information gained from this approach with the patient. Laplante-Lévesque, Hickson, and Worrall (2012) describe this as facilitating the matching of evidence with patient preferences and recommend three approaches: joint goal setting, shared decision making, and the use of decision aids. The overall aim is to involve the patient in the rehabilitation process as much as possible, cognizant of the evidence from systematic reviews that greater patient-centeredness is associated with improved intervention adherence, better outcomes, and higher levels of satisfaction (Lewin, Skea, Entwistle, Zwarenstein, & Dick, 2009; Swift, Callahan, & Vollmer, 2011). An extension to patient-centered care is family-centered care, which encourages a collaborative partnership between the clinician, the patient, and the patient’s family (Ekberg et al., 2015). As shown in other chapters in this book, in the context of family-centered care, the term *family* can be used to describe someone with whom the patient has a significant relationship, be it biological, legal, emotional, or other. In audiologic rehabilitation, practicing family-centered care means the clinician can address the communication difficulties associated with hearing impairment for both the patient and their family, who are often a patient’s frequent communication partners. Research has shown that family involvement in hear-

ing rehabilitation leads to greater uptake of services and interventions and better rehabilitation outcomes (Wallhagen, 2010; Scarinci et al., 2013; Meyer et al., 2015) and that family play a key role in making decisions about the uptake of hearing interventions (Laplante-Lévesque et al., 2010).

Joint goal setting at the start of the EBP process, using tools such as the Client-Oriented Scale of Improvement (COSI) (Dillon, James, & Ginis, 1997), the Goal Attainment Scaling (GAS) (Turner-Stokes, 2009) or, if involving family, the Goal Sharing for Partners Strategy (GPS) (Preminger & Lind, 2012) will assist the clinician to know the research evidence they need to focus on when using EBP and what outcomes are most important to the patient and their communication partners. In the case introduced at the start of the chapter, the patient wants to improve his communication, but it would be important to explore this more with him and his communication partners, and determine the situations in which they want to improve and what would be a good outcome for them. Would it be feeling less fatigued at the end of a day, understanding more of what is said, or having fewer conversation breakdowns around the home? His answers may serve to refine the PICO question. It may become “For patients with acquired mild sensorineural hearing impairment (patient), is hearing aid fitting (intervention) effective compared to no intervention (comparison) for improving speech understanding and reducing listening effort (outcomes)?”

Shared decision making is central to Step 4 of EBP, relating the evidence to the patient. Laplante-Lévesque, Hickson, and Worrall (2012, p. 48) provide some tips for shared decision making based on Epstein, Alper, and Quill (2004) and these can be further tailored if family members are involved:

- Prepare patients and any family members for shared decision making by stating early that they will be invited to be involved in decisions.
- Try to understand the patient and family’s experiences and expectations; e.g., ask what they expect would happen at the clinic.
- Communicate information at a level appropriate to patients and families. Be prepared to simplify explanations, use nontechnical language, convey quantitative information in a qualitative way, and use diagrams to aid comprehension.
- Communicate the values of your workplace to your patients and their families.

- Ensure patients and family members feel comfortable to ask questions.
- Build a partnership with the patient and their family members.
- Describe the evidence about interventions to the patient and family, including any uncertainties.
- Take into account the values and preferences of your patients and families when discussing the options.
- Check that patients and family members understand the information provided and the options available to them.
- Give patients time to go away and think about options. Many will want to involve their family if they were not at the appointment.

Decision aids are a very useful tool for relaying information to patients as they “describe the evidence surrounding a health condition, the intervention options, their associated benefits and limitations” (Laplante-Lévesque, Hickson, & Worrall, 2012, p. 51). An example of a decision aid that presented evidence-based options to adults with hearing impairment seeking help for the first time, along with a description of the adults’ response to this approach can be found in Laplante-Lévesque, Hickson, and Worrall (2010). There is growing interest in the use of decision aids in audiology; for example, Pryce et al. (2018) describe the development of a decision aid for patients with tinnitus.

EBP Step 5: Audit the Process

The final step is about reflecting on the EBP process itself and asking relevant questions:

- Were steps 1 to 4 undertaken in an appropriate manner?
- Was the PICO question correctly targeted to the needs of the patient?
- Was the literature search conducted efficiently?
- Was there sufficient high-quality evidence to answer the clinical question?
- Was the information provided to the patient and his or her family in a way that could be understood?

It is also important to evaluate outcomes, as well as the process itself. Based on the evidence, and in dis-

cussion with the audiologist, the patient may choose to trial hearing aids. It is then important to follow up with the patient and determine if the intended outcomes were achieved. In this case, did the use of hearing aids improve hearing and communication in everyday activities? And if not, why not? Reported research benefits are one thing; individual outcomes another. Further information about evaluating and implementing EBP can be found in Wong and Hickson (2012b).

Limitations of EBP

Although EBP is a worthy aim, Kent (2006) reminds us in his paper discussing EBP in speech-language pathology and audiology that “EBP is very much a work in progress, not a polished perfection” (p. 268). Worrall and Bennett (2001) list some of the barriers to EBP for speech language pathologists and many are equally applicable to audiology. Possible barriers are: (a) access and ability to use databases for searching the scientific literature; (b) the fact that not all relevant literature is in some databases; (c) a lack of evidence in some topic areas; (d) the fact that the level of evidence is not high in some areas; and (e) the fact that the evidence does not always match the reality of clinical services. This last point relates to the fact that participants included in research studies are often highly selected cases (e.g., inclusion criteria mean that they have no other impairments, English as a first language, are willing to participate in research) and the treatment regimes used in research may be unrealistic in clinical settings where resources are limited. In reviewing the uptake of EBP in audiology, Moodie et al. (2011) found that despite the commitment of audiologist associations to promote the use of evidence-based principles among their members, gaps still existed between the organizations’ best practice recommendations and clinical practice. A recent Australian study (Boisvert et al., 2017), however, found an increasing awareness of both EBP and patient-centered care for clinical decision making in audiology. Although this is promising, the study also highlighted that little is known about the extent of EBP implementation in day-to-day audiologic practice. The next section of this chapter focuses on a review of the high-level evidence to date related to the specific PICO question about the benefits of hearing aids for adults with mild hearing impairment and is an example of applying evidence-based approaches to decisions for an individual patient.

Research on Hearing Aid Benefit for Adults With Mild Hearing Impairment

The starting points for this review of the existing high-quality evidence are two systematic reviews by Ferguson et al. (2017) and Johnson et al. (2016) investigating hearing aid outcomes for adults with mild hearing impairment. The review by Ferguson et al. (2017) was a Cochrane Review of the research evidence about hearing aids for adults with mild to moderate hearing impairment, while Johnson et al. (2016) systematically reviewed research studies that investigated hearing aid benefit in adults solely with mild degrees of hearing impairment. Both studies also conducted meta-analyses. A meta-analysis is a special case of a systematic review in which data are statistically combined from multiple studies to calculate a summary measure of the effect of the interventions. It is particularly useful when research in a field is characterized by limited participant numbers in individual studies, as is often the case in audiology.

Systematic Reviews of Hearing Aid Outcomes for Mild Hearing Impairment

The Ferguson et al. (2017) Cochrane Review included studies if they used an RCT or crossover design, and in the case of the latter, used the results from the first treatment period. The studies included were with participants 18 years and older with either qualitatively described mild or moderate hearing impairment or a measured average pure-tone threshold between 26 to 40 dB HL (mild) or 41 to 60 dB HL (moderate). The intervention of interest was acoustic hearing aids (bone conduction or implantable hearing aids were excluded), irrespective of hearing aid style or technology. The outcomes included in the review were hearing-specific health-related quality of life, health-related quality of life, and listening ability, all as measured by self-report questionnaires.

Searches were conducted on a number of databases including PubMed, Ovid MedLine, Google Scholar, and the Cochrane Register of Studies Online, and clinical trial registries such as ClinicalTrials.gov and the World Health Organization Platform (<http://apps.who.int/trialsearch/>). Of the 2,840 studies initially identified after removing duplicates, five studies were eligible for inclusion in the review, all of which used an RCT design.

In these five studies, hearing aid outcomes were obtained between 6 weeks and 6 months after the hearing aid fitting, for a total of 825 participants with a mean age between 69 and 83 years. Three studies used the Hearing Handicap Inventory for the Elderly (HHIE) (Ventry & Weinstein, 1982) to measure hearing-specific health-related quality of life. Of these, two also reported measuring health-related quality of life, one using the World Health Organization Disability Assessment Scale II (WHO-DAS II) (World Health Organization, 2001) and the other the Self-Evaluation of Life Function (SELF) (Linn & Linn, 1984). Listening ability was measured in three studies: one using the Profile of Hearing Aid Performance (PHAP) (Cox & Gilmore, 1990); another study using its abbreviated version, the APHAB (Cox & Alexander, 1995); and the third a study-specific hearing ability scale.

Of the five studies included in the Ferguson et al. (2017) Cochrane Review, three had sufficient quantitative data to allow a meta-analysis and indicated that hearing aids had a large beneficial effect for hearing-specific health-related quality of life and listening ability, and a small beneficial effect for general health-related quality of life. The review also found there was too little information to estimate the risk of adverse effects, such as pain or noise-induced hearing loss from hearing aids. Overall, the reviewers concluded that the quality of evidence was moderate as, despite all being RCTs, many of the studies were at risk of bias due to a lack of blinding (of participants and/or researchers), a lack of generalizability (many participants were male military veterans) or due to the short space of time between fitting and outcome assessment (6 weeks).

Ferguson et al. (2017) concluded, “if the goals and needs of an individual with hearing loss are to improve their listening abilities and, in doing so, improve their participation with others in everyday life and their health-related quality of life more generally, then hearing aids are an appropriate intervention” (p. 22). Pertinent to this chapter’s sample clinical question, Ferguson et al. (2017) attempted to compare hearing aid benefit for adults with mild versus moderate hearing impairment but found that such a subgroup analysis was not possible in their systematic review as the data were not available. They concluded that more research was required and future studies should measure hearing aid benefit separately for different degrees of hearing impairment, as well as age groups, genders, and types of hearing aids.

The systematic review by Johnson et al. (2016) also recognized the paucity of evidence showing the benefits of hearing aids specifically for adults with

mild hearing impairment. The review sought to answer two questions: (a) do adults with mild sensorineural hearing impairment obtain benefit from hearing aids in terms of reduction of activity limitations and participation restrictions, improvement in health-related quality of life, and satisfaction with hearing aids? and (b) was there evidence for clinicians to recommend hearing aids for adults with mild hearing impairment? The criteria for studies to be included in the review were: at least Level 4 evidence based on Cox’s (2005) guidelines (see Table 27–1); participants must be 18 years or older with four-frequency average pure-tone threshold of equal to or less than 45 dB HL in the better ear; the intervention must be air-conduction hearing aids of any style; and outcome measures that assess activity limitation, participation restriction, satisfaction, or health-related quality of life.

Of the 972 references identified in the search, 10 studies from 1984 to 2012 with variable research designs and different outcome measures met the review inclusion criteria. One of the ten studies was at Level 2 of evidence, with the remaining at Level 3 or 4 (see Table 27–1). Participant age groups also varied, but were typically older than 45 years. The outcomes measures used in the 10 studies were self-report questionnaires used both pre- and postfitting or only postfitting. Measures of hearing aid benefit varied across studies and included the PHAP and APHAB, the Hearing Aid Performance Inventory and its shortened version (HAPI and SHAPI) (Walden et al. 1984), the short form of the HHIE (HHIE-S) (Ventry & Weinstein, 1983), the Satisfaction With Amplification in Daily Life (SADL) (Cox & Alexander, 1999), and some study-specific questionnaires.

All 10 studies included in the Johnson et al. (2016) systematic review reported significant benefit from hearing aids for adults with mild hearing impairment. Johnson and colleagues aimed to quantify this benefit and for this purpose conducted a meta-analysis. Five of the 10 studies reviewed provided sufficient data to calculate the effect size (or degree of benefit) for hearing aids. After correcting for possible publication bias, the meta-analysis revealed a small to medium effect size, providing evidence that adults with mild hearing impairment can benefit from using hearing aids.

Summary of Research

In summary, there is a small body of high-quality evidence to support hearing aids as an effective treatment for adults with mild hearing impairment.

In relation to the PICO question in this chapter, the patient could be advised that the evidence indicates that hearing aids are likely to improve his hearing and communication in everyday life compared to not wearing hearing aids.

New Directions in Hearing Aid Benefit Research

Applying EBP in the context of the PICO question about a patient with mild hearing impairment helped to identify a number of key areas for future research related to this important clinical population. Mild hearing impairment is the most prevalent degree of hearing impairment, and with the number of adults over 65 years tripling between 2010 and 2050, mild hearing impairment will increasingly affect older adults in the future (World Health Organization, 2011). Despite this, there is evidence that there is still some hesitance among clinicians to provide hearing aids for patients with mild hearing impairment (Timmer, Hickson, & Launer, 2015).

Clearly, there is a need for more research evidence regarding the benefit of hearing aids for adults, specifically for those with mild hearing impairment. In addition, with new and varied service delivery models emerging in audiology, it is vital that researchers provide extensive detail about the content and nature of the intervention itself. Many different approaches can be taken to the fitting of hearing aids, and it would therefore be useful to investigate which aspects of the fitting and follow-up process are most beneficial. For example, what are the effects of the interaction between patient and clinician, such as the giving of instructions and provision of counseling during the hearing aid fitting process? A study by Humes et al. (2017) compared two models of service delivery: audiology best-practice and a model in which the patient self-selected a preprogrammed hearing aid. The hearing aids provided for the two groups were identical. While reported hearing aid benefit was similar for the two groups, the study showed that the service provided by the clinician had an effect on the patient's self-reported hearing aid satisfaction and intention to purchase hearing aids after the trial. Further research is necessary to address the relative benefits of different models of clinician service and involvement in the rehabilitation process. In the case of adults with mild hearing impairment, the option of different service delivery models may also affect their help-seeking behavior.

While the clinician's model of service may affect hearing aid outcomes, so could other, nonaudiologic factors. Meyer et al. (2014) investigated this aspect with a group of 307 adults with hearing impairment, grouped as having consulted a hearing care professional or not, and being successful hearing aid users or not. The study showed that factors such as attitudinal beliefs (e.g., positive attitude to hearing aids and a belief in the ability to manage a hearing aid) and external cues to action (e.g., support from significant others) were significant factors in predicting help seeking and successful hearing aid use.

Building on the knowledge that motivation and a belief in the intervention can be a factor in the success of that intervention, more research using a "patient preference trial" design could be beneficial. This design is a type of clinical trial where participants are first asked if they want to be randomized or not. If the answer is yes, they are randomized to the treatment options. If the answer is no, they are then asked for their treatment preference and they proceed with that treatment. It is possible then to compare the randomized and the preferred groups and to measure the preference effect, which is the difference in outcome for those who chose a particular intervention compared to those who were randomly allocated to it (McPherson, Britton, & Wennberg, 1997; Wingham, Dalal, Sweeney, & Evans, 2006). Such an approach recognizes the relationship between outcomes and a person's motivation; that is, people who are more motivated to try a particular intervention are more likely to achieve better results. This is certainly the case with hearing aid fitting (Hickson et al., 1999; Ridgway, Hickson, & Lind, 2015).

New Approaches to Building an Evidence Base

In hearing aid research, outcomes are often measured with objective and subjective means. It would be beneficial to determine if the current measures and methodologies used to gather evidence about hearing aid outcomes are appropriate. Objective means such as real ear measures or other verification methods provide the clinician with a picture of how well the hearing aids are performing when in situ in the ear, relative to a variable such as target gain. Hearing aid benefit or satisfaction is most commonly measured in research as well as clinical practice by subjective means with the use of self-reporting questionnaires. In research, effect sizes (e.g., degree of benefit or

satisfaction) for audiologic rehabilitation interventions are typically in the low range. This could be either because the intervention options provide little benefit or there is a need for more appropriate outcome tools. Such new tools may be more sensitive to the effects of the treatment, particularly when they embrace the benefits of patient-centered care and include individualized measures, rather than the generalized situations described in current self-report questionnaires. A further drawback of current self-report questionnaires is that all such measures require input from the patient based on his or her memory and experience of certain listening situations. One approach gaining momentum in hearing research is the use of Ecological Momentary Assessment (EMA), which captures information from the patient in real time, in the patient's natural sound environment (Timmer, Hickson, & Launer, 2018b). Also called *experiential sampling*, EMA allows for the gathering of data about current listening experiences in listening situations that are relevant to the individual. For example, in one study, 29 older adults (mean age = 69 years) with mild hearing impairment completed a survey on a smartphone several times per day over 14 days (Timmer, Hickson, & Launer, 2017a). The survey contained questions about the listening activity, the acoustic environment and the participants' self-rated hearing performance for aspects such as speech understanding and listening effort. The 1,128 surveys collected showed that although these adults with mild hearing impairment reported good speech understanding in more than 90% of their listening situations, they were expending considerable effort to reach that performance in two-thirds of these situations.

A follow-up study by Timmer, Hickson, and Launer (2018a) with 10 adults with an average age of 70 years and mild hearing impairment showed that EMA could also measure hearing aid benefit. The participants filled in a similar survey to the previous study, 3 times per day, over 4 weeks, either wearing hearing aids (baseline and withdrawal phases) or not wearing hearing aids (intervention phase). A comparison of the survey answers from the baseline and intervention phases showed that speech understanding and listening enjoyment were improved, and listening effort and feeling hampered by hearing difficulties were reduced, when participants were wearing hearing aids.

The use of EMA has contributed to the evidence base relevant for this chapter's PICO question related to mild hearing impairment in adults and the benefits of providing hearing aids. In the future, it is envisaged that EMA will evolve further into a tool for patient-centered EBP. Sidani, Epstein, and Miranda (2006) describe a patient-centered evidence-based

approach, which consists of (a) identifying alternative treatment options on the basis of the best evidence available, (b) consulting with the patient to elicit his or her preference among the options, and (c) accounting for patient's preference in providing treatment (p. 118). As discussed earlier in this chapter, relaying research information and encouraging patients to express their treatment preferences is Step 4 in the five steps of applying EBP. This could be through the use of decision aids (Charles, Gafni, Whelan, & O'Brien, 2005; Laplante-Lévesque, Hickson, & Worrall, 2012). Alternatively, a clinical EMA tool could be a means of supporting a patient-centered EBP-approach. Such a tool may be a smartphone application (app) used by patients to monitor their own performance with and without hearing aids in different real-world listening environments and thus collect evidence of their individual outcomes. The data from this monitoring could then be relayed back to patients and clinicians so that they can make informed decisions.

Finally, it is worth remembering that audiologic rehabilitation is a clinical activity. It, therefore, seems particularly important that there is ongoing clinical research about the effectiveness of rehabilitation so that the evidence base continues to build. There are two major avenues for this to occur. First, clinicians could regularly measure the outcomes of audiologic rehabilitation they are conducting and report these at conferences and in journal articles for their peers. Randomized controlled trials of different interventions may not be clinically feasible, but lower levels of evidence from cohort studies and cross-sectional surveys are still extremely valuable. Simple outcome measures, such as the self-reported measures described in this chapter, could be readily used for this purpose. Second, researchers and clinicians could work together to collect and disseminate audiologic rehabilitation research. Such collaborations are beneficial to both groups. They afford the researchers the opportunity to understand the PICO questions of concern to patients and clinicians and, for clinicians, it can be an opportunity to gain practical experience about the research process itself and to learn what is needed for quality research to be undertaken. There are always more questions to be asked and we all have a role to play in finding the answers.

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